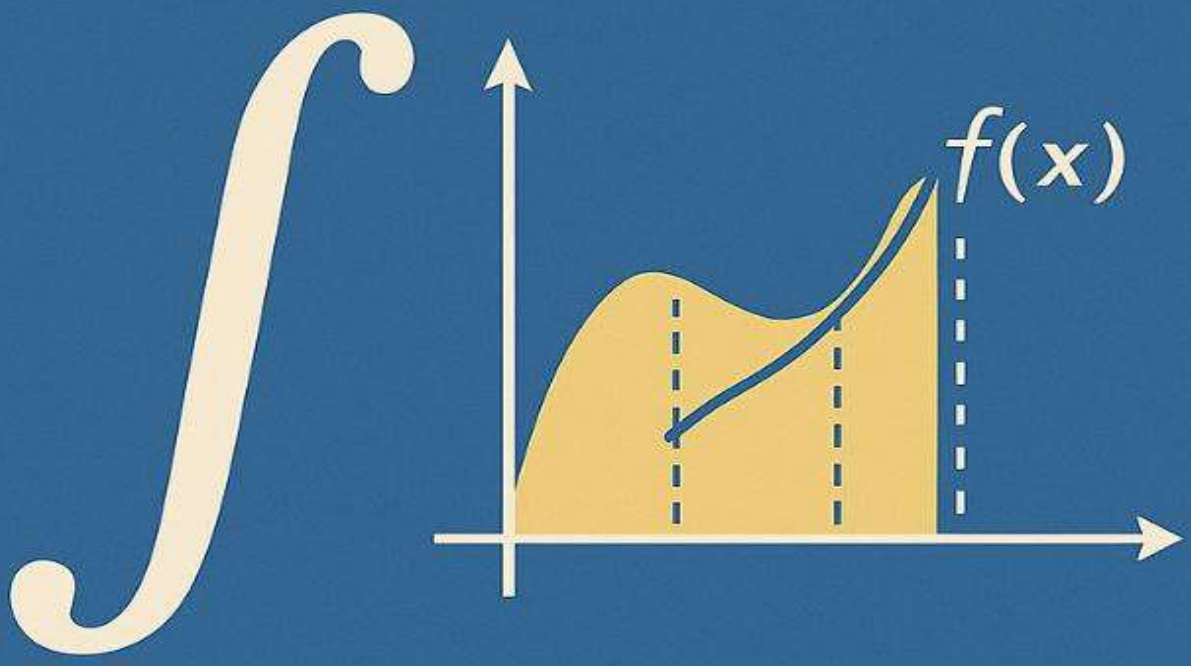
INTEGRAL BOOK



500 Exercises with Answer

រៀបរៀង និងនិពន្ធដោយ ជាត់ តារិន

មាតិកា

I. អារម្ភកថា

II. ព្រឹត្តិការប្រវត្តិសាស្ត្រនៃការគណនាអាំងតេក្រាល

III. ទំនាក់ទំនងសំខាន់ៗ និងរូបមន្តផ្សេងៗ

IV. លំហាត់ និងចម្លើយ៥០០លំហាត់

V. លំហាត់ស្រាវជ្រាវបន្ថែមនឹងចម្លើយ

អារម្ភកថា

សួស្តីប្រិយមិត្ត អ្នកសិក្សាទាំងអស់គ្នាជាទីស្នេហា។ សៀវភៅលំហាត់ INTEGRAL ដែលខ្ញុំបាទបានរៀបរៀងឡើងក្នុងគោលបំណងទុកជាឯកសារសម្រាប់ស្វ័យសិក្សា និងពង្រីកចំណេះដឹងបន្ថែមលើមុខវិទ្យាគណិតវិទ្យា ហើយត្រូវបានកែប្រែជាច្រើនលើកច្រើនសារមកហើយ។

កម្រងលំហាត់ដែលមាននៅក្នុងសៀវភៅនេះ ជាលំហាត់ដែលខ្ញុំបាទបានដកស្រង់នឹងប្រមូលចេញពីឯកសារខ្មែរ និងបរទេសជាច្រើនផងដែរ។ ម្យ៉ាងទៀតសូមប្រិយមិត្តអ្នកអានមេត្តាអធ្យាស្រ័យរាល់កំហុសខុសឆ្គងនានាដែលកើតឡើងដោយអចេតនា ហើយខ្ញុំបាទក៏សូមស្វាគមន៍រាល់មតិរិះគន់នានាក្នុងន័យស្ថាបនាពីបណ្តាសំណាក់លោកគ្រូ អ្នកគ្រូ និងមិត្តសិក្សាទាំងអស់ដោយក្តីរីករាយ។ មួយវិញទៀតនៅក្នុងការគណនាលំហាត់ INTEGRAL មានវិធីសាស្ត្រ ឬក្បួននៅក្នុងការដោះស្រាយជាច្រើន តែខ្ញុំបាទលើកយកវិធីសាស្ត្រខ្លះៗមកបង្ហាញតែប៉ុណ្ណោះ។

ជាចុងក្រោយ ខ្ញុំបាទសូមជូនពរដល់មិត្តអ្នកអានទាំងអស់គ្នាឲ្យមានសុខភាពល្អ ទទួលបានជោគជ័យក្នុងការសិក្សានិងសូមសម្រេចរាល់បំណងប្រាថ្នា ។

ព្រឹត្តិការប្រវត្តិសាស្ត្រនៃការគណនាអាំងតេក្រាល

១. ប្រវត្តិនៃការគណនាអាំងតេក្រាល

ការគណនាអាំងតេក្រាលជាវិធីសាស្ត្រដើម្បីរកផ្ទៃក្រឡានិងមាឌសូលីតដោយបំបែកជាបំណែកតូចៗដែលរកឃើញដោយតារាវិទូជនជាតិក្រិចបុរាណគឺលោក Eudoxus (ca . 370 BC) ហើយត្រូវបានពង្រីកបន្ថែមដោយលោក Archimedes (3rd century BC) និងជនជាតិចិនគឺលោក Liu Hui (3rd century AD) ដើម្បីគណនាផ្ទៃក្រឡាប៉ារ៉ាបូលនិងផ្ទៃក្រឡារង្វង់។ លុះដល់សតវត្សទី ៥ គណិត

វិទូជនជាតិចិនពីរូបគឺលោក Zu Geng និង Zu ChongZhi បានប្រើអាំងតេក្រាលនេះដើម្បីគណនាមាឌ របស់ស្វែរ ។

នៅសតវត្សទី ១៦ លោក Cavalieri បានរកឃើញនូវការគណនាអាំងតេក្រាល x^n ហើយលោក Wallis បានរកឃើញបន្ថែមលើស្វ័យគុណ n នៃ x ។ លុះនៅសតវត្សទី ១៧ គឺលោក Isaac Newton និងលោក Gottfried Leibniz ក៏បានស្រាយបំភ្លឺនូវទំនាក់ទំនងរវាងអាំងតេក្រាល និងដេរីវេហើយដាក់ បញ្ចូលដេរីវេជាប្រមាណវិធីច្រាសគ្នានឹងអាំងតេក្រាល (អាំងតេក្រាលមិនកំណត់) ។

២. ប្រវត្តិនៃសញ្ញាអាំងតេក្រាល

លោក Isaac Newton បានប្រើរបាបញ្ឈរនៅពីលើអថេរជាសញ្ញាអាំងតេក្រាលប៉ុន្តែសញ្ញានេះ ច្រឡំទៅនឹងសញ្ញាដេរីវេ $(x'ox)$ ក៏គេឈប់ប្រើសញ្ញានេះទៅ។ លុះមកដល់ឆ្នាំ ១៦៧៥ លោក Gottfried Leibniz បានប្រើសញ្ញា $\int$ តាងឱ្យអាំងតេក្រាលមិនកំណត់។ បន្ទាប់មកលោក Joseph Fourier ក៏បានរកឃើញ(បង្កើត)សញ្ញាអាំងតេក្រាលកំណត់ ។

សរសេរកំឡុងពេលរៀនសៅគរុកោសល្យ (14, 03, 2025)

មិនលើកទឹកចិត្តឱ្យស្រឡាំងភ្នែកនឹងស្នាដៃគ្រូទេ!!!

ទំនាក់ទំនងសំខាន់ៗ និងរូបមន្តផ្សេងៗ

$$\oplus\ e^u=\sum_{n=0}^{\infty}\left(\frac{x^n}{n!}\right)=1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\frac{x^4}{4!}+\cdots$$

$$\oplus\ \log(x)=\sum_{n=1}^{\infty}\left(\frac{(-1)^{n+1}x^n}{n}\right)=x-\frac{x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}+\cdots$$

$$\oplus\left(x+y\right)^z=\sum_{i=0}^{z}\binom{z}{i}x^{z-i}y^i=\binom{z}{0}x^z+\binom{z}{1}x^{z-1}y+\cdots+\binom{z}{z-1}x.y^{z-1}+\binom{z}{z}y^z$$

$$\oplus\ \frac{1}{1-x}=\sum_{n=0}^{\infty}\left(x^n\right)=1+x+x^2+x^3+\cdots\qquad ,|x|<1$$

$$\oplus\ \frac{1}{(1-x)^n}=\sum_{k=0}^{\infty}x^kC(n+k-1,k)=1+xC(n,1)+x^2C(n+1,2)+\cdots\qquad ,|x|<1$$

$$\oplus\ \log(1-x)=-\sum_{n=1}^{\infty}\left(\frac{x^n}{n}\right)=-x-\frac{x^2}{2}-\frac{x^3}{3}-\cdots$$

$$\oplus\ \log\left(\frac{1+x}{1-x}\right)=2\sum_{n=0}^{\infty}\left(\frac{x^{2n+1}}{2n+1}\right)=2\left(x+\frac{x^3}{3}+\frac{x^5}{5}+\frac{x^7}{7}+\cdots\right)$$

$$\oplus\ \log\big(\cos(x)\big)=\log\left(\frac{1}{2}\right)-\sum_{n=1}^{\infty}\left(\frac{(-1)^n}{n}\cos(2nx)\right)=\sum_{n=1}^{\infty}\left(\frac{(1-2^{2n})\mathrm{B}_{2n}}{2n(2n)!}\cdot x^{2n}\right)\quad ,|x|<\frac{\pi}{2}$$

$$\oplus\ \log\big(\sin(x)\big)=\log\left(\frac{1}{2}\right)-\sum_{n=1}^{\infty}\left(\frac{\cos(2nx)}{n}\right)=\sum_{n=1}^{\infty}\left(\frac{(1-2^{2n-1})\mathrm{B}_{2n}}{2n(2n)!}\left(x-\frac{\pi}{2}\right)^{2n}\right)\quad ,\ x\in(0,\pi)$$

$$\oplus\ \sin(x)=\sum_{n=0}^{\infty}\left(\frac{(-1)^n x^{2n+1}}{(2n+1)!}\right)=x-\frac{x^3}{3!}+\frac{x^5}{5!}-\frac{x^7}{7!}+\cdots$$

$$\oplus\ \cos(x)=\sum_{n=0}^{\infty}\left(\frac{(-1)^n x^{2n}}{(2n)!}\right)=1-\frac{x^2}{2!}+\frac{x^4}{4!}-\frac{x^6}{6!}+\cdots$$

$$\oplus\ \tan(x)=\sum_{n=1}^{\infty}\left(\frac{B_{2n}(-4)^n(1-4^n)x^{2n-1}}{(2n)!}\right)=x+\frac{x^3}{3}+\frac{2x^5}{15}+\cdots\quad ,|x|<\frac{\pi}{2}$$

$$\oplus\ \sec(x)=\sum_{n=1}^{\infty}\left(\frac{(-1)^n x^{2n}E_{2n}}{(2n)!}\right)=1+\frac{x^2}{2}+\frac{5x^4}{24}+\ldots\quad ,|x|<\frac{\pi}{2}$$

$$\oplus\ G=\sum_{n=0}^{\infty}\left(\frac{(-1)^n}{(2n+1)^2}\right)=1-\frac{1}{3^2}+\frac{1}{5^2}-\frac{1}{7^2}+\ldots\qquad \text{“}G\text{”}\ \textit{Catalan's constant}$$

$$\oplus\ \mathbf{B}_n=\sum_{n=0}^{\infty}\left(\frac{(-1)^n}{(2n+1)^n}\right)\qquad \text{“}\textit{Bernoulli numbers}\text{”}$$

$$\oplus W(x)=\sum_{n=1}^{\infty}\left(\frac{(-n)^{n-1}}{n!}.x^n\right)=x-x^2+\frac{3x^3}{2}-\frac{16x^4}{6}+\ldots\qquad \text{“}W\text{”}\ \text{Lambert function}$$

$$\oplus n!!=\begin{cases} n(n-2)...(6)(4)(2) & ,If:``n''even \\ n(n-2)...(5)(3)(1) & ,If:``n''odd \end{cases} \quad ;Ex:8!!=8\times6\times4\times2 \quad ;Ex:7!!=7\times5\times3\times1$$

$$\oplus\ \eta(s)=\sum_{n=1}^{\infty}\left(\frac{(-1)^{n-1}}{n^s}\right)=1-\frac{1}{2^s}+\frac{1}{3^s}-\frac{1}{4^s}+\cdots$$

$$\oplus\ \zeta(s)=\sum_{n=1}^{\infty}\left(\frac{1}{n^s}\right)=1+\frac{1}{2^s}+\frac{1}{3^s}+\frac{1}{4^s}+\cdots$$

$$\oplus\ \eta(s)=(1-2^{1-s})\zeta(s)$$

$$\oplus\Gamma(n)=\int_0^{\infty}x^{n-1}e^{-x}dx$$

$$\oplus\Gamma(n)\Gamma(1-n)=\pi\csc(n\pi)$$

$$\oplus\Gamma(n+1)=n\Gamma(n)$$

$$\oplus\ \psi(1+x)=\psi(x)+\frac{1}{x}$$

$$\oplus\psi(x)=\frac{\Gamma'(x)}{\Gamma(x)}=-\gamma+H_{(x-1)}$$

$$\oplus\Gamma'(x+1)=x!\left(-\gamma+H_{(x)}\right)$$

$$\oplus Li_2(1)=\zeta(2)=\frac{\pi^2}{6}$$

$$\oplus Li_s(n)+Li_s(-n)=2^{1-s}Li_s(n^2)$$

$$\oplus\Gamma(-n)=-\frac{1}{n}\Gamma(-n+1)$$

$$\oplus n!!=\prod_{k=1}^{\frac{n}{2}}(2k)=n(n-2)(n-4)...(4)(2)$$

$$\oplus n!!=\prod_{k=1}^{\frac{n+1}{2}}(2k-1)=n(n-2)(n-4)...(3)(1)$$

$$\oplus\int_a^b f'(x)dx=f(b)-f(a)$$

$$\oplus\int_a^b f(x)dx=\int_a^b f(b+a-x)dx$$

$$\oplus\int_0^{2a}f(x)dx=\int_0^a\left(f(x)+f(2a-x)\right)dx$$

$$\oplus\int_{-a}^{a}f(x)dx=\int_0^a\left(f(x)+f(-x)\right)dx$$

$$\oplus\Gamma(s)\zeta(s)=\int_0^{+\infty}\frac{x^{s-1}}{e^x-1}dx$$

$$\oplus\int_0^{+\infty}x^ae^{-bx}dx=\frac{\Gamma(a+1)}{b^{a+1}}$$

$$\oplus\int_0^{+\infty}e^{-x^2}dx=\Gamma\left(\frac{1}{2}\right)=\left(-\frac{1}{2}\right)!=\sqrt{\pi}$$

$$\oplus\ \varphi=\frac{1+\sqrt{5}}{2}$$

$$\oplus\Gamma\left(\frac{1}{2}-n\right)=\frac{(-2)^n}{(2n-1)!!}\sqrt{\pi}$$

$$\oplus\Gamma\left(\frac{1}{2}+n\right)=\frac{(2n-1)!!}{2^n}\sqrt{\pi}$$

$\oplus \Gamma(n+1)=n!$

$\oplus \Gamma(n)\Gamma\left(\frac{1}{2}+n\right)=2^{1-2n}\sqrt{\pi}\Gamma(2n)$

$\oplus B(x,y)=B(y,x)=\frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}$

$\oplus B(y,z)=\int_0^1 \frac{x^{y-1}+x^{z-1}}{(1+x)^{y+z}}dx$

$\oplus B(x,y)=\int_0^1 z^{x-1}(1-z)^{y-1}dz=2\int_0^{\frac{\pi}{2}}\sin^{2x-1}(z)\cos^{2y-1}(z)\,dz=\int_0^{\infty}\frac{z^{x-1}}{(1+z)^{x+y}}dz=\int_0^{\infty}\frac{z^{y-1}}{(1+z)^{x+y}}dz$

$\oplus \int_0^{n\pi}\left|\sin(x)\right|dx=\int_0^{n\pi}\left|\cos(x)\right|dx=n\int_0^{\pi}\left|\sin(x)\right|dx=n\int_0^{\pi}\left|\cos(x)\right|dx=2n \quad ,\forall n\in\mathbb{N}$

$$\oplus \int_0^{\frac{\pi}{2}}\sin^n(x)dx=\int_0^{\frac{\pi}{2}}\cos^n(x)dx=\begin{cases}\frac{\pi}{2}\times\frac{(n-1)!!}{n!!} & ,If:``n''even \quad ;Ex:6!!=6\times4\times2\\ \frac{(n-1)!!}{n!!} & ,If:``n''odd \quad ;Ex:5!!=5\times3\times1\end{cases}$$

$\oplus \{x\}=x-\lfloor x\rfloor \quad ,\{x\}:is\,the\,fraction\,part\,function$

$\oplus \cot^{-1}(x)=\tan\left(\frac{1}{x}\right)$

$\oplus \tan^{-1}(x)=\cot\left(\frac{1}{x}\right)$

$\oplus \tan^{-1}(x)+\cot^{-1}(x)=\frac{\pi}{2} \quad ,x>0$

$\oplus \tan^{-1}(x)+\cot^{-1}(x)=-\frac{\pi}{2} \quad ,x<0$

$\oplus \sin^{-1}(x)+\cos^{-1}(x)=\frac{\pi}{2}$

$\oplus \sin^{-1}(-x)=-\sin^{-1}(x)$

$\oplus \sin(-x)=-\sin(x)$

$\oplus \sin^{-1}(-x)=-\sin^{-1}(x)$

$\oplus \cos^{-1}(-x)=\pi-\cos^{-1}(x)$

$\oplus \cos(-x)=\cos(x)$

$\oplus \tan^{-1}(-x)=-\tan^{-1}(x)$

$\oplus \tan(-x)=-\tan(x)$

$\oplus \sec(x)=\frac{1}{\cos(x)}$

$\oplus \csc(x)=\frac{1}{\sin(x)}$

$\oplus If:t=\tan\left(\frac{x}{2}\right) \;;\; That:\ \sin(x)=\frac{2t}{1+t^2} \quad And \quad \cos(x)=\frac{1-t^2}{1+t^2}$

$\oplus \lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{mn} f\left(\frac{k}{n}\right) = \int_0^m f(x)dx$ $\oplus \lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{(m-1)n} f\left(\frac{k}{n}\right) = \int_0^{m-1} f(x)dx$

$\oplus \int_0^{\infty} \frac{\sin(x)}{x} f(x)\,dx = \int_0^{\frac{\pi}{2}} f(x)dx \;, If: f(x) = f(\pi \pm x)$

$\otimes Min\{x,y\} = \begin{cases} x & When \;\; x \le y \\ y & When \;\; y \le x \end{cases}$ On the space (a,b)

$\otimes Max\{x,y\} = \begin{cases} x & When \;\; x \ge y \\ y & When \;\; y \ge x \end{cases}$ On the space (a ,b)

$\oplus\; Li_n(z) = \sum_{k=1}^{\infty} \left(\frac{z^k}{k^z}\right) \quad , x < 1$ " **Polylogarithm Function** "

$\oplus\; Ti_n(z) = \sum_{k=1}^{\infty} \left(\frac{(-1)^k z^{2k+1}}{(2k+1)^{n+1}}\right) \quad , |x| \le 1$ " **Inverse Tangent Integral Function** "

$\oplus\; \Phi(z,s,q) = \sum_{k=1}^{\infty} \left(\frac{z^k}{(k+q)^s}\right) \quad , \quad |z| < 1 \; ; \; \mathrm{Re}(s) > 1 \; ; \; q > 0$ " **Lerch Transcendent** "

លំហាត់ និងចម្លើយ៥០០លំហាត់

001. Calculate Integral $I=\int_{-1}^{0}\frac{x(x+2)}{\log(x+1)}dx$

Solution; $I=\int_{-1}^{0}\frac{x(x+2)}{\log(x+1)}dx$

$$=\int_{-1}^{0}\frac{(x+1)^2-1}{\log(x+1)}dx$$

Let : $u=x+1\Rightarrow du=dx$ **; If:** $x\in(-1,0)\Rightarrow u\in(0,1)$

That : $I=\int_{0}^{1}\frac{u^2-1}{\log(u)}du$

$$\Rightarrow I(t)=\int_{0}^{1}\frac{u^t-1}{\log(u)}dx$$

$$\Leftrightarrow I'(t)=\int_{0}^{1}\frac{\partial}{\partial t}\left(\frac{u^t-1}{\log(u)}\right)du$$

Note : $\frac{d}{dt}(u^t)=u^t\log(u)$

$$=\int_{0}^{1}u^t du=\frac{1}{t+1}$$

Equivalent : $I(t)=\log(t+1)+C$

If : $t=0\Rightarrow I(0)=0=\log(0+1)+C\Rightarrow C=0$

If : $t=2\Rightarrow I(2)=I=\log(2+1)+C\equiv I=\log(3)$

SO, $\boxed{\int_{-1}^{0}\frac{x(x+2)}{\log(x+1)}dx=\log(3)}$

002. Calculate Integral $I=\int_{-\infty}^{+\infty}\frac{\cos(x)}{x^2+1}dx$

Solution; $I=\int_{-1}^{0}\frac{x(x+2)}{\log(x+1)}dx$

They got it $I(t)=\int_{-\infty}^{+\infty}\frac{\cos(tx)}{x^2+1}dx$

That $I'(t)=\int_{-\infty}^{+\infty}\frac{\partial}{\partial t}\left(\frac{\cos(tx)}{x^2+1}\right)dx$

$$=\int_{-\infty}^{+\infty}\left(\frac{x\sin(tx)}{x^2+1}\right)dx$$

$$=\int_{-\infty}^{+\infty}\left(\frac{x^2\sin(tx)}{x(x^2+1)}\right)dx$$

$$=-\int_{-\infty}^{+\infty}\left(\frac{\sin(tx)}{x}\right)dx+\int_{-\infty}^{+\infty}\left(\frac{\sin(tx)}{x(x^2+1)}\right)dx$$

$$=-\pi+\int_{-\infty}^{+\infty}\left(\frac{\sin(tx)}{x(x^2+1)}\right)dx \quad (*)$$

Then: $I''(t)=\int_{-\infty}^{+\infty}\frac{\partial}{\partial t}\left(\frac{\sin(tx)}{x(x^2+1)}\right)dx$

$$=\int_{-\infty}^{+\infty}\left(\frac{\cos(tx)}{x^2+1}\right)dx$$

$$=J(t)$$

Equivalent $I''(t)-J(t)=0$ **(According to the differential equation)**

They got it $I(t)=me^{t}+ne^{-t}$

Equivalent $I'(t)=me^{t}-ne^{-t}$

But : $I(0)=\int_{-\infty}^{+\infty}\frac{1}{x^2+1}dx=\pi=m+n$ (1)

Take (*) **Then** $I'(0)=-\pi=m-n$ (2)

Take : (1) & (2) : $\begin{cases}\pi=m+n\\-\pi=m-n\end{cases}$ **That** $m=0,n=\pi$

That : $I(t)=0+\pi e^{-t}$ **; But :** $I(1)=I$ **They got it :** $I(1)=I=\pi e^{-1}$

SO, $\boxed{\int_{-\infty}^{+\infty}\frac{\cos(x)}{x^2+1}dx=\frac{\pi}{e}}$

003. Calculate Integral $I=\int_{-45^\circ}^{+45^\circ}\frac{\tan^2(x)}{1+2030^x}dx$

Solution; $I=\int_{-45^\circ}^{+45^\circ}\frac{\tan^2(x)}{1+2030^x}dx \quad (1)$

$$=\int_{-45^\circ}^{+45^\circ}\frac{\tan^2(-x)}{1+2030^{(-x)}}dx$$

Because; $\int_a^b f(x)\,dx=\int_a^b f(a+b-x)\,dx$

$$=\int_{-45^\circ}^{+45^\circ}\frac{2030^x\tan^2(x)}{1+2030^x}dx \quad (2)$$

Take : $(1)+(2)$ **: They have** $2I=\int_{-45^\circ}^{+45^\circ}\frac{\tan^2(x)}{1+2030^x}dx+\int_{-45^\circ}^{+45^\circ}\frac{2030^x\tan^2(x)}{1+2030^x}dx$

Equivalent

$$I=\frac{1}{2}\int_{-45^\circ}^{+45^\circ}\tan^2(x)dx$$

$$=\frac{1}{2}\int_{-45^\circ}^{+45^\circ}\left(1+\tan^2(x)-1\right)dx$$

$$=\frac{1}{2}\int_{-45^\circ}^{+45^\circ}\left[\left(\tan(x)\right)'-(x)'\right]dx$$

Note : $\int_a^b f'(x)dx=f(x)\Big|_a^b$

$$=\frac{1}{2}\left[\tan(x)-x\right]_{-45^\circ}^{+45^\circ}=\frac{4-\pi}{4}$$

SO, $\boxed{\int_{-45^\circ}^{+45^\circ}\frac{\tan^2(x)}{1+2030^x}dx=\frac{4-\pi}{4}}$

004. Calculate Integral $I=\int_{-\infty}^{+\infty}\left(\frac{\sin(x)}{x}\right)dx$

Solution; $I=\underbrace{\int_{-\infty}^{+\infty}\frac{\sin(x)}{x}dx}_{is\ an\ even\ function}$

That : $I=2\int_0^{+\infty}\frac{\sin(x)}{x}dx$

$$\Rightarrow I(t)=2\int_0^{+\infty}\frac{\sin(x)e^{-tx}}{x}dx$$

$$\Leftrightarrow I'(t)=2\int_0^{+\infty}\frac{\partial}{\partial t}\left(\frac{\sin(x)e^{-tx}}{x}\right)dx$$

$$=-2\int_0^{+\infty}e^{-tx}\sin(x)dx$$

$$= -2\left[\frac{t\sin(x)+\cos(x)}{t^2+1}e^{-tx}\right]_0^{+\infty}$$

$$= -2\left(\frac{1}{t^2+1}\right)$$

Equivalent : $I(t) = -2\arctan(t) + C$

If : $t = +\infty \Rightarrow I(+\infty) = 0 = -2\arctan(+\infty) + C \equiv C = \pi$

If : $t = 0 \Rightarrow I(0) = I = 0 + C \equiv I = \pi$

$SO,$ $\boxed{\int_{-\infty}^{+\infty}\frac{\sin(x)}{x}dx = \pi}$

005. Calculate Integral $I = \int_0^1 \frac{x^n - 1}{\log(x)}dx \quad , n \geq 0$

Solution; $I = \int_0^1 \frac{x^n - 1}{\log(x)}dx \quad , n \geq 0$

That $I(t) = \int_0^1 \frac{x^t - 1}{\log(x)}dx$

$$\Rightarrow I'(t) = \int_0^1 \frac{\partial}{\partial t}\left(\frac{x^t - 1}{\log(x)}\right)dx$$

$$= \int_0^1 \left(\frac{x^t \log(x)}{\log(x)}\right)dx$$

$$= \frac{1}{t+1}$$

Equivalent $I(t) = \log(t+1) + C$

If : $I(0) = 0 = \log(0+1) + C \Rightarrow C = 0$

If : $I(n) = I = \log(n+1)$

$SO,$ $\boxed{\int_0^1 \frac{x^n - 1}{\log(x)}dx = \log(n+1)}$

006. Calculate Integral $I=\frac{1}{2\pi}\int_0^{2\pi}\frac{1}{n+\cos(x)}dx$

Solution; $I=\frac{1}{2\pi}\int_0^{2\pi}\frac{1}{n+\cos(x)}dx$

$$=\frac{2}{2\pi}\int_0^{\pi}\frac{1}{n+\cos(x)}dx$$

$$=\frac{1}{\pi}\int_0^{\pi}\frac{1}{n+\cos(x)}dx$$

$$Use:\begin{cases} f(2\pi-x)=f(x) \\ \int_0^{2a} f(x)dx=2\int_0^{a} f(x)dx \end{cases}$$

Let : $t=\tan\left(\frac{x}{2}\right)\Rightarrow dx=\frac{2}{1+t^2}dt$ **; If :** $x\in(0,\pi)\Rightarrow t\in(0,\infty)$

Note: $\cos(x)=\frac{1-t}{1+t}$

That $I=\frac{1}{\pi}\int_0^{\infty}\left(\frac{1}{n+\frac{1-t^2}{1+t^2}}\right)\left(\frac{2}{1+t^2}\right)dt$

$$=\frac{2}{\pi}\int_0^{\infty}\frac{1}{n(1+t^2)+1-t^2}dt$$

$$=\frac{2}{\pi(n-1)}\int_0^{\infty}\frac{1}{\left(\frac{n+1}{n-1}\right)+t^2}dt$$

$$=\frac{2}{\pi(n-1)}\cdot\sqrt{\frac{n-1}{n+1}}\left[\arctan\left(t\sqrt{\frac{n-1}{n+1}}\right)\right]_0^{\infty}$$

$$=\frac{2}{\pi}\cdot\frac{1}{\sqrt{n^2-1}}\cdot\left[\arctan(\infty)-\arctan(0)\right]$$

$$=\frac{1}{\sqrt{n^2-1}}$$

SO, $\boxed{\frac{1}{2\pi}\int_0^{2\pi}\frac{1}{n+\cos(x)}dx=\frac{1}{\sqrt{n^2-1}}}$

007. Calculate Integral $I=\frac{1}{2\pi}\int_0^{2\pi}\frac{1}{n+\cos(x)}dx$

Solution; $I=\int_0^{+\infty}\frac{1}{x^4+x^2+1}dx \quad (*)$

Let : $x=\frac{1}{t}\Rightarrow dx=-\frac{1}{t^2}dt$ **; If :** $x\in(0,+\infty)\Rightarrow t\in(+\infty,0)$

That $I=-\int_0^{+\infty}\frac{1}{\frac{1}{t^4}+\frac{1}{t^2}+1}\times\frac{1}{t^2}dx=\int_0^{+\infty}\frac{t^2}{t^4+t^2+1}dx \quad (**)$

Take $(*)+(**)$ **They Have :** $2I=\int_0^{+\infty}\frac{t^2+1}{t^4+t^2+1}dx$

Note : $\int_a^b f(x)dx=\int_a^b f(t)dt$

$$=\int_0^{+\infty}\frac{1+\frac{1}{t^2}}{\left(t-\frac{1}{t}\right)+3}dt=\int_0^{+\infty}\frac{\left(t-\frac{1}{t}\right)'}{\left(t-\frac{1}{t}\right)^2+3}dt$$

$$=\frac{1}{\sqrt{3}}\left[\arctan\left(\frac{t-t^{-1}}{\sqrt{3}}\right)\right]_0^{+\infty}=\frac{\pi}{\sqrt{3}}$$

$$\Rightarrow I=\frac{\pi}{2\sqrt{3}}$$

SO, $\boxed{\int_0^{+\infty}\frac{1}{x^4+x^2+1}dx=\frac{\pi}{2\sqrt{3}}}$

008. Calculate Integral $I=\int_0^{\frac{\pi}{2}}\log\left(\sin(x)\right)dx$

Solution; $I=\int_0^{\frac{\pi}{2}}\log\left(\sin(x)\right)dx \quad (*)$

That $I=\int_0^{\frac{\pi}{2}}\log\left(\sin(\frac{\pi}{2}-x)\right)dx$

Note : $\int_0^a f(x)\,dx=\int_0^a f(a-x)\,dx$

$$=\int_0^{\frac{\pi}{2}}\log\left(\cos(x)\right)dx \quad (**)$$

Take $(*)+(**)$ **They Have :** $2I=\int_0^{\frac{\pi}{2}}\log\left(\sin(x)\cos(x)\right)dx$

$$= \int_0^{\frac{\pi}{2}} \log\left(\frac{1}{2}\sin(2x)\right)dx$$

$$= \int_0^{\frac{\pi}{2}} \log\left(\sin(2x)\right)dx - \frac{\pi}{2}\log(2)$$

$$2J + \frac{\pi}{2}\log(2) = \int_0^{\frac{\pi}{2}} \log\left(\sin(2x)\right)dx$$

Let : $t = 2x \Rightarrow dx = \frac{1}{2}dt$, **If** : $x \in \left(0, \frac{\pi}{2}\right) \Rightarrow t \in (0, \pi)$

Equivalent; $2J + \frac{\pi}{2} = \frac{1}{2}\int_0^{\pi} \log\left(\sin(t)\right)dt$

$$= \frac{1}{2} \cdot 2\int_0^{\frac{\pi}{2}} \log\left(\sin(t)\right)dt \qquad \textbf{Use:} \begin{cases} f(2a-t) = f(t) \\ \int_0^{2a} f(t)\,dt = 2\int_0^{a} f(t)\,dt \end{cases}$$

$$2J + \frac{\pi}{2}\log(2) = J \Rightarrow J = -\frac{\pi}{2}\log(2)$$

$SO,$ $\boxed{\int_0^{\frac{\pi}{2}} \log\left(\sin(x)\right)dx = \int_0^{\frac{\pi}{2}} \log\left(\cos(x)\right)dx = -\frac{\pi}{2}\log(2)}$

009. Calculate Integral $I = \int_0^{+\infty} \frac{\log(x^2+1)}{(x^2+1)}dx$

Solution; $I = \int_0^{+\infty} \frac{\log(x^2+1)}{(x^2+1)}dx$

Let : $x = \tan(u) \Rightarrow dx = \left(\tan^2(u)+1\right)du$, **If** : $x \in (0, +\infty) \Rightarrow t \in \left(0, \frac{\pi}{2}\right)$

That $I = \int_0^{\frac{\pi}{2}} \frac{\log\left(\tan^2(u)+1\right)}{\left(\tan^2(u)+1\right)} \cdot \left(\tan^2(u)+1\right)du$

$$= \int_0^{\frac{\pi}{2}} \log\left(\tan^2(x)+1\right)du = -2\int_0^{\frac{\pi}{2}} \log\left(\cos(u)\right)du = -2\left(-\frac{\pi}{2}\log(2)\right) = \pi\log(2)$$

$SO,$ $\boxed{\int_0^{+\infty} \frac{\log(x^2+1)}{(x^2+1)}dx = \pi\log(2)}$

010. Calculate Integral $I=\int_0^1 \frac{\log(x+1)}{(x^2+1)}du$

Solution; $I=\int_0^1 \frac{\log(x+1)}{(x^2+1)}du$

Let: $x=\tan(u)\Rightarrow dx=(\tan^2(u)+1)du$ **; If :** $x\in(0,1)\Rightarrow\in\left(0,\frac{\pi}{4}\right)$

That $I=\int_0^{\frac{\pi}{4}} \frac{\log(\tan(u)+1)}{(\tan^2(u)+1)}.(\tan^2(u)+1)dx$

$$=\int_0^{\frac{\pi}{4}}\log(\tan(u)+1)du=\int_0^{\frac{\pi}{4}}\log\left(\tan\left(\frac{\pi}{4}-u\right)+1\right)du$$

$$=\int_0^{\frac{\pi}{4}}\log\left(\frac{1-\tan(u)}{1+\tan(u)}+1\right)du=\int_0^{\frac{\pi}{4}}\log\left(\frac{1-\tan(u)+1+\tan(u)}{1+\tan(u)}\right)du$$

$$=\int_0^{\frac{\pi}{4}}\log\left(\frac{2}{1+\tan(u)}\right)du=\int_0^{\frac{\pi}{4}}\log(2)du-\int_0^{\frac{\pi}{4}}\log(\tan(u)+1)du$$

Equivalent; $I=\frac{\pi}{4}\log(2)-J\Rightarrow J=\frac{\pi}{8}\log(2)$

SO, $\boxed{\int_0^1 \frac{\log(x+1)}{(x^2+1)}dx=\frac{\pi}{8}\log(2)}$

011. Calculate Integral $I=\int_0^{\frac{\pi}{2}}\frac{x}{\tan(x)}dx$

Method 1

Solution; $I=\int_0^{\frac{\pi}{2}}\frac{x}{\tan(x)}dx$

$$=\int_0^{\frac{\pi}{2}}\frac{\arctan(\tan(x))}{\tan(x)}dx$$

That $I(a)=\int_0^{\frac{\pi}{2}}\frac{\arctan(a\tan(x))}{\tan(x)}dx$

$$\Rightarrow I'(a)=\int_0^{\frac{\pi}{2}}\frac{\partial}{\partial a}\left(\frac{\arctan\left(a\tan(x)\right)}{\tan(x)}\right)dx$$

$$=\int_0^{\frac{\pi}{2}}\frac{\tan(x)}{\tan(x)\left[1+\left(a\tan(x)\right)^2\right]}dx$$

$$=\int_0^{\frac{\pi}{2}}\frac{1}{\left(1+\tan^2(x)\right)\left[1+\left(a\tan(x)\right)^2\right]}d\left(\tan(x)\right)$$

$$=\int_0^{\frac{\pi}{2}}\left(\frac{\frac{a^2}{a^2-1}}{1+\left(a\tan(x)\right)^2}-\frac{\frac{1}{a^2-1}}{1+\tan^2(x)}\right)d\left(\tan(x)\right)$$

$$=\frac{1}{a^2+1}\left[a\left(\arctan\left(a\tan(x)\right)\right)-\arctan\left(\tan(x)\right)\right]_0^{\frac{\pi}{2}}$$

$$=\frac{1}{a^2+1}\left(a\frac{\pi}{2}-\frac{\pi}{2}\right)=\frac{\pi}{2}\times\frac{1}{a+1}$$

That $I(a)=\frac{\pi}{2}\log(a+1)+C$

If : $a=0\Rightarrow I(0)=0=C$ **That** $C=0$

If : $a=1\Rightarrow I(1)=I=\frac{\pi}{2}\log(2)$

Method 2

Solution; $I=\int_0^{\frac{\pi}{2}}\frac{x}{\tan(x)}dx$

$Let: u=x\Rightarrow du=dx\ , dv=\int\frac{1}{\tan(x)}dx=\log\left|\sin(x)\right|dx$

They got it $I=x\log\left|\sin(x)\right|\Big|_0^{\frac{\pi}{2}}-\int_0^{\frac{\pi}{2}}\log\left|\sin(x)\right|dx$

$$=0-\left(-\frac{\pi}{2}\log(2)\right)=\frac{\pi}{2}\log(2)$$

$SO,$ $\boxed{\int_0^{\frac{\pi}{2}}\frac{x}{\tan(x)}dx=\frac{\pi}{2}\log(2)}$

012. Calculate Integral $I=\int_0^{2\pi}\left(\left|\sin(x)\right|+\left|\cos(x)\right|\right)dx$

Method 1

Solution; $I=\int_0^{2\pi}\left(\left|\sin(x)\right|+\left|\cos(x)\right|\right)dx$

$$=\int_0^{2\pi}\left(\left|\sin(x)\right|\right)dx+\int_0^{2\pi}\left(\left|\cos(x)\right|\right)dx$$

For $\int_0^{2\pi}\left|\sin(x)\right|dx=\int_0^{\pi}\sin(x)\,dx-\int_{\pi}^{2\pi}\sin(x)\,dx$

$$=-\left[\cos(x)\right]_0^{\pi}+\left[\cos(x)\right]_{\pi}^{2\pi}$$

$$=-(-1-0)+(1-0)=4$$

For $\int_0^{2\pi}\left|\cos(x)\right|dx=\int_0^{\frac{\pi}{2}}\cos(x)\,dx-\int_{\frac{\pi}{2}}^{\pi}\cos(x)\,dx-\int_{\pi}^{\frac{3\pi}{2}}\cos(x)\,dx+\int_{\frac{3\pi}{2}}^{2\pi}\cos(x)\,dx$

$$=\left[\sin(x)\right]_0^{\frac{\pi}{2}}-\left[\sin(x)\right]_{\frac{\pi}{2}}^{\pi}-\left[\sin n(x)\right]_{\pi}^{\frac{3\pi}{2}}+\left[\sin(x)\right]_{\frac{3\pi}{2}}^{2\pi}$$

$$=(1-0)-(0-1)-(-1-0)+(0+1)=4$$

In short $I=4+4=8$

Method 2

Solution; $I=\int_0^{2\pi}\left(\left|\sin(x)\right|+\left|\cos(x)\right|\right)dx$

$$=\int_0^{2\pi}\left(\left|\sin(x)\right|\right)dx+\int_0^{2\pi}\left(\left|\cos(x)\right|\right)dx$$

For $\int_0^{2\pi}\left|\sin(x)\right|dx=\ 2\times 2=4$

For $\int_0^{2\pi}\left|\cos(x)\right|dx=2\times 2=4$

Note: $\begin{cases}\int_0^{n\pi}\left|\sin(x)\right|dx=n\int_0^{\pi}\left|\sin(x)\right|dx=2n\\ \int_0^{n\pi}\left|\cos(x)\right|dx=n\int_0^{\pi}\left|\cos(x)\right|dx=2n\end{cases}$

In short $I=4+4=8$

SO, $\boxed{\int_0^{2\pi}\left(\left|\sin(x)\right|+\left|\cos(x)\right|\right)dx=8}$

013. Calculate Integral $I=\int_{-1}^{+1}x\arcsin(x)\arccos(x)dx$

Solution; $I=\int_{-1}^{+1} x\arcsin(x)\arccos(x)dx$

$$=\int_{-1}^{+1}(-x)\arcsin(-x)\arccos(-x)dx$$

Note: $\begin{cases}\arcsin(-x)=-\arcsin(x)\\ \arccos(-x)=\pi-\arccos(x)\end{cases}$

They got it $I=\int_{-1}^{+1} x\arcsin(x)\big(\pi-\arccos(x)\big)dx$

$$=\pi\int_{-1}^{+1} x\arcsin(x)dx-\underbrace{\int_{-1}^{+1} x\arcsin(x)\arccos(x)dx}_{I}$$

That $I=\frac{\pi}{2}\int_{-1}^{+1} x\arcsin(x)dx=\pi\underbrace{\int_{0}^{1} x\arcsin(x)dx}_{\text{is an even function}}$

Let: $u=\arcsin(x)\Rightarrow du=\frac{1}{\sqrt{1-x^2}}dx$, $dv=x\Rightarrow v=\frac{x^2}{2}$

They got it $I=\frac{\pi}{2}\left[x^2\arcsin(x)\Big|_0^1-\int_0^1\frac{x^2}{\sqrt{1-x^2}}dx\right]$

$$=\frac{\pi}{2}\left(\frac{\pi}{2}+\int_0^1\frac{1-x^2-1}{\sqrt{1-x^2}}dx\right)=\frac{\pi}{2}\left(\frac{\pi}{2}+\int_0^1\sqrt{1-x^2}dx-\int_0^1\frac{1}{\sqrt{1-x^2}}dx\right)$$

$$=\frac{\pi}{2}\left(\frac{\pi}{2}+\int_0^1\sqrt{1-x^2}dx-\arcsin(x)\Big|_0^1\right)=\frac{\pi}{2}\int_0^1\sqrt{1-x^2}dx$$

Let : $x=\sin(t)\Rightarrow dx=\cos(t)dt$ **; If :** $x\in(0,1)\Rightarrow t\in\left(0,\frac{\pi}{2}\right)$

That $I=\frac{\pi}{2}\int_0^{\frac{\pi}{2}}\sqrt{1-\sin^2(t)}\cos(t)\,dt$

$$=\frac{\pi}{2}\int_0^{\frac{\pi}{2}}\big|\cos(t)\big|\cos(t)\,dt \quad \textbf{Which } \forall t\in\left(0,\frac{\pi}{2}\right)\Rightarrow\big|\cos(t)\big|=\cos(x)$$

They got it $I=\int_0^{\frac{\pi}{2}}\cos^2(t)dt=\frac{\pi}{4}$

$SO,$ $\boxed{\int_{-1}^{+1} x\arcsin(x)\arccos(x)dx=\frac{\pi^2}{8}}$

014. Calculate Integral $I=\int_0^{\frac{\pi}{4}}\frac{\log\left(\sqrt{1-\tan^2(x)}\right)}{\sqrt{1-\tan^2(x)}}\left(1+\tan^2(x)\right)dx$

Solution; $I=\int_0^{\frac{\pi}{4}}\frac{\log\left(\sqrt{1-\tan^2(x)}\right)}{\sqrt{1-\tan^2(x)}}\left(1+\tan^2(x)\right)dx$

Let : $\tan(x)=\sin(t)\Rightarrow\left(1+\tan^2(x)\right)dx=\cos(t)dt$ **; If** : $x\in\left(0,\frac{\pi}{4}\right)\Rightarrow t\in\left(0,\frac{\pi}{2}\right)$

They got it $I=\int_0^{\frac{\pi}{2}}\frac{\log\left(\sqrt{1-\sin^2(t)}\right)}{\sqrt{1-\sin^2(t)}}\cdot\cos(t)dt$

$$=\int_0^{\frac{\pi}{2}}\frac{\log\left(\sqrt{1-\sin^2(t)}\right)}{\left|\cos(t)\right|}\cdot\cos(t)dt \quad \textbf{Which } \forall t\in\left(0,\frac{\pi}{2}\right)\Rightarrow\left|\cos(t)\right|=\cos(x)$$

$$=\int_0^{\frac{\pi}{2}}\log\left(\cos(t)\right)dt=-\frac{\pi}{2}\log(2)$$

SO, $\boxed{\int_0^{\frac{\pi}{4}}\frac{\log\left(\sqrt{1-\tan^2(x)}\right)}{\sqrt{1-\tan^2(x)}}\left(1+\tan^2(x)\right)=-\frac{\pi}{2}\log(2)}$

015. Calculate Integral $I=\int_0^1(1-x)^5\,dx$

Solution; $I=\int_0^1(1-x)^5\,dx$

$$=\int_0^1\left(\sum_{k=0}^{5}C_5^k(1)^{5-k}(-x)^k\right)dx$$

Because: $(a+b)^n=\sum_{k=0}^{n}C_n^k(a)^{n-k}(b)^k$

$$=\sum_{k=0}^{5}\left((-1)^kC_5^k\int_0^1(x)^k\,dx\right)$$

$$=\sum_{k=0}^{5}\frac{(-1)^kC_5^k}{(k+1)}=\frac{1}{6}$$

SO, $\boxed{\int_0^1(x-1)^5\,dx=\frac{1}{6}}$

016. Calculate Integral $I=\int_0^{+\infty}\frac{x\tan^{-1}(x)}{x^4+x^2+1}dx$

Solution; $I=\int_0^{+\infty}\frac{x\tan^{-1}(x)}{x^4+x^2+1}dx \quad (*)$

Let : $x=\frac{1}{t}\Rightarrow dx=-\frac{1}{t^2}dt$ **; If :** $x\in(0,+\infty)\Rightarrow t\in(+\infty,0)$

They got it $I=-\int_{+\infty}^0\frac{\frac{1}{t}\tan^{-1}\left(\frac{1}{t}\right)}{\frac{1}{t^4}+\frac{1}{t^2}+1}\times\frac{1}{t^2}dt=\int_0^{+\infty}\frac{t\tan^{-1}\left(\frac{1}{t}\right)}{t^4+t^2+1}dt \quad (**)$

Take $(*)+(**)$ **That Have :** $2I=\int_0^{+\infty}\frac{t\left[\tan^{-1}(t)+\tan^{-1}\left(\frac{1}{t}\right)\right]}{t^4+t^2+1}dt$

$$=\frac{\pi}{4}\int_0^{+\infty}\frac{t}{t^4+t^2+1}dt=\frac{\pi}{8}\int_0^{+\infty}\frac{1}{t^4+t^2+1}d\left(t^2\right)$$

$$=\frac{\pi}{4\sqrt{3}}\arctan\left(\frac{2}{\sqrt{3}}t^2+\frac{1}{2}\right)\Bigg|_0^{+\infty}$$

$$=\frac{\pi^2}{12\sqrt{3}}$$

$SO,$ $\boxed{\int_0^{+\infty}\frac{x\tan^{-1}(x)}{x^4+x^2+1}dx=\frac{\pi^2}{12\sqrt{3}}}$

017. Calculate Integral $I=\int_2^4\frac{C_x^1\cdot C_x^3\cdot C_x^5}{C_x^2\cdot C_x^4\cdot C_x^6}dx$

Solution; $I=\int_2^4\frac{C_x^1\cdot C_x^3\cdot C_x^5}{C_x^2\cdot C_x^4\cdot C_x^6}dx$

They got it $I=\int_2^4\frac{\frac{x!}{(x-1)!.1!}.\frac{x!}{(x-3)!.3!}.\frac{x!}{(x-5)!.5!}}{\frac{x!}{(x-2)!.2!}.\frac{x!}{(x-4)!.4!}.\frac{x!}{(x-6)!.6!}}dx$

$$= 48\int_2^4 \frac{x}{x(x-1)} \cdot \frac{x(x-1)(x-2)}{x(x-1)(x-2)(x-3)} \cdot \frac{x(x-1)(x-2)(x-3)(x-4)}{x(x-1)(x-2)(x-3)(x-4)(x-5)} dx$$

$$= 48\int_2^4 \frac{1}{(x-1)(x-3)(x-5)} dx$$

Let: $t = x-3 \Rightarrow dt = dx, (x-1 = t+2, x-5 = t-2)$; **If**: $x \in (2,4) \Rightarrow t \in (-1,1)$

They got it $I = 48\int_{-1}^1 \frac{1}{t(t+2)(t-2)} dt$

$$= 48 \underbrace{\int_{-1}^1 \frac{1}{t(t^2-4)} dt}_{Is\ odd\ function} = 0$$

$SO,$ $\boxed{I = \int_2^4 \frac{C_x^1 \cdot C_x^3 \cdot C_x^5}{C_x^2 \cdot C_x^4 \cdot C_x^6} dx = 0}$

018. Calculate Integral $I = \int_0^1 \left(\frac{\log(x)}{x^2-1} \right) dx$

Solution; $I = \int_0^1 \left(\frac{\log(x)}{x^2-1} \right) dx$

We Have : $\frac{1}{1-x^2} = \sum_{k=0}^{\infty} x^{2k} \Rightarrow \frac{1}{x^2-1} = -\sum_{k=0}^{\infty} x^{2k}$

That $I = -\sum_{k=0}^{\infty} \left(\int_0^1 x^{2k} \log(x) dx \right)$

Let : $u = \log(x) \Rightarrow du = \frac{1}{x} dx$ **And :** $dv = x^{2k} dx \Leftrightarrow v = \frac{x^{2k+1}}{2k+1}$

They got it $I = -\sum_{k=0}^{\infty} \left[\underbrace{\frac{x^{2k+1}}{2k+1} \log(x) \Big|_0^1}_{0} - \frac{1}{2k+1} \int_0^1 \frac{x^{2k+1}}{x} dx \right]$

$$= \sum_{k=0}^{\infty} \left[\frac{1}{2k+1} \times \frac{x^{2k+1}}{2k+1} \Big|_0^1 \right]$$

$$= \sum_{k=0}^{\infty} \left[\frac{1}{(2k+1)^2} \right]$$

$$=\left(\frac{1}{1^2}+\frac{1}{3^2}+\frac{1}{5^2}+\cdots\right)$$

$$=\left(1+\frac{1}{2^2}+\frac{1}{3^2}+\frac{1}{4^2}+\cdots\right)-\frac{1}{4}\left(1+\frac{1}{2^2}+\frac{1}{3^2}+\frac{1}{4^2}+\cdots\right)$$

$$=\frac{3}{4}\left(1+\frac{1}{2^2}+\frac{1}{3^2}+\frac{1}{4^2}+\cdots\right)$$

$$=\frac{3}{4}\zeta(2)=\frac{3}{4}\cdot\frac{\pi^2}{6}=\frac{\pi^2}{8}$$

SO, $\boxed{\int_0^1 \frac{\log(x)}{x^2-1}dx=\frac{\pi^2}{8}}$

019. Calculate Integral $I=\int_0^{\pi}\frac{\sin^{2030}(x)}{\sin^{2030}(x)+\cos^{2030}(x)}dx$

Solution; $I=\int_0^{\pi}\frac{\sin^{2030}(x)}{\sin^{2030}(x)+\cos^{2030}(x)}dx$

$$=2\int_0^{\frac{\pi}{2}}\frac{\sin^{2030}(x)}{\sin^{2030}(x)+\cos^{2030}(x)}dx \quad (*)$$

$$=2\int_0^{\frac{\pi}{2}}\frac{\sin^{2030}\left(\frac{\pi}{2}-x\right)}{\sin^{2030}\left(\frac{\pi}{2}-x\right)+\cos^{2030}\left(\frac{\pi}{2}-x\right)}dx$$

$$=2\int_0^{\frac{\pi}{2}}\frac{\cos^{2030}(x)}{\sin^{2030}(x)+\cos^{2030}(x)}dx \quad (**)$$

That $(*)+(**)$ **That Have :** $2I=2\int_0^{\frac{\pi}{2}}\frac{\sin^{2030}(x)+\cos^{2030}(x)}{\sin^{2030}(x)+\cos^{2030}(x)}dx$

$$\Rightarrow I=x\Big|_0^{\pi/2}=\frac{\pi}{2}$$

SO, $\boxed{\int_0^{\pi}\frac{\sin^{2030}(x)}{\sin^{2030}(x)+\cos^{2030}(x)}dx=\frac{\pi}{2}}$

020. Calculate Integral $I=\int_{-\frac{\pi}{2}}^{+\frac{\pi}{2}}\frac{\cos(x)}{e^{\frac{1}{x}}+1}dx$

Solution; $I=\int_{-\frac{\pi}{2}}^{+\frac{\pi}{2}}\frac{\cos(x)}{e^{\frac{1}{x}}+1}dx \quad (*)$

$$=\int_{-\frac{\pi}{2}}^{+\frac{\pi}{2}}\frac{\cos(-x)}{e^{-\frac{1}{x}}+1}dx$$

Use : $\int_{-a}^{+a}f(x)dx=\int_{-a}^{+a}f(-x)dx$

$$=\int_{-\frac{\pi}{2}}^{+\frac{\pi}{2}}\frac{e^{\frac{1}{x}}\cos(x)}{e^{\frac{1}{x}}+1}dx \quad (**)$$

Take $(*)+(**)$ **That Have :** $2I=\int_{-\frac{\pi}{2}}^{+\frac{\pi}{2}}\frac{e^{\frac{1}{x}}\cos(x)+\cos(x)}{e^{\frac{1}{x}}+1}dx=2$

They got it $I=1$

SO, $\boxed{\int_{-\frac{\pi}{2}}^{+\frac{\pi}{2}}\frac{\cos(x)}{e^{\frac{1}{x}}+1}dx=1}$

021. Calculate Integral $I=\int_{0}^{\frac{\pi}{2}}\sqrt{\sin(x)-\sin^2(x)}dx$

Solution; $I=\int_{0}^{\frac{\pi}{2}}\sqrt{\sin(x)-\sin^2(x)}dx$

$$=\int_{0}^{\frac{\pi}{2}}\sqrt{\sin(x)}\sqrt{1-\sin(x)}dx$$

$$=\int_{0}^{\frac{\pi}{2}}\frac{\sqrt{\sin(x)}}{\sqrt{1+\sin(x)}}\cos(x)dx$$

Let : $t=\sin(x)\Rightarrow dt=\cos(x)dx$ **; If :** $x\in\left(0,\frac{\pi}{2}\right)\Rightarrow t\in(0,1)$

They got it $I=\int_{0}^{1}\frac{\sqrt{t}}{\sqrt{1+t}}dt$

Let : $y=\sqrt{\frac{t}{t+1}}\Leftrightarrow t=\frac{y^2}{1-y^2}\Rightarrow dt=d\left(\frac{y^2}{1-y^2}\right)$ **; If :** $t\in(0,1)\Rightarrow y\in\left(0,\frac{\sqrt{2}}{2}\right)$

They got it $I=\int_0^{\frac{\sqrt{2}}{2}} yd\left(\frac{y^2}{1-y^2}\right)$

Let : $u=y\Rightarrow du=dy$ **And** $dv=d\left(\frac{y^2}{1-y^2}\right)\Rightarrow v=\frac{y^2}{1-y^2}$

They got it $I=y\cdot\frac{y^2}{1-y^2}\Big|_0^{\frac{\sqrt{2}}{2}}-\int_0^{\frac{\sqrt{2}}{2}}\frac{y^2}{1-y^2}dy$

$$=\frac{\sqrt{2}}{2}+\int_0^{\frac{\sqrt{2}}{2}}\frac{1-y^2+1}{1-y^2}dy$$

$$=\frac{\sqrt{2}}{2}+\left[y-\tanh^{-1}(y)\right]_0^{\frac{\sqrt{2}}{2}}$$

$$=\sqrt{2}-\log\left(\sqrt{2}+1\right)$$

SO, $\boxed{\int_0^{\frac{\pi}{2}}\sqrt{\sin(x)-\sin^2(x)}dx=\sqrt{2}-\log\left(\sqrt{2}+1\right)}$

022. Calculate Integral $I=\int_0^{\frac{\pi}{4}}e^x\left(\tan^2(x)+\log\left(\cos(x)\right)\right)dx$

Solution; $I=\int_0^{\frac{\pi}{4}}e^x\left(\tan^2(x)+\log\left(\cos(x)\right)\right)dx$

$$=\int_0^{\frac{\pi}{4}}e^x\left(\sec^2(x)-1+\log\left(\cos(x)\right)\right)dx$$

$$=\underbrace{-\int_0^{\frac{\pi}{4}}e^x dx}_{I_1}+\underbrace{\int_0^{\frac{\pi}{4}}e^x\sec^2(x)dx}_{I_2}+\underbrace{\int_0^{\frac{\pi}{4}}e^x\log(\cos(x))dx}_{I_3}$$

For $-\int_0^{\frac{\pi}{4}}e^x dx=1-e^{\frac{\pi}{4}}$

For : $I_2=\int_0^{\frac{\pi}{4}}e^x\sec^2(x)dx$

Let : $u=e^x\Rightarrow du=e^x dx$ **And** $dv=\sec^2(x)dx\Rightarrow v=\tan(x)$

They got it : $I_2=\left[e^x\tan(x)\right]_0^{\frac{\pi}{4}}-\int_0^{\frac{\pi}{4}}e^x\tan(x)dx$

For : $I_3=\int_0^{\frac{\pi}{4}} e^x \log(\cos(x))dx$

Let : $u=\log\left(\cos(x)\right)\Rightarrow du=-\tan(x)dx$ **And** $dv=e^x dx\Rightarrow v=e^x$

They got it $I_3=\left[e^x \log\left(\cos(x)\right)\right]_0^{\frac{\pi}{4}}+\int_0^{\frac{\pi}{4}} e^x \tan(x)dx$

That $I=1-e^{\frac{\pi}{4}}+\left(\left[e^x \tan(x)\right]_0^{\frac{\pi}{4}}-\int_0^{\frac{\pi}{4}} e^x \tan(x)dx\right)+\left(\left[e^x \log\left(\cos(x)\right)\right]_0^{\frac{\pi}{4}}+\int_0^{\frac{\pi}{4}} e^x \tan(x)dx\right)$

$$=1-e^{\frac{\pi}{4}}+e^{\frac{\pi}{4}}+e^{\frac{\pi}{4}}\log\left(\frac{\sqrt{2}}{2}\right)$$

$$=1-\frac{1}{2}e^{\frac{\pi}{4}}\log(2)$$

SO, $\boxed{\int_0^{\frac{\pi}{4}} e^x\left(\tan^2(x)+\log\left(\cos(x)\right)\right)dx=1-\frac{1}{2}e^{\frac{\pi}{4}}\log(2)}$

023. Calculate Integral $I=\int_0^{+\infty}\frac{x^2+x}{\left(e^x+x+1\right)^2}dx$

Solution; $I=\int_0^{+\infty}\frac{x^2+x}{\left(e^x+x+1\right)^2}dx$

$$=\int_0^{+\infty}\frac{(x+1)e^{-x}}{\left(1+(x+1)e^{-x}\right)^2}xe^{-x}dx$$

Let : $u=(x+1)e^{-x}\Rightarrow du=-xe^{-x}dx$ **; If** : $x\in(0,+\infty)\Rightarrow u\in(1,0)$

They got it $I=-\int_1^0\frac{u}{(1+u)^2}du$

$$=\int_0^1\frac{(1+u)-1}{(1+u)^2}du$$

$$=\log(2)-\frac{1}{2}$$

SO, $\boxed{\int_0^{+\infty}\frac{x^2+x}{\left(e^x+x+1\right)^2}dx=\log(2)-\frac{1}{2}}$

024. Calculate Integral $I=\int_0^{\frac{\pi}{2}}\frac{x}{\sin(x)+\cos(x)}dx$

Solution; $I=\int_0^{\frac{\pi}{2}}\frac{x}{\sin(x)+\cos(x)}dx$

$$=\int_0^{\frac{\pi}{2}}\frac{\left(\frac{\pi}{2}-x\right)}{\sin\left(\frac{\pi}{2}-x\right)+\cos\left(\frac{\pi}{2}-x\right)}dx$$

$$=\frac{\pi}{2}\int_0^{\frac{\pi}{2}}\frac{1}{\sin(x)+\cos(x)}dx-\underbrace{\int_0^{\frac{\pi}{2}}\frac{x}{\sin(x)+\cos(x)}dx}_{I}$$

That $I=\frac{\pi}{4}\int_0^{\frac{\pi}{2}}\frac{1}{\sin(x)+\cos(x)}dx$

Let: $u=\tan\left(\frac{x}{2}\right)\Rightarrow dx=\frac{2}{1+u^2}du$ **; If** $:x\in\left(0,\frac{\pi}{2}\right)\Rightarrow u\in(0,1)$

Which: $\begin{cases}\cos(x)=\frac{1-u^2}{1+u^2}\\ \sin(x)=\frac{2u}{1+u^2}\end{cases}$

They got it $I=\frac{\pi}{4}\int_0^1\frac{1}{\frac{2u}{1+u^2}+\frac{1-u^2}{1+u^2}}\times\frac{2}{1+u^2}du$

$$=\frac{\pi}{2}\int_0^1\frac{1}{1+2u-u^2}du$$

$$=\frac{\pi}{2}\times\left(\frac{1}{2\sqrt{2}}\log\left(\frac{\sqrt{2}+u}{\sqrt{2}-u}\right)\right)\Bigg|_0^1$$

$$=\frac{\pi\sqrt{2}}{8}\log\left(\frac{\sqrt{2}+1}{\sqrt{2}-1}\right)$$

SO, $\boxed{\int_0^{\frac{\pi}{2}}\frac{x}{\sin(x)+\cos(x)}dx=\frac{\pi\sqrt{2}}{4}\log\left(\sqrt{2}+1\right)}$

025. Calculate Integral $I=\int_0^{\pi}\frac{x}{\cos^2(x)+9\sin^2(x)}dx$

Solution; $I=\int_0^{\pi}\frac{x}{\cos^2(x)+9\sin^2(x)}dx$

$$=\int_0^{\pi}\frac{\pi-x}{\cos^2(\pi-x)+9\sin^2(\pi-x)}dx$$

They got it $I=\frac{\pi}{2}\int_0^{\pi}\frac{1}{\cos^2(x)+9\sin^2(x)}dx$

Let: $u=\tan\left(\frac{x}{2}\right)\Rightarrow dx=\frac{2}{1+u^2}du$ **; If** $:x\in(0,\pi)\Rightarrow u\in(0,+\infty)$

That $I=\frac{\pi}{2}\int_0^{+\infty}\frac{1}{\left(\frac{1-u^2}{1+u^2}\right)^2+9\left(\frac{2u}{1+u^2}\right)^2}\times\frac{2}{1+u^2}du$

$$=\pi\int_0^{+\infty}\frac{1+u^2}{\left(u-u^{-1}\right)^2+36}du$$

Let $:v=u-u^{-1}\Rightarrow dv=(1+u^{-2})du$ **; if** $:u\in(0,+\infty)\Rightarrow v\in(-\infty,+\infty)$

They got it $=\pi\int_{-\infty}^{+\infty}\frac{v}{v^2+36}dv=\frac{\pi}{6}\tan^{-1}\left(\frac{v}{6}\right)\Big|_{-\infty}^{+\infty}=\frac{\pi^2}{6}$

SO, $\boxed{\int_0^{\pi}\frac{x}{\cos^2(x)+9\sin^2(x)}dx=\frac{\pi^2}{6}}$

026. Calculate Integral $I=\int_0^1\frac{1}{\sqrt[3]{(1+x)^2(1-x)}}dx$

Solution; $I=\int_0^1\frac{1}{\sqrt[3]{(1+x)^2(1-x)}}dx$

$$=\int_0^1\frac{1}{(1+x)\left(\frac{1-x}{1+x}\right)^{\frac{1}{3}}}dx$$

Let : $y^3=\left(\frac{1-x}{1+x}\right)\Rightarrow 3y^2dy=-\frac{2}{(1+x)^2}dx$ **;If :** $x\in(0,1)\Rightarrow y\in(1,0)$ **Which**

$$\frac{1}{1+x}=\frac{2}{1+y^3}$$

They got it $I=-3\int_1^0\frac{y^2}{(1+y^3)y}dx=3\int_0^1\frac{y}{(1+y)(y^2-y+1)}dx$

$$=\int_0^1\frac{(1+y)^2-(y^2-y+1)}{(1+y)(y^2-y+1)}dx$$

$$=\underbrace{\int_0^1\frac{y+1}{y^2-y+1}dx}_{I_1}-\underbrace{\int_0^1\frac{1}{y+1}dx}_{I_2}$$

For $I_2=\int_0^1\frac{1}{y+1}dx=\log(2)$

For $I_1=\frac{1}{2}\int_0^1\frac{2y-1+3}{y^2-y+1}dx$

$$=\frac{1}{2}\int_0^1\frac{2y-1}{y^2-y+1}dx+\frac{3}{2}\int_0^1\frac{1}{y^2-y+1}dx$$

$$=\underbrace{\frac{1}{2}\log\left(y^2-y+1\right)\Big|_0^1}_{0}+\underbrace{\sqrt{3}\arctan\left(\frac{2x-1}{\sqrt{3}}\right)\Big|_0^1}_{\pi/\sqrt{3}}$$

$$=\frac{\pi}{\sqrt{3}}$$

They got it $I=\frac{\pi}{\sqrt{3}}-\log(2)$

SO, $\boxed{\int_0^1\frac{1}{\sqrt[3]{(1+x)^2(1-x)}}dx=\frac{\pi}{\sqrt{3}}-\log(2)}$

027. Calculate Integral $I=\int_0^1\log\left(\Gamma(x)\right)dx$

Solution; $I=\int_0^1\log\left(\Gamma(x)\right)dx$ (!)

$$=\int_0^1\log\left(\Gamma(1-x)\right)dx \quad (!!)$$

Note: $\log(y)+\log(z)=\log(yz)$

Take $(!)+(!!)$: $2I=\int_0^1 \log\left(\Gamma(x)\right)dx+\int_0^1 \log\left(\Gamma(1-x)\right)dx$

They got it $I=\frac{1}{2}\int_0^1 \log\left(\Gamma(x)\Gamma(1-x)\right)dx$

Note: $\Gamma(1-x)\Gamma(x)=\pi\csc(\pi x)$

$$=\frac{1}{2}\int_0^1 \log\left(\frac{\pi}{\sin(\pi x)}\right)dx$$

$$=\frac{1}{2}\left(\int_0^1 \log\left(\pi\right)dx-\int_0^1 \log\left(\sin(\pi x)\right)dx\right)$$

$$=\frac{1}{2}\left(\log(\pi)-I_1\right)\quad (*)$$

Take $I_1=\int_0^1 \log\left(\sin(\pi x)\right)dx$

Let : $t=\pi x\Rightarrow dx=\frac{1}{\pi}dt$ **; If** : $x\in(0,1)\Rightarrow t\in(0,\pi)$

That $I_1=\frac{1}{\pi}\int_0^{\pi} \log\left(\sin(t)\right)dt$

$$=\frac{2}{\pi}\int_0^{\frac{\pi}{2}} \log\left(\sin(t)\right)dt=\frac{2}{\pi}\left(-\frac{\pi}{2}\log(2)\right)=-\log(2)$$

Take $(*)$ **They Have** $I=\frac{1}{2}\left(\log(\pi)+\log(2)\right)=\frac{\log(2\pi)}{2}$

$SO,$ $\boxed{\int_0^1 \log\left(\Gamma(x)\right)dx=\frac{\log(2\pi)}{2}}$

028. Calculate Integral $I=\int_0^1\left[3^x W\left(3^x x\log(3)\right)\right]dx$

Solution; $I=\int_0^1\left[3^x W\left(3^x x\log(3)\right)\right]dx$

$$=\int_0^1\left[W\left(x\log(3)e^{x\log(3)}\right)e^{x\log(3)}\right]dx$$

$$=\log(3)\int_0^1 x3^x dx$$

Let : $u=x\Rightarrow du=dx$ **And** $dv=3^x dx\Rightarrow v=\int 3^x dx=\frac{3^x}{\log(3)}$

They got it $I=\left.\frac{x3^x}{\log(3)}\right|_0^1-\int_0^1\frac{3^x}{\log(3)}dx=\frac{3\log(3)-2}{\log^2(3)}$

SO, $\boxed{\int_0^1\left[3^xW\left(3^x x\log(3)\right)\right]dx=\frac{3\log(3)-2}{\log^2(3)}}$

029. Calculate Integral $I=\int_0^1\Gamma\left(1+\frac{x}{2}\right)\Gamma\left(1-\frac{x}{2}\right)dx$

Solution; $I=\int_0^1\Gamma\left(1+\frac{x}{2}\right)\Gamma\left(1-\frac{x}{2}\right)dx$

$$=\int_0^1\frac{x}{2}\Gamma\left(\frac{x}{2}\right)\Gamma\left(1-\frac{x}{2}\right)dx=\int_0^1\left(\frac{x}{2}\cdot\pi\csc\left(\frac{x\pi}{2}\right)\right)dx$$

Let : $t=\frac{x\pi}{2}\Rightarrow dx=\frac{2}{\pi}dt$ **; If :** $x\in(0,1)\Rightarrow t\in\left(0,\frac{\pi}{2}\right)$

They got it $I=\frac{2}{\pi}\int_0^{\frac{\pi}{2}}t\csc(t)dt$

Let : $u=t\Rightarrow du=dt$ **And** $dv=\csc(t)dt\Rightarrow v=-\log\left(\csc(t)+\cot(t)\right)$

That $I=\frac{2}{\pi}\left[\underbrace{-t\log\left(\csc(t)+\cot(t)\right)\Big|_0^{\frac{\pi}{2}}}_{0}+\int_0^{\frac{\pi}{2}}\log\left(\csc(t)+\cot(t)\right)dt\right]$

$$=\frac{2}{\pi}\int_0^{\frac{\pi}{2}}\log\left(\frac{1+\cos(t)}{\sin(t)}\right)dt=\frac{4}{\pi}\int_0^{\frac{\pi}{2}}\log\left(\cot\left(\frac{t}{2}\right)\right)\frac{dt}{2}$$

Let : $y=\frac{t}{2}\Rightarrow dy=\frac{dt}{2}$ **; if :** $t\in\left(0,\frac{\pi}{2}\right)\Rightarrow y\in\left(0,\frac{\pi}{4}\right)$

That $I=\frac{4}{\pi}\int_0^{\frac{\pi}{4}}\log\left(\cot\left(y\right)\right)dy=\frac{4}{\pi}G$

SO, $\boxed{\int_0^1\Gamma\left(1+\frac{x}{2}\right)\Gamma\left(1-\frac{x}{2}\right)dx=\frac{4}{\pi}G}$

030. Calculate Integral $I=\int_0^1 \frac{1}{1+x+x^2+x^3+\cdots}dx$

Solution; $I=\int_0^1 \frac{1}{1+x+x^2+x^3+\cdots}dx$

$$=\int_0^1 \frac{1}{1+\frac{x}{1-x}}dx=\int_0^1 (1-x)dx=\frac{1}{2}$$

Note: $u_1+u_2+u_3+\ldots+u_n=\frac{u_1}{1-q}$, $|q|<1$

SO, $\boxed{\int_0^1 \frac{1}{1+x+x^2+x^3+\cdots}dx=\frac{1}{2}}$

031. Calculate Integral $I=\int_0^{\pi} \frac{1}{(1+\sin(x))^2}dx$

Solution; $I=\int_0^{\pi} \frac{1}{(1+\sin(x))^2}dx$

Let: $y=\tan\left(\frac{x}{2}\right)\Rightarrow dx=\frac{2}{1+y^2}dy$ **; If** $: x\in(0,\pi)\Rightarrow y\in(0,\infty)$

They got it $I=\int_0^{\infty} \frac{1}{\left(1+\frac{2y}{1+y^2}\right)^2}\times\frac{2}{1+y^2}dy=2\int_0^{\infty}\frac{1+y^2}{(1+y)^4}dy$

Let $: u=1+y\Leftrightarrow y=u-1\Rightarrow dy=du$ **; If** $: y\in(0,\infty)\Rightarrow u\in(1,\infty)$

That $I=2\int_1^{\infty}\frac{1+(u-1)^2}{(u)^4}du=2\int_1^{\infty}\left(\frac{1}{u^2}-\frac{2}{u^3}+\frac{2}{u^4}\right)du=\frac{4}{3}$

SO, $\boxed{\int_0^{\pi} \frac{1}{(1+\sin(x))^2}dx=\frac{4}{3}}$

032. Calculate Integral $I=\int_{-1}^0 \frac{1}{\sqrt{e^x}+\sqrt{e^{3x}}+\sqrt{e^{5x}}+\sqrt{e^{7x}}+\cdots}dx$

Solution; $I=\int_{-1}^0 \frac{1}{\sqrt{e^x}+\sqrt{e^{3x}}+\sqrt{e^{5x}}+\sqrt{e^{7x}}+\cdots}dx$

$$=\int_{-1}^{0}\frac{e^{-\frac{1}{2}x}}{1+e^{x}+e^{2x}+e^{3x}+\cdots}dx$$

$$=\int_{-1}^{0}\frac{e^{-\frac{1}{2}x}}{\frac{1}{1-e^{x}}}dx=\int_{-1}^{0}\left(e^{-\frac{1}{2}x}-e^{\frac{1}{2}x}\right)dx=\frac{2(\sqrt{e}-e)}{e}$$

SO, $\boxed{\int_{-1}^{0}\frac{1}{\sqrt{e^{x}}+\sqrt{e^{3x}}+\sqrt{e^{5x}}+\sqrt{e^{7x}}+\cdots}dx=\frac{2(\sqrt{e}-e)}{e}}$

033. Calculate Integral $I=\int_{0}^{1}\frac{(1-x^{2})}{(1+x^{2})\sqrt{1+x^{4}}}dx$

Solution; $I=\int_{0}^{1}\frac{(1-x^{2})}{(1+x^{2})\sqrt{1+x^{4}}}dx$

$$=-\int_{0}^{1}\frac{1}{(x+x^{-1})\sqrt{x^{2}+x^{-2}}}\times\left(1-x^{-2}\right)dx$$

$$=-\int_{0}^{1}\frac{1}{(x+x^{-1})\sqrt{(x+x^{-1})^{2}-2}}\times\left(1-x^{-2}\right)dx$$

Let : $\sqrt{2}\sec(t)=x+x^{-1}\Rightarrow\sqrt{2}\sec(t)\tan(t)dt=(1-x^{-2})dx$ **; If :** $t\in(0,1)\Rightarrow y\in\left(\frac{\pi}{2},\frac{\pi}{4}\right)$

They got it $I=-\int_{\frac{\pi}{2}}^{\frac{\pi}{4}}\frac{\sqrt{2}\sec(t)\tan(t)}{\sqrt{2}\sec(t)\sqrt{2\sec^{2}(t)-2}}dt$

$$=\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}\frac{\sqrt{2}\sec(t)\tan(t)}{\sqrt{2}.\sqrt{2}\sec(t)\tan(t)}dt=\frac{\pi}{4\sqrt{2}}$$

SO, $\boxed{\int_{0}^{1}\frac{(1-x^{2})}{(1+x^{2})\sqrt{1+x^{4}}}dx=\frac{\pi}{4\sqrt{2}}}$

034. Calculate Integral $I=\int_{0}^{+\infty}\frac{x^{n-1}}{e^{x}-1}dx$

Solution; $I=\int_{0}^{+\infty}\frac{x^{n-1}}{e^{x}-1}dx$

$$= \int_0^{+\infty} \frac{x^{n-1}e^{-x}}{1-e^{-x}}dx$$

By : $\frac{1}{1-a} = \sum_{i=0}^{\infty}\left(a^i\right)$

That $I = \int_0^{+\infty}\left[x^{n-1}e^{-x}\sum_{i=0}^{\infty}\left(e^{-xi}\right)\right]dx$

$$= \sum_{i=0}^{\infty}\left(\int_0^{+\infty} x^{n-1}e^{-x(i+1)}dx\right)$$

Let : $y=(i+1)x \Rightarrow dx = \frac{1}{i+1}dy$; **If** : $x\in(0,+\infty) \Rightarrow y\in(0,+\infty)$

They got it $I = \sum_{i=0}^{\infty}\left[\int_0^{+\infty}\left(\frac{y}{i+1}\right)^{n-1}\frac{e^{-y}}{(i+1)}dy\right]$

$$= \sum_{i=0}^{\infty}\left[\frac{1}{(i+1)^n}\int_0^{+\infty} y^{n-1}e^{-y}dy\right]$$

$$= \sum_{i=0}^{\infty}\frac{\Gamma(n)}{(i+1)^n} = \zeta(n)\Gamma(n)$$

SO, $\boxed{\int_0^{+\infty}\frac{x^{n-1}}{e^x-1}dx = \zeta(n)\Gamma(n)}$

035. Calculate Integral $I = \int_0^{\frac{\pi}{2}}\frac{\cos(x)\sqrt{\cos(x)}-\sin(x)\sqrt{\sin(x)}}{\sin(x)\sqrt{\cos(x)}+\cos(x)\sqrt{\sin(x)}}dx$

Solution; $I = \int_0^{\frac{\pi}{2}}\frac{\cos(x)\sqrt{\cos(x)}-\sin(x)\sqrt{\sin(x)}}{\sin(x)\sqrt{\cos(x)}+\cos(x)\sqrt{\sin(x)}}dx$

$$= \int_0^{\frac{\pi}{2}}\frac{\cos\left(\frac{\pi}{2}-x\right)\sqrt{\cos\left(\frac{\pi}{2}-x\right)}-\sin\left(\frac{\pi}{2}-x\right)\sqrt{\sin\left(\frac{\pi}{2}-x\right)}}{\sin\left(\frac{\pi}{2}-x\right)\sqrt{\cos\left(\frac{\pi}{2}-x\right)}+\cos\left(\frac{\pi}{2}-x\right)\sqrt{\sin\left(\frac{\pi}{2}-x\right)}}dx$$

$$= \int_0^{\frac{\pi}{2}}\frac{\sin(x)\sqrt{\sin(x)}-\cos(x)\sqrt{\cos(x)}}{\sin(x)\sqrt{\cos(x)}+\cos(x)\sqrt{\sin(x)}}dx$$

$$= \underbrace{-\int_0^{\frac{\pi}{2}} \frac{\cos(x)\sqrt{\cos(x)} - \sin(x)\sqrt{\sin(x)}}{\sin(x)\sqrt{\cos(x)} + \cos(x)\sqrt{\sin(x)}}\,dx}_{I}$$

They got it $I = -I \Rightarrow I = 0$

SO, $\boxed{\int_0^{\frac{\pi}{2}} \frac{\cos(x)\sqrt{\cos(x)} - \sin(x)\sqrt{\sin(x)}}{\sin(x)\sqrt{\cos(x)} + \cos(x)\sqrt{\sin(x)}}\,dx = 0}$

036. Calculate Integral $I = \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{1}{\sin^3(x)\cos^5(x)}\,dx$

Solution; $I = \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{1}{\sin^3(x)\cos^5(x)}\,dx$

$$= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{1}{\sin^3(x)\cos^3(x)} \cdot \sec^2(x)dx$$

$$= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \left(\frac{\sin^2(x)+\cos^2(x)}{\sin(x)\cos(x)} \right)^3 d\left(\tan(x)\right)$$

$$= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \left(\tan(x) + \frac{1}{\tan(x)} \right)^3 d\left(\tan(x)\right)$$

$$= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \left[\tan^3(x) + 3\tan(x) + \frac{3}{\tan(x)} + \frac{1}{\tan^3(x)} \right] d\left(\tan(x)\right)$$

$$= \left[\frac{\tan^4(x)}{4} + \frac{3\tan^2(x)}{2} + 3\log\left(\tan(x)\right) - \frac{1}{2\tan^2(x)} \right]_{\frac{\pi}{4}}^{\frac{\pi}{3}}$$

$$= \frac{16}{3} + 3\log\sqrt{3}$$

SO, $\boxed{\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{1}{\sin^3(x)\cos^5(x)}\,dx = \frac{16}{3} + 3\log\sqrt{3}}$

037. Calculate Integral $I=\int_0^{+\infty}\frac{nx^{n-1}}{(x^2+1)^{\frac{n+2}{2}}}dx \quad ,(n>0)$

Solution; $I=\int_0^{+\infty}\frac{nx^{n-1}}{(x^2+1)^{\frac{n+2}{2}}}dx \quad ,(n>0)$

Let : $x=\tan(y)\Rightarrow dx=\sec^2(y)dy$ **; If :** $x\in(0,+\infty)\Rightarrow y\in\left(0,\frac{\pi}{2}\right)$

They got it $I=\int_0^{+\infty}\frac{n\tan^{n-1}(y)}{(1+\tan^2(y))^{\frac{n+2}{2}}}.\sec^2(y)dy$

$$=n\int_0^{+\infty}\frac{\tan^{n-1}(y)}{\sec^n(y)}dy=\frac{n}{2}\int_0^{+\infty}2\sin^{n-1}(y)\cos(y)dy$$

$$=\frac{n}{2}\int_0^{+\infty}2\sin^{2\left(\frac{n}{2}\right)-1}(y)\cos^{2\times1-1}(y)dy=\frac{n}{2}B\left(\frac{n}{2},1\right)$$

$$=\frac{n}{2}\times\frac{\Gamma\left(\frac{n}{2}\right)\Gamma(1)}{\Gamma\left(\frac{n}{2}+1\right)}=\frac{n}{2}\times\frac{\Gamma\left(\frac{n}{2}\right)}{\frac{n}{2}\Gamma\left(\frac{n}{2}\right)}=1$$

Note: $\Gamma(n+1)=n\Gamma(n)$

SO, $\boxed{\int_0^{+\infty}\frac{nx^{n-1}}{(x^2+1)^{\frac{n+2}{2}}}dx=1}$

038. Calculate Integral $I=\int_1^2(x+1)^2e^{\frac{x^2-1}{x}}dx$

Solution; $I=\int_1^2(x+1)^2e^{\frac{x^2-1}{x}}dx$

$$=\int_1^2\left(2xe^{x-\frac{1}{x}}+x^2(1+\frac{1}{x^2})e^{x-\frac{1}{x}}\right)dx$$

$$=\int_1^2\left(x^2e^{x-\frac{1}{x}}\right)'dx=4e^{3/2}-1$$

SO, $\boxed{\int_1^2(x+1)^2e^{\frac{x^2-1}{x}}dx=4e^{3/2}-1}$

039. Calculate Integral $I=\int_1^2 x^x\left(1+\log(x)\right)dx$

Solution; $I=\int_1^2 x^x\left(1+\log(x)\right)dx$

$$=\int_1^2 e^{x\log(x)}\left(1+\log(x)\right)dx$$

$$=\int_1^2 e^{x\log(x)}d\left(x\log(x)\right)$$

$$=e^{x\log(x)}\Big|_1^2=3$$

SO, $\boxed{\int_1^2 x^x\left(1+\log(x)\right)dx=3}$

040. Calculate Integral $I=\sum_{n=0}^{\infty}\left[\int_{2n}^{2n+1}\left(\frac{\cos(x)}{x}-\frac{\sin(x)}{x^2}\right)dx\right]+\sum_{n=1}^{\infty}\left[\int_{2n}^{2n-1}\left(\frac{\cos(x)}{x}-\frac{\sin(x)}{x^2}\right)dx\right]$

Solution; $I=\sum_{n=0}^{\infty}\left[\int_{2n}^{2n+1}\left(\frac{\cos(x)}{x}-\frac{\sin(x)}{x^2}\right)dx\right]+\sum_{n=1}^{\infty}\left[\int_{2n}^{2n-1}\left(\frac{\cos(x)}{x}-\frac{\sin(x)}{x^2}\right)dx\right]$

We Have $\left(\frac{\sin(x)}{x}\right)'=\left(\frac{\cos(x)}{x}-\frac{\sin(x)}{x^2}\right)$

That $I=\sum_{n=0}^{\infty}\left[\int_{2n}^{2n+1}\left(\frac{\sin(x)}{x}\right)'dx\right]+\sum_{n=1}^{\infty}\left[\int_{2n-1}^{2n}\left(\frac{\sin(x)}{x}\right)'dx\right]$

$$=\left[\int_0^1\left(\frac{\sin(x)}{x}\right)'dx+\int_2^3\left(\frac{\sin(x)}{x}\right)'dx+\cdots\right]+\left[\int_1^2\left(\frac{\sin(x)}{x}\right)'dx+\int_3^4\left(\frac{\sin(x)}{x}\right)'dx+\cdots\right]$$

$$=\int_0^1\left(\frac{\sin(x)}{x}\right)'dx+\int_1^2\left(\frac{\sin(x)}{x}\right)'dx+\int_2^3\left(\frac{\sin(x)}{x}\right)'dx+\int_3^4\left(\frac{\sin(x)}{x}\right)'dx+\cdots$$

$$=\int_0^{\infty}\left(\frac{\sin(x)}{x}\right)'dx=\lim_{x\to\infty}\frac{\sin(x)}{x}-\lim_{x\to 0}\frac{\sin(x)}{x}=-1$$

SO, $\boxed{\sum_{n=0}^{\infty}\left[\int_{2n}^{2n+1}\left(\frac{\cos(x)}{x}-\frac{\sin(x)}{x^2}\right)dx\right]+\sum_{n=1}^{\infty}\left[\int_{2n}^{2n-1}\left(\frac{\cos(x)}{x}-\frac{\sin(x)}{x^2}\right)dx\right]=-1}$

041. Calculate Integral $I=\int_0^1(-1)^x e^{\frac{\pi}{2}i}dx \quad ,\left(i^2=-1\right)$

Solution; $I=\int_0^1(-1)^x e^{\frac{\pi}{2}i}dx \quad ,\left(i^2=-1\right)$

By; $0+i=\cos\left(\frac{\pi}{2}\right)+i\sin\left(\frac{\pi}{2}\right)=e^{\frac{\pi}{2}i}$ **And** $(-1)^x=e^{ik\pi x} \quad \forall k\in(1,3,5,7,...)$

They got it $I=\int_0^1 e^{ik\pi x}\cdot idx=i\int_0^1 e^{ik\pi x}dx$

$$=i\times\frac{1}{ik\pi}e^{ik\pi x}\Big|_0^1=\frac{1}{k\pi}(e^{ik\pi}-1)$$

$$=\frac{1}{k\pi}\left[(-1)^k-1\right]=-\frac{2}{\pi k}$$

$SO,$ $\boxed{\int_0^1(-1)^x e^{\frac{\pi}{2}i}dx=-\frac{2}{k\pi}}$

042. Calculate Integral $I=\int_0^1 x^x dx$

Solution; $I=\int_0^1 x^x dx$

By : $e^x=\sum_{n=0}^{\infty}\left(\frac{x^n}{n!}\right)$ **That :** $x^x=e^{x\log(x)}=\sum_{n=0}^{\infty}\left(\frac{\left(x\log(x)\right)^n}{n!}\right)$

They got it $I=\sum_{n=0}^{\infty}\left(\int_0^1\frac{\left(x\log(x)\right)^n}{n!}\right)dx$

$$=\sum_{n=0}^{\infty}\left(\frac{1}{n!}\int_0^1 x^n\log^n(x)dx\right)$$

Let : $t=-\log(x)\Leftrightarrow x=e^{-t}\Rightarrow dx=-e^{-t}dt$ **; If :** $x\in(0,1)\Rightarrow t\in(\infty,0)$

That $J=-\sum_{n=0}^{\infty}\left(\frac{1}{n!}\int_{\infty}^0 e^{-tn}(-t)^n e^{-t}dt\right)$

$$=\sum_{n=0}^{\infty}\left(\frac{(-1)^n}{n!}\int_0^{\infty}e^{-t(n+1)}(t)^n dt\right)$$

Let : $u=t(n+1)\Leftrightarrow\frac{u}{n+1}=t\Rightarrow\frac{du}{n+1}=dt$ **; If :** $t\in(0,\infty)\Rightarrow u\in(0,\infty)$

That $I=\sum_{n=0}^{\infty}\left(\frac{(-1)^n}{n!}\int_0^{\infty}e^{-u}\left(\frac{u}{n+1}\right)^n\times\frac{du}{n+1}\right)$

$$=\sum_{n=0}^{\infty}\left(\frac{(-1)^n}{(n+1)^{n+1}n!}\int_0^{\infty}u^ne^{-u}du\right)$$

$$=\sum_{n=0}^{\infty}\left(\frac{(-1)^n}{(n+1)^{n+1}n!}\cdot\Gamma(n+1)\right)$$ **Note:** $\Gamma(n+1)=n!$

$$=\sum_{n=0}^{\infty}\left(\frac{(-1)^n}{(n+1)^{n+1}}\right)$$

$$=1-\frac{1}{2^2}+\frac{1}{3^3}-\frac{1}{4^4}+\cdots$$

SO, $\boxed{\int_0^1 x^x dx=1-\frac{1}{2^2}+\frac{1}{3^3}-\frac{1}{4^4}+\cdots}$

043. Calculate Integral $I=\int_0^{\infty}\left\lfloor ne^{-x}\right\rfloor dx\ ,n\in\mathbb{N}$

Solution; $I=\int_0^{\infty}\left\lfloor ne^{-x}\right\rfloor dx\ ,n\in\mathbb{N}$

Let : $y=ne^{-x}\Rightarrow dy=-ne^{-x}dx\Leftrightarrow dx=-\frac{1}{y}dy$ **; If :** $x\in(0,\infty)\Rightarrow y\in(n,0)$

They got it $I=-\int_n^0\frac{\lfloor y\rfloor}{y}dy=\int_0^n\frac{\lfloor y\rfloor}{y}dy$

$$=\int_0^1\frac{\lfloor y\rfloor}{y}dy+\int_1^2\frac{\lfloor y\rfloor}{y}dy+\int_2^3\frac{\lfloor y\rfloor}{y}dy+\ldots+\int_{n-1}^n\frac{\lfloor y\rfloor}{y}dy$$

$$=\int_0^1\frac{0}{y}dy+\int_1^2\frac{1}{y}dy+\int_2^3\frac{2}{y}dy+\ldots+\int_{n-1}^n\frac{n-1}{y}dy$$

$$=(\log(2)-\log(1))+2(\log(3)-\log(2))+\ldots+(n-1)(\log(n)-\log(n-1))$$

$$=\log\left(\frac{1}{(n-1)!}\right)+\log(n^{n-1})=\log\left(\frac{n^{n-1}}{(n-1)!}\right)$$

SO, $\boxed{\int_0^{\infty}\left\lfloor ne^{-x}\right\rfloor dx=\log\left(\frac{n^{n-1}}{\Gamma(n)}\right)}$

044. Calculate Integral $I=\int_1^e \frac{\log^2(x)}{x^3}dx$

Solution; $I=\int_1^e \frac{\log^2(x)}{x^3}dx=\int_1^e x^{-3}\log^2(x)dx$

Let : $y=\log(x)\Rightarrow x=e^y \Rightarrow dx=e^y dy$ **; If :** $x\in(1,e)\Rightarrow y\in(0,1)$

They got it $I=\int_0^1 e^{-3y}y^2e^y dy=\int_0^1 e^{-2y}y^2dy$

$$=\left(-\frac{1}{2}y^2-\frac{1}{2}y-\frac{1}{4}\right)e^{-2y}\Big|_0^1=\frac{1}{4}-\frac{5}{4}e^{-2}$$

SO, $\boxed{\int_1^e \frac{\log^2(x)}{x^3}dx=\frac{e^{-2}}{4}\left(e^2-5\right)}$

045. Calculate Integral $I=\int_0^{\frac{\pi}{2}} \frac{\sin^2(x)}{\sin(x)+\cos(x)}dx$

Solution; $I=\int_0^{\frac{\pi}{2}} \frac{\sin^2(x)}{\sin(x)+\cos(x)}dx \quad (!)$

$$=\int_0^{\frac{\pi}{2}} \frac{\sin^2\left(\frac{\pi}{2}-x\right)}{\sin\left(\frac{\pi}{2}-x\right)+\cos\left(\frac{\pi}{2}-x\right)}dx$$

$$=\int_0^{\frac{\pi}{2}} \frac{\cos^2(x)}{\sin(x)+\cos(x)}dx \quad (!!)$$

Take $(!)+(!!)$ **That Have :** $2I=\int_0^{\frac{\pi}{2}} \frac{\sin^2(x)+\cos^2(x)}{\sin(x)+\cos(x)}dx$

They got it $I=\frac{1}{2}\int_0^{\frac{\pi}{2}} \frac{1}{\sin(x)+\cos(x)}dx$

$$= \frac{1}{2\sqrt{2}} \int_0^{\frac{\pi}{2}} \frac{1}{\cos(45^\circ)\cos(x) + \sin(45^\circ)\sin(x)} dx$$

$$= \frac{1}{2\sqrt{2}} \int_0^{\frac{\pi}{2}} \frac{1}{\cos\left(x - \frac{\pi}{4}\right)} dx$$

$$= \frac{1}{2\sqrt{2}} \int_0^{\frac{\pi}{2}} \sec\left(x - \frac{\pi}{4}\right) dx$$

$$= \frac{1}{2\sqrt{2}} \left[\log\left| \sec\left(x - \frac{\pi}{4}\right) + \tan\left(x - \frac{\pi}{4}\right) \right| \right]_0^{\frac{\pi}{2}}$$

$$= \frac{1}{2\sqrt{2}} \left[\log\left(\sqrt{2}+1\right) - \log\left(\sqrt{2}-1\right) \right]$$

$$= \frac{\sqrt{2}}{2} \log\left(\sqrt{2}+1\right)$$

SO, $\boxed{\int_0^{\frac{\pi}{2}} \frac{\sin^2(x)}{\sin(x)+\cos(x)} dx = \frac{\sqrt{2}}{2} \log\left(\sqrt{2}+1\right)}$

046. Calculate Integral $I = \int_0^{\frac{\pi}{2}} \sqrt{\tan(x)} dx$

Method 1

Solution; $I = \int_0^{\frac{\pi}{2}} \sqrt{\tan(x)} dx \quad (1)$

$$= \int_0^{\frac{\pi}{2}} \sqrt{\tan\left(\frac{\pi}{2} - x\right)} dx$$

$$= \int_0^{\frac{\pi}{2}} \sqrt{\cot(x)} dx \quad (2)$$

Take $(1)+(2)$ **That Have :** $2I = \int_0^{\frac{\pi}{2}} \left(\sqrt{\tan(x)} + \sqrt{\cot(x)} \right) dx$

$$= \sqrt{2}\int_0^{\frac{\pi}{2}} \left(\frac{\sin(x)+\cos(x)}{\sqrt{1-\left(\sin(x)-\cos(x)\right)^2}} \right) dx$$

$$= \sqrt{2}\arcsin\left(\sin(x)-\cos(x)\right)\Big|_0^{\frac{\pi}{2}}$$

$$= \pi\sqrt{2}$$

They got it $I = \dfrac{\pi\sqrt{2}}{2}$

Method 2

Solution; $I = \int_0^{\frac{\pi}{2}} \sqrt{\tan(x)}dx = \int_0^{\frac{\pi}{2}} \sin^{\frac{1}{2}}(x)\cos^{-\frac{1}{2}}(x)dx$

$$= \int_0^{\frac{\pi}{2}} \sin^{2\left(\frac{3}{4}\right)-1}(x)\cos^{2\left(\frac{1}{4}\right)-1}(x)dx = \frac{1}{2}B\left(\frac{1}{4},\frac{3}{4}\right)$$

$$= \frac{1}{2}\times\frac{\Gamma\left(1/4\right)\Gamma\left(3/4\right)}{\Gamma\left(1/4+3/4\right)} = \frac{\pi}{2}\csc\left(\frac{\pi}{4}\right)$$

$$= \frac{\pi}{\sqrt{2}}$$

SO, $\boxed{\int_0^{\frac{\pi}{2}} \sqrt{\tan(x)}dx = \frac{\pi}{\sqrt{2}}}$

047. Calculate Integral $I = \int_0^1 \dfrac{\log(1-x^2)}{(1+x^2)}dx$

Solution; $I = \int_0^1 \dfrac{\log(1-x^2)}{(1+x^2)}dx$

Let : $x = \tan(u) \Rightarrow dx = \left(1+\tan^2(u)\right)du$ **; If** : $x \in (1,0) \Rightarrow u \in \left(\dfrac{\pi}{4},0\right)$

They got it $I = \int_0^{\frac{\pi}{4}} \dfrac{\log\left(1-\tan^2(u)\right)}{\left(1+\tan^2(u)\right)} \times \left(1+\tan^2(u)\right)du$

$$= \int_0^{\frac{\pi}{4}} \log\left(\frac{\cos^2(u)-\sin^2(u)}{\cos^2(u)}\right)du$$

$$= \int_0^{\frac{\pi}{4}} \log\big(\cos(2u)\big)du - 2\int_0^{\frac{\pi}{4}} \log\big(\cos(u)\big)du$$

$$= \frac{1}{2}\underbrace{\int_0^{\frac{\pi}{4}} \log\big(\cos(2u)\big)d(2u)}_{I_2} - 2\underbrace{\int_0^{\frac{\pi}{4}} \log\big(\cos(u)\big)du}_{I_1}$$

For : $I_1 = \int_0^{\frac{\pi}{4}} \log\big(\cos(u)\big)du = -\frac{\pi}{4}\log(2) + \frac{1}{2}G$

For : $I_2 = \underbrace{\int_0^{\frac{\pi}{4}} \log\big(\cos(2u)\big)d(2u)}_{Let: t=2u}$

$$= \int_0^{\frac{\pi}{2}} \log\big(\cos(t)\big)dt = -\frac{\pi}{2}\log(2)$$

That $I = \frac{1}{2}\left(-\frac{\pi}{2}\log(2)\right) - 2\left(-\frac{\pi}{4}\log(2) + \frac{1}{2}G\right)$

$$= \frac{\pi}{4}\log(2) - G$$

SO, $\boxed{\int_0^1 \frac{\log(1-x^2)}{(1+x^2)}dx = \frac{\pi}{4}\log(2) - G}$

048. Calculate Integral $I = \int_0^{\frac{\pi}{4}} \log\big(\cos(x)\big)dx$

Solution; $I = \int_0^{\frac{\pi}{4}} \log\big(\cos(x)\big)dx$

Take $J = \int_0^{\frac{\pi}{4}} \log\big(\sin(x)\big)dx$

Take $J + K = \int_0^{\frac{\pi}{4}} \big[\log\big(\sin(2x)\big) - \log(2)\big]dx$

$$= -\frac{\pi}{4}\log(2) + \frac{1}{2}\int_0^{\frac{\pi}{4}} \log\big(\sin(2x)\big)d(2x)$$

$$= -\frac{\pi}{4}\log(2) + \frac{1}{2}\left(-\frac{\pi}{2}\log(2)\right) = -\frac{\pi}{2}\log(2)$$

Take $J - K = \int_0^{\frac{\pi}{4}} \log\big(\cot(x)\big)dx = -\int_0^{\frac{\pi}{4}} \log\big(\tan(x)\big)dx$

Let: $y=\tan(x)\Rightarrow dx=\dfrac{1}{1+y^2}dy$ **; If :** $x\in\left(0,\dfrac{\pi}{4}\right)\Rightarrow y\in(0,1)$

They got it $J-K=-\int_0^1\dfrac{\log(y)}{1+y^2}dy$

$$=-\int_0^1\left(\sum_{n=0}^{\infty}(-1)^n y^{2n}\log(y)\right)dy$$

$$=-\sum_{n=0}^{\infty}\left((-1)^n\int_0^1 y^{2n}\log(y)\,dy\right)$$

Let: $u=\log(y)\Rightarrow dt=\dfrac{1}{y}dy$ **And** $dv=y^{2n}dy\Rightarrow v=\dfrac{y^{2n+1}}{2n+1}$

That $I-J=-\sum_{n=0}^{\infty}\left[(-1)^n\left(\underbrace{\dfrac{\log(y)y^{2n+1}}{2n+1}\Big|_0^1}_{0}-\int_0^1\dfrac{y^{2n}}{2n+1}dy\right)\right]$

$$=\sum_{n=0}^{\infty}\left((-1)^n\int_0^1\frac{y^{2n}}{2n+1}dy\right)=\sum_{n=0}^{\infty}\frac{(-1)^n}{(2n+1)^2}=G$$

Take $(I+J)+(I-J)=-\dfrac{\pi}{2}\log(2)+G$

That $I=\dfrac{1}{2}\left(G-\dfrac{\pi}{2}\log(2)\right)$

$SO,$ $\boxed{\int_0^{\frac{\pi}{4}}\log\left(\cos(x)\right)dx=\frac{1}{2}\left(G-\frac{\pi}{2}\log(2)\right)}$

049. Calculate Integral $I=\int_0^1\dfrac{\tan^{-1}(x)}{\left(1+x^2\right)^2}dx$

Solution; $I=\int_0^1\dfrac{\tan^{-1}(x)}{\left(1+x^2\right)^2}dx$

Let : $x=\tan(u)\Rightarrow dx=\left(1+\tan^2(u)\right)du$ **; If :** $x\in(0,1)\Rightarrow u\in\left(0,\dfrac{\pi}{4}\right)$

They got it $I=\int_0^{\frac{\pi}{4}}\frac{u}{\left(1+\tan^2(u)\right)^2}\left(1+\tan^2(u)\right)du$

$$=\int_0^{\frac{\pi}{4}}\frac{u}{\left(1+\tan^2(u)\right)}du$$

$$=\frac{1}{2}\int_0^{\frac{\pi}{4}}u\left(1+\cos(2u)\right)du$$

$$=\frac{1}{2}\int_0^{\frac{\pi}{4}}udu+\frac{1}{2}\underbrace{\int_0^{\frac{\pi}{4}}u\cos(2u)du}_{\text{Use Integral By Parts}}$$

$$=\frac{1}{4}u^2\Big|_0^{\frac{\pi}{4}}+\frac{1}{2}\left[\frac{1}{2}u\sin(2u)+\frac{1}{4}\cos(2u)\right]\Big|_0^{\frac{\pi}{4}}$$

$$=\frac{\pi^2}{64}+\frac{1}{2}\left(\frac{\pi}{8}-\frac{1}{4}\right)$$

SO, $\boxed{\int_0^1\frac{\tan^{-1}(x)}{\left(1+x^2\right)^2}dx=\frac{\pi^2}{64}+\frac{\pi}{16}-\frac{1}{8}}$

050. Calculate Integral $I=\int_0^1\frac{\log\left(\frac{1}{x^2}\right)}{\left(1+x^2\right)^2}dx$

Solution; $I=\int_0^1\frac{\log\left(\frac{1}{x^2}\right)}{\left(1+x^2\right)^2}dx$

Let : $y=-\log(x)\Leftrightarrow x=e^{-y}\Rightarrow dx=-e^{y}dy$ **; If** : $x\in(0,1)\Rightarrow y\in(\infty,0)$

They got it $I=-2\int_\infty^0\frac{ye^{-y}}{\left(1+e^{-2y}\right)^2}dy$

$$= 2\int_0^{\infty}\left[y\times\frac{e^{-y}}{\left(1+e^{-2y}\right)^2}\right]dy$$

$$= 2\int_0^{\infty}\left(y\sum_{n=0}^{\infty}(-1)^n(n+1)e^{-(2n+1)y}\right)dy$$

$$= 2\sum_{n=0}^{\infty}\left((-1)^n(n+1)\int_0^{\infty} ye^{-(2n+1)y}dy\right)$$

$$= 2\sum_{n=0}^{\infty}\left((-1)^n(n+1)\frac{\Gamma(2)}{(2n+1)^2}\right)$$

$$= 2\sum_{n=0}^{\infty}\left((-1)^n\frac{(n+1)}{(2n+1)^2}\right)$$

$$= \left(\sum_{n=0}^{\infty}\frac{(-1)^n}{(2n+1)}+\sum_{n=0}^{\infty}\frac{(-1)^n}{(2n+1)^2}\right)$$

$$= \left(\frac{\pi}{4}+G\right)$$

$$SO,\ \boxed{\int_0^1\frac{\log\left(1/x^2\right)}{\left(1+x^2\right)^2}dx=\left(\frac{\pi}{4}+G\right)}$$

051. Calculate Integral $I=\int_0^{\infty}\frac{1}{1+x+x^2+x^3+x^4+x^5}dx$

Solution; $I=\int_0^{\infty}\frac{1}{1+x+x^2+x^3+x^4+x^5}dx$ (!)

Let : $x=\frac{1}{t}\Rightarrow dx=-\frac{1}{t^2}dt$ **; If :** $x\in(0,\infty)\Rightarrow t\in(\infty,0)$

They got it $=-\int_{\infty}^0\frac{1}{1+\frac{1}{t}+\frac{1}{t^2}+\frac{1}{t^3}+\frac{1}{t^4}+\frac{1}{t^5}}\times\frac{1}{t^2}dt\quad(x=t)$

$$=\int_0^{\infty}\frac{x^3}{1+x+x^2+x^3+x^4+x^5}dx\quad(!!)$$

Take $(!)+(!!)$ **We Have** : $2I=\int_0^{\infty}\frac{x^3+1}{1+x+x^2+x^3+x^4+x^5}dx$

$$=\int_0^{\infty}\frac{(x^3+1)}{(1+x^3)(1+x+x^2)}dx$$

$$=\int_0^{\infty}\frac{1}{(1+x+x^2)}dx$$

$$=\int_0^{\infty}\frac{1}{\left(x+\frac{1}{2}\right)^2+\frac{3}{4}}dx$$

Let : $\left(x+\frac{1}{2}\right)=\frac{\sqrt{3}}{2}\tan(u)\Rightarrow dx=\frac{\sqrt{3}}{2}\left(1+\tan^2(u)\right)du$; **If** : $x\in(0,\infty)\Rightarrow u\in\left(\frac{\pi}{6},\frac{\pi}{2}\right)$

They got it $2I=\int_{\frac{\pi}{6}}^{\frac{\pi}{2}}\frac{\frac{\sqrt{3}}{2}\left(1+\tan^2(u)\right)}{\frac{3}{4}\tan^2(u)+\frac{3}{4}}du$

That $I=\frac{\sqrt{3}}{3}\left(\frac{\pi}{2}-\frac{\pi}{6}\right)=\frac{\pi}{3\sqrt{3}}$

SO, $\boxed{\int_0^{\infty}\frac{1}{1+x+x^2+x^3+x^4+x^5}dx=\frac{\pi}{3\sqrt{3}}}$

052. Calculate Integral $I=\int_0^1 x\left\lfloor\frac{1}{x}\right\rfloor dx$

Solution; $I=\int_0^1 x\left\lfloor\frac{1}{x}\right\rfloor dx$

Let : $u=\frac{1}{x}\Rightarrow dx=-\frac{1}{u^2}du$; **If** : $x\in(0,1)\Rightarrow u\in(\infty,1)$

They got it $I=-\int_{\infty}^1\left(\frac{\lfloor u\rfloor}{u^3}\right)du=\lim_{n\to\infty}\int_1^n\left(\frac{\lfloor u\rfloor}{u^3}\right)du$

By: $\forall k\in\mathbb{Z},x\in\mathbb{R}\ :k\le x\le k+1\Rightarrow\lfloor x\rfloor=k$

That $=\lim_{n\to\infty}\left(\int_1^2\frac{\lfloor u\rfloor}{u^3}du+\int_2^3\frac{\lfloor u\rfloor}{u^3}du+\int_3^4\frac{\lfloor u\rfloor}{u^3}du+\ldots+\int_{n-1}^n\frac{\lfloor u\rfloor}{u^3}du\right)$

$$=\lim_{n\to\infty}\left(\int_1^2\frac{1}{u^3}du+\int_2^3\frac{2}{u^3}du+\int_3^4\frac{3}{u^3}du+...+\int_{n-1}^n\frac{n-1}{u^3}du\right)$$

$$=-\frac{1}{2}\lim_{n\to\infty}\left(\frac{1}{u^2}\Big|_1^2+\frac{2}{u^2}\Big|_2^3+\frac{3}{u^2}\Big|_3^4+...+\frac{n-1}{u^2}\Big|_{n-1}^n\right)$$

$$=-\frac{1}{2}\lim_{n\to\infty}\left(\frac{1}{2^2}-\frac{1}{1^2}+\frac{2}{3^2}-\frac{2}{2^2}+\frac{3}{4^2}-\frac{3}{3^2}+...+\frac{n-1}{n^2}-\frac{n-1}{(n-1)^2}\right)$$

$$=-\frac{1}{2}\lim_{n\to\infty}\left(-\frac{1}{1^2}-\frac{1}{2^2}-\frac{1}{3^2}-\frac{1}{4^2}-...-\frac{1}{(n-1)^2}-\frac{1}{n^2}+\frac{1}{n}\right)$$

$$=\frac{1}{2}\lim_{n\to\infty}\left(\frac{1}{1^2}+\frac{1}{2^2}+\frac{1}{3^2}+...+\frac{1}{(n-1)^2}+\frac{1}{n^2}-\frac{1}{n}\right)$$

$$=\frac{1}{2}\zeta(2)-\frac{1}{2}\lim_{n\to\infty}\frac{1}{n}$$

$$=\frac{1}{2}\times\frac{\pi^2}{6}-0$$

$$=\frac{\pi^2}{12}$$

SO, $\boxed{\int_0^1 x\left\lfloor\frac{1}{x}\right\rfloor dx=\frac{\pi^2}{12}}$

053. Calculate Integral $I=\int_0^1\sin\left(\sqrt{-\log(x)}\right)dx$

Solution; $I=\int_0^1\sin\left(\sqrt{-\log(x)}\right)dx$

Let : $u=\sqrt{-\log(x)}\Rightarrow x=e^{-u^2}\Rightarrow dx=-2ue^{-u^2}du$ **; If :** $x\in(0,1)\Rightarrow u\in(\infty,0)$

They got it $=-2\int_\infty^0 u\sin(u)e^{-u^2}du$

$$=2\int_0^\infty u\sin(u)e^{-u^2}du$$

$$=\underbrace{-\sin(u)e^{-u^2}\Big|_0^\infty}_{0}+\int_0^\infty\cos(u)e^{-u^2}du$$

$$=\int_0^\infty\cos(u)e^{-u^2}du$$

That $I(a)=\int_0^{\infty}\cos(au)e^{-u^2}du$

$$\Rightarrow I'(a)=\frac{a}{2}\int_0^{\infty}\sin(au)(-2u)e^{-u^2}du$$

$$=\frac{a}{2}\left[\underbrace{\sin(u)e^{-u^2}\Big|_0^{\infty}}_{0}-\int_0^{\infty}\cos(au)e^{-u^2}du\right]$$

$$=-\frac{a}{2}\left(\int_0^{\infty}\cos(au)e^{-u^2}du\right)$$

Equivalent $I'(a)=-\frac{a}{2}I(a)\Leftrightarrow I'(a)+\frac{a}{2}I(a)=0$ **That** $I(a)=C\cdot e^{-\frac{a^2}{4}}$

If : $a=1\Rightarrow I(1)=I=C\cdot e^{-\frac{1}{4}}$

If: $a=0\Rightarrow I(0)=\int_0^{\infty}e^{-u^2}du=\frac{\sqrt{\pi}}{2}$

They got it $I=\frac{\sqrt{\pi}}{2}e^{-\frac{1}{4}}$

SO, $\boxed{\int_0^1\sin\left(\sqrt{-\log(x)}\right)dx=\frac{\sqrt{\pi}}{2}e^{-\frac{1}{4}}}$

054. Calculate Integral $I=\int_0^1\frac{\log(x)}{1+x}dx$

Solution; $I=\int_0^1\frac{\log(x)}{1+x}dx$

They got it $I=\int_0^1\left(\sum_{n=0}^{\infty}(-1)^n x^n\log(x)\right)dx$ **Because :** $\frac{1}{1+x}=\sum_{n=0}^{\infty}\left((-1)^n x^n\right)$

$$=\sum_{n=0}^{\infty}(-1)^n\int_0^1 x^n\log(x)\,dx$$

Let : $u=\log(x)\Rightarrow du=\frac{1}{x}dx$ **And** $dv=x^n dx\Rightarrow v=\frac{x^{n+1}}{n+1}$

That $I=\sum_{n=0}^{\infty}\left[(-1)^n\left(\underbrace{\frac{x^{n+1}\log(x)}{n+1}\Big|_0^1}_{0}-\int_0^1\frac{x^{n+1}}{n+1}\times\frac{1}{x}dx\right)\right]$

$$=-\sum_{n=0}^{\infty}\left[\frac{(-1)^n}{n+1}\left(\int_0^1 x^n dx\right)\right]$$

$$=-\sum_{n=0}^{\infty}\frac{(-1)^n}{(n+1)^2}=-\left(\frac{1}{1^2}-\frac{1}{2^2}+\frac{1}{3^2}-\frac{1}{4^2}+\frac{1}{5^2}-\frac{1}{6^2}+\ldots\right)$$

$$=-\left[\left(1+\frac{1}{2^2}+\frac{1}{3^2}+\frac{1}{4^2}+\frac{1}{5^2}+\ldots\right)-\frac{2}{2^2}\left(1+\frac{1}{2^2}+\frac{1}{3^2}+\frac{1}{4^2}+\frac{1}{5^2}+\ldots\right)\right]$$

$$=-\frac{1}{2}\left(1+\frac{1}{2^2}+\frac{1}{3^2}+\frac{1}{4^2}+\frac{1}{5^2}+\ldots\right)$$

$$=-\frac{1}{2}\zeta(2)=-\frac{\pi^2}{12}$$

SO, $\boxed{\int_0^1\frac{\log(x)}{1+x}dx=-\frac{\pi^2}{12}}$

Note : $\eta(2)=\sum_{n=0}^{\infty}\frac{(-1)^n}{(n+1)^2}=(1-2^{1-2})\zeta(2)=\frac{1}{2}\times\frac{\pi^2}{6}=\frac{\pi^2}{12}$

055. Calculate Integral $I=\int_1^e\frac{\log\left[\Gamma\left(1-\log(x)\right)\right]}{x}dx$

Solution; $I=\int_1^e\frac{\log\left[\Gamma\left(1-\log(x)\right)\right]}{x}dx$

Let : $y=\log(x)\Rightarrow dy=\frac{1}{x}dx$ **; If :** $x\in(1,e)\Rightarrow y\in(0,1)$

They got it $I=\int_0^1\log\left(\Gamma(1-y)\right)dy\quad(1)$

$$=\int_0^1\log\left(\Gamma(y)\right)dy\qquad(2)$$

Take $(1)+(2)$ **We Have** $2I=\int_0^1\log\left(\Gamma(1-y)\right)dy+\int_0^1\log\left(\Gamma(y)\right)dy$

$$= \int_0^1 \log\left(\Gamma(y)\Gamma(1-y)\right)dy$$

$$= \int_0^1 \log\left(\frac{\pi}{\sin(\pi y)}\right)dy$$

$$= \int_0^1 \log(\pi)\,dy - \int_0^1 \log\left(\sin(\pi y)\right)dy$$

$$= \log(\pi) - \int_0^1 \log\left(\sin(\pi y)\right)dy \quad (*)$$

Take : $I_1 = \int_0^1 \log\left(\sin(\pi y)\right)dy$

Let : $u = \pi y \Rightarrow \frac{du}{\pi} = dy$ **; If :** $y \in (0,1) \Rightarrow u \in (0,\pi)$

They got it $I = \frac{1}{\pi}\int_0^{\pi} \log\left(\sin(u)\right)du$

$$= \frac{2}{\pi}\int_0^{\frac{\pi}{2}} \log\left(\sin(u)\right)du$$

$$= \frac{2}{\pi}\left(-\frac{\pi}{2}\log(2)\right) = -\log(2)$$

Take (*) **We have** $2I = \log(\pi) + \log(2)$ **That** $I = \log\sqrt{2\pi}$

SO, $\boxed{\int_1^e \frac{\log\left[\Gamma\left(1-\log(x)\right)\right]}{x}dx = \log\sqrt{2\pi}}$

056. Calculate Integral $I = \int_0^1 \frac{1}{x\sqrt{1-x^2}}\log\left(\frac{2+x}{2-x}\right)dx$

Solution; $I = \int_0^1 \frac{1}{x\sqrt{1-x^2}}\log\left(\frac{2+x}{2-x}\right)dx$

They got it $I(a) = \int_0^1 \frac{1}{x\sqrt{1-x^2}}\log\left(\frac{a+x}{a-x}\right)dx$

That $I'(a)=\int_0^1 \frac{1}{x\sqrt{1-x^2}}\left(\frac{1}{a+x}-\frac{1}{a-x}\right)dx$

$$=\int_0^1 \frac{1}{x\sqrt{1-x^2}}\left(\frac{-2x}{(a+x)(a-x)}\right)dx$$

$$=-2\int_0^1 \frac{1}{\left(a^2-x^2\right)\sqrt{1-x^2}}dx$$

Let : $x=\sin(y)\Rightarrow dx=\cos(y)dy$; **If** : $x\in(0,1)\Rightarrow y\in\left(0,\frac{\pi}{2}\right)$

They got it $I=-2\int_0^{\frac{\pi}{2}} \frac{\cos(y)}{\left(a^2-\sin^2(y)\right)\sqrt{1-\sin^2(y)}}dy$

$$=-2\int_0^{\frac{\pi}{2}} \frac{1}{\left(a^2-\sin^2(y)\right)}dy$$

$$=-2\int_0^{\frac{\pi}{2}} \frac{\sec^2(y)}{\left(a^2\left(1+\tan^2(y)\right)-\tan^2(y)\right)}dy$$

$$=\frac{-2}{a^2-1}\int_0^{\frac{\pi}{2}} \frac{1}{\left(\frac{a^2}{a^2-1}+\tan^2(y)\right)}d(\tan y)$$

$$=\frac{-2}{1-a^2}\left[\frac{\sqrt{1-a^2}}{a}\tan^{-1}\left(\frac{\sqrt{1-a^2}}{a}\tan(y)\right)\right]_0^{\frac{\pi}{2}}$$

$$=\frac{-2}{(1-a^2)}\times\frac{\sqrt{(1-a^2)}}{a}\times\frac{\pi}{2}$$

$$=\frac{-\pi}{a\sqrt{1-a^2}}$$

Equivalent $\int I'(a)da=\int \frac{-\pi}{a\sqrt{1-a^2}}da\Rightarrow I(a)=-\pi\sec^{-1}(a)+C$

If : $a=2\Rightarrow I(2)=I=-\pi\sec^{-1}(2)+c=-\frac{\pi^2}{3}+C$

If : $a=\infty \Rightarrow I(\infty)=0=-\pi\sec^{-1}(\infty)+C$ **That** $C=\frac{\pi^2}{2}$

They got it $I=-\frac{\pi^2}{3}+\frac{\pi^2}{2}=\frac{\pi^2}{6}$

SO, $\boxed{\int_0^1 \frac{1}{x\sqrt{1-x^2}}\log\left(\frac{2+x}{2-x}\right)dx=\frac{\pi^2}{6}}$

057. Calculate Integral $I=\int_0^\infty \frac{\sqrt{x}}{(x+9)^2}dx$

Solution; $I=\int_0^\infty \frac{\sqrt{x}}{(x+9)^2}dx$

Let : $x=9y\Rightarrow dx=9dy$ **; If** : $x\in(0,\infty)\Rightarrow y\in(0,\infty)$

They got it $I=\frac{1}{81}\int_0^\infty \frac{\sqrt{9y}}{(y+1)^2}9dy$

$$=\frac{1}{3}\int_0^\infty \frac{\sqrt{y}}{(y+1)^2}dy$$

$$=\frac{1}{3}\int_0^\infty \frac{y^{\frac{3}{2}-1}}{(y+1)^{\frac{3}{2}+\frac{1}{2}}}dy$$

$$=\frac{1}{3}B\left(\frac{3}{2},\frac{1}{2}\right)=\frac{1}{3}\times\frac{\Gamma\left(\frac{3}{2}\right)\Gamma\left(\frac{1}{2}\right)}{\Gamma(2)}=\frac{1}{6}\Gamma^2\left(\frac{1}{2}\right)=\frac{\pi}{6}$$

Note $\Gamma\left(\frac{1}{2}\right)=\sqrt{\pi}$

SO, $\boxed{\int_0^\infty \frac{\sqrt{x}}{(x+9)^2}dx=\frac{\pi}{6}}$

058. Calculate Integral $I=\int_0^1 \frac{\pi-4\tan^{-1}(x)}{1-x^2}dx$

Solution; $I=\int_0^1 \frac{\pi-4\tan^{-1}(x)}{1-x^2}dx$

$$=4\int_0^1 \frac{\tan^{-1}(1)-\tan^{-1}(x)}{1-x^2}dx$$

Use : $\tan^{-1}(x)-\tan^{-1}(y)=\tan^{-1}\left(\frac{x-y}{1+xy}\right)$

$$=4\int_0^1 \frac{\tan^{-1}\left(\frac{1-x}{1+x}\right)}{1-x^2}dx$$

Let : $y=\frac{1-x}{1+x} \Leftrightarrow x=\frac{1-y}{1+y} \Rightarrow dx=-\frac{2}{(1+y)^2}dy$ **; If :** $x\in(0,1)\Rightarrow y\in(1,0)$

They got it $I=-4\int_1^0 \frac{\tan^{-1}(y)}{1-\left(\frac{1-y}{1+y}\right)^2}\times\frac{2}{(1+y)^2}dy$

$$=8\int_0^1 \frac{\tan^{-1}(y)}{(1+y)^2-(1-y)^2}dy$$

$$=2\int_0^1 \frac{\tan^{-1}(y)}{y}dy$$

By : $\tan^{-1}(x)=\sum_{n=0}^{\infty}\frac{(-1)^n x^{2n+1}}{(2n+1)}$

$$=2\int_0^1 \left(\frac{1}{y}\sum_{n=0}^{\infty}\frac{(-1)^n x^{2n+1}}{(2n+1)}\right)dy$$

$$=2\sum_{n=0}^{\infty}\left(\frac{(-1)^n}{(2n+1)}\int_0^1 \frac{y^{2n+1}}{y}dy\right)$$

$$=2\sum_{n=0}^{\infty}\left(\frac{(-1)^n}{(2n+1)^2}y^{2n+1}\Big|_0^1\right)$$

$$=2\sum_{n=0}^{\infty}\frac{(-1)^n}{(2n+1)^2}=2G$$

SO, $\boxed{\int_0^1 \frac{\pi-4\tan^{-1}(x)}{1-x^2}dx=2G}$

059. Calculate Integral $I=\int_0^1 \frac{x^p\log(x)}{x-1}dx$

Solution; $I=\int_0^1 \frac{x^p\log(x)}{x-1}dx$

$$=-\sum_{n=0}^{\infty}\int_0^1 x^n x^p\log(x)dx$$

Note : $\frac{1}{x-1}=-\frac{1}{1-x}=-\sum_{n=0}^{\infty}\left(x^n\right)$

They got it $I=-\sum_{n=0}^{\infty}\left(\int_0^1 x^{n+p}\log(x)dx\right)$ **(Use Partial Integral)**

That $I=-\sum_{n=0}^{\infty}\left[\underbrace{\frac{x^{n+p+1}\log(x)}{n+p+1}\Big|_0^1}_{0}-\int_0^1\frac{x^{n+p+1}}{n+p+1}\times\frac{1}{x}dx\right]$

$$=\frac{1}{n+p+1}\int_0^1\left(x^{n+p}\right)dx=\sum_{n=0}^{\infty}\left(\frac{1}{(n+p+1)^2}\right)$$

$SO,$ $\boxed{\int_0^1\frac{x^p\log(x)}{x-1}dx=\frac{1}{(p+1)^2}+\frac{1}{(p+2)^2}+\frac{1}{(p+3)^2}+\cdots}$

060. Calculate Integral $I=\int_0^1\frac{1}{\sqrt{x(1-x)}}dx$

Solution; $I=\int_0^1\frac{1}{\sqrt{x(1-x)}}dx$

$$=\int_0^1 x^{-\frac{1}{2}}(1-x)^{-\frac{1}{2}}dx$$

$$=\int_0^1 x^{\frac{1}{2}-1}(1-x)^{\frac{1}{2}-1}dx=B\left(\frac{1}{2},\frac{1}{2}\right)$$

$$=\frac{\Gamma\left(\frac{1}{2}\right)\Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{1}{2}+\frac{1}{2}\right)}=\frac{\Gamma\left(\frac{1}{2}\right)\Gamma\left(\frac{1}{2}\right)}{\Gamma(1)}=\pi$$

$SO,$ $\boxed{\int_0^1\frac{1}{\sqrt{x(1-x)}}dx=\pi}$

061. Calculate Integral $I=\int_0^n \frac{\log(x)}{x^2+n^2}dx$

Solution; $I=\int_0^n \frac{\log(x)}{x^2+n^2}dx$

Let : $x=nt \Rightarrow dx=ndt$ **; If :** $x\in(0,n)\Rightarrow t\in(0,1)$

They got it $I=n\int_0^1 \frac{\log(nt)}{(nt)^2+n^2}dt$

$$=\int_0^1 \frac{\log(n)}{t^2+1}dt+\int_0^1 \frac{\log(t)}{t^2+1}dt$$

$$=\log(n)\tan^{-1}(t)\Big|_0^1+\int_0^1\left(\sum_{m=0}^{\infty}(-1)^m t^{2m}\log(t)\right)dt$$

$$=\frac{\pi\log(n)}{4}+\sum_{m=0}^{\infty}\left((-1)^m\underbrace{\int_0^1 t^{2m}\log(t)\,dt}_{Use\ Partial\ Integral}\right)$$

$$=\frac{\pi\log(n)}{4}+\sum_{m=0}^{\infty}\left[(-1)^m\left(\underbrace{\frac{t^{2m+1}\log(t)}{2m+1}\Big|_0^1}_{0}-\int_0^1\frac{t^{2m+1}}{2m+1}\times\frac{1}{t}dt\right)\right]$$

$$=\frac{\pi\log(n)}{4}-\sum_{m=0}^{\infty}\left[(-1)^m\left(\int_0^1\frac{t^{2m}}{2m+1}dt\right)\right]$$

$$=\frac{\pi\log(n)}{4}-\sum_{m=0}^{\infty}\left[(-1)^m\left(\frac{t^{2m+1}}{(2m+1)^2}\Big|_0^1\right)\right]$$

$$=\frac{\pi\log(n)}{4}-\sum_{m=0}^{\infty}\frac{(-1)^m}{(2m+1)^2}$$

$$=\frac{\pi\log(n)}{4}-G$$

SO, $\boxed{\int_0^n \frac{\log(x)}{x^2+n^2}dx=\frac{\pi\log(n)}{4}-G}$

062. Calculate Integral $I=\int_0^1 \frac{1}{\sqrt{1-x^3}}dx$

Solution; $I=\int_0^1 \frac{1}{\sqrt{1-x^3}}dx$

Let : $x=y^{\frac{1}{3}} \Rightarrow dx=\frac{1}{3}y^{-\frac{2}{3}}dy$ **; If :** $x\in(0,1)\Rightarrow y\in(0,1)$

They got it $I=\frac{1}{3}\int_0^1 \frac{y^{-\frac{2}{3}}}{\sqrt{1-y}}dy=\frac{1}{3}\int_0^1 y^{\frac{1}{3}-1}(1-y)^{\frac{1}{2}-1}dy$

$$=\frac{1}{3}B\left(\frac{1}{3},\frac{1}{2}\right)=\frac{1}{3}\times\frac{\Gamma\left(\frac{1}{2}\right)\Gamma\left(\frac{1}{3}\right)}{\Gamma\left(\frac{1}{2}+\frac{1}{3}\right)}=\frac{\sqrt{\pi}}{3}\times\frac{\Gamma\left(\frac{1}{3}\right)}{\Gamma\left(\frac{5}{6}\right)}$$

SO, $\boxed{\int_0^1 \frac{1}{\sqrt{1-x^3}}dx=\frac{\sqrt{\pi}}{3}\times\frac{\Gamma\left(\frac{1}{3}\right)}{\Gamma\left(\frac{5}{6}\right)}}$

063. Calculate Integral $I=\int_0^{\frac{\pi}{2}} \frac{1}{\sqrt{1+\sin(x)}}dx$

Solution; $I=\int_0^{\frac{\pi}{2}} \frac{1}{\sqrt{1+\sin(x)}}dx$

$$=\int_0^{\frac{\pi}{2}} \frac{1}{\sqrt{1+\sin\left(\frac{\pi}{2}-x\right)}}dx=\int_0^{\frac{\pi}{2}} \frac{1}{\sqrt{1+\cos(x)}}dx=\int_0^{\frac{\pi}{2}} \frac{1}{\sqrt{2\cos^2\left(\frac{x}{2}\right)}}dx$$

$$=\frac{1}{\sqrt{2}}\int_0^{\frac{\pi}{2}}\sec\left(\frac{x}{2}\right)dx=\frac{2}{\sqrt{2}}\left[\log\left|\sec\left(\frac{x}{2}\right)+\tan\left(\frac{x}{2}\right)\right|\right]_0^{\frac{\pi}{2}}=\sqrt{2}\log\left(\sqrt{2}+1\right)$$

SO, $\boxed{\int_0^{\frac{\pi}{2}} \frac{1}{\sqrt{1+\sin(x)}}dx=\sqrt{2}\log\left(\sqrt{2}+1\right)}$

064. Calculate Integral $I=\int_0^{\infty}\frac{\tan^{-1}(2x)}{\sqrt[3]{x^2}}dx$

Solution; $I=\int_0^{\infty}\frac{\tan^{-1}(2x)}{\sqrt[3]{x^2}}dx$

Let : $u=\tan^{-1}(2x)\Rightarrow du=\frac{2}{1+(2x)^2}dx$ **And** $dv=\int\frac{1}{\sqrt[3]{x^2}}dx\Rightarrow v=-2x^{-\frac{1}{2}}$

They got it $I=-2x^{-\frac{1}{2}}\tan^{-1}(2x)\Big|_0^{\infty}+\int_0^{\infty}\frac{4}{\left(1+4x^2\right)\sqrt{x}}dx$

$$=\int_0^{\infty}\frac{4}{\left(1+4x^2\right)\sqrt{x}}dx$$

Let : $y=\sqrt{x}\Rightarrow dx=2ydy$ **; If :** $x\in(0,\infty)\Rightarrow y\in(0,\infty)$

That $I=\int_0^{\infty}\frac{4}{\left(1+4y^4\right)y}\times 2ydy$

$$=8\int_0^{\infty}\frac{1}{\left(1+4y^4\right)}dy$$

Let : $t=4y^4\Rightarrow y=\frac{1}{\sqrt{2}}t^{\frac{1}{4}}\Rightarrow dy=\frac{1}{4\sqrt{2}}t^{\frac{1}{4}-1}dt$ **; If :** $y\in(0,\infty)\Rightarrow t\in(0,\infty)$

They got it $I=\frac{8}{4\sqrt{2}}\int_0^{\infty}\frac{t^{\frac{1}{4}-1}}{(1+t)^{\frac{1}{4}+\frac{3}{4}}}dy$

$$=\sqrt{2}B\left(\frac{1}{4},\frac{3}{4}\right)=\sqrt{2}\frac{\Gamma\left(\frac{1}{4}\right)\Gamma\left(\frac{3}{4}\right)}{\Gamma\left(\frac{1}{4}+\frac{3}{4}\right)}$$

$$=\sqrt{2}\Gamma\left(\frac{1}{4}\right)\Gamma\left(1-\frac{1}{4}\right)=\frac{\pi\sqrt{2}}{\sin\left(\frac{\pi}{4}\right)}=2\pi$$

SO, $\boxed{\int_0^{\infty}\frac{\tan^{-1}(2x)}{\sqrt[3]{x^2}}dx=2\pi}$

065. Calculate Integral $I=\int_0^{2\pi}\lfloor 2023\sin(x)\rfloor dx$

Solution; $I=\int_0^{2\pi}\lfloor 2023\sin(x)\rfloor dx$

By : $\begin{cases}0\le x\le\dfrac{\pi}{2}\Rightarrow \sin\left(0+x\right)=\sin(x)\\ \dfrac{\pi}{2}\le x\le\pi\Rightarrow \sin\left(\dfrac{\pi}{2}+x\right)=\cos(x)\end{cases}$ **And** $\begin{cases}\pi\le x\le\dfrac{3\pi}{2}\Rightarrow \sin\left(\pi+x\right)=-\sin(x)\\ \dfrac{3\pi}{2}\le x\le 2\pi\Rightarrow \sin\left(\dfrac{3\pi}{2}+x\right)=-\cos(x)\end{cases}$

That $I=\int_0^{\frac{\pi}{2}}\lfloor 2023\sin(x)\rfloor dx+\int_0^{\frac{\pi}{2}}\lfloor 2023\cos(x)\rfloor dx+\int_0^{\frac{\pi}{2}}\lfloor -2023\sin(x)\rfloor dx+\int_0^{\frac{\pi}{2}}\lfloor -2023\cos(x)\rfloor dx$

$=\int_0^{\frac{\pi}{2}}\lfloor 2023\sin(x)\rfloor dx+\int_0^{\frac{\pi}{2}}\lfloor 2023\cos(x)\rfloor dx$

$-\int_0^{\frac{\pi}{2}}\left(\lfloor 2023\sin(x)\rfloor+1\right)dx-\int_0^{\frac{\pi}{2}}\left(\lfloor 2023\cos(x)\rfloor+1\right)dx$

$=\int_0^{\frac{\pi}{2}}\lfloor 2023\sin(x)\rfloor dx-\int_0^{\frac{\pi}{2}}\lfloor 2023\sin(x)\rfloor dx+\int_0^{\frac{\pi}{2}}\lfloor 2023\cos(x)\rfloor dx$

$-\int_0^{\frac{\pi}{2}}\lfloor 2023\cos(x)\rfloor dx-\int_0^{\frac{\pi}{2}}2\,dx$

$=-\int_0^{\frac{\pi}{2}}2dx=-\pi$

SO, $\boxed{\int_0^{2\pi}\lfloor 2023\sin(x)\rfloor dx=-\pi}$

066. Calculate Integral $I=\int_0^{n}\dfrac{\log(x+1)}{x}dx$

Solution; $I=\int_0^{n}\dfrac{\log(x+1)}{x}dx$

Let : $x=-y\Rightarrow dx=-dy$ **; If :** $x\in(0,n)\Rightarrow y\in(0,-n)$

They got it $I=-\int_0^{-n}\dfrac{\log(1-y)}{-y}dx$

$=\int_0^{-n}\dfrac{\log(1-y)}{y}dx=-Li_2(-n)$

Note : $Li_2(n)=\int_0^{n}\dfrac{\log(-x+1)}{-x}dx$

SO, $\boxed{\int_0^{n}\dfrac{\log(x+1)}{x}dx=-Li_2(-n)}$

067. Calculate Integral $I=\int_1^2 \frac{\log(x+1)-\log(2)}{(x^2-1)}dx$

Solution; $I=\int_1^2 \frac{\log(x+1)-\log(2)}{(x^2-1)}dx$

$$=-\int_1^2 \frac{\log\left(1-\frac{x-1}{x+1}\right)}{\left(\frac{x-1}{x+1}\right)(x+1)^2}dx$$

Let : $u=\frac{x-1}{x+1}\Rightarrow \frac{1}{2}du=\frac{1}{(x+1)^2}dx$ **; If :** $x\in(1,2)\Rightarrow u\in\left(0,\frac{1}{3}\right)$

They got it $=-\frac{1}{2}\int_0^{\frac{1}{3}}\frac{\log(1-u)}{u}du=\frac{1}{2}Li_2\left(\frac{1}{3}\right)$

SO, $\boxed{\int_1^2 \frac{\log(x+1)-\log(2)}{(x^2-1)}dx=\frac{1}{2}Li_2\left(\frac{1}{3}\right)}$

068. Calculate Integral $I=\int_e^\infty x^{1-\log(x)}dx$

Solution; $I=\int_e^\infty x^{1-\log(x)}dx$

$$=\int_e^\infty e^{(1-\log(x))\log(x)}dx$$

Let : $x=e^t\Rightarrow dx=e^t dt$ **; If** $x\in(e,\infty)\Rightarrow t\in(1,\infty)$

They got it $I=\int_1^\infty e^{(1-t)t}e^t dt$

$$=e\int_1^\infty e^{-(t-1)^2}dt$$

Let : $u=t-1\Rightarrow du=dt$ **; If :** $x\in(e,\infty)\Rightarrow t\in(0,\infty)$

That $I=e\int_0^\infty e^{-u^2}du=\frac{e\sqrt{\pi}}{2}$

SO, $\boxed{\int_e^\infty x^{1-\log(x)}dx=\frac{e\sqrt{\pi}}{2}}$

069. Calculate Integral $I=\int_0^1 \lfloor x \rfloor^{-1}\,dx$

Solution; $I=\int_0^1 \lfloor x \rfloor^{-1}\,dx$

They got it $I=\int_{\frac{1}{2}}^{1} 1^{-1}dx+\int_{\frac{1}{3}}^{\frac{1}{2}} 2^{-1}dx+\int_{\frac{1}{4}}^{\frac{1}{3}} 3^{-1}dx+\int_{\frac{1}{5}}^{\frac{1}{4}} 4^{-1}dx+\cdots$

$$=\frac{1}{1}\left(1-\frac{1}{2}\right)+\frac{1}{2}\left(\frac{1}{2}-\frac{1}{3}\right)+\frac{1}{3}\left(\frac{1}{3}-\frac{1}{4}\right)+\frac{1}{4}\left(\frac{1}{4}-\frac{1}{5}\right)+\cdots$$

$$=1-\frac{1}{1\times 2}+\frac{1}{2^2}-\frac{1}{2\times 3}+\frac{1}{3^2}-\frac{1}{3\times 4}+\frac{1}{4^2}-\frac{1}{4\times 5}+\cdots$$

$$=\left(1+\frac{1}{2^2}+\frac{1}{3^2}+\frac{1}{4^2}+\cdots\right)-\lim_{n\to\infty}\left(\frac{1}{1\times 2}+\frac{1}{2\times 3}+\frac{1}{3\times 4}+\cdots+\frac{1}{n(n+1)}\right)$$

$$=\zeta(2)-\lim_{n\to\infty}\left(1-\frac{1}{(n+1)}\right)=\frac{\pi^2}{6}-1$$

SO, $\boxed{\int_0^1 \lfloor x \rfloor^{-1}\,dx=\frac{\pi^2}{6}-1}$

070. Calculate Integral $I=\int_0^1 \frac{x^{\pi}-x^2}{x\log(x)}\,dx$

Solution; $I=\int_0^1 \frac{x^{\pi}-x^2}{x\log(x)}\,dx$

$$=\int_0^1 \frac{x^{\pi-1}-x}{\log(x)}\,dx$$

$$=\int_0^1 \frac{x^{\pi-1}-1}{\log(x)}\,dx-\int_0^1 \frac{x^{1}-1}{\log(x)}\,dx$$

$$=\log\left((\pi-1)+1\right)-\log(1+1)$$

$$=\log\left(\frac{\pi}{2}\right)$$

Note: $\int_0^1 \frac{x^n-1}{\log(x)}\,dx=\log(n+1)$

SO, $\boxed{\int_0^1 \frac{x^{\pi}-x^2}{x\log(x)}\,dx=\log\left(\frac{\pi}{2}\right)}$

071. Calculate Integral $I=\int_0^{\frac{\pi}{2}}\frac{\sin(x)+\cos(x)}{\sin^5(x)+\cos^5(x)}dx$

Solution; $I=\int_0^{\frac{\pi}{2}}\frac{\sin(x)+\cos(x)}{\sin^5(x)+\cos^5(x)}dx$

$$=\int_0^{\frac{\pi}{2}}\frac{\sin\left(\frac{\pi}{2}-x\right)+\cos(x)}{\sin^5\left(\frac{\pi}{2}-x\right)+\cos^5\left(\frac{\pi}{2}-x\right)}dx$$

$$=2\int_0^{\frac{\pi}{2}}\frac{\cos(x)}{\sin^5(x)+\cos^5(x)}dx$$

$$=2\int_0^{\frac{\pi}{2}}\frac{\tan^2(x)+1}{\tan^5(x)+1}\sec^2(x)dx$$

Let : $u=\tan(x)\Rightarrow du=\sec^2(x)dx$ **; If** : $x\in\left(0,\frac{\pi}{2}\right)\Rightarrow u\in(0,\infty)$

They got it $I=2\int_0^{\infty}\frac{u^2+1}{u^5+1}du=2\int_0^{\infty}\frac{u^2}{u^5+1}du+2\int_0^{\infty}\frac{1}{u^5+1}du$

Let: $y=u^5\Rightarrow u=y^{\frac{1}{5}}\Rightarrow du=\frac{1}{5}y^{\frac{1}{5}-1}dy$ **; If** : $x\in(0,\infty)\Rightarrow u\in(0,\infty)$

That $I=\frac{2}{5}\int_0^{\infty}\frac{y^{\frac{3}{5}-1}}{y+1}dy+\frac{2}{5}\int_0^{\infty}\frac{y^{\frac{1}{5}-1}}{y+1}dy$

$$=\frac{2\pi}{5}\left[B\left(\frac{3}{5},\frac{2}{5}\right)+B\left(\frac{1}{5},\frac{4}{5}\right)\right]=\frac{2\pi}{5}\left[\Gamma\left(\frac{3}{5}\right)\Gamma\left(1-\frac{3}{5}\right)+\Gamma\left(\frac{1}{5}\right)\Gamma\left(1-\frac{1}{5}\right)\right]$$

$$=\frac{2\pi}{5}\left(\frac{1}{\sin\left(\frac{3\pi}{5}\right)}+\frac{1}{\sin\left(\frac{\pi}{5}\right)}\right)=\frac{2\pi}{5}\left(\frac{1}{\frac{1}{2}\sqrt{\frac{5+\sqrt{5}}{2}}}+\frac{1}{\frac{1}{2}\sqrt{\frac{5-\sqrt{5}}{2}}}\right)$$

SO, $\boxed{\int_0^{\frac{\pi}{2}}\frac{\sin(x)+\cos(x)}{\sin^5(x)+\cos^5(x)}dx=\frac{2\sqrt{2}\pi}{5}\left(\sqrt{1-\frac{1}{\sqrt{5}}}+\sqrt{1+\frac{1}{\sqrt{5}}}\right)}$

072. Calculate Integral $I=\int_1^e x^{\log(x)-1}\log(x)dx$

Solution; $I=\int_1^e x^{\log(x)-1}\log(x)dx$

$$=\int_1^e x^{\log(x)}\frac{\log(x)}{x}dx=\int_1^e e^{\log(x^{\log(x)})}\frac{\log(x)}{x}dx$$

$$=\int_1^e e^{\log(x)\times\log(x)}\frac{\log(x)}{x}dx=\int_1^e e^{\log^2(x)}\frac{\log(x)}{x}dx$$

$$=\frac{1}{2}\int_1^e e^{\log^2(x)}\frac{2\log(x)}{x}dx=\frac{1}{2}\int_1^e e^{\log^2(x)}d\left(\log^2(x)\right)$$

$$=\frac{1}{2}e^{\log^2(x)}\Big|_1^e=\frac{e-1}{2}$$

SO, $\boxed{\int_1^e x^{\log(x)-1}\log(x)dx=\frac{e-1}{2}}$

073. Calculate Integral $I=\int_0^1\sqrt{\frac{1}{x}\log\left(\frac{1}{x}\right)}dx$

Solution; $I=\int_0^1\sqrt{\frac{1}{x}\log\left(\frac{1}{x}\right)}dx$

Let : $t=\log\left(\frac{1}{x}\right)\Leftrightarrow x=e^{-t}\Rightarrow dx=-e^{-t}dt$ **; If :** $x\in(0,1)\Rightarrow t\in(\infty,0)$

They got it $I=-\int_\infty^0\sqrt{te^t}e^{-t}dt=\int_0^\infty\sqrt{te^t}e^{-t}dt=\int_0^\infty t^{\frac{1}{2}}e^{-\frac{1}{2}t}dt$

Let : $\frac{1}{2}t=y\Leftrightarrow dt=2dy$ **; If :** $t\in(0,\infty)\Rightarrow y\in(0,\infty)$

That $I=2\int_0^\infty(2y)^{\frac{1}{2}}e^{-y}dy=2\sqrt{2}\int_0^\infty y^{\left(\frac{1}{2}+1\right)-1}e^{-y}dy$

$$=2\sqrt{2}\Gamma\left(1+\frac{1}{2}\right)=2\sqrt{2}\cdot\frac{1}{2}\cdot\Gamma\left(\frac{1}{2}\right)$$

$$=\sqrt{2}\sqrt{\pi}=\sqrt{2\pi}$$

SO, $\boxed{\int_0^1\sqrt{\frac{1}{x}\log\left(\frac{1}{x}\right)}dx=\sqrt{2\pi}}$

074. Calculate Integral $I=\int_0^{\pi}\sqrt[\log(x)]{x}dx$

Solution; $I=\int_0^{\pi}\sqrt[\log(x)]{x}dx$

$$=\int_0^{\pi}x^{\frac{1}{\log(x)}}dx=\int_0^{\pi}e^{\log\left(x^{\frac{1}{\log(x)}}\right)}dx$$

$$=\int_0^{\pi}e^{\frac{1}{\log(x)}\cdot\log(x)}dx=\int_0^{\pi}edx=\pi e$$

SO, $\boxed{\int_0^{1}\sqrt[\log(x)]{x}dx=\pi e}$

075. Calculate Integral $I=\int_0^1\frac{\log(x)}{\sqrt{x}(x-1)}dx$

Solution; $I=\int_0^1\frac{\log(x)}{\sqrt{x}(x-1)}dx$

Let : $u=\sqrt{x}\Rightarrow x=u^2\Rightarrow dx=2udu$ **; If :** $x\in(0,1)\Rightarrow u\in(0,1)$

They got it $I=2\int_0^1\frac{\log(u^2)}{u(u^2-1)}udu$

$$=-4\int_0^1\frac{\log(u)}{1-u^2}du$$

Note: $\frac{1}{1-u^2}=\sum_{n=0}^{\infty}\left(u^{2n}\right)$

That $I=-4\int_0^1\left(\sum_{n=0}^{\infty}u^{2n}\log(u)\right)du$

$$=-4\sum_{n=0}^{\infty}\left(\underbrace{\int_0^1u^{2n}\log(u)du}_{\text{Use Integral By Parts}}\right)$$

$$=-4\sum_{n=0}^{\infty}\left[\underbrace{\frac{u^{2n+1}\log(u)}{2n+1}\Big|_0^1}_{0}-\int_0^1\frac{u^{2n+1}}{2n+1}\times\frac{1}{u}du\right]$$

$$=4\sum_{n=0}^{\infty}\left(\int_0^1\frac{u^{2n}}{2n+1}du\right)$$

$$= 4\sum_{n=0}^{\infty}\frac{1}{(2n+1)^2}$$

$$= 4\left(\frac{1}{1^2}+\frac{1}{3^2}+\frac{1}{5^2}+...\right)$$

$$= 4\left[\left(1+\frac{1}{2^2}+\frac{1}{3^2}+\frac{1}{4^2}+...\right)-\frac{1}{2^2}\left(1+\frac{1}{2^2}+\frac{1}{3^2}+\frac{1}{4^2}+...\right)\right]$$

$$= 4\left(\zeta(2)-\frac{1}{4}\zeta(2)\right)=\frac{\pi^2}{2}$$

SO, $\boxed{\int_0^1\frac{\log(x)}{\sqrt{x}(x-1)}dx=\frac{\pi^2}{2}}$

076. Calculate Integral $I=\int_0^{\pi}x\sin^4(x)dx$

Solution; $I=\int_0^{\pi}x\sin^4(x)dx$

$$=\int_0^{\pi}(\pi-x)\sin^4(\pi-x)dx$$

$$=\pi\int_0^{\pi}\sin^4(x)dx-\underbrace{\int_0^{\pi}x\sin^4(x)dx}_{I}$$

They got it $2I=\frac{\pi}{4}\int_0^{\pi}\left(1-\cos(2x)\right)^2dx$

That $I=\frac{\pi}{8}\int_0^{\pi}\left(1-2\cos(2x)+\cos^2(2x)\right)dx$

$$=\frac{\pi}{8}\int_0^{\pi}\left(1-2\cos(2x)+\frac{1}{2}\left(1+\cos(4x)\right)\right)dx$$

$$=\frac{\pi}{8}\left(\frac{3x}{2}-\sin(x)+\frac{1}{8}\sin(4x)\right)\Big|_0^{\pi}$$

$$=\frac{3\pi^2}{16}$$

SO, $\boxed{\int_0^{\pi}x\sin^4(x)dx=\frac{3\pi^2}{16}}$

077. Calculate Integral $I=\int_{\frac{1}{2025}}^{2025}\frac{x^2+1}{x^2+x^{2025}}dx$

Solution; $I=\int_{\frac{1}{2025}}^{2025}\frac{x^2+1}{x^2+x^{2025}}dx$

$$=\int_{\frac{1}{2025}}^{2025}\left(\frac{x^2+x^{2025}+1-x^{2025}}{x^2+x^{2025}}\right)dx$$

$$=\int_{\frac{1}{2025}}^{2025}\left(1+\frac{1-x^{2025}}{x^2+x^{2025}}\right)dx$$

$$=2025-\frac{1}{2025}+\underbrace{\int_{\frac{1}{2025}}^{2025}\left(\frac{1-x^{2025}}{x^2+x^{2025}}\right)dx}_{J}\quad (*)$$

That : $J=\int_{\frac{1}{2025}}^{2025}\left(\frac{1-x^{2025}}{x^2+x^{2025}}\right)dx$

Let : $x=\frac{1}{u}\Rightarrow dx=-\frac{1}{u^2}du$ **; If :** $x\in\left(2025,\frac{1}{2025}\right)\Rightarrow u\in\left(\frac{1}{2025},2025\right)$

They got it $J=-\int_{2025}^{\frac{1}{2025}}\left(\frac{1-\frac{1}{x^{2025}}}{\frac{1}{x^2}+\frac{1}{x^{2025}}}\right)\times\frac{1}{u^2}du$

$$=\int_{\frac{1}{2025}}^{2025}\left(\frac{x^{2025}-1}{x^2+x^{2025}}\right)du$$

$$=\underbrace{-\int_{\frac{1}{2025}}^{2025}\left(\frac{1-x^{2025}}{x^2+x^{2025}}\right)du}_{-J}$$

That $J=-J\Rightarrow J=0$

Take $(*)$ **We Have** $I=2025-\frac{1}{2025}+0=\frac{2024\times2026}{2025}$

$$SO,\ \boxed{\int_{\frac{1}{2025}}^{2025}\frac{x^2+1}{x^2+x^{2025}}dx=\frac{2024\times2026}{2025}}$$

078. Calculate Integral $I=\int_0^{e^e} x^{2e-\log(xe)}dx$

Solution; $I=\int_0^{e^e} x^{2e-\log(xe)}dx$

$$=\int_0^{e^e} e^{(2e-\log(x))\log(x)}\frac{1}{x}dx=\int_0^{e^e} e^{-\log^2(x)+2e\log(x)}\frac{1}{x}dx$$

$$=e^{e^2}\int_0^{e^e} e^{-e^2+2e\log(x)-\log^2(x)}\frac{1}{x}dx=e^{e^2}\int_0^{e^e} e^{-(e-\log(x))^2}\frac{1}{x}dx$$

Let: $u=e-\log(x)\Rightarrow du=-\frac{1}{x}dx$ **; If :** $x\in\left(0,e^e\right)\Rightarrow u\in(\infty,0)$

They got it $I=-e^{e^2}\int_\infty^0 e^{-u^2}du$

$$=e^{e^2}\int_0^\infty e^{-u^2}du=\frac{\sqrt{\pi}}{2}e^{e^2}$$

SO, $\boxed{\int_0^{e^e} x^{2e-\log(xe)}dx=\frac{\sqrt{\pi}}{2}e^{e^2}}$

079. Calculate Integral $I=\int_{-1}^{+1}\frac{1}{\sqrt[3]{1+x}-\sqrt[3]{1-x}}dx$

Solution; $I=\int_{-1}^{+1}\frac{1}{\sqrt[3]{1+x}-\sqrt[3]{1-x}}dx$

$$=\int_{-1}^{+1}\frac{1}{\sqrt[3]{1+(-x)}-\sqrt[3]{1-(-x)}}dx$$

$$=\int_{-1}^{+1}\frac{1}{\sqrt[3]{1-x}-\sqrt[3]{1+x}}dx$$

$$=-\int_{-1}^{+1}\frac{1}{\sqrt[3]{1+x}-\sqrt[3]{1-x}}dx$$

Equivalent $I=-I$ **That** $I=0$

SO, $\boxed{\int_{-1}^{+1}\frac{1}{\sqrt[3]{1+x}-\sqrt[3]{1-x}}dx=0}$

080. Calculate Integral $I=\int_1^{\infty}\frac{1}{x^n(x^2+1)}dx$

Solution; $I=\int_1^{\infty}\frac{1}{x^n(x^2+1)}dx$

Let : $x=\frac{1}{u}\Rightarrow dx=-\frac{1}{u^2}du$ **; If :** $x\in(1,\infty)\Rightarrow u\in(1,0)$

They got it $I=-\int_1^0\frac{1}{\frac{1}{u^n}\left(\frac{1}{u^2}+1\right)}\times\frac{1}{u^2}du$

$$=\int_0^1\frac{u^n}{(u^2+1)}du$$

$$=\int_0^1\left(u^n\sum_{m=0}^{\infty}\left((-1)^m u^{2m}\right)\right)du$$

$$=\sum_{m=0}^{\infty}\left((-1)^m\int_0^1 u^{2m+n}du\right)$$

$$=\sum_{m=0}^{\infty}\frac{(-1)^m}{(2m+n+1)}$$

$SO,$ $\boxed{\int_1^{\infty}\frac{1}{x^n(x^2+1)}dx=\sum_{m=0}^{\infty}\frac{(-1)^m}{(2m+n+1)}}$

081. Calculate Integral $I=\int_{-\infty}^{+\infty}\frac{\arctan^2(x)}{x^2}dx$

Solution; $I=\int_{-\infty}^{+\infty}\frac{\arctan^2(x)}{x^2}dx$

$$=2\int_0^{+\infty}\frac{\arctan^2(x)}{x^2}dx$$

Let : $x=\tan(y)\Rightarrow dx=\sec^2(y)dy$ **; If :** $x\in(0,+\infty)\Rightarrow u\in\left(0,\frac{\pi}{2}\right)$

They got it $I=2\int_0^{\frac{\pi}{2}}\frac{y^2\sec^2(y)}{\tan^2(y)}dy$

$$= 2\int_0^{\frac{\pi}{2}} y^2 \csc^2(y)\,dy \quad \textbf{Use Partial Integral}$$

$$= 2\left[y^2 \cot(y)\Big|_0^{\frac{\pi}{2}} + 2\int_0^{\frac{\pi}{2}} y\cot(y)\,dy \right]$$

$$= 2\left[0 + 2\int_0^{\frac{\pi}{2}} y\cot(y)\,dy \right]$$

$$= 4\int_0^{\frac{\pi}{2}} y\cot(y)\,dy \quad \textbf{Use Partial Integral}$$

$$= 4\left[y\log\big(\sin(y)\big)\Big|_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} \log\big(\sin(y)\big)\,dy \right]$$

$$= 4\left[0 - \int_0^{\frac{\pi}{2}} \log\big(\sin(y)\big)\,dy \right]$$

$$= 4\left(\frac{\pi}{2}\log(2) \right) = \pi\log(4)$$

SO, $\boxed{\int_{-\infty}^{+\infty} \frac{\arctan^2(x)}{x^2}\,dx = \pi\log(4)}$

082. Calculate Integral $I = \int_0^{+\infty} e^{-\lfloor x \rfloor}dx$

Solution; $I = \int_0^{+\infty} e^{-\lfloor x \rfloor}dx$

$$= \int_0^1 e^{-\lfloor x \rfloor}dx + \int_1^2 e^{-\lfloor x \rfloor}dx + \int_2^3 e^{-\lfloor x \rfloor}dx + \int_3^4 e^{-\lfloor x \rfloor}dx + \cdots$$

$$= \int_0^1 e^{-0}dx + \int_1^2 e^{-1}dx + \int_2^3 e^{-2}dx + \int_3^4 e^{-3}dx + \cdots$$

$$= 1 + e^{-1} + e^{-2} + e^{-3} + \cdots$$

$$= \frac{1}{1-\frac{1}{e}} = \frac{e}{e-1}$$

SO, $\boxed{\int_0^{+\infty} e^{-\lfloor x \rfloor}dx = \frac{e}{e-1}}$

083. Calculate Integral $I=\int_0^{2\pi}\left(\sin(x)+\cos(x)\right)^{11}dx$

Solution; $I=\int_0^{2\pi}\left(\sin(x)+\cos(x)\right)^{11}dx$

Note: $\int_0^{2a}f(x)dx=\int_0^{a}f(x)dx+\int_0^{a}f(2a-x)dx$

$$=\int_0^{\pi}\left(\sin(x)+\cos(x)\right)^{11}dx+\int_0^{\pi}\left(\sin(2\pi-x)+\cos(2\pi-x)\right)^{11}dx$$

$$=\int_0^{\pi}\left(\sin(x)+\cos(x)\right)^{11}dx+\int_0^{\pi}\left(-\sin(x)+\cos(x)\right)^{11}dx$$

$$=\int_0^{\pi}\left(\sin(x)+\cos(x)\right)^{11}dx+\int_0^{\pi}\left(-\sin(\pi-x)+\cos(\pi-x)\right)^{11}dx$$

$$=\int_0^{\pi}\left(\sin(x)+\cos(x)\right)^{11}dx+\int_0^{\pi}\left(-\sin(x)-\cos(x)\right)^{11}dx$$

$$=\int_0^{\pi}\left(\sin(x)+\cos(x)\right)^{11}dx-\int_0^{\pi}\left(\sin(x)+\cos(x)\right)^{11}dx$$

$$=0$$

SO, $\boxed{\int_0^{2\pi}\left(\sin(x)+\cos(x)\right)^{11}dx=0}$

084. Calculate Integral $I=\int_{-\infty}^{+\infty}\frac{1}{1+x^4}dx$

Method 1

Solution; $I=\int_{-\infty}^{+\infty}\frac{1}{1+x^4}dx=2\int_0^{+\infty}\frac{1}{1+x^4}dx$

Let $x=y^{\frac{1}{4}}\Rightarrow dx=\frac{1}{4}y^{\frac{1}{4}-1}dy$ **; If** $x\in(0,\infty)\Rightarrow y\in(0,\infty)$

They got it $I=2\cdot\frac{1}{4}\int_0^{+\infty}\frac{y^{\frac{1}{4}-1}}{1+y}dy$

$$=\frac{1}{2}B\left(\frac{1}{4},\frac{3}{4}\right)=\frac{1}{2}\Gamma\left(\frac{1}{4}\right)\Gamma\left(\frac{3}{4}\right)$$

$$=\frac{1}{2}\Gamma\left(\frac{1}{4}\right)\Gamma\left(1-\frac{1}{4}\right)=\frac{\pi}{2}\csc\left(\frac{\pi}{4}\right)$$

$$=\frac{\pi}{\sqrt{2}}$$

SO, $\boxed{\int_{-\infty}^{+\infty}\frac{1}{1+x^4}dx=\frac{\pi}{\sqrt{2}}}$

Method 2

Solution; $I = 2\int_0^{+\infty} \frac{1}{1+x^4}\,dx$

Let : $x = \frac{1}{t} \Rightarrow dx = -\frac{1}{t^2}\,dt$ **; If :** $x \in (0,+\infty) \Rightarrow t \in (+\infty, 0)$

They got it $I = -2\int_{+\infty}^{0} \frac{1}{1+\frac{1}{t^4}}\left(\frac{1}{t^2}\right)dt$

$$= 2\int_0^{+\infty} \frac{1}{t^2+t^{-2}}\,dt$$

$$= \int_0^{+\infty} \frac{1-t^{-2}+1+t^{-2}}{t^2+t^{-2}}\,dt$$

$$= \int_0^{+\infty} \frac{1-t^{-2}}{t^2+t^{-2}}\,dt + \int_0^{+\infty} \frac{1+t^{-2}}{t^2+t^{-2}}\,dt$$

$$= \int_0^{+\infty} \frac{d(t+t^{-1})}{(t+t^{-1})^2-2} + \int_0^{+\infty} \frac{d(t-t^{-1})}{(t-t^{-1})^2+2}$$

$$= \frac{1}{2\sqrt{2}} \log\left(\frac{t+t^{-1}-\sqrt{2}}{t+t^{-1}+\sqrt{2}}\right)\Bigg|_0^{+\infty} + \frac{1}{\sqrt{2}} \tan^{-1}\left(\frac{t+t^{-1}}{\sqrt{2}}\right)\Bigg|_0^{+\infty}$$

$$= \frac{1}{2\sqrt{2}} \log\left(\frac{t^2-\sqrt{2}t+1}{t^2+\sqrt{2}t+1}\right)\Bigg|_0^{+\infty} + \frac{1}{\sqrt{2}} \tan^{-1}\left(\frac{t^2+1}{t\sqrt{2}}\right)\Bigg|_0^{+\infty}$$

$$= \frac{1}{2\sqrt{2}}(0-0) + \frac{1}{\sqrt{2}}\left[\frac{\pi}{2} - \tan^{-1}\left(\frac{\sqrt{2}}{2}\right)\right]$$

$$= \frac{1}{\sqrt{2}}\left[\frac{\pi}{2} - \tan^{-1}\left(\frac{\sqrt{2}}{2}\right)\right]$$

SO, $\boxed{\int_{-\infty}^{+\infty} \frac{1}{1+x^4}\,dx = \frac{1}{\sqrt{2}}\left[\frac{\pi}{2} - \tan^{-1}\left(\frac{\sqrt{2}}{2}\right)\right]}$

085. Calculate Integral $I=\int_1^2 \frac{\log(x)}{x^2-2x+2}dx$

Solution; $I=\int_1^2 \frac{\log(x)}{x^2-2x+2}dx=\int_1^2 \frac{\log(x)}{(x-1)^2+1}dx$

Let : $x-1=\tan(y)\Rightarrow dx=\left(\tan^2(y)+1\right)dy$; **If** : $x\in(1,2)\Rightarrow u\in\left(0,\frac{\pi}{4}\right)$

They got it $I=\int_0^{\frac{\pi}{4}} \frac{\log\left(1+\tan(y)\right)}{\tan^2(y)+1}\times\left(\tan^2(y)+1\right)dy$

$$=\int_0^{\frac{\pi}{4}}\log\left(1+\tan(y)\right)dy$$

Use : $\int_0^a f(x)\,dx=\int_0^a f(a-x)\,dx$

$$=\int_0^{\frac{\pi}{4}}\log\left(1+\tan\left(\frac{\pi}{4}-y\right)\right)dy$$

$$=\int_0^{\frac{\pi}{4}}\log\left(1+\frac{\tan\left(\frac{\pi}{4}\right)-\tan(y)}{1+\tan\left(\frac{\pi}{4}\right)\tan(y)}\right)dy$$

Note: $\tan(a-b)=\frac{\tan(a)-\tan(b)}{1-\tan(a)\tan(b)}$

$$=\int_0^{\frac{\pi}{4}}\log\left(1+\frac{1-\tan(y)}{1+\tan(y)}\right)dy$$

$$=\int_0^{\frac{\pi}{4}}\log\left(\frac{1+\tan(y)+1-\tan(y)}{1+\tan(y)}\right)dy$$

$$=\int_0^{\frac{\pi}{4}}\log\left(\frac{2}{1+\tan(y)}\right)dy$$

Note: $\log\left(\frac{a}{b}\right)=\log(a)-\log(b)$

$$=\int_0^{\frac{\pi}{4}}\left(\log(2)-\log\left(1+\tan(y)\right)\right)dy$$

$$=\int_0^{\frac{\pi}{4}}\log(2)\,dy-\underbrace{\int_1^{\frac{\pi}{4}}\log\left(1+\tan(y)\right)dy}_{I}$$

That $I=\frac{\pi}{4}\log(2)-I\Rightarrow I=\frac{\pi}{8}\log(2)$

SO, $\boxed{\int_1^2 \frac{\log(x)}{x^2-2x+2}dx=\frac{\pi}{8}\log(2)}$

086. Calculate Integral $I=\int_{-1}^{+1}\frac{e^x-1}{e^x+1}dx$

Solution; $I=\int_{-1}^{+1}\frac{e^x-1}{e^x+1}dx$

$$=\int_{-1}^{+1}\frac{e^{-x}-1}{e^{-x}+1}dx$$

$$=\int_{-1}^{+1}\frac{e^{-x}-1}{e^{-x}+1}\times\frac{e^x}{e^x}dx$$

$$=\int_{-1}^{+1}\frac{1-e^x}{e^{-x}+1}dx$$

$$=\underbrace{-\int_{-1}^{+1}\frac{e^x-1}{e^{-x}+1}dx}_{-I}$$

Equivalent $I=-I$ **That** $I=0$

$SO,$ $\boxed{\int_{-1}^{+1}\frac{e^x-1}{e^x+1}dx=0}$

087. Calculate Integral $I=\int_{-1}^{+1}\log\left(\frac{1-x}{1+x}\right)dx$

Solution; $I=\int_{-1}^{+1}\log\left(\frac{1-x}{1+x}\right)dx$

$$=\int_{-1}^{+1}\log\left(\frac{1-(-x)}{1+(-x)}\right)dx$$

$$=\int_{-1}^{+1}\log\left(\frac{1+x}{1-x}\right)dx$$

$$=\int_{-1}^{+1}\log\left(\frac{1-x}{1+x}\right)^{-1}dx$$

$$=\underbrace{-\int_{-1}^{+1}\log\left(\frac{1-x}{1+x}\right)dx}_{-I}$$

Equivalent $I=-I$ **That** $I=0$

$SO,$ $\boxed{\int_{-1}^{+1}\log\left(\frac{1-x}{1+x}\right)dx=0}$

088. Calculate Integral $I=\int_1^{\infty}\frac{\left(\frac{x-1}{x+1}\right)}{(x+1)(x-1)}\log^2\left(\sqrt{\frac{x+1}{x-1}}\right)dx$

Solution; $I=\int_1^{\infty}\frac{\left(\frac{x-1}{x+1}\right)}{(x+1)(x-1)}\log^2\left(\sqrt{\frac{x+1}{x-1}}\right)dx$

$$=\int_1^{\infty}\frac{1}{(x+1)^2}\log^2\left(\frac{x-1}{x+1}\right)^{-\frac{1}{2}}dx$$

$$=\frac{1}{4}\int_1^{\infty}\frac{1}{(x+1)^2}\log^2\left(\frac{x-1}{x+1}\right)dx$$

Let: $t=\frac{x-1}{x+1}\Rightarrow dt=\frac{\begin{bmatrix}1 & -1\\ 1 & 1\end{bmatrix}}{(x+1)^2}dx\Leftrightarrow\frac{dt}{2}=\frac{1}{(x+1)^2}dx$ **; If :** $x\in(1,\infty)\Rightarrow t\in(0,1)$

They got it $I=\frac{1}{2\times4}\int_0^1\log^2(t)\,dt$

$$=\frac{1}{2\times4}\left[t\log^2(t)\Big|_0^1-2\int_0^1\log(t)\,dt\right]$$

$$=\frac{1}{2\times4}\left[0-2\int_0^1\log(t)\,dt\right]$$

$$=-\frac{1}{4}\left[\int_0^1\log(t)\,dt\right]$$

$$=-\frac{1}{4}\left[\left(t\log(t)\Big|_0^1-\int_0^1 dt\right)\right]$$

$$=-\frac{1}{4}\left(0-x\Big|_0^1\right)$$

$$=-\frac{1}{4}(-1+0)=\frac{1}{4}$$

SO, $\boxed{\int_1^{\infty}\frac{\left(\frac{x-1}{x+1}\right)}{(x+1)(x-1)}\log^2\left(\sqrt{\frac{x+1}{x-1}}\right)dx=\frac{1}{4}}$

089. Calculate Integral $I=\int_0^{\infty}\frac{x^n}{e^x-1}dx$

Solution; $I=\int_0^{\infty}\frac{x^n}{e^x-1}dx$

$$=\int_0^{\infty}\frac{x^n e^{-x}}{1-e^{-x}}dx$$

Note : $\frac{1}{1-x}=\sum_{k=0}^{\infty}\left(x^k\right)$

$$=\sum_{m=0}^{\infty}\left(\int_0^{\infty}x^n e^{-(m+1)x}dx\right)$$

Let : $u=(m+1)x\Rightarrow\frac{du}{(m+1)}=dx$ **; If :** $x\in(0,\infty)\Rightarrow t\in(0,\infty)$

They got it $I=\sum_{m=0}^{\infty}\left(\int_0^{\infty}\left(\frac{u}{m+1}\right)^n e^{-u}\frac{du}{(m+1)}\right)$

$$=\sum_{m=0}^{\infty}\left(\frac{1}{(m+1)^{n+1}}\int_0^{\infty}u^n e^{-u}du\right)$$

$$=\sum_{m=0}^{\infty}\frac{\Gamma(n+1)}{(m+1)^{n+1}}$$

$$=\Gamma(n+1)\sum_{m=0}^{\infty}\frac{1}{(m+1)^{n+1}}$$

$$=\Gamma(n+1)\zeta(n+1)$$

SO, $\boxed{\int_0^{\infty}\frac{x^n}{e^x-1}dx=\Gamma(n+1)\zeta(n+1)}$

090. Calculate Integral $I=\int_0^1\frac{\log^2(x)}{x^2-1}dx$

Solution; $I=\int_0^1\frac{\log^2(x)}{x^2-1}dx$

$$=-\int_0^1\frac{\log^2(x)}{1-x^2}dx$$

$$=-\sum_{n=0}^{\infty}\left(\int_0^1 x^{2n}\log^2(x)dx\right)$$

$$=-\sum_{n=0}^{\infty}\left(\underbrace{\frac{x^{2n+1}\log^2(x)}{(2n+1)}\Big|_0^1}_{0}-2\int_0^1\frac{x^{2n+1}\log(x)}{(2n+1)x}dx\right)\quad\textbf{Use Partial Integral}$$

$$=2\sum_{n=0}^{\infty}\left[\int_0^1\frac{x^{2n}\log(x)}{(2n+1)}dx\right]$$

$$=2\sum_{n=0}^{\infty}\left(\underbrace{\frac{x^{2n+1}\log^2(x)}{(2n+1)^2}\Big|_0^1}_{0}-\int_0^1\frac{x^{2n+1}}{(2n+1)^2x}dx\right)\quad\textbf{Use Partial Integral}$$

$$=-2\sum_{n=0}^{\infty}\left(\int_0^1\frac{x^{2n}}{(2n+1)^2}dx\right)$$

$$=-2\sum_{n=0}^{\infty}\left(\frac{1}{(2n+1)^3}\right)$$

$$=-2\left(1+\frac{1}{3^3}+\frac{1}{5^3}+\frac{1}{7^3}\ldots\right)$$

$$=-2\left[\left(1+\frac{1}{2^3}+\frac{1}{3^3}+\frac{1}{4^3}+\frac{1}{5^3}\ldots\right)-\frac{1}{2^3}\left(1+\frac{1}{2^3}+\frac{1}{4^3}+\frac{1}{6^3}+\frac{1}{8^3}\ldots\right)\right]$$

$$=-2\left(\zeta(3)-\frac{1}{8}\zeta(3)\right)=-\frac{7\zeta(3)}{4}$$

$SO,\ \boxed{\int_0^1\frac{\log^2(x)}{x^2-1}dx=-\frac{7\zeta(3)}{4}}$

091. Calculate Integral $I=\int_0^{e-1}\frac{x}{(x+1)\log(x+1)}dx$

Solution; $I=\int_0^{e-1}\frac{x}{(x+1)\log(x+1)}dx$

Let : $u=\log(x+1)\Leftrightarrow x=e^u-1\Rightarrow dx=e^u du$ **; If** : $x\in(0,e-1)\Rightarrow u\in(0,1)$

They got it $I=\int_0^1\frac{(e^u-1)e^u}{ue^u}du$

We Have : $e^u=1+u+\frac{u^2}{2!}+\frac{u^3}{3!}+\cdots$ **; That** : $e^u-1=u+\frac{u^2}{2!}+\frac{u^3}{3!}+\cdots=\sum_{n=1}^{\infty}\frac{u^n}{n!}$

That $I=\sum_{n=1}^{\infty}\left(\frac{1}{n!}\int_0^1\frac{u^n}{u}du\right)$

$$=\sum_{n=1}^{\infty}\frac{1}{n!}\times\frac{u^n}{n}\Big|_0^1$$

$$=\sum_{n=1}^{\infty}\left(\frac{1}{n.n!}\right)$$

SO, $\boxed{\int_0^{e-1}\frac{x}{\log(x+1)}dx=\sum_{n=1}^{\infty}\left(\frac{1}{n.n!}\right)}$

092. Calculate Integral $I=\int_1^{2024}\lfloor\log_{43}(x)\rfloor dx$

Solution; $I=\int_1^{2024}\lfloor\log_{43}(x)\rfloor dx$

By : $\lfloor\log_{43}(43)\rfloor=1,\lfloor\log_{43}(43^2)\rfloor=2$ **And** $n\le\lfloor n\rfloor\le n+1\Rightarrow\lfloor n\rfloor=n$

Example : $\lfloor 1.5\rfloor=1,\lfloor -1.5\rfloor=-2$

That $I=\int_1^{43}\lfloor\log_{43}(x)\rfloor dx+\int_{43}^{43^2}\lfloor\log_{43}(x)\rfloor dx+\int_{43^2}^{2024}\lfloor\log_{43}(x)\rfloor dx$

But : $\lfloor\log_{43}(x)\rfloor=\begin{cases}0 & When\ 1\le x\le 43\\ 1 & When\ 43\le x\le 43^2\\ 2 & When\ 43^2\le x\le 2024\end{cases}$

They got it $=\int_1^{43}0dx+\int_{43}^{43^2}1dx+\int_{43^2}^{2024}2dx$

$$=43^2-43+2(2024-43^2)$$

$$=2156$$

SO, $\boxed{\int_1^{2024}\lfloor\log_{43}(x)\rfloor dx=2156}$

093. Calculate Integral $I=\int_0^{\frac{\pi}{2}}\frac{x+\sin(x)}{1+\cos(x)}dx$

Solution; $I=\int_0^{\frac{\pi}{2}}\frac{x+\sin(x)}{1+\cos(x)}dx$

$$=\int_0^{\frac{\pi}{2}}\frac{x+2\sin\left(x/2\right)\cos\left(x/2\right)}{2\cos^2\left(x/2\right)}dx$$

Note: $\begin{cases}\sin(x)=2\sin\left(x/2\right)\cos\left(x/2\right)\\1+\cos(x)=2\cos^2\left(x/2\right)\end{cases}$

$$=\int_0^{\frac{\pi}{2}}\frac{x}{2\cos^2\left(x/2\right)}dx+\int_0^{\frac{\pi}{2}}\frac{\sin\left(x/2\right)}{\cos\left(x/2\right)}dx$$

$$=\frac{1}{2}\int_0^{\frac{\pi}{2}}x\sec^2(x/2)dx+\int_0^{\frac{\pi}{2}}\frac{\sin\left(x/2\right)}{\cos\left(x/2\right)}dx$$

$$=\int_0^{\frac{\pi}{2}}(x/2)\sec^2(x/2)dx+\int_0^{\frac{\pi}{2}}\tan(x/2)dx$$

Let $:u=x/2\Rightarrow du=2dx$ **; If** $:x\in\left(0,\frac{\pi}{2}\right)\Rightarrow u\in\left(0,\frac{\pi}{4}\right)$

They got it $I=2\int_0^{\frac{\pi}{4}}u\sec^2(u)du+2\int_0^{\frac{\pi}{4}}\tan(u)du$

$$=2\int_0^{\frac{\pi}{4}}\left[u\sec^2(u)+\tan(u)\right]du$$

$$=2\int_0^{\frac{\pi}{4}}\left[u\left(\tan(u)\right)'+u'\tan(u)\right]du$$

Note : $u'v+v'u=(uv)'$

$$=2\int_0^{\frac{\pi}{4}}\left(u\tan(u)\right)'dt$$

$$=2\left(u\tan(u)\right)\Big|_0^{\frac{\pi}{4}}=\frac{\pi}{2}$$

SO, $\boxed{\int_0^{\frac{\pi}{2}}\frac{x+\sin(x)}{1+\cos(x)}dx=\frac{\pi}{2}}$

Note : $\sin(2x)=2\sin(x)\cos(x)$

$1-\cos(2x)=2\sin^2(x)$

$1+\cos(2x)=2\cos^2(x)$

094. Calculate Integral $I=\int_{-\infty}^{1}\frac{\log(x)-\log(x+1)}{x(x+1)}dx$

Solution; $I=\int_{-\infty}^{1}\frac{\log(x)-\log(x+1)}{x(x+1)}dx$

$$=-\int_{-\infty}^{1}\frac{\log\left(\frac{x+1}{x}\right)}{x^2\left(\frac{x+1}{x}\right)}dx=\int_{1}^{-\infty}\frac{\log\left(1+\frac{1}{x}\right)}{x^2\left(1+\frac{1}{x}\right)}dx$$

Let : $u=\log\left(1+\frac{1}{x}\right)\Rightarrow du=x^2\left(1+\frac{1}{x}\right)$ **; If :** $x\in(1,-\infty)\Rightarrow u\in(\log(2),0)$

They got it $I=\int_{\log(2)}^{0}udu=\frac{u^2}{2}\Big|_{\log(2)}^{0}=-\frac{1}{2}\log^2(2)$

SO, $\boxed{\int_{-\infty}^{1}\frac{\log(x)-\log(x+1)}{x(x+1)}dx=-\frac{1}{2}\log^2(2)}$

095. Calculate Integral $I=\int_{0}^{1}\sqrt{1-x^{\pi}}\,dx$

Solution; $I=\int_{0}^{1}\sqrt{1-x^{\pi}}\,dx$

Let : $u=x^{\pi}\Leftrightarrow x=u^{\frac{1}{\pi}}\Rightarrow dx=\frac{1}{\pi}u^{\frac{1}{\pi}-1}du$ **; If :** $x\in(0,1)\Rightarrow u\in(0,1)$

They got it $I=\frac{1}{\pi}\int_{0}^{1}u^{\frac{1}{\pi}-1}\sqrt{1-u}\,du$

$$=\frac{1}{\pi}\int_{0}^{1}u^{\frac{1}{\pi}-1}(1-u)^{\frac{3}{2}-1}du=\frac{1}{\pi}B\left(\frac{1}{\pi},\frac{3}{2}\right)$$

$$=\frac{1}{\pi}\frac{\Gamma\left(1/\pi\right)\Gamma\left(3/2\right)}{\Gamma\left(1/\pi+3/2\right)}=\frac{\sqrt{\pi}}{(\pi+2)}\times\frac{\Gamma\left(1/\pi\right)}{\Gamma\left(1/\pi+1/2\right)}$$

SO, $\boxed{\int_{0}^{1}\sqrt{1-x^{\pi}}\,dx=\frac{\sqrt{\pi}}{(\pi+2)}\times\frac{\Gamma(1/\pi)}{\Gamma((\pi+2)/2\pi)}}$

096. Calculate Integral $I=\int_0^{\frac{\pi}{2}}\frac{\sin(x)+1}{\sin(x)+\cos(x)+1}dx$

Solution; $I=\int_0^{\frac{\pi}{2}}\frac{\sin(x)+1}{\sin(x)+\cos(x)+1}dx$

$$=\int_0^{\frac{\pi}{2}}\left(1-\frac{\cos(x)}{\sin(x)+\cos(x)+1}\right)dx$$

$$=2\pi-\int_0^{\frac{\pi}{2}}\left(\frac{\cos(x)}{\sin(x)+\cos(x)+1}\right)dx$$

Let : $u=\tan\left(\frac{x}{2}\right)\Rightarrow dx=\frac{2}{1+u^2}du$ **; If** : $x\in\left(0,\frac{\pi}{2}\right)\Rightarrow u\in(0,1)$

They got it $I=\frac{\pi}{2}-\int_0^1\left(\frac{\frac{1-u^2}{1+u^2}}{\frac{2u}{1+u^2}+\frac{1-u^2}{1+u^2}+1}\right)du$

$$=\frac{\pi}{2}-\frac{1}{2}\int_0^1\left(\frac{1-u^2}{1+u}\right)du$$

$$=\frac{\pi}{2}-\frac{1}{2}\int_0^1(1-u)\,du=\frac{2\pi-1}{4}$$

SO, $\boxed{\int_0^{\frac{\pi}{2}}\frac{\sin(x)+1}{\sin(x)+\cos(x)+1}dx=\frac{2\pi-1}{4}}$

097. Calculate Integral $I=\int_0^{\infty}\frac{x\log(x)}{x^4+1}dx$

Solution; $I=\int_0^{\infty}\frac{x\log(x)}{x^4+1}dx$

$$=\int_0^1\frac{x\log(x)}{x^4+1}dx+\underbrace{\int_1^{\infty}\frac{x\log(x)}{x^4+1}dx}_{J}\quad(*)$$

For : $J=\int_1^{\infty}\frac{x\log(x)}{x^4+1}dx$

Let : $x=\frac{1}{u}\Rightarrow dx=-\frac{1}{u^2}du$; **If** : $x\in(1,\infty)\Rightarrow u\in(1,0)$

They got it $J=-\int_1^0\frac{u^{-1}\log(u^{-1})}{u^{-4}+1}u^{-2}du$

$=-\int_0^1\frac{u\log(u)}{u^4+1}du$ **Note** : $\int_a^b f(x)dx=\int_a^b f(u)du$

$=-\int_0^1\frac{x\log(x)}{x^4+1}dx$

Take (*) **We Have** : $I=\int_0^1\frac{x\log(x)}{x^4+1}dx-\int_0^1\frac{x\log(x)}{x^4+1}dx=0$

SO, $\boxed{\int_0^\infty\frac{x\log(x)}{x^4+1}dx=0}$

098. Calculate Integral $I=\int_{-\frac{\pi}{2}}^{+\frac{\pi}{2}}\frac{\sin^2(x)}{1+e^x}dx$

Solution; $I=\int_{-\frac{\pi}{2}}^{+\frac{\pi}{2}}\frac{\sin^2(x)}{1+e^x}dx$ (*)

$=\int_{-\frac{\pi}{2}}^{+\frac{\pi}{2}}\frac{\sin^2(-x)}{1+e^{-x}}dx$

$=\int_{-\frac{\pi}{2}}^{+\frac{\pi}{2}}\frac{e^x\sin^2(x)}{1+e^x}dx$ (**)

Take (*)+(**) **We Have** : $2I=\int_{-\frac{\pi}{2}}^{+\frac{\pi}{2}}\frac{\sin^2(x)}{1+e^x}dx+\int_{-\frac{\pi}{2}}^{+\frac{\pi}{2}}\frac{e^x\sin^2(x)}{1+e^x}dx$

$=2\int_{-\frac{\pi}{2}}^{+\frac{\pi}{2}}\sin^2(x)dx$

$=2\int_0^{\frac{\pi}{2}}\sin^2(x)dx=2\cdot\frac{\pi}{2}.\frac{1}{2}=\frac{\pi}{2}$

By : $2I=\frac{\pi}{2}$ **That** $I=\frac{\pi}{4}$

SO, $\boxed{\int_{-\frac{\pi}{2}}^{+\frac{\pi}{2}}\frac{\sin^2(x)}{1+e^x}dx=\frac{\pi}{4}}$

099. Calculate Integral $I=\int_0^9 \frac{x+\frac{x+...}{1+...}}{1+\frac{x+...}{1+...}}dx$

Solution; $I=\int_0^9 \frac{x+\frac{x+...}{1+...}}{1+\frac{x+...}{1+...}}dx$

Let : $u=\frac{x+\frac{x+...}{1+...}}{1+\frac{x+...}{1+...}}=\frac{x+u}{1+u}\Rightarrow x=u^2\Rightarrow dx=2udu$ **; If :** $x\in(0,9)\Rightarrow u\in(0,3)$

They got it $I=2\int_0^3 u^2dx=18$

SO, $\boxed{I=18}$

100. Calculate Integral $I=\int_0^{2\pi}\frac{1}{e^{\sin(x)}+1}dx$

Solution; $I=\int_0^{2\pi}\frac{1}{e^{\sin(x)}+1}dx \quad (*)$

$$=\int_0^{2\pi}\frac{1}{e^{\sin(2\pi-x)}+1}dx$$

$$=\int_0^{2\pi}\frac{1}{e^{-\sin(x)}+1}dx$$

$$=\int_0^{2\pi}\frac{e^{\sin(x)}}{e^{\sin(x)}+1}dx \quad (**)$$

Take $(*)+(**)$ **They Have :** $2I=\int_0^{2\pi}\frac{1}{e^{\sin(x)}+1}dx+\int_0^{2\pi}\frac{e^{\sin(x)}}{e^{\sin(x)}+1}dx$

$$=\int_0^{2\pi}\frac{e^{\sin(x)}+1}{e^{\sin(x)}+1}dx$$

$$=2\pi$$

By : $2I=2\pi$ **That** $I=\pi$

SO, $\boxed{\int_0^{2\pi}\frac{1}{e^{\sin(x)}+1}dx=\pi}$

101. Calculate Integral $I=\int_0^1 \frac{\sqrt{x(1-x)}}{1+x}dx$

Solution; $I=\int_0^1 \frac{\sqrt{x(1-x)}}{1+x}dx$

Let : $x=\sin^2(u)\Rightarrow dx=2\sin(u)\cos(u)du$ **; If** : $x\in(0,1)\Rightarrow u\in\left(0,\frac{\pi}{2}\right)$

They got it $I=2\int_0^{\frac{\pi}{2}} \frac{\sqrt{\sin^2(u)\left(1-\sin^2(u)\right)}}{1+\sin^2(u)}\times\sin(u)\cos(u)du$

$$=2\int_0^{\frac{\pi}{2}} \frac{\sin^2(u)\cos^2(u)}{1+\sin^2(u)}du$$

$$=2\int_0^{\frac{\pi}{2}} \left(\cos^2(u)-\frac{\cos^2(u)}{1+\sin^2(u)}\right)du$$

$$=\int_0^{\frac{\pi}{2}} 2\cos^2(u)du-2\int_0^{\frac{\pi}{2}} \frac{1}{\sec^2(u)+\tan^2(u)}du$$

$$=\int_0^{\frac{\pi}{2}} \left(1+\cos(2u)\right)du-2\int_0^{\frac{\pi}{2}} \frac{\sec^2(u)}{\left(1+2\tan^2(u)\right)\left(1+\tan^2(u)\right)}du$$

$$=\left(u+\sin(u)\cos(u)\right)\Big|_0^{\frac{\pi}{2}}-2\int_0^{\frac{\pi}{2}} \frac{1}{\left(1+2\tan^2(u)\right)\left(1+\tan^2(u)\right)}d\left(\tan(u)\right)$$

$$=\frac{\pi}{2}-2\int_0^{\frac{\pi}{2}}\left[\frac{2}{1+2\tan^2(u)}-\frac{1}{1+\tan^2(u)}\right]d\left(\tan(u)\right)$$

$$=\frac{\pi}{2}-2\left(\frac{2}{\sqrt{2}}\tan^{-1}\left(\sqrt{2}\tan(u)\right)-\tan^{-1}\left(\tan(u)\right)\right)\Big|_0^{\frac{\pi}{2}}$$

$$=\frac{3\pi}{2}-\sqrt{2}\pi$$

SO, $\boxed{\int_0^1 \frac{\sqrt{x(1-x)}}{1+x}dx=\left(\frac{3}{2}-\sqrt{2}\right)\pi}$

102. Calculate Integral $I=\int_0^{\frac{\pi}{2}}\frac{\sin(x)+\cos(x)}{\sqrt{\sin(2x)}}dx$

Solution; $I=\int_0^{\frac{\pi}{2}}\frac{\sin(x)+\cos(x)}{\sqrt{\sin(2x)}}dx$

$$=\int_0^{\frac{\pi}{2}}\frac{\sin(x)+\cos(x)}{\sqrt{1-\left(1-2\sin(x)\cos(x)\right)}}dx$$

$$=\int_0^{\frac{\pi}{2}}\frac{\sin(x)+\cos(x)}{\sqrt{1-\left(\sin(x)-\cos(x)\right)^2}}dx$$

Let : $u=\sin(x)-\cos(x)\Rightarrow du=\left(\sin(x)+\cos(x)\right)dx$ **; If :** $x\in\left(0,\frac{\pi}{2}\right)\Rightarrow u\in(-1,1)$

They got it $I=\underbrace{\int_{-1}^{1}\frac{1}{\sqrt{1-u^2}}du}_{is\ an\ even\ funtion}$

$$=2\int_0^1\frac{1}{\sqrt{1-u^2}}du$$

Let : $u=\sin(t)\Rightarrow du=\cos(t)dt$ **; If :** $u\in(0,1)\Rightarrow t\in\left(0,\frac{\pi}{2}\right)$

That $I=2\int_0^{\frac{\pi}{2}}\frac{\cos(t)}{\sqrt{1-\sin^2(t)}}dt$

$$=2\int_0^{\frac{\pi}{2}}\frac{\cos(t)}{\left|\cos(t)\right|}dt$$

$$=2\int_0^{\frac{\pi}{2}}\frac{\cos(t)}{\cos(x)}dt$$

$$=\pi$$

Note : $\forall t\in\left(0,\frac{\pi}{2}\right)\Rightarrow\left|\cos(t)\right|=\cos(t)$

SO, $\boxed{\int_0^{\frac{\pi}{2}}\frac{\sin(x)+\cos(x)}{\sqrt{\sin(2x)}}dx=\pi}$

103. Calculate Integral $I=\int_0^{\frac{\pi}{2}}\frac{\cos(x)}{\left(1+\sqrt{\sin(2x)}\right)^2}dx$

Solution; $I=\int_0^{\frac{\pi}{2}}\frac{\cos(x)}{\left(1+\sqrt{\sin(2x)}\right)^2}dx \quad (1)$

$$=\int_0^{\frac{\pi}{2}}\frac{\cos\left(\frac{\pi}{2}-x\right)}{\left(1+\sqrt{\sin\left(\pi-2x\right)}\right)^2}dx$$

$$=\int_0^{\frac{\pi}{2}}\frac{\sin(x)}{\left(1+\sqrt{\sin(2x)}\right)^2}dx \quad (2)$$

Take $(1)+(2)$ **They Have :** $2I=\int_0^{\frac{\pi}{2}}\frac{\cos(x)+\sin(x)}{\left(1+\sqrt{\sin(2x)}\right)^2}dx$

$$=\int_0^{\frac{\pi}{2}}\frac{\cos(x)+\sin(x)}{\left(1+\sqrt{1-\left(\sin(x)-\cos(x)\right)^2}\right)^2}dx$$

Let: $z=\sin(x)-\cos(x)\Rightarrow dz=\left(\cos(x)+\sin(x)\right)dz$ **; If :** $x\in\left(0,\frac{\pi}{2}\right)\Rightarrow z\in(-1,1)$

They got it $2I=\int_{-1}^{1}\frac{1}{\left(1+\sqrt{1-z^2}\right)^2}dz$

Let : $z=\sin(u)\Rightarrow dz=\cos(u)du$ **; If :** $z\in(-1,1)\Rightarrow u\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$

They got it : $2I=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\frac{\cos(u)}{\left(1+\sqrt{1-\sin^2(u)}\right)^2}du$

That $I=\int_0^{\frac{\pi}{2}}\frac{\cos(u)}{\left(1+\cos(u)\right)^2}du$

Note : $\forall u\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)\Rightarrow\sqrt{1-\sin^2(u)}=\cos(u)$

Let: $y=\tan\left(\frac{u}{2}\right)\Rightarrow du=\frac{2}{1+y^2}dy$ **; If :** $u\in\left(0,\frac{\pi}{2}\right)\Rightarrow y\in(0,1)$

They got it $I=\int_0^1\frac{\frac{1-y^2}{1+y^2}}{\left(1+\frac{1-y^2}{1+y^2}\right)^2}\times\frac{2}{1+y^2}dy$

$$=\frac{1}{2}\int_0^1(1-y^2)\,dy=\frac{1}{3}$$

SO, $\boxed{\int_0^{\frac{\pi}{2}}\frac{\cos(x)}{\left(1+\sqrt{\sin(2x)}\right)^2}dx=\frac{1}{3}}$

104. Calculate Integral $I=\int_1^{\sqrt[4]{2}}\frac{x^8-1}{x(x^8+1)}dx$

Solution; $I=\int_1^{\sqrt[4]{2}}\frac{x^8-1}{x(x^8+1)}dx$

$$=\int_1^{\sqrt[4]{2}}\frac{x^8-1}{x(x^8+1)}\times\frac{x^{-5}}{x^{-5}}dx$$

$$=\int_1^{\sqrt[4]{2}}\frac{x^3-x^{-5}}{x^4+x^{-4}}dx$$

$$=\frac{1}{4}\int_1^{\sqrt[4]{2}}\frac{4x^3-4x^{-5}}{x^4+x^{-4}}dx$$

Let : $u=x^4+x^{-4}\Rightarrow du=4x^3-4x^{-5}dx$ **; If :** $x\in\left(1,\sqrt[4]{2}\right)\Rightarrow u\in\left(2,\frac{5}{2}\right)$

They got it $I=\frac{1}{4}\int_2^{\frac{5}{2}}\frac{1}{u}du$

$$=\frac{1}{4}\log\left(\frac{5}{4}\right)$$

SO, $\boxed{\int_1^{\sqrt[4]{2}}\frac{x^8-1}{x(x^8+1)}dx=\frac{1}{4}\log\left(\frac{5}{4}\right)}$

105. Calculate Integral $I=\int_{-1}^{+1}\frac{1}{\sqrt{1+x}+\sqrt{1-x}+2}dx$

Solution; $I=\int_{-1}^{+1}\frac{1}{\sqrt{1+x}+\sqrt{1-x}+2}dx$

Let : $x=\cos(2u)\Rightarrow dx=-2\sin(2u)du$ **; If :** $x\in(-1,1)\Rightarrow u\in\left(\frac{\pi}{2},0\right)$

They got it $I=\int_{\frac{\pi}{2}}^{0}\frac{-2\sin(2u)}{\sqrt{1+\cos(2u)}+\sqrt{1-\cos(2u)}+2}du$

$$=\int_{0}^{\frac{\pi}{2}}\frac{\sin(2u)}{\frac{\sqrt{2}}{2}\cos(u)+\frac{\sqrt{2}}{2}\sin(u)+1}du$$

$$=\int_{0}^{\frac{\pi}{2}}\frac{\sin(2u)}{\cos\left(\frac{\pi}{4}-u\right)+1}du$$

Let : $t=\frac{\pi}{4}-u\Rightarrow du=-dt$, $u=\frac{\pi}{4}-t$ **; If :** $u\in\left(\frac{\pi}{2},0\right)\Rightarrow t\in\left(-\frac{\pi}{4},\frac{\pi}{4}\right)$

They got it $I=-\int_{+\frac{\pi}{4}}^{-\frac{\pi}{4}}\frac{\sin\left(\pi/2-2t\right)}{\cos(t)+1}dt$

$$=2\int_{0}^{\frac{\pi}{4}}\frac{\cos(2t)}{\cos(t)+1}dt$$

$$=2\int_{0}^{\frac{\pi}{4}}\frac{\left(1-2\sin^2(t)\right)\left(1-\cos(t)\right)}{\sin^2(t)}dt$$

$$=2\int_{0}^{\frac{\pi}{4}}\left(\csc^2(t)-\csc(t)\cot(t)+2\cos(t)-2\right)dt$$

$$=2\left(-\sin(t)+\csc(t)+2\sin(t)-2t\right)\Big|_{0}^{\pi/4}$$

$$=4\sqrt{2}-2-\pi$$

SO, $\boxed{\int_{-1}^{+1}\frac{1}{\sqrt{1+x}+\sqrt{1-x}+2}dx=4\sqrt{2}-2-\pi}$

106. Calculate Integral $I=\int_0^{\infty}\frac{x^2}{x^4+1}dx$

Solution; $I=\int_0^{\infty}\frac{x^2}{x^4+1}dx \quad (*)$

Let : $x=\frac{1}{u}\Rightarrow dx=-\frac{1}{u^2}du$ **; If :** $x\in(0,\infty)\Rightarrow u\in(\infty,0)$

They got it $I=-\int_{\infty}^{0}\frac{\frac{1}{u^2}}{\frac{1}{u^4}+1}\times\frac{1}{u^2}du=\int_0^{\infty}\frac{1}{u^4+1}du$

$$=\int_0^{\infty}\frac{1}{x^4+1}dx \quad (**)$$

Take $(*)+(**)$ **They Have :** $2I=\int_0^{\infty}\frac{x^2+1}{x^4+1}dx$

$$=\int_0^{\infty}\frac{1+x^{-2}}{x^2+x^{-2}}dx$$

$$=\int_0^{\infty}\frac{1+x^{-2}}{(x-x^{-1})^2+2}dx$$

$$=\int_0^{\infty}\frac{d(x-x^{-1})}{(x-x^{-1})^2+2}$$

$$=\frac{1}{\sqrt{2}}\arctan\left(\frac{x-x^{-1}}{\sqrt{2}}\right)\Bigg|_0^{\infty}$$

$$=\frac{1}{\sqrt{2}}\left[\lim_{x\to\infty}\arctan\left(\frac{x^2-1}{\sqrt{2}x}\right)-\lim_{x\to 0}\arctan\left(\frac{x^2-1}{\sqrt{2}x}\right)\right]$$

$$=\frac{1}{\sqrt{2}}\left[\frac{\pi}{2}+\frac{\pi}{2}\right]$$

They got it $I=\frac{\pi}{2\sqrt{2}}$

SO, $\boxed{\int_0^{\infty}\frac{x^2}{x^4+1}dx=\frac{\pi}{2\sqrt{2}}}$

107. Calculate Integral $I=\int_0^{\infty}\frac{x^2}{(x^4+1)^2}dx$

Solution; $I=\int_0^{\infty}\frac{x^2}{(x^4+1)^2}dx$

$$=\frac{1}{4}\int_0^{\infty}\frac{4x^3}{x(x^4+1)^2}dx$$

Let : $u=\frac{1}{x}\Rightarrow du=-\frac{1}{x^2}dx$ **And** $v=\int\frac{4x^3}{(x^4+1)^2}dx=-\frac{1}{(x^4+1)}$

They got it $I=\frac{1}{4}\left(-\frac{1}{x(x^4+1)}\Big|_0^{\infty}-\int_0^{\infty}\frac{1}{x^2(x^4+1)}dx\right)$

$$=\frac{1}{4}\left(-\frac{1}{x(x^4+1)}\Big|_0^{\infty}-\int_0^{\infty}\frac{(x^4+1)-x^4}{x^2(x^4+1)}dx\right)$$

$$=\frac{1}{4}\left(-\frac{1}{x(x^4+1)}\Big|_0^{\infty}-\int_0^{\infty}\frac{1}{x^2}dx+\int_0^{\infty}\frac{x^2}{x^4+1}dx\right)$$

$$=\frac{1}{4}\left(-\frac{1}{x(x^4+1)}\Big|_0^{\infty}+\frac{1}{x}\Big|_0^{\infty}+\frac{\pi}{2\sqrt{2}}\right)$$

$$=\frac{1}{4}\left(-\frac{1}{x(x^4+1)}\Big|_0^{\infty}+\frac{x^4+1}{x(x^4+1)}\Big|_0^{\infty}+\frac{\pi}{2\sqrt{2}}\right)$$

$$=\frac{1}{4}\left(\frac{-1+x^4+1}{x(x^4+1)}\Big|_0^{\infty}+\frac{\pi}{2\sqrt{2}}\right)$$

$$=\frac{1}{4}\left(\frac{x^3}{x^4+1}\Big|_0^{\infty}+\frac{\pi}{2\sqrt{2}}\right)=\frac{\pi}{8\sqrt{2}}$$

SO, $\boxed{\int_0^{\infty}\frac{x^2}{(x^4+1)^2}dx=\frac{\pi}{8\sqrt{2}}}$

108. Calculate Integral $I=\int_0^{1013\pi}\left|\sin(1013x)\right|dx$

Solution; $I=\int_0^{1013\pi}\left|\sin(1013x)\right|dx$

Let : $u=1013x\Rightarrow dx=\frac{1}{1013}du$ **; If :** $x\in(0,1013\pi)\Rightarrow u\in(0,1013^2\pi)$

They got it $I=\frac{1}{1013}\int_0^{1013^2\pi}\left|\sin(u)\right|du$

$$=\frac{1013^2}{1013}\int_0^{\pi}\left|\sin(u)\right|du$$

$$=1013\int_0^{\pi}\sin(u)du$$

$$=-1013\left(\cos(u)\right)\Big|_0^{\pi}$$

$$=2026$$

Note : $\int_0^{n\pi}\left|\sin(x)\right|dx=n\int_0^{\pi}\left|\sin(x)\right|dx=2n$

SO, $\boxed{\int_0^{1013\pi}\left|\sin(1013x)\right|dx=2026}$

109. Calculate Integral $I=\int_0^1\frac{x^4(x^2-1)}{(2x^3+1)^3}dx$

Solution; $I=\int_0^1\frac{x^4(x^2-1)}{(2x^3+1)^3}dx$

$$=\int_0^1\frac{x^4(x^2-1)}{x^6(2x+x^{-2})^3}dx$$

$$=\frac{1}{2}\int_0^1\frac{(2-2x^{-3})}{(2x+x^{-2})^3}dx$$

$$=\frac{1}{2}\int_0^1\frac{d(2x+x^{-2})}{(2x+x^{-2})^3}$$

$$=-\frac{1}{4}\times\frac{1}{(2x+x^{-2})^2}\Big|_0^1=-\frac{1}{36}$$

SO, $\boxed{\int_0^1\frac{x^4(x^2-1)}{(2x^3+1)^3}dx=-\frac{1}{36}}$

110. Calculate Integral $I=\int_{-1}^{+1}\frac{(x+\sqrt{x^2+1})^3}{\sqrt{x^2+1}}dx$

Solution; $I=\int_{-1}^{+1}\frac{(x+\sqrt{x^2+1})^3}{\sqrt{x^2+1}}dx$

$$=\int_{-1}^{+1}(x+\sqrt{x^2+1})^2\frac{(x+\sqrt{x^2+1})}{\sqrt{x^2+1}}dx$$

$$=\int_{-1}^{+1}(x+\sqrt{x^2+1})^2\left(1+\frac{x}{\sqrt{x^2+1}}\right)dx$$

$$=\int_{-1}^{+1}(x+\sqrt{x^2+1})^2d(x+\sqrt{x^2+1})$$

$$=\frac{1}{3}\times(x+\sqrt{x^2+1})^3\Big|_{-1}^{+1}=\frac{14}{3}$$

SO, $\boxed{\int_{-1}^{+1}\frac{(x+\sqrt{x^2+1})^3}{\sqrt{x^2+1}}dx=\frac{14}{3}}$

111. Calculate Integral $I=\int_{-\pi}^{\pi}\cos(x)\log\left(\frac{1-x}{1+x}\right)dx$

Solution; $I=\int_{-\pi}^{\pi}\cos(x)\log\left(\frac{1-x}{1+x}\right)dx$

$$=\int_{-\pi}^{\pi}\cos(-x)\log\left(\frac{1-(-x)}{1+(-x)}\right)dx$$

$$=\int_{-\pi}^{\pi}\cos(x)\log\left(\frac{1+x}{1-x}\right)dx$$

$$=\underbrace{-\int_{-\pi}^{\pi}\cos(x)\log\left(\frac{1-x}{1+x}\right)dx}_{-I}$$

They got it $I=0$

SO, $\boxed{\int_{-\pi}^{\pi}\cos(x)\log\left(\frac{1-x}{1+x}\right)dx=0}$

112. Calculate Integral $I=\int_0^1 \frac{\sqrt{x}}{(x^2+1)\sqrt{1-x^2}}dx$

Solution; $I=\int_0^1 \frac{\sqrt{x}}{(x^2+1)\sqrt{1-x^2}}dx$

Let: $x=\tan(u)\Rightarrow dx=\left(\tan^2(u)+1\right)du$ **; If** : $x\in(0,1)\Rightarrow u\in\left(0,\frac{\pi}{4}\right)$

They got it $I=\int_0^{\frac{\pi}{4}} \frac{\sqrt{\tan(u)}\left(\tan^2(u)+1\right)}{\left(\tan^2(u)+1\right)\sqrt{1-\tan^2(u)}}du$

$$=\int_0^{\frac{\pi}{4}} \frac{\sqrt{\tan(u)}}{\sqrt{1-\tan^2(u)}}\times\frac{\sqrt{\cos^2(u)}}{\sqrt{\cos^2(u)}}du$$

$$=\int_0^{\frac{\pi}{4}} \frac{\sqrt{\sin(u)\cos(u)}}{\sqrt{\cos^2(u)-\sin^2(u)}}du$$

$$=\frac{1}{\sqrt{2}}\int_0^{\frac{\pi}{4}} \frac{\sqrt{\sin(2u)}}{\sqrt{\cos(2u)}}du$$

$$=\frac{1}{\sqrt{2}}\int_0^{\frac{\pi}{4}} \sin^{\frac{1}{2}}(2u)\cos^{-\frac{1}{2}}(2u)du$$

Let : $2u=y\Rightarrow du=\frac{1}{2}dy$ **; If** : $u\in\left(0,\frac{\pi}{4}\right)\Rightarrow y\in\left(0,\frac{\pi}{2}\right)$

That $I=\frac{1}{2\sqrt{2}}\int_0^{\frac{\pi}{2}} \sin^{\frac{1}{2}}(y)\cos^{-\frac{1}{2}}(y)dy$

$$=\frac{1}{2\sqrt{2}}\int_0^{\frac{\pi}{2}} \sin^{2\left(\frac{3}{4}\right)-1}(y)\cos^{2\left(\frac{1}{4}\right)-1}(y)dy$$

$$=\frac{1}{4\sqrt{2}}B\left(\frac{3}{4},\frac{1}{4}\right)$$

$$=\frac{1}{4\sqrt{2}}\times\frac{\Gamma\left(\frac{3}{4}\right)\Gamma\left(\frac{1}{4}\right)}{\Gamma\left(\frac{3}{4}+\frac{1}{4}\right)}$$

$$=\frac{1}{4\sqrt{2}}\Gamma\left(\frac{3}{4}\right)\Gamma\left(\frac{1}{4}\right)=\frac{\pi}{4}$$

$$SO,\ \boxed{\int_0^1 \frac{\sqrt{x}}{(x^2+1)\sqrt{1-x^2}}dx=\frac{\pi}{4}}$$

113. Calculate Integral $I=\int_0^\infty \frac{1}{(1+x^\varphi)^\varphi}dx \quad ,\varphi=\frac{1+\sqrt{5}}{2}$

Solution; $I=\int_0^\infty \frac{1}{(1+x^\varphi)^\varphi}dx$

Let : $u=x^\varphi \Leftrightarrow x=u^{\frac{1}{\varphi}} \Rightarrow dx=\frac{1}{\varphi}u^{\frac{1}{\varphi}-1}du$ **; If :** $x\in(0,\infty)\Rightarrow u\in(0,\infty)$

They got it $I=\frac{1}{\varphi}\int_0^\infty \frac{u^{\frac{1}{\varphi}-1}}{(1+u)^\varphi}du$

$$=\frac{1}{\varphi}\int_0^\infty \frac{u^{(\varphi-1)-1}}{(1+u)^{(\varphi-1)+1}}du$$

$$=\frac{1}{\varphi}B(\varphi-1,1)$$

$$=\frac{1}{\varphi}\times\frac{\Gamma(\varphi-1)\Gamma(1)}{\Gamma(\varphi-1+1)}$$

$$=\frac{1}{\varphi}\times\frac{(\varphi-2)!}{(\varphi-2)!(x-1)}$$

$$=\frac{1}{\varphi(\varphi-1)}$$

$$=\frac{(\varphi-1)}{(\varphi-1)}$$

$$=1$$

Note $\varphi=\frac{1+\sqrt{5}}{2}$ **That** $\frac{1}{\varphi}=\varphi-1$

$$SO,\ \boxed{\int_0^\infty \frac{1}{(1+x^\varphi)^\varphi}dx=1}$$

114. Calculate Integral $I=\int_{-\frac{3\pi}{4}}^{\frac{\pi}{4}}\sqrt{1+\sin(2x)}\,dx$

Solution; $I=\int_{-\frac{3\pi}{4}}^{\frac{\pi}{4}}\sqrt{1+\sin(2x)}\,dx$

$$=\int_{-\frac{3\pi}{4}}^{\frac{\pi}{4}}\sqrt{\left(\sin(x)+\cos(x)\right)^2}\,dx$$

$$=\int_{-\frac{3\pi}{4}}^{\frac{\pi}{4}}\left|\sin(x)+\cos(x)\right|dx$$

$$=\sqrt{2}\int_{-\frac{3\pi}{4}}^{\frac{\pi}{4}}\left|\sin\left(\frac{\pi}{4}+x\right)\right|dx$$

Let $:u=\frac{\pi}{4}+x\Rightarrow du=dx$ **; If** $:x\in\left(-\frac{3\pi}{4},\frac{\pi}{4}\right)\Rightarrow u\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$

They got it $I=\sqrt{2}\underbrace{\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left|\sin(u)\right|du}_{Is\,an\,even\,function}$

$$=2\sqrt{2}\int_{0}^{\frac{\pi}{2}}\left|\sin(u)\right|du$$

$$=2\sqrt{2}\left(\frac{1}{2}\cdot 2\right)=2\sqrt{2}$$

SO, $\boxed{\int_{-\frac{3\pi}{4}}^{\frac{\pi}{4}}\sqrt{1+\sin(2x)}\,dx=2\sqrt{2}}$

115. Calculate Integral $I=\int_0^{\infty}\frac{e^{-2x}\sin(3x)}{x}dx$

Solution; $I=\int_0^{\infty}\frac{e^{-2x}\sin(3x)}{x}dx$

They got it $I(a)=\int_0^{\infty}\frac{e^{-ax}\sin(3x)}{x}dx$

That $I'(a)=-\int_0^{\infty}\frac{xe^{-ax}\sin(3x)}{x}dx$

$$=-\int_0^{\infty}e^{-ax}\sin(3x)\,dx$$

$$=-\left[\left(-\frac{1}{3}e^{-ax}\cos(3x)-\frac{a}{9}e^{-ax}\sin(3x)\right)\Big|_0^{\infty}-\frac{a^2}{9}\int_0^{\infty}e^{-ax}\sin(3x)\,dx\right]$$

$$=\left(\frac{1}{3}e^{-ax}\cos(3x)+\frac{a}{9}e^{-ax}\sin(3x)\right)\Big|_0^{\infty}+\frac{a^2}{9}J'(a)$$

Equivalent $I'(a)-\frac{a^2}{9}I'(a)=-\frac{1}{3}$

They got it $I(a)=\frac{3}{a^2-9}$

That $I(a)=\int\frac{3}{a^2-9}da$

$$=\frac{1}{2}\log\left|\frac{a-3}{a+3}\right|+C$$

If : $a=2\Rightarrow I(2)=I=\frac{1}{2}\log\left|\frac{1}{5}\right|+C$

If : $a=\infty\Rightarrow I(\infty)=0=\lim_{a\to\infty}\left(\frac{1}{2}\log\left|\frac{a-3}{a+3}\right|+C\right)$

Equivalent $0=0+C\Rightarrow C=0$

They got it $: I=-\frac{1}{2}\log(5)$

SO, $\boxed{\int_0^{\infty}\frac{e^{-2x}\sin(3x)}{x}dx=-\frac{1}{2}\log(5)}$

116. Calculate Integral $I=\int_0^1\frac{\log(x+1)\log(x)}{x}dx$

Solution; $I=\int_0^1\frac{\log(x+1)\log(x)}{x}dx$

$$=\int_0^1\left(\frac{\log(x)}{x}\sum_{n=0}^{\infty}\frac{(-1)^n x^{n+1}}{n+1}\right)dx$$

$$=\sum_{n=0}^{\infty}\left(\frac{(-1)^n}{n+1}\int_0^1 x^n\log(x)\,dx\right)$$

$$=\sum_{n=0}^{\infty}\left[\frac{(-1)^n}{n+1}\left(\underbrace{\frac{x^{n+1}\log(x)}{n+1}\Big|_0^1}_{0}-\int_0^1\frac{x^n}{n+1}dx\right)\right]$$ **(Use Partial Integral)**

$$=-\sum_{n=0}^{\infty}\frac{(-1)^n}{(n+1)^3}$$

$$=-\left(1-\frac{1}{2^3}+\frac{1}{3^3}-\frac{1}{4^3}+\cdots\right)$$

$$=-\left[\left(1+\frac{1}{2^3}+\frac{1}{3^3}+\frac{1}{4^3}+\cdots\right)-\frac{1}{2^2}\left(1+\frac{1}{2^3}+\frac{1}{3^3}+\frac{1}{4^3}+\cdots\right)\right]$$

$$=-\frac{3}{4}\left(1+\frac{1}{2^3}+\frac{1}{3^3}+\frac{1}{4^3}+\cdots\right)$$

$$=-\frac{3}{4}\zeta(3)$$

SO, $\boxed{\int_0^1\frac{\log(x+1)\log(x)}{x}dx=-\frac{3}{4}\zeta(3)}$

117. Calculate Integral $I=\int_0^1\frac{x^e-x^\pi}{\log(x)}dx$

Solution; $I=\int_0^1\frac{x^e-x^\pi}{\log(x)}dx$

$$I=\int_0^1\frac{(x^e-1)-(x^\pi-1)}{\log(x)}dx$$

$$=\int_0^1\frac{x^e-1}{\log(x)}dx-\int_0^1\frac{x^\pi-1}{\log(x)}dx$$

$$=\log(e+1)-\log(\pi+1)$$

$$=\log\left(\frac{e+1}{\pi+1}\right)$$

Note : $\int_0^1\frac{x^n-1}{\log(x)}dx=\log(n+1)$

SO, $\boxed{\int_0^1\frac{x^e-x^\pi}{\log(x)}dx=\log\left(\frac{e+1}{\pi+1}\right)}$

118. Calculate Integral $I=\int_{e^{-1}}^{e^3}\frac{\sqrt{1+\log(x)}}{x\log(x)}dx$

Solution; $I=\int_{e^{-1}}^{e^3}\frac{\sqrt{1+\log(x)}}{x\log(x)}dx$

Let : $u^2=1+\log(x)\Leftrightarrow\log(x)=u^2-1\Rightarrow 2udu=\frac{1}{x}dx$ **; If :** $x\in\left(e^{-1},e^3\right)\Rightarrow u\in(0,2)$

They got it $I=2\int_0^2\frac{u^2}{u^2-1}du$

$$=2\int_0^2\left(1-\frac{1}{u^2-1}\right)du$$

$$=2\left(u-\frac{1}{2}\log\left|\frac{u-1}{u+1}\right|\right)\Bigg|_0^2=4-\log(3)$$

SO, $\boxed{\int_{e^{-1}}^{e^3}\frac{\sqrt{1+\log(x)}}{x\log(x)}dx=4-\log(3)}$

119. Calculate Integral $I=\int_0^{\pi}\sqrt{\frac{1+\cos(2x)}{3}}dx$

Solution; $I=\int_0^{\pi}\sqrt{\frac{1+\cos(2x)}{3}}dx$

$$=\frac{1}{\sqrt{3}}\int_0^{\pi}\sqrt{2\cos^2(x)}\,dx$$

$$=\frac{\sqrt{2}}{\sqrt{3}}\int_0^{\pi}\left|\cos(x)\right|dx$$

$$=\frac{\sqrt{6}}{3}\left(\int_0^{\frac{\pi}{2}}\cos(x)\,dx-\int_{\frac{\pi}{2}}^{\pi}\cos(x)\,dx\right)$$

$$=\frac{2\sqrt{6}}{3}$$

SO, $\boxed{\int_0^{\pi}\sqrt{\frac{1+\cos(2x)}{3}}dx=\frac{2\sqrt{6}}{3}}$

120. Calculate Integral $I=\int_0^1\left(x\log(x)\right)^n dx$

Solution; $I=\int_0^1\left(x\log(x)\right)^n dx$

Let : $u=-\log(x)\Rightarrow x=e^{-u}\Rightarrow dx=-e^{-u}du$ **; If :** $x\in(0,1)\Rightarrow u\in(\infty,0)$

They got it $I=-\int_\infty^0 e^{-nu}\left(-u\right)^n e^{-u}du$

$$=(-1)^n\int_0^\infty u^n e^{-(n+1)u}du$$

Let : $t=(n+1)u\Leftrightarrow u=\frac{1}{n+1}t\Rightarrow du=\frac{1}{n+1}dt$ **; If :** $u\in(0,\infty)\Rightarrow t\in(0,\infty)$

They got it $I=(-1)^n\int_0^\infty\left(\frac{t}{n+1}\right)^n\frac{e^{-t}}{n+1}dt$

$$=\frac{(-1)^n}{(n+1)^{n+1}}\int_0^\infty t^n e^{-t}dt$$

$$=\frac{(-1)^n\Gamma(n+1)}{(n+1)^{(n+1)}}=\frac{(-1)^n n!}{(n+1)^{(n+1)}}$$

$SO,$ $\boxed{\int_0^1\left(x\log(x)\right)^n dx=\frac{(-1)^n n!}{(n+1)^{(n+1)}}}$

121. Calculate Integral $I=\int_{-1}^1 x^{\frac{x}{\log(x)}}dx$

Solution; $I=\int_{-1}^1 x^{\frac{x}{\log(x)}}dx$

$$=\int_{-1}^1 e^{\log\left(x^{\frac{x}{\log(x)}}\right)}dx$$

$$=\int_{-1}^1 e^{\frac{x}{\log(x)}\log(x)}dx$$

$$=\int_{-1}^1 e^x dx=\frac{(e-1)(e+1)}{e}$$

$SO,$ $\boxed{\int_{-1}^1 x^{\frac{x}{\log(x)}}dx=\frac{(e-1)(e+1)}{e}}$

122. Calculate Integral $I=\int_0^{\frac{\pi}{2}}\frac{1}{\cos(x)-\sin(x)}dx$

Solution; $I=\int_0^{\frac{\pi}{2}}\frac{1}{\cos(x)-\sin(x)}dx$

$$=\int_0^{\frac{\pi}{2}}\frac{1}{\cos\left(\frac{\pi}{2}-x\right)-\sin\left(\frac{\pi}{2}-x\right)}dx$$

$$=\int_0^{\frac{\pi}{2}}\frac{1}{\sin(x)-\cos(x)}dx$$

$$=\underbrace{-\int_0^{\frac{\pi}{2}}\frac{1}{\cos(x)-\sin(x)}dx}_{-I}$$

They got it $I=-I$ **That** $I=0$

SO, $\boxed{\int_0^{\frac{\pi}{2}}\frac{1}{\cos(x)-\sin(x)}dx=0}$

123. Calculate Integral $I=\int_{\pi}^{e}x^{\frac{1-2x}{x}}\log\left(\frac{x}{e}\right)dx$

Solution; $I=\int_{\pi}^{e}x^{\frac{1-2x}{x}}\log\left(\frac{x}{e}\right)dx$

$$=\int_{\pi}^{e}x^{\frac{1}{x}-2}\log\left(\frac{x}{e}\right)dx=\int_{e}^{\pi}x^{\frac{1}{x}}\frac{\log\left(\frac{x}{e}\right)}{x^2}dx$$

$$=\int_{e}^{\pi}e^{\frac{\log(x)}{x}}\frac{\log\left(\frac{x}{e}\right)}{x^2}dx=\int_{e}^{\pi}e^{\frac{\log(x)}{x}}d\left(\frac{\log(x)}{x}\right)$$

$$=e^{\frac{\log(x)}{x}}\Big|_{e}^{\pi}=\sqrt[\pi]{\pi}-\sqrt[e]{e}$$

SO, $\boxed{\int_{\pi}^{e}x^{\frac{1-2x}{x}}\log\left(\frac{x}{e}\right)dx=\sqrt[\pi]{\pi}-\sqrt[e]{e}}$

124. Calculate Integral $I=\int_0^{\infty}\left(x-\frac{x^2}{2}+\frac{x^5}{2\times4}-\frac{x^7}{2\times4\times6}+...\right)\left(1+\frac{x^2}{2^2}+\frac{x^4}{2^2\times4^2}+...\right)dx$

Solution; $I=\int_0^{\infty}\left(x-\frac{x^2}{2}+\frac{x^5}{2\times4}-\frac{x^7}{2\times4\times6}+...\right)\left(1+\frac{x^2}{2^2}+\frac{x^4}{2^2\times4^2}+...\right)dx$

We Have $x-\frac{x^2}{2}+\frac{x^5}{2\times4}-\frac{x^7}{2\times4\times6}+...=\sum_{n=0}^{\infty}\frac{(-1)^n x^{2n+1}}{2^n\times n!}=x\sum_{n=0}^{\infty}\frac{\left(-\frac{x^2}{2}\right)^n}{n!}=xe^{-\frac{x^2}{2}}$

And $1+\frac{x^2}{2^2}+\frac{x^4}{2^2\times4^2}+\frac{x^6}{2^2\times4^2\times6^2}+...=\sum_{m=0}^{\infty}\frac{x^{2m}}{2^{2m}\times(m!)^2}$

Equivalent $I=\int_0^{\infty}\left(xe^{-\frac{x^2}{2}}\right)\left(\sum_{m=0}^{\infty}\frac{(x^2)^m}{2^{2m}\times(m!)^2}\right)dx$

$$=\int_0^{\infty}\left(e^{-\frac{x^2}{2}}\sum_{m=0}^{\infty}\frac{(x^2)^m}{2^{2m}\times(m!)^2}\right)xdx$$

Let $:u=\frac{x^2}{2}\Rightarrow du=xdx$ **; If** $:x\in(0,\infty)\Rightarrow u\in(0,\infty)$

They got it $I=\int_0^{\infty}\left(e^{-u}\sum_{m=0}^{\infty}\frac{(2u)^m}{2^{2m}\times(m!)^2}\right)du$

$$=\sum_{m=0}^{\infty}\left(\frac{2^m}{2^{2m}\times(m!)^2}\int_0^{\infty}e^{-u}u^m du\right)$$

$$=\sum_{m=0}^{\infty}\frac{m!}{2^m\times(m!)^2}$$

$$=\sum_{m=0}^{\infty}\frac{1}{2^m\times m!}$$

$$=\sum_{m=0}^{\infty}\frac{\left(\frac{1}{2}\right)^m}{m!}=e^{1/2}$$

SO, $\boxed{\int_0^{\infty}\left(x-\frac{x^2}{2}+\frac{x^5}{2\times4}-\frac{x^7}{2\times4\times6}+...\right)\left(1+\frac{x^2}{2^2}+\frac{x^4}{2^2\times4^2}+\frac{x^6}{2^2\times4^2\times6^2}+...\right)dx=\sqrt{e}}$

125. Calculate Integral $I=\int_0^1 \log\big((x-1)!\big)dx$

Solution; $I=\int_0^1 \log\big((x-1)!\big)dx$

$$=\int_0^1 \log\big(\Gamma(x)\big)dx \qquad (!)$$

$$=\int_0^1 \log\big(\Gamma(x-1)\big)dx \quad (!!)$$

Take $(!)+(!!)$ **They Have :** $2I=\int_0^1 \log\big(\Gamma(x)\Gamma(1-x)\big)dx$

$$2I=\int_0^1 \log\left(\frac{\pi}{\sin(\pi x)}\right)dx$$

$$=\int_0^1 \log\big(\pi\big)dx-\int_0^1 \log\big(\sin(\pi x)\big)dx$$

$$=\log(\pi)-J \qquad (!!!)$$

For : $J=\int_0^1 \log\big(\sin(\pi x)\big)dx$

Let : $t=\pi x\Rightarrow dx=\frac{1}{\pi}dt$ **; If :** $x\in(0,1)\Rightarrow t\in(0,\pi)$

They got it $J=\frac{1}{\pi}\int_0^{\pi}\log\big(\sin(t)\big)dt$

$$=\frac{2}{\pi}\int_0^{\frac{\pi}{2}}\log\big(\sin(t)\big)dt$$

$$=\frac{2}{\pi}\left(-\frac{\pi}{2}\log(2)\right)$$

$$=-\log(2)$$

Note : $\begin{cases}\int_0^{2a} f(x)dx=2\int_0^a f(x)dx\\ f(2a-x)=f(x)\end{cases}$

Take $(!!!)$ **They have** $2I=\log(\pi)-\big(-\log(2)\big)$

$$=\log(2\pi)$$

They got it $I=\frac{\log(2\pi)}{2}$

$SO,$ $\boxed{\int_0^1 \log\big((x-1)!\big)dx=\frac{\log(2\pi)}{2}}$

126. Calculate Integral $I=\int_{-\infty}^{\infty}\Gamma(1+ix)\Gamma(1-ix)\,dx$

Solution; $I=\int_{-\infty}^{\infty}\Gamma(1+ix)\Gamma(1-ix)\,dx$

$$=\int_{-\infty}^{\infty}ix\Gamma(ix)\Gamma(1-ix)\,dx=\int_{-\infty}^{\infty}ix\frac{\pi}{\sin(i\pi x)}dx$$

Note : $\begin{cases}\Gamma(1+n)=n\Gamma(n)\\ \Gamma(n)\Gamma(1-n)=\dfrac{\pi}{\sin(n\pi)}\end{cases}$

$$=\int_{-\infty}^{\infty}\frac{ix\pi}{\dfrac{e^{i(i\pi x)}-e^{-i(i\pi x)}}{2i}}dx=-\int_{-\infty}^{\infty}\frac{2\pi x}{e^{-\pi x}-e^{\pi x}}dx$$

$$=2\pi\int_{-\infty}^{\infty}\frac{x}{e^{\pi x}-e^{-\pi x}}dx=4\pi\int_{0}^{\infty}\frac{xe^{-\pi x}}{1-e^{-2\pi x}}dx$$

Let : $u=\pi x\Rightarrow du=\pi dx$ **; If :** $x\in(-\infty,\infty)\Rightarrow u\in(-\infty,\infty)$

They got it $I=\dfrac{4}{\pi}\int_{0}^{\infty}\dfrac{ue^{-u}}{1-e^{-2u}}du$

$$=\frac{4}{\pi}\int_{0}^{\infty}\left(ue^{-u}\sum_{n=0}^{\infty}e^{-2un}\right)du=\frac{4}{\pi}\sum_{n=0}^{\infty}\left(\int_{0}^{\infty}ue^{-(2n+1)u}du\right)$$

Let : $t=(2n+1)u\Rightarrow du=\dfrac{dt}{(2n+1)}$ **; If :** $u\in(0,\infty)\Rightarrow t\in(0,\infty)$

That $I=\dfrac{4}{\pi}\sum_{n=0}^{\infty}\left(\dfrac{1}{(2n+1)^2}\int_{0}^{+\infty}te^{-t}\right)dt$

$$=\frac{4}{\pi}\sum_{n=0}^{\infty}\frac{\Gamma(2)}{(2n+1)^2}$$

$$=\frac{4}{\pi}\left(1+\frac{1}{3^2}+\frac{1}{5^2}+\ldots\right)$$

$$=\frac{4}{\pi}\left[\left(1+\frac{1}{2^2}+\frac{1}{3^2}+\ldots\right)-\frac{1}{4}\left(1+\frac{1}{2^2}+\frac{1}{3^2}+\ldots\right)\right]$$

$$=\frac{4}{\pi}\left(\zeta(2)-\frac{1}{4}\times\zeta(2)\right)=\frac{\pi}{2}$$

SO, $\boxed{\int_{-\infty}^{\infty}\Gamma(1+ix)\Gamma(1-ix)\,dx=\frac{\pi}{2}}$

127. Calculate Integral $I=\int_0^{\infty}\log\left(1+\frac{1}{x^2}+\frac{1}{x^4}\right)dx$

Solution; $I=\int_0^{\infty}\log\left(1+\frac{1}{x^2}+\frac{1}{x^4}\right)dx$

Let : $x=\frac{1}{u}\Rightarrow dx=-\frac{1}{u^2}du$ **; If** : $x\in(0,\infty)\Rightarrow u\in(\infty,0)$

They got it $I=-\int_{\infty}^{0}\frac{\log\left(1+u^2+u^4\right)}{u^2}du$

$$=\underbrace{\int_0^{\infty}\frac{\log\left(1+u^2+u^4\right)}{u^2}du}_{\text{Use Partial Integral}}$$

$$=-\frac{\log\left(1+u^2+u^4\right)}{u}\Bigg|_0^{\infty}+\int_0^{\infty}\frac{2u+4u^3}{u(1+u^2+u^4)}du$$

$$=-\left[\lim_{x\to\infty}\frac{\log\left(1+u^2+u^4\right)}{u}-\lim_{x\to 0}\frac{\log\left(1+u^2+u^4\right)}{u}\right]+\int_0^{\infty}\frac{2+4u^2}{1+u^2+u^4}du$$

$$=-[0-0]+\int_0^{\infty}\frac{2+4u^2}{1+u^2+u^4}du$$

$$=\int_0^{\infty}\frac{2+4u^2}{1+u^2+u^4}du\quad(!)$$

Let : $u=\frac{1}{t}\Rightarrow du=-\frac{1}{t^2}dt$ **; If** : $u\in(0,\infty)\Rightarrow t\in(\infty,0)$

That $I=-\int_{\infty}^{0}\frac{2+4\cdot\frac{1}{t^2}}{1+\frac{1}{t^2}+\frac{1}{t^4}}\times\frac{1}{t^2}dt$

$$=\int_0^{\infty}\frac{2x^2+4}{1+x^2+x^4}dx\quad(!!)$$

Take $(!)+(!!)$ **They Have** $2I=6\int_0^{\infty}\frac{1+x^2}{1+x^2+x^4}dx$

That $I=3\int_0^{\infty}\dfrac{\left(1+\dfrac{1}{x^2}\right)}{\left(x-\dfrac{1}{x^2}\right)^2+3}dx$

$$=3\int_0^{\infty}\frac{\left(1+\frac{1}{x}\right)'}{\left(x-\frac{1}{x}\right)^2+3}dx$$

$$=\sqrt{3}\arctan\left(\frac{1}{\sqrt{3}}\left(x-\frac{1}{x}\right)\right)\Bigg|_0^{\infty}$$

$$=\sqrt{3}\arctan\left(\frac{x^2-1}{\sqrt{3}x}\right)\Bigg|_0^{\infty}=\sqrt{3}\pi$$

SO, $\boxed{\int_0^{\infty}\log\left(1+\frac{1}{x^2}+\frac{1}{x^4}\right)dx=\sqrt{3}\pi}$

128. Calculate Integral $I=\int_0^{45^\circ}\dfrac{1}{\cos(x)+\sin(x)}dx$

Solution; $I=\int_0^{45^\circ}\dfrac{1}{\cos(x)+\sin(x)}dx$

$$=\frac{1}{\sqrt{2}}\int_0^{45^\circ}\frac{1}{\cos(x)\cos(45^\circ)+\sin(x)\sin(45^\circ)}dx$$

$$=\frac{1}{\sqrt{2}}\int_0^{45^\circ}\frac{1}{\cos(x-45^\circ)}dx$$

$$=\frac{1}{\sqrt{2}}\log\left|\sec(x-45^\circ)+\tan(x-45^\circ)\right|\Bigg|_0^{45^\circ}$$

$$=\frac{1}{\sqrt{2}}\log(\sqrt{2}+1)$$

SO, $\boxed{\int_0^{45^\circ}\frac{1}{\cos(x)+\sin(x)}dx=\frac{1}{\sqrt{2}}\log(\sqrt{2}+1)}$

129. Calculate Integral $I=\int_0^{\infty}\frac{x\sqrt{x}}{e^{2x}-1}dx$

Solution; $I=\int_0^{\infty}\frac{x\sqrt{x}}{e^{2x}-1}dx$

$$=\int_0^{\infty}\frac{x^{\frac{3}{2}}e^{-2x}}{1-e^{-2x}}dx$$

Note : $\frac{1}{1-u}=\sum_{k=0}^{\infty}(u)\ ,|u|<1$

$$=\int_0^{\infty}\left(x^{\frac{3}{2}}e^{-2x}\sum_{n=0}^{\infty}e^{-2xn}\right)dx=\sum_{n=0}^{\infty}\left(\int_0^{\infty}x^{\frac{3}{2}}e^{-2(n+1)x}dx\right)$$

Let : $t=2(n+1)x\Leftrightarrow x=\frac{t}{2(n+1)}\Rightarrow dx=\frac{1}{2(n+1)}dt$ **; If :** $x\in(0,\infty)\Rightarrow t\in(0,\infty)$

They got it $I=\sum_{n=0}^{\infty}\left(\int_0^{\infty}\left(\frac{t}{2(n+1)}\right)^{\frac{3}{2}}\frac{e^{-t}}{2(n+1)}dt\right)$

$$=\sum_{n=0}^{\infty}\left(\frac{1}{\left(2(n+1)\right)^{\frac{5}{2}}}\int_0^{\infty}t^{\frac{3}{2}}e^{-t}dx\right)$$

$$=\frac{1}{4\sqrt{2}}\sum_{n=0}^{\infty}\left(\frac{1}{(n+1)^{\frac{5}{2}}}\int_0^{\infty}t^{\left(1+\frac{3}{2}\right)-1}e^{-t}dx\right)$$

$$=\frac{1}{4\sqrt{2}}\zeta\left(\frac{5}{2}\right)\Gamma\left(1+\frac{3}{2}\right)$$

$$=\frac{1}{4\sqrt{2}}\times\frac{3}{2}\zeta\left(\frac{5}{2}\right)\Gamma\left(1+\frac{1}{2}\right)$$

$$=\frac{3}{16}\times\sqrt{\frac{\pi}{2}}\zeta\left(\frac{5}{2}\right)$$

SO, $\boxed{\int_0^{\infty}\frac{x\sqrt{x}}{e^{2x}-1}dx=\frac{3}{16}\times\sqrt{\frac{\pi}{2}}\zeta\left(\frac{5}{2}\right)}$

130. Calculate Integral $I=\int_0^1 \frac{x^8+1}{x^{10}+1}dx$

Solution; $I=\int_0^1 \frac{x^8+1}{x^{10}+1}dx \qquad (1)$

Let : $x=\frac{1}{u}\Rightarrow dx=-\frac{1}{u^2}du$ **; If :** $x\in(0,1)\Rightarrow u\in(\infty,1)$

They got it $I=-\int_\infty^1 \frac{\frac{1}{u^8}+1}{\frac{1}{u^{10}}+1}\cdot\frac{1}{u^2}du=\int_1^\infty \frac{u^8+1}{u^{10}+1}du \qquad (2)$

Take $(1)+(2)$ **Bring** $2I=\int_0^1 \frac{u^8+1}{u^{10}+1}du+\int_1^\infty \frac{u^8+1}{u^{10}+1}du$

$$=\int_0^\infty \frac{u^8+1}{u^{10}+1}du$$

Note : $\begin{cases}\int_a^b f(x)dx+\int_b^c f(x)dx=\int_a^c f(x)dx\\ a\geq b\geq c\end{cases}$

Let : $t=u^{10}\Leftrightarrow u=t^{\frac{1}{10}}\Rightarrow du=\frac{1}{10}t^{\frac{1}{10}-1}dt$ **; If :** $u\in(0,\infty)\Rightarrow t\in(0,\infty)$

They got it $2I=\frac{1}{10}\int_0^\infty \frac{t^{\frac{8}{10}}+1}{t+1}\times t^{\frac{1}{10}-1}dt$

$$=\frac{1}{10}\left(\int_0^\infty \frac{t^{\frac{9}{10}-1}}{t+1}dt+\int_0^\infty \frac{t^{\frac{1}{10}-1}}{t+1}dt\right)$$

$$=\frac{1}{20}\left(\int_0^\infty \frac{t^{\frac{9}{10}-1}}{(t+1)^{\frac{9}{10}+\frac{1}{10}}}dt+\int_0^\infty \frac{t^{\frac{1}{10}-1}}{(t+1)^{\frac{1}{10}+\frac{9}{10}}}dt\right)$$

$$=\frac{1}{10}\left(B\left(\frac{9}{10},\frac{1}{10}\right)+B\left(\frac{1}{10},\frac{9}{10}\right)\right)$$

$$=\frac{\Gamma\left(\frac{1}{10}\right)\Gamma\left(1-\frac{1}{10}\right)}{5}$$

$$= \frac{\pi}{5\sin\left(\frac{\pi}{10}\right)} = \frac{(\sqrt{5}+1)\pi}{5}$$

Bring $I = \frac{(1+\sqrt{5})\pi}{10}$

SO, $\boxed{\int_0^1 \frac{x^8+1}{x^{10}+1}dx = \frac{(1+\sqrt{5})\pi}{10}}$

131. Calculate Integral $I = \int_0^\infty \log\left(x+\frac{1}{x}\right)\frac{1}{(x^2+1)}dx$

Solution; $I = \int_0^\infty \log\left(x+\frac{1}{x}\right)\frac{1}{(x^2+1)}dx$

Let : $x = \tan(u) \Rightarrow dx = \left(1+\tan^2(u)\right)du$ **; If** : $x \in (0,\infty) \Rightarrow u \in \left(0,\frac{\pi}{2}\right)$

They got it $I = \int_0^{\frac{\pi}{2}} \log\left(\tan(u)+\frac{1}{\tan(u)}\right)\frac{1}{(\tan^2(u)+1)}\left(1+\tan^2(u)\right)du$

$$= \int_0^{\frac{\pi}{2}} \log\left(\frac{\tan^2(u)+1}{\tan(u)}\right)du$$

$$= \int_0^{\frac{\pi}{2}} \log\left(\tan^2(u)+1\right)du - \underbrace{\int_0^{\frac{\pi}{2}} \log\left(\tan(u)\right)du}_{0}$$

$$= \int_0^{\frac{\pi}{2}} \log\left(\frac{1}{\cos^2(u)}\right)du$$

$$= -2\int_0^{\frac{\pi}{2}} \log\left(\cos(u)\right)du$$

$$= -2\left(-\frac{\pi}{2}\log(2)\right) = \pi\log(2)$$

SO, $\boxed{\int_0^\infty \log\left(x+\frac{1}{x}\right)\frac{1}{(x^2+1)}dx = \pi\log(2)}$

132. Calculate Integral $I=\int_{-\infty}^{\infty}e^{-(x-x^{-1})^2}(x+x^{-2})dx$

Solution; $I=\int_{-\infty}^{\infty}e^{-(x-x^{-1})^2}(x+x^{-2})dx$

$$=\underbrace{\int_{-\infty}^{0}e^{-(x-x^{-1})^2}(x+x^{-2})dx}_{K}+\int_{0}^{\infty}e^{-(x-x^{-1})^2}(x+x^{-2})dx \quad (*)$$

Take : $K=\int_{-\infty}^{0}e^{-(x-x^{-1})^2}(x+x^{-2})dx$

Let : $x=-\frac{1}{u}\Rightarrow dx=\frac{1}{u^2}du$ **; If :** $x\in(-\infty,0)\Rightarrow u\in(0,\infty)$

Bring $K=\int_{0}^{\infty}e^{-(u-u^{-1})^2}(-\frac{1}{u}+u^2)\frac{1}{u^2}du$

$$=\int_{0}^{\infty}e^{-(u-u^{-1})^2}(-\frac{1}{u^3}+1)du$$

$$=\int_{0}^{\infty}e^{-(u-u^{-1})^2}(1-u^{-3})du$$

Take (*) **That :** $I=\int_{0}^{\infty}e^{-(u-u^{-1})^2}(1-u^{-3})du+\int_{0}^{\infty}e^{-(x-x^{-1})^2}(x+x^{-2})dx$

$$=\int_{0}^{\infty}e^{-(x-x^{-1})^2}(1-x^{-3}+x+x^{-2})dx$$

$$=\int_{0}^{\infty}e^{-(x-x^{-1})^2}\left(x-x^{-1}+1\right)\left(1+x^{-2}\right)dx$$

Let: $u=x-x^{-1}\Rightarrow du=(1+x^{-2})du$ **; If :** $x\in(0,\infty)\Rightarrow u\in(-\infty,\infty)$

They got it $I=\int_{-\infty}^{\infty}e^{-u^2}\left(u+1\right)du$

$$=\underbrace{\int_{-\infty}^{\infty}ue^{-u^2}du}_{Is\,an\,odd\ function}+\int_{-\infty}^{\infty}e^{-u^2}du$$

$$=0+2\int_{0}^{\infty}e^{-u^2}du$$

$$=2\cdot\frac{\sqrt{\pi}}{2}$$

$$=\sqrt{\pi}$$

SO, $\boxed{\int_{-\infty}^{\infty}e^{-(x-x^{-1})^2}(x+x^{-2})dx=\sqrt{\pi}}$

133. Calculate Integral $I=\int_0^{\infty}\frac{x\log(x)}{(x^2+1)^2}dx$

Solution; $I=\int_0^{\infty}\frac{x\log(x)}{(x^2+1)^2}dx$

Let: $x=\tan(u)\Rightarrow dx=\sec^2(u)du$ **; If :** $x\in(0,\infty)\Rightarrow u\in\left(0,\frac{\pi}{2}\right)$

They got it $I=\int_0^{\frac{\pi}{2}}\frac{\tan(u)\log(\tan(u))}{(\tan^2(u)+1)^2}\times\sec^2(u)\,du$

$$=\int_0^{\frac{\pi}{2}}\frac{\tan(u)\log(\tan(u))}{\sec^4(u)}\times\sec^2(u)\,du$$

$$=\int_0^{\frac{\pi}{2}}\frac{\tan(u)\log(\tan(u))}{\sec^2(u)}du$$

$$=\int_0^{\frac{\pi}{2}}\frac{\sin(u)}{\cos(x)}.\cos^2(u)\log(\tan(u))du$$

$$=\frac{1}{2}\int_0^{\frac{\pi}{2}}2\sin(u)\cos(u)\log(\tan(u))du$$

$$=\frac{1}{2}\int_0^{\frac{\pi}{2}}\sin(2u)\log(\tan(u))du$$

Note : $\int_0^{2a}f(x)dx=\int_0^{a}\left(f(x)+f(2a-x)\right)dx$

$$=\frac{1}{2}\int_0^{\frac{\pi}{4}}\left(\sin(2u)\log(\tan(u))+\sin 2\left(\frac{\pi}{2}-u\right)\log\left(\tan\left(\frac{\pi}{2}-u\right)\right)\right)du$$

$$=\frac{1}{2}\int_0^{\frac{\pi}{4}}\left(\sin(2u)\log(\tan(u))+\sin\left(\pi-2u\right)\log\left(\cot(u)\right)\right)du$$

$$=\frac{1}{2}\int_0^{\frac{\pi}{4}}\left(\sin(2u)\log(\tan(u))+\sin(2u)\log(\tan(u)^{-1}\right)du$$

$$=\frac{1}{2}\int_0^{\frac{\pi}{4}}\left(\sin(2u)\log(\tan(u))-\sin(2u)\log(\tan(u)\right)du$$

$$=0$$

SO, $\boxed{\int_0^{\infty}\frac{x\log(x)}{(x^2+1)^2}dx=0}$

134. Calculate Integral $I=\int_0^1 x^{-x}dx$

Solution; $I=\int_0^1 x^{-x}dx$

$$=\int_0^1 e^{-x\log(x)}dx$$

By Contact : $e^x=\sum_{n=0}^{\infty}\left(\frac{x^n}{n!}\right)$ **That :** $e^{-x\log(x)}=\sum_{n=0}^{\infty}\left(\frac{\left(-x\log(x)\right)^n}{n!}\right)$

They got it $I=\sum_{n=0}^{\infty}\left(\int_0^1\frac{\left(-x\log(x)\right)^n}{n!}dx\right)$

$$=\sum_{n=0}^{\infty}\left(\frac{(-1)^n}{n!}\int_0^1 x^n\log^n(x)dx\right)$$

Let: $u=t(n+1)\Leftrightarrow\frac{u}{n+1}=t\Rightarrow\frac{du}{n+1}=dt$ **; If :** $t\in(0,\infty)\Rightarrow u\in(0,\infty)$

That $I=\sum_{n=0}^{\infty}\left(\frac{1}{n!}\int_0^{\infty}e^{-u}\frac{u^n}{(n+1)^n}\times\frac{du}{n+1}\right)$

$$=\sum_{n=0}^{\infty}\left(\frac{1}{(n+1)^{n+1}\Gamma(n+1)}\int_0^{\infty}u^ne^{-u}du\right)$$

$$=\sum_{n=0}^{\infty}\left(\frac{1}{(n+1)^{n+1}\Gamma(n+1)}\cdot\Gamma(n+1)\right)$$

$$=\sum_{n=0}^{\infty}\left(\frac{1}{(n+1)^{n+1}}\right)$$

$$=\sum_{n=1}^{\infty}\left(\frac{1}{n^n}\right)$$

$$=1+\frac{1}{2^2}+\frac{1}{3^3}+\frac{1}{4^4}+\ldots$$

SO, $\boxed{\int_0^1 x^{-x}dx=\sum_{n=1}^{\infty}\left(\frac{1}{n^n}\right)}$

135. Calculate Integral $I=\int_0^{90^\circ}\sin^3(2x)\cos(x)dx$

Solution; $I=\int_0^{90^\circ}\sin^3(2x)\cos(x)dx$

$$=\int_0^{90^\circ}\left(2\sin(x)\cos(x)\right)^3\cos(x)dx$$

$$=8\int_0^{90^\circ}\sin^3(x)\cos^4(x)dx$$

$$=8\int_0^{90^\circ}\sin^{2(2)-1}(x)\cos^{2\left(\frac{5}{2}\right)-1}(x)dx$$

$$=4B\left(2,\frac{5}{2}\right)=\frac{4\Gamma(2)\Gamma\left(\frac{5}{2}\right)}{\Gamma\left(2+\frac{5}{2}\right)}=\frac{16}{35}$$

SO, $\boxed{\int_0^{90^\circ}\sin^3(2x)\cos(x)dx=\frac{16}{35}}$

136. Calculate Integral $I=\int_0^{\frac{\pi}{2}}\sin^4(x)\cos^5(x)dx$

Solution; $I=\int_0^{\frac{\pi}{2}}\sin^4(x)\cos^5(x)dx$

$$=\int_0^{\frac{\pi}{2}}\sin^{2\left(\frac{5}{2}\right)-1}(x)\cos^{2(3)-1}(x)dx$$

$$=\frac{1}{2}B\left(\frac{5}{2},3\right)$$

$$=\frac{1}{2}.\frac{\Gamma\left(\frac{5}{2}\right)\Gamma(3)}{\Gamma\left(3+\frac{5}{2}\right)}=\frac{8}{315}$$

SO, $\boxed{\int_0^{\frac{\pi}{2}}\sin^4(x)\cos^5(x)dx=\frac{8}{315}}$

137. Calculate Integral $I=\int_0^{\frac{\pi}{2}}\frac{\sqrt[3]{\tan(x)}}{\left(\sin(x)+\cos(x)\right)^2}dx$

Solution; $I=\int_0^{\frac{\pi}{2}}\frac{\sqrt[3]{\tan(x)}}{\left(\sin(x)+\cos(x)\right)^2}dx$

$$=\int_0^{\frac{\pi}{2}}\frac{\sqrt[3]{\tan(x)}}{\left(1+\tan(x)\right)^2\cos^2(x)}dx$$

$$=\int_0^{\frac{\pi}{2}}\frac{\sqrt[3]{\tan(x)}}{\left(1+\tan(x)\right)^2}d\left(\tan(x)\right)$$

Let : $u=\tan(x)\Rightarrow du=d\left(\tan(x)\right)$ **; If** : $x\in\left(0,\frac{\pi}{2}\right)\Rightarrow u\in(0,\infty)$

They got it $I=\int_0^{\infty}\frac{\sqrt[3]{u}}{\left(1+u\right)^2}du$

$$=\int_0^{\infty}\frac{u^{\frac{4}{3}-1}}{\left(1+u\right)^{\frac{4}{3}+\frac{2}{3}}}du$$

$$=B\left(\frac{4}{3},\frac{2}{3}\right)$$

$$=\frac{\Gamma\left(\frac{4}{3}\right)\Gamma\left(\frac{2}{3}\right)}{\Gamma\left(\frac{4}{3}+\frac{2}{3}\right)}$$

$$=\Gamma\left(1+\frac{1}{3}\right)\Gamma\left(\frac{2}{3}\right)$$

$$=\frac{\pi}{3}\cdot\csc\left(\frac{\pi}{3}\right)=\frac{2\pi}{3\sqrt{3}}$$

SO, $\boxed{\int_0^{\frac{\pi}{2}}\frac{\sqrt[3]{\tan(x)}}{\left(\sin(x)+\cos(x)\right)^2}dx=\frac{2\pi}{3\sqrt{3}}}$

138. Calculate Integral $I=\int_0^{90^\circ}\frac{1}{5\cos^2(x)+4\sin^2(x)-3}dx$

Solution; $I=\int_0^{90^\circ}\frac{1}{5\cos^2(x)+4\sin^2(x)-3}dx$

$$=\int_0^{90^\circ}\frac{1}{5\cos^2(x)+4\sin^2(x)-3}\times\frac{\sec^2(x)}{\sec^2(x)}dx$$

$$=\int_0^{90^\circ}\frac{\sec^2(x)}{5+4\tan^2(x)-3\sec^2(x)}dx$$

$$=\int_0^{90^\circ}\frac{1}{5+4\tan^2(x)-3\left(1+\tan^2(x)\right)}d(\tan(x))$$

$$=\int_0^{90^\circ}\frac{1}{2+\tan^2(x)}d(\tan(x))$$

$$=\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{\tan(x)}{\sqrt{2}}\right)\Bigg|_0^{90^\circ}$$

$$=\frac{1}{\sqrt{2}}\left(\frac{\pi}{2}-0\right)=\frac{\pi}{2\sqrt{2}}$$

SO, $\boxed{\int_0^{90^\circ}\frac{1}{5\cos^2(x)+4\sin^2(x)-3}dx=\frac{\pi}{2\sqrt{2}}}$

139. Calculate Integral $I=\int_0^1\frac{1}{\sqrt{x}(1+\sqrt{x})\sqrt{1-x}}dx$

Solution; $I=\int_0^1\frac{1}{\sqrt{x}(1+\sqrt{x})\sqrt{1-x}}dx$

$$=2\int_0^1\frac{1}{(1+\sqrt{x})\sqrt{(1-\sqrt{x})(1+\sqrt{x})}}d(\sqrt{x})$$

$$=2\int_0^1\frac{1}{(1+\sqrt{x})^2\sqrt{\frac{2}{(1+\sqrt{x})}-1}}d(\sqrt{x})$$

$$= 2\int_0^1 \frac{1}{(1+\sqrt{x})^2\sqrt{\frac{2}{(1+\sqrt{x})}-1}}\, d(1+\sqrt{x})$$

$$= -\int_0^1 \frac{1}{\sqrt{\frac{2}{(1+\sqrt{x})}-1}}\, d\left(\frac{2}{1+\sqrt{x}}\right)$$

$$= -2\sqrt{\frac{2}{(1+\sqrt{x})}-1}\Bigg|_0^1 = 2$$

$SO,$ $\boxed{\int_0^1 \frac{1}{\sqrt{x}(1+\sqrt{x})\sqrt{1-x}}dx = 2}$

140. Calculate Integral $I = \int_0^{\pi} \frac{1}{1+e^{\tan(x)}}\, dx$

Solution; $I = \int_0^{\pi} \frac{1}{1+e^{\tan(x)}}\, dx \ \ (1)$

$$= \int_0^{\pi} \frac{1}{1+e^{\tan(\pi-x)}}\, dx$$

$$= \int_0^{\pi} \frac{1}{1+e^{-\tan(x)}}\, dx$$

$$= \int_0^{\pi} \frac{e^{\tan(x)}}{1+e^{\tan(x)}}\, dx \ \ (2)$$

Take $(1)+(2)$ **That :** $2I = \int_0^{\pi} \frac{e^{\tan(x)}}{1+e^{\tan(x)}}\, dx + \int_0^{\pi} \frac{1}{1+e^{\tan(x)}}\, dx$

$$= \int_0^{\pi} \frac{1+e^{\tan(x)}}{1+e^{\tan(x)}}\, dx = \pi$$

They got it $I = \frac{\pi}{2}$

$SO,$ $\boxed{\int_0^{\pi} \frac{1}{1+e^{\tan(x)}}\, dx = \frac{\pi}{2}}$

141. Calculate Integral $I=\int_0^1 \frac{x^{\frac{1}{\log(2x)}-1}}{\log^2(2x)}dx$

Solution; $I=\int_0^1 \frac{x^{\frac{1}{\log(2x)}-1}}{\log^2(2x)}dx$

$$=\int_0^1 \frac{(2x)^{\frac{1}{\log(2x)}}\times(2)^{-\frac{1}{\log(2x)}}}{x\log^2(2x)}dx$$

$$=\int_0^1 \frac{e\times(2)^{-\frac{1}{\log(2x)}}}{x\log^2(2x)}dx$$

$$=e\int_0^1 \frac{(2)^{-\frac{1}{\log(2x)}}}{x\log^2(2x)}dx$$

Let : $u=-\frac{1}{\log(2x)}\Rightarrow du=\frac{1}{x\log^2(2x)}dx$ **; If :** $x\in(0,1)\Rightarrow u\in\left(0,-\frac{1}{\log(2)}\right)$

They got it $I=e\int_0^1 2^u du$

$$=e\int_0^1 e^{u\log(2)}du$$

$$=\frac{e}{\log(2)}\int_0^1 \log(2)e^{u\log(2)}du$$

$$=\frac{e}{\log(2)}\int_0^1 e^{u\log(2)}d\left(u\log(2)\right)$$

$$=\frac{2^u e}{\log(2)}\Bigg|_0^{-\frac{1}{\log(2)}}$$

$$=\frac{1-e}{\log(2)}$$

SO, $\boxed{\int_0^1 \frac{x^{\frac{1}{\log(2x)}-1}}{\log^2(2x)}dx=\frac{1-e}{\log(2)}}$

142. Calculate Integral $I=\int_e^{\pi}\sqrt{x-e}\cdot\sqrt{\pi-x}\,dx$

Solution; $I=\int_e^{\pi}\sqrt{x-e}\cdot\sqrt{\pi-x}\,dx$

$$=\int_e^{\pi}(\pi-x)\sqrt{\frac{x-e}{\pi-x}}\,dx$$

Let : $u=\frac{x-e}{\pi-x}\Leftrightarrow x=\frac{\pi u+e}{u+1}\Rightarrow dx=\frac{\pi+e}{(u+1)^2}du$ **; If :** $x\in(e,\pi)\Rightarrow u\in(0,\infty)$

They got it $I=\int_0^{\infty}\left(\pi-\frac{\pi u+e}{u+1}\right)\sqrt{u}\times\frac{\pi-e}{(u+1)^2}du$

$$=(\pi-e)^2\int_0^{\infty}\frac{u^{\frac{1}{2}}}{(u+1)^3}du$$

$$=(\pi-e)^2\int_0^{\infty}\frac{u^{\frac{3}{2}-1}}{(u+1)^{\frac{3}{2}+\frac{3}{2}}}du$$

$$=(\pi-e)^2B\left(\frac{3}{2},\frac{3}{2}\right)$$

$$=\frac{(\pi-e)^2\Gamma^2\left(\frac{3}{2}\right)}{\Gamma(3)}=\frac{\pi(\pi-e)^2}{8}$$

SO, $\boxed{\int_e^{\pi}\sqrt{x-e}\sqrt{\pi-x}\,dx=\frac{\pi(\pi-e)^2}{8}}$

143. Calculate Integral $I=\int_{-\infty}^{\infty}\frac{1}{x^{12}+1}dx$

Solution; $I=\underbrace{\int_{-\infty}^{\infty}\frac{1}{x^{12}+1}dx}_{\text{is an even function}}$

$$=2\int_0^{\infty}\frac{1}{x^{12}+1}dx$$

Let : $x=t^{\frac{1}{12}}\Rightarrow dx=\frac{1}{12}t^{\frac{1}{12}-1}dt$ **; If :** $x\in(0,\infty)\Rightarrow t\in(0,\infty)$

They got it $I=2\times\frac{1}{12}\int_0^{\infty}\frac{t^{\frac{1}{12}-1}}{(1+t)}dx$

$$=\frac{1}{6}\int_0^{\infty}\frac{t^{\frac{1}{12}-1}}{(1+t)^{\frac{1}{12}+\frac{11}{12}}}dx$$

$$=\frac{1}{6}.B\left(\frac{1}{12},\frac{11}{12}\right)$$

$$=\frac{1}{6}.\frac{\Gamma\left(\frac{1}{12}\right)\Gamma\left(\frac{11}{12}\right)}{\Gamma\left(\frac{1}{12}+\frac{11}{12}\right)}$$

$$=\frac{1}{6}.\frac{\pi}{\sin\left(\frac{\pi}{12}\right)}=\frac{\pi(\sqrt{6}+\sqrt{2})}{6}$$

SO, $\boxed{\int_{-\infty}^{\infty}\frac{1}{x^{12}+1}dx=\frac{\pi(\sqrt{6}+\sqrt{2})}{6}}$

144. Calculate Integral $I=\int_0^1\frac{\sin^{-1}(x)}{x}dx$

Solution; $I=\int_0^1\frac{\sin^{-1}(x)}{x}dx$

Let : $x=\sin(u)\Rightarrow dx=\cos(u)du$ **; If :** $x\in(0,1)\Rightarrow u\in\left(0,\frac{\pi}{2}\right)$

They got it $I=\int_0^{\frac{\pi}{2}}\frac{\sin^{-1}(\sin(u))}{\sin(u)}\times\cos(u)du$

$$=\int_0^{\frac{\pi}{2}}\frac{u}{\tan(u)}du=\frac{\pi}{2}\log(2)$$

SO, $\boxed{\int_0^1\frac{\sin^{-1}(x)}{x}dx=\frac{\pi}{2}\log(2)}$

145. Calculate Integral $I=\int_0^2\left|(1-x)\log(x)\right|dx$

Solution; $I=\int_0^2\left|(1-x)\log(x)\right|dx$

$=\int_0^2\left|(x-1)\log(x)\right|dx$ **Note** : $|a\pm b|=|b\pm a|$

If : $x=1\Rightarrow\left|(x-1)\log(x)\right|=0$ **That** : $\begin{cases}0<x<1\Rightarrow(x-1)\log(x)>0\\1<x<2\Rightarrow(x-1)\log(x)>0\end{cases}$

They got it $I=\underbrace{\int_0^2(x-1)\log(x)\,dx}_{\text{Use partial integral}}$

$=\left(\frac{x^2}{2}-x\right)\log(x)\Big|_0^2-\int_0^2\left(\frac{x^2}{2}-x\right).\frac{1}{x}dx$

$=\lim_{x\to2}\left(\frac{x^2}{2}-x\right)\log(x)-\lim_{x\to0^+}\left(\frac{x^2}{2}-x\right)\log(x)-\int_0^2\left(\frac{x}{2}-1\right)dx$

$=0\times\log(2)-\lim_{x\to0^+}\left(\frac{x^2-2x}{2}\right)\log(x)-\int_0^2\left(\frac{x}{2}-1\right)dx$

$=-\lim_{x\to0^+}\frac{\left(\log(x)\right)}{\left(\frac{2}{x^2-2x}\right)}+\left(x-\frac{x^2}{4}\right)\Big|_0^2$

$=-\lim_{x\to0^+}\frac{d\left(\log(x)\right)}{d\left(\frac{2}{x^2-2x}\right)}+1$

$=1+\lim_{x\to0^+}\frac{(x^2-2x)^2}{4x(x-1)}$

$=1+\lim_{x\to0^+}\frac{x^2(x-2)^2}{4x(x-1)}$

$=1$

SO, $\boxed{\int_0^2\left|(1-x)\log(x)\right|dx=1}$

146. Calculate Integral $I=\int_{e^{-1}}^{e^2}\left|\frac{\log(x)}{x}\right|dx$

Solution; $I=\int_{e^{-1}}^{e^2}\left|\frac{\log(x)}{x}\right|dx$

By : $x=1\Rightarrow\frac{\log(x)}{x}=0$ **That :** $\begin{cases} e^{-1}<x<1\Rightarrow\dfrac{\log(x)}{x}<0 \\ 1<x<e^2\Rightarrow\dfrac{\log(x)}{x}>0 \end{cases}$

They got it $I=\int_{e^{-1}}^{1}\left|\frac{\log(x)}{x}\right|dx+\int_{1}^{e^2}\left|\frac{\log(x)}{x}\right|dx$

$$=-\int_{e^{-1}}^{1}\frac{\log(x)}{x}dx+\int_{1}^{e^2}\frac{\log(x)}{x}dx$$

$$=-\frac{1}{2}\left(\log^2(x)\right)\Big|_{e^{-1}}^{1}+\frac{1}{2}\left(\log^2(x)\right)\Big|_{1}^{e^2}$$

$$=-\frac{1}{2}\left(0-(-1)^2\right)+\frac{1}{2}(4-0)=\frac{5}{2}$$

SO, $\boxed{\int_{e^{-1}}^{e^2}\left|\frac{\log(x)}{x}\right|dx=\frac{5}{2}}$

147. Calculate Integral $I=\int_{0}^{\frac{\pi}{2}}\lfloor 1+\sin(x)\rfloor dx$

Solution; $I=\int_{0}^{\frac{\pi}{2}}\lfloor 1+\sin(x)\rfloor dx$

$$=\int_{0}^{\frac{\pi}{2}}\left(\lfloor\sin(x)\rfloor+1\right)dx$$

Note : $\begin{cases} 0\le\sin(x)\le 1\Rightarrow\lfloor\sin(x)\rfloor=0 \\ \lfloor 1+\alpha\rfloor=\lfloor\alpha\rfloor+1 \end{cases}$

$$=\int_{0}^{\frac{\pi}{2}}(0+1)dx$$

$$=\frac{\pi}{2}$$

SO, $\boxed{\int_{0}^{\frac{\pi}{2}}\lfloor 1+\sin(x)\rfloor dx=\frac{\pi}{2}}$

148. Calculate Integral $I=\int_1^e \frac{x-1}{x^2-\log(x^x)}dx$

Solution; $I=\int_1^e \frac{x-1}{x(x-\log(x))}dx$

$$=\int_1^e \frac{1-\frac{1}{x}}{x-\log(x)}dx$$

$$=\int_1^e \frac{d\left(x-\log(x)\right)}{\left(x-\log(x)\right)}$$

$$=\log\left|x-\log(x)\right|_1^e$$

$$=\log\left|e-\log(e)\right|-\log\left|1-\log(1)\right|$$

$$=\log(e-1)$$

SO, $\boxed{\int_1^e \frac{x-1}{x^2-\log(x^x)}dx=\log(e-1)}$

149. Calculate Integral $I=\int_{-1}^1 \sqrt{\frac{1-x}{1+x}}dx$

Solution; $I=\int_{-1}^1 \sqrt{\frac{1-x}{1+x}}dx$

$$=\int_{-1}^1 \frac{1-x}{\sqrt{1-x^2}}dx$$

$$=\int_{-1}^1 \frac{1}{\sqrt{1-x^2}}dx-\underbrace{\int_{-1}^1 \frac{x}{\sqrt{1-x^2}}dx}_{is\ an\ odd\ function}$$

$$=\int_{-1}^1 \left(\arcsin(x)\right)'dx-0$$

$$=2\arcsin(1)$$

$$=\pi$$

SO, $\boxed{\int_{-1}^1 \sqrt{\frac{1-x}{1+x}}dx=\pi}$

150. Calculate Integral $I=\int_1^2\sqrt{\frac{x-1}{2-x}}\,dx$

Solution; $I=\int_1^2\sqrt{\frac{x-1}{2-x}}\,dx$

Let : $u=\sqrt{2-x}\Leftrightarrow x=2-u^2\Rightarrow dx=-2udu$ **; If :** $x\in(1,2)\Rightarrow u\in(1,0)$

They got it $I=-2\int_1^0\frac{\sqrt{1-u^2}}{u}\times udu$

$=2\int_0^1\sqrt{1-u^2}\,du$

Let : $u=\sin(t)\Rightarrow du=\cos(t)dt$ **; If :** $u\in(0,1)\Rightarrow t\in\left(0,\frac{\pi}{2}\right)$

They got it $I=2\int_0^{\frac{\pi}{2}}\sqrt{1-\sin^2(t)}\cos(t)\,dt$

$=2\int_0^{\frac{\pi}{2}}|\cos(t)|\cos(t)\,dt$

$=2\int_0^{\frac{\pi}{2}}\cos^2(t)\,dt=\frac{\pi}{2}$

SO, $\boxed{\int_1^2\sqrt{\frac{x-1}{2-x}}\,dx=\frac{\pi}{2}}$

151. Calculate Integral $I=\int_{-1}^1 x\sqrt{x^2}\,dx$

Solution; $I=\int_{-1}^1 x\sqrt{x^2}\,dx$

$=\int_{-1}^1 x|x|\,dx$

$=\int_{-1}^1(-x)|-x|\,dx$

$=-\int_{-1}^1 x|x|\,dx$

Bring $K=-K$ **That** $K=0$

SO, $\boxed{\int_{-1}^1 x|x|\,dx=0}$

152. Calculate Integral $I=\int_0^{\infty}\frac{\sqrt{x}}{(x+1)^2}dx$

Solution; $I=\int_0^{\infty}\frac{\sqrt{x}}{(x+1)^2}dx$

$$=\int_0^{\infty}\frac{x^{(1+\frac{1}{2})-1}}{(x+1)^{(1+\frac{1}{2})+\frac{1}{2}}}dx$$

$$=B\left(1+\frac{1}{2},\frac{1}{2}\right)=\frac{\Gamma\left(1+\frac{1}{2}\right)\Gamma\left(\frac{1}{2}\right)}{\Gamma\left(1+\frac{1}{2}+\frac{1}{2}\right)}$$

$$=\frac{1}{2}\Gamma\left(\frac{1}{2}\right)=\frac{\sqrt{\pi}}{2}$$

SO, $\boxed{\int_0^{\infty}\frac{\sqrt{x}}{(x+1)^2}dx=\frac{\sqrt{\pi}}{2}}$

153. Calculate Integral $I=\int_0^{\infty}\frac{x}{1+x^3}dx$

Solution; $I=\int_0^{\infty}\frac{x}{1+x^3}dx$

Let : $x=t^{\frac{1}{3}}\Rightarrow dx=\frac{1}{3}t^{\frac{1}{3}-1}dt$ **; If :** $x\in(0,\infty)\Rightarrow t\in(0,\infty)$

They got it $I=\int_0^{\infty}\frac{t^{\frac{1}{3}}}{(1+t)}\times\frac{1}{3}t^{\frac{1}{3}-1}dt$

$$=\frac{1}{3}\int_0^{\infty}\frac{t^{\frac{2}{3}-1}}{(1+t)^{\frac{2}{3}+\frac{1}{3}}}dt=B\left(\frac{2}{3},\frac{1}{3}\right)$$

$$=\frac{1}{3}\left(\pi\csc\left(\frac{\pi}{3}\right)\right)=\frac{2\pi}{3\sqrt{3}}$$

SO, $\boxed{\int_0^{\infty}\frac{x}{1+x^3}dx=\frac{2\pi}{3\sqrt{3}}}$

154. Calculate Integral $I=\int_0^{\infty}\frac{x^6+1}{x^{12}+1}dx$

Solution; $I=\int_0^{\infty}\frac{x^6+1}{x^{12}+1}dx$

Let : $x=t^{\frac{1}{12}}\Rightarrow dx=\frac{1}{12}t^{\frac{1}{12}-1}dt$ **; If :** $x\in(0,\infty)\Rightarrow t\in(0,\infty)$

They got it $I=\int_0^{\infty}\frac{t^{\frac{6}{12}}+1}{t+1}\times\frac{1}{12}t^{\frac{1}{12}-1}dt$

$$=\frac{1}{12}\int_0^{\infty}\left(\frac{t^{\frac{7}{12}-1}}{t+1}+\frac{t^{\frac{1}{12}-1}}{t+1}\right)dt$$

$$=\frac{1}{12}\int_0^{\infty}\left(\frac{t^{\frac{7}{12}-1}}{(t+1)^{\frac{7}{12}+\frac{5}{12}}}+\frac{t^{\frac{1}{12}-1}}{(t+1)^{\frac{1}{12}+\frac{11}{12}}}\right)dt$$

$$=\frac{1}{12}\left[B\left(\frac{7}{12},\frac{5}{2}\right)+B\left(\frac{1}{12},\frac{11}{12}\right)\right]$$

$$=\frac{1}{12}\left[\frac{\Gamma\left(\frac{7}{12}\right)\Gamma\left(\frac{5}{12}\right)}{\Gamma\left(\frac{7}{12}+\frac{5}{12}\right)}+\frac{\Gamma\left(\frac{1}{12}\right)\Gamma\left(\frac{11}{12}\right)}{\Gamma\left(\frac{1}{12}+\frac{11}{12}\right)}\right]$$

$$=\frac{1}{12}\left[\Gamma\left(\frac{7}{12}\right)\Gamma\left(1-\frac{7}{12}\right)+\Gamma\left(\frac{1}{12}\right)\Gamma\left(1-\frac{1}{12}\right)\right]$$

$$=\frac{1}{12}\left[\frac{\pi}{\sin\left(\frac{7\pi}{12}\right)}+\frac{\pi}{\sin\left(\frac{\pi}{12}\right)}\right]$$

$$=\frac{\pi}{12}\left[\frac{1}{\cos\left(\frac{\pi}{12}\right)}+\frac{1}{\sin\left(\frac{\pi}{12}\right)}\right]$$

$$=\frac{\pi}{12}\left(\frac{1}{\frac{\sqrt{2+\sqrt{3}}}{2}}+\frac{1}{\frac{\sqrt{2-\sqrt{3}}}{2}}\right)$$

$$=\frac{\pi}{\sqrt{6}}$$

Note : $\begin{cases}\cos\left(\frac{\pi}{12}\right)=\frac{\sqrt{2+\sqrt{3}}}{2}\\ \sin\left(\frac{\pi}{12}\right)=\frac{\sqrt{2-\sqrt{3}}}{2}\end{cases}$

SO, $\boxed{\int_0^{\infty}\frac{x^6+1}{x^{12}+1}dx=\frac{\pi}{\sqrt{6}}}$

155. Calculate Integral $I=\int_0^{\pi/2}\tan^n(x)dx$

Solution; $I=\int_0^{\pi/2}\tan^n(x)dx$

$$=\int_0^{\pi/2}\sin^n(x)\cos^{-n}(x)dx$$

$$=\int_0^{\pi/2}\sin^{2\left(\frac{1+n}{2}\right)-1}(x)\cos^{2\left(\frac{1-n}{2}\right)-1}(x)dx$$

$$=\frac{1}{2}B\left(\frac{1+n}{2},\frac{1-n}{2}\right)$$

$$=\frac{1}{2}\times\frac{\Gamma\left(\frac{1+n}{2}\right)\Gamma\left(\frac{1-n}{2}\right)}{\left(\frac{1+n}{2}+\frac{1-n}{2}\right)}$$

$$=\frac{1}{2}\times\Gamma\left(\frac{1+n}{2}\right)\Gamma\left(1-\frac{1+n}{2}\right)$$

$$=\frac{\pi}{2}\csc\left(\frac{1+n}{2}\right)$$

SO, $\boxed{\int_0^{\pi/2}\tan^n(x)dx=\frac{\pi}{2}\csc\left(\frac{1+n}{2}\right)}$

156. Calculate Integral $I=\int_0^{\frac{\pi}{2}}\frac{x\sin(x)\cos(x)}{\sin^4(x)+\cos^4(x)}dx$

Solution; $I=\int_0^{\frac{\pi}{2}}\frac{x\sin(x)\cos(x)}{\sin^4(x)+\cos^4(x)}dx$

$$=\int_0^{\frac{\pi}{2}}\frac{\left(\frac{\pi}{2}-x\right)\sin\left(\frac{\pi}{2}-x\right)\cos\left(\frac{\pi}{2}-x\right)}{\sin^4\left(\frac{\pi}{2}-x\right)+\cos^4\left(\frac{\pi}{2}-x\right)}dx$$

$$=\frac{\pi}{2}\int_0^{\frac{\pi}{2}}\frac{\sin(x)\cos(x)}{\sin^4(x)+\cos^4(x)}dx-\underbrace{\int_0^{\frac{\pi}{2}}\frac{x\sin(x)\cos(x)}{\sin^4(x)+\cos^4(x)}dx}_{I}$$

Bring $I=\frac{\pi}{2}\int_0^{\frac{\pi}{2}}\frac{\sin(x)\cos(x)}{\sin^4(x)+\cos^4(x)}dx$

$$=\frac{\pi}{4}\int_0^{\frac{\pi}{2}}\frac{\tan(x)}{1+\tan^4(x)}\times\frac{1}{\cos^2(x)}dx$$

$$=\frac{\pi}{4}\int_0^{\frac{\pi}{2}}\frac{\tan(x)}{1+\tan^4(x)}d\left(\tan(x)\right)$$

$$=\frac{\pi}{8}\int_0^{\frac{\pi}{2}}\frac{2\tan(x)}{1+\tan^4(x)}d\left(\tan(x)\right)$$

$$=\frac{\pi}{8}\int_0^{\frac{\pi}{2}}\frac{1}{1+\tan^2(x)}d\left(\tan^2(x)\right)$$

$$=\frac{\pi}{8}\left(\arctan\left(\tan^2(x)\right)\right)\Big|_0^{\frac{\pi}{2}}$$

$$=\frac{\pi}{8}\left(\arctan(\infty)-\arctan(0)\right)$$

$$=\frac{\pi}{8}\left(\frac{\pi}{2}-0\right)=\frac{\pi^2}{16}$$

SO, $\boxed{\int_0^{\frac{\pi}{2}}\frac{x\sin(x)\cos(x)}{\sin^4(x)+\cos^4(x)}dx=\frac{\pi^2}{16}}$

157. Calculate Integral $I=\int_0^{\frac{\pi}{2}}\frac{\sin^2(x)}{1+\sin(x)\cos(x)}dx$

Solution; $I=\int_0^{\frac{\pi}{2}}\frac{\sin^2(x)}{1+\sin(x)\cos(x)}dx \quad (*)$

$$=\int_0^{\frac{\pi}{2}}\frac{\sin^2\left(\frac{\pi}{2}-x\right)}{1+\sin\left(\frac{\pi}{2}-x\right)\cos\left(\frac{\pi}{2}-x\right)}dx$$

$$=\int_0^{\frac{\pi}{2}}\frac{\cos^2(x)}{1+\sin(x)\cos(x)}dx \quad (**)$$

Take $(*)+(**)$ **That Have :** $2I=\int_0^{\frac{\pi}{2}}\frac{\sin^2(x)}{1+\sin(x)\cos(x)}dx+\int_0^{\frac{\pi}{2}}\frac{\cos^2(x)}{1+\sin(x)\cos(x)}dx$

Bring $I=\int_0^{\frac{\pi}{2}}\frac{1}{2+2\sin(x)\cos(x)}dx=\int_0^{\frac{\pi}{2}}\frac{1}{2+\sin(2x)}dx$

Let : $t=2x\Rightarrow dx=\frac{1}{2}dt$ **; If :** $x\in\left(0,\frac{\pi}{2}\right)\Rightarrow t\in(0,\pi)$

They got it $I=\int_0^{\pi}\frac{1}{2+\sin(t)}dt$

Let : $y=\tan\left(\frac{t}{2}\right)\Rightarrow dt=\frac{2}{1+y^2}dy$ **; If :** $x\in(0,\pi)\Rightarrow t\in(0,\infty)$

That $I=\int_0^{\infty}\frac{1}{2+\frac{2y}{1+y^2}}\times\frac{2}{1+y^2}dy$

$$=\int_0^{\infty}\frac{1}{y^2+t+1}dy=\int_0^{\infty}\frac{1}{(2y+1)^2+3}dy$$

$$=\frac{1}{\sqrt{3}}\tan^{-1}\left(\frac{2y+1}{\sqrt{3}}\right)\Bigg|_0^{\infty}=\frac{\pi}{3\sqrt{3}}$$

SO, $\boxed{\int_0^{\frac{\pi}{2}}\frac{\sin^2(x)}{1+\sin(x)\cos(x)}dx=\frac{\pi}{3\sqrt{3}}}$

158. Calculate Integral $I=\int_0^{45^\circ}\tan(x)\log\big(\tan(x)\big)dx$

Solution; $I=\int_0^{45^\circ}\tan(x)\log\big(\tan(x)\big)dx$

Let : $u=\tan(x)\Leftrightarrow x=\tan^{-1}(t)\Rightarrow dx=\dfrac{1}{1+t^2}dt$ **; If** : $x\in\big(0,45^\circ\big)\Rightarrow t\in(0,1)$

Bring $I=\int_0^1\dfrac{t\log(t)}{1+t^2}dt$

Let : $u=\log(t)\Rightarrow du=\dfrac{1}{t}dt$ **And** : $v=\int\dfrac{t}{1+t^2}dt=\dfrac{1}{2}\log(1+t^2)$

They got it $I=\underbrace{\frac{1}{2}.\log(t)\log(t^2+1)\Big|_0^1}_{0}-\frac{1}{2}\int_0^1\frac{\log(1+t^2)}{t}dt$

$$=-\frac{1}{2}\int_0^1\frac{1}{t}\left(\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}\cdot t^{2n}\right)dt$$

Note : $\log(1+x^2)=\sum_{n=1}^{\infty}\left(\dfrac{(-1)^{n+1}x^{2n}}{n}\right)$

$$=-\frac{1}{2}\sum_{n=1}^{\infty}\left(\frac{(-1)^{n+1}}{n}\int_0^1 x^{2n-1}dt\right)$$

$$=-\frac{1}{2}\sum_{n=1}^{\infty}\left(\frac{(-1)^{n+1}}{n}\times\frac{x^{2n}}{2n}\Big|_0^1\right)$$

$$=-\frac{1}{4}\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n^2}$$

$$=-\frac{1}{4}\eta(2)$$

$$=-\frac{1}{4}\times\frac{\pi^2}{12}$$

$$=-\frac{\pi^2}{48}$$

$SO,$ $\boxed{\int_0^{45^\circ}\tan(x)\log\big(\tan(x)\big)dx=-\dfrac{\pi^2}{48}}$

159. Calculate Integral $I=\int_0^1 \frac{1}{\sqrt{-\log(x)}}dx$

Solution; $I=\int_0^1 \frac{1}{\sqrt{-\log(x)}}dx$

Let : $t=-\log(x)\Rightarrow dx=-e^{-t}dt$ **; If :** $x\in(0,1)\Rightarrow t\in(0,\infty)$

Implies That $J=-\int_\infty^0 \frac{e^{-t}}{\sqrt{t}}dt$

$$=\int_0^\infty t^{-\frac{1}{2}}e^{-t}dt$$

$$=\int_0^\infty t^{\frac{1}{2}-1}e^{-t}dt$$

$$=\Gamma\left(\frac{1}{2}\right)=\left(-\frac{1}{2}\right)!$$

$$=\sqrt{\pi}$$

SO, $\boxed{\int_0^1 \frac{1}{\sqrt{-\log(x)}}dx=\sqrt{\pi}}$

160. Calculate Integral $I=\int_0^\infty \frac{\log(x)}{x^2-x+1}dx$

Solution; $I=\int_0^\infty \frac{\log(x)}{x^2-x+1}dx$

Let : $x=\frac{1}{t}\Rightarrow dx=-\frac{1}{t^2}dt$ **; If :** $x\in(0,\infty)\Rightarrow t\in(\infty,0)$

Implies that $I=\int_\infty^0 \frac{\log(1/t)}{\left(\frac{1}{t}\right)^2-\left(\frac{1}{t}\right)+1}\left(-\frac{1}{t^2}\right)dt$

$$=-\int_0^\infty \frac{\log(t)}{t^2-t+1}dt=-I$$

Note : $\int_0^a f(x)dx=\int_0^a f(t)dt$

Equivalent $I=-I\Rightarrow I=0$

SO, $\boxed{\int_0^\infty \frac{\log(x)}{x^2-x+1}dx=0}$

161. Calculate Integral $I=\int_0^{\infty}\frac{(2-x)^{2024}}{(2+x)^{2026}}dx$

Solution; $I=\int_0^{\infty}\frac{(2-x)^{2024}}{(2+x)^{2026}}dx$

$$=-\frac{1}{4}\int_0^{\infty}\left(\frac{2-x}{2+x}\right)^{2024}\times\frac{-4}{(2+x)^2}dx$$

Let : $t=\frac{2-x}{2+x}\Rightarrow dt=\frac{\begin{bmatrix}-1 & 2\\ 1 & 2\end{bmatrix}}{(2+x)^2}dx=\frac{-4}{(2+x)^2}dx$ **; If** : $x\in(0,\infty)\Rightarrow t\in(1,-1)$

Implies That $I=-\frac{1}{4}\int_1^{-1}(t)^{2024}\,dt$

$$=\frac{1}{2}\int_0^{1}(t)^{2024}\,dt$$

$$=\frac{1}{4050}$$

SO, $\boxed{\int_0^{\infty}\frac{(2-x)^{2024}}{(2+x)^{2026}}dx=\frac{1}{4050}}$

162. Calculate Integral $I=\int_0^1\left(\log\left(\frac{1}{x}\right)\right)^{n-1}dx$

Solution; $I=\int_0^1\left(\log\left(\frac{1}{x}\right)\right)^{n-1}dx$

Let : $t=\log\left(\frac{1}{x}\right)\Leftrightarrow x=e^{-t}\Rightarrow dx=-e^{-t}dt$ **;** **If** : $x\in(0,1)\Rightarrow t\in(0,\infty)$

Implies That $I=-\int_{\infty}^{0}t^{n-1}e^{-t}dt$

$$=\int_0^{\infty}t^{n-1}e^{-t}dt$$

$$=\Gamma(n)=(n-1)!$$

SO, $\boxed{\int_0^1\left(\log\left(\frac{1}{x}\right)\right)^{n-1}dx=\Gamma(n)}$

163. Calculate Integral $I=\int_0^1 \frac{e^x-1}{x}dx$

Method 1

Solution; $I=\int_0^1 \frac{e^x-1}{x}dx$

By : $e^x=\sum_{n=0}^{\infty}\left(\frac{x^n}{n!}\right)\Rightarrow e^x-1=\sum_{n=1}^{\infty}\left(\frac{x^n}{n!}\right)$

Implies That $I=\int_0^1\left(\frac{1}{x}\sum_{n=1}^{\infty}\left(\frac{x^n}{n!}\right)\right)dx=\sum_{n=1}^{\infty}\left(\frac{1}{n!}\int_0^1 x^{n-1}dx\right)$

$$=\sum_{n=1}^{\infty}\left(\frac{1}{n!}\right)\frac{x^n}{n}\Big|_0^1=\sum_{n=1}^{\infty}\left(\frac{1}{n.n!}\right)$$

Method 2

By $I=\int_0^1 \frac{e^x-1}{x}dx$

Implies That $I(a)=\int_0^1 \frac{e^{ax}-1}{x}dx$

$$\Rightarrow I'(a)=\int_0^1 e^{ax}dx=\frac{e^a-1}{a}$$

Equivalent $I(a)=\int\frac{e^a-1}{a}da=\sum_{n=1}^{\infty}\left(\frac{a^n}{n.n!}\right)+C$

If : $a=0\Rightarrow I(0)=0=\sum_{n=1}^{\infty}\left(\frac{1}{n!}\right)\frac{0^n}{n}+C\equiv C=0$

If : $a=1\Rightarrow I(1)=I=\sum_{n=1}^{\infty}\left(\frac{1}{n!}\right)\frac{1^n}{n}+C$ **But** $C=0$

Implies That : $I=\sum_{n=1}^{\infty}\left(\frac{1}{n.n!}\right)$

SO, $\boxed{\int_0^1 \frac{e^x-1}{x}dx=\sum_{n=1}^{\infty}\left(\frac{1}{n.n!}\right)}$

164. Calculate Integral $I=\int_0^{\infty} x^{-\log(x)}\log(x^x)dx$

Solution; $I=\int_0^{\infty} x^{-\log(x)}\log(x^x)dx$

$$=\int_0^{\infty} x^{-\log(x)+1}\log(x)dx$$

If : $t=\log(x)\Rightarrow dx=e^t dt$ **; If** : $x\in(0,\infty)\Rightarrow t\in(-\infty,\infty)$

Implies That $I=\int_{-\infty}^{\infty}(e^t)^{-t+1}t.e^t dt=e\int_{-\infty}^{\infty}e^{-(t-1)^2}tdt$

$$=e\int_{-\infty}^{\infty}e^{-(t-1)^2}[(t-1)+1]dt$$

$$=e\int_{-\infty}^{\infty}(t-1)e^{-(t-1)^2}dt+e\int_{-\infty}^{\infty}e^{-(t-1)^2}dt$$

$$=\frac{e}{2}\int_{-\infty}^{\infty}e^{-(t-1)^2}d\left((t-1)^2\right)+e\int_{-\infty}^{\infty}e^{-(t-1)^2}d(t-1)$$

$$=\frac{e}{2}e^{-(t-1)^2}\Big|_{-\infty}^{\infty}+e\sqrt{\pi}$$

$$=e\sqrt{\pi}$$

SO, $\boxed{\int_0^{\infty} x^{-\log(x)}\log(x^x)dx=e\sqrt{\pi}}$

165. Calculate Integral $I=\int_0^{\infty}\frac{1}{1+x^n}dx$

Solution; $I=\int_0^{\infty}\frac{1}{1+x^n}dx$

Let : $x=t^{\frac{1}{n}}\Rightarrow dx=\frac{1}{n}t^{\frac{1}{n}-1}dt$ **; If** : $x\in(0,\infty)\Rightarrow t\in(0,\infty)$

Implies $I=\frac{1}{n}\int_0^{\infty}\frac{t^{\frac{1}{n}-1}}{(1+t)}dt=\frac{1}{n}\int_0^{\infty}\frac{t^{\frac{1}{n}-1}}{(1+t)^{\frac{1}{n}+\frac{n-1}{n}}}dt$

$$=\frac{1}{n}B\left(\frac{1}{n},\frac{n-1}{n}\right)=\frac{1}{n}\times\pi\csc\left(\frac{\pi}{n}\right)$$

SO, $\boxed{\int_0^{\infty}\frac{1}{1+x^n}dx=\frac{\pi}{n}\csc\left(\frac{\pi}{n}\right)}$

166. Calculate Integral $I=\int_0^1 \frac{\tan^{-1}(x)}{x+1}$

Solution; $I=\int_0^1 \frac{\tan^{-1}(x)}{x+1}$

Let : $u=\tan^{-1}(x)\Rightarrow du=\frac{1}{x^2+1}dx$ **; And** $v=\int\frac{1}{x+1}dx=\log(x+1)$

Implies That $I=\tan^{-1}(x)\log(x+1)\Big|_0^1-\int_0^1\frac{\log(x+1)}{x^2+1}dx$

$$=\frac{\pi}{4}\log(2)-\int_0^1\frac{\log(x+1)}{x^2+1}dx \quad (*)$$

Take : $J=\int_0^1\frac{\log(x+1)}{x^2+1}dx$

Let : $x=\tan(u)\Rightarrow dx=\sec^2(u)du$ **; If :** $x\in(0,1)\Rightarrow u\in\left(0,\frac{\pi}{4}\right)$

Implies That : $J=\int_0^{\frac{\pi}{4}}\frac{\log\left(\tan(u)+1\right)}{\tan^2(u)+1}\sec^2(u)\,du$

$$=\int_0^{\frac{\pi}{4}}\log\left(\tan(u)+1\right)du$$

$$=\int_0^{\frac{\pi}{4}}\log\left(\tan\left(\frac{\pi}{4}-u\right)+1\right)du$$

$$=\int_0^{\frac{\pi}{4}}\log\left(\frac{1-\tan(u)}{1+\tan(u)}+1\right)du$$

$$=\int_0^{\frac{\pi}{4}}\log\left(2\right)du-\int_0^{\frac{\pi}{4}}\log\left(\tan(u)+1\right)du$$

Equivalent : $J=\frac{\pi}{4}\log(2)-J\Rightarrow J=\frac{\pi}{8}\log(2)$

Take $(*)$ **They Have :** $I=\frac{\pi}{4}\log(2)-\frac{\pi}{8}\log(2)=\frac{\pi}{8}\log(2)$

$SO,$ $\boxed{\int_0^1\frac{\tan^{-1}(x)}{x+1}dx=\frac{\pi}{8}\log(2)}$

167. Calculate Integral $I=\int_0^{\log(2)}\sqrt{\frac{e^x-1}{e^x+1}}\,dx$

Solution; $I=\int_0^{\log(2)}\sqrt{\frac{e^x-1}{e^x+1}}\,dx$

Let : $e^x=\sec(2u)\Rightarrow dx=-2\tan(2u)du$ **; If :** $x\in\left(0,\log(2)\right)\Rightarrow u\in\left(0,\frac{\pi}{6}\right)$

Implies That $I=-2\int_0^{\frac{\pi}{6}}\sqrt{\frac{\sec(2u)-1}{\sec(2u)+1}}\tan(2u)du$

$$=-2\int_0^{\frac{\pi}{6}}\sqrt{\frac{1-\cos(2u)}{1+\cos(2u)}}\tan(2u)du=-2\int_0^{\frac{\pi}{6}}\sqrt{\tan^2(u)}\tan(2u)du$$

$$=-2\int_0^{\frac{\pi}{3}}\tan(u)\tan(2u)du=-4\int_0^{\frac{\pi}{3}}\frac{\sin^2(u)}{\cos^2(u)-\sin^2(u)}du$$

$$=-4\int_0^{\frac{\pi}{3}}\frac{\tan^2(u)}{\left(1-\tan^2(u)\right)\left(1+\tan^2(u)\right)}\left(1+\tan^2(u)\right)du$$

$$=4\int_0^{\frac{\pi}{3}}\frac{\tan^2(u)}{\left(\tan^2(u)-1\right)\left(1+\tan^2(u)\right)}d\left(\tan(u)\right)$$

$$=2\int_0^{\frac{\pi}{3}}\left(\frac{1}{\left(\tan^2(u)-1\right)}+\frac{1}{\left(1+\tan^2(u)\right)}\right)d\left(\tan(u)\right)$$

$$=2\int_0^{\frac{\pi}{3}}\left(\frac{1}{\left(\tan^2(u)-1\right)}\right)d\left(\tan(u)\right)+2\int_0^{\frac{\pi}{3}}\frac{1}{\left(1+\tan^2(u)\right)}d\left(\tan(u)\right)$$

$$=\log\left|\frac{\tan(u)-1}{\tan(u)+1}\right|\Bigg|_0^{\frac{\pi}{3}}+2\tan^{-1}\left(\tan(u)\right)\Big|_0^{\frac{\pi}{3}}$$

$$=\log\left(\frac{\sqrt{3}-1}{\sqrt{3}+1}\right)+\frac{2\pi}{3}$$

SO, $\boxed{\int_0^{\log(2)}\sqrt{\frac{e^x-1}{e^x+1}}\,dx=\frac{2\pi}{3}+\log\left(\frac{\sqrt{3}-1}{\sqrt{3}+1}\right)}$

168. Calculate Integral $I=\int_0^1 \frac{\eta(x)}{\zeta(x)}dx$

Solution; $I=\int_0^1 \frac{\eta(x)}{\zeta(x)}dx$

By : $\eta(x)=(1-2^{1-x})\zeta(x)\Rightarrow \frac{\eta(x)}{\zeta(x)}=(1-2^{1-x})$

We obtain $I=\int_0^1 (1-2^{1-x})dx$

$$=\int_0^1 (1-2e^{-x\log 2})dx$$

$$=1-2\int_0^1 e^{-x\log 2}dx$$

$$=1+\frac{2}{\log(2)}\int_0^1 (-x\log 2)'e^{-x\log 2}dx$$

$$=1+\frac{2}{\log(2)}(-x\log 2)\Big|_0^1$$

$$=1-\log_2(e)$$

$SO,$ $\boxed{\int_0^1 \frac{\eta(x)}{\zeta(x)}dx=1-\log_2(e)}$

169. Calculate Integral $I=\int_{-1}^{\infty}\frac{9x+4}{4x^5+3x^2+x}dx$

Solution; $I=\int_{-1}^{\infty}\frac{9x+4}{4x^5+3x^2+x}dx$

$$=\int_{-1}^{\infty}\frac{9x+4}{4x^5+3x^2+x}\times\frac{x^{-5}}{x^{-5}}dx$$

$$=\int_{-1}^{\infty}\frac{9x^{-4}+4x^{-5}}{4+3x^{-3}+x^{-4}}dx$$

Let : $u=4+3x^{-3}+x^{-4}\Rightarrow -du=(9x^{-4}+4x^{-5})dx$ **; If :** $x\in(-1,\infty)\Rightarrow u\in(2,1)$

Then $I=-\int_2^1 \frac{1}{u}du=\log(2)$

$SO,$ $\boxed{\int_{-1}^{\infty}\frac{9x+4}{4x^5+3x^2+x}dx=\log(2)}$

170. Calculate Integral $I=\int_0^{\infty}\frac{x^n}{e^x+1}dx$

Solution; $I=\int_0^{\infty}\frac{x^n}{e^x+1}dx$

$$=\int_0^{\infty}\frac{x^n e^{-x}}{1+e^{-x}}dx$$

$$=\int_0^{\infty}x^n e^{-x}\sum_{m=0}^{\infty}(-1)^m e^{-mx}dx$$

$$=\sum_{m=0}^{\infty}\left[(-1)^m\int_0^{\infty}x^n e^{-x}\times e^{-mx}dx\right]$$

$$=\sum_{m=0}^{\infty}\left[(-1)^m\int_0^{\infty}x^n e^{-(m+1)x}dx\right]$$

Let : $y=(m+1)x\Rightarrow dx=\frac{dy}{m+1}$ **; If :** $x\in(0,\infty)\Rightarrow y\in(0,\infty)$

Implies That $I=\sum_{m=0}^{\infty}\left[(-1)^m\int_0^{\infty}\left(\frac{y}{m+1}\right)^n e^{-y}\frac{dy}{m+1}\right]$

$$=\sum_{m=0}^{\infty}\frac{(-1)^m}{(m+1)^{n+1}}\int_0^{\infty}(y)^n e^{-y}dy$$

$$=\Gamma(n+1)\sum_{m=0}^{\infty}\frac{(-1)^m}{(m+1)^{n+1}}$$

$$=\Gamma(n+1)\left(1-\frac{1}{2^{n+1}}+\frac{1}{3^{n+1}}-\frac{1}{4^{n+1}}+\frac{1}{5^{n+1}}-\frac{1}{6^{n+1}}+\ldots\right)$$

$$=\Gamma(n+1)\left[\left(1+\frac{1}{2^{n+1}}+\frac{1}{3^{n+1}}+\frac{1}{4^{n+1}}+\ldots\right)-\frac{2}{2^{n+1}}\left(1+\frac{1}{2^{n+1}}+\frac{1}{3^{n+1}}+\frac{1}{4^{n+1}}+\ldots\right)\right]$$

$$=\Gamma(n+1)\left[\zeta(n+1)-\frac{1}{2^n}\zeta(n+1)\right]$$

$$=\frac{2^n-1}{2^n}\Gamma(n+1)\zeta(n+1)$$

SO, $\boxed{\int_0^{\infty}\frac{x^n}{e^x+1}dx=(1-2^{-n})\Gamma(n+1)\zeta(n+1)}$

171. Calculate Integral $I=\int_0^{\infty}\frac{x^n}{e^x-1}dx$

Solution; $I=\int_0^{\infty}\frac{x^n}{e^x-1}dx$

$$=\int_0^{\infty}\frac{x^n e^{-x}}{1-e^{-x}}dx$$

$$=\int_0^{\infty}x^n e^{-x}\sum_{m=0}^{\infty}e^{-mx}dx$$

$$=\sum_{m=0}^{\infty}\left(\int_0^{\infty}x^n e^{-x}\times e^{-mx}dx\right)$$

$$=\sum_{m=0}^{\infty}\left(\int_0^{\infty}x^n e^{-(m+1)x}dx\right)$$

Let : $y=(m+1)x\Rightarrow dx=\frac{dy}{m+1}$ **; If :** $x\in(0,\infty)\Rightarrow y\in(0,\infty)$

Implies That $I=\sum_{m=0}^{\infty}\left[\int_0^{\infty}\left(\frac{y}{m+1}\right)^n e^{-y}\frac{dy}{m+1}\right]$

$$=\sum_{m=0}^{\infty}\frac{1}{(m+1)^{n+1}}\int_0^{\infty}(y)^n e^{-y}dy$$

$$=\Gamma(n+1)\sum_{m=0}^{\infty}\frac{1}{(m+1)^{n+1}}$$

$$=\Gamma(n+1)\sum_{m=1}^{\infty}\frac{1}{(m)^{n+1}}$$

$$=\Gamma(n+1)\zeta(n+1)$$

SO, $\boxed{\int_0^{\infty}\frac{x^n}{e^x-1}dx=\Gamma(n+1)\zeta(n+1)}$

172. Calculate Integral $I=\int_0^{\infty}\frac{x^n}{x+\frac{x^2}{2!}+\frac{x^3}{3!}+\frac{x^4}{4!}+\cdots}dx$

Solution; $I=\int_0^{\infty}\frac{x^n}{x+\frac{x^2}{2!}+\frac{x^3}{3!}+\frac{x^4}{4!}+\cdots}dx$

By : $e^x=1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\frac{x^4}{4!}+\cdots \Leftrightarrow e^x-1=x+\frac{x^2}{2!}+\frac{x^3}{3!}+\frac{x^4}{4!}+\cdots$

Implies That $I=\int_0^{\infty}\frac{x^n}{e^x-1}dx$

$$=\int_0^{\infty}\frac{x^{(n+1)-1}}{e^x-1}dx$$

$$=\Gamma(n+1)\zeta(n+1)$$

SO, $\boxed{\int_0^{\infty}\frac{x^n}{x+\frac{x^2}{2!}+\frac{x^3}{3!}+\frac{x^4}{4!}+\cdots}dx=\Gamma(n+1)\zeta(n+1)}$

173. Calculate Integral $I=\int_0^{\infty}\frac{x^n}{1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\frac{x^4}{4!}+\cdots}dx$

Solution; $I=\int_0^{\infty}\frac{x^n}{1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\frac{x^4}{4!}+\cdots}dx$

By : $e^x=1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\frac{x^4}{4!}+\cdots$

Implies That $I=\int_0^{\infty}\frac{x^n}{e^x}dx$

$$=\int_0^{\infty}x^ne^{-x}dx=\Gamma(n+1)=n!$$

SO, $\boxed{\int_0^{\infty}\frac{x^n}{1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\ldots+\frac{x^n}{n!}}dx=n!}$

174. Calculate Integral $I=\int_0^1 \frac{\log(x)}{1+x}dx$

Solution; $I=\int_0^1 \frac{\log(x)}{1+x}dx$

$$=\int_0^1\left(\log(x)\sum_{n=0}^{\infty}(-1)^n x^n\right)dx$$

$$=\sum_{n=0}^{\infty}\left[(-1)^n\underbrace{\int_0^1 x^n\log(x)dx}_{(Use\ partial\ integral)}\right]$$

We obtain $I=\sum_{n=0}^{\infty}\left[\frac{(-1)^n}{n+1}\left(\underbrace{x^{n+1}\log(x)\Big|_0^1}_{0}-\int_0^1 x^{n+1}\times\frac{1}{x}dx\right)\right]$

$$=-\sum_{n=0}^{\infty}\frac{(-1)^n}{n+1}\left(\int_0^1 x^n dx\right)$$

$$=-\sum_{n=0}^{\infty}\frac{(-1)^n}{(n+1)^2}$$

$$=-\eta(2)=-\frac{\pi^2}{12}$$

SO, $\boxed{\int_0^1 \frac{\log(x)}{1+x}dx=-\frac{\pi^2}{12}}$

175. Calculate Integral $I=\int_0^1 \frac{\log(x)}{1-x}dx$

Solution; $I=\int_0^1 \frac{\log(x)}{1-x}dx$

$$=\int_0^1\left(\log(x)\sum_{n=0}^{\infty}x^n\right)dx$$

$$=\sum_{n=0}^{\infty}\left(\underbrace{\int_0^1 x^n\log(x)dx}_{(Use\ partial\ integral)}\right)$$

$$=\sum_{n=0}^{\infty}\frac{1}{n+1}\left(\underbrace{x^{n+1}\log(x)\Big|_0^1}_{0}-\int_0^1 x^{n+1}\times\frac{1}{x}dx\right)$$

$$=-\sum_{n=0}^{\infty}\frac{1}{n+1}\left(\int_0^1 x^n dx\right)$$

$$=-\sum_{n=0}^{\infty}\frac{1}{(n+1)^2}$$

$$=-\zeta(2)=-\frac{\pi^2}{6}$$

SO, $\boxed{\int_0^1\frac{\log(x)}{1+x}dx=-\frac{\pi^2}{12}}$

176. Calculate Integral $I=\int_0^{\infty}\log\left(\frac{e^x+1}{e^x-1}\right)dx$

Solution; $I=\int_0^{\infty}\log\left(\frac{e^x+1}{e^x-1}\right)dx$

$$=\int_0^{\infty}\log\left(\frac{1+e^{-x}}{1-e^{-x}}\right)dx$$

$$=\int_0^{\infty}\left[\log\left(1+e^{-x}\right)-\log\left(1-e^{-x}\right)\right]dx$$

$$=\int_0^{\infty}\sum_{n=1}^{\infty}\left(\frac{(-1)^{n+1}e^{-nx}}{n}\right)dx-\int_0^{\infty}\left(-\sum_{n=1}^{\infty}\frac{e^{-nx}}{n}\right)dx$$

$$=\sum_{n=1}^{\infty}\left(\frac{(-1)^{n+1}}{n}\int_0^{\infty}e^{-nx}dx\right)+\left(\sum_{n=1}^{\infty}\frac{1}{n}\int_0^{\infty}e^{-nx}dx\right)$$

$$=\sum_{n=1}^{\infty}\left(\frac{(-1)^{n+1}}{n^2}\right)+\left(\sum_{n=1}^{\infty}\frac{1}{n^2}\right)$$

$$=\eta(2)+\zeta(2)=\frac{\pi^2}{4}$$

SO, $\boxed{\int_0^{\infty}\log\left(\frac{e^x+1}{e^x-1}\right)dx=\frac{\pi^2}{4}}$

177. Calculate Integral $I=\int_{\frac{\pi}{2}}^{\pi}\log^2\left(1+(e-1)\sin(x)\right)\sin(2x)dx$

Solution; $I=\int_{\frac{\pi}{2}}^{\pi}\log^2\left(1+(e-1)\sin(x)\right)\sin(2x)dx$

$$=2\int_{\frac{\pi}{2}}^{\pi}\log^2\left(1+(e-1)\sin(x)\right)\sin(x)\cos(x)dx$$

Let $:y=1+(e-1)\sin(x)\Rightarrow\cos(x)dx=\frac{1}{e-1}dy$ **; If** $:x\in(\frac{\pi}{2},\pi)\Rightarrow y\in(e,1)$

We obtain $I=2\int_e^1\left(\frac{y-1}{e-1}\right)\log^2(y)\frac{dy}{e-1}$

$$=-\frac{2}{(e-1)^2}\int_1^e(y-1)\log^2(y)dy$$

$$=-\frac{2}{(e-1)^2}\times\frac{e^2-4e+7}{4}=-\frac{e^2-4e+7}{2(e-1)^2}$$

SO, $\boxed{\int_{\frac{\pi}{2}}^{\pi}\log^2\left(1+(e-1)\sin(x)\right)\sin(2x)dx=-\frac{e^2-4e+7}{2(e-1)^2}}$

178. Calculate Integral $I=\int_0^1\frac{(1+x^2)x^3}{(1+x)^{10}}dx$

Solution; $I=\int_0^1\frac{(1+x^2)x^3}{(1+x)^{10}}dx$

$$=\int_0^1\frac{x^3+x^5}{(1+x)^{10}}dx=\int_0^1\frac{x^{4-1}+x^{6-1}}{(1+x)^{4+6}}dx$$

$$=B(4,6)=\frac{\Gamma(4)\Gamma(6)}{\Gamma(10)}$$

$$=\frac{3!5!}{9!}=\frac{1}{504}$$

SO, $\boxed{SO,\ \int_0^1\frac{x^3(1+x^2)}{(1+x)^{10}}dx=\frac{1}{504}}$

179. Calculate Integral $I=\int_0^{\pi}\frac{x\sin^2(x)}{1+\cos^2(x)}dx$

Solution; $I=\int_0^{\pi}\frac{x\sin^2(x)}{1+\cos^2(x)}dx$

$$=\int_0^{\pi}\frac{(\pi-x)\sin^2(\pi-x)}{1+\cos^2(\pi-x)}dx$$

Use : $\int_0^a f(x)dx=\int_0^a f(a-x)dx$

$$=\pi\int_0^{\pi}\frac{\sin^2(x)}{1+\cos^2(x)}dx-K$$

Implies That $I=\frac{\pi}{2}\int_0^{\pi}\frac{\sin^2(x)}{1+\cos^2(x)}dx$

Equivalent $I=\frac{\pi}{2}\times 2\int_0^{\frac{\pi}{2}}\frac{\sin^2(x)}{1+\cos^2(x)}dx$

Use : $\begin{cases}\int_0^{2a} f(x)dx=2\int_0^a f(x)dx\\ f(2a-x)=f(x)\end{cases}$

$$=\pi\int_0^{\frac{\pi}{2}}\frac{\tan^2(x)}{\left(2+\tan^2(x)\right)\left(1+\tan^2(x)\right)}\left(1+\tan^2(x)\right)dx$$

Let : $u=\tan(x)\Rightarrow du=\left(1+\tan^2(x)\right)dx$ **; If :** $x\in(0,\frac{\pi}{2})\Rightarrow u\in(0,\infty)$

We get $I=\pi\int_0^{\infty}\frac{u^2}{\left(2+u^2\right)\left(1+u^2\right)}du$

$$=\pi\int_0^{\infty}\left[\frac{2}{\left(2+u^2\right)}-\frac{1}{\left(1+u^2\right)}\right]du$$

$$=\pi\left(\frac{2}{\sqrt{2}}\tan^{-1}\left(\frac{u}{\sqrt{2}}\right)-\tan^{-1}(u)\right)\Big|_0^{\infty}$$

$$=\pi\left(\frac{2}{\sqrt{2}}\times\frac{\pi}{2}-\frac{\pi}{2}\right)=\left(\frac{\sqrt{2}-1}{2}\right)\pi^2$$

SO, $\boxed{\int_0^{\pi}\frac{x\sin^2(x)}{1+\cos^2(x)}dx=\left(\frac{\sqrt{2}-1}{2}\right)\pi^2}$

180. Calculate Integral $I=\int_0^{\infty}\frac{x}{x^8+2x^4+1}dx$

Solution; $I=\int_0^{\infty}\frac{x}{x^8+2x^4+1}dx$

$$=\frac{1}{2}\int_0^{\infty}\frac{1}{(x^4+1)^2}d(x^2)$$

$$=\frac{1}{2}\int_0^{\infty}\frac{1}{\left((x^2)^2+1\right)^2}d(x^2)$$

Let : $x^2=t^{\frac{1}{2}}\Rightarrow d(x^2)=\frac{1}{2}t^{\frac{1}{2}-1}dt$ **; If :** $x\in(0,\infty)\Rightarrow t\in(0,\infty)$

We obtain $I=\frac{1}{2}\times\frac{1}{2}\int_0^{\infty}\frac{t^{\frac{1}{2}-1}}{(t+1)^2}d(t)$

$$=\frac{1}{4}\int_0^{\infty}\frac{t^{\frac{1}{2}-1}}{(t+1)^{\frac{1}{2}+\frac{3}{2}}}d(t)$$

$$=\frac{1}{4}B\left(\frac{1}{2},\frac{3}{2}\right)$$

$$=\frac{1}{4}.\frac{\Gamma\left(\frac{1}{2}\right)\Gamma\left(\frac{3}{2}\right)}{\Gamma\left(\frac{1}{2}+\frac{3}{2}\right)}$$

$$=\frac{1}{4}.\frac{\Gamma\left(\frac{1}{2}\right)\frac{1}{2}\Gamma\left(\frac{1}{2}\right)}{1!}$$

$$=\frac{1}{8}\Gamma^2\left(\frac{1}{2}\right)$$

$$=\frac{1}{8}\left(\sqrt{\pi}\right)^2=\frac{\pi}{8}$$

$SO,\ \boxed{\int_0^{\infty}\frac{x}{x^8+2x^4+1}dx=\frac{\pi}{8}}$

181. Calculate Integral $I=\int_0^1 \frac{\sin\left(\sqrt[x]{\log(x)}\right)}{\log(x)}dx$

Solution; $I=\int_0^1 \frac{\sin\left(\sqrt[x]{\log(x)}\right)}{\log(x)}dx$

$$=\int_0^1 \frac{\sin\left(\log(x)\right)}{x\log(x)}dx$$

Let : $t=-\log(x)\Leftrightarrow x=e^{-t}\Rightarrow dx=-e^{-t}dt$ **; If :** $x\in(0,1)\Rightarrow t\in(\infty,0)$

We obtain $I=-\int_\infty^0 \frac{\sin(t)}{(t)e^{-t}}\times e^{-t}dt$

$$=\int_0^\infty \frac{\sin(t)}{(t)}dt=\frac{\pi}{2}$$

$SO,$ $\boxed{\int_0^1 \frac{\sin\left(\sqrt[x]{\log(x)}\right)}{\log(x)}dx=\frac{\pi}{2}}$

182. Calculate Integral $I=\int_0^2 \sqrt{x+\sqrt{x+\sqrt{x+\ldots}}}dx$

Solution; $I=\int_0^2 \sqrt{x+\sqrt{x+\sqrt{x+\ldots}}}dx$

Let : $y_{(y>0\forall x\in(0,2))}=\sqrt{x+\sqrt{x+\sqrt{x+\cdots}}}$ **Equivalent** $y=\sqrt{x+y}$

Implies That $y^2-y-x=0\Rightarrow y=\frac{1\pm\sqrt{1+4x}}{2}=\begin{cases}y=\frac{1+\sqrt{1+4x}}{2}\\ y=\frac{1-\sqrt{1+4x}}{2}\end{cases}$ (Do not take)

We Get $I=\int_0^2 \frac{1+\sqrt{1+4x}}{2}$

$$=\int_0^2 \frac{1}{2}dx+\frac{1}{2}\int_0^2 \sqrt{1+4x}dx=\frac{19}{6}$$

$SO,$ $\boxed{\int_0^2 \sqrt{x+\sqrt{x+\sqrt{x+\ldots}}}dx=\frac{19}{6}}$

183. Calculate Integral $I=\int_0^{\pi}\frac{x\sin(x)}{\cos(2x)+1}dx$

Solution; $I=\int_0^{\pi}\frac{x\sin(x)}{\cos(2x)+1}dx$

$$=\frac{1}{2}\int_0^{\pi}\frac{x\sin(x)}{\cos^2(x)}dx$$

Let : $u=x\Rightarrow du=dx$; **If** : $v=\int\frac{\sin(x)}{\cos^2(x)}dx=\frac{1}{\cos(x)}=\sec(x)$

We Get $I=\frac{1}{2}\times\frac{x}{\cos(x)}\Big|_0^{\pi}-\frac{1}{2}\int_0^{\pi}\sec(x)\,dx$

$$=-\frac{\pi}{2}-\underbrace{\log\left|\sec(x)+\tan(x)\right\|_0^{\pi}}_{0}=-\frac{\pi}{2}$$

SO, $\boxed{\int_0^{\pi}\frac{x\sin(x)}{\cos(2x)+1}dx=-\frac{\pi}{2}}$

184. Calculate Integral $I=\int_0^{\pi}x\sin^6(x)dx$

Solution; $I=\int_0^{\pi}x\sin^6(x)dx$

$$=\int_0^{\pi}(\pi-x)\sin^6(\pi-x)dx$$

$$=\int_0^{\pi}\pi\sin^6(x)dx-\int_0^{\pi}x\sin^6(x)dx$$

Then $2I=\pi\int_0^{\pi}\sin^6(x)dx$

Use: $\begin{cases}\int_0^{2a}f(x)dx=2\int_0^{a}f(x)dx\\ f(2a-x)=f(x)\end{cases}$

Implies That $I=\frac{\pi}{2}\times 2\int_0^{\frac{\pi}{2}}\sin^6(x)dx$

$$=\pi\left(\frac{\pi}{2}\times\frac{5\times3\times1}{6\times4\times2}\right)=\frac{5\pi^2}{32}$$

SO, $\boxed{\int_0^{\pi}x\sin^6(x)dx=\frac{5\pi^2}{32}}$

185. Calculate Integral $I=\int_0^{\infty}\left(\frac{\log(x)}{1+x}\right)^2 dx$

Solution; $I=\int_0^{\infty}\left(\frac{\log(x)}{1+x}\right)^2 dx$

$$=\int_0^1\left(\frac{\log(x)}{1+x}\right)^2 dx+\underbrace{\int_1^{\infty}\left(\frac{\log(x)}{1+x}\right)^2 dx}_{I'}$$

For : $I'=\int_1^{\infty}\left(\frac{\log(x)}{1+x}\right)^2 dx$

Let : $t=\frac{1}{x}\Rightarrow dx=-\frac{1}{t^2}dt$; **If** : $x\in(1,\infty)\Rightarrow t\in(1,0)$

Implies That : $I'=-\int_1^0\left(\frac{\log(t^{-1})}{1+t^{-1}}\right)^2\times\frac{1}{t^2}dx$

$$=\int_0^1\left(\frac{\log(t)}{1+t}\right)^2 dx=\int_0^1\left(\frac{\log(x)}{1+x}\right)^2 dx$$

We Get $I=\int_0^1\left(\frac{\log(x)}{1+x}\right)^2 dx+\int_0^1\left(\frac{\log(x)}{1+x}\right)^2 dx$

$$=2\int_0^1\left(\frac{\log(x)}{1+x}\right)^2 dx$$

Let : $t=-\log(x)\Leftrightarrow x=e^{-t}\Rightarrow dx=-e^{-t}dt$; **If** : $x\in(0,1)\Rightarrow t\in(\infty,0)$

We obtain $I=-2\int_{\infty}^0\left(\frac{-t}{1+e^{-t}}\right)^2 e^{-t}dt$

$$=2\int_0^{\infty}t^2\frac{e^{-t}}{(1+e^{-t})^2}dt$$

$$=2\int_0^{\infty}t^2\sum_{n=1}^{\infty}(-1)^{n-1}ne^{-nt}dt$$

$$=2\sum_{n=1}^{\infty}\left((-1)^{n-1}n\int_0^{\infty}t^2e^{-nt}dt\right)$$

Let : $y=nt\equiv t=\frac{y}{n}\Rightarrow dt=\frac{dy}{n}$; **If** : $t\in(0,\infty)\Rightarrow y\in(0,\infty)$

Implies That $I=2\sum_{n=1}^{\infty}\left((-1)^{n-1}n\int_0^{\infty}\left(\frac{y}{n}\right)^2e^{-y}\frac{dy}{n}\right)$

$$=2\sum_{n=1}^{\infty}\left(\frac{(-1)^{n-1}}{n^2}\int_0^{\infty}(y)^2e^{-y}dy\right)$$

$$=2\sum_{n=1}^{\infty}\left(\frac{(-1)^{n-1}}{n^2}\times 2!\right)$$

$$=4\eta(2)=\frac{\pi^2}{3}$$

SO, $\boxed{\int_0^{\infty}\left(\frac{\log(x)}{1+x}\right)^2dx=\frac{\pi^2}{3}}$

186. Calculate Integral $I=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left(\sin(x)\sin(2x)\right)^2dx$

Solution; $I=\underbrace{\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left(\sin(x)\sin(2x)\right)^2dx}_{is\,an\,even\,function}$

$$=2\int_0^{\frac{\pi}{2}}4\cos^2(x)\sin^4(x)dx$$

$$=8\int_0^{\frac{\pi}{2}}\left(1-\sin^2(x)\right)\sin^4(x)dx$$

$$=8\left[\int_0^{\frac{\pi}{2}}\sin^4(x)dx-\int_0^{\frac{\pi}{2}}\sin^6(x)dx\right]$$

$$=8\times\frac{\pi}{2}\left(\frac{3\times1}{4\times2}-\frac{5\times3\times1}{6\times4\times2}\right)=\frac{\pi}{4}$$

SO, $\boxed{\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left(\sin(x)\sin(2x)\right)^2dx=\frac{\pi}{4}}$

187. Calculate Integral $I=\int_0^{2\pi}\frac{1}{2+\cos(x)}dx$

Solution; $I=\int_0^{2\pi}\frac{1}{2+\cos(x)}dx$

$$=\int_0^{2\pi}\frac{1}{1+2\cos^2\left(\frac{x}{2}\right)}dx$$

$$=2\int_0^{2\pi}\frac{1}{1+2\cos^2\left(\frac{x}{2}\right)}d\left(\frac{x}{2}\right)$$

$$=2\int_0^{\pi}\frac{1}{1+2\cos^2(x)}dx$$

$$=2\int_0^{\frac{\pi}{2}}\frac{1}{\sin^2(x)+\cos^2(x)+2\cos^2(x)}d(x)$$

$$=4\int_0^{\frac{\pi}{2}}\frac{1}{\sin^2(x)+3\cos^2(x)}d(x)$$

$$=4\int_0^{\frac{\pi}{2}}\frac{1}{\tan^2(x)+3}\times\frac{1}{\cos^2(x)}d(x)$$

$$=4\int_0^{\frac{\pi}{2}}\frac{1}{\tan^2(x)+3}d(\tan(x))$$

$$=\frac{4}{\sqrt{3}}\int_0^{\frac{\pi}{2}}\frac{1}{1+\left(\frac{\tan(x)}{\sqrt{3}}\right)^2}d\left(\frac{\tan(x)}{\sqrt{3}}\right)$$

$$=\frac{4}{\sqrt{3}}\tan^{-1}\left(\frac{\tan(x)}{\sqrt{3}}\right)\Bigg|_0^{\frac{\pi}{2}}$$

$$=\frac{4}{\sqrt{3}}\times\frac{\pi}{2}=\frac{2\pi}{\sqrt{3}}$$

SO, $\boxed{\int_0^{2\pi}\frac{1}{2+\cos(x)}dx=\frac{2\pi}{\sqrt{3}}}$

188. Calculate Integral $I=\int_0^3\sqrt{\frac{x}{3-x}}\,dx$

Method 1

Solution; $I=\int_0^3\sqrt{\frac{x}{3-x}}\,dx$

$$=\int_0^3 x^{\frac{1}{2}}(3-x)^{-\frac{1}{2}}dx$$

Let : $x=3u\Rightarrow dx=3du$ **; If** : $x\in(0,3)\Rightarrow u\in(0,1)$

Implies That : $I=\int_0^1(3u)^{\frac{1}{2}}(3-3u)^{-\frac{1}{2}}3du$

$$=3\int_0^1(u)^{\left(1+\frac{1}{2}\right)-1}(1-u)^{\frac{1}{2}-1}du$$

$$=3\times\frac{\Gamma\left(1+\frac{1}{2}\right)\Gamma\left(\frac{1}{2}\right)}{\Gamma\left(1+\frac{1}{2}+\frac{1}{2}\right)}=\frac{3\pi}{2}$$

Method 2

Let : $x=3\sin^2(u)\Rightarrow dx=6\sin(u)\cos(u)du$ **; If** : $x\in(0,3)\Rightarrow u\in\left(0,\frac{\pi}{2}\right)$

Implies That : $I=6\int_0^{\frac{\pi}{2}}\sqrt{\frac{3\sin^2(u)}{3-3\sin^2(u)}}\times\sin(u)\cos(u)du$

$$=6\int_0^{\frac{\pi}{2}}\left|\tan(u)\right|\times\sin(u)\cos(u)du$$

$$=6\int_0^{\frac{\pi}{2}}\frac{\sin(u)}{\cos(u)}\times\sin(u)\cos(u)du$$

$$=6\int_0^{\frac{\pi}{2}}\sin^2(u)\,du=\frac{3\pi}{2}$$

SO, $\boxed{\int_0^3\sqrt{\frac{x}{3-x}}\,dx=\frac{3\pi}{2}}$

189. Calculate Integral $I=\int_0^{\infty}\frac{1}{(x^9+1)(x^2+1)}\,dx$

Solution; $I=\int_0^{\infty}\frac{1}{(x^9+1)(x^2+1)}\,dx \quad (*)$

Let : $x=\frac{1}{u}=u^{-1}\Rightarrow dx=-\frac{1}{u^2}du$ **; If:** $x\in(0,\infty)\Rightarrow u\in(\infty,0)$

Implies That $I=-\int_{\infty}^{0}\frac{1}{(u^{-9}+1)(u^{-2}+1)}\times\frac{1}{u^2}du$

$$=\int_0^{\infty}\frac{u^9}{(u^9+1)(u^2+1)}du \quad (**)$$

Take $(*)+(**)$ **They Have :** $2I=\int_0^{\infty}\frac{u^9+1}{(u^9+1)(u^2+1)}du=\tan^{-1}(u)\Big|_0^{\infty}=\frac{\pi}{2}$

Implies That $I=\frac{\pi}{4}$

SO, $\boxed{\int_0^{\infty}\frac{1}{(x^9+1)(x^2+1)}dx=\frac{\pi}{4}}$

190. Calculate Integral $I=\int_0^{\infty}2^{-3x^2}dx$

Solution; $I=\int_0^{\infty}2^{-3x^2}dx$

$$=\int_0^{\infty}e^{-3x^2\log(2)}dx$$

Let : $u=3x^2\log(2)\Rightarrow dx=\frac{1}{2\sqrt{3\log(2)}}u^{\frac{1}{2}-1}du$ **; If :** $x\in(0,\infty)\Rightarrow u\in(0,\infty)$

We obtain $I=\frac{1}{2\sqrt{3\log(2)}}\int_0^{\infty}u^{\frac{1}{2}-1}e^{-u}du=\frac{\Gamma\left(\frac{1}{2}\right)}{2\sqrt{3\log(2)}}=\frac{\sqrt{\pi}}{2\sqrt{3\log(2)}}$

SO, $\boxed{\int_0^{\infty}2^{-3x^2}dx=\frac{\sqrt{\pi}}{2\sqrt{3\log(2)}}}$

191. Calculate Integral $I=\int_0^{\infty}\frac{1}{e^{\log^2(x)-89\log(x)+2025}}dx$

Solution; $I=\int_0^{\infty}\frac{1}{e^{\log^2(x)-89\log(x)+2025}}dx$

$$=\int_0^{\infty}\frac{1}{e^{\log(x)}e^{\log^2(x)-90\log(x)+2025}}dx$$

$$=\int_0^{\infty}\frac{e^{-\left(\log^2(x)-90\log(x)+2025\right)}}{x}dx$$

$$=\int_0^{\infty}\frac{e^{-\left(\log(x)-45\right)^2}}{x}dx$$

Let : $u=\log(x)-45\Rightarrow du=\frac{1}{x}dx$ **; If:** $x\in(0,\infty)\Rightarrow u\in(-\infty,\infty)$

We obtain $I=\int_{-\infty}^{\infty}e^{-u^2}\,du=\sqrt{\pi}$

$SO,$ $\boxed{\int_0^{\infty}\frac{1}{e^{\log^2(x)-89\log(x)+2025}}dx=\sqrt{\pi}}$

192. Calculate Integral $I=\int_0^{\pi}\cos(x)\cos(2x)\cos(3x)...\cos(2024x)\cos(2025x)dx$

Solution; $I=\int_0^{\pi}\cos(x)\cos(2x)\cos(3x)...\cos(2024x)\cos(2025x)dx$

$$=\int_0^{\pi}\cos(\pi-x)\cos(2\pi-2x)\cos(3\pi-3x)...\cos(2025\pi-2025x)dx$$

$$=\int_0^{\pi}\left(-\cos(x)\right)\cos(2x)\left(-\cos(3x)\right)...\left(-\cos(2025x)\right)dx$$

$$=(-1)^{\frac{2025\times2026}{2}}\int_0^{\pi}\cos(x)\cos(2x)\cos(3x)...\cos(2024x)\cos(2025x)dx$$

$$=(-1)^{2025\times1013}\int_0^{\pi}\cos(x)\cos(2x)\cos(3x)...\cos(2024x)\cos(2025x)dx$$

$$=-I$$

Equivalent $I=-I$

We Get $I=0$

$SO,$ $\boxed{\int_0^{\pi}\cos(x)\cos(2x)\cos(3x)...\cos(2024x)\cos(2025x)dx=0}$

193. Calculate Integral $I=\int_0^1\left(\frac{1}{x}-\frac{1}{1+\sqrt{x-1}}\right)dx$

Solution; $I=\int_0^1\left(\frac{1}{x}-\frac{1}{1+\sqrt{x-1}}\right)dx$

$$=\int_0^1\left(\frac{1}{x}-\frac{1-\sqrt{x-1}}{1^2-(\sqrt{x-1})^2}\right)dx$$

$$=\int_0^1\left(\frac{1}{x}-\frac{1-\sqrt{x-1}}{1-|x-1|}\right)dx$$

$$=\int_0^1\left(\frac{1}{x}-\frac{1-\sqrt{x-1}}{1+x-1}\right)dx$$

$$=\int_0^1\left(\frac{\sqrt{x-1}}{x}\right)dx$$

Let : $u^2=x-1\equiv x=u^2+1\Rightarrow dx=2udu$ **; If :** $x\in(0,1)\Rightarrow u\in(-1,0)$

We obtain $I=2\int_{-1}^0\left(\frac{\sqrt{u^2}}{u^2+1}\right)udu$

$$=-2\int_{-1}^0\left(\frac{u^2}{u^2+1}\right)du$$

Let : $u=-y\Rightarrow -du=dy$ **; If :** $u\in(-1,0)\Rightarrow u\in(1,0)$

Implies That $I=2\int_1^0\left(\frac{(-y)^2}{(-y)^2+1}\right)dy$

$$=-2\int_0^1\left(1-\frac{1}{y^2+1}\right)dy$$

$$=-2\left(y-\tan^{-1}(y)\right)\Big|_0^1$$

$$=-2\left((1-\frac{\pi}{4})-(0-0)\right)=\frac{\pi}{2}-2$$

SO, $\boxed{\int_0^1\left(\frac{1}{x}-\frac{1}{1+\sqrt{x-1}}\right)dx=\frac{\pi}{2}-2}$

194. Calculate Integral $I=\int_0^1\left(\frac{1}{x+\sqrt{1-x^2}}\right)dx$

Solution; $I=\int_0^1\left(\frac{1}{x+\sqrt{1-x^2}}\right)dx \quad (1)$

Let : $t=\sqrt{1-x^2} \Leftrightarrow x=\sqrt{1-t^2} \Rightarrow dx=-\frac{t}{\sqrt{1-t^2}}dt$; **If** : $x\in(0,1)\Rightarrow t\in(1,0)$

We Get $I=-\int_1^0\left(\frac{1}{t+\sqrt{1-t^2}}\right)\frac{t}{\sqrt{1-t^2}}dt$

$$=\int_0^1\left(\frac{1}{t+\sqrt{1-t^2}}\right)\frac{t}{\sqrt{1-t^2}}dt$$

$$=\int_0^1\left(\frac{1}{x+\sqrt{1-x^2}}\right)\frac{x}{\sqrt{1-x^2}}dx \quad (2)$$

Take $(1)+(2)$ **Then** : $2I=\int_0^1\left(\frac{1}{x+\sqrt{1-x^2}}\right)dx+\int_0^1\left(\frac{1}{x+\sqrt{1-x^2}}\right)\frac{x}{\sqrt{1-x^2}}dx$

$$=\int_0^1\left(\frac{1}{x+\sqrt{1-x^2}}+\left(\frac{1}{x+\sqrt{1-x^2}}\right)\frac{x}{\sqrt{1-x^2}}\right)dx$$

$$=\int_0^1\left(\frac{x+\sqrt{1-x^2}}{x+\sqrt{1-x^2}}\right)\frac{1}{\sqrt{1-x^2}}dx$$

$$=\int_0^1\frac{1}{\sqrt{1-x^2}}dx$$

$$=\sin^{-1}(x)\Big|_0^1=\frac{\pi}{2}$$

We Get $I=\frac{\pi}{2}$

SO, $\boxed{\int_0^1\left(\frac{1}{x+\sqrt{1-x^2}}\right)dx=\frac{\pi}{4}}$

195. Calculate Integral $I=\int_0^{\sqrt[6]{2}}\frac{x^5}{\left(x^2+1\right)\left(x^2+x+1\right)\left(x^4-x^2+1\right)\left(x^4-x^3+x-1\right)}dx$

Solution; $I=\int_0^{\sqrt[6]{2}}\frac{x^5}{\left(x^2+1\right)\left(x^2+x+1\right)\left(x^4-x^2+1\right)\left(x^4-x^3+x-1\right)}dx$

$$=\int_0^{\sqrt[6]{2}}\frac{x^5}{\underbrace{\left(x^2+1\right)\left(x^4-x^2+1\right)}_{x^6+1}\underbrace{\left(x^2+x+1\right)\left(x^4-x^3+x-1\right)}_{x^6-1}}dx$$

$$=\int_0^{\sqrt[6]{2}}\frac{x^5}{\left(x^6+1\right)\left(x^6-1\right)}dx$$

$$=\frac{1}{6}\int_0^{\sqrt[6]{2}}\frac{1}{\left(x^6+1\right)\left(x^6-1\right)}d(x^6)$$

$$=\frac{1}{12}\int_0^{\sqrt[6]{2}}\left(\frac{1}{\left(x^6-1\right)}-\frac{1}{\left(x^6+1\right)}\right)d(x^6)$$

$$=\frac{1}{12}\log\left|\frac{x^6-1}{x^6+1}\right|\Bigg|_0^{\sqrt[6]{2}}=-\frac{1}{12}\log(3)$$

$SO,$ $\boxed{\int_0^{\sqrt[6]{2}}\frac{x^5}{\left(x^2+1\right)\left(x^2+x+1\right)\left(x^4-x^2+1\right)\left(x^4-x^3+x-1\right)}dx=-\frac{1}{12}\log(3)}$

196. Calculate Integral $I=\int_{-\infty}^{+\infty}\frac{x^2e^x}{(1+e^x)^2}dx$

Solution; $I=\underbrace{\int_{-\infty}^{+\infty}\frac{x^2e^x}{(1+e^x)^2}dx}_{is\,even\,function}$

$$=2\int_0^{+\infty}\frac{x^2e^x}{(1+e^x)^2}dx$$

$$=\underbrace{2\int_0^{+\infty}\frac{x^2}{(1+e^x)^2}d(1+e^x)}_{Use\ partial\ integral}$$

Implies That $I=2\left(\underbrace{-\frac{x^2}{1+e^x}\Big|_0^{+\infty}}_{0}-2\int_0^{+\infty}\frac{x}{1+e^x}dx\right)$

$$=4\int_0^{+\infty}\frac{x}{1+e^x}dx$$

$$=4\int_0^{+\infty}\frac{xe^{-x}}{1+e^{-x}}dx$$

$$=4\int_0^{+\infty}\left(xe^{-x}\sum_{n=0}^{\infty}(-1)^n e^{-nx}\right)dx$$

$$=4\sum_{n=0}^{\infty}\left((-1)^n\int_0^{+\infty}\left(xe^{-x}e^{-nx}\right)dx\right)$$

$$=4\sum_{n=0}^{\infty}\left((-1)^n\int_0^{+\infty}xe^{-(n+1)x}dx\right)$$

Let : $x=\frac{u}{n+1}\Rightarrow dx=\frac{1}{n+1}du$; **If** : $x\in(0,+\infty)\Rightarrow u\in(0,+\infty)$

We obtain $I=4\sum_{n=0}^{\infty}\left((-1)^n\int_0^{+\infty}\frac{u}{n+1}e^{-u}\frac{du}{n+1}\right)$

$$=4\sum_{n=0}^{\infty}\left(\frac{(-1)^n}{(n+1)^2}\int_0^{+\infty}ue^{-u}du\right)$$

$$=4\sum_{n=0}^{\infty}\left(\frac{(-1)^n}{(n+1)^2}\Gamma(2)\right)$$

$$=4\sum_{n=0}^{\infty}\left(\frac{(-1)^n}{(n+1)^2}\right)=4\eta(2)$$

$$=4\times\frac{\pi^2}{12}=\frac{\pi^2}{3}$$

SO, $\boxed{\int_{-\infty}^{+\infty}\frac{x^2e^x}{(1+e^x)^2}dx=\frac{\pi^2}{3}}$

197. Calculate Integral $I=\int_0^1 \frac{\sqrt{x^2-2x+1}}{\sqrt[3]{x^3-3x^2+3x-1}}dx$

Solution; $I=\int_0^1 \frac{\sqrt{x^2-2x+1}}{\sqrt[3]{x^3-3x^2+3x-1}}dx$

$$=\int_0^1 \frac{\sqrt{(x-1)^2}}{\sqrt[3]{(x-1)^3}}dx$$

$$=\int_0^1 \frac{|x-1|}{(x-1)}dx$$

$$=-\int_0^1 \frac{(x-1)}{(x-1)}dx=-1$$

$SO,$ $\boxed{\int_0^1 \frac{\sqrt{x^2-2x+1}}{\sqrt[3]{x^3-3x^2+3x-1}}dx=-1}$

198. Calculate Integral $I=\int_{-1}^1 \frac{x^{46}+2025}{x\left(45-x^{45}\right)}dx$

Solution; $I=\int_{-1}^1 \frac{x^{46}+2025}{x\left(45-x^{45}\right)}dx$

$$=-\int_{-1}^1 \frac{x^{46}+2025}{x\left(x^{46}-45\right)}\times\frac{x^{-45}}{x^{-45}}dx$$

$$=-\int_{-1}^1 \frac{1+2025x^{-46}}{\left(x-45x^{-45}\right)}dx$$

Let : $u=x-45x^{-45}\Rightarrow du=\left(1+2025x^{-46}\right)dx$; **If** : $x\in(-1,1)\Rightarrow u\in(46,-44)$

We obtain $I=-\int_{46}^{-44}\frac{1}{u}du=\log|u|\Big|_{-44}^{46}=\log\left(\frac{23}{22}\right)$

$SO,$ $\boxed{\int_{-1}^1 \frac{x^{46}+2025}{x\left(45-x^{45}\right)}dx=\log\left(\frac{23}{22}\right)}$

199. Calculate Integral $I=\int_1^2 x^{2x}\left(\log(x)+1\right)dx$

Solution; $I=\int_1^2 x^{2x}\left(\log(x)+1\right)dx$

$$=\int_1^2 e^{2x\log(x)}\left(\log(x)+1\right)dx$$
$$=\frac{1}{2}\int_1^2 e^{2x\log(x)}\left(2\log(x)+2\right)dx$$
$$=\frac{1}{2}\int_1^2 e^{2x\log(x)}d\left(2x\log(x)\right)$$
$$=\frac{1}{2}e^{2x\log(x)}\Big|_1^2$$
$$=\frac{1}{2}(15)$$

$SO,$ $\boxed{\int_1^2 x^{2x}\left(\log(x)+1\right)dx=\frac{15}{2}}$

200. Calculate Integral $I=\int_0^1\left(x+e+1\right)x^e e^x\,dx$

Solution; $I=\int_0^1\left(x+e+1\right)x^e e^x\,dx$

$$=\int_0^1\left(x+(e+1)\right)x^e e^x\,dx$$
$$=\int_0^1\left(x^{e+1}e^x+(e+1)x^e e^x\right)dx$$
$$=\int_0^1\left((e^x)'x^{e+1}+(x^{e+1})'e^x\right)dx$$
$$=\int_0^1\left(x^{e+1}e^x\right)'dx$$
$$=x^{e+1}e^x\Big|_0^1$$
$$=e$$

$SO,$ $\boxed{\int_0^1\left(x+e+1\right)x^e e^x\,dx=e}$

201. Calculate Integral $I=\int_{-\infty}^{\infty}\frac{x^2+1}{x^8+1}dx$

Solution; $I=\underbrace{\int_{-\infty}^{\infty}\frac{x^2+1}{x^8+1}dx}_{is\ an\ even\ function}$

$$=2\int_0^{\infty}\frac{x^2+1}{x^8+1}dx$$

Let : $x=t^{\frac{1}{8}}\Rightarrow dx=\frac{1}{8}t^{\frac{1}{8}-1}dt$ **; If :** $x\in(0,\infty)\Rightarrow t\in(0,\infty)$

We obtain $I=2\int_0^{\infty}\frac{t^{\frac{2}{8}}+1}{t+1}\times\frac{1}{8}t^{\frac{1}{8}-1}dt$

$$=\frac{1}{4}\int_0^{\infty}\left(\frac{t^{\frac{3}{8}-1}}{t+1}+\frac{t^{\frac{1}{8}-1}+1}{t+1}\right)dt$$

$$=\frac{1}{4}\int_0^{\infty}\left(\frac{t^{\frac{3}{8}-1}}{(t+1)^{\frac{3}{8}+\frac{5}{8}}}+\frac{t^{\frac{1}{8}-1}+1}{(t+1)^{\frac{1}{8}+\frac{7}{8}}}\right)dt$$

$$=\frac{1}{4}\left(\frac{\Gamma\left(\frac{3}{8}\right)\Gamma\left(\frac{5}{8}\right)}{\Gamma\left(\frac{3}{8}+\frac{5}{8}\right)}+\frac{\Gamma\left(\frac{1}{8}\right)\Gamma\left(\frac{7}{8}\right)}{\Gamma\left(\frac{1}{8}+\frac{7}{8}\right)}\right)$$

$$=\frac{1}{4}\left[\Gamma\left(\frac{3}{8}\right)\Gamma\left(1-\frac{3}{8}\right)+\Gamma\left(\frac{1}{8}\right)\Gamma\left(1-\frac{1}{8}\right)\right]$$

$$=\frac{1}{4}\left[\frac{\pi}{\sin\left(\frac{3\pi}{8}\right)}+\frac{\pi}{\sin\left(\frac{\pi}{8}\right)}\right]=\frac{\pi}{2}\left(\sqrt{2+\sqrt{2}}\right)$$

SO, $\boxed{\int_{-\infty}^{\infty}\frac{x^2+1}{x^8+1}dx=\frac{\pi}{2}\left(\sqrt{2+\sqrt{2}}\right)}$

202. Calculate Integral $I=\int_0^{\infty}\frac{x}{e^{\pi x}-1}dx$

Solution; $I=\int_0^{\infty}\frac{x}{e^{\pi x}-1}dx$

Let : $t=\pi x\Rightarrow dx=\frac{1}{\pi}dt$; **If** : $x\in(0,\infty)\Rightarrow t\in(0,\infty)$

We obtain $I=\frac{1}{\pi^2}\int_0^{\infty}\frac{t}{e^t-1}dt$

$$=\frac{1}{\pi^2}\int_0^{\infty}\frac{te^{-t}}{(e^t-1)e^{-t}}dt$$

$$=\frac{1}{\pi^2}\int_0^{\infty}\frac{te^{-t}}{1-e^{-t}}dt$$

$$=\frac{1}{\pi^2}\int_0^{\infty}\left(te^{-t}\sum_{n=0}^{\infty}e^{-nt}\right)dt$$

$$=\frac{1}{\pi^2}\sum_{n=0}^{\infty}\left[\int_0^{\infty}te^{-(n+1)t}dt\right]$$

Let : $u=(n+1)t\Rightarrow dt=\frac{1}{n+1}du$; **If** : $x\in(0,\infty)\Rightarrow t\in(0,\infty)$

Implies That $I=\frac{1}{\pi^2}\sum_{n=0}^{\infty}\left[\int_0^{\infty}\left(\frac{u}{n+1}\right)e^{-u}\left(\frac{1}{n+1}\right)du\right]$

$$=\frac{1}{\pi^2}\sum_{n=0}^{\infty}\frac{1}{(n+1)^2}\left[\int_0^{\infty}ue^{-u}du\right]$$

$$=\frac{1}{\pi^2}\sum_{n=0}^{\infty}\frac{\Gamma(2)}{(n+1)^2}=\frac{1}{\pi^2}\zeta(2)$$

$$=\frac{1}{\pi^2}\times\frac{\pi^2}{6}=\frac{1}{6}$$

SO, $\boxed{\int_0^{\infty}\frac{x}{e^{\pi x}-1}dx=\frac{1}{6}}$

Uniquely : $I=\frac{1}{\pi^2}\int_0^{\infty}\frac{t}{e^t-1}dt=\frac{1}{\pi^2}\int_0^{\infty}\frac{t^{2-1}}{e^t-1}dt=\frac{1}{\pi^2}\zeta(2)\Gamma(2)=\frac{1}{6}$

203. Calculate Integral $I=\int_0^{\infty}\frac{x^4}{(1+x^3)^2}dx$

Solution; $I=\int_0^{\infty}\frac{x^4}{(1+x^3)^2}dx$

Let : $t=x^3\Rightarrow dx=\frac{1}{3}t^{1/3-1}dt$ **; If :** $x\in(0\infty)\Rightarrow t\in(0,\infty)$

Implies That $I=\frac{1}{3}\int_0^{\infty}\frac{t^{\frac{4}{3}}}{(1+t)^2}\times t^{\frac{1}{3}-1}dt$

$$=\frac{1}{3}\int_0^{\infty}\frac{t^{\frac{5}{3}-1}}{(1+t)^{\frac{5}{3}+\frac{1}{3}}}dt$$

$$=\frac{1}{3}\times\frac{\Gamma\left(\frac{5}{3}\right)\Gamma\left(\frac{1}{3}\right)}{\Gamma(2)}$$

$$=\frac{1}{3}\times\frac{2}{3}\Gamma\left(1-\frac{1}{3}\right)\Gamma\left(\frac{1}{3}\right)$$

$$=\frac{2\pi}{9}\csc\left(\frac{\pi}{3}\right)=\frac{4\pi}{9\sqrt{3}}$$

$SO,$ $\boxed{\int_0^{\infty}\frac{x^4}{(1+x^3)^2}dx=\frac{4\pi}{9\sqrt{3}}}$

204. Calculate Integral $I=\int_0^{\infty}\frac{\sqrt{x}}{e^{\sqrt{x}}}dx$

Solution; $I=\int_0^{\infty}\frac{\sqrt{x}}{e^{\sqrt{x}}}dx$

Let : $x=t^2\Rightarrow dx=2tdt$ **; If :** $x\in(0,\infty)\Rightarrow u\in(0,\infty)$

We obtain $I=\int_0^{\infty}\frac{t}{e^t}\times 2tdt=2\int_0^{\infty}t^2e^{-t}dt=2.2!=4$

$SO,$ $\boxed{\int_0^{\infty}\frac{\sqrt{x}}{e^{\sqrt{x}}}dx=4}$

205. Calculate Integral $I=\int_0^{\infty}\frac{x}{e^x+e^{-x}}dx$

Solution; $I=\int_0^{\infty}\frac{x}{e^x+e^{-x}}dx$

$$=\int_0^{\infty}\frac{xe^{-x}}{1+e^{-2x}}dx$$

$$=\int_0^{\infty}\left(xe^{-x}\sum_{n=0}^{\infty}(-1)^n e^{-2nx}\right)dx$$

$$=\sum_{n=0}^{\infty}\left((-1)^n\int_0^{\infty}xe^{-x}e^{-2nx}dx\right)$$

$$=\sum_{n=0}^{\infty}\left((-1)^n\underbrace{\int_0^{\infty}xe^{-(2n+1)x}dx}_{Use\ partial\ integral}\right)$$

$$=\sum_{n=0}^{\infty}(-1)^n\times\frac{1}{(2n+1)^2}$$

$$=\sum_{n=0}^{\infty}\frac{(-1)^n}{(2n+1)^2}=G$$

$SO,$ $\boxed{\int_0^{\infty}\frac{x}{e^x+e^{-x}}dx=G}$

206. Calculate Integral $I=\int_0^1\log(x)\log(1-x)dx$

Solution; $I=\int_0^1\log(x)\log(1-x)dx$

$$=-\int_0^1\log(x)\sum_{n=0}^{\infty}\frac{x^{n+1}}{n+1}dx$$

$$=-\sum_{n=0}^{\infty}\frac{1}{n+1}\underbrace{\int_0^1 x^{n+1}\log(x)dx}_{(Use\ partial\ integral)}$$

$$=-\sum_{n=0}^{\infty}\frac{1}{n+1}\left(\underbrace{\frac{x^{n+2}\log(x)}{n+2}\Big|_0^1}_{0}-\int_0^1\frac{x^{n+2}}{n+2}\times\frac{1}{x}dx\right)$$

$$=\sum_{n=0}^{\infty}\frac{1}{(n+1)(n+2)^2}$$

$$=\sum_{n=0}^{\infty}\frac{1}{(n+1)(n+2)^2}$$

$$=\sum_{n=1}^{\infty}\frac{1}{(n)(n+1)^2}$$

$$=\sum_{n=1}^{\infty}\left(\frac{1}{n}-\frac{1}{n+1}-\frac{1}{(n+1)^2}\right)$$

$$=\sum_{n=1}^{\infty}\left(\frac{1}{n}-\frac{1}{n+1}\right)-\sum_{n=1}^{\infty}\left(\frac{1}{(n+1)^2}\right)$$

$$=\lim_{m\to\infty}\sum_{n=1}^{m}\left(\frac{1}{n}-\frac{1}{n+1}\right)-\left(\sum_{n=0}^{\infty}\left(\frac{1}{(n+1)^2}\right)-1\right)$$

$$=\lim_{m\to\infty}\left(1-\frac{1}{m+1}\right)-\left(\zeta(2)-1\right)$$

$$=1-\left(\zeta(2)-1\right)=2-\zeta(2)$$

$SO,\ \boxed{\int_0^1\log(x)\log(1-x)dx=2-\zeta(2)}$

207. Calculate Integral $I=\int_0^1\cos\left(\log(x)\right)dx$

Solution; $I=\int_0^1\cos\left(\log(x)\right)dx$

Assume That $J=\int_0^1\sin\left(\log(x)\right)dx$

Take $I+iJ=\int_0^1\left[\cos\left(\log(x)\right)+i\sin\left(\log(x)\right)\right]dx$

$$=\int_0^1 e^{i\log(x)}dx=\int_0^1 x^i dx$$

$$=\frac{1}{1+i}=\frac{1}{2}-\frac{1}{2}i$$

Equivalent $I+iJ=\frac{1}{2}-\frac{1}{2}i$ **We Get :** $I=\frac{1}{2}$

$SO,\ \boxed{\int_0^1\cos\left(\log(x)\right)dx=\frac{1}{2}}$

208. Calculate Integral $I=\int_0^{\infty}\frac{\log(x+x^{-1})}{x^2+1}dx$

Solution; $I=\int_0^{\infty}\frac{\log(x+x^{-1})}{x^2+1}dx$

$$=\int_0^{\infty}\frac{\log(x^2+1)-\log(x)}{x^2+1}dx$$

$$=\int_0^{\infty}\frac{\log(x^2+1)}{x^2+1}dx-\int_0^{\infty}\frac{\log(x)}{x^2+1}dx$$

Let : $x=\tan(u)\Rightarrow dx=\sec^2(u)du$ **; If** : $x\in(0,\infty)\Rightarrow u\in\left(0,\frac{\pi}{2}\right)$

We obtain $I=\int_0^{\frac{\pi}{2}}\frac{\log(\tan^2(u)+1)}{\tan^2(u)+1}\times\sec^2(u)du$

$$=\int_0^{\frac{\pi}{2}}\log(\tan^2(u)+1)\,du-\int_0^{\frac{\pi}{2}}\log(\tan(u))\,du$$

$$=\int_0^{\frac{\pi}{2}}\log(\sec^2(u))\,du-\int_0^{\frac{\pi}{2}}\log\left(\frac{\sin(u)}{\cos(u)}\right)du$$

$$=2\int_0^{\frac{\pi}{2}}\log(\cos^{-1}(u))\,du-\int_0^{\frac{\pi}{2}}\log\left(\frac{\sin(u)}{\cos(u)}\right)du$$

$$=-2\int_0^{\frac{\pi}{2}}\log(\cos(u))\,du-\int_0^{\frac{\pi}{2}}\log(\sin(u))\,du+\int_0^{\frac{\pi}{2}}\log(\cos(u))\,du$$

$$=-\int_0^{\frac{\pi}{2}}\log(\cos(u))\,du-\int_0^{\frac{\pi}{2}}\log\left(\sin(\frac{\pi}{2}-u)\right)du$$

$$=-\int_0^{\frac{\pi}{2}}\log(\cos(u))\,du-\int_0^{\frac{\pi}{2}}\log(\cos(u))\,du$$

$$=-2\int_0^{\frac{\pi}{2}}\log(\cos(u))\,du$$

$$=\pi\log(2)$$

SO, $\boxed{\int_0^{\infty}\frac{\log(x+x^{-1})}{x^2+1}dx=\pi\log(2)}$

209. Calculate Integral $I=\int_0^{\frac{\pi}{2}}\frac{1}{\sqrt{\tan^4(x)+1}}dx$

Solution; $I=\int_0^{\frac{\pi}{2}}\frac{1}{\sqrt{\tan^4(x)+1}}dx \quad (*)$

$$=\int_0^{\frac{\pi}{2}}\frac{1}{\sqrt{\tan^4\left(\frac{\pi}{2}-x\right)+1}}dx$$

$$=\int_0^{\frac{\pi}{2}}\frac{1}{\sqrt{\frac{1}{\tan^4(x)}+1}}dx$$

$$=\int_0^{\frac{\pi}{2}}\frac{\tan^2(x)}{\sqrt{\tan^4(x)+1}}dx \quad (**)$$

Take $(*)+(**)$ **They Have :** $2I=\int_0^{\frac{\pi}{2}}\frac{\tan^2(x)+1}{\sqrt{\tan^4(x)+1}}dx$

Implies That $I=\frac{1}{2}\int_0^{\frac{\pi}{2}}\frac{1}{\sqrt{\tan^4(x)+1}}d\left(\tan(x)\right)$

Let : $y=\tan^4(x)+1\Rightarrow d\left(\tan(x)\right)=\frac{1}{4}(y-1)^{\frac{1}{4}-1}dy$ **; If :** $x\in(0,\frac{\pi}{2})\Rightarrow y\in(1,\infty)$

We Get $I=\frac{1}{2\times4}\int_1^{\infty}\frac{(y-1)^{\frac{1}{4}-1}}{\sqrt{y}}dy$

$$=\frac{1}{2\times4}\int_1^{\infty}y^{-\frac{1}{2}}(y-1)^{\frac{1}{4}-1}dy$$

Let : $y=\frac{1}{u}\Rightarrow dy=-\frac{1}{u^2}du$ **; If :** $y\in(1,\infty)\Rightarrow u\in(1,0)$

Implies That $I=-\frac{1}{2\times4}\int_1^0\left(u^{-1}\right)^{-\frac{1}{2}}(u^{-1}-1)^{\frac{1}{4}-1}dy$

$$=\frac{1}{8}\int_0^1u^{\frac{1}{4}-1}(1-u)^{\frac{1}{4}-1}du$$

$$=\frac{1}{8}B\left(\frac{1}{4},\frac{1}{4}\right)$$

$$=\frac{1}{8}\times\frac{\Gamma\left(\frac{1}{4}\right)\Gamma\left(\frac{1}{4}\right)}{\Gamma\left(\frac{1}{2}\right)}$$

$$=\frac{\Gamma^2\left(\frac{1}{4}\right)}{8\sqrt{\pi}}$$

SO, $\boxed{\int_0^{\frac{\pi}{2}}\frac{1}{\sqrt{\tan^4(x)+1}}dx=\frac{\Gamma^2\left(1/4\right)}{8\sqrt{\pi}}}$

210. Calculate Integral $I=\int_1^e x^{\log(x)-1}\log(x^2)\,dx$

Solution; $I=\int_1^e x^{\log(x)-1}\log(x^2)\,dx$

$$=\int_1^e 2x^{\log(x)}\log(x)\frac{dx}{x}$$

$$=\int_1^e 2\left(e^{\log(x)}\right)^{\log(x)}\log(x)\frac{dx}{x}$$

$$=\int_1^e 2e^{\log^2(x)}\log(x)\frac{dx}{x}$$

$$=\int_1^e 2\log(x)e^{\log^2(x)}d\left(\log(x)\right)$$

$$=\int_1^e e^{\log^2(x)}d\left(\log^2(x)\right)$$

$$=e^{\log^2(x)}\Big|_1^e$$

$$=x^{\log(x)}\Big|_1^e=e-1$$

SO, $\boxed{\int_1^e x^{\log(x)-1}\log(x^2)\,dx=e-1}$

211. Calculate Integral $I=\int_0^1 \frac{\log^3(1-x)}{x}dx$

Solution; $I=\int_0^1 \frac{\log^3(1-x)}{x}dx$

$$=\int_0^1 \frac{\log^3(x)}{1-x}dx$$

Let : $x=e^{-u} \Rightarrow dx=-e^{-u}du$ **; If :** $x\in(0,1)\Rightarrow u\in(\infty,0)$

We obtain $I=-\int_\infty^0 \frac{e^{-u}\log^3(e^{-u})}{1-e^{-u}}du$

$$=-\int_0^\infty \frac{u^3}{e^u-1}du$$

$$=-\zeta(4)\Gamma(4)$$

$$=-\frac{\pi^4}{90}\times 6=-\frac{\pi^4}{15}$$

SO, $\boxed{\int_0^1 \frac{\log^3(1-x)}{x}dx=-\frac{\pi^4}{15}}$

212. Calculate Integral $I=\int_0^\infty \frac{x^2+1}{x^4-x^2+1}dx$

Solution; $I=\int_0^\infty \frac{x^2+1}{x^4-x^2+1}dx$

$$=\int_0^\infty \frac{1+x^{-2}}{(x-x^{-1})^2+1}dx$$

Let : $u=x-x^{-1} \Rightarrow du=(1+x^{-2})dx$ **; If :** $x\in(1,\infty)\Rightarrow u\in(0,\infty)$

Implies That $I=\int_0^\infty \frac{1}{u^2+1}dx=\frac{\pi}{2}$

SO, $\boxed{\int_0^\infty \frac{x^2+1}{x^4-x^2+1}dx=\frac{\pi}{2}}$

213. Calculate Integral $I=\int_1^{\infty}\frac{1}{e^x\sqrt{x-1}}dx$

Solution; $I=\int_1^{\infty}\frac{1}{e^x\sqrt{x-1}}dx$

$$=\int_1^{\infty}e^{-x}(x-1)^{-\frac{1}{2}}dx$$

$$=\frac{1}{e}\int_1^{\infty}(x-1)^{-\frac{1}{2}}e^{-(x-1)}dx$$

Let : $u=x-1\Rightarrow du=dx$ **; If :** $x\in(1,\infty)\Rightarrow u\in(0,\infty)$

We obtain $I=\frac{1}{e}\int_0^{\infty}u^{-\frac{1}{2}}e^{-u}dx$

$$=\frac{1}{e}\left(-\frac{1}{2}\right)!=\frac{\sqrt{\pi}}{e}$$

$SO,$ $\boxed{\int_1^{\infty}\frac{1}{e^x\sqrt{x-1}}dx=\frac{\sqrt{\pi}}{e}}$

214. Calculate Integral $I=\int_0^{\infty}e^{-x^n}dx$

Solution; $I=\int_0^{\infty}e^{-x^n}dx$

Let : $x=u^{\frac{1}{n}}\Rightarrow dx=\frac{1}{n}u^{\frac{1}{n}-1}du$ **; If :** $x\in(0,\infty)\Rightarrow u\in(0,\infty)$

We Get $I=\frac{1}{n}\int_0^{\infty}e^{-u}u^{\frac{1}{n}-1}du$

$$=\frac{1}{n}\Gamma\left(\frac{1}{n}\right)$$

$$=\Gamma\left(1+\frac{1}{n}\right)$$

$SO,$ $\boxed{\int_0^{\infty}e^{-x^n}dx=\Gamma\left(\frac{n+1}{n}\right)}$

215. Calculate Integral $I=\int_0^{\infty} e^{-ax^n}dx$

Solution; $I=\int_0^{\infty} e^{-ax^n}dx$

Let : $x=\left(\frac{t}{a}\right)^{\frac{1}{n}} \Rightarrow dx=\frac{1}{n\sqrt[n]{a}}t^{\frac{1}{n}-1}dt$ **; If :** $x\in(0,\infty)\Rightarrow t\in(0,\infty)$

Implies That $I=\frac{1}{n\sqrt[n]{a}}\int_0^{\infty} t^{\frac{1}{n}-1}e^{-t}dt$

$$=\frac{1}{n\sqrt[n]{a}}\Gamma\left(\frac{1}{n}\right)$$

SO, $\boxed{\int_0^{\infty} e^{-x^n}dx=\frac{1}{n\sqrt[n]{a}}\Gamma\left(\frac{1}{n}\right)}$

216. Calculate Integral $I=\int_0^{\frac{\pi}{2}}\sqrt{\sin(x)}dx$

Solution; $I=\int_0^{\frac{\pi}{2}}\sqrt{\sin(x)}dx$

$$=\int_0^{\frac{\pi}{2}}\sin^{\frac{1}{2}}(x)\,dx=\int_0^{\frac{\pi}{2}}\sin^{2\left(\frac{3}{4}\right)-1}\cos^{2\left(\frac{1}{2}\right)-1}dx$$

$$=\frac{1}{2}B\left(\frac{3}{4},\frac{1}{2}\right)=\frac{1}{2}\times\frac{\Gamma\left(\frac{3}{4}\right)\Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{3}{4}+\frac{1}{2}\right)}$$

$$=\frac{\sqrt{\pi}}{2}\times\frac{\Gamma\left(\frac{3}{4}\right)}{\Gamma\left(\frac{5}{4}\right)}=2\sqrt{\pi}\times\frac{\Gamma\left(\frac{3}{4}\right)}{\Gamma\left(\frac{1}{4}\right)}$$

SO, $\boxed{\int_0^{\frac{\pi}{2}}\sqrt{\sin(x)}dx=2\sqrt{\pi}\times\frac{\Gamma\left(\frac{3}{4}\right)}{\Gamma\left(\frac{1}{4}\right)}}$

217. Calculate Integral $I=\int_{-1}^{1}(1-x^2)^n\,dx$

Solution; $I=\underbrace{\int_{-1}^{1}(1-x^2)^n\,dx}_{is\ an\ even\ function}$

$$=2\int_0^1(1-x^2)^n\,dx$$

Let : $t=x^2\Rightarrow dx=\frac{1}{2}t^{\frac{1}{2}-1}dt$ **; If :** $x\in(0,1)\Rightarrow t\in(0,1)$

Implies That $I=2\times\frac{1}{2}\int_0^1(1-t)^n t^{\frac{1}{2}-1}dt$

$$=\int_0^1 t^{\frac{1}{2}-1}(1-t)^{(n+1)-1}dt$$

$$=B\left(\frac{1}{2},n+1\right)=\frac{\Gamma\left(1/2\right)\Gamma(n+1)}{\Gamma\left(1/2+1+n\right)}$$

$$=\frac{\sqrt{\pi}.n!}{\left(1/2+n\right)\Gamma\left(1/2+n\right)}=\frac{2\sqrt{\pi}.n!}{(2n+1)\Gamma\left(n+\frac{1}{2}\right)}$$

$$=\frac{2\sqrt{\pi}.n!\Gamma(n)}{(2n+1)\Gamma(n)\Gamma\left(n+1/2\right)}=\frac{2\sqrt{\pi}.n!(n-1)!}{(2n+1)2^{1-2n}\sqrt{\pi}\,\Gamma(2n)}$$

$$=\frac{n!(n-1)!}{(2n+1)2^{-2n}\Gamma(2n)}=\frac{2^{2n}(2n)n!(n-1)!}{(2n+1)(2n)(2n-1)!}$$

$$=\frac{2^{2n+1}n!(n)!}{(2n+1)(2n)!}=\frac{2^{2n+1}}{(2n+1)\frac{(2n)!}{n!(n)!}}$$

$$=\frac{2^{2n+1}}{(2n+1)\frac{(2n)!}{n!(n)!}}=\frac{2^{2n+1}}{(2n+1)C_{2n}^{n}}$$

SO, $\boxed{\int_{-1}^{1}(1-x^2)^n\,dx=\frac{2^{2n+1}}{(2n+1)C_{2n}^{n}}}$

218. Calculate Integral $I=\int_0^{\pi}\log\left(\left|\sin(x)\right|\right)dx$

Solution; $I=\int_0^{\pi}\log\left(\left|\sin(x)\right|\right)dx$

$$=2\int_0^{\frac{\pi}{2}}\log\left(\left|\sin(x)\right|\right)dx$$

Use : $\int_0^{2a}f(x)dx=2\int_0^{a}f(x)dx\ ;f(2a-x)=f(x)$

$$=2\int_0^{\frac{\pi}{2}}\log\left(\sin(x)\right)dx$$

$$=2\left(-\frac{\pi}{2}\log(2)\right)=-\pi\log(2)$$

Likewise $J=\int_0^{\pi}\log\left(\left|\cos(x)\right|\right)dx$

$$=2\int_0^{\frac{\pi}{2}}\log\left(\left|\cos(x)\right|\right)dx$$

$$=2\left(-\frac{\pi}{2}\log(2)\right)=-\pi\log(2)$$

SO, $\boxed{\int_0^{\pi}\log\left(\left|\sin(x)\right|\right)dx=\int_0^{\pi}\log\left(\left|\cos(x)\right|\right)dx=-\pi\log(2)}$

219. Calculate Integral $I=\int_0^{\pi}\log\left(\left|\tan(x)\right|\right)dx$

Solution; $I=\int_0^{\pi}\log\left(\left|\tan(x)\right|\right)dx$

$$=2\int_0^{\frac{\pi}{2}}\log\left(\left|\tan(x)\right|\right)dx$$

$$=2\left(\int_0^{\frac{\pi}{2}}\log\left(\left|\sin(x)\right|\right)dx-\int_0^{\frac{\pi}{2}}\log\left(\left|\cos(x)\right|\right)dx\right)$$

$$=2\left(\int_0^{\frac{\pi}{2}}\log\left(\sin(x)\right)dx-\int_0^{\frac{\pi}{2}}\log\left(\cos(x)\right)dx\right)$$

$$=2\left(-\frac{\pi}{2}\log(2)+\frac{\pi}{2}\log(2)\right)=0$$

SO, $\boxed{\int_0^{\pi}\log\left(\left|\tan(x)\right|\right)dx=0}$

220. Calculate Integral $I=\int_0^{\pi}\left(\sqrt[\pi]{i^i}\right)^{-2x}dx$

Solution; $I=\int_0^{\pi}\left(\sqrt[\pi]{i^i}\right)^{-2x}dx$

We Have $\left(\sqrt[\pi]{i^i}\right)^{-2}=\left(i^{\frac{i}{\pi}}\right)^{-2}=i^{-\frac{2}{\pi}i}$ **But** $i=e^{\frac{\pi}{2}i}$

Implies That $\left(\sqrt[\pi]{i^i}\right)^{-2}=\left(e^{\frac{\pi}{2}i}\right)^{-\frac{2}{\pi}i}=e$ **Then** $\left(\sqrt[\pi]{i^i}\right)^{-2x}=e^x$

We obtain $I=\int_0^{\pi}e^x\,dx=e^{\pi}-1$

$SO,\ \boxed{\int_0^{\pi}\left(\sqrt[\pi]{i^i}\right)^{-2x}dx=e^{\pi}-1}$

221. Calculate Integral $I=\int_0^{\infty}i^{ix^2}dx\ \ ,i=\sqrt{-1}$

Solution; $I=\int_0^{\infty}i^{ix^2}dx$

By $i=e^{\frac{\pi}{2}i}$ **Then** $i^i=e^{\frac{\pi}{2}i^2}=e^{-\frac{\pi}{2}}$

We obtain $I=\int_0^{\infty}e^{-\frac{\pi}{2}x^2}dx$

Let : $y=\frac{\pi}{2}x^2\Leftrightarrow x=\frac{\sqrt{2}}{\sqrt{\pi}}\sqrt{y}\Rightarrow dx=\frac{\sqrt{2}}{2\sqrt{\pi}}\times\frac{1}{\sqrt{y}}dy$ **; If :** $x\in(0,\infty)\Rightarrow y\in(0,\infty)$

Implies That $I=\frac{\sqrt{2}}{2\sqrt{\pi}}\int_0^{\infty}e^{-y}\times\frac{1}{\sqrt{y}}dy$

$$=\frac{\sqrt{2}}{2\sqrt{\pi}}\int_0^{\infty}y^{\frac{1}{2}-1}e^{-y}dy$$

$$=\frac{\sqrt{2}}{2\sqrt{\pi}}\Gamma\left(\frac{1}{2}\right)=\frac{\sqrt{2}}{2}$$

$SO,\ \boxed{\int_0^{\infty}i^{ix^2}dx=\frac{\sqrt{2}}{2}}$

222. Calculate Integral $I=\int_0^1 \frac{\log^3(1-x^2)}{x}dx$

Solution; $I=\int_0^1 \frac{\log^3(1-x^2)}{x}dx$

$$=\frac{1}{2}\int_0^1 \frac{\log^3(1-x^2)}{x^2}d(x^2)$$

$$=\frac{1}{2}\int_0^1 \frac{\log^3\left(1-(1-x^2)\right)}{1-x^2}d(x^2)$$

$$=\frac{1}{2}\int_0^1 \frac{\log^3(x^2)}{1-x^2}d(x^2)$$

Let : $x^2=e^{-u}\Rightarrow d(x^2)=-e^{-u}du$ **; If :** $x^2\in(0,1)\Rightarrow u\in(\infty,0)$

Implies That $I=-\frac{1}{2}\int_\infty^0 \frac{\log^3(e^{-u})e^{-u}}{1-e^{-u}}du$

$$=-\frac{1}{2}\int_0^\infty \frac{u^{4-1}}{e^u-1}du$$

$$=-\frac{1}{2}\Gamma(4)\zeta(4)=-\frac{\pi^4}{30}$$

SO, $\boxed{\int_0^1 \frac{\log^3(1-x^2)}{x}dx=-\frac{\pi^4}{30}}$

223. Calculate Integral $I=\int_0^\infty e^{-x^2}dx$

Solution; $I=\int_0^\infty e^{-x^2}dx$

$$=\int_0^\infty x^{-1}e^{-x^2}xdx$$

$$=\frac{1}{2}\int_0^\infty (x^2)^{-\frac{1}{2}}e^{-x^2}d(x^2)$$

$$=\frac{1}{2}\Gamma\left(\frac{1}{2}\right)=\frac{\sqrt{\pi}}{2}$$

SO, $\boxed{\int_0^\infty e^{-x^2}dx=\frac{\sqrt{\pi}}{2}}$

224. Calculate Integral $I=\int_0^1 \log\left(1-x^4\right)dx$

Solution; $I=\int_0^1 \log\left(1-x^4\right)dx$

$$=\int_0^1 \log\left((1+x^2)(1-x^2)\right)dx$$

$$=\int_0^1 \log\left((1+x^2)(1-x)(1+x)\right)dx$$

$$=\underbrace{\int_0^1 \log\left(1+x\right)dx}_{J_1}+\underbrace{\int_0^1 \log\left(1-x\right)dx}_{J_2}+\underbrace{\int_0^1 \log\left(1+x^2\right)dx}_{J_3}$$

For : $J_1=\int_0^1 \log\left(1+x\right)dx$, $\begin{cases} Let: x+1=t\Rightarrow dx=dt \\ If: x\in(0,1)\Rightarrow t\in(1,2) \end{cases}$

We obtain : $J_1=\int_1^2 \log(t)\,dt=t\left(\log(t)-1\right)\Big|_1^2=\log(4)-1$

For: $J_2=\int_0^1 \log\left(1-x\right)dx$, $\begin{cases} Let: 1-x=t\Rightarrow dx=-dt \\ If: x\in(0,1)\Rightarrow t\in(1,0) \end{cases}$

We obtain : $J_2=\int_0^1 \log(t)\,dt=t\left(\log(t)-1\right)\Big|_0^1=-1$

For : $J_3=\int_0^1 \log\left(1+x^2\right)dx$, $\begin{cases} Let: u=\log\left(1+x^2\right)\Rightarrow du=\dfrac{2x}{1+x^2}dx \\ dv=dx\Rightarrow v=x \end{cases}$

We obtain : $J_3=x\log\left(1+x^2\right)\Big|_0^1-2\int_0^1\frac{x^2}{1+x^2}dx$

$$=\log(2)-2\int_0^1\left(\frac{1}{1+x^2}-1\right)dx$$

$$=\log(2)+2\left(\tan^{-1}(x)-x\right)\Big|_0^1$$

$$=\log(2)+2\left(\frac{\pi}{4}-1\right)$$

Take : $(*)$ $: J=\left(\log(4)-1\right)-1+\left(\log(2)+\frac{\pi}{2}-2\right)=\frac{\pi}{2}+3\log(2)-4$

$SO,$ $\boxed{\int_0^1 \log\left(1-x^4\right)dx=\frac{\pi}{2}+3\log(2)-4}$

225. Calculate Integral $I=\int_1^{\infty}\left(\left(\frac{x}{x+1}\right)^2\left(\frac{x-1}{x+1}\right)\left(\frac{1}{x+1}\right)\right)^2 dx$

Solution; $I=\int_1^{\infty}\left(\left(\frac{x}{x+1}\right)^2\left(\frac{x-1}{x+1}\right)\left(\frac{1}{x+1}\right)\right)^2 dx$

$$=\int_1^{\infty}\left(\frac{x}{x+1}\right)^4\left(\frac{x-1}{x+1}\right)^2\left(\frac{1}{x+1}\right)^2 dx$$

$$=\frac{1}{16}\int_1^{\infty}\left(\frac{2x}{x+1}\right)^4\left(\frac{x-1}{x+1}\right)^2\left(\frac{1}{x+1}\right)^2 dx$$

$$=\frac{1}{16}\int_1^{\infty}\left(1-\frac{x-1}{x+1}\right)^4\left(\frac{x-1}{x+1}\right)^2\left(\frac{1}{x+1}\right)^2 dx$$

Let : $y=\frac{x-1}{x+1}\Rightarrow dy=\frac{1}{(x+1)^2}dx$ **; If :** $x\in(1,\infty)\Rightarrow y\in(0,1)$

We obtain $I=\frac{1}{16}\int_0^1(1-y)^4(y)^2dy$

$$=\frac{1}{16}B(5,3)=\frac{1}{560}$$

SO, $\boxed{\int_1^{\infty}\left(\left(\frac{x}{x+1}\right)^2\left(\frac{x-1}{x+1}\right)\left(\frac{1}{x+1}\right)\right)^2 dx=\frac{1}{560}}$

226. Calculate Integral $I=\int_0^{\infty}\frac{\sin^2(x)}{x^2(x^2+1)}dx$

Solution; $I=\int_0^{\infty}\frac{\sin^2(x)}{x^2(x^2+1)}dx$

$$=\int_0^{\infty}\frac{\left((x^2+1)-x^2\right)\sin^2(x)}{x^2(x^2+1)}dx$$

$$=\int_0^{\infty}\frac{\sin^2(x)}{x^2}dx-\int_0^{\infty}\frac{\sin^2(x)}{x^2+1}dx$$

$$= \frac{\pi}{2} - I \quad (*)$$

For : $J = \int_0^{\infty} \frac{\sin^2(x)}{x^2+1} dx$ **Suppose** : $K = \int_0^{\infty} \frac{\cos^2(x)}{x^2+1} dx$

Take : $K + J = \int_0^{\infty} \frac{\sin^2(x) + \cos^2(x)}{x^2+1} dx$

$$= \int_0^{\infty} \frac{1}{x^2+1} dx$$

$$= \tan^{-1}(x)\Big|_0^{\infty} = \frac{\pi}{2}$$

Take : $K - J = \int_0^{\infty} \frac{\cos^2(x) - \sin^2(x)}{x^2+1} dx$

$$= \int_0^{\infty} \frac{\cos(2x)}{x^2+1} dx = \frac{\pi}{2e^2}$$

Use : $\int_0^{\infty} \frac{\cos(ax)}{x^2+1} dx = \frac{\pi}{2e^a}$

Implies That : $(K+J) - (K-J) = \frac{\pi}{2} - \frac{\pi}{2e^2}$

We obtain $K = \frac{\pi}{4} - \frac{\pi e^{-2}}{4}$

According to the relation $(*)$ **Then** $I = \frac{\pi}{2} - \frac{\pi}{4} + \frac{\pi e^{-2}}{4} = \frac{\pi}{4}\left(e^{-2}+1\right)$

$SO,$ $\boxed{\int_0^{\infty} \frac{\sin^2(x)}{x^2(x^2+1)} dx = \frac{\pi}{4}\left(e^{-2}+1\right)}$

227. Calculate Integral $I = \int_1^{\infty} \left(\frac{\log^2(x)}{x(x+1)} \right) dx$

Solution; $I = \int_1^{\infty} \left(\frac{\log^2(x)}{x(x+1)} \right) dx$

Let : $x = e^y \Rightarrow dx = e^y dy$ **; If** : $x \in (1,\infty) \Rightarrow y \in (0,\infty)$

We obtain $I = \int_0^{\infty} \frac{y^2 e^y}{e^y(e^y+1)} dy$

$$= \int_0^{\infty} \frac{y^2}{(1+e^y)} dy$$

$$= \int_0^\infty \frac{y^2 e^{-y}}{(1+e^{-y})} dy$$

$$= \int_0^\infty y^2 e^{-y} \left(1 + \sum_{n=1}^{\infty} (-1)^n e^{-ny}\right) dy$$

$$= \int_0^\infty y^2 e^{-y} dy + \sum_{n=1}^{\infty} \left((-1)^n \int_0^\infty y^2 e^{-(n+1)y} dy\right)$$

$$= \Gamma(3) + \sum_{n=1}^{\infty} \left((-1)^n \int_0^\infty y^2 e^{-(n+1)y} dy\right)$$

Take : $t = (n+1)y \equiv y = \frac{t}{n+1} \Rightarrow dy = \frac{1}{n+1} dt$ **; If** : $y \in (0,\infty) \Rightarrow t \in (0,\infty)$

We obtain $I = \Gamma(3) + \sum_{n=1}^{\infty} \left[(-1)^n \int_0^\infty \left(\frac{t}{n+1}\right)^2 e^{-t} \frac{dt}{n+1} \right]$

$$= \Gamma(3) + \sum_{n=1}^{\infty} \left(\frac{(-1)^n}{(n+1)^3} \int_0^\infty (t)^2 e^{-t} dt \right)$$

$$= \Gamma(3) + \sum_{n=1}^{\infty} \left(\frac{(-1)^n}{(n+1)^3} \Gamma(3) \right)$$

$$= 2\left(1 + \sum_{n=1}^{\infty} \frac{(-1)^n}{(n+1)^3}\right)$$

$$= -2 \sum_{n=0}^{\infty} \frac{(-1)^{n+1}}{(n+1)^3}$$

$$= -2\eta(3)$$

$$= (1 - 2^{1-3} \zeta(3)$$

$$= -\frac{3}{4} \zeta(3)$$

SO, $\boxed{\int_1^\infty \left(\frac{\log^2(x)}{x(x+1)}\right) dx = -\frac{3}{4}\zeta(3)}$

228. Calculate Integral $I=\int_{-1}^{1} x\tan(x)\tan\left(\frac{1}{x}\right)dx$

Solution; $I=\int_{-1}^{1} x\tan(x)\tan\left(\frac{1}{x}\right)dx$

$$=\int_{-1}^{1} x\tan^{-1}(x)\cot^{-1}(x)dx$$

$$=\int_{-1}^{1} (-x)\tan^{-1}(-x)\cot^{-1}(-x)dx$$

By: $\begin{cases}\tan^{-1}\left(\frac{1}{x}\right)=\cot^{-1}(x)\ ;\ \cot^{-1}(-x)=\pi-\cos^{-1}(x)\\ \tan^{-1}(-x)=-\tan^{-1}(x)\ ;\ \tan^{-1}(-x)=-\tan^{-1}(x)\end{cases}$

Implies That $I=\int_{-1}^{1} x\tan^{-1}(x)\left(\pi-\cot^{-1}(x)\right)dx$

$$=\pi\int_{-1}^{1} x\tan^{-1}(x)dx-\int_{-1}^{1} x\tan^{-1}(x)\cot^{-1}(x)dx$$

We Get $I=\frac{\pi}{2}\underbrace{\int_{-1}^{1} x\tan^{-1}(x)dx}_{is\ an\ even\ function}$

$$=\pi\int_{0}^{1} x\tan^{-1}(x)dx$$

Let : $u=\tan^{-1}(x)\Rightarrow du=\frac{1}{1+x^2}dx$ **; And** $v=\int xdx=\frac{1}{2}x^2$

We obtain $I=\pi\left(\frac{1}{2}x^2\tan^{-1}(x)\Big|_0^1-\frac{1}{2}\int_0^1\frac{x^2}{1+x^2}dx\right)$

$$=\frac{\pi}{2}\left(\frac{\pi}{4}-\int_0^1\frac{1+x^2-1}{1+x^2}dx\right)$$

$$=\frac{\pi}{2}\left(\frac{\pi}{4}-\int_0^1\left(1-\frac{1}{1+x^2}\right)dx\right)$$

$$=\frac{\pi(\pi-2)}{4}$$

SO, $\boxed{\int_{-1}^{1} x\tan(x)\tan\left(\frac{1}{x}\right)dx=\frac{\pi(\pi-2)}{4}}$

229. Calculate Integral $I=\int_0^{\infty}\frac{\log(2e^x-1)}{e^x-1}dx$

Solution; $I=\int_0^{\infty}\frac{\log(2e^x-1)}{e^x-1}dx$

Let : $t=e^x\Rightarrow dx=\frac{1}{t}dt$ **; If :** $x\in(0,\infty)\Rightarrow t\in(1,\infty)$

We obtain $I=\int_1^{\infty}\frac{\log(2t-1)}{t(t-1)}dt$

Let : $t=\frac{1}{y}\Rightarrow dt=-\frac{1}{y^2}dy$ **; If :** $t\in(1,\infty)\Rightarrow y\in(1,0)$

Implies That $I=-\int_1^0\frac{\log\left(\frac{2}{y}-1\right)}{\frac{1}{y}\left(\frac{1}{y}-1\right)}\times\frac{1}{y^2}dy$

$$=\int_0^1\frac{\log(2-y)-\log(y)}{(1-y)}dy$$

$$=\int_0^1\frac{\log(1+y)-\log(1-y)}{y}dy$$

$$=\int_0^1\frac{\left(y-\frac{y^2}{2}+\frac{y^3}{3}-\frac{y^4}{4}+\cdots\right)-\left(-y-\frac{y^2}{2}-\frac{y^3}{3}-\frac{y^4}{4}-\cdots\right)}{y}dy$$

$$=2\int_0^1\left(1+\frac{y^2}{3}+\frac{y^4}{5}+\frac{y^6}{7}+\cdots\right)dy$$

$$=2\left(1+\frac{1}{3^2}+\frac{1}{5^2}+\frac{1}{7^2}+\cdots\right)$$

$$=2\sum_{n=0}^{\infty}\frac{1}{(2n+1)^2}=\frac{\pi^2}{4}$$

SO, $\boxed{\int_0^{\infty}\frac{\log(2e^x-1)}{e^x-1}dx=\frac{\pi^2}{4}}$

230. Calculate Integral $I=\int_0^{\pi}\frac{x\sin^{2026}(x)}{\cos^{2026}(x)+\sin^{2026}(x)}dx$

Solution; $I=\int_0^{\pi}\frac{x\sin^{2026}(x)}{\cos^{2026}(x)+\sin^{2026}(x)}dx$

$$=\int_0^{\pi}\frac{(\pi-x)\sin^{2026}(\pi-x)}{\cos^{2026}(\pi-x)+\sin^{2026}(\pi-x)}dx$$

$$=\int_0^{\pi}\frac{\pi\sin^{2026}(x)}{\cos^{2026}(x)+\sin^{2026}(x)}dx-\underbrace{\int_0^{\pi}\frac{x\sin^{2026}(x)}{\cos^{2026}(x)+\sin^{2026}(x)}dx}_{I}$$

We obtain $I=\frac{\pi}{2}\int_0^{\pi}\frac{\sin^{2026}(x)}{\cos^{2026}(x)+\sin^{2026}(x)}dx$

$$=\pi\int_0^{\frac{\pi}{2}}\frac{\sin^{2026}(x)}{\cos^{2026}(x)+\sin^{2026}(x)}dx\quad(1)$$

$$=\pi\int_0^{\frac{\pi}{2}}\frac{\sin^{2026}\left(\frac{\pi}{2}-x\right)}{\cos^{2026}\left(\frac{\pi}{2}-x\right)+\sin^{2026}\left(\frac{\pi}{2}-x\right)}dx\ \mathbf{n}$$

$$=\pi\int_0^{\frac{\pi}{2}}\frac{\cos^{2026}(x)}{\cos^{2026}(x)+\sin^{2026}(x)}dx\quad(2)$$

Take $(1)+(2)$ **Then** : $2I=\pi\int_0^{\frac{\pi}{2}}\frac{\sin^{2026}(x)}{\cos^{2026}(x)+\sin^{2026}(x)}dx+\pi\int_0^{\frac{\pi}{2}}\frac{\cos^{2026}(x)}{\cos^{2026}(x)+\sin^{2026}(x)}dx$

$$=\pi\int_0^{\frac{\pi}{2}}\frac{\cos^{2026}(x)+\sin^{2026}(x)}{\cos^{2026}(x)+\sin^{2026}(x)}dx$$

$$=\frac{\pi^2}{2}$$

We obtain $I=\frac{\pi^2}{4}$

SO, $\boxed{\int_0^{\pi}\frac{x\sin^{2026}(x)}{\cos^{2026}(x)+\sin^{2026}(x)}dx=\frac{\pi^2}{4}}$

231. Calculate Integral $I=\int_0^1 \frac{1-x+\log(x^x)}{\log^2(x^{\sqrt{x}})}dx$

Solution; $I=\int_0^1 \frac{1-x+\log(x^x)}{\log^2(x^{\sqrt{x}})}dx$

$$=\int_0^1 \frac{1-x+x\log(x)}{x\log^2(x)}dx$$

$$=\int_0^1 \frac{1/x-1+\log(x)}{\log^2(x)}dx$$

$$=\int_0^1 \frac{(x)'\log(x)-x\big(\log(x)\big)'}{\log^2(x)}dx+\int_0^1 \frac{1/x}{\log^2(x)}dx$$

$$=\int_0^1 d\left(\frac{x}{\log(x)}\right)-\int_0^1 d\left(\frac{1}{\log(x)}\right)$$

$$=\lim_{x\to 1}\frac{x-1}{\log(x)}-\lim_{x\to 0}\frac{x-1}{\log(x)}$$

$$=1-0=1$$

$SO,$ $\boxed{\int_0^1 \frac{1-x+\log(x^x)}{\log^2(x^{\sqrt{x}})}dx=1}$

232. Calculate Integral $I=\int_0^{1/2}\frac{\log[(1-x)x]}{x-1}dx$

Solution; $I=\int_0^{1/2}\frac{\log[(1-x)x]}{x-1}dx$

$$=-\int_0^{1/2}\frac{\log[(1-x)x]}{1-x}dx$$

Let : $y=1-x\Rightarrow dx=-dy$ **; If :** $x\in\left(0,1/2\right)\Rightarrow y\in\left(1,1/2\right)$

We obtain $I=\int_1^{1/2}\frac{\log[(1-y)y]}{y}dy$

$$=\int_1^{1/2}\frac{\log(y)}{y}dy+\int_1^{1/2}\frac{\log(1-y)}{y}dy$$

$$= \frac{1}{2}\log^2(y)\Big|_1^{1/2} - Li_2(y)\Big|_1^{1/2}$$

$$= \frac{1}{2}\log^2(2) - Li_2\left(\frac{1}{2}\right) + Li_2(1)$$

$$= \frac{1}{2}\log^2(2) - \left(\frac{\pi^2}{12} - \frac{1}{2}\log^2(2)\right) + \frac{\pi^2}{6}$$

$$= \log^2(2) + \frac{\pi^2}{12}$$

SO, $\boxed{\int_0^{1/2} \frac{\log[(1-x)x]}{x-1}\,dx = \log^2(2) + \frac{\pi^2}{12}}$

233. Calculate Integral $I = \int_0^{\frac{\pi}{4}} \log\left(\cot(x) - 1\right) dx$

Solution; $I = \int_0^{\frac{\pi}{4}} \log\left(\cot(x) - 1\right) dx$

$$= \int_0^{\frac{\pi}{4}} \log\left(\cot\left(\frac{\pi}{4} - x\right) - 1\right) dx$$

$$= \int_0^{\frac{\pi}{4}} \log\left(\frac{\cot(x)+1}{\cot(x)-1} - 1\right) dx$$

$$= \int_0^{\frac{\pi}{4}} \log\left(\frac{2}{\cot(x)-1}\right) dx$$

$$= \int_0^{\frac{\pi}{4}} \log(2)\,dx - \int_0^{\frac{\pi}{4}} \log\left(\cot(x) - 1\right) dx$$

Equivalent $J = \frac{\pi}{4}\log(2) - J$

We obtain $J = \frac{\pi}{8}\log(2)$

SO, $\boxed{\int_0^{\frac{\pi}{4}} \log\left(\cot(x) - 1\right) dx = \frac{\pi}{8}\log(2)}$

234. Calculate Integral $I=\int_0^{\infty}\left(\frac{\log(x)}{1+x}\right)^2 dx$

Solution; $I=\int_0^{\infty}\left(\frac{\log(x)}{1+x}\right)^2 dx$

$$=\int_0^1\left(\frac{\log(x)}{1+x}\right)^2 dx+\int_1^{\infty}\left(\frac{\log(x)}{1+x}\right)^2 dx \quad (*)$$

For: $J=\int_1^{\infty}\left(\frac{\log(x)}{1+x}\right)^2 dx$

Let : $x=\frac{1}{u}\Rightarrow dx=-\frac{1}{u^2}du$ **; If :** $x\in(1,\infty)\Rightarrow u\in(1,0)$

Implies That : $J=-\int_1^0\left(\frac{-\log(u)}{1+\frac{1}{u}}\right)^2\frac{1}{u^2}du=\int_0^1\left(\frac{\log(u)}{1+u}\right)^2 du$

Take (*) Then $I=2\int_0^1\left(\frac{\log(x)}{1+x}\right)^2 dx$

$$=2\int_0^1\frac{\log^2(x)}{(1+x)^2}dx$$

$$=2\int_0^1\log^2(x)\left(\sum_{n=0}^{\infty}(-1)^{n-1}nx^{n-1}\right)dx$$

Note: $\frac{1}{(1+x)^2}=\sum_{n=0}^{\infty}(-1)^{n-1}nx^{n-1}$

$$=2\sum_{n=0}^{\infty}\left(n.(-1)^{n-1}\underbrace{\int_0^1 x^{n-1}\log^2(x)dx}_{\text{Use partial integral}}\right)$$

$$=4\sum_{n=0}^{\infty}n.(-1)^{n-1}\left(\frac{1}{n^3}\right)$$

Note: $\eta(s)=(1-2^{1-s})\zeta(s)$

$$=4\sum_{n=0}^{\infty}\left(\frac{(-1)^{n-1}}{n^2}\right)=4\eta(2)=\frac{\pi^2}{3}$$

SO, $\boxed{\int_0^{\infty}\left(\frac{\log(x)}{1+x}\right)^2 dx=\frac{\pi^2}{3}}$

235. Calculate Integral $I=\int_0^{\frac{\pi}{4}}\frac{x}{\big(\sin(x)+\cos(x)\big)\cos(x)}dx$

Solution; $I=\int_0^{\frac{\pi}{4}}\frac{x}{\big(\sin(x)+\cos(x)\big)\cos(x)}dx$

$$=\int_0^{\frac{\pi}{4}}\frac{\left(\frac{\pi}{4}-x\right)}{\left(\sin\left(\frac{\pi}{4}-x\right)+\cos\left(\frac{\pi}{4}-x\right)\right)\cos\left(\frac{\pi}{4}-x\right)}dx$$

$$=\int_0^{\frac{\pi}{4}}\frac{\left(\frac{\pi}{4}-x\right)}{\big(\cos(x)+\sin(x)\big)\cos(x)}dx$$

$$=\frac{\pi}{4}\int_0^{\frac{\pi}{4}}\frac{1}{\big(\cos(x)+\sin(x)\big)\cos(x)}dx-\underbrace{\int_0^{\frac{\pi}{4}}\frac{x}{\big(\sin(x)+\cos(x)\big)\cos(x)}dx}_{I}$$

Implies That $I=\frac{\pi}{8}\int_0^{\frac{\pi}{4}}\frac{1}{[\sin(x)+\cos(x)]\cos(x)}dx$

$$=\int_0^{\frac{\pi}{4}}\frac{1}{\tan(x)+1}\times\frac{1}{\cos^2(x)}dx$$

$$=\frac{\pi}{8}\int_0^{\frac{\pi}{4}}\frac{1}{\tan(x)+1}d\big(\tan(x)\big)$$

$$=\frac{\pi}{8}\log\big|\tan(x)+1\big|\Big|_0^{\frac{\pi}{4}}$$

$$=\frac{\pi}{8}\log(2)$$

SO, $\boxed{\int_0^{\frac{\pi}{4}}\frac{x}{\big(\sin(x)+\cos(x)\big)\cos(x)}dx=\frac{\pi}{8}\log(2)}$

236. Calculate Integral $I=\int_1^2 \frac{\log(x-1)}{x(2-x)}dx$

Solution; $I=\int_1^2 \frac{\log(x-1)}{x(2-x)}dx$

$$=-\int_1^2 \frac{\log(x-1)}{x^2-2x}dx$$

$$=-\int_1^2 \frac{\log(x-1)}{1+(x-1)^2}dx$$

Let : $u=x-1\Rightarrow du=dx$; **If** : $x\in(1,2)\Rightarrow u\in(0,1)$

Implies That $I=-\int_0^1 \frac{\log(u)}{1+u^2}du$

Method 1 : $I=-\int_0^1 \frac{\log(u)}{1+u^2}du$

$$=-\int_0^1 \log(u)\sum_{n=0}^{\infty}(-1)^n u^{2n}du$$

$$=-\sum_{n=0}^{\infty}\left((-1)^n \underbrace{\int_0^1 u^{2n}\log(u)du}_{Use\ partial\ integral}\right)$$

$$=-\sum_{n=0}^{\infty}(-1)^n\left(\underbrace{\frac{u^{2n+1}\log(u)}{2n+1}\Big|_0^1}_{0}-\int_0^1 \frac{u^{2n+1}}{2n+1}\times\frac{1}{u}du\right)$$

$$=\sum_{n=0}^{\infty}\frac{(-1)^n}{(2n+1)^2}=G$$

Method 2 : $I=-\int_0^1 \frac{\log(u)}{1+u^2}du$

Let : $u=\tan(y)\Rightarrow du=\sec^2(y)dy$; **If** : $u\in(0,1)\Rightarrow y\in\left(0,\frac{\pi}{4}\right)$

We obtain $I=-\int_0^{\frac{\pi}{4}} \frac{\log(\tan(y))}{1+\tan^2(y)}\times\sec^2(y)dy$

$$= -\int_0^{\frac{\pi}{4}} \log(\tan(y))dy = G$$

Method 3 : $I = -\int_0^1 \frac{\log(u)}{1+u^2}du$

Let : $u = \cot(y) \Rightarrow du = -\csc^2(y)dy$; **If** : $u \in (0,1) \Rightarrow y \in \left(0, \frac{\pi}{4}\right)$

We Get $I = \int_0^{\frac{\pi}{4}} \frac{\log(\cot(y))}{1+\cot^2(y)} \times \csc^2(y)dy$

$$= \int_0^{\frac{\pi}{4}} \log(\cot(y))dy = G$$

SO, $\boxed{\int_1^2 \frac{\log(x-1)}{x(2-x)}dx = G}$

237. Calculate Integral $I = \int_0^1 \frac{\log(x^2+1)}{x}dx$

Solution; $I = \int_0^1 \frac{\log(x^2+1)}{x}dx$

$$= \int_0^1 \frac{1}{x}\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}x^{2n}dx$$

$$= \sum_{n=1}^{\infty}\left(\frac{(-1)^{n+1}}{n}\int_0^1 x^{2n}dx\right)$$

$$= \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n(2n+1)}$$

SO, $\boxed{\int_0^1 \frac{\log(x^2+1)}{x}dx = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n(2n+1)}}$

Note: $\log(x+1) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}x^n \Rightarrow \log(x^2+1) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}x^{2n}$

238. Calculate Integral $I=\int_0^1 \frac{\log(x+1)}{x}dx$

Solution; $I=\int_0^1 \frac{\log(x+1)}{x}dx$

$$=\int_0^1 \frac{1}{x}\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}x^n dx$$

$$=\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}\int_0^1 \frac{1}{x}x^n dx$$

$$=\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}\int_0^1 x^{n-1}dx$$

$$=\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n^2}=\eta(2)=\frac{\pi^2}{12}$$

$SO,$ $\boxed{\int_0^1 \frac{\log(x+1)}{x}dx=\frac{\pi^2}{12}}$

239. Calculate Integral $I=\int_1^e (x-1)\log^2(x)dx$

Solution; $I=\int_1^e (x-1)\log^2(x)dx$

Let $:u=\log(x)\Leftrightarrow x=e^u\Rightarrow dx=e^u du$ **; If** $: x\in(1,e)\Rightarrow u\in(0,1)$

Implies That $I=\int_0^1 u^2(e^u-1)e^u dx$

$$=\int_0^1 u^2e^{2u}dx-\int_0^1 u^2e^u dx \quad \textbf{(Use Partial Integral)}$$

$$=\left(\frac{1}{2}u^2-\frac{1}{2}u+\frac{1}{4}\right)e^{2u}\Big|_0^1-\left(u^2-2u+2\right)e^{2u}\Big|_0^1$$

$$=\frac{e^2-4e+7}{4}$$

$SO,$ $\boxed{\int_1^e (x-1)\log^2(x)dx=\frac{e^2-4e+7}{4}}$

240. Calculate Integral $I=\int_0^{\frac{\pi}{2}}\frac{\sin^2(x)}{\sin(x)+\cos(x)}dx$

Solution; $I=\int_0^{\frac{\pi}{2}}\frac{\sin^2(x)}{\sin(x)+\cos(x)}dx$ (*)

$$=\int_0^{\frac{\pi}{2}}\frac{\sin^2\left(\frac{\pi}{2}-x\right)}{\sin\left(\frac{\pi}{2}-x\right)+\cos\left(\frac{\pi}{2}-x\right)}dx$$

$$=\int_0^{\frac{\pi}{2}}\frac{\cos^2(x)}{\sin(x)+\cos(x)}dx \quad (**)$$

Take $(*)+(**)$ **Then** : $2I=\int_0^{\frac{\pi}{2}}\frac{\sin^2(x)}{\sin(x)+\cos(x)}dx+\int_0^{\frac{\pi}{2}}\frac{\cos^2(x)}{\sin(x)+\cos(x)}dx$

Implies That $I=\frac{1}{2}\int_0^{\frac{\pi}{2}}\frac{1}{\sin(x)+\cos(x)}dx$

Let : $y=\tan\left(\frac{x}{2}\right)\Rightarrow dx=\frac{2}{1+y^2}dy$ **; If** : $x\in\left(0,\frac{\pi}{2}\right)\Rightarrow y\in(0,1)$

But : $\sin(x)=\frac{2y}{1+y^2}$ **And** : $\cos(x)=\frac{1-y^2}{1+y^2}$

We obtain $I=\frac{1}{2}\int_0^1\frac{1}{\frac{2y}{1+y^2}+\frac{1-y^2}{1+y^2}}\times\frac{2}{1+y^2}dy$

$$=2\int_0^1\frac{1}{2-(y-1)^2}dy$$

$$=2\times\frac{1}{\sqrt{2}}\sin^{-1}\left(\frac{y-1}{\sqrt{2}}\right)\Bigg|_0^1=\sqrt{2}\sin^{-1}\left(\sin\left(\frac{\pi}{4}\right)\right)=\frac{\pi}{2\sqrt{2}}$$

SO, $\boxed{\int_0^{\frac{\pi}{2}}\frac{\sin^2(x)}{\sin(x)+\cos(x)}dx=\frac{\pi}{2\sqrt{2}}}$

241. Calculate Integral $I=\int_0^{\pi} x\cos^4(x)\sin^5(x)dx$

Solution; $I=\int_0^{\pi} x\cos^4(x)\sin^5(x)dx$

$$=\int_0^{\pi}(\pi-x)\cos^4(\pi-x)\sin^5(\pi-x)dx$$

$$=\int_0^{\pi}(\pi-x)\cos^4(x)\sin^5(x)dx$$

$$=\pi\int_0^{\pi}\cos^4(x)\sin^5(x)dx-\underbrace{\int_0^{\pi} x\cos^4(x)\sin^5(x)dx}_{I}$$

Implies That $I=\frac{\pi}{2}\int_0^{\pi}\cos^4(x)\sin^5(x)dx$

$$=\frac{\pi}{2}\times 2\int_0^{\frac{\pi}{2}}\cos^4(x)\sin^5(x)dx$$

Use : $\begin{cases}\int_0^{2a} f(x)dx=2\int_0^{a} f(x)dx\\ f(2a-x)=f(x)\end{cases}$

$$=\pi\int_0^{\frac{\pi}{2}}\cos^{2(\frac{5}{2})-1}(x)\sin^{2(3)-1}(x)dx$$

$$=\pi\int_0^{\frac{\pi}{2}}\cos^{2(\frac{5}{2})-1}(x)\sin^{2(3)-1}(x)dx$$

$$=\frac{\pi}{2}B\left(5/2,3\right)$$

$$=\frac{\pi}{2}\times\frac{\Gamma\left(5/2\right)\Gamma(3)}{\Gamma\left(3+5/2\right)}$$

$$=\frac{\pi\Gamma\left(5/2\right)}{\left(2+5/2\right)\left(1+5/2\right)\left(5/2\right)\Gamma\left(5/2\right)}$$

$$=\frac{8\pi}{315}$$

$SO,$ $\boxed{\int_0^{\pi} x\cos^4(x)\sin^5(x)dx=\frac{8\pi}{315}}$

242. Calculate Integral $I=\int_0^{\frac{\pi}{2}}\sin^2(x)\sin^2(2x)dx$

Solution; $I=\int_0^{\frac{\pi}{2}}\sin^2(x)\sin^2(2x)dx \quad (*)$

$$=\int_0^{\frac{\pi}{2}}\sin^2\left(\frac{\pi}{2}-x\right)\sin^2\left(\pi-2x\right)dx$$

$$=\int_0^{\frac{\pi}{2}}\cos^2(x)\sin^2(2x)dx \quad (**)$$

Take $(*)+(**)$ **Then :** $2I=\int_0^{\frac{\pi}{2}}\sin^2(x)\sin^2(2x)dx+\int_0^{\frac{\pi}{2}}\cos^2(x)\sin^2(2x)dx$

$$=\int_0^{\frac{\pi}{2}}\sin^2(2x)dx$$

$$=\frac{1}{2}\int_0^{\frac{\pi}{2}}\left(1-\cos(4x)\right)dx=\frac{\pi}{4}$$

$SO,\ \boxed{\int_0^{\frac{\pi}{2}}\sin^2(x)\sin^2(2x)dx=\frac{\pi}{4}}$

243. Calculate Integral $I=\int_1^e\left(\frac{\tan^{-1}(x)}{x}+\frac{\log(x)}{x^2+1}\right)dx$

Solution; $I=\int_1^e\left(\frac{\tan^{-1}(x)}{x}+\frac{\log(x)}{x^2+1}\right)dx$

$$=\int_1^e\left[\left(\log(x)\right)'\tan^{-1}(x)+\left(\tan^{-1}(x)\right)'\log(x)\right]dx$$

$$=\int_1^e\left[\left(\log(x)\tan^{-1}(x)\right)'\right]dx$$

$$=\log(x)\tan^{-1}(x)\Big|_1^e$$

$$=\tan^{-1}(e)=\tan^{-1}(e)$$

$SO,\ \boxed{\int_1^e\left(\frac{\tan^{-1}(x)}{x}+\frac{\log(x)}{x^2+1}\right)dx=\tan^{-1}(e)}$

244. Calculate Integral $I=\int_0^{\infty}\frac{1}{\sqrt{e^{2x}+e^{x}+1}}dx$

Solution; $I=\int_0^{\infty}\frac{1}{\sqrt{e^{2x}+e^{x}+1}}dx$

Let : $y=e^{x}\Rightarrow dx=\frac{1}{y}dy$ **; If :** $x\in(0,\infty)\Rightarrow y\in(1,\infty)$

We obtain $I=\int_1^{\infty}\frac{1}{y\sqrt{y^{2}+y+1}}dy$

Let : $t=\frac{1}{y}\Rightarrow dy=-\frac{1}{t^{2}}dt$ **; If :** $y\in(1,\infty)\Rightarrow t\in(1,0)$

We Get $I=-\int_1^{0}\frac{1}{\frac{1}{t}\sqrt{\frac{1}{t^{2}}+\frac{1}{t}+1}}\frac{1}{t^{2}}dt$

$$=\int_0^{1}\frac{1}{\sqrt{\left(t+\frac{1}{2}\right)^{2}+\frac{3}{4}}}dt$$

Let : $t+\frac{1}{2}=\sqrt{\frac{3}{4}}\tan(u)\Rightarrow dt=\sqrt{\frac{3}{4}}\sec^{2}(u)du$ **; If :** $t\in(0,1)\Rightarrow u\in\left(\frac{\pi}{6},\frac{\pi}{3}\right)$

Implies That $I=\sqrt{\frac{3}{4}}\int_{\frac{\pi}{6}}^{\frac{\pi}{3}}\frac{1}{\sqrt{\frac{3}{4}\left(\tan^{2}(u)+1\right)}}\sec^{2}(u)du$

$$=\int_{\frac{\pi}{6}}^{\frac{\pi}{3}}\frac{1}{\sqrt{\sec^{2}(u)}}\sec^{2}(u)du$$

$$=\log\left(\sec(u)+\tan(u)\right)\Big|_{\frac{\pi}{6}}^{\frac{\pi}{3}}$$

$$=\log\left(1+\frac{2\sqrt{3}}{3}\right)$$

SO, $\boxed{\int_0^{\infty}\frac{1}{\sqrt{e^{2x}+e^{x}+1}}dx=\log\left(1+\frac{2\sqrt{3}}{3}\right)}$

245. Calculate Integral $I=\int_0^{\infty}\log\left(\frac{e^x+1}{e^x-1}\right)dx$

Solution; $I=\int_0^{\infty}\log\left(\frac{e^x+1}{e^x-1}\right)dx$

$$=\int_0^{\infty}\log\left(\frac{1+e^{-x}}{1-e^{-x}}\right)dx$$

Let : $u=\log\left(\frac{1+e^{-x}}{1-e^{-x}}\right)\Rightarrow du=-\frac{2e^{-x}}{1-e^{-2x}}dx$ **; And** $dv=dx\Rightarrow v=x$

We obtain $I=\underbrace{x\log\left(\frac{1+e^{-x}}{1-e^{-x}}\right)\Big|_0^{\infty}}_{0}+\int_0^{\infty}\frac{2xe^{-x}}{1-e^{-2x}}dx$

$$=\int_0^{\infty}\left(2xe^{-x}\sum_{n=0}^{\infty}e^{-2nx}dx\right)$$

$$=2\sum_{n=0}^{\infty}\left(\int_0^{\infty}xe^{-(2n+1)x}dx\right)$$

Let : $z=(2n+1)x\equiv x=\frac{z}{(2n+1)}\Rightarrow dx=\frac{1}{(2n+1)}dz$ **; If :** $x\in(0,\infty)\Rightarrow z\in(0,\infty)$

Implies That $I=2\sum_{n=0}^{\infty}\left[\int_0^{\infty}\left(\frac{z}{(2n+1)}\right)e^{-z}\times\frac{1}{(2n+1)}dz\right]$

$$=2\sum_{n=0}^{\infty}\left(\frac{1}{(2n+1)^2}\int_0^{\infty}ze^{-z}dz\right)$$

$$=2\sum_{n=0}^{\infty}\left(\frac{1}{(2n+1)^2}\times\Gamma(2)\right)$$

$$=\frac{\pi^2}{4}$$

Note : $\sum_{n=0}^{\infty}\frac{1}{(2n+1)^2}=\frac{\pi^2}{8}$

SO, $\boxed{\int_0^{\infty}\log\left(\frac{e^x+1}{e^x-1}\right)dx=\frac{\pi^2}{4}}$

246. Calculate Integral $I=\int_0^1 \frac{\log(x)}{\sqrt{x}(x-1)}dx$

Solution; $I=\int_0^1 \frac{\log(x)}{\sqrt{x}(x-1)}dx$

Let : $x=t^2 \Rightarrow dx=2tdt$ **; If :** $x\in(0,1)\Rightarrow t\in(0,1)$

We Get $I=\int_0^1 \frac{\log(t^2)}{t(t^2-1)}\times 2tdt$

$$=-4\int_0^1 \frac{\log(t)}{(1-t^2)}dt$$

$$=\underbrace{-4\sum_{n=0}^{\infty}\left(\int_0^1 t^{2n}\log(t)dt\right)}_{(Use\ partial\ integral)}$$

$$=-4\sum_{n=0}^{\infty}\left(\underbrace{\frac{t^{2n+1}\log(t)}{2n+1}\Big|_0^1}_{0}-\int_0^1 \frac{t^{2n+1}}{2n+1}\times\frac{1}{t}dt\right)$$

$$=4\sum_{n=0}^{\infty}\left(\frac{t^{2n+1}\Big|_0^1}{(2n+1)^2}\right)$$

$$=4\sum_{n=0}^{\infty}\left(\frac{1}{(2n+1)^2}\right)$$

$$=4\left(\frac{1}{1^2}+\frac{1}{3^2}+\frac{1}{5^2}+\cdots\right)$$

$$=4\left[\left(1+\frac{1}{2^2}+\frac{1}{3^2}+\frac{1}{4^2}+\cdots\right)-\frac{1}{2^2}\left(1+\frac{1}{2^2}+\frac{1}{3^2}+\frac{1}{4^2}+\cdots\right)\right]$$

$$=4\left(\zeta(2)-\frac{1}{4}\zeta(2)\right)=\frac{\pi^2}{2}$$

SO, $\boxed{\int_0^1 \frac{\log(x)}{\sqrt{x}(x-1)}dx=\frac{\pi^2}{2}}$

247. Calculate Integral $I=\int_0^{\infty}\frac{\log(x)}{\sqrt{x}(x-1)}dx$

Solution; $I=\int_0^{\infty}\frac{\log(x)}{\sqrt{x}(x-1)}dx$

$$=\int_0^1\frac{\log(x)}{\sqrt{x}(x-1)}dx+\underbrace{\int_1^{\infty}\frac{\log(x)}{\sqrt{x}(x-1)}dx}_{I_1}$$

For : $I_1=\int_1^{\infty}\frac{\log(x)}{\sqrt{x}(x-1)}dx$

Let : $y=\frac{1}{x}\Rightarrow dx=-\frac{1}{y^2}dy$ **; If :** $x\in(1,\infty)\Rightarrow y\in(1,0)$

We obtain : $I_1=-\int_1^0\frac{\log\left(1/y\right)}{\sqrt{1/y}\left(1/y-1\right)}\cdot\frac{1}{y^2}dy$

$$=\int_0^1\frac{\log(y)}{\sqrt{y}(y-1)}dy$$

$$=\int_0^1\frac{\log(x)}{\sqrt{x}(x-1)}dx$$

Implies That : $I=\int_0^1\frac{\log(x)}{\sqrt{x}(x-1)}dx+\int_0^1\frac{\log(x)}{\sqrt{x}(x-1)}dx$

$$=2\int_0^1\frac{\log(x)}{\sqrt{x}(x-1)}dx$$

$$=2\left(\frac{\pi^2}{2}\right)=\pi^2$$

SO, $\boxed{\int_0^{\infty}\frac{\log(x)}{\sqrt{x}(x-1)}dx=\pi^2}$

248. Calculate Integral $I=\int_0^{\infty}\frac{\log(x)}{x^2+y^2}dx$

Solution; $I=\int_0^{\infty}\frac{\log(x)}{x^2+y^2}dx$

Let : $x=yz\Rightarrow dx=ydz$ **; If :** $x\in(0,\infty)\Rightarrow z\in(0,\infty)$

We Get $I=\int_0^{\infty}\frac{\log(yz)}{y^2z^2+y^2}ydz$

$$=\frac{1}{y}\int_0^{\infty}\frac{\log(y)+\log(z)}{z^2+1}dz$$

$$=\frac{1}{y}\underbrace{\int_0^{\infty}\frac{\log(y)}{z^2+1}dz}_{J}+\frac{1}{y}\underbrace{\int_0^{\infty}\frac{\log(z)}{z^2+1}dz}_{k}\quad(*)$$

For : $J=\frac{1}{y}\int_0^{\infty}\frac{\log(y)}{z^2+1}dz$

$$=\frac{\log(y)}{y}\tan^{-1}(x)\Big|_0^{\infty}=\frac{\pi}{2y}\log(y)$$

For : $K=\frac{1}{y}\int_0^{\infty}\frac{\log(z)}{z^2+1}dz$

$$=\frac{1}{y}\int_0^{1}\frac{\log(z)}{z^2+1}dz+\frac{1}{y}\underbrace{\int_1^{\infty}\frac{\log(z)}{z^2+1}dz}_{L}$$

For : $L=\int_1^{\infty}\frac{\log(z)}{z^2+1}dz$

Let : $z=\frac{1}{t}\Rightarrow dz=-\frac{1}{t^2}dt$ **; If :** $z\in(1,\infty)\Rightarrow z\in(1,0)$

Implies That : $L=-\int_1^{0}\frac{\log(t^{-1})}{t^{-2}+1}\times\frac{1}{t^2}dt=\int_0^{1}\frac{\log(t)}{t^2+1}dt$

We obtain : $K=\frac{1}{y}\int_0^{1}\frac{\log(z)}{z^2+1}dz-\frac{1}{y}\int_0^{1}\frac{\log(t)}{t^2+1}dt=0$

Take $(*)$ **When** : $I=\frac{\pi}{2y}\log(y)+\frac{1}{y}\times 0=\frac{\pi}{2y}\log(y)$

$SO,\ \boxed{\int_0^{\infty}\frac{\log(x)}{x^2+y^2}dx=\frac{\pi}{2y}\log(y)}$

249. Calculate Integral $I=\int_0^1\frac{\log(x)}{\sqrt{1-x^2}}dx$

Solution; $I=\int_0^1\frac{\log(x)}{\sqrt{1-x^2}}dx$

Let : $x=\sin(u)\Rightarrow dx=\cos(u)du$ **; If** : $x\in(0,1)\Rightarrow z\in\left(0,\frac{\pi}{2}\right)$

We obtain $I=\int_0^{\frac{\pi}{2}}\frac{\log(\sin(u))}{\sqrt{1-\sin^2(u)}}\times\cos(u)du$

$$=\int_0^{\frac{\pi}{2}}\frac{\log(\sin(u))}{|\cos(u)|}\times\cos(u)du$$

$$=\int_0^{\frac{\pi}{2}}\frac{\log(\sin(u))}{\cos(u)}\times\cos(u)du$$

$$=-\frac{\pi}{2}\log(2)$$

$SO,\ \boxed{\int_0^1\frac{\log(x)}{\sqrt{1-x^2}}dx=-\frac{\pi}{2}\log(2)}$

250. Calculate Integral $I=\int_1^{\infty}\frac{\log^3(x)}{x^2(x-1)}dx$

Solution; $I=\int_1^{\infty}\frac{\log^3(x)}{x^2(x-1)}dx$

Let : $t=\log(x)\Leftrightarrow x=e^t\Rightarrow dx=e^tdt$ **; If** : $x\in(1,\infty)\Rightarrow t\in(0,\infty)$

Implies That $I=\int_0^{\infty}\frac{t^3}{e^{2t}(e^t-1)}\times e^tdt$

$$=\int_0^{\infty}\frac{t^3e^{-2t}}{(1-e^{-t})}dt$$

$$= \int_0^{\infty} \left(t^3 e^{-2t} \sum_{n=0}^{\infty} e^{-nt} \right) dt$$

$$= \sum_{n=0}^{\infty} \left(\int_0^{\infty} t^3 e^{-(n+2)t} dt \right)$$

Let : $y = (n+2)t \Leftrightarrow t = \dfrac{y}{(n+2)} \Rightarrow dt = \dfrac{1}{(n+2)} dy$ **; If :** $t \in (0,\infty) \Rightarrow y \in (0,\infty)$

We obtain $I = \sum_{n=0}^{\infty} \left(\int_0^{\infty} \left(\dfrac{y}{n+2} \right)^3 e^{-y} \times \dfrac{1}{(n+2)} dy \right)$

$$= \sum_{n=0}^{\infty} \left(\frac{1}{(n+2)^4} \int_0^{\infty} (y)^3 e^{-y} dy \right)$$

$$= \sum_{n=0}^{\infty} \left(\frac{\Gamma(4)}{(n+2)^4} \right)$$

$$= 6 \left[\sum_{n=0}^{\infty} \left(\frac{1}{(n+1)^4} \right) - 1 \right] = 6(\zeta(4) - 1)$$

SO, $\boxed{\int_1^{\infty} \dfrac{\log^3(x)}{x^2(x-1)} dx = 6(\zeta(4)-1)}$

251. Calculate Integral $I = \int_{-1}^{1} \dfrac{x^4 - 1}{x(x^4-5)(x^5-5x+1)} dx$

Solution; $I = \int_{-1}^{1} \dfrac{x^4-1}{x(x^4-5)(x^5-5x+1)} dx$

$$= \frac{1}{5} \int_{-1}^{1} \frac{5(x^4-1)}{(x^5-5x)(x^5-5x+1)} dx$$

Let : $x^5 - 5x = t \Rightarrow 5(x^4-1)dx = dt$ **; If :** $x \in (-1,1) \Rightarrow t \in (4,-4)$

Implies That $I = \dfrac{1}{5} \int_4^{-4} \dfrac{1}{t(t+1)} dx$

$$= \frac{1}{5} \log \left| \frac{t}{t+1} \right|_4^{-4} = \frac{1}{5} \log \left(\frac{5}{3} \right)$$

SO, $\boxed{\int_{-1}^{1} \dfrac{x^4-1}{x(x^4-5)(x^5-5x+1)} dx = \dfrac{1}{5} \log \left(\dfrac{5}{3} \right)}$

252. Calculate Integral $I=\int_0^{\frac{\pi}{4}}\sec^2(x)\tan^{-1}\left(\frac{\sqrt{\cos(2x)}}{\sin(x)}\right)dx$

Solution; $I=\int_0^{\frac{\pi}{4}}\sec^2(x)\tan^{-1}\left(\frac{\sqrt{\cos(2x)}}{\sin(x)}\right)dx$

$$=\int_0^{\frac{\pi}{4}}\sec^2(x)\tan^{-1}\left(\frac{\sqrt{\cos^2(x)-\sin^2(x)}}{\sin(x)}\right)dx$$

$$=\int_0^{\frac{\pi}{4}}\sec^2(x)\tan^{-1}\left(\frac{\sqrt{1-\tan^2(x)}}{\tan(x)}\right)dx$$

Let : $\cos(u)=\tan(x)\Rightarrow\sec^2(x)dx=-\sin(u)du$ **; If** : $x\in\left(0,\frac{\pi}{4}\right)\Rightarrow u\in\left(\frac{\pi}{2},0\right)$

We obtain $I=-\int_{\frac{\pi}{2}}^0\tan^{-1}\left(\frac{\sqrt{1-\cos^2(u)}}{\cos(u)}\right)\sin(u)du$

$$=\int_0^{\frac{\pi}{2}}\tan^{-1}\left(\frac{|\sin(u)|}{\cos(u)}\right)\sin(u)du$$

$$=\int_0^{\frac{\pi}{2}}\tan^{-1}\big(\tan(u)\big)\sin(u)du$$

$$=\underbrace{\int_0^{\frac{\pi}{2}}u\sin(u)du}_{Use\ partial\ integral}$$ **Use Partial Integral**

$$=\big(-u\cos(u)+\sin(x)\big)\Big|_0^{\frac{\pi}{2}}$$

$$=1$$

SO, $\boxed{\int_0^{\frac{\pi}{4}}\sec^2(x)\tan^{-1}\left(\frac{\sqrt{\cos(2x)}}{\sin(x)}\right)dx=1}$

253. Calculate Integral $I=\int_{\frac{1}{2}}^{2}\sqrt{\log^2(x)}dx$

Solution; $I=\int_{\frac{1}{2}}^{2}\sqrt{\log^2(x)}dx$

$$=\int_{\frac{1}{2}}^{2}\left|\log(x)\right|dx$$

$$=\int_{\frac{1}{2}}^{1}\left|\log(x)\right|dx+\int_{1}^{2}\left|\log(x)\right|dx$$

$$=-\int_{\frac{1}{2}}^{1}\log(x)dx+\int_{1}^{2}\log(x)dx$$

$$=-x\big(\log(x)-1\big)\Big|_{\frac{1}{2}}^{1}+x\big(\log(x)-1\big)\Big|_{1}^{2}$$

$$=\frac{1}{2}\big(\log(2)-1\big)$$

SO, $\boxed{\int_{\frac{1}{2}}^{2}\sqrt{\log^2(x)}dx=\frac{1}{2}\big(\log(2)-1\big)}$

254. Calculate Integral $I=\int_{1}^{\infty}\frac{1}{\sqrt{x}(1+x)}dx$

Solution; $I=\int_{1}^{\infty}\frac{1}{\sqrt{x}(1+x)}dx$

Let : $x=u^2\Rightarrow dx=2udu$ **; If :** $x\in(1,\infty)\Rightarrow u\in(1,\infty)$

We obtain $I=2\int_{1}^{\infty}\frac{u}{\sqrt{u^2}(1+u^2)}du$

$$=2\int_{1}^{\infty}\frac{1}{(1+u^2)}du$$

$$=\frac{\pi}{2}$$

SO, $\boxed{\int_{1}^{\infty}\frac{1}{\sqrt{x}(1+x)}dx=\frac{\pi}{2}}$

255. Calculate Integral $I=\int_0^{\infty}\frac{x^2e^{-\frac{x}{2}}}{e^x-1}dx$

Solution; $I=\int_0^{\infty}\frac{x^2e^{-\frac{x}{2}}}{e^x-1}dx$

Let : $x=2y\Rightarrow dx=2dy$ **; If :** $x\in(0,\infty)\Rightarrow y\in(0,\infty)$

We obtain $I=2\int_0^{\infty}\frac{(2y)^2e^{-y}}{e^{2y}-1}dy$

$$=8\int_0^{\infty}\frac{y^2e^{-3y}}{1-e^{-2y}}dy$$

$$=8\int_0^{\infty}\left(y^2e^{-3y}\sum_{n=0}^{\infty}e^{-2ny}\right)dy$$

$$=8\sum_{n=0}^{\infty}\left(\int_0^{\infty}y^2e^{-(2n+3)y}\,dy\right)$$

Let : $y=\frac{z}{(2n+3)}\Rightarrow dy=\frac{1}{(2n+3)}dz$ **; If :** $y\in(0,\infty)\Rightarrow z\in(0,\infty)$

We obtain $I=8\sum_{n=0}^{\infty}\left(\int_0^{\infty}\left(\frac{z}{(2n+3)}\right)^2\frac{e^{-z}}{(2n+3)}dz\right)$

$$=8\sum_{n=0}^{\infty}\left(\frac{1}{(2n+3)^3}\int_0^{\infty}z^2e^{-z}dz\right)$$

$$=16\sum_{n=0}^{\infty}\left(\frac{1}{(2n+3)^3}\right)$$

$$=-16+16\sum_{n=0}^{\infty}\left(\frac{1}{(2n+1)^3}\right)$$

$$=-16+16\left(1+\frac{1}{5^3}+\frac{1}{7^3}+\ldots\right)=-16+14\zeta(3)$$

$SO,$ $\boxed{\int_0^{\infty}\frac{x^2e^{-\frac{x}{2}}}{e^x-1}dx=14\zeta(3)-16}$

256. Calculate Integral $I=\int_0^4\binom{x}{5}^{-1}dx$

Solution; $I=\int_0^4\binom{x}{5}^{-1}dx$

By : $\binom{x}{5}=\frac{x!}{(x-5).5!}$ **When :** $\binom{x}{5}^{-1}=\frac{(x-5).5!}{x!}=\frac{5!}{x(x-1)(x-2)(x-3)(x-4)}$

We obtain $I=\int_0^4\frac{5!}{x(x-1)(x-2)(x-3)(x-4)}dx$

Let : $t=x-2\Rightarrow dx=dt$ **; If :** $x\in(0,4)\Rightarrow t\in(-2,2)$

We Get $I=\int_{-2}^{2}\frac{5!}{(t+2)(t+1)(t)(t-1)(t-2)}dt$

$$=\underbrace{\int_{-2}^{2}\frac{5!}{(t)(t^2-1)(t^2-4)}dt}_{is\,an\,odd\,function}=0$$

SO, $\boxed{\int_0^4\binom{x}{5}^{-1}dx=0}$

257. Calculate Integral $I=\int_0^1 x^2(x-1)^3dx$

Solution; $I=\int_0^1 x^2(x-1)^3dx$

$$=-\int_0^1 x^{3-1}(1-x)^{4-1}dx$$
$$=-B(3,4)$$
$$=-\frac{1}{60}$$

SO, $\boxed{\int_0^1 x^2(x-1)^3dx=-\frac{1}{60}}$

258. Calculate Integral $I=\int_{-\pi}^{2\pi}\left(\tan^{-1}(x)+\tan^{-1}\left(\frac{1}{x}\right)\right)dx$

Solution; $I=\int_{-\pi}^{2\pi}\left(\tan^{-1}(x)+\tan^{-1}\left(\frac{1}{x}\right)\right)dx$

$$=\int_{-\pi}^{0}\left(\tan^{-1}(x)+\tan^{-1}\left(\frac{1}{x}\right)\right)dx+\int_{0}^{2\pi}\left(\tan^{-1}(x)+\tan^{-1}\left(\frac{1}{x}\right)\right)dx$$

$$=\int_{-\pi}^{0}\left(-\frac{\pi}{2}\right)dx+\int_{0}^{2\pi}\left(\frac{\pi}{2}\right)dx$$

$$=\frac{\pi^2}{2}$$

Note: $\tan^{-1}(x)+\cot^{-1}(x)=\tan^{-1}(x)+\tan^{-1}\left(\frac{1}{x}\right)=\begin{cases}\frac{\pi}{2} & ,x>0\\ -\frac{\pi}{2} & ,x<0\end{cases}$

SO, $\boxed{\int_{-\pi}^{2\pi}\left(\tan^{-1}(x)+\tan^{-1}\left(\frac{1}{x}\right)\right)=\frac{\pi^2}{2}}$

259. Calculate Integral $I=\int_{1}^{\int_{1}^{\cdots^{(2x)dx}}(2x)dx}(2x)dx$

Solution; $I=\int_{1}^{\int_{1}^{\cdots^{(2x)dx}}(2x)dx}(2x)dx$

$$=\int_{1}^{I}(2x)dx$$

$$=x^2\Big|_1^I=I^2-1$$

Equivalent $I=I^2-1$ **That** $I=\frac{1\pm\sqrt{5}}{2}$

We Get $I=\frac{1+\sqrt{5}}{2}$

SO, $\boxed{\int_{1}^{\int_{1}^{\cdots^{(2x)dx}}(2x)dx}(2x)dx=\frac{1+\sqrt{5}}{2}}$

260. Calculate Integral $I=\int_0^{\frac{\pi}{4}}\frac{\sin(x)+\cos(x)}{9+16\sin(2x)}dx$

Solution; $I=\int_0^{\frac{\pi}{4}}\frac{\sin(x)+\cos(x)}{9+16\sin(2x)}dx$

$$=\int_0^{\frac{\pi}{4}}\frac{\sin(x)+\cos(x)}{9+16\left[1-\left(\sin(x)-\cos(x)\right)^2\right]}dx$$

Let : $y=\sin(x)-\cos(x)\Rightarrow dy=\left(\sin(x)+\cos(x)\right)dx$ **; If :** $x\in\left(0,\frac{\pi}{4}\right)\Rightarrow y\in(-1,0)$

We obtain $I=\int_{-1}^{0}\frac{1}{9+16\left(1-y^2\right)}dx$

$$=\int_{-1}^{0}\frac{1}{25-16y^2}dx$$

$$=\frac{1}{16}\int_{-1}^{0}\frac{1}{(5/4)^2-y^2}dx$$

$$=\frac{1}{40}\log\left(\frac{5+4y}{5-4y}\right)\Bigg|_{-1}^{0}$$

$$=\frac{3}{40}\log(3)$$

SO, $\boxed{\int_0^{\frac{\pi}{4}}\frac{\sin(x)+\cos(x)}{9+16\sin(2x)}dx=\frac{3}{40}\log(3)}$

261. Calculate Integral $I=\int_0^{\infty}\frac{\log(x)}{x^2+2x+4}dx$

Solution; $I=\int_0^{\infty}\frac{\log(x)}{x^2+2x+4}dx$

Let : $x=\frac{4}{u}\Rightarrow dx=-\frac{4}{u^2}du$ **; If :** $x\in(0,\infty)\Rightarrow u\in(\infty,0)$

Implies That $I=-\int_{\infty}^{0}\frac{\log\left(4/u\right)}{\left(4/u\right)^2+2\left(4/u\right)+4}\times\frac{4}{u^2}du$

$$= \int_0^{\infty} \frac{\log(4) - \log(u)}{u^2 + 2u + 4} du$$

$$= \int_0^{\infty} \frac{\log(4)}{u^2 + 2u + 4} du - \int_0^{\infty} \frac{\log(u)}{u^2 + 2u + 4} du$$

We obtain $I = \frac{\log(4)}{2} \int_0^{\infty} \frac{1}{u^2 + 2u + 4} du$

$$= \log(2) \int_0^{\infty} \frac{1}{(u+1)^2 + 3} du$$

$$= \frac{\pi \log(2)}{3\sqrt{3}}$$

SO, $\boxed{\int_0^{\infty} \frac{\log(x)}{x^2 + 2x + 4} dx = \frac{\pi \log(2)}{3\sqrt{3}}}$

262. Calculate Integral $I = \int_{-\pi}^{\pi} \frac{x\left(\sin(x) + 1\right)}{\cos^2(x) + 1} dx$

Solution; $I = \int_{-\pi}^{\pi} \frac{x\left(\sin(x) + 1\right)}{\cos^2(x) + 1} dx$

$$= \underbrace{\int_{-\pi}^{\pi} \frac{x \sin(x)}{\cos^2(x) + 1} dx}_{is\ an\ even\ function} + \underbrace{\int_{-\pi}^{\pi} \frac{x}{\cos^2(x) + 1} dx}_{is\ an\ odd\ function}$$

$$= 2\int_0^{\pi} \frac{x \sin(x)}{\cos^2(x) + 1} dx$$

$$= 2\int_0^{\pi} \frac{(\pi - x)\sin(\pi - x)}{\cos^2(\pi - x) + 1} dx$$

$$= 2\int_0^{\pi} \frac{\pi \sin(x)}{\cos^2(x) + 1} dx - 2\int_0^{\pi} \frac{x \sin(x)}{\cos^2(x) + 1} dx$$

We Get $I = -\pi \int_0^{\pi} \frac{1}{\cos^2(x) + 1} d\left(\cos(x)\right) = -\pi\left(-\frac{\pi}{2}\right) = \frac{\pi^2}{2}$

SO, $\boxed{\int_{-\pi}^{\pi} \frac{2x\left(\sin(x) + 1\right)}{\cos^2(x) + 1} dx = \frac{\pi^2}{2}}$

263. Calculate Integral $I=\int_{-1}^{1}\frac{1}{8^x+1}dx$

Solution; $I=\int_{-1}^{1}\frac{1}{8^x+1}dx$

$$=\int_{-1}^{1}\left(1-\frac{8^x}{8^x+1}\right)dx$$

$$=2-\frac{1}{\log(8)}\int_{-1}^{1}\left(\frac{\left(8^x+1\right)'}{8^x+1}\right)dx$$

$$=2-\frac{1}{\log(8)}\log\left(8^x+1\right)\Big|_{-1}^{1}$$

$$=2-1=1$$

$SO,$ $\boxed{\int_{-1}^{1}\frac{1}{8^x+1}dx=1}$

264. Calculate Integral $I=\int_{0}^{1}\sqrt[2025]{x}\left(1-\sqrt[2025]{x}\right)^{2026}dx$

Solution; $I=\int_{0}^{1}\sqrt[2025]{x}\left(1-\sqrt[2025]{x}\right)^{2026}dx$

Let : $x=u^{2025}\Rightarrow dx=2025u^{2024}du$ **; If** : $x\in(0,1)\Rightarrow u\in(0,1)$

Implies That $I=2025\int_{0}^{1}u\left(1-u\right)^{2026}u^{2024}du$

$$=2025\int_{0}^{1}u^{2026-1}\left(1-u\right)^{2027-1}du$$

$$=2025B\left(2026,2027\right)$$

$$=\frac{2025}{2026}\cdot\frac{(2026!)^2}{4052!}$$

$SO,$ $\boxed{\int_{0}^{1}\sqrt[2025]{x}\left(1-\sqrt[2025]{x}\right)^{2026}dx=\frac{2025}{2026}\cdot\frac{(2026!)^2}{4052!}}$

265. Calculate Integral $I=\int_1^e \frac{1-\log(x)}{x^2}dx$

Method 1

Solution; $I=\int_1^e \frac{1-\log(x)}{x^2}dx$

$$=\int_1^e \frac{x\left(\log(x)\right)'-(x)'\log(x)}{x^2}dx$$

Use : $\left(\frac{u}{v}\right)'=\frac{u'v-v'u}{v^2}$

We obtain $I=\int_1^e \left(\frac{\log(x)}{x}\right)'dx$

$$=\left(\frac{\log(x)}{x}\right)\Big|_1^e=e^{-1}$$

Method 2

Solution; $I=\int_1^e \frac{1-\log(x)}{x^2}dx$

$$=\int_1^e \frac{1+\log\left(\frac{1}{x}\right)}{x^2}dx$$

Let : $u=\frac{1}{x}\Rightarrow -du=\frac{1}{x^2}dx$**; If :** $x\in(e,1)\Rightarrow u\in(e^{-1},0)$

Implies That $I=-\int_0^{e^{-1}}\left(1+\log(u)\right)du$

$$=-\int_0^{e^{-1}}\left(x\left(\log(x)\right)'+(x)'\log(x)\right)du$$

Use: $(uv)'=u'v+v'u$

$$=-\left(x\log(x)\right)\Big|_0^{e^{-1}}$$

$$=e^{-1}$$

SO, $\boxed{\int_1^e \frac{1-\log(x)}{x^2}dx=e^{-1}}$

266. Calculate Integral $I=\int_{-1}^{1}\log\left(x+\sqrt{1+x^2}\right)dx$

Solution; $I=\int_{-1}^{1}\log\left(x+\sqrt{1+x^2}\right)dx \quad (*)$

$$=\int_{-1}^{1}\log\left(\sqrt{1+x^2}-x\right)dx \quad (**)$$

Use : $\int_{-a}^{a}f(x)dx=\int_{-a}^{a}f(-x)dx$

Take $(*)+(**)$ **Then :** $2I=\int_{-1}^{1}\log\left(x+\sqrt{1+x^2}\right)dx+\int_{-1}^{1}\log\left(\sqrt{1+x^2}-x\right)dx$

$$=\int_{-1}^{1}\log\left[\left(x+\sqrt{1+x^2}\right)\left(\sqrt{1+x^2}-x\right)\right]dx$$

$$=\int_{-1}^{1}\log(1)\,dx=0$$

Implies That $I=0$

SO, $\boxed{\int_{-1}^{1}\log\left(x+\sqrt{1+x^2}\right)dx=0}$

267. Calculate Integral $I=\int_{0}^{\pi}\left(1+\sin(x)\right)^2\sin^5(x)\,dx$

Solution; $I=\int_{0}^{\pi}\left(1+\sin(x)\right)^2\sin^5(x)\,dx$

$$=\int_{0}^{\frac{\pi}{2}}\left[\left(1+\sin(x)\right)^2\sin^5(x)+\left(1+\sin(\pi-x)\right)^2\sin^5(\pi-x)\right]dx$$

$$=2\int_{0}^{\frac{\pi}{2}}\left(1+\sin(x)\right)^2\sin^5(x)\,dx$$

$$=2\int_{0}^{\frac{\pi}{2}}\left(\sin^5(x)+2\sin^6(x)+\sin^7(x)\right)dx$$

$$=2\left(\frac{4!!}{5!!}+2\frac{\pi}{2}\times\frac{5!!}{6!!}+\frac{6!!}{7!!}\right)$$

$$=\frac{208}{105}+\frac{5\pi}{16}$$

SO, $\boxed{\int_{0}^{\pi}\left(1+\sin(x)\right)^2\sin^5(x)\,dx=\frac{5\pi}{16}+\frac{208}{105}}$

268. Calculate Integral $I=\int_0^1\sqrt{1-x^2}\,dx$

Solution; $I=\int_0^1\sqrt{1-x^2}\,dx$

Let : $x=t^{\frac{1}{2}}\Rightarrow dx=\frac{1}{2}t^{\frac{1}{2}-1}dt$ **; If :** $x\in(0,1)\Rightarrow t\in(0,1)$

Implies That $I=\frac{1}{2}\int_0^1\sqrt{1-t}\times t^{\frac{1}{2}-1}dt$

$$=\frac{1}{2}\int_0^1 t^{\frac{1}{2}-1}(1-t)^{\left(\frac{1}{2}+1\right)-1}dt$$

$$=\frac{1}{2}B\left(\frac{1}{2},1+\frac{1}{2}\right)$$

$$=\frac{1}{2}\times\frac{\Gamma\left(1/2\right)\Gamma\left(1+1/2\right)}{\Gamma\left(1/2+1+1/2\right)}$$

$$=\frac{1}{2}\times\frac{\frac{1}{2}\Gamma^2\left(1/2\right)}{\Gamma(2)}=\frac{\pi}{4}\qquad;\Gamma\left(\frac{1}{2}\right)=\sqrt{\pi}$$

SO, $\boxed{\int_0^1\sqrt{1-x^2}\,dx=\frac{\pi}{4}}$

269. Calculate Integral $I=\int_0^1 x^6(1-x^4)^{\frac{1}{4}}dx$

Solution; $I=\int_0^1 x^6(1-x^4)^{\frac{1}{4}}dx$

Let : $x=u^{\frac{1}{4}}\Rightarrow dx=\frac{1}{4}u^{\frac{1}{4}-1}du$ **; If :** $x\in(0,1)\Rightarrow u\in(0,1)$

We obtain $I=\frac{1}{4}\int_0^1 u^{\frac{6}{4}}(1-u)^{\frac{1}{4}}\times u^{\frac{1}{4}-1}dx$

$$=\frac{1}{4}\int_0^1 u^{\frac{7}{4}-1}(1-u)^{\frac{5}{4}-1}dx$$

$$=\frac{1}{4}B\left(\frac{7}{4},\frac{5}{4}\right)$$

$$=\frac{1}{4}\times\frac{\Gamma\left(1+1-\frac{1}{4}\right)\Gamma\left(1+\frac{1}{4}\right)}{\Gamma(3)}$$

$$=\frac{1}{8}\times\frac{3}{4}\Gamma\left(1-\frac{1}{4}\right)\frac{1}{4}\Gamma\left(\frac{1}{4}\right)$$

$$=\frac{3}{128}\times\pi\csc\left(\frac{\pi}{4}\right)=\frac{3\pi}{64\sqrt{2}}$$

$SO,$ $\boxed{\int_0^1 x^6(1-x^4)^{\frac{1}{4}}dx=\frac{3\pi}{64\sqrt{2}}}$

270. Calculate Integral $I=\int_0^\infty \frac{x}{x^8+2x^4+1}dx$

Solution; $I=\int_0^\infty \frac{x}{x^8+2x^4+1}dx$

$$=\frac{1}{2}\int_0^\infty \frac{1}{(x^4+1)^2}d(x^2)$$

Let : $x^2=\sqrt{u}\Rightarrow d(x^2)=\frac{1}{2\sqrt{u}}du$; **If** : $x\in(0,\infty)\Rightarrow u\in(0,\infty)$

Implies That $I=\frac{1}{4}\int_0^\infty \frac{1}{(u+1)^2\sqrt{u}}du$

$$=\frac{1}{4}\int_0^\infty \frac{u^{\frac{1}{2}-1}}{(u+1)^{\frac{1}{2}+\frac{3}{2}}}du$$

$$=\frac{1}{4}B\left(\frac{1}{2},\frac{3}{2}\right)=\frac{\pi}{8}$$

$SO,$ $\boxed{\int_0^\infty \frac{x}{x^8+2x^4+1}dx=\frac{\pi}{8}}$

271. Calculate Integral $I=\int_1^e \log^{-1}(x)\,dx$

Solution; $I=\int_1^e \log^{-1}(x)\,dx$

Let : $x=e^u \Rightarrow dx=e^u du$ **; If :** $x\in(e,1)\Rightarrow u\in(1,0)$

Implies That $I=\int_0^1 u^{-1}e^u\,du$

$$=\int_0^1\left(u^{-1}\sum_{n=0}^{\infty}\frac{u^n}{n!}\right)du$$

$$=\sum_{n=0}^{\infty}\left(\frac{1}{n!}\int_0^1 u^{n-1}\,du\right)=\sum_{n=0}^{\infty}\left(\frac{1}{n\times n!}\right)$$

SO, $\boxed{\int_1^e \log^{-1}(x)\,dx=\sum_{n=0}^{\infty}\left(\frac{1}{n\times n!}\right)}$

272. Calculate Integral $I=\int_0^{\infty}\frac{x^{\pi}}{\pi^x}dx$

Solution; $I=\int_0^{\infty}\frac{x^{\pi}}{\pi^x}dx$

Let : $\pi^x=e^y\equiv x=\frac{y}{\log(\pi)}\Rightarrow dx=\frac{1}{\log(\pi)}dy$ **;** **If :** $x\in(0,\infty)\Rightarrow y\in(0,\infty)$

We obtain $I=\int_0^{\infty}\frac{\left(\frac{y}{\log(\pi)}\right)^{\pi}}{e^y}\times\frac{1}{\log(\pi)}dy$

$$=\frac{1}{\log^{\pi+1}(\pi)}\int_0^{\infty}y^{\pi}e^{-x}\,dy$$

$$=\frac{\Gamma(\pi+1)}{\log^{\pi+1}(\pi)}$$

SO, $\boxed{\int_0^{\infty}\frac{x^{\pi}}{\pi^x}dx=\frac{\Gamma(\pi+1)}{\log^{\pi+1}(\pi)}}$

273. Calculate Integral $I=\int_0^{\pi}\frac{xe^x}{(x+\pi)^{\pi+1}}dx$

Solution; $I=\int_0^{\pi}\frac{xe^x}{(x+\pi)^{\pi+1}}dx$

$$=\int_0^{\pi}e^x\frac{(x+\pi)-\pi}{(x+\pi)^{\pi+1}}dx$$

$$=\int_0^{\pi}e^x\frac{(x+\pi)-\pi}{(x+\pi)^{\pi+1}}\times\frac{(x+\pi)^{\pi-1}}{(x+\pi)^{\pi-1}}dx$$

$$=\int_0^{\pi}e^x\frac{(x+\pi)^{\pi}-\pi(x+\pi)^{\pi-1}}{(x+\pi)^{2\pi}}dx$$

$$=\int_0^{\pi}\left(\frac{(e^x)'(x+\pi)^{\pi}-\left((x+\pi)^{\pi}\right)'e^x}{(x+\pi)^{2\pi}}\right)dx$$

$$=\int_0^{\pi}\left(\frac{e^x}{(x+\pi)^{\pi}}\right)'dx$$

$$=\frac{e^x}{(x+\pi)^{\pi}}\Bigg|_0^{\pi}=\frac{e^{\pi}-2^{\pi}}{(2\pi)^{\pi}}$$

$SO,$ $\boxed{\int_0^{\pi}\frac{xe^x}{(x+\pi)^{\pi+1}}dx=\frac{e^{\pi}-2^{\pi}}{(2\pi)^{\pi}}}$

274. Calculate Integral $I=\int_0^{a}x^4\sqrt{a^2-x^2}dx$

Solution; $I=\int_0^{a}x^4\sqrt{a^2-x^2}dx$

$$=\int_0^{a}x^4\left(a^2-x^2\right)^{\frac{1}{2}}dx$$

Let : $x=ay^{\frac{1}{2}}\Rightarrow dx=\frac{a}{2}y^{\frac{1}{2}-1}dy$ **; If :** $x\in(0,a)\Rightarrow y\in(0,1)$

We obtain $I=\frac{a}{2}\int_0^1\left(ay^{\frac{1}{2}}\right)^4\left(a^2-a^2y\right)^{\frac{1}{2}}y^{\frac{1}{2}-1}dy$

$$= \frac{a^6}{2}\int_0^1 y^2.y^{\frac{1}{2}-1}(1-y)^{\frac{1}{2}}dy$$

$$= \frac{a^6}{2}\int_0^1 y^{\left(2+\frac{1}{2}\right)-1}(1-y)^{\left(1+\frac{1}{2}\right)-1}dy$$

$$= \frac{a^6}{2}B\left(2+\frac{1}{2},1+\frac{1}{2}\right)$$

$$= \frac{a^6}{2}\times\frac{\Gamma\left(2+\frac{1}{2}\right)\Gamma\left(1+\frac{1}{2}\right)}{\Gamma\left(2+\frac{1}{2}+1+\frac{1}{2}\right)} = \frac{\pi a^6}{32}$$

SO, $\boxed{\int_0^a x^4\sqrt{a^2-x^2}\,dx = \frac{\pi a^6}{32}}$

275. Calculate Integral $I = \int_0^1 (1-x^2)^4\,dx$

Solution; $I = \int_0^1 (1-x^2)^4\,dx$

Let : $x = y^{\frac{1}{2}} \Rightarrow dx = \frac{1}{2}y^{\frac{1}{2}-1}\,dy$ **; If :** $x \in (0,1) \Rightarrow y \in (0,1)$

We obtain $I = \frac{1}{2}\int_0^1 (1-y)^4.y^{\frac{1}{2}-1}\,dy$

$$= \frac{1}{2}\int_0^1 y^{\frac{1}{2}-1}(1-y)^{3-1}dy$$

$$= \frac{1}{2}B\left(\frac{1}{2},3\right) = \frac{1}{2}\times\frac{\Gamma(3)\Gamma\left(1/2\right)}{\Gamma\left(3+1/2\right)}$$

$$= \frac{8}{(4+1)(2+1)} = \frac{8}{15}$$

SO, $\boxed{\int_0^1 (1-x^2)^4\,dx = \frac{8}{15}}$

276. Calculate Integral $I=\int_0^1 \frac{\sqrt{1-x}}{x\sqrt{x}}dx$

Solution; $I=\int_0^1 \frac{\sqrt{1-x}}{x\sqrt{x}}dx$

$$=\int_0^1 x^{-\frac{3}{2}}(1-x)^{\frac{1}{2}}dx$$

$$=\int_0^1 x^{-\frac{1}{2}-1}(1-x)^{\frac{3}{2}-1}dx$$

$$=\frac{\Gamma\left(-\frac{1}{2}\right)\Gamma\left(\frac{3}{2}\right)}{\Gamma\left(\frac{3}{2}-\frac{1}{2}\right)}=\frac{-2\Gamma\left(\frac{1}{2}\right)\cdot\frac{1}{2}\cdot\Gamma\left(\frac{1}{2}\right)}{\Gamma(1)}$$

$$=-\Gamma^2\left(\frac{1}{2}\right)=-\pi$$

Note : $\Gamma(-n)=-\frac{1}{n}\Gamma(-n+1)$

SO, $\boxed{\int_0^1 \frac{\sqrt{1-x}}{x\sqrt{x}}dx=-\pi}$

277. Calculate Integral $I=\int_0^\infty \frac{1}{x\sqrt{x}\sqrt{1+x}}dx$

Solution; $I=\int_0^\infty \frac{1}{x\sqrt{x}\sqrt{1+x}}dx$

$$=\int_0^\infty \frac{x^{-\frac{3}{2}}}{(1+x)^{\frac{1}{2}}}dx$$

$$=\int_0^\infty \frac{x^{-\frac{1}{2}-1}}{(1+x)^{-\frac{1}{2}+1}}dx=\frac{\Gamma\left(-\frac{1}{2}\right)\Gamma(1)}{\Gamma\left(1-\frac{1}{2}\right)}$$

$$=\frac{-2\Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{1}{2}\right)}=-2$$

SO, $\boxed{\int_0^\infty \frac{1}{x\sqrt{x}\sqrt{1+x}}dx=-2}$

278 Calculate Integral $I=\int_0^{\infty}\frac{1}{(1+x^2)(1+x^n)}dx$

Method 1

Solution; $I=\int_0^{\infty}\frac{1}{(1+x^2)(1+x^n)}dx$

Let : $x=\tan(u)\Rightarrow dx=\left(1+\tan^2(u)\right)du$ **; If** : $x\in(0,\infty)\Rightarrow u\in\left(0,\frac{\pi}{2}\right)$

We obtain $I=\int_0^{\frac{\pi}{2}}\frac{\left(1+\tan^2(u)\right)}{(1+\tan^2(u))(1+\tan^n(u))}du$

$$=\int_0^{\frac{\pi}{2}}\frac{1}{(1+\tan^n(u))}du \qquad (*)$$

$$=\int_0^{\frac{\pi}{2}}\frac{1}{\left(1+\tan^n(\frac{\pi}{2}-u)\right)}du$$

$$=\int_0^{\frac{\pi}{2}}\frac{\tan^n(u)}{\left(1+\tan^n(u)\right)}du \qquad (**)$$

Take $(*)+(**)$ **Then** : $2K=\int_0^{\frac{\pi}{2}}\frac{1}{\left(1+\tan^n(u)\right)}du+\int_0^{\frac{\pi}{2}}\frac{\tan^n(u)}{\left(1+\tan^n(u)\right)}du$

$$=\int_0^{\frac{\pi}{2}}\frac{\left(1+\tan^n(u)\right)}{\left(1+\tan^n(u)\right)}du=\frac{\pi}{2}$$

Implies That $I=\frac{\pi}{4}$

Method 2

Solution; $I=\int_0^{\infty}\frac{1}{(1+x^2)(1+x^n)}dx \quad (*)$

Let : $x=t^{-1}\Rightarrow dx=-t^{-2}dt$ **; If** : $x\in(0,\infty)\Rightarrow u\in(\infty,0)$

We Get $I=-\int_{\infty}^{0}\frac{t^{-2}}{(1+t^{-2})(1+t^{-n})}dt$

$$=\int_{0}^{\infty}\frac{t^{n}}{(1+t^{2})(1+t^{n})}dt \quad (**)$$

Take $(*)+(**)$ **Then** : $2I=\int_{0}^{\infty}\frac{1}{(1+t^{2})(1+t^{n})}dt+\int_{0}^{\infty}\frac{t^{n}}{(1+t^{2})(1+t^{n})}dt$

$$=\int_{0}^{\infty}\frac{1}{(1+t^{2})}dt=\tan^{-1}(t)\Big|_{0}^{\infty}=\frac{\pi}{2}$$

Implies That $I=\frac{\pi}{4}$

$SO,$ $\boxed{\int_{0}^{\infty}\frac{1}{(1+x^{2})(1+x^{n})}dx=\frac{\pi}{4}}$

279. Calculate Integral $I=\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\frac{\tan^{2}(x)}{e^{x}+1}dx$

Solution; $I=\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\frac{\tan^{2}(x)}{e^{x}+1}dx \qquad (1)$

$$=\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\frac{\tan^{2}(-x)}{e^{-x}+1}dx$$

$$=\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\frac{e^{x}\tan^{2}(x)}{e^{x}+1}dx \quad (2)$$

Take $(1)+(2)$ **Then** : $2I=\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\frac{\tan^{2}(x)}{e^{x}+1}dx+\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\frac{e^{x}\tan^{2}(x)}{e^{x}+1}dx$

$$=\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\tan^{2}(x)\,dx=2\left(1-\frac{\pi}{4}\right)$$

We obtain $I=1-\frac{\pi}{4}$

$SO,$ $\boxed{\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\frac{\tan^{2}(x)}{e^{x}+1}dx=1-\frac{\pi}{4}}$

280. Calculate Integral $I=\int_0^{\frac{\pi}{2}}\frac{\sin(x)}{\left(\sin(x)+\cos(x)\right)^5}dx$

Solution; $I=\int_0^{\frac{\pi}{2}}\frac{\sin(x)}{\left(\sin(x)+\cos(x)\right)^5}dx \quad (1)$

$$=\int_0^{\frac{\pi}{2}}\frac{\cos(x)}{\left(\sin(x)+\cos(x)\right)^5}dx \quad (2)$$

Take $(1)+(2)$ **That** : $2I=\int_0^{\frac{\pi}{2}}\frac{\sin(x)}{\left(\sin(x)+\cos(x)\right)^5}dx+\int_0^{\frac{\pi}{2}}\frac{\cos(x)}{\left(\sin(x)+\cos(x)\right)^5}dx$

$$=\int_0^{\frac{\pi}{2}}\frac{1}{\left(\sin(x)+\cos(x)\right)^4}dx$$

$$=\frac{1}{4}\int_0^{\frac{\pi}{2}}\frac{1}{\cos^4\left(x-\frac{\pi}{4}\right)}dx$$

Let : $y=x-\frac{\pi}{4}\Rightarrow dy=dx$; **If** : $x\in\left(0,\frac{\pi}{2}\right)\Rightarrow y\in\left(-\frac{\pi}{4},\frac{\pi}{4}\right)$

We obtain $2I=\frac{1}{4}\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\frac{1}{\cos^4(y)}dx$

$$=\frac{1}{2}\int_0^{\frac{\pi}{4}}\sec^4(x)\,dx$$

$$=\frac{1}{2}\int_0^{\frac{\pi}{4}}\sec^2(x)\,d\left(\tan(x)\right)$$

$$=\frac{1}{2}\int_0^{\frac{\pi}{4}}\left(1+\tan^2(x)\right)d\left(\tan(x)\right)$$

$$=\frac{1}{2}\left(\tan(x)+\frac{1}{4}\tan^3(x)\right)\Big|_0^{\frac{\pi}{4}}=\frac{2}{3}$$

We Get $I=\frac{1}{3}$

$SO,$ $\boxed{\int_0^{\frac{\pi}{2}}\frac{\sin(x)}{\left(\sin(x)+\cos(x)\right)^5}dx=\frac{1}{3}}$

281. Calculate Integral $I=\int_0^{2\pi}\frac{x^2\sin(x)}{1+\cos^2(x)}dx$

Solution; $I=\int_0^{2\pi}\frac{x^2\sin(x)}{1+\cos^2(x)}dx$

Let : $x=\pi+u\Rightarrow dx=du$ **; If :** $x\in(0,2\pi)\Rightarrow u\in(-\pi,\pi)$

Implies That $I=\int_{-\pi}^{\pi}\frac{(\pi+u)^2\sin(\pi+u)}{1+\cos^2(\pi+u)}du$

$$=-\int_{-\pi}^{\pi}\frac{(\pi^2+2u\pi+u^2)\sin(u)}{1+\cos^2(u)}du$$

$$=\underbrace{-\int_{-\pi}^{\pi}\frac{\pi^2\sin(u)}{1+\cos^2(u)}du}_{is\,odd\ function}\underbrace{-\int_{-\pi}^{\pi}\frac{2u\pi\sin(u)}{1+\cos^2(u)}du}_{is\,even\ function}\underbrace{-\int_{-\pi}^{\pi}\frac{u^2\sin(u)}{1+\cos^2(u)}du}_{is\,odd\ function}$$

$$=-4\pi\int_0^{\pi}\frac{u\sin(u)}{1+\cos^2(u)}du\qquad(!)$$

$$=-4\pi\int_0^{\pi}\frac{(\pi-u)\sin(u)}{1+\cos^2(u)}du\quad(!!)$$

Take $(!)+(!!)$ **Then :** $2I=-4\pi\int_0^{\pi}\frac{u\sin(u)}{1+\cos^2(u)}du-4\pi\int_0^{\pi}\frac{(\pi-u)\sin(u)}{1+\cos^2(u)}du$

$$=-4\pi^2\int_0^{\pi}\frac{\sin(u)}{1+\cos^2(u)}du$$

$$=4\pi^2\tan^{-1}\left(\cos(x)\right)\Big|_0^{\pi}$$

$$=-2\pi^3$$

We Get $I=-\pi^3$

$$SO,\ \boxed{\int_0^{2\pi}\frac{x^2\sin(x)}{1+\cos^2(x)}dx=-\pi^3}$$

282. Calculate Integral $I=\int_0^e \left(x\log(x)+\log(x)+1\right)d(e^x)$

Solution; $I=\int_0^e \left(x\log(x)+\log(x)+1\right)d(e^x)$

$$=\int_0^e \left(x\log(x)+\log(x)+1\right)(e^x)'dx$$
$$=\int_0^e \left(x\log(x)+\log(x)+1\right)e^x dx$$
$$=\int_0^e \left(xe^x\log(x)+e^x\log(x)+e^x\right)dx$$
$$=\int_0^e \left(x(e^x)'\log(x)+(x)'e^x\log(x)+x(\log(x))'e^x\right)dx$$
$$=\int_0^e \left(xe^x\log(x)\right)'dx$$
$$=xe^x\log(x)\Big|_0^e=e^{e+1}$$

SO, $\boxed{\int_0^e \left(x\log(x)+\log(x)+1\right)d(e^x)=e^{e+1}}$

283. Calculate Integral $I=\int_0^{\pi}\sin(3x)d\left(\cos(x)\right)$

Solution; $I=\int_0^{\pi}\sin(3x)d\left(\cos(x)\right)$

$$=\int_0^{\pi}\left(3\sin(x)-4\sin^3(x)\right)\left(\cos(x)\right)'dx$$
$$=\int_0^{\pi}\left(4\sin^3(x)-3\sin(x)\right)\sin(x)dx$$
$$=\int_0^{\pi}\left(4\sin^4(x)-3\sin^2(x)\right)dx$$
$$=2\int_0^{\frac{\pi}{2}}\left(4\sin^4(x)-3\sin^2(x)\right)dx$$
$$=2\left(4\times\frac{\pi}{2}\times\frac{3\times1}{4\times2}-3\times\frac{\pi}{2}\times\frac{1}{2}\right)=0$$

SO, $\boxed{\int_0^{\pi}\sin(3x)d\left(\cos(x)\right)=0}$

284. Calculate Integral $I=\int_0^{\frac{\pi}{2}}\sin^2(x)\log(\tan(x))\,dx$

Solution; $I=\int_0^{\frac{\pi}{2}}\sin^2(x)\log(\tan(x))\,dx \qquad (*)$

$$=\int_0^{\frac{\pi}{2}}\cos^2(x)\log\big(\cot(x)\big)\,dx$$

$$=\int_0^{\frac{\pi}{2}}\cos^2(x)\log\Big(\big[\tan(x)\big]^{-1}\Big)\,dx$$

$$=-\int_0^{\frac{\pi}{2}}\cos^2(x)\log\big(\tan(x)\big)\,dx \quad (**)$$

Take $(*)+(**)$ **Then :** $2I=\int_0^{\frac{\pi}{2}}\Big(\sin^2(x)-\cos^2(x)\Big)\log(\tan(x))dx$

$$=-\int_0^{\frac{\pi}{2}}\Big(\cos^2(x)-\sin^2(x)\Big)\log(\tan(x)dx$$

$$=\underbrace{-\int_0^{\frac{\pi}{2}}\cos(2x)\log(\tan(x))dx}_{Use\ Partinl\ Integral}$$

$$=\underbrace{-\frac{\sin(2x)\log(\tan)}{2}\Big|_0^{\frac{\pi}{2}}}_{0}+\frac{1}{2}\int_0^{\frac{\pi}{2}}\sin(2x)\left(\frac{\sec^2(x)}{\tan(x)}\right)dx$$

$$=\frac{1}{2}\int_0^{\frac{\pi}{2}}2\sin(x)\cos(x)\left(\frac{\sec^2(x)}{\tan(x)}\right)dx$$

$$=\int_0^{\frac{\pi}{2}}\sin(x)\cos(x)\left(\frac{1}{\sin(x)\cos(x)}\right)dx$$

$$=\int_0^{\frac{\pi}{2}}(1)\,dx=\frac{\pi}{2}$$

We obtain $I=\frac{\pi}{4}$

$SO,$ $\boxed{\int_0^{\frac{\pi}{2}}\sin^2(x)\log(\tan(x))dx=\frac{\pi}{4}}$

285. Calculate Integral $I=\int_0^1 \frac{x-1}{(x+1)^3}e^x dx$

Solution; $I=\int_0^1 \frac{x-1}{(x+1)^3}e^x dx$

$$=\int_0^1\left(\frac{1}{(x+1)^2}-\frac{2}{(x+1)^3}\right)e^x dx$$

$$=\int_0^1\left[\frac{1}{(x+1)^2}\left(e^x\right)'+\left(\frac{1}{(x+1)^2}\right)'e^x\right]dx$$

$$=\int_0^1\left(\frac{e^x}{(x+1)^2}\right)'dx=\frac{e-4}{4}$$

$SO,$ $\boxed{\int_0^1 \frac{x-1}{(x+1)^3}e^x dx=\frac{e-4}{4}}$

286. Calculate Integral $I=\int_1^2 \frac{\sqrt{x-1}\tan^{-1}\left(\sqrt{x-1}\right)}{x}dx$

Solution; $I=\int_1^2 \frac{\sqrt{x-1}\tan^{-1}\left(\sqrt{x-1}\right)}{x}dx$

Let : $\sqrt{x-1}=\tan(u)\equiv x=\sec^2(u)\Rightarrow dx=2\tan(u)\sec^2(u)du$ **; If :** $x\in(1,2)\Rightarrow t\in(0,\frac{\pi}{4})$

We obtain $I=2\int_0^{\frac{\pi}{4}}\frac{\tan^2(u)\tan^{-1}\left(\tan(u)\right)\sec^2(u)}{\sec^2(u)}du$

$$=2\underbrace{\int_0^{\frac{\pi}{4}}u\tan^2(u)du}_{Use\ partial\ integral}$$

$$=2\left[u\left(\tan(u)-u\right)+\frac{u^2}{2}+\log\left(\cos(u)\right)\right]_0^{\frac{\pi}{4}}=\frac{\pi}{2}-\frac{\pi^2}{16}-\log(2)$$

$SO,$ $\boxed{\int_1^2 \frac{\sqrt{x-1}\tan^{-1}\left(\sqrt{x-1}\right)}{x}dx=\frac{\pi}{2}-\frac{\pi^2}{16}-\log(2)}$

287. Calculate Integral $I=\int_1^{\infty}\left(\frac{\log(x)}{x}\right)^{n+m}dx$

Solution; $I=\int_1^{\infty}\left(\frac{\log(x)}{x}\right)^{n+m}dx$

Let : $u=\log(x)\Leftrightarrow x=e^u\Rightarrow dx=e^u du$ **; If** : $x\in(1,\infty)\Rightarrow u\in(0,\infty)$

We obtain $I=\int_0^{\infty}\left(\frac{u}{e^u}\right)^{n+m}e^u du$

$$=\int_0^{\infty}u^{n+m}e^{-(n+m-1)u}du$$

Let : $y=(n+m-1)u\Rightarrow du=\frac{1}{n+m-1}dy$ **;** **If** : $u\in(0,\infty)\Rightarrow y\in(0,\infty)$

Implies That $I=\int_0^{\infty}\left(\frac{y}{n+m-1}\right)^{n+m}\left(\frac{e^{-y}}{n+m-1}\right)dy$

$$=\left(\frac{1}{n+m-1}\right)^{n+m+1}\int_0^{\infty}y^{n+m}e^{-y}dy$$

$$=\left(\frac{1}{n+m-1}\right)^{n+m+1}\int_0^{\infty}y^{(n+m+1)-1}e^{-y}dy$$

$$=\left(\frac{1}{n+m-1}\right)^{n+m+1}\Gamma(n+m+1)$$

$$=\left(\frac{1}{n+m-1}\right)^{n+m+1}(n+m)!$$

$$=\frac{(n+m)!}{(n+m-1)^{n+m+1}}$$

SO, $\boxed{\int_1^{\infty}\left(\frac{\log(x)}{x}\right)^{n+m}dx=\frac{(n+m)!}{(n+m-1)^{n+m+1}}}$

288. Calculate Integral $I=\int_0^{\infty}(-1)^{ix^2}dx\ ,i=\sqrt{-1}$

Solution; $I=\int_0^{\infty}(-1)^{ix^2}dx$

$$=\int_0^{\infty}(e^{\pi i})^{ix^2}dx \qquad \textbf{Note}: -1=\cos(\pi)+i\sin(\pi)=e^{\pi i}$$

$$=\int_0^{\infty}e^{-\pi x^2}dx$$

Let : $t=\pi x^2 \Leftrightarrow x=\frac{1}{\sqrt{\pi}}\sqrt{t} \Rightarrow dx=\frac{t^{-\frac{1}{2}}}{2\sqrt{\pi}}dt$; **If** : $x\in(0,\infty)\Rightarrow t\in(0,\infty)$

We obtain $I=\int_0^{\infty}e^{-t}\times\frac{t^{-\frac{1}{2}}}{2\sqrt{\pi}}dt$

$$=\frac{1}{2\sqrt{\pi}}\int_0^{\infty}t^{-\frac{1}{2}}e^{-t}dt$$

$$=\frac{1}{2\sqrt{\pi}}\Gamma\left(\frac{1}{2}\right)=\frac{1}{2}$$

$SO,$ $\boxed{\int_0^{\infty}(-1)^{ix^2}dx=\frac{1}{2}}$

289. Calculate Integral $I=\int_0^{\infty}\frac{1}{x^4+2x^2+1}dx$

Solution; $I=\int_0^{\infty}\frac{1}{x^4+2x^2+1}dx$

$$=\int_0^{\infty}\frac{1}{(x^2+1)^2}dx$$

Let : $x=\tan(u)\Rightarrow dx=\left(\tan^2(u)+1\right)du$; **If** : $x\in(0,\infty)\Rightarrow u\in\left(0,\frac{\pi}{2}\right)$

We obtain $I=\int_0^{\frac{\pi}{2}}\frac{\left(\tan^2(u)+1\right)}{(\tan^2(u)+1)^2}du$

$$=\int_0^{\frac{\pi}{2}}\frac{1}{\left(\tan^2(u)+1\right)}du$$

$$= \int_0^{\frac{\pi}{2}} \cos^2(u)\,du$$

$$= \frac{\pi}{2} \times \frac{1}{2} = \frac{\pi}{4}$$

SO, $\boxed{\int_0^{\infty} \frac{1}{x^4 + 2x^2 + 1}\,dx = \frac{\pi}{4}}$

290. Calculate Integral $I = \int_0^1 \frac{1}{\sqrt{1-x^n}}\,dx$

Solution; $I = \int_0^1 \frac{1}{\sqrt{1-x^n}}\,dx$

$$= \int_0^1 (1-x^n)^{-\frac{1}{2}}\,dx$$

Let : $x = u^{\frac{1}{n}} \Rightarrow dx = \frac{1}{n}u^{\frac{1}{n}-1}\,du$ **; If :** $x \in (0,1) \Rightarrow u \in (0,1)$

We obtain $I = \frac{1}{n}\int_0^1 (1-u)^{-\frac{1}{2}} u^{\frac{1}{n}-1}\,du$

$$= \frac{1}{n}\int_0^1 u^{\frac{1}{n}-1}(1-u)^{\frac{1}{2}-1}\,du$$

$$= B\left(\frac{1}{n}, \frac{1}{2}\right)$$

$$= \frac{\sqrt{\pi}}{n} \times \frac{\Gamma\left(\frac{1}{n}\right)}{\Gamma\left(\frac{1}{n} + \frac{1}{2}\right)}$$

SO, $\boxed{\int_0^1 \frac{1}{\sqrt{1-x^n}}\,dx = \frac{\sqrt{\pi}}{n} \times \frac{\Gamma\left(\frac{1}{n}\right)}{\Gamma\left(\frac{1}{n} + \frac{1}{2}\right)}}$

291. Calculate Integral $I=\int_1^{\infty}\frac{1}{x\sqrt{x^n-1}}dx$

Solution; $I=\int_1^{\infty}\frac{1}{x\sqrt{x^n-1}}dx$

$$=\int_1^{\infty}\frac{x^{n-1}}{x^n\sqrt{x^n-1}}dx$$

Let : $u^2=x^n-1\equiv x^n=u^2+1\Rightarrow x^{n-1}dx=\frac{2u}{n}du$ **; If :** $x\in(1,\infty)\Rightarrow u\in(0,\infty)$

Implies That $I=\frac{2}{n}\int_0^{\infty}\frac{u}{(u^2+1)u}du$

$$=\frac{2}{n}\int_0^{\infty}\frac{1}{(u^2+1)}du$$

$$=\frac{2}{n}\times\frac{\pi}{2}=\frac{\pi}{n}$$

SO, $\boxed{\int_1^{\infty}\frac{1}{x\sqrt{x^n-1}}dx=\frac{\pi}{n}}$

292. Calculate Integral $I=\int_0^{\infty}x^{m-1}e^{-x^n}dx$

Solution; $I=\int_0^{\infty}x^{m-1}e^{-x^n}dx$

Let : $x=u^{\frac{1}{n}}\Rightarrow dx=\frac{1}{n}u^{\frac{1}{n}-1}du$ **; If :** $x\in(0,\infty)\Rightarrow u\in(0,\infty)$

We obtain $I=\frac{1}{n}\int_0^{\infty}u^{\frac{m-1}{n}}.e^{-u}.u^{\frac{1}{n}-1}dx$

$$=\frac{1}{n}\int_0^{\infty}u^{\frac{m}{n}-1}e^{-u}dx=\frac{1}{n}\Gamma\left(\frac{m}{n}\right)$$

SO, $\boxed{\int_0^{\infty}x^{m-1}e^{-x^n}dx=\frac{1}{n}\Gamma\left(\frac{m}{n}\right)}$

293. Calculate Integral $I=\int_0^{\infty}\frac{5x^2}{1+x^{10}}dx$

Solution; $I=\int_0^{\infty}\frac{5x^2}{1+x^{10}}dx$

Let : $x=u^{\frac{1}{10}}\Rightarrow dx=\frac{1}{10}u^{\frac{1}{10}-1}du$ **;** **If :** $x\in(0,\infty)\Rightarrow u\in(0,\infty)$

We obtain $I=\frac{1}{10}\int_0^{\infty}\frac{5u^{\frac{2}{10}}}{(1+u)}\times u^{\frac{1}{10}-1}du$

$$=\frac{1}{2}\int_0^{\infty}\frac{u^{\frac{3}{10}-1}}{(1+u)^{\frac{3}{10}+\frac{7}{10}}}du$$

$$=\frac{1}{2}B\left(\frac{3}{10},\frac{7}{10}\right)=\frac{1}{2}\cdot\pi\csc\left(\frac{3\pi}{10}\right)=\frac{\pi}{\varphi}$$

$SO,$ $\boxed{\int_0^{\infty}\frac{5x^2}{1+x^{10}}dx=\frac{\pi}{\varphi}}$

294. Calculate Integral $I=\int_{\log\left(\frac{1}{\pi}\right)}^{\log(\pi)}\left(\sin(e^x)-\sin(e^{-x})\right)dx$

Solution; $I=\int_{\log\left(\frac{1}{\pi}\right)}^{\log(\pi)}\left(\sin(e^x)-\sin(e^{-x})\right)dx$

$$=\int_{-\log(\pi)}^{\log(\pi)}\left(\sin(e^{-x})-\sin(e^{-(-x)})\right)dx$$

$$=\underbrace{\int_{-\log(\pi)}^{\log(\pi)}\left(\sin(e^{-x})-\sin(e^{x})\right)dx}_{-I}$$

Then $I=-I$ **That** $I=0$

$SO,$ $\boxed{\int_{\log\left(\frac{1}{\pi}\right)}^{\log(\pi)}\left(\sin(e^x)-\sin(e^{-x})\right)dx=0}$

295. Calculate Integral $I=\int_0^1 \frac{x}{\sqrt[n]{1-x}}d(x^2)$

Solution; $I=\int_0^1 \frac{x}{\sqrt[n]{1-x}}d(x^2)$

$$=\int_0^1 x(1-x)^{\frac{1}{n}}(x^2)'dx$$

$$=2\int_0^1 x^2(1-x)^{\frac{1}{n}}dx$$

$$=2\int_0^1 x^{3-1}(1-x)^{\left(1+\frac{1}{n}\right)-1}dx$$

$$=2B\left(1+\frac{1}{n},3\right)$$

$$=2\times\frac{\Gamma\left(1+\frac{1}{n}\right)\Gamma(3)}{\Gamma\left(4+\frac{1}{n}\right)}$$

$$=\frac{4\cdot\frac{1}{n}\Gamma\left(\frac{1}{n}\right)}{\left(3+\frac{1}{n}\right)\left(2+\frac{1}{n}\right)\left(1+\frac{1}{n}\right)\left(\frac{1}{n}\right)\Gamma\left(\frac{1}{n}\right)}$$

$$=\frac{4n^3}{(3n+1)(2n+1)(n+1)}$$

SO, $\boxed{\int_0^1 \frac{x}{\sqrt[n]{1-x}}d(x^2)=\frac{4n^3}{(3n+1)(2n+1)(n+1)}}$

296. Calculate Integral $I=\int_0^1 x^n\sqrt{1-x}dx$

Solution; $I=\int_0^1 x^n\sqrt{1-x}dx$

$$=\int_0^1 x^n(1-x)^{\frac{1}{2}}dx$$

$$= \int_0^1 x^{(n+1)-1}(1-x)^{\frac{3}{2}-1}dx$$

$$= B\left(n+1,\frac{3}{2}\right) = \frac{\Gamma(n+1)\Gamma(1+\frac{1}{2})}{\Gamma(n+1+1+\frac{1}{2})}$$

$$= \frac{\frac{1}{2}n!\Gamma(\frac{1}{2})}{\left(n+1+\frac{1}{2}\right)\left(n+\frac{1}{2}\right)\Gamma(n+\frac{1}{2})} = \frac{2\pi n!}{(2n+3)(2n+1)\Gamma(n+\frac{1}{2})}$$

$$= \frac{2\pi n!}{(2n+3)(2n+1)\frac{(2n-1)!!}{2^n}\sqrt{\pi}} = \frac{2^{n+1}\sqrt{\pi}n!}{(2n+3)(2n+1)(2n-1)!!}$$

$$SO, \boxed{\int_0^1 x^n\sqrt{1-x}dx = \frac{2^{n+1}\sqrt{\pi}n!}{(2n+3)(2n+1)(2n-1)!!}}$$

297. Calculate Integral $I = \int_0^\infty \frac{\sin^5(x)}{x}dx$

Solution; $I = \int_0^\infty \frac{\sin^5(x)}{x}dx$

$$= \int_0^\infty \frac{\sin(x)}{x}\sin^4(x)dx$$

Let : $f(x) = \sin^4(x)$ **Then :** $\begin{cases} f(\pi+x) = \sin^4(\pi+x) = f(x) \\ f(\pi-x) = \sin^4(\pi-x) = f(x) \end{cases}$

We obtain $I = \int_0^{\frac{\pi}{2}} \sin^4(x)dx$

$$= \frac{\pi}{2}\times\frac{3\times1}{4\times2} = \frac{3\pi}{16}$$

$$SO, \boxed{\int_0^\infty \frac{\sin^5(x)}{x}dx = \frac{3\pi}{16}}$$

298. Calculate Integral $I=\int_1^{\infty}\left(\frac{\log(x)}{x}\right)^{2025}dx$

Solution; $I=\int_1^{\infty}\left(\frac{\log(x)}{x}\right)^{2025}dx$

$$=\int_1^{\infty}x^{-2025}\log^{2025}(x)dx$$

Let : $x=e^u\Rightarrow dx=e^u du$; **If** : $x\in(1,\infty)\Rightarrow u\in(0,\infty)$

We obtain $I=\int_0^{\infty}e^{-2025u}u^{2025}e^u du$

$$=\int_0^{\infty}u^{2025}e^{-2024u}du$$

Let: $u=\frac{y}{2024}\Rightarrow du=\frac{1}{2024}dy$; **If** : $x\in(0,\infty)\Rightarrow y\in(0,\infty)$

Implies That $I=\frac{1}{2024}\int_0^{\infty}\left(\frac{y}{2024}\right)^{2025}e^{-y}dy$

$$=\frac{1}{2024^{2026}}\int_0^{\infty}y^{2025}e^{-y}dy$$

$$=\frac{2025!}{2024^{2026}}$$

SO, $\boxed{\int_1^{\infty}\left(\frac{\log(x)}{x}\right)^{2025}dx=\frac{2025!}{2024^{2026}}}$

299. Calculate Integral $I=\int_0^1 x\left(-\log(x)\right)^3dx$

Solution; $I=\int_0^1 x\left(-\log(x)\right)^3dx$

Let : $u=-\log(x)\Leftrightarrow x=e^{-u}\Rightarrow dx=-e^{-u}du$; **If** : $x\in(0,1)\Rightarrow u\in(\infty,0)$

We obtain $I=-\int_{\infty}^0 u^3e^{-u}\times e^{-u}du$

$$= \int_0^{\infty} u^3 e^{-2u} du$$

Let : $y = 2u \Rightarrow du = \frac{dy}{2}$; **If** : $u \in (0,\infty) \Rightarrow y \in (0,\infty)$

Implies That $I = \frac{1}{2}\int_0^{\infty} \left(\frac{y}{2}\right)^3 e^{-y} dy$

$$= \frac{1}{16}\int_0^{\infty} y^3 e^{-y} dy$$

$$= \frac{1}{16} \times 3!$$

$$= \frac{3}{8}$$

SO, $\boxed{\int_0^1 x\left(-\log(x)\right)^3 dx = \frac{3}{8}}$

300. Calculate Integral $I = \int_0^{\pi} \log\left(\left|\tan(x)\right|\right) dx$

Solution; $I = \int_0^{\pi} \log\left(\left|\tan(x)\right|\right) dx$

$$= 2\int_0^{\frac{\pi}{2}} \log\left(\left|\tan(x)\right|\right) dx$$

Use: $\int_0^{2a} f(x)dx = 2\int_0^{a} f(x)dx$, $f(2a-x) = f(x)$

$$= 2\left[\int_0^{\frac{\pi}{2}} \log\left(\left|\sin(x)\right|\right) dx - \int_0^{\frac{\pi}{2}} \log\left(\left|\cos(x)\right|\right) dx\right]$$

$$= 2\left[\int_0^{\frac{\pi}{2}} \log\left(\sin(x)\right) dx - \int_0^{\frac{\pi}{2}} \log\left(\cos(x)\right) dx\right]$$

$$= 2\left[\left(-\frac{\pi}{2}\log(2)\right) - \left(-\frac{\pi}{2}\log(2)\right)\right] = 0$$

SO, $\boxed{\int_0^{\pi} \log\left(\left|\tan(x)\right|\right) dx = 0}$

301. Calculate Integral $I=\int_0^{\pi}\log\left(\left|\sin(x)\right|\right)dx$

Solution; $I=\int_0^{\pi}\log\left(\left|\sin(x)\right|\right)dx$

$$=2\int_0^{\frac{\pi}{2}}\log\left(\left|\sin(x)\right|\right)dx$$

Use: $\int_0^{2a}f(x)dx=2\int_0^{a}f(x)dx\ ,\ f(2a-x)=f(x)$

$$=2\int_0^{\frac{\pi}{2}}\log\left(\sin(x)\right)dx$$

$$=2\left(-\frac{\pi}{2}\log(2)\right)$$

$$=-\pi\log(2)$$

$SO,$ $\boxed{\int_0^{\pi}\log\left(\left|\sin(x)\right|\right)dx=\pi\log\left(\frac{1}{2}\right)}$

302. Calculate Integral $I=\int_0^{\pi}\log\left(\left|\cos(x)\right|\right)dx$

Solution; $I=\int_0^{\pi}\log\left(\left|\cos(x)\right|\right)dx$

$$=2\int_0^{\frac{\pi}{2}}\log\left(\left|\cos(x)\right|\right)dx$$

$$=2\int_0^{\frac{\pi}{2}}\log\left(\cos(x)\right)dx$$

$$=2\left(-\frac{\pi}{2}\log(2)\right)$$

$$=-\pi\log(2)$$

$SO,$ $\boxed{\int_0^{\pi}\log\left(\left|\cos(x)\right|\right)dx=\pi\log\left(\frac{1}{2}\right)}$

Note : $\int_0^{2a}f(x)dx=2\int_0^{a}f(x)dx\ ,\ f(2a-x)=f(x)\ \equiv\ \int_0^{2a}\left(f(x)+f(2a-x)\right)dx$

303. Calculate Integral $I=\int_1^e\left[\left(\frac{x}{e}\right)^x+\left(\frac{e}{x}\right)^x\right]\log(x)dx$

Solution; $I=\int_1^e\left[\left(\frac{x}{e}\right)^x+\left(\frac{e}{x}\right)^x\right]\log(x)dx$

$$I=\int_1^e\left[\left(\frac{x}{e}\right)^x+\frac{1}{\left(\frac{x}{e}\right)^x}\right]\log(x)dx$$

Let : $u=\left(\frac{x}{e}\right)^x\Leftrightarrow\log(u)=x\log(x)-x\Rightarrow\frac{1}{u}du=\log(x)dx$ **; If :** $x\in(1,e)\Rightarrow u\in\left(\frac{1}{e},1\right)$

We obtain $I=\int_{\frac{1}{e}}^1\left(u+\frac{1}{u}\right)\frac{1}{u}du$

$$=\left(u-\frac{1}{u}\right)\Big|_{\frac{1}{e}}^1=\frac{(e-1)(e+1)}{e}$$

SO, $\boxed{\int_1^e\left[\left(\frac{x}{e}\right)^x+\left(\frac{e}{x}\right)^x\right]\log(x)dx=\frac{(e-1)(e+1)}{e}}$

304. Calculate Integral $I=\int_0^\infty\frac{\sin(x)}{x+x\cos^2(x)}dx$

Solution; $I=\int_0^\infty\frac{\sin(x)}{x+x\cos^2(x)}dx$

$$=\int_0^\infty\frac{\sin(x)}{x}\times\frac{1}{1+\cos^2(x)}dx$$

Let : $f(x)=\frac{1}{1+\cos^2(x)}$ **Then :** $f(\pi\pm x)=\frac{1}{1+\cos^2(\pi\pm x)}=f(x)$

We obtain $I=\int_0^{\frac{\pi}{2}}\frac{1}{1+\cos^2(x)}dx$

$$=\int_0^{\frac{\pi}{2}}\frac{\sec^2(x)}{\left(1+\cos^2(x)\right)\sec^2(x)}dx$$

$$= \int_0^{\frac{\pi}{2}} \frac{\sec^2(x)}{2+\tan^2(x)}dx$$

Let : $\tan(x) = \sqrt{2}t \Rightarrow \sec^2(x)dx = \sqrt{2}dt$; **If** : $x \in \left(0, \frac{\pi}{2}\right) \Rightarrow t \in (0, \infty)$

Implies That $I = \sqrt{2}\int_0^{\infty} \frac{1}{2+2t^2}dt$

$$= \frac{\sqrt{2}}{2}\tan^{-1}(t)\Big|_0^{\infty} = \frac{\pi}{2\sqrt{2}}$$

SO, $\boxed{\int_0^{\infty} \frac{\sin(x)}{x+x\cos^2(x)}dx = \frac{\pi}{2\sqrt{2}}}$

305. Calculate Integral $I = \int_0^{\infty} x^2 e^{-x}\cos(x)dx$

Solution; $I = \int_0^{\infty} x^2 e^{-x}\cos(x)dx$

Suppose $J = \int_0^{\infty} x^2 e^{-x}\sin(-x)dx$

Take $I + iJ = \int_0^{\infty} x^2 e^{-x}e^{-xi}dx = \int_0^{\infty} x^2 e^{-(i+1)x}dx$

Let : $t = (i+1)x \Rightarrow dx = \frac{1}{i+1}dt$; **If** : $x \in (0, \infty) \Rightarrow t \in (0, \infty)$

We obtain $I + iJ = \int_0^{\infty} \left(\frac{t}{i+1}\right)^2 e^{-t}\frac{1}{i+1}dt$

$$= \frac{1}{(i+1)^3}\int_0^{\infty} t^2 e^{-t}dx$$

$$= \frac{\Gamma(3)}{(2i)(i+1)} = -\frac{1}{2} - \frac{1}{2}i$$

Pullable $I = -\frac{1}{2}$

SO, $\boxed{\int_0^{\infty} x^2 e^{-x}\cos(x)dx = -\frac{1}{2}}$

306. Calculate Integral $I=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left[(x)!+(x-1)!\right](-x)!dx$

Solution; $I=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left[(x)!+(x-1)!\right](-x)!dx$

$$=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left[\Gamma(x+1)+\Gamma(x)\right]\Gamma(1-x)dx$$

$$=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left(\Gamma(x+1)\Gamma(1-x)+\Gamma(x)\Gamma(1-x)\right)dx$$

$$=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left(x\Gamma(x)\Gamma(1-x)+\Gamma(x)\Gamma(1-x)\right)dx$$

$$=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left(x+1\right)\Gamma(x)\Gamma(1-x)dx$$

$$=\pi\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\frac{(x+1)}{\sin(x)}dx$$

$$=\pi\underbrace{\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\frac{x}{\sin(x)}dx}_{is\ an\ even\ function}+\pi\underbrace{\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\frac{1}{\sin(x)}dx}_{is\ an\ odd\ function}$$

$$=2\pi\int_{0}^{\frac{\pi}{2}}\frac{x}{\sin(x)}dx$$

Let $u=x\Rightarrow du=dx$ **And** $v=\int\frac{1}{\sin(x)}dx=\log\left(\frac{1-\cos(x)}{\sin(x)}\right)=\log\left(\tan\left(\frac{x}{2}\right)\right)$

We obtain $I=\underbrace{2\pi x\log\left(\frac{1-\cos(x)}{\sin(x)}\right)\Big|_{0}^{\frac{\pi}{2}}}_{0}-2\pi\int_{0}^{\frac{\pi}{2}}\log\left(\frac{1-\cos(x)}{\sin(x)}\right)dx$

$$=-2\pi\int_{0}^{\frac{\pi}{2}}\log\left(\tan\left(\frac{x}{2}\right)\right)dx$$

Let : $u=\frac{x}{2}\Rightarrow dx=2du$; **If** : $x\in\left(0,\frac{\pi}{2}\right)\Rightarrow u\in\left(0,\frac{\pi}{4}\right)$

Implies That $I=-4\pi\int_0^{\frac{\pi}{4}}\log\left(\tan(u)\right)dx$

$$=4\pi\left(-\int_0^{\frac{\pi}{4}}\log\left(\tan(u)\right)dx\right)$$

$$=4\pi G \quad \textbf{Where “ G ” Catalan's Constant}$$

$$SO,\ \boxed{\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left[(x)!+(x-1)!\right](-x)!dx=4\pi G}$$

307. Calculate Integral $I=\int_0^{\pi}\frac{\sin(x)}{\sin(x)+\cos(x)}dx$

Solution; $I=\int_0^{\pi}\frac{\sin(x)}{\sin(x)+\cos(x)}dx$

Let : $f(x)=\sin(x)+\cos(x)\Rightarrow f'(x)=\cos(x)-\sin(x)$

If : $x\to 0\Rightarrow f(0)=1$, *If* : $x\to\pi\Rightarrow f(\pi)=-1$

We obtain : $\frac{1}{2}\left(f(x)-f'(x)\right)=\sin(x)$

Implies That $I=\frac{1}{2}\int_0^{\pi}\frac{f(x)-f'(x)}{f(x)}dx$

$$=\frac{1}{2}\left(x-\log\left|f(x)\right|\right)\Big|_0^{\pi}$$

$$=\frac{1}{2}\left(\pi-\log\left|\frac{-1}{1}\right|\right)=\frac{\pi}{2}$$

$$SO,\ \boxed{\int_0^{\pi}\frac{\sin(x)}{\sin(x)+\cos(x)}dx=\frac{\pi}{2}}$$

308. Calculate Integral $I=\int_0^{\frac{\pi}{2}}\left(1-\sin(x)+\sin^2(x)-\sin^3(x)+\cdots\right)dx$

Solution; $I=\int_0^{\frac{\pi}{2}}\left(1-\sin(x)+\sin^2(x)-\sin^3(x)+\cdots\right)dx$

Implies That $I=\int_0^{\frac{\pi}{2}}\frac{1}{1+\sin(x)}dx$

Use: $\frac{1}{1+x}=\sum_{n=0}^{\infty}(-x)^n=1-x+x^2-x^3+\cdots$

$$=\int_0^{\frac{\pi}{2}}\frac{1-\sin(x)}{1-\sin^2(x)}dx$$

$$=\int_0^{\frac{\pi}{2}}\frac{1-\sin(x)}{\cos^2(x)}dx$$

$$=\int_0^{\frac{\pi}{2}}\left(\frac{1}{\cos^2(x)}-\frac{\sin(x)}{\cos^2(x)}\right)dx$$

$$=\int_0^{\frac{\pi}{2}}\left[\left(\tan(x)\right)'-\left(\frac{1}{\cos(x)}\right)'\right]dx$$

$$=\left(\tan(x)-\frac{1}{\cos(x)}\right)\Bigg|_0^{\frac{\pi}{2}}$$

$$=\lim_{x\to\frac{\pi}{2}}\left(\tan(x)-\frac{1}{\cos(x)}\right)-\lim_{x\to 0}\left(\tan(x)-\frac{1}{\cos(x)}\right)$$

$$=\underbrace{\lim_{x\to\frac{\pi}{2}}\left(\frac{\sin(x)-1}{\cos(x)}\right)}_{\text{Use Lopital}}-\left(\tan(0)-\frac{1}{\cos(0)}\right)$$

$$=-\lim_{x\to\frac{\pi}{2}}\left(\frac{\cos(x)}{\sin(x)}\right)-(0-1)$$

$$=-0-(0-1)=1$$

SO, $\boxed{\int_0^{\frac{\pi}{2}}\left(1-\sin(x)+\sin^2(x)-\sin^3(x)+\cdots\right)dx=1}$

309. Calculate Integral $I=\int_0^{\frac{\pi}{2}}\frac{\sin(x)-\cos(x)}{e^x-\sin(x)}dx$

Solution; $I=\int_0^{\frac{\pi}{2}}\frac{\sin(x)-\cos(x)}{e^x-\sin(x)}dx$

Let : $f(x)=e^x-\sin(x)\Rightarrow f'(x)=e^x-\cos(x)$

If $:x\in\left(0,\frac{\pi}{2}\right)\Rightarrow f(0)=1,\ f\left(\frac{\pi}{2}\right)=e^{\frac{\pi}{2}}-1$

Then : $f'(x)-f(x)=\sin(x)-\cos(x)$

We obtain $I=\int_0^{\frac{\pi}{2}}\frac{f'(x)-f(x)}{f(x)}dx$

$$=\left(\log\left|f(x)\right|-x\right)\Big|_0^{\frac{\pi}{2}}$$

$$=-\frac{\pi}{2}+\log\left(e^{\frac{\pi}{2}}-1\right)$$

$SO,$ $\boxed{\int_0^{\frac{\pi}{2}}\frac{\sin(x)-\cos(x)}{e^x-\sin(x)}dx=-\frac{\pi}{2}+\log\left(\sqrt{e^\pi}-1\right)}$

310. Calculate Integral $I=\int_0^1\tan^{-1}\left(\sec(x)+\tan(x)\right)dx$

Solution; $I=\int_0^1\tan^{-1}\left(\sec(x)+\tan(x)\right)dx$

$$=\int_0^1\tan^{-1}\left(\frac{1+\sin(x)}{\cos(x)}\right)dx$$

$$=\int_0^1\tan^{-1}\left(\frac{1-\cos\left(\frac{\pi}{2}+x\right)}{\sin\left(\frac{\pi}{2}+x\right)}\right)dx$$

$$=\int_0^1 \tan^{-1}\left(\frac{2\sin^2\left(\frac{\pi}{4}+\frac{x}{2}\right)}{2\sin\left(\frac{\pi}{4}+\frac{x}{2}\right)\cos\left(\frac{\pi}{4}+\frac{x}{2}\right)}\right)dx$$

$$=\int_0^1 \tan^{-1}\left(\tan\left(\frac{\pi}{4}+\frac{x}{2}\right)\right)dx$$

$$=\int_0^1\left(\frac{\pi}{4}+\frac{x}{2}\right)dx=\frac{3\pi}{4}$$

SO, $\boxed{\int_0^1 \tan^{-1}\left(\sec(x)+\tan(x)\right)dx=\frac{3\pi}{4}}$

311. Calculate Integral $I=\int_0^{\pi}\frac{x+\sin(x)-\cos(x)-1}{e^x+\sin(x)+x}dx$

Solution; $I=\int_0^{\pi}\frac{x+\sin(x)-\cos(x)-1}{e^x+\sin(x)+x}dx$

Let : $f(x)=e^x+\sin(x)+x \Rightarrow f'(x)=e^x+\cos(x)+1$

If : $x\to 0 \Rightarrow f(0)=1$, *If :* $x\to\pi \Rightarrow f(\pi)=e^{\pi}+\pi$

Then : $f(x)-f'(x)=x+\sin(x)-\cos(x)-1$

Implies That $I=\int_0^{\pi}\frac{f(x)-f'(x)}{f(x)}dx$

$$=\left(x-\log\left|f(x)\right|\right)\Big|_0^{\pi}$$

$$=\pi-\log\left|\frac{e^{\pi}+\pi}{1}\right|$$

$$=\pi-\log\left(e^{\pi}+\pi\right)$$

SO, $\boxed{\int_0^{\pi}\frac{x+\sin(x)-\cos(x)-1}{e^x+\sin(x)+x}dx=\pi-\log\left(e^{\pi}+\pi\right)}$

312. Calculate Integral $I=\int_0^{\pi}\frac{\sin(x)+\frac{x}{2}}{e^x+\sin(x)+\cos(x)+x+1}dx$

Solution; $I=\int_0^{\pi}\frac{\sin(x)+\frac{x}{2}}{e^x+\sin(x)+\cos(x)+x+1}dx$

Let : $f(x)=e^x+\sin(x)+\cos(x)+x+1\Rightarrow f'(x)=e^x+\cos(x)-\sin(x)+1$

If $x\in(0,\pi)\Rightarrow f(0)=3\ ;f(\pi)=e^{\pi}+\pi$

Then : $f(x)-f'(x)=2\sin(x)+x\Leftrightarrow\frac{1}{2}\left(f(x)-f'(x)\right)=\sin(x)+\frac{x}{2}$

We obtain $I=\frac{1}{2}\int_0^{\pi}\frac{f(x)-f'(x)}{f(x)}dx$

$$=\frac{1}{2}\left(x-\log\left|f(x)\right|\right)\Big|_0^{\pi}$$

$$=\frac{1}{2}\left(\pi-\log\left(\frac{e^{\pi}+\pi}{3}\right)\right)$$

SO, $\boxed{\int_0^{\pi}\frac{\sin(x)+\frac{x}{2}}{e^x+\sin(x)+\cos(x)+x+1}dx=\frac{1}{2}\left(\pi-\log\left(\frac{e^{\pi}+\pi}{3}\right)\right)}$

313. Calculate Integral $I=\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\left(\frac{\cos^3(x)+\sin^3(x)}{\cos^4(x)+\sin^4(x)}\right)^2dx$

Solution; $I=\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\left(\frac{\cos^3(x)+\sin^3(x)}{\cos^4(x)+\sin^4(x)}\right)^2dx$

$$=\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\sec^2(x)\left(\frac{1+\tan^3(x)}{1+\tan^4(x)}\right)^2dx$$

Let : $t=\tan(x)\Rightarrow dt=\sec^2(x)dx$ **; If :** $x\in\left(-\frac{\pi}{4},\frac{\pi}{4}\right)\Rightarrow t\in(-1,1)$

We obtain $I=\int_{-1}^{1}\left(\frac{1+t^3}{1+t^4}\right)^2 dt$

$$=\underbrace{\int_{-1}^{0}\left(\frac{1+t^3}{1+t^4}\right)^2 dt}_{I'}+\int_{0}^{1}\left(\frac{1+t^3}{1+t^4}\right)^2 dt \quad (1)$$

For: $I'=\int_{-1}^{0}\left(\frac{1+t^3}{1+t^4}\right)^2 dx$

Let : $t=-u\Rightarrow dt=-du$ **; If :** $x\in(-1,0)\Rightarrow t\in(1,0)$

Implies That : $I'=-\int_{1}^{0}\left(\frac{1+(-u)^3}{1+(-u)^4}\right)^2 du$

$$=\int_{0}^{1}\left(\frac{1-t^3}{1+t^4}\right)^2 dt$$

Take (1) **Then :** $I=\int_{0}^{1}\left(\frac{1-t^3}{1+t^4}\right)^2 dt+\int_{0}^{1}\left(\frac{1+t^3}{1+t^4}\right)^2 dt$

$$=2\int_{0}^{1}\frac{1+t^6}{(1+t^4)^2}dt$$

Let : $t=\frac{1}{z}\Rightarrow dt=-\frac{1}{z^2}dz$ **; If :** $t\in(0,1)\Rightarrow z\in(\infty,1)$

We Get $I=-2\int_{\infty}^{1}\frac{1+\frac{1}{z^6}}{\left(1+\frac{1}{z^4}\right)^2}\times\frac{1}{z^2}dz$

$$=\underbrace{-\frac{1}{2}\int_{1}^{\infty}\frac{z^6+1}{z^3}\times\frac{-4z^3}{(1+z^4)^2}dz}_{\text{Use partial integral}}$$

We obtain $I=\underbrace{-\frac{1}{2}\left(\frac{z^6+1}{z^3}\right)\left(\frac{1}{1+z^4}\right)\Big|_{1}^{\infty}}_{1/2}+\frac{1}{2}\int_{1}^{\infty}\left(\frac{1}{1+z^4}\right)\left(\frac{z^3(6z^5)-(z^6+1)(3z^2)}{z^6}\right)dz$

$$=\frac{1}{2}+\frac{3}{2}\int_1^{\infty}\left(\frac{z^6-1}{z^4(1+z^4)}\right)dz$$

$$=\frac{1}{2}+\frac{3}{2}\int_1^{\infty}\left(\frac{z^2+1}{z^4+1}-\frac{1}{z^4}\right)dz$$

$$=\frac{1}{2}+\frac{3}{2}\int_1^{\infty}\left(\frac{1+z^{-2}}{(z-z^{-1})^2+2}\right)dz-\underbrace{\frac{3}{2}\int_1^{\infty}\left(\frac{1}{z^4}\right)dz}_{\frac{1}{2}}$$

$$=\frac{3}{2}\int_1^{\infty}\frac{d(z+z^{-1})}{(z-z^{-1})^2+\sqrt{2^2}}$$

$$=\frac{3}{2\sqrt{2}}\tan^{-1}\left(\frac{z-z^{-1}}{\sqrt{2}}\right)\Bigg|_1^{\infty}=\frac{3\pi}{4\sqrt{2}}$$

SO, $\boxed{\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\left(\frac{\cos^3(x)+\sin^3(x)}{\cos^4(x)+\sin^4(x)}\right)^2dx=\frac{3\pi}{4\sqrt{2}}}$

Or in another Way $I=2\int_0^1\frac{1+t^6}{(1+t^4)^2}dt$

Let $t=y^{\frac{1}{4}}\Rightarrow dt=\frac{1}{4}y^{\frac{1}{4}-1}dy$ **; If** $x\in(0,1)\Rightarrow y\in(0,1)$

Implies That $I=\frac{1}{2}\int_0^1\frac{1+y^{\frac{6}{4}}}{(1+y)^2}\cdot y^{\frac{1}{4}-1}dt$

$$=\frac{1}{2}\int_0^1\left(\frac{y^{\frac{1}{4}-1}+y^{\frac{7}{4}-1}}{(1+y)^{\frac{1}{4}+\frac{7}{4}}}\right)dt=\frac{1}{2}B\left(\frac{1}{4},\frac{7}{4}\right)$$

$$=\frac{1}{2}\cdot\frac{3}{4}\Gamma\left(\frac{1}{4}\right)\Gamma\left(1-\frac{1}{4}\right)=\frac{\pi}{8}\csc\left(\frac{\pi}{4}\right)$$

$$=\frac{3\pi}{4\sqrt{2}}\quad\textbf{Same Answer}$$

SO, $\boxed{\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\left(\frac{\cos^3(x)+\sin^3(x)}{\cos^4(x)+\sin^4(x)}\right)^2dx=\frac{3\pi}{4\sqrt{2}}}$

314. Calculate Integral $I=\int_1^e \frac{1+x^{2024}}{x+x^{2025}}dx$

Solution; $I=\int_1^e \frac{1+x^{2024}}{x+x^{2025}}dx$

$$=\int_1^e \frac{1}{x+x^{2025}}dx+\frac{1}{2025}\int_1^e \frac{1+2025x^{2024}-1}{x+x^{2025}}dx$$

$$=\int_1^e \frac{1}{x+x^{2025}}dx-\frac{1}{2025}\int_1^e \frac{1}{x+x^{2025}}dx+\frac{1}{2025}\int_1^e \frac{1+2025x^{2024}}{x+x^{2025}}dx$$

$$=\frac{2024}{2025}\int_1^e \frac{1}{x+x^{2025}}dx+\frac{1}{2025}\int_1^e \frac{1+2025x^{2024}}{x+x^{2025}}dx$$

$$=\frac{2024}{2025}\int_1^e \frac{x^{-2025}}{(1+x^{-2024})}dx+\frac{1}{2025}\int_1^e \frac{1+2025x^{2024}}{x+x^{2025}}dx$$

$$=-\frac{1}{2025}\int_1^e \frac{\left(1+x^{-2024}\right)'}{(1+x^{-2024})}dx+\frac{1}{2025}\int_1^e \frac{\left(x+x^{2025}\right)'}{x+x^{2025}}dx$$

$$=\frac{1}{2025}\left[\log\left(x+x^{2025}\right)-\log\left(1+x^{-2024}\right)\right]\Big|_1^e$$

$$=\frac{1}{2025}\left[\left(\log(e+e^{2025})-\log(1+e^{-2024})\right)-\left(\log(2)-\log(2)\right)\right]$$

$$=\frac{1}{2025}\left[\log(e)+\log(1+e^{2024})-\log(1+e^{2024})+\log(e^{2024})\right]$$

$$=\frac{1}{2025}\left[1+2024\right]$$

$$=1$$

SO, $\boxed{\int_1^e \frac{1+x^{2024}}{x+x^{2025}}dx=1}$

315. Calculate Integral $I=\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\frac{2x^{2025}-x^{2023}+1}{\cos(x)}dx$

Solution; $I=\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\frac{2x^{2025}-x^{2023}+1}{\cos(x)}dx$

$$=\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\frac{2(-x)^{2025}-(-x)^{2023}+1}{\cos(-x)}dx$$

$$=-\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\frac{2x^{2025}-x^{2023}-1}{\cos(x)}dx$$

$$=-\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\frac{2x^{2025}-x^{2023}+1-2}{\cos(x)}dx$$

$$=-\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\frac{2x^{2025}-x^{2023}+1}{\cos(x)}dx+2\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\frac{1}{\cos(x)}dx$$

Then $I=-I+2\log\left(\sec(x)+\tan(x)\right)\Big|_{-\frac{\pi}{4}}^{\frac{\pi}{4}}$

Equivalent $2I=2\left[\log\left(\sqrt{2}+1\right)-\log\left(\sqrt{2}-1\right)\right]$

$$=2\log\left(\sqrt{2}+1\right)$$

$SO,$ $\boxed{\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\frac{2x^3-x+1}{\cos(x)}dx=2\log\left(\sqrt{2}+1\right)}$

316. Calculate Integral $I=\int_0^{\frac{\pi}{2}}\frac{1+\sin(3x)}{1+2\sin(x)}dx$

Solution; $I=\int_0^{\frac{\pi}{2}}\frac{1+\sin(3x)}{1+2\sin(x)}dx$

$$=\int_0^{\frac{\pi}{2}}\frac{1+3\sin(x)-4\sin^3(x)}{1+2\sin(x)}dx$$

$$=\int_0^{\frac{\pi}{2}}\frac{1+2\sin(x)+\sin(x)-4\sin^3(x)}{1+2\sin(x)}dx$$

$$= \int_0^{\frac{\pi}{2}} \left(1 + \frac{\sin(x)\left(1-2\sin(x)\right)\left(1+2\sin(x)\right)}{1+2\sin(x)} \right) dx$$

$$= \int_0^{\frac{\pi}{2}} \left(1+\sin(x)-2\sin^2(x)\right) dx$$

$$= \int_0^{\frac{\pi}{2}} (1) dx + \int_0^{\frac{\pi}{2}} \left(\sin(x)\right) dx - 2\int_0^{\frac{\pi}{2}} \left(\sin^2(x)\right) dx$$

$$= \frac{\pi}{2} + 1 - 2 \times \frac{\pi}{2} \times \frac{1}{2} = 1$$

$SO,$ $\boxed{\int_0^{\frac{\pi}{2}} \frac{1+\sin(3x)}{1+2\sin(x)} dx = 1}$

317. Calculate Integral $I = \int_0^{2\pi} \arccos\left(\sin(x)\right) dx$

Solution; $I = \int_0^{2\pi} \arccos\left(\sin(x)\right) dx$

By $: \arccos(t) + \arcsin(t) = \frac{\pi}{2} \equiv \arccos(t) = \frac{\pi}{2} - \arcsin(t)$

If $: x = \sin(x) \Rightarrow \arccos(\sin(x)) = \frac{\pi}{2} - \arcsin(\sin(x))$

We obtain $I = \int_0^{2\pi} \left(\frac{\pi}{2} - \arcsin(\sin(x)) \right) dx$

$$= \int_0^{2\pi} \frac{\pi}{2} dx - \int_0^{2\pi} \arcsin(\sin(x)) dx$$

$$= \pi^2 - \int_0^{\pi} \left[\arcsin(\sin(x)) + \arcsin(\sin(2\pi - x))\right] dx$$

$$= \pi^2 - \underbrace{\int_0^{\pi} \left[\arcsin(\sin(x)) - \arcsin(\sin(x))\right] dx}_{0}$$

$$= \pi^2$$

$SO,$ $\boxed{\int_0^{2\pi} \arccos\left(\sin(x)\right) dx = \pi^2}$

318. Calculate Integral $I=\int_0^1 x^{n-1}\log(1-x)\,dx$

Solution; $I=\int_0^1 x^{n-1}\log(1-x)\,dx$

$$=\frac{1}{n}\int_0^1\left(x^n-1\right)'\log(1-x)\,dx$$

Let : $u=\log(1-x)\Rightarrow du=\frac{1}{1-x}dx$ **And** $dv=\left(x^n-1\right)'dx\Rightarrow v=(x^n-1)$

We obtain $I=\underbrace{\frac{1}{n}(x^n-1)\log(1-x)\Big|_0^1}_{0}-\frac{1}{n}\int_0^1\frac{x^n-1}{1-x}dx$

$$=\frac{1}{n}\int_0^1\frac{x^n-1}{x-1}dx$$

$$=\frac{1}{n}\int_0^1(x^{n-1}+x^{n-2}+\ldots+x+1)dx=\frac{1}{n}\int_0^1\left(\sum_{k=1}^{n}x^{k-1}\right)dx$$

$$=\frac{1}{n}\sum_{k=1}^{n}\left(\int_0^1 x^{k-1}\right)dx=\frac{1}{n}\sum_{k=1}^{n}\left(\frac{1}{k}\right)dx=\frac{H_n}{n}$$

SO, $\boxed{\int_0^1 x^{n-1}\log(1-x)\,dx=\frac{H_n}{n}}$

319. Calculate Integral $I=\int_{-\pi}^{\pi}\frac{x^2}{1+\sin(x)+\sqrt{1+\sin^2(x)}}dx$

Solution; $I=\int_{-\pi}^{\pi}\frac{x^2}{1+\sin(x)+\sqrt{1+\sin^2(x)}}dx$

$$=\underbrace{\int_{-\pi}^{0}\frac{x^2}{1+\sin(x)+\sqrt{1+\sin^2(x)}}dx}_{J}+\int_0^{\pi}\frac{x^2}{1+\sin(x)+\sqrt{1+\sin^2(x)}}dx\quad(*)$$

$$For: J=\int_{-\pi}^{0}\frac{x^2}{1+\sin(x)+\sqrt{1+\sin^2(x)}}dx$$

Let : $x=-u\Rightarrow dx=-du$; **If** : $x\in(-\pi,0)\Rightarrow u\in(\pi,0)$

Implies That : $J=-\int_{\pi}^{0}\frac{(-u)^2}{1+\sin(-u)+\sqrt{1+\sin^2(-u)}}du$

$$=\int_{0}^{\pi}\frac{x^2}{1-\sin(x)+\sqrt{1+\sin^2(x)}}dx$$

Take (*) Then : $I=\int_{0}^{\pi}\frac{x^2}{1-\sin(x)+\sqrt{1+\sin^2(x)}}dx+\int_{0}^{\pi}\frac{x^2}{1+\sin(x)+\sqrt{1+\sin^2(x)}}dx$

$$=\int_{0}^{\pi}\frac{x^2\left(2+2\sqrt{1+\sin^2(x)}\right)}{\left(1+\sqrt{1+\sin^2(x)}\right)^2-\sin^2(x)}dx$$

$$=\int_{0}^{\pi}\frac{x^2\left(2+2\sqrt{1+\sin^2(x)}\right)}{1+2\sqrt{1+\sin^2(x)}+1+\sin^2(x)-\sin^2(x)}dx$$

$$=\int_{0}^{\pi}\frac{x^2\left(2+2\sqrt{1+\sin^2(x)}\right)}{\left(2+2\sqrt{1+\sin^2(x)}\right)}dx$$

$$=\int_{0}^{\pi}x^2dx=\frac{\pi^3}{3}$$

SO, $\boxed{\int_{-\pi}^{\pi}\frac{x^2}{1+\sin(x)+\sqrt{1+\sin^2(x)}}dx=\frac{\pi^3}{3}}$

320. Calculate Integral $I=\int_{0}^{\pi}e^x\sin(x)dx$

Solution; $I=\int_{0}^{\pi}e^x\sin(x)dx$

Suppose $J=\int_{0}^{\pi}e^x\cos(x)dx$

That $J+iI=\int_{0}^{\pi}\left(e^x\cos(x)+ie^x\sin(x)\right)dx$

$$=\int_{0}^{\pi}e^x\left(\cos(x)+i\sin(x)\right)dx$$

$$= \int_0^{\pi} e^{(i+1)x} dx$$

$$= \frac{1}{1+i} e^{(i+1)x} \Big|_0^{\pi}$$

$$= \frac{1}{2}(1-i)\left(e^{i\pi+\pi} - 1\right)$$

$$= -\frac{1}{2}\left(e^{\pi} + 1\right) + \frac{1}{2}\left(e^{\pi} + 1\right)i$$

Pullable $I = \frac{1}{2}\left(e^{\pi} + 1\right)$

SO, $\boxed{\int_0^{\pi} e^x \sin(x) dx = \frac{1}{2}\left(e^{\pi} + 1\right)}$

321. Calculate Integral $I = \int_0^{\infty} \frac{\sin^{2n+1}(x)}{x} dx \quad , \forall n \in \mathbb{N}$

Solution; $I = \int_0^{\infty} \frac{\sin^{2n+1}(x)}{x} dx$

$$= \int_0^{\infty} \frac{\sin(x)}{x} \times \sin^{2n}(x) dx$$

By : $f(x) = \sin^{2n}(x) \equiv f(x) = f(\pi - x) = f(\pi + x)$

We obtain $I = \int_0^{\infty} \frac{\sin(x)}{x} \times \sin^{2n}(x) dx$

$$= \int_0^{\frac{\pi}{2}} \sin^{2n}(x) dx$$

$$= \frac{\pi}{2} \cdot \frac{(2n-1)!!}{(2n)!!}$$

$$= \frac{\pi}{2} \cdot \frac{(2n-1)(2n-3)(2n-5)...(3)(1)}{(2n)(2n-2)...(4)(2)}$$

SO, $\boxed{\int_0^{\infty} \frac{\sin^{2n+1}(x)}{x} dx = \frac{\pi}{2} \times \frac{(2n-1)(2n-3)(2n-5)...(3)(1)}{(2n)(2n-2)...(4)(2)}}$

322. Calculate Integral $I=\int_0^{\frac{\pi}{2}}\frac{\cos(x)}{\sin^5(x)+\cos^5(x)}dx$

Solution; $I=\int_0^{\frac{\pi}{2}}\frac{\cos(x)}{\sin^5(x)+\cos^5(x)}dx$

$$=\int_0^{\frac{\pi}{2}}\frac{1}{\left(\tan^5(x)+1\right)\cos^4(x)}dx$$

$$=\int_0^{\frac{\pi}{2}}\frac{\tan^2(x)+1}{\left(\tan^5(x)+1\right)\cos^2(x)}dx$$

$$=\int_0^{\frac{\pi}{2}}\frac{\tan^2(x)}{\left(\tan^5(x)+1\right)}d\left(\tan(x)\right)+\int_0^{\frac{\pi}{2}}\frac{1}{\left(\tan^5(x)+1\right)}d\left(\tan(x)\right)$$

Let : $\tan^5(x)=u\Leftrightarrow\tan(x)=u^{\frac{1}{5}}\Rightarrow d\left(\tan(x)\right)=\frac{1}{5}u^{\frac{1}{5}-1}du$ **; If :** $x\in\left(0,\frac{\pi}{2}\right)\Rightarrow u\in(0,\infty)$

We obtain $I=\frac{1}{5}\int_0^{\infty}\frac{u^{\frac{2}{5}}}{(u+1)}\times u^{\frac{1}{5}-1}du+\frac{1}{5}\int_0^{\infty}\frac{1}{(u+1)}\times u^{\frac{1}{5}-1}du$

$$=\frac{1}{5}\int_0^{\infty}\frac{u^{\frac{3}{5}-1}}{(u+1)^{\frac{3}{5}+\frac{2}{5}}}du+\frac{1}{5}\int_0^{\infty}\frac{u^{\frac{1}{5}-1}}{(u+1)^{\frac{1}{5}+\frac{4}{5}}}du$$

$$=\frac{1}{5}\left[B\left(\frac{3}{5},\frac{2}{5}\right)+B\left(\frac{1}{5},\frac{4}{5}\right)\right]$$

$$=\frac{1}{5}\left[\Gamma\left(\frac{3}{5}\right)\Gamma\left(1-\frac{3}{5}\right)+\Gamma\left(\frac{1}{5}\right)\Gamma\left(1-\frac{1}{5}\right)\right]$$

$$=\frac{1}{5}\left(\pi\csc\left(\frac{3\pi}{5}\right)+\pi\csc\left(\frac{\pi}{5}\right)\right)$$

$$=\frac{2\pi\sqrt{2}}{5}\left(\frac{1}{\sqrt{5-\sqrt{5}}}+\frac{1}{\sqrt{5+\sqrt{5}}}\right)$$

SO, $\boxed{\int_0^{\frac{\pi}{2}}\frac{\cos(x)}{\sin^5(x)+\cos^5(x)}dx=\frac{2\pi\sqrt{2}}{5}\left(\frac{1}{\sqrt{5-\sqrt{5}}}+\frac{1}{\sqrt{5+\sqrt{5}}}\right)}$

323. Calculate Integral $I=\int_0^{\frac{\pi}{4}}\sqrt{\cos^4(x)-\frac{1}{4}\sin^2(2x)}\,dx$

Solution; $I=\int_0^{\frac{\pi}{4}}\sqrt{\cos^4(x)-\frac{1}{4}\sin^2(2x)}\,dx$

$$=\int_0^{\frac{\pi}{4}}\sqrt{\cos^4(x)-\sin^2(x)\cos^2(x)}\,dx$$

$$=\int_0^{\frac{\pi}{4}}\left|\cos(x)\right|\sqrt{\cos^2(x)-\sin^2(x)}\,dx$$

$$=\int_0^{\frac{\pi}{4}}\cos(x)\sqrt{1-\sin^2(x)-\sin^2(x)}\,dx$$

$$=\int_0^{\frac{\pi}{4}}\cos(x)\sqrt{1-2\sin^2(x)}\,dx$$

Let : $u=\sqrt{2}\sin(x)dx\Rightarrow\cos(x)dx=\frac{1}{\sqrt{2}}du$ **; If :** $x\in\left(0,\frac{\pi}{4}\right)\Rightarrow u\in(0,1)$

We obtain $I=\frac{1}{\sqrt{2}}\int_0^1\sqrt{1-u^2}\,du$

Let : $u=\sin(y)\Rightarrow du=\cos(y)dy$ **; If :** $x\in(0,1)\Rightarrow y\in\left(0,\frac{\pi}{2}\right)$

We Get $I=\frac{1}{\sqrt{2}}\int_0^{\frac{\pi}{2}}\sqrt{1-\sin^2(y)}\cos(y)\,dy$

$$=\frac{1}{\sqrt{2}}\int_0^{\frac{\pi}{2}}\cos^2(y)\,dy$$

$$=\frac{1}{\sqrt{2}}\times\frac{\pi}{2}\times\frac{1}{2}=\frac{\pi}{4\sqrt{2}}$$

SO, $\boxed{\int_0^{\frac{\pi}{4}}\sqrt{\cos^4(x)-\frac{1}{4}\sin^2(2x)}\,dx=\frac{\pi}{4\sqrt{2}}}$

324. Calculate Integral $I=\int_0^1 \sin^2\left(\log(x)\right)dx$

Solution; $I=\int_0^1 \sin^2\left(\log(x)\right)dx$

$$=\frac{1}{2}\int_0^1\left[1-\cos\left(2\log(x)\right)\right]dx$$

$$=\frac{1}{2}-\frac{1}{2}\int_0^1\cos\left(2\log(x)\right)dx$$

Let : $u=\log(x)\Leftrightarrow x=e^u\Rightarrow dx=e^u du$; **If** : $x\in(0,1)\Rightarrow u\in(-\infty,0)$

Implies That $I=\frac{1}{2}-\frac{1}{2}\underbrace{\int_{-\infty}^0\cos\left(2u\right))e^u du}_{Use\ partial\ integral}$

$$=\frac{1}{2}-\frac{1}{2}\times\frac{1}{5}\left(e^u\left(2\sin(2x)+\cos(2x)\right)\right)\Big|_{-\infty}^0$$

$$=\frac{1}{2}-\frac{1}{2}\times\frac{1}{5}=\frac{2}{5}$$

$SO,$ $\boxed{\int_0^1 \sin^2\left(\log(x)\right)dx=\frac{2}{5}}$

325. Calculate Integral $I=\int_0^1 x^{-1}\log^2(x)\sqrt{1-\log(x)}\,dx$

Solution; $I=\int_0^1 x^{-1}\log^2(x)\sqrt{1-\log(x)}\,dx$

Let : $u=-\log(x)\Rightarrow -du=x^{-1}dx$; **If** : $x\in(0,1)\Rightarrow u\in(\infty,0)$

We Get $I=-\int_\infty^0 u^2\sqrt{1+u}\,du$

$$=\int_0^\infty u^2(1+u)^{\frac{1}{2}}\,du$$

$$=\int_0^1\frac{u^{3-1}}{(1+u)^{3-\frac{7}{2}}}du$$

$$=B\left(3,-\frac{7}{2}\right)$$

$$=\frac{\Gamma(3)\Gamma\left(-\frac{7}{2}\right)}{\Gamma\left(3-\frac{7}{2}\right)}$$

$$=\frac{2\left(-\frac{2}{7}\right)\left(-\frac{2}{5}\right)\left(-\frac{2}{3}\right)\Gamma\left(-\frac{1}{2}\right)}{\Gamma\left(-\frac{1}{2}\right)}$$

$$=2\left(-\frac{2}{7}\right)\left(-\frac{2}{5}\right)\left(-\frac{2}{3}\right)=-\frac{16}{105}$$

$SO,$ $\boxed{\int_0^1 x^{-1}\log^2(x)\sqrt{1-\log(x)}\,dx=-\frac{16}{105}}$

326. Calculate Integral $I=\int_e^{\pi}\frac{\log\left(\frac{x}{\pi}\right)}{\log^{\log(e\pi)}(x)}dx$

Solution; $I=\int_e^{\pi}\frac{\log\left(\frac{x}{\pi}\right)}{\log^{\log(e\pi)}(x)}dx$

$$=\int_e^{\pi}\frac{\log(x)-\log(\pi)}{\log^{1+\log(\pi)}(x)}dx$$

$$=\int_e^{\pi}\left(\frac{\log(x)}{\log^{1+\log(\pi)}(x)}-\frac{\log(\pi)}{\log^{1+\log(\pi)}(x)}\right)d(x)$$

$$=\int_e^{\pi}\left(\frac{1}{\log^{\log(\pi)}(x)}-\frac{\log(\pi)}{\log^{1+\log(\pi)}(x)}\right)d(x)$$

Method 1

We Have $I=\int_e^{\pi}\left(\frac{1}{\log^{\log(\pi)}(x)}-\frac{\log(\pi)}{\log^{1+\log(\pi)}(x)}\right)d(x)$

$$= \int_e^{\pi} \left(\frac{(x)'\log^{\log(\pi)}(x)}{\left(\log^{\log(\pi)}(x)\right)^2} - \frac{x\left(\frac{\log(\pi)\log^{\log(\pi)}(x)}{x}\right)}{\left(\log^{\log(\pi)}(x)\right)^2} \right) d(x)$$

$$= \int_e^{\pi} \left(\frac{(x)'\log^{\log(\pi)}(x) - x\left(\log^{\log(\pi)}(x)\right)'}{\left(\log^{\log(\pi)}(x)\right)^2} \right) d(x)$$

$$= \int_e^{\pi} \left(\frac{x}{\log^{\log(\pi)}(x)} \right)' d(x)$$

$$= \frac{x}{\log^{\log(\pi)}(x)}\Bigg|_e^{\pi} = \frac{\pi}{(\log\pi)^{\log(\pi)}} - e$$

Method 2

We Have $I = \int_e^{\pi} \left(\frac{1}{\log^{\log(\pi)}(x)} - \frac{\log(\pi)}{\log^{1+\log(\pi)}(x)} \right) d(x)$

$$= \underbrace{\int_e^{\pi} \left(\frac{1}{\log^{\log(\pi)}(x)} \right) d(x)}_{K'} - \int_e^{\pi} \left(\frac{\log(\pi)}{\log^{1+\log(\pi)}(x)} \right) d(x) \quad (*)$$

For : $I' = \int_e^{\pi} \left(\frac{1}{\log^{\log(\pi)}(x)} \right) d(x)$

Let : $u = \frac{1}{\log^{\log(\pi)}(x)} \Rightarrow du = -\frac{\log(\pi)\log^{\log(\pi)-1}(x)}{x\log^{2\log(\pi)}(x)} dx$ **And** $dv = dx \Rightarrow v = x$

We obtain $I' = \frac{x}{\log^{\log(\pi)}(x)}\Bigg|_e^{\pi} + \int_e^{\pi} \left(\frac{\log(\pi)\log^{\log(\pi)-1}(x)}{x\log^{2\log(\pi)}(x)}(x) \right) d(x)$

$$= \frac{\pi}{(\log\pi)^{\log(\pi)}} - e + \int_e^{\pi} \left(\frac{\log(\pi)}{\log^{1+\log(\pi)}(x)} \right) d(x)$$

Take (*) **Then** $I = \dfrac{\pi}{(\log\pi)^{\log(\pi)}} - e + \int_e^{\pi}\left(\dfrac{\log(\pi)}{\log^{1+\log(\pi)}(x)}\right)d(x) - \int_e^{\pi}\left(\dfrac{\log(\pi)}{\log^{1+\log(\pi)}(x)}\right)d(x)$

$$= \frac{\pi}{(\log\pi)^{\log(\pi)}} - e$$

SO, $\boxed{\int_e^{\pi}\dfrac{\log\left(\dfrac{x}{\pi}\right)}{\log^{\log(e\pi)}(x)}dx = \dfrac{\pi}{(\log\pi)^{\log(\pi)}} - e}$

327. Calculate Integral $I = \int_1^{\infty}\dfrac{1}{(x+1)\sqrt{(x+1)(x-1)}}dx$

Solution; $I = \int_1^{\infty}\dfrac{1}{(x+1)\sqrt{(x+1)(x-1)}}dx$

$$= \int_1^{\infty}\frac{1}{(x+1)\sqrt{(x+1)^2\times\left(\dfrac{x-1}{x+1}\right)}}dx$$

$$= \int_1^{\infty}\frac{1}{(x+1)^2\sqrt{\left(\dfrac{x-1}{x+1}\right)}}dx$$

Let : $u = \left(\dfrac{x-1}{x+1}\right) \Rightarrow \dfrac{1}{2}du = \dfrac{1}{(x+1)^2}dx$; **If** : $x\in(1,\infty) \Rightarrow u\in(0,1)$

We Get $I = \dfrac{1}{2}\int_0^1\dfrac{1}{\sqrt{u}}du$

$$= \sqrt{u}\Big|_0^1 = 1$$

SO, $\boxed{\int_1^{\infty}\dfrac{1}{(x+1)\sqrt{(x+1)(x-1)}}dx = 1}$

328. Calculate Integral $I=\int_1^e \frac{x-1}{x\left(1+\log(x^x)\right)}dx$

Solution; $I=\int_1^e \frac{x-1}{x\left(1+\log(x^x)\right)}dx$

$$=\int_1^e \frac{x^{-1}-x^{-2}}{\left(x^{-1}+\log(x)\right)}dx$$

Let : $u=\log(x)+x^{-1}\Rightarrow du=(x^{-1}-x^{-2})dx$ **; If** : $x\in(1,e)\Rightarrow u\in(1,e^{-1}+1)$

We obtain $I=\int_1^{e^{-1}+1}\frac{1}{(u)}du$

$$=\log(e^{-1}+1)$$

SO, $\boxed{\int_1^e \frac{x-1}{x\left(1+\log(x^x)\right)}dx=\log(e^{-1}+1)}$

329. Calculate Integral $I=\int_0^{\frac{\pi}{2}}\sin(2x)\cos\left(\cos(x)\right)dx$

Solution; $I=\int_0^{\frac{\pi}{2}}\sin(2x)\cos\left(\cos(x)\right)dx$

$$=-2\int_0^{\frac{\pi}{2}}\cos(x)\cos\left(\cos(x)\right)d\left(\cos(x)\right)$$

Let : $\cos(x)=y\Rightarrow d\left(\cos(x)\right)=dy$ **; If** : $x\in\left(0,\frac{\pi}{2}\right)\Rightarrow y\in(1,0)$

We Get $I=-2\int_1^0 y\cos(y)dy$

$$=2\left[y\sin(y)\Big|_0^1-\int_0^1\sin(y)\,dy\right]$$

$$=2\left[\sin(1)+\cos(y)\Big|_0^1\right]=2\left[\sin(1)+\cos(1)-1\right]$$

SO, $\boxed{\int_0^{\frac{\pi}{2}}\sin(2x)\cos\left(\cos(x)\right)dx=2\left(\sin(1)+\cos(1)-1\right)}$

330. Calculate Integral $I=\int_0^{\infty}\frac{x^{m-1}}{e^x+e^{-x}}dx$

Solution; $I=\int_0^{\infty}\frac{x^{m-1}}{e^x+e^{-x}}dx$

$$=\int_0^{\infty}\frac{x^{m-1}e^{-x}}{1+e^{-2x}}dx$$

$$=\int_0^{\infty}\left((x^{m-1}e^{-x})\sum_{n=0}^{\infty}(-1)^n e^{-2nx}\right)dx$$

$$=\sum_{n=0}^{\infty}\left((-1)^n\int_0^{\infty}x^{m-1}e^{-(2n+1)x}\,dx\right)$$

Let : $x=\frac{z}{(2n+1)}\Rightarrow dx=\frac{1}{(2n+1)}dz$ **; If :** $x\in(0,\infty)\Rightarrow z\in(0,\infty)$

We Get $I=\sum_{n=0}^{\infty}\left((-1)^n\int_0^{\infty}\left(\frac{z}{(2n+1)}\right)^{m-1}e^{-z}\frac{dz}{(2n+1)}\right)$

$$=\sum_{n=0}^{\infty}\left(\frac{(-1)^n}{(2n+1)^m}\int_0^{\infty}z^{(m+1)-1}e^{-z}\,dz\right)$$

$$=\sum_{n=0}^{\infty}\left(\frac{(-1)^n}{(2n+1)^m}\Gamma(m+1)\right)$$

$$=\Gamma(m+1)\sum_{n=0}^{\infty}\left(\frac{(-1)^n}{(2n+1)^m}\right)$$

$$=m!B_m$$

Note : $B_m=\sum_{n=0}^{\infty}\left(\frac{(-1)^n}{(2n+1)^m}\right)$

SO, $\boxed{\int_0^{\infty}\frac{x^{m-1}}{e^x+e^{-x}}dx=m!B_m}$

331. Calculate Integral $I=\int_0^{\infty}\frac{\sqrt{x}}{x^2+1}dx$

Solution; $I=\int_0^{\infty}\frac{\sqrt{x}}{x^2+1}dx$

Let : $x=y^{\frac{1}{2}}\Rightarrow dx=\frac{1}{2}y^{\frac{1}{2}-1}dy$ **; If :** $x\in(0,\infty)\Rightarrow y\in(0,\infty)$

We obtain $I=\frac{1}{2}\int_0^{\infty}\frac{\sqrt{y^{\frac{1}{2}}}}{y+1}.y^{\frac{1}{2}-1}dy$

$$=\frac{1}{2}\int_0^{\infty}\frac{y^{\frac{3}{4}-1}}{(y+1)^{\frac{3}{4}+\frac{1}{4}}}dy=\frac{1}{2}B\left(\frac{3}{4},\frac{1}{4}\right)$$

$$=\frac{1}{2}\times\frac{\Gamma\left(\frac{3}{4}\right)\Gamma\left(\frac{1}{4}\right)}{\Gamma\left(\frac{3}{4}+\frac{1}{4}\right)}=\frac{1}{2}\Gamma\left(1-\frac{1}{4}\right)\Gamma\left(\frac{1}{4}\right)$$

$$=\frac{\pi}{2}\csc\left(\frac{\pi}{4}\right)=\frac{\pi}{\sqrt{2}}$$

SO, $\boxed{\int_0^{\infty}\frac{\sqrt{x}}{x^2+1}dx=\frac{\pi}{\sqrt{2}}}$

332. Calculate Integral $I=\int_0^{\frac{\pi}{2}}\left(1-\sin(x)+\sin^2(x)-\sin^3(x)+\cdots\right)dx$

Solution; $I=\int_0^{\frac{\pi}{2}}\left(1-\sin(x)+\sin^2(x)-\sin^3(x)+\cdots\right)dx$

By : $\forall x\in(0,1)$ **Then** $\frac{1}{1+x}=\sum_{n=0}^{\infty}(-x)^n=1-x+x^2-x^3+\cdots$

We obtain $I=\int_0^{\frac{\pi}{2}}\frac{1}{1+\sin(x)}dx$

$$=\int_0^{\frac{\pi}{2}}\frac{1-\sin(x)}{\left(1+\sin(x)\right)\left(1-\sin(x)\right)}dx$$

$$=\int_0^{\frac{\pi}{2}}\frac{1-\sin(x)}{\cos^2(x)}dx$$

$$=\int_0^{\frac{\pi}{2}}\left[\left(\tan(x)\right)'-\left(\frac{1}{\cos(x)}\right)'\right]dx$$

$$=\left(\tan(x)-\frac{1}{\cos(x)}\right)\Bigg|_0^{\frac{\pi}{2}}$$

$$=\lim_{x\to\frac{\pi}{2}}\left(\tan(x)-\frac{1}{\cos(x)}\right)-\lim_{x\to 0}\left(\tan(x)-\frac{1}{\cos(x)}\right)$$

$$=\underbrace{\lim_{x\to\frac{\pi}{2}}\left(\frac{\sin(x)-1}{\cos(x)}\right)}_{Use\ Lopital}-\left(\tan(0)-\frac{1}{\cos(0)}\right)$$

$$=-\lim_{x\to\frac{\pi}{2}}\left(\frac{\cos(x)}{\sin(x)}\right)-(0-1)$$

$$=-0-(0-1)=1$$

$SO,\ \boxed{\int_0^{\frac{\pi}{2}}\left(1-\sin(x)+\sin^2(x)-\sin^3(x)+\cdots\right)dx=1}$

333. Calculate Integral $I=\int_{-2}^{2}\left[x^{2025}\cos\left(\frac{x}{2026}\right)+\frac{1}{2}\right]\sqrt{4-x^2}\,dx$

Solution; $I=\int_{-2}^{2}\left[x^{2025}\cos\left(\frac{x}{2026}\right)+\frac{1}{2}\right]\sqrt{4-x^2}\,dx$

$$=\underbrace{\int_{-2}^{2}x^{2025}\cos\left(\frac{x}{2026}\right)\sqrt{4-x^2}\,dx}_{is\ an\ odd\ function}+\frac{1}{2}\underbrace{\int_{-2}^{2}\sqrt{4-x^2}\,dx}_{is\ an\ even\ function}$$

$$=\int_0^2\sqrt{4-x^2}\,dx$$

Let : $x=2\sin(u)\Rightarrow dx=2\cos(u)du$; **If** : $x\in(0,2)\Rightarrow u\in\left(0,\frac{\pi}{2}\right)$

We obtain $I = 2\int_0^{\frac{\pi}{2}} \sqrt{4-4\sin^2(u)}\cos(u)du$

$$= 4\int_0^{\frac{\pi}{2}} \sqrt{1-\sin^2(u)}\cos(u)du$$

$$= 4\int_0^{\frac{\pi}{2}} \left|\cos(u)\right|\cos(u)du$$

$$= 4\int_0^{\frac{\pi}{2}} \cos^2(u)du$$

$$= 4\cdot\frac{\pi}{2}\cdot\frac{1}{2} = \pi$$

SO, $\boxed{\int_{-2}^{2}\left[x^{2025}\cos\left(\frac{x}{2026}\right)+\frac{1}{2}\right]\sqrt{4-x^2}\,dx = \pi}$

334. Calculate Integral $I = \int_0^1 \tan^{-1}\left(\sec(x)+\tan(x)\right)dx$

Solution; $I = \int_0^1 \tan^{-1}\left(\sec(x)+\tan(x)\right)dx$

$$= \int_0^1 \tan^{-1}\left(\frac{1+\sin(x)}{\cos(x)}\right)dx$$

$$= \int_0^1 \tan^{-1}\left(\frac{1-\cos\left(\pi/2+x\right)}{\sin\left(\pi/2+x\right)}\right)dx$$

$$= \int_0^1 \tan^{-1}\left(\frac{2\sin^2\left(\pi/4+x/2\right)}{2\sin\left(\pi/4+x/2\right)\cos\left(\pi/4+x/2\right)}\right)dx$$

$$= \int_0^1 \tan^{-1}\left(\tan\left(\pi/4+x/2\right)\right)dx$$

$$= \int_0^1 \left(\pi/4+x/2\right)dx$$

$$= \frac{3\pi}{4}$$

SO, $\boxed{\int_0^1 \tan^{-1}\left(\sec(x)+\tan(x)\right)dx = \frac{3\pi}{4}}$

335. Calculate Integral $I=\int_0^1 \frac{1}{x^2+1}dx$

Solution; $I=\int_0^1 \frac{1}{x^2+1}dx$

Suppose : $J=\int \frac{1}{x^2+1}dx$

$$=\frac{1}{2i}\int\left(\frac{1}{x-i}-\frac{1}{x+i}\right)dx$$

$$=\frac{1}{2i}\log\left(\frac{x-i}{x+1}\right)+C$$

If : $Z=a+bi$ **That :** $Z=|r|e^{i\tan^{-1}\left(\frac{b}{a}\right)}$ **Then :** $\begin{cases} x-i=\sqrt{x^2+1}e^{i\tan^{-1}\left(-\frac{1}{x}\right)} \\ x+i=\sqrt{x^2+1}e^{i\tan^{-1}\left(\frac{1}{x}\right)} \end{cases}$

We obtain : $J=\frac{1}{2i}\log\left(\frac{\sqrt{x^2+1}e^{i\tan^{-1}\left(-\frac{1}{x}\right)}}{\sqrt{x^2+1}e^{i\tan^{-1}\left(\frac{1}{x}\right)}}\right)+C$

$$=\frac{1}{2i}\log\left(e^{-i\tan^{-1}\left(\frac{1}{x}\right)}\times e^{-i\tan^{-1}\left(\frac{1}{x}\right)}\right)+C$$

$$=\frac{1}{2i}\left(-2i\tan^{-1}\left(\frac{1}{x}\right)\right)+C$$

$$=-\tan^{-1}\left(\frac{1}{x}\right)+$$

$$=\tan^{-1}(x)+C$$

Implies That : $I=\int_0^1 \frac{1}{x^2+1}dx$

$$=\tan^{-1}(x)\Big|_0^1=\frac{\pi}{4}$$

$SO,$ $\boxed{\int_0^1 \frac{1}{x^2+1}dx=\frac{\pi}{4}}$

336. Calculate Integral $I=\int_0^{\frac{\pi}{2}}\frac{\log\left(1+\sin(x)\right)}{\left(1+\sin(x)\right)}dx$

Solution; $I=\int_0^{\frac{\pi}{2}}\frac{\log\left(1+\sin(x)\right)}{\left(1+\sin(x)\right)}dx$

$$=\int_0^{\frac{\pi}{2}}\frac{\log\left(1+\sin\left(\frac{\pi}{2}-x\right)\right)}{\left(1+\sin\left(\frac{\pi}{2}-x\right)\right)}dx$$

$$=\int_0^{\frac{\pi}{2}}\frac{\log\left(1+\cos(x)\right)}{\left(1+\cos(x)\right)}dx$$

$$=\int_0^{\frac{\pi}{2}}\frac{\log\left(2\cos^2\left(\frac{x}{2}\right)\right)}{\left(2\cos^2\left(\frac{x}{2}\right)\right)}dx$$

Let : $t=\frac{x}{2}\Rightarrow 2dt=dx$ **; If** : $x\in\left(0,\frac{\pi}{2}\right)\Rightarrow t\in\left(0,\frac{\pi}{4}\right)$

We obtain $I=\int_0^{\frac{\pi}{4}}\frac{\log(2)+2\log\left(\cos(t)\right)}{\cos^2(t)}dx$

Let : $u=\log(2)+2\log\left(\cos(t)\right)\Rightarrow du=-2\tan(u)$ **And** $dv=\frac{1}{\cos^2(u)}du\Rightarrow v=\tan(u)$

Implies That $I=\underbrace{\left[\log(2)+2\log\left(\cos(u)\right)\right]\tan(u)\Big|_0^{\frac{\pi}{4}}}_{0}+2\int_0^{\frac{\pi}{4}}\tan^2(u)\,dx$

$$=2\int_0^{\frac{\pi}{4}}\left(\tan(u)-u\right)'dx$$

$$=2\left(\tan(u)-u\right)\Big|_0^{\frac{\pi}{4}}=2-\frac{\pi}{2}$$

SO, $\boxed{\int_0^{\frac{\pi}{2}}\frac{\log\left(1+\sin(x)\right)}{\left(1+\sin(x)\right)}dx=2-\frac{\pi}{2}}$

337. Calculate Integral $I=\int_0^{\pi}\frac{x}{a^2\cos^2(x)+b^2\sin^2(x)}dx$

Solution; $I=\int_0^{\pi}\frac{x}{a^2\cos^2(x)+b^2\sin^2(x)}dx$

$$=\int_0^{\pi}\frac{(\pi-x)}{a^2\cos^2(\pi-x)+b^2\sin^2(\pi-x)}dx$$

$$=\int_0^{\pi}\frac{\pi}{a^2\cos^2(x)+b^2\sin^2(x)}dx-\underbrace{\int_0^{\pi}\frac{x}{a^2\cos^2(x)+b^2\sin^2(x)}dx}_{J}$$

We Get : $2I=\int_0^{\pi}\frac{\pi}{a^2\cos^2(x)+b^2\sin^2(x)}dx$

$$=2\int_0^{\frac{\pi}{2}}\frac{\pi}{a^2+b^2\tan^2(x)}\times\frac{1}{\cos^2(x)}dx$$

Then $I=\pi\int_0^{\frac{\pi}{2}}\frac{1}{a^2+b^2\tan^2(x)}d\left(\tan(x)\right)$

$$=\frac{\pi}{ab}\tan^{-1}\left(\frac{b}{a}\tan(x)\right)\Big|_0^{\frac{\pi}{2}}=\frac{\pi^2}{2ab}$$

SO, $\boxed{\int_0^{\pi}\frac{x}{a^2\cos^2(x)+b^2\sin^2(x)}dx=\frac{\pi^2}{2ab}}$

338. Calculate Integral $I=\int_0^1 x\sin(\pi x)dx$

Solution; $I=\int_0^1 x\sin(\pi x)dx$

$$=\frac{1}{\pi}\int_0^1 x\sin(\pi x)d(\pi x)$$

$$=\frac{1}{\pi^2}\int_0^1(\pi x)\sin(\pi x)d(\pi x)$$

$$= \frac{1}{\pi^2}\int_0^1 (\pi - \pi x)\sin(\pi - \pi x)d(\pi x)$$

$$= \frac{1}{\pi^2}\int_0^1 (\pi - \pi x)\sin(\pi x)d(\pi x)$$

$$= \frac{1}{\pi^2}\int_0^1 \pi\sin(\pi x)d(\pi x) - \frac{1}{\pi^2}\int_0^1 (\pi x)\sin(\pi x)d(\pi x)$$

$$= \frac{1}{\pi^2}\int_0^1 \pi\sin(\pi x)d(\pi x) - \underbrace{\frac{1}{\pi}\int_0^1 x\sin(\pi x)d(\pi x)}_{I}$$

Implies That $2I = \frac{1}{\pi}\int_0^1 \sin(\pi x)d(\pi x)$

$$= -\frac{1}{\pi}\cos(\pi x)\Big|_0^1 = \frac{2}{\pi} \quad \textbf{Then} \quad I = \frac{1}{\pi}$$

SO, $\boxed{\int_0^1 x\sin(\pi x)dx = \frac{1}{\pi}}$

339. Calculate Integral $I = \int_0^1 x^{m-1}\log(1-x)dx$

Method 1

Solution; $I = \int_0^1 x^{m-1}\log(1-x)dx$

$$= \int_0^1 \left[x^{m-1}\left(-\sum_{n=1}^{\infty}\frac{x^n}{n} \right) \right] dx$$

$$= -\sum_{n=1}^{\infty}\left(\frac{1}{n}\int_0^1 x^{m-1}x^n dx \right)$$

$$= -\sum_{n=1}^{\infty}\left(\frac{1}{n}\int_0^1 x^{m+n-1}dx \right)$$

$$= -\sum_{n=1}^{\infty}\left(\frac{1}{n(n+m)} \right)$$

$$= -\frac{1}{m}\sum_{n=1}^{\infty}\left(\frac{1}{n} - \frac{1}{n+m} \right) = -\frac{1}{m}\left(\psi(m+1) + \gamma \right)$$

Method 2

Solution; $I=\underbrace{\int_0^1 x^{m-1}\log(1-x)dx}_{Use\ partial\ integral}$

We Get $I=\dfrac{x^m\log(1-x)}{m}\Big|_0^1+\dfrac{1}{m}\int_0^1\dfrac{x^m}{1-x}dx$

$$=\frac{x^m\log(1-x)}{m}\Big|_0^1-\frac{1}{m}\int_0^1\frac{1-x^m-1}{1-x}dx$$

$$=\underbrace{\frac{(x^m-1)\log(1-x)}{m}\Big|_0^1}_{0}-\frac{1}{m}\int_0^1\frac{1-x^m}{1-x}dx$$

$$=-\frac{1}{m}\left(\psi(m+1)+\gamma\right)$$

$SO,$ $\boxed{\int_0^1 x^{m-1}\log(1-x)dx=-\frac{1}{m}\left(\psi(m+1)+\gamma\right)}$

Note : $\int_0^1\frac{1-x^m}{1-x}dx=\sum_{n=1}^{\infty}\left(\frac{1}{n}-\frac{1}{n+m}\right)=\left(\psi(m+1)+\gamma\right)\quad;\quad\psi(x+1)=-\gamma+\sum_{n=1}^{\infty}\left(\frac{1}{n}-\frac{1}{n+m}\right)$

340. Calculate Integral $I=\int_0^{\infty}W\left(\frac{1}{x^n}\right)dx$

Solution; $I=\int_0^{\infty}W\left(\frac{1}{x^n}\right)dx$

Let : $u=\frac{1}{x^n}\Leftrightarrow x=u^{-\frac{1}{n}}\Rightarrow dx=-\frac{1}{n}u^{-\frac{1}{n}-1}du$; **If :** $x\in(0,\infty)\Rightarrow u\in(\infty,0)$

Implies That $I=-\frac{1}{n}\int_{\infty}^0 u^{-\frac{1}{n}-1}W(u)\,du$

$$=\frac{1}{n}\int_0^{\infty}u^{-\frac{1}{n}-1}W(u)\,du$$

Let : $t = W(u) \Leftrightarrow u = te^t \Rightarrow du = (t+1)e^t dt$; **If** : $u \in (0,\infty) \Rightarrow t \in (0,\infty)$

We Get $I = \frac{1}{n}\int_0^\infty \left[t\left(te^t\right)^{-\frac{1}{n}-1}(t+1)e^t \right] dt$

$$= \frac{1}{n}\int_0^\infty \left[\left(t^{1-\frac{1}{n}} + t^{-\frac{1}{n}} \right) e^{-\frac{t}{n}} \right] dt$$

Let : $\frac{t}{n} = z \Leftrightarrow t = nz \Rightarrow dt = ndz$; **If** : $t \in (0,\infty) \Rightarrow z \in (0,\infty)$

Implies That $I = \frac{1}{n}\int_0^\infty \left[\left((nz)^{1-\frac{1}{n}} + (nz)^{-\frac{1}{n}} \right) e^{-z} \right] ndz$

$$= \int_0^\infty (nz)^{1-\frac{1}{n}} e^{-z} dz + \int_0^\infty (nz)^{-\frac{1}{n}} e^{-z} dz$$

$$= n^{\left(1-\frac{1}{n}\right)} \int_0^\infty z^{\left(2-\frac{1}{n}\right)-1} e^{-z} dz + n^{-\frac{1}{n}} \int_0^\infty z^{\left(1-\frac{1}{n}\right)-1} e^{-z} dz$$

$$= n^{\left(1-\frac{1}{n}\right)} \Gamma\left(2-\frac{1}{n}\right) + n^{-\frac{1}{n}} \Gamma\left(1-\frac{1}{n}\right)$$

$$= \left[n^{\left(1-\frac{1}{n}\right)} \left(1-\frac{1}{n}\right) \Gamma\left(1-\frac{1}{n}\right) + n^{-\frac{1}{n}} \Gamma\left(1-\frac{1}{n}\right) \right]$$

$$= \left[n^{\left(1-\frac{1}{n}\right)} \left(1-\frac{1}{n}\right) + n^{-\frac{1}{n}} \right] \Gamma\left(1-\frac{1}{n}\right)$$

$$= \left[n^{\left(1-\frac{1}{n}\right)} - n^{-\frac{1}{n}} + n^{-\frac{1}{n}} \right] \Gamma\left(1-\frac{1}{n}\right)$$

$$= n^{\left(1-\frac{1}{n}\right)} \Gamma\left(1-\frac{1}{n}\right)$$

SO, $\boxed{\int_0^\infty W\left(\frac{1}{x^n}\right) dx = n^{\left(1-\frac{1}{n}\right)} \Gamma\left(1-\frac{1}{n}\right)}$

341. Calculate Integral $I=\int_0^{\infty}\frac{W(x)}{x\sqrt{x}}dx$

Solution; $I=\int_0^{\infty}\frac{W(x)}{x\sqrt{x}}dx$

Let : $x=ue^u \Rightarrow dx=(u+1)e^u du$; **If** : $x\in(0,\infty)\Rightarrow u\in(0,\infty)$

We obtain $I=\int_0^{\infty}\frac{W(ue^u)}{ue^u\sqrt{ue^u}}\times(u+1)e^u du$

$$=\int_0^{\infty}\frac{u(u+1)e^u}{ue^u\sqrt{ue^u}}du$$

$$=\int_0^{\infty}\frac{(u+1)}{\sqrt{ue^u}}du$$

$$=\int_0^{\infty}\left(u^{\frac{1}{2}}+u^{-\frac{1}{2}}\right)e^{-\frac{1}{2}u}du$$

Let : $y=\frac{1}{2}u\Leftrightarrow u=2y\Rightarrow 2dy=du$; **If:** $x\in(0,\infty)\Rightarrow u\in(0,\infty)$

We Get $I=2\int_0^{\infty}\left((2y)^{\frac{1}{2}}+(2y)^{-\frac{1}{2}}\right)e^{-y}dy$

$$=2\sqrt{2}\int_0^{\infty}y^{\left(1+\frac{1}{2}\right)-1}e^{-y}dy+\frac{2}{\sqrt{2}}\int_0^{\infty}y^{\frac{1}{2}-1}e^{-y}dy$$

$$=2\sqrt{2}\Gamma\left(1+\frac{1}{2}\right)+\sqrt{2}\Gamma\left(\frac{1}{2}\right)$$

$$=2\sqrt{2}\times\frac{1}{2}\Gamma\left(\frac{1}{2}\right)+\sqrt{2}\Gamma\left(\frac{1}{2}\right)$$

$$=2\sqrt{2}\Gamma\left(\frac{1}{2}\right)=2\sqrt{2\pi}$$

SO, $\boxed{\int_0^{\infty}\frac{W(x)}{x\sqrt{x}}dx=2\sqrt{2\pi}}$

342. Calculate Integral $I=\int_0^{\infty}\left(W(x)e^{-W(x)}\right)^n dx$

Solution; $I=\int_0^{\infty}\left(W(x)e^{-W(x)}\right)^n dx$

Let : $x=ye^y \Rightarrow dx=(y+1)e^y dy$ **; If** : $x\in(0,\infty)\Rightarrow y\in(0,\infty)$ **Note** : $W(xe^x)=x$

We obtain $I=\int_0^{\infty}\left(ye^{-y}\right)^n(y+1)e^y dy$

$$=\int_0^{\infty}(y^{n+1}+y^n)e^{-(n-1)y}dy$$

Let : $y=\frac{z}{(n+1)}\Rightarrow dy=dy=\frac{1}{(n+1)}dz$ **; If** : $y\in(0,\infty)\Rightarrow z\in(0,\infty)$

We Get $I=\int_0^{\infty}\left[\left(\frac{z}{(n+1)}\right)^{n+1}+\left(\frac{z}{(n+1)}\right)^n\right]e^{-z}\frac{1}{(n+1)}dz$

$$=\int_0^{\infty}\left(\frac{1}{(n+1)^{n+2}}z^{n+1}+\frac{1}{(n+1)^{n+1}}z^n\right)e^{-z}dz$$

$$=\frac{1}{(n+1)^{n+2}}\int_0^{\infty}\left(z^{n+1}e^{-z}\right)dz+\frac{1}{(n+1)^{n+1}}\int_0^{\infty}\left(z^n e^{-z}\right)dz$$

$$=\frac{\Gamma(n+2)}{(n+1)^{n+2}}+\frac{\Gamma(n+1)}{(n+1)^{n+1}}$$

$$=\frac{(n+1)\Gamma(n+1)}{(n+1)^{n+2}}+\frac{\Gamma(n+1)}{(n+1)^{n+1}}$$

$$=\frac{2\Gamma(n+1)}{(n+1)^{n+1}}$$

SO, $\boxed{\int_0^{\infty}\left(W(x)e^{-W(x)}\right)^n dx=\frac{2\Gamma(n+1)}{(n+1)^{n+1}}}$

343. Calculate Integral $I=\int_0^{\pi}\sin^5(x)\left(1-\cos(x)\right)^3\,dx$

Solution; $I=\int_0^{\pi}\sin^5(x)\left(1-\cos(x)\right)^3\,dx$

Let $:x=y+\dfrac{\pi}{2}\Rightarrow dx=dy$ **; If** $:x\in(0,\pi)\Rightarrow y\in\left(-\dfrac{\pi}{2},\dfrac{\pi}{2}\right)$

Implies That $I=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\sin^5\left(\dfrac{\pi}{2}+y\right)\left[1-\cos\left(\dfrac{\pi}{2}+y\right)\right]^3\,dy$

$$=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\cos^5(y)\left(1-\sin(y)\right)^3\,dy$$

$$=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\cos^5(y)\left(1+3\sin(y)+3\sin^2(y)+\sin^3(y)\right)dy$$

$$=\underbrace{\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\cos^5(y)\left(1+3\sin^2(y)\right)dy}_{\text{is an even function}}+\underbrace{\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\cos^5(y)\left(3\sin(y)+\sin^3(y)\right)dy}_{\text{is an odd function}}$$

$$=2\int_0^{\frac{\pi}{2}}\cos^5(y)\left(1+3\sin^2(y)\right)dy$$

$$=2\int_0^{\frac{\pi}{2}}\cos^5(y)\left(4-3\cos^2(y)\right)dy$$

$$=8\int_0^{\frac{\pi}{2}}\cos^5(y)dy-6\int_0^{\frac{\pi}{2}}\cos^7(y)dy$$

$$=8\cdot\frac{4\times2}{5\times3\times1}-6\cdot\frac{6\times4\times2}{7\times5\times3\times1}$$

$$=\frac{32}{21}$$

SO, $\boxed{\int_0^{\pi}\sin^5(x)\left(1-\cos(x)\right)^3\,dx=\frac{32}{21}}$

344. Calculate Integral $I=\int_0^{\frac{\pi}{2}}\frac{\sin(x)-\cos(x)}{e^x-\sin(x)}dx$

Solution; $I=\int_0^{\frac{\pi}{2}}\frac{\sin(x)-\cos(x)}{e^x-\sin(x)}dx$

Let: $f(x)=e^x-\sin(x)\Rightarrow f'(x)=e^x-\cos(x)$

If : $x\in\left(0,\frac{\pi}{2}\right)\Rightarrow f(0)=1,\ f\left(\frac{\pi}{2}\right)=e^{\frac{\pi}{2}}-1$

Then : $f'(x)-f(x)=\sin(x)-\cos(x)$

We Get $I=\int_0^{\frac{\pi}{2}}\frac{f'(x)-f(x)}{f(x)}dx=\int_0^{\frac{\pi}{2}}\left(\frac{f'(x)}{f(x)}-1\right)dx$

$$=\left(\log|f(x)|-x\right)\Big|_0^{\frac{\pi}{2}}$$

$$=\left(\log\left(e^{\frac{\pi}{2}}-1\right)-\frac{\pi}{2}\right)-\left(\log|1|-0\right)=-\frac{\pi}{2}+\log\left(e^{\frac{\pi}{2}}-1\right)$$

SO, $\boxed{\int_0^{\frac{\pi}{2}}\frac{\sin(x)-\cos(x)}{e^x-\sin(x)}dx=-\frac{\pi}{2}+\log\left|e^{\frac{\pi}{2}}-1\right|}$

345. Calculate Integral $I=\int_0^{\pi}\sec(x)\log\left(1+\frac{1}{2}\cos(x)\right)dx$

Solution; $I=\int_0^{\pi}\sec(x)\log\left(1+\frac{1}{2}\cos(x)\right)dx$

Then $I(a)=\int_0^{\pi}\sec(x)\log\left(1+a\cos(x)\right)dx$ **Which** $I(0)=0$ **And** $I\left(\frac{1}{2}\right)=I$

That $I'(a)=\int_0^{\pi}\frac{\sec(x)\cos(x)}{1+a\cos(x)}dx$

$$= \int_0^{\pi} \frac{1}{1+a\cos(x)}\,dx$$

Let : $t = \tan\left(\frac{x}{2}\right) \Rightarrow dx = \frac{2}{1+t^2}\,dt$; **If** : $x \in (0,\pi) \Rightarrow t \in (0,\infty)$ **Note** : $\cos(x) = \frac{1-t^2}{t^2+1}$

We Get $I'(a) = \int_0^{\infty} \frac{1}{1+a\frac{1-t^2}{t^2+1}} \times \frac{2}{1+t^2}\,dt$

$$= 2\int_0^{\infty} \frac{1}{(1+t^2)+a(1-t^2)}\,dt$$

$$= 2\int_0^{\infty} \frac{1}{(1+a)+(1-a)t^2}\,dt$$

$$= \frac{2}{1-a}\int_0^{\infty} \frac{1}{\sqrt{\left(\frac{1+a}{1-a}\right)^2 + t^2}}\,dt$$

$$= \frac{2}{1-a} \times \frac{\sqrt{1-a}}{\sqrt{1+a}} \tan^{-1}\left(\frac{\sqrt{1-a}}{\sqrt{1+a}}t\right)\Bigg|_0^{\infty}$$

$$= \frac{\pi}{\sqrt{1-a^2}}$$

Implies That $I(a) = \int \frac{\pi}{\sqrt{1-a^2}}\,da$

$$= \pi\sin^{-1}(a) + C$$

If : $a = 0 \Rightarrow I(0) = 0 = \pi\sin^{-1}(0) + C \Rightarrow C = 0$

If : $a = \frac{1}{2} \Rightarrow I\left(\frac{1}{2}\right) = I = \pi\sin^{-1}\left(\frac{1}{2}\right) + 0 = \pi \times \frac{\pi}{6} = \frac{\pi^2}{6}$

$SO,$ $\boxed{\int_0^{\pi} \sec(x)\log\left(1+\frac{1}{2}\cos(x)\right)dx = \frac{\pi^2}{6}}$

346. Calculate Integral $I=\int_0^{\infty}\frac{1-e^{\pi x^2}}{x^2}dx$

Solution; $I=\int_0^{\infty}\frac{1-e^{\pi x^2}}{x^2}dx$

Then $I(a)=\int_0^{\infty}\frac{1-e^{-ax^2}}{x^2}dx$ **Which** $I(0)=0$ **And** $I(\pi)=I$

That $I'(a)=\int_0^{\infty}\frac{x^2e^{-ax^2}}{x^2}dx$

$$=\int_0^{\infty}e^{-ax^2}\,dx$$

Let : $u=\sqrt{a}x\Rightarrow du=\sqrt{a}dx$ **; If :** $x\in(0,\infty)\Rightarrow u\in(0,\infty)$

We Get $I'(a)=\frac{1}{\sqrt{a}}\int_0^{\infty}e^{-u^2}\,du$

$$=\frac{1}{\sqrt{a}}\times\frac{\sqrt{\pi}}{2}$$

Implies That $I(a)=\frac{\sqrt{\pi}}{2}\int\left(\frac{1}{\sqrt{a}}\right)da$

$$=\sqrt{\pi a}+C$$

If : $a=0\Rightarrow I(0)=0=\sqrt{\pi\times 0}+C\Rightarrow C=0$

If : $a=\pi\Rightarrow I(\pi)=I=\sqrt{\pi\times\pi}+0\Rightarrow I=\pi$

SO, $\boxed{\int_0^{\infty}\frac{1-e^{-\pi x^2}}{x^2}dx=\pi}$

347. Calculate Integral $I=\int_0^{\infty} x^{n-1}\log(1+x)dx$

Solution; $I=\int_0^{\infty} x^{n-1}\log(1+x)dx$

Then $I(a)=\int_0^{\infty} x^{n-1}\log(1+ax)dx$ **Which** $I(0)=0$ **And** $I(1)=I$

That $I'(a)=\int_0^{\infty} x^{n-1}\frac{x}{(1+ax)}dx$

$$=\int_0^{\infty}\frac{x^n}{(1+ax)}dx$$

Let : $x=\frac{u}{a}\Rightarrow dx=\frac{1}{a}du$ **; If** : $x\in(0,\infty)\Rightarrow u\in(0,\infty)$

We obtain $I'(a)=\frac{1}{a}\int_0^{\infty}\frac{\left(\frac{u}{a}\right)^n}{(1+u)}du$

$$=\frac{1}{a^{n+1}}\int_0^{\infty}\frac{u^{(n+1)-1}}{(1+u)^{(n+1)-n}}du$$

$$=-\frac{\pi}{a^{n+1}\sin(\pi n)}$$

We get $I(a)=-\frac{\pi}{\sin(\pi n)}\int a^{-(n+1)}da$

$$=\frac{\pi}{n\sin(\pi n)}a^{-n}+C$$

If : $a=0\Rightarrow J(0)=0=0+C\Rightarrow C=0$

If *If* : $a=1\Rightarrow J(1)=J=\frac{\pi}{n\sin(\pi n)}(1)^{-n}+0=\frac{\pi}{n\sin(\pi n)}$

SO, $\boxed{\int_0^{\infty} x^{n-1}\log(1+x)dx=\frac{\pi}{n}\csc(n\pi)}$

348. Calculate Integral $I=\int_0^{\infty}\frac{e^{-\pi x}-e^{-ex}}{x}dx$

Solution; $I=\int_0^{\infty}\frac{e^{-\pi x}-e^{-ex}}{x}dx$

$$=\int_0^{\infty}\frac{1-e^{-(e-\pi)x}}{x}e^{-\pi x}dx$$

Then $I(a)=\int_0^{\infty}\frac{1-e^{-(e-\pi)x}}{x}e^{-ax}dx$ **Which** $I(\infty)=0$ **And** $I(\pi)=I$

That $I'(a)=\int_0^{\infty}\frac{\left(1-e^{-(e-\pi)x}\right)(-x)e^{-ax}}{x}dx$

$$=\int_0^{\infty}\left(1-e^{-(e-\pi)x}\right)e^{-ax}dx$$

$$=\int_0^{\infty}e^{-(e-\pi+a)x}dx-\int_0^{\infty}e^{-ax}dx$$

$$=\frac{1}{e-\pi+a}-\frac{1}{a}$$

We obtain $I(a)=\int\left(\frac{1}{e-\pi+a}-\frac{1}{a}\right)da$

$$=\log\left(\frac{a+e-\pi}{a}\right)+C$$

If : $a=\infty\Rightarrow I(\infty)=0=\lim_{a\to\infty}\left[\log\left(\frac{\pi+e-\pi}{\pi}\right)+C\right]=0+C\Rightarrow C=0$

If : $a=\pi\Rightarrow I(\pi)=I=\log\left(\frac{\pi+e-\pi}{\pi}\right)+0=\log\left(\frac{e}{\pi}\right)$

We Get : $I=\log\left(\frac{e}{\pi}\right)$

$SO,$ $\boxed{\int_0^{\infty}\frac{e^{-\pi x}-e^{-ex}}{x}dx=\log\left(\frac{e}{\pi}\right)}$

349. Calculate Integral $I=\int_0^{\pi}\frac{\log\left(1-\sin(x)\right)}{\sin(x)}dx$

Solution; $I=\int_0^{\pi}\frac{\log\left(1-\sin(x)\right)}{\sin(x)}dx$

Then $I(a)=\int_0^{\pi}\frac{\log\left(1-\sin(a)\sin(x)\right)}{\sin(x)}dx$ **Which** $I(0)=0$ **And** $I\left(\frac{\pi}{2}\right)=I$

That $I'(a)=-\int_0^{\pi}\frac{\cos(a)\sin(x)}{\left(1-\sin(a)\sin(x)\right)\sin(x)}dx$

$$=-\cos(a)\int_0^{\pi}\frac{1}{1-\sin(a)\sin(x)}dx$$

Let : $y=\tan\left(\frac{x}{2}\right)\Rightarrow dx=\frac{2}{1+y^2}dy$ **; If :** $x\in(0,\pi)\Rightarrow y\in(0,\infty)$ **Note :** $\sin(x)=\frac{2y}{1+y^2}$

We obtain $I'(a)=\cos(a)\int_0^{\infty}\frac{1}{1-\sin(a)\times\frac{2y}{1+y^2}}\times\frac{2}{1+y^2}dy$

$$=-2\cos(a)\int_0^{\infty}\frac{1}{\cos^2(y)+(y-\sin(x))^2}dy$$

$$=-2\cos(a)\left[\frac{1}{\cos(a)}\tan^{-1}\left(\frac{y-\sin(a)}{\cos(a)}\right)\right]_0^{\infty}$$

$$=-2\left(\frac{\pi}{2}+a\right)$$

We Get $I(a)=-2\int\left(\frac{\pi}{2}+a\right)da=-(\pi a+a^2)+C$

If : $a=0\Rightarrow I(0)=0=0+C\Rightarrow C=0$

If : $a=\frac{\pi}{2}\Rightarrow I\left(\frac{\pi}{2}\right)=I=-\left(\frac{\pi^2}{2}+\frac{\pi^2}{4}\right)+0=-\frac{3\pi^2}{4}$

SO, $\boxed{\int_0^{\pi}\frac{\log\left(1-\sin(x)\right)}{\sin(x)}dx=-\frac{3\pi^2}{4}}$

350. Calculate Integral $I=\int_0^{\infty}\frac{\tan^{-1}(ex)-\tan^{-1}(x)}{x}dx$

Solution; $I=\int_0^{\infty}\frac{\tan^{-1}(ex)-\tan^{-1}(x)}{x}dx$

Then $I(a)=\int_0^{\infty}\frac{\tan^{-1}(ax)-\tan^{-1}(x)}{x}dx$ **Which** $I(0)=0$ **And** $I(1)=I$

That $I'(a)=\int_0^{\infty}\frac{1}{x}\left(\frac{x}{1+(ax)^2}\right)dx$

$$=\int_0^{\infty}\left(\frac{1}{1+(ax)^2}\right)dx=\frac{\pi}{2a}$$

Implies That $I(a)=\frac{\pi}{2}\log(a)+C$

If : $a=1\Rightarrow I(1)=\frac{\pi}{2}\log(0)+C=0\Rightarrow C=0$

If : $a=e\Rightarrow I(e)=I=\frac{\pi}{2}\log(e)+0=\frac{\pi}{2}$

SO, $\boxed{\int_0^{\infty}\frac{\tan^{-1}(ex)-\tan^{-1}(x)}{x}dx=\frac{\pi}{2}}$

351. Calculate Integral $I=\int_1^{\infty}\frac{x-1}{x^4\log(x)}dx$

Solution; $I=\int_1^{\infty}\frac{x-1}{x^4\log(x)}dx$

Let : $u=\log(x)\Leftrightarrow x=e^u\Rightarrow x=e^u du$ **; If :** $x\in(1,\infty)\Rightarrow u\in(0,\infty)$

Implies That $I=\int_0^{\infty}\frac{e^u-1}{ue^{4u}}\times e^u du$

$$=\int_0^{\infty}\frac{1-e^{-u}}{u}e^{-2u}du$$

Then $I(a)=\int_0^{\infty}\frac{1-e^{-u}}{u}e^{-au}du$ **Which** $I(\infty)=0$ **And** $I(2)=I$

That $I'(a)=-\int_0^{\infty}\left(1-e^{-u}\right)e^{-au}du$

$$=\int_0^{\infty}e^{-(a+1)u}du-\int_0^{\infty}e^{-au}du$$

$$=-\frac{1}{a+1}e^{-(a+1)u}\Big|_0^{\infty}+\frac{1}{a}e^{-au}\Big|_0^{\infty}$$

$$=\frac{1}{a+1}-\frac{1}{a}$$

Implies That $I(a)=\int\left(\frac{1}{a+1}-\frac{1}{a}\right)da$

$$=\log\left|\frac{a+1}{a}\right|+C$$

If : $a=\infty\Rightarrow I(\infty)=0=0+C\Rightarrow C=0$

If: $a=2\Rightarrow I(2)=I=\log\left|\frac{2+1}{2}\right|+C$ **That :** $I=\log\left(\frac{3}{2}\right)$

SO, $\boxed{\int_1^{\infty}\frac{x-1}{x^4\log(x)}dx=\log\left(\frac{3}{2}\right)}$

352. Calculate Integral $I=\int_0^{\infty}\frac{e^{-2x}\cos(x)-e^{-3x}}{x}dx$

Solution; $I=\int_0^{\infty}\frac{e^{-2x}\cos(x)-e^{-3x}}{x}dx$

$$=\int_0^{\infty}\frac{\cos(x)-e^{-x}}{x}e^{-2x}dx$$

Then $I(a)=\int_0^{\infty}\frac{\cos(x)-e^{-x}}{x}e^{-ax}dx$

That $I'(a) = -\int_0^{\infty} \left(\cos(x) - e^{-x}\right) e^{-ax} dx$

We obtain $I'(a) = \underbrace{\int_0^{\infty} e^{-(a+1)x} dx}_{K_1} - \underbrace{\int_0^{\infty} \cos(x) e^{-ax} dx}_{K_2}$

For : $K_1 = \int_0^{\infty} e^{-(a+1)x} dx$

$$= -\frac{1}{a+1} e^{-(a+1)x} \Big|_0^{\infty}$$

$$= \frac{1}{a+1}$$

For : $K_2 = \underbrace{\int_0^{\infty} \cos(x) e^{-ax} dx}_{(Use\ partial\ integral)}$

$$= \frac{1}{a^2+1} \left[e^{-ax} \left(\sin(x) - a\cos(x)\right) \right]_0^{\infty}$$

$$= \frac{a}{a^2+1}$$

Implies That : $I'(a) = \frac{1}{a+1} - \frac{a}{a^2+1}$

We Get $I(a) = \log(a+1) - \frac{1}{2}\log(a^2+1) + C$

$$= \log\left(\frac{a+1}{\sqrt{a^2+1}}\right) + C$$

If : $a = \infty \Rightarrow I(\infty) = 0 = \lim_{a\to\infty} \left[\log\left(\frac{a+1}{\sqrt{a^2+1}}\right) + C \right] \equiv 0 = 0 + C \Rightarrow C = 0$

If : $a = 2 \Rightarrow I(2) = I = \log\left(\frac{2+1}{\sqrt{2^2+1}}\right) + 0 = \log\left(\frac{3}{\sqrt{5}}\right)$

$SO,$ $\boxed{\int_0^{\infty} \frac{e^{-2x}\cos(x) - e^{-3x}}{x} dx = \log\left(\frac{3}{\sqrt{5}}\right)}$

353. Calculate Integral $I=\int_{-1}^{1}\left|3^x-2^x\right|dx$

Solution; $I=\int_{-1}^{1}\left|3^x-2^x\right|dx$

$$=\int_{-1}^{0}\left|3^x-2^x\right|dx+\int_{0}^{1}\left|3^x-2^x\right|dx$$

By $\forall x\in(-1,0)$ **Then** $3^x-2^x\le 0$ **We Get :** $\int_{-1}^{0}\left|3^x-2^x\right|dx=-\int_{-1}^{0}\left(3^x-2^x\right)dx$

And $\forall x\in(0,1)$ **Then** $3^x-2^x\ge 0$ **We Get :** $\int_{0}^{1}\left|3^x-2^x\right|dx=\int_{0}^{1}\left(3^x-2^x\right)dx$

Implies That $I=-\int_{-1}^{0}\left(3^x-2^x\right)dx+\int_{0}^{1}\left(3^x-2^x\right)dx$

$$=\left(\frac{2^x}{\log(2)}-\frac{3^x}{\log(3)}\right)\Bigg|_{-1}^{0}+\left(\frac{3^x}{\log(3)}-\frac{2^x}{\log(2)}\right)\Bigg|_{0}^{1}$$

$$=\left(\frac{3^{-1}+1}{\log(3)}\right)-\left(\frac{2^{-1}}{\log(2)}\right)$$

SO, $\boxed{\int_{-1}^{1}\left|3^x-2^x\right|dx=\left(\frac{3^{-1}+1}{\log(3)}\right)-\left(\frac{2^{-1}}{\log(2)}\right)}$

354. Calculate Integral $I=\int_{0}^{4}\frac{|x-1|}{|x-2|+|x-3|}dx$

Solution; $I=\int_{0}^{4}\frac{|x-1|}{|x-2|+|x-3|}dx$

$$=\underbrace{\int_{0}^{1}\frac{|x-1|}{|x-2|+|x-3|}dx}_{I_1}+\underbrace{\int_{1}^{2}\frac{|x-1|}{|x-2|+|x-3|}dx}_{I_2}+\underbrace{\int_{2}^{3}\frac{|x-1|}{|x-2|+|x-3|}dx}_{I_3}+\underbrace{\int_{3}^{4}\frac{|x-1|}{|x-2|+|x-3|}dx}_{I_4}$$

By : $\forall x\in(0,1)\Rightarrow|x-1|=-(x-1)\ ;\ |x-2|=-(x-2)\ ;\ |x-3|=-(x-3)$

We obtain $I_1=\int_{0}^{1}\frac{-(x-1)}{-(x-2)-(x-3)}dx$

$$=\int_{0}^{1}\frac{x-1}{2x-5}dx=\left(\frac{1}{2}+\frac{3}{4}\log\left(\frac{3}{5}\right)\right)$$

By : $\forall x \in (1,2) \Rightarrow |x-1| = (x-1)$; $|x-2| = -(x-2)$; $|x-3| = -(x-3)$

We obtain $I_2 = \int_1^2 \frac{(x-1)}{-(x-2)-(x-3)}\,dx$

$$= \int_1^2 \frac{x-1}{-2x+5}\,dx = \left(-\frac{1}{2}+\frac{3}{4}\log(3)\right)$$

By : $\forall x \in (2,3) \Rightarrow |x-1| = (x-1)$; $|x-2| = (x-2)$; $|x-3| = -(x-3)$

We obtain $I_3 = \int_2^3 \frac{(x-1)}{(x-2)-(x-3)}\,dx$

$$= \int_2^3 \frac{x-1}{1}\,dx = \left(\frac{3}{2}\right)$$

By : $\forall x \in (3,4) \Rightarrow |x-1| = (x-1)$; $|x-2| = (x-2)$; $|x-3| = (x-3)$

We obtain $I_4 = \int_3^4 \frac{(x-1)}{(x-2)-(x-3)}\,dx$

$$= \int_3^4 \frac{x-1}{2x-5}\,dx = \left(\frac{1}{2}+\frac{3}{4}\log(3)\right)$$

Implies That $I = I_1 + I_2 + I_3 + I_4$

$$= \left(\frac{1}{2}+\frac{3}{4}\log\left(\frac{3}{5}\right)\right)+\left(-\frac{1}{2}+\frac{3}{4}\log(3)\right)+\left(\frac{3}{2}\right)+\left(\frac{1}{2}+\frac{3}{4}\log(3)\right)$$

$$= 2+\frac{3}{4}\log\left(\frac{27}{5}\right)$$

SO, $\boxed{\int_0^4 \frac{|x-1|}{|x-2|+|x-3|}\,dx = 2+\frac{3}{4}\log\left(\frac{27}{5}\right)}$

355. Calculate Integral $I=\int_0^{\pi}\frac{|x-\pi|}{|\pi-x|+|x|}dx$

Solution; $I=\int_0^{\pi}\frac{|x-\pi|}{|\pi-x|+|x|}dx \quad (!)$

$$=\int_0^{\pi}\frac{|\pi-x|}{|\pi-x|+|x|}dx$$

$$=\int_0^{\pi}\frac{|x|}{|x|+|\pi-x|}dx \quad (!!)$$

Use : $\int_0^a f(x)dx=\int_0^a f(a-x)dx$

Take $(!)+(!!)$ **Then :** $2I=\int_0^{\pi}\frac{|\pi-x|}{|\pi-x|+|x|}dx+\int_0^{\pi}\frac{|x|}{|x|+|\pi-x|}dx$

$$=\int_0^{\pi}\frac{|\pi-x|+|x|}{|\pi-x|+|x|}dx=\pi$$

We obtain $I=\frac{\pi}{2}$

SO, $\boxed{\int_0^{\pi}\frac{|x-\pi|}{|\pi-x|+|x|}dx=\frac{\pi}{2}}$

356. Calculate Integral $I=\int_0^{\sqrt{2}}\lfloor x^2\rfloor dx$

Solution; $I=\int_0^{\sqrt{2}}\lfloor x^2\rfloor dx$

Let : $x=\sqrt{t}\Rightarrow dx=\frac{1}{2\sqrt{t}}dt$ **; If :** $x\in(0,\sqrt{2})\Rightarrow t\in(0,2)$

We obtain $I=\frac{1}{2}\int_0^2\frac{\lfloor t\rfloor}{\sqrt{t}}dx$

$$=\frac{1}{2}\int_0^1\frac{\lfloor t\rfloor}{\sqrt{t}}dx+\frac{1}{2}\int_1^2\frac{\lfloor t\rfloor}{\sqrt{t}}dx$$

$$=\frac{1}{2}\int_0^1\frac{0}{\sqrt{t}}dx+\frac{1}{2}\int_1^2\frac{1}{\sqrt{t}}dx=\sqrt{2}-1$$

SO, $\boxed{\int_0^{\sqrt{2}}\lfloor x^2\rfloor dx=\sqrt{2}-1}$

357. Calculate Integral $I=\int_0^{45}\lfloor 45x\rfloor dx$

Method 1

Solution; $I=\int_0^{45}\lfloor 45x\rfloor dx$

Let : $t=45x\Rightarrow dx=\frac{1}{45}dt$ **; If** : $x\in(0,45)\Rightarrow t\in(0,2025)$

We obtain $I=\frac{1}{45}\int_0^{2025}\lfloor t\rfloor dt$

$$=\frac{1}{45}\left(\int_0^1\lfloor t\rfloor dt+\int_1^2\lfloor t\rfloor dt+\int_2^3\lfloor t\rfloor dt+...+\int_{2024}^{2025}\lfloor t\rfloor dt\right)$$
$$=\frac{1}{45}\left(\int_0^1 0dt+\int_1^2 1dt+\int_2^3 2dt+...+\int_{2024}^{2025}2024dt\right)$$
$$=\frac{1}{45}(1+2+3+...+2024)$$
$$=\frac{1}{45}\cdot\frac{2024\times 2025}{2}$$
$$=45540$$

Method 2

We Get $I=\frac{1}{45}\int_0^{2025}\lfloor t\rfloor dt$

$$=\frac{1}{45}\sum_{k=0}^{2024}\left(\int_k^{k+1}ndx\right)$$
$$=\frac{1}{45}\sum_{k=0}^{2024}(n)$$
$$=\frac{1}{45}\cdot\frac{2024\times 2025}{2}$$
$$=25325$$

$SO,$ $\boxed{\int_0^{45}\lfloor 45x\rfloor dx=25325}$

358. Calculate Integral $I=\int_1^2 \lfloor x^2-x \rfloor dx$

Solution; $I=\int_1^2 \lfloor x^2-x \rfloor dx$

Let : $t=x^2-x \Rightarrow dx=\dfrac{1}{\sqrt{4t+1}}dt$ **; If :** $x\in(1,2)\Rightarrow t\in(0,2)$

We obtain $I=\int_0^2 \dfrac{\lfloor t \rfloor}{\sqrt{4t+1}}dt$

$$=\int_0^1 \frac{\lfloor t \rfloor}{\sqrt{4t+1}}dt+\int_1^2 \frac{\lfloor t \rfloor}{\sqrt{4t+1}}dt$$
$$=\int_0^1 \frac{0}{\sqrt{4t+1}}dt+\int_1^2 \frac{1}{\sqrt{4t+1}}dt$$
$$=\int_1^2 \frac{1}{\sqrt{4t+1}}dt=\frac{1}{2}\left(3-\sqrt{5}\right)$$

$SO,$ $\boxed{\int_1^2 \lfloor x^2-x \rfloor dx=\frac{1}{2}\left(3-\sqrt{5}\right)}$

359. Calculate Integral $I=\int_0^2 \sin\left(\lfloor x+1 \rfloor\right)dx$

Solution; $I=\int_0^2 \sin\left(\lfloor x+1 \rfloor\right)dx$

$$=\int_0^2 \sin\left(\lfloor x \rfloor+1\right)dx$$

Because: $\lfloor x+m \rfloor=\lfloor x \rfloor+m \ ,m\in\mathbb{Z}$

$$=\int_0^1 \sin\left(\lfloor x \rfloor+1\right)dx+\int_1^2 \sin\left(\lfloor x \rfloor+1\right)dx$$
$$=\int_0^1 \sin\left(0+1\right)dx+\int_1^2 \sin\left(1+1\right)dx$$
$$=\sin(1)+\sin(2)=2\sin\left(\frac{3}{2}\right)\cos\left(\frac{1}{2}\right)$$

$SO,$ $\boxed{\int_0^2 \sin\left(\lfloor x+1 \rfloor\right)dx=2\sin\left(\frac{3}{2}\right)\cos\left(\frac{1}{2}\right)}$

360. Calculate Integral $I=\int_0^{\frac{\pi}{2}}\lfloor\cos(2x)\rfloor dx$

Solution; $I=\int_0^{\frac{\pi}{2}}\lfloor\cos(2x)\rfloor dx$

Let : $t=\cos(2x)\Rightarrow dx=-\frac{1}{2\sqrt{1-t^2}}dt$; **If** : $x\in\left(0,\frac{\pi}{2}\right)\Rightarrow t\in(1,-1)$

We Get $I=-\int_1^{-1}\frac{\lfloor t\rfloor}{2\sqrt{1-t^2}}dx$

$$=\int_{-1}^{0}\frac{\lfloor t\rfloor}{2\sqrt{1-t^2}}dx+\int_0^1\frac{\lfloor t\rfloor}{2\sqrt{1-t^2}}dx$$

$$=\int_{-1}^{0}\frac{-1}{2\sqrt{1-t^2}}dx+\int_0^1\frac{0}{2\sqrt{1-t^2}}dx$$

$$=-\frac{1}{2}\sin^{-1}(t)\Big|_{-1}^{0}=-\frac{\pi}{4}$$

SO, $\boxed{\int_0^{\frac{\pi}{2}}\lfloor\cos(2x)\rfloor dx=-\frac{\pi}{4}}$

361. Calculate Integral $I=\int_{\sqrt{2}}^{\sqrt{3}}\frac{\lfloor x^2\rfloor}{\lfloor x^2-1\rfloor}dx$

Solution; $I=\int_{\sqrt{2}}^{\sqrt{3}}\frac{\lfloor x^2\rfloor}{\lfloor x^2-1\rfloor}dx$

Let : $t=x^2-1\Rightarrow dx=\frac{1}{2\sqrt{t+1}}dt$; **If** : $x\in\left(\sqrt{2},\sqrt{3}\right)\Rightarrow t\in(1,2)$

We obtain $I=\int_1^2\frac{\lfloor t+1\rfloor}{\lfloor t\rfloor}\cdot\frac{1}{2\sqrt{t+1}}dt$

$$=\int_1^2\frac{2}{1}\times\frac{1}{2\sqrt{t+1}}dt$$

$$=2\sqrt{t+1}\Big|_1^2=\frac{2}{\sqrt{3}-\sqrt{2}}$$

SO, $\boxed{\int_{\sqrt{2}}^{\sqrt{3}}\frac{\lfloor x^2\rfloor}{\lfloor x^2-1\rfloor}dx=\frac{2}{\sqrt{3}-\sqrt{2}}}$

362. Calculate Integral $I=\int_{-2}^{2}\frac{\sin(x)}{\left\lfloor\frac{x}{\pi}\right\rfloor+2}dx$

Solution; $I=\int_{-2}^{2}\frac{\sin(x)}{\left\lfloor\frac{x}{\pi}\right\rfloor+2}dx$

Let : $t=\frac{x}{\pi}\Rightarrow dx=\pi dt$ **; If** : $x\in(-2,2)\Rightarrow t\in\left(-\frac{2}{\pi},\frac{2}{\pi}\right)$

We obtain $I=\pi\int_{-\frac{2}{\pi}}^{\frac{2}{\pi}}\frac{\sin(\pi t)}{\lfloor t\rfloor+2}dt$

$$=\pi\int_{-\frac{2}{\pi}}^{0}\frac{\sin(\pi t)}{\lfloor t\rfloor+2}dt+\pi\int_{0}^{\frac{2}{\pi}}\frac{\sin(\pi t)}{\lfloor t\rfloor+2}dt$$

$$=\pi\int_{-\frac{2}{\pi}}^{0}\frac{\sin(\pi t)}{-1+2}dt+\pi\int_{0}^{\frac{2}{\pi}}\frac{\sin(\pi t)}{0+2}dt$$

$$=\pi\int_{-\frac{2}{\pi}}^{0}\sin(\pi t)dt+\frac{\pi}{2}\int_{0}^{\frac{2}{\pi}}\sin(\pi t)dt$$

$$=\frac{1}{2}(\cos(2)-1)$$

SO, $\boxed{\int_{-2}^{2}\frac{\sin(x)}{\lfloor x/\pi\rfloor+2}dx=\frac{1}{2}(\cos(2)-1)}$

363. Calculate Integral $I=\int_{e-1}^{e^2-1}\lfloor\log(x+1)\rfloor dx$

Solution; $I=\int_{e-1}^{e^2-1}\lfloor\log(x+1)\rfloor dx$

Let : $t=\log(x+1)\Rightarrow dx=e^t dt$ **; If** : $x\in(e-1,e^2-1)\Rightarrow t\in(1,2)$

We obtain $I=\int_{1}^{2}\lfloor t\rfloor e^t dt$

$$=e(e-1)$$

SO, $\boxed{\int_{e-1}^{e^2-1}\lfloor\log(x+1)\rfloor dx=e(e-1)}$

364. Calculate Integral $I=\int_{-2}^{2}\frac{\left\lfloor x/\pi \right\rfloor}{\left\lfloor x/\pi \right\rfloor+\frac{1}{2}}dx$

Solution; $I=\int_{-2}^{2}\frac{\left\lfloor x/\pi \right\rfloor}{\left\lfloor x/\pi \right\rfloor+\frac{1}{2}}dx$

Let : $t=\frac{x}{\pi}\Rightarrow dx=\pi dt$ **; If** : $x\in(-2,2)\Rightarrow t\in\left(-\frac{2}{\pi},\frac{2}{\pi}\right)$

We obtain $I=\pi\int_{-\frac{2}{\pi}}^{\frac{2}{\pi}}\frac{\lfloor t\rfloor}{\lfloor t\rfloor+\frac{1}{2}}dx$

$$=\pi\int_{-\frac{2}{\pi}}^{0}\frac{\lfloor t\rfloor}{\lfloor t\rfloor+\frac{1}{2}}dt+\pi\int_{0}^{\frac{2}{\pi}}\frac{\lfloor t\rfloor}{\lfloor t\rfloor+\frac{1}{2}}dt$$

$$=\pi\int_{-\frac{2}{\pi}}^{0}\frac{-1}{-1+\frac{1}{2}}dt+\pi\int_{0}^{\frac{2}{\pi}}\frac{0}{0+\frac{1}{2}}dt$$

$$=\pi\int_{-\frac{2}{\pi}}^{0}2dt=4$$

$SO,$ $\boxed{\int_{-2}^{2}\frac{\left\lfloor x/\pi \right\rfloor}{\left\lfloor x/\pi \right\rfloor+\frac{1}{2}}dx=4}$

365. Calculate Integral $I=\int_{0}^{\log(3)}\left\lfloor e^{x}+1\right\rfloor dx$

Solution; $I=\int_{0}^{\log(3)}\left\lfloor e^{x}+1\right\rfloor dx$

$$=\int_{0}^{\log(3)}\left(\left\lfloor e^{x}\right\rfloor+1\right)dx$$

$$= \log(3) + \int_0^{\log(3)} \lfloor e^x \rfloor dx$$

Let : $t = e^x \Rightarrow dx = \frac{1}{t}dt$; **If** : $x \in (0, \log(3)) \Rightarrow t \in (1,3)$

We obtain $I = \log(3) + \int_1^3 \frac{\lfloor t \rfloor}{t} dt$

$$= \log(3) + \int_1^2 \frac{\lfloor t \rfloor}{t} dt + \int_2^3 \frac{\lfloor t \rfloor}{t} dt$$

$$= \log(3) + \int_1^2 \frac{1}{t} dt + \int_2^3 \frac{2}{t} dt$$

$$= \log\left(\frac{27}{2}\right)$$

SO, $\boxed{\int_0^{\log(3)} \lfloor e^x + 1 \rfloor dx = \log\left(\frac{27}{2}\right)}$

366. Calculate Integral $I = \int_{-2}^{2} \frac{\lfloor x \rfloor}{|x+1|} dx$

Solution; $I = \int_{-2}^{2} \frac{\lfloor x \rfloor}{|x+1|} dx$

$$= \int_{-2}^{-1} \frac{\lfloor x \rfloor}{|x|+1} dx + \int_{-1}^{0} \frac{\lfloor x \rfloor}{|x|+1} dx + \int_{0}^{1} \frac{\lfloor x \rfloor}{|x|+1} dx + \int_{0}^{2} \frac{\lfloor x \rfloor}{|x|+1} dx$$

$$= \int_{-2}^{-1} \frac{-2}{-x+1} dx + \int_{-1}^{0} \frac{-1}{-x+1} dx + \int_{0}^{1} \frac{0}{x+1} dx + \int_{1}^{2} \frac{1}{x+1} dx$$

$$= 2\log|x-1|\Big|_{-2}^{-1} + \log|x-1|\Big|_{-1}^{0} dx + 0 + \log|x+1|\Big|_{1}^{2}$$

$$= -\log(3)$$

SO, $\boxed{\int_{-2}^{2} \frac{\lfloor x \rfloor}{|x+1|} dx = \log\left(\frac{1}{3}\right)}$

367. Calculate Integral $I=\int_{-2}^{2}\frac{\lceil x\rceil}{|x+1|}dx$

Solution; $I=\int_{-2}^{2}\frac{\lceil x\rceil}{|x+1|}dx$

$$=\int_{-2}^{-1}\frac{\lceil x\rceil}{|x|+1}dx+\int_{-1}^{0}\frac{\lceil x\rceil}{|x|+1}dx+\int_{0}^{1}\frac{\lceil x\rceil}{|x|+1}dx+\int_{0}^{2}\frac{\lceil x\rceil}{|x|+1}dx$$

$$=\int_{-2}^{-1}\frac{-1}{-x+1}dx+\int_{-1}^{0}\frac{0}{-x+1}dx+\int_{0}^{1}\frac{1}{x+1}dx+\int_{1}^{2}\frac{2}{x+1}dx$$

$$=\log|x-1|\Big|_{-2}^{-1}+0+\log|x+1|\Big|_{0}^{1}dx+\log|x+1|\Big|_{1}^{2}$$

$$=\log(2)$$

$SO,$ $\boxed{\int_{-2}^{2}\frac{\lceil x\rceil}{|x+1|}dx=\log(2)}$

368. Calculate Integral $I=\int_{0}^{\frac{\pi}{2}}\log\left(\lfloor 1+\sin(x)\rfloor\right)dx$

Solution; $I=\int_{0}^{\frac{\pi}{2}}\log\left(\lfloor 1+\sin(x)\rfloor\right)dx$

$$=\int_{0}^{\frac{\pi}{2}}\log\left(1+\lfloor \sin(x)\rfloor\right)dx$$

By : $\sin(x)\in\left(0,\frac{\pi}{2}\right)$ **Then** $0<\sin(x)<1$ **That** $\lfloor \sin(x)\rfloor=0$

We obtain $I=\int_{0}^{\frac{\pi}{2}}\log(1+0)dx=0$

$SO,$ $\boxed{\int_{0}^{\frac{\pi}{2}}\log\left(\lfloor \sin(x)+1\rfloor\right)dx=0}$

369. Calculate Integral $I=\int_0^2 \frac{\lfloor x^2 \rfloor}{\lfloor x^2-4x+4 \rfloor+\lfloor x^2 \rfloor}dx$

Solution; $I=\int_0^2 \frac{\lfloor x^2 \rfloor}{\lfloor x^2-4x+4 \rfloor+\lfloor x^2 \rfloor}dx$

$$=\int_0^2 \frac{\lfloor x^2 \rfloor}{\lfloor (2-x)^2 \rfloor+\lfloor x^2 \rfloor}dx \qquad (1)$$

$$=\int_0^2 \frac{\lfloor (2-x)^2 \rfloor}{\lfloor (x)^2 \rfloor+\lfloor (2-x)^2 \rfloor}dx \qquad (2)$$

Use: $\int_0^a f(x)dx=\int_0^a f(a-x)dx$

Take $(1)+(2)$ **Then :** $2I=\int_0^2 \frac{\lfloor x^2 \rfloor}{\lfloor (2-x)^2 \rfloor+\lfloor x^2 \rfloor}dx+\int_0^2 \frac{\lfloor (2-x)^2 \rfloor}{\lfloor (2-x)^2 \rfloor+\lfloor (x)^2 \rfloor}dx$

$$=\int_0^2 \frac{\lfloor (2-x)^2 \rfloor+\lfloor (x)^2 \rfloor}{\lfloor (2-x)^2 \rfloor+\lfloor (x)^2 \rfloor}dx=2$$

We Get $I=1$

$SO,$ $\boxed{\int_0^2 \frac{\lfloor x^2 \rfloor}{\lfloor x^2-4x+4 \rfloor+\lfloor x^2 \rfloor}dx=1}$

370. Calculate Integral $I=\int_0^1 \frac{1}{1+\left\lfloor \frac{1}{x} \right\rfloor}dx$

Solution; $I=\int_0^1 \frac{1}{1+\left\lfloor \frac{1}{x} \right\rfloor}dx$

Let : $t=\frac{1}{x}\Rightarrow dx=-\frac{1}{t^2}dt$ **; If :** $x\in(0,1)\Rightarrow t\in(\infty,1)$

We obtain $I=-\int_{\infty}^{1}\frac{1}{1+\lfloor t\rfloor}\times\frac{1}{t^2}dx$

$$=\int_{1}^{\infty}\frac{1}{1+\lfloor t\rfloor}\times\frac{1}{t^2}dx$$

$$=\sum_{n=1}^{\infty}\int_{n}^{n+1}\frac{1}{1+n}\times\frac{1}{t^2}dx$$

$$=\sum_{n=1}^{\infty}\left(\frac{1}{1+n}\right)\left(\frac{1}{n}-\frac{1}{n+1}\right)$$

$$=\sum_{n=1}^{\infty}\left(\frac{1}{n(1+n)}\right)-\sum_{n=1}^{\infty}\left(\frac{1}{(1+n)^2}\right)$$

$$=1-\left(\zeta(2)-1\right)$$

$$=2-\zeta(2)$$

SO, $\boxed{\int_{0}^{1}\frac{1}{1+\left\lfloor 1/x\right\rfloor}dx=2-\zeta(2)}$

371. Calculate Integral $I=\int_{e}^{e^2}\lfloor\log(x)\rfloor^{\lfloor\log(x)+1\rfloor}dx$

Solution; $I=\int_{e}^{e^2}\lfloor\log(x)\rfloor^{\lfloor\log(x)+1\rfloor}dx$

Let : $t=\log(x)\Rightarrow dx=e^t dt$ **; If** : $x\in\left(e,e^2\right)\Rightarrow t\in(1,2)$

We obtain $I=\int_{1}^{2}\lfloor t\rfloor^{\lfloor t+1\rfloor}e^t dt$

By $\forall t\in[1,2]$ **Then** $\lfloor t\rfloor=1$ **And** $\lfloor t+1\rfloor=2$

We Get $I=\int_{1}^{2}1^2e^t dt=e(e-1)$

SO, $\boxed{\int_{e}^{e^2}\lfloor\log(x)\rfloor^{\lfloor\log(x)+1\rfloor}dx=e(e-1)}$

372. Calculate Integral $I=\int_0^3 \frac{1}{x^2+1+\lfloor x\rfloor(\lfloor x\rfloor-2x)}dx$

Solution; $I=\int_0^3 \frac{1}{x^2+1+\lfloor x\rfloor(\lfloor x\rfloor-2x)}dx$

$$=\underbrace{\int_0^1 \frac{1}{x^2+1+\lfloor x\rfloor(\lfloor x\rfloor-2x)}dx}_{I_1}+\underbrace{\int_1^2 \frac{1}{x^2+1+\lfloor x\rfloor(\lfloor x\rfloor-2x)}dx}_{I_2}+\underbrace{\int_2^3 \frac{1}{x^2+1+\lfloor x\rfloor(\lfloor x\rfloor-2x)}dx}_{I_3}$$

By : $\forall x\in(0,1)$ **Then** $\lfloor x\rfloor=0$

We Get $I_1=\int_0^1 \frac{1}{x^2+1+0(0-2x)}dx$

$$=\int_0^1 \frac{1}{x^2+1}dx=\tan^{-1}(x)\Big|_0^1=\frac{\pi}{4}$$

By : $\forall x\in(1,2)$ **Then** $\lfloor x\rfloor=1$

We Get $I_2=\int_1^2 \frac{1}{x^2+1+1(1-2x)}dx$

$$=\int_1^2 \frac{1}{(x-1)^2+1}dx=\tan^{-1}(x-1)\Big|_1^2=\frac{\pi}{4}$$

By : $\forall x\in(1,2)$ **Then** $\lfloor x\rfloor=2$

We Get $I_3=\int_2^3 \frac{1}{x^2+1+2(2-2x)}dx$

$$=\int_2^3 \frac{1}{(x-2)^2+1}dx=\tan^{-1}(x-2)\Big|_2^3=\frac{\pi}{4}$$

Implies That $I=I_1+I_2+I_3$

$$=\frac{\pi}{4}+\frac{\pi}{4}+\frac{\pi}{4}=\frac{3\pi}{4}$$

SO, $\boxed{\int_0^3 \frac{1}{x^2+1+\lfloor x\rfloor(\lfloor x\rfloor-2x)}dx=\frac{3\pi}{4}}$

373. Calculate Integral $I=\int_0^{\infty}\lfloor x\rfloor e^{-x}dx$

Solution; $I=\int_0^{\infty}\lfloor x\rfloor e^{-x}dx$

$$=\lim_{n\to\infty}\int_0^{n}\lfloor x\rfloor e^{-x}dx$$

$$=\lim_{n\to\infty}\left(\int_0^{1}\lfloor x\rfloor e^{-x}dx+\int_1^{2}\lfloor x\rfloor e^{-x}dx+\int_2^{3}\lfloor x\rfloor e^{-x}dx+\ldots+\int_{n-1}^{n}\lfloor x\rfloor e^{-x}dx\right)$$

$$=\lim_{n\to\infty}\left(\int_0^{1}0.e^{-x}dx+\int_1^{2}1.e^{-x}dx+\int_2^{3}2.e^{-x}dx\ldots+\int_{n-1}^{n}(n-1)e^{-x}dx\right)$$

$$=\lim_{n\to\infty}\left(-e^{-x}\Big|_0^{1}-2e^{-x}\Big|_1^{2}-3e^{-x}\Big|_2^{3}-\ldots-(n-1)e^{-x}\Big|_{n-1}^{n}\right)$$

$$=\lim_{n\to\infty}\left(-\left(e^{-2}-e^{-1}\right)-\left(2e^{-2}-2e^{-3}\right)-\ldots-\left((n-1)e^{-n}-(n-1)e^{-n+1}\right)\right)$$

$$=\lim_{n\to\infty}\left(e^{-1}+e^{-2}+e^{-3}\ldots+e^{-n+1}+e^{-n}-ne^{-n}\right)$$

$$=\lim_{n\to\infty}\left(e^{-1}\frac{e^{-n}-1}{e^{-1}-1}-ne^{-n}\right)$$

$$=\lim_{n\to\infty}\left(\frac{e^{-n}-1-(1-e)ne^{-n}}{1-e}\right)=\frac{1}{e-1}$$

$SO,$ $\boxed{\int_0^{\infty}\lfloor x\rfloor e^{-x}dx=\frac{1}{e-1}}$

374. Calculate Integral $I=\int_0^{2025}\left(x^2-\lfloor x\rfloor\lceil x\rceil\right)dx$

Solution; $I=\int_0^{2025}\left(x^2-\lfloor x\rfloor\lceil x\rceil\right)dx$

$$=\int_0^{2025}\left(x^2\right)dx-\int_0^{2025}\lfloor x\rfloor\lceil x\rceil dx$$

$$=\frac{2025^3}{3}-\int_0^{2025}\lfloor x\rfloor\lceil x\rceil dx$$

$$=\frac{2025^3}{3}-\sum_{k=0}^{2024}\int_k^{k+1}k(k+1)\,dx$$

$$= \frac{2025^3}{3} - \lim_{n\to 2024}\sum_{k=0}^{n} k(k+1)$$

$$= \lim_{n\to 2024}\left(\frac{(n+1)^3}{3}\right) - \lim_{n\to 2024}\left(\frac{n\times(n+1)\times(2n+1)}{6} + \frac{n\times(n+1)}{2}\right)$$

$$= \lim_{n\to 2024}\left(\frac{(n+1)^3}{3} - \frac{2n\times(n+1)\times(n+2)}{6}\right)$$

$$= \lim_{n\to 2024}\left(\frac{(n+1)^3 - n\times(n+1)\times(n+2)}{3}\right)$$

$$= \lim_{n\to 2024}\left(\frac{(n+1)\left((n+1)^2 - n\times(n+2)\right)}{3}\right)$$

$$= \lim_{n\to 2024}\left(\frac{(n+1)\left(n^2+2n+1-n^2-2n\right)}{3}\right)$$

$$= \lim_{n\to 2024}\left(\frac{(n+1)}{3}\right) = \frac{2025}{3}$$

$$SO,\ \boxed{\int_0^{2022}\left(x^2 - \lfloor x\rfloor\lceil x\rceil\right)dx = \frac{2022}{3}}$$

375. Calculate Integral $I = \int_0^1 \left\lfloor \frac{1}{\sqrt{x}} \right\rfloor dx$

Solution; $I = \int_0^1 \left\lfloor \frac{1}{\sqrt{x}} \right\rfloor dx$

Let : $y = \frac{1}{\sqrt{x}} \Rightarrow dx = -\frac{2}{y^3}dy$ **; If :** $x \in (0,1) \Rightarrow y \in (\infty,1)$

Implies That $I = -2\int_\infty^1 \frac{\lfloor y\rfloor}{y^3}dy$

$$= 2\int_1^\infty \frac{\lfloor y\rfloor}{y^3}dy$$

$$= 2\sum_{n=1}^{\infty}\left(\int_n^{n+1}\frac{n}{y^3}dy\right)$$

$$=-\sum_{n=1}^{\infty} n\left(\frac{1}{y^2}\right)\Bigg|_{n}^{n+1}$$

$$=\sum_{n=1}^{\infty} n\left(\frac{1}{n^2}-\frac{1}{(n+1)^2}\right)$$

$$=2\sum_{n=1}^{\infty}\left(\frac{1}{(n+1)^2}\right)+\sum_{n=1}^{\infty}\left(\frac{1}{n(n+1)^2}\right)$$

$$=2\sum_{n=1}^{\infty}\left(\frac{1}{(n+1)^2}\right)+\sum_{n=1}^{\infty}\left(\frac{1}{n}-\frac{1}{n+1}-\frac{1}{(n+1)^2}\right)$$

$$=\sum_{n=1}^{\infty}\left(\frac{1}{(n+1)^2}\right)+\sum_{n=1}^{\infty}\left(\frac{1}{n}-\frac{1}{n+1}\right)$$

$$=\sum_{n=1}^{\infty}\left(\frac{1}{(n+1)^2}\right)+\lim_{m\to\infty}\sum_{n=1}^{m}\left(\frac{1}{n}-\frac{1}{n+1}\right)$$

$$=\zeta(2)-1+\lim_{m\to\infty}\left(\frac{1}{1}-\frac{1}{2}+\frac{1}{2}-\frac{1}{3}+\frac{1}{3}-\frac{1}{4}+\ldots\frac{1}{m}-\frac{1}{m+1}\right)$$

$$=\zeta(2)-1+\lim_{m\to\infty}\left(\frac{1}{1}-\frac{1}{m+1}\right)$$

$$=\zeta(2)=\frac{\pi^2}{6}$$

SO, $\boxed{\int_0^1\left\lfloor\frac{1}{\sqrt{x}}\right\rfloor dx=\frac{\pi^2}{6}}$

376. Calculate Integral $I=\int_1^{\infty} a^{-\lfloor \log_b(x)\rfloor}dx$

Solution; $I=\int_1^{\infty} a^{-\lfloor \log_b(x)\rfloor}dx$

Let : $u=\log_b(x)\equiv x=b^u \Rightarrow dx=b^u\log(b)du$ **; If** : $x\in(1,\infty)\Rightarrow u\in(0,\infty)$

We Get $I=\int_0^{\infty} a^{-\lfloor u\rfloor}b^u\log(b)du$

$$= \sum_{n=0}^{\infty}\left(\int_{n}^{n+1} a^{-n} b^{u} \log(b) du\right)$$

$$= \sum_{n=0}^{\infty}\left(a^{-n}\int_{n}^{n+1} b^{u} \log(b) du\right)$$

$$= \sum_{n=0}^{\infty}\left[a^{-n}\left(b^{n+1}-b^{n}\right)\right]$$

$$= (b-1)\sum_{n=0}^{\infty} a^{-n} b^{n}$$

$$= (b-1)\sum_{n=0}^{\infty}\left(\frac{b}{a}\right)^{n}$$

$$= (b-1)\left(\frac{1}{1-\frac{b}{a}}\right) = \frac{a(b-1)}{(a-b)}$$

$SO,$ $\boxed{\int_{1}^{\infty} a^{-\lfloor \log_b(x) \rfloor} dx = \frac{a(b-1)}{(a-b)}}$

377. Calculate Integral $I = \int_{0}^{1} \{x\}^{x} dx$

Solution; $I = \int_{0}^{1} \{x\}^{x} dx$

$$= \int_{0}^{1} \left(x - \lfloor x \rfloor\right)^{x} dx$$

$$= \int_{0}^{1} \left(x-0\right)^{x} dx$$

$$= \int_{0}^{1} x^{x} dx$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^{n}}{(n+1)^{n+1}} = 1 - \frac{1}{2^{2}} + \frac{1}{3^{3}} - \frac{1}{4^{4}} + \cdots$$

$SO,$ $\boxed{\int_{0}^{1} \{x\}^{x} dx = 1 - \frac{1}{2^{2}} + \frac{1}{3^{3}} - \frac{1}{4^{4}} + \cdots}$

378. Calculate Integral $I=\int_0^3 \{x\}^{\lfloor x\rfloor}dx$

Solution; $I=\int_0^3 \{x\}^{\lfloor x\rfloor}dx$

$$=\int_0^3 \left(x-\lfloor x\rfloor\right)^{\lfloor x\rfloor}dx \quad \textbf{Note } \{x\}=x-\lfloor x\rfloor \ ; \ \{x\} \textbf{ Is that fraction part function}$$

$$=\int_0^1 \left(x-\lfloor x\rfloor\right)^{\lfloor x\rfloor}dx+\int_1^2 \left(x-\lfloor x\rfloor\right)^{\lfloor x\rfloor}dx+\int_2^3 \left(x-\lfloor x\rfloor\right)^{\lfloor x\rfloor}dx$$

$$=\int_0^1 (x-0)^0dx+\int_1^2 (x-1)^1dx+\int_2^3 (x-2)^2dx$$

$$=\int_0^1 1dx+\int_1^2 (x-1)\,dx+\int_2^3 (x-2)^2dx$$

$$=\frac{11}{6}$$

SO, $\boxed{\int_0^3 \{x\}^{\lfloor x\rfloor}dx=\frac{11}{6}}$

379. Calculate Integral $I=\int_0^\infty e^{-\lfloor x\rfloor(1+\{x\})}dx$

Solution; $I=\int_0^\infty e^{-\lfloor x\rfloor(1+\{x\})}dx$

$$=\int_0^1 e^{-\lfloor x\rfloor(1+\{x\})}dx+\int_1^\infty e^{-\lfloor x\rfloor(1+\{x\})}dx$$

$$=\int_0^1 e^0dx+\sum_{n=1}^{\infty}\left(\int_n^{n+1} e^{\left(n^2-n-nx\right)}dx\right)$$

$$=1+\sum_{n=1}^{\infty} e^{\left(n^2-n\right)}\int_n^{n+1} e^{-nx}dx$$

$$=1-\sum_{n=1}^{\infty} e^{\left(n^2-n\right)}\frac{\left(e^{-(n^2+n)}-e^{-n^2}\right)}{n}$$

$$=1+\sum_{n=1}^{\infty}\frac{\left(e^{-n}-e^{-2n}\right)}{n}$$

$$=1+\sum_{n=1}^{\infty}\left(\frac{e^{-n}}{n}\right)-\sum_{n=1}^{\infty}\left(\frac{e^{-2n}}{n}\right)$$

Note : $\sum_{n=1}^{\infty}\left(\frac{x^n}{n}\right)=-\log(1-x)$

We Get $I=1-\log(1-e^{-1})+\log(1-e^{-2})$

$$=1-\log\left(\frac{e-1}{e}\right)+\log\left(\frac{e^2-1}{e^2}\right)$$

$$=\log(e+1)$$

SO, $\boxed{\int_0^{\infty}e^{-\lfloor x\rfloor(1+\{x\})}dx=\log(e+1)}$

380. Calculate Integral $I=\int_1^{\infty}\frac{1}{\lfloor x\rfloor\lceil x\rceil}dx$

Solution; $I=\int_1^{\infty}\frac{1}{\lfloor x\rfloor\lceil x\rceil}dx$

$$=\lim_{n\to\infty}\int_1^{n}\frac{1}{\lfloor x\rfloor\lceil x\rceil}dx$$

$$=\lim_{n\to\infty}\left(\int_1^{2}\frac{1}{\lfloor x\rfloor\lceil x\rceil}dx+\int_2^{3}\frac{1}{\lfloor x\rfloor\lceil x\rceil}dx+\ldots+\int_{n-1}^{n}\frac{1}{\lfloor x\rfloor\lceil x\rceil}dx\right)$$

$$=\lim_{n\to\infty}\left(\int_1^{2}\frac{1}{1\times 2}dx+\int_2^{3}\frac{1}{2\times 3}dx+\ldots+\int_{n-1}^{n}\frac{1}{(n-1)\times n}dx\right)$$

$$=\lim_{n\to\infty}\left(\frac{1}{1\times 2}+\frac{1}{2\times 3}+\frac{1}{3\times 4}+\ldots+\frac{1}{(n-1)\times n}\right)$$

$$=\lim_{n\to\infty}\left(\frac{1}{1}-\frac{1}{2}+\frac{1}{2}-\frac{1}{3}+\frac{1}{3}-\frac{1}{4}+\ldots+\frac{1}{n-1}-\frac{1}{n}\right)$$

$$=\lim_{n\to\infty}\left(\frac{1}{1}-\frac{1}{n}\right)=1$$

SO, $\boxed{\int_1^{\infty}\frac{1}{\lfloor x\rfloor\lceil x\rceil}dx=1}$

381. Calculate Integral $I=\int_1^{2026}\frac{\left|\lfloor -x\rfloor\right|}{\lfloor x\rfloor}dx$

Solution; $I=\int_1^{2026}\frac{\left|\lfloor -x\rfloor\right|}{\lfloor x\rfloor}dx$

$$=\int_1^{2026}\frac{\left|-\lfloor x\rfloor-1\right|}{\lfloor x\rfloor}dx$$

Note : $\lfloor -x\rfloor=-\lfloor x\rfloor-1$

$$=\int_1^{2026}\frac{\lfloor x\rfloor+1}{\lfloor x\rfloor}dx$$

$$=2025+\int_1^{2026}\frac{1}{\lfloor x\rfloor}dx$$

$$=2025+\sum_{k=1}^{2025}\left(\int_k^{k+1}\frac{1}{k}dx\right)$$

$$=2025+\sum_{k=1}^{2025}\left(\frac{1}{k}\right)$$

$$=2025+H_{2025}$$

SO, $\boxed{\int_1^{2026}\frac{\left|\lfloor -x\rfloor\right|}{\lfloor x\rfloor}dx=2025+H_{2025}}$

382. Calculate Integral $I=\int_1^{\infty}\frac{\{x\}}{x\lfloor x\rfloor}dx$

Method 1

Solution; $I=\int_1^{\infty}\frac{\{x\}}{x\lfloor x\rfloor}dx$

$$=\int_1^{\infty}\frac{x-\lfloor x\rfloor}{x\lfloor x\rfloor}dx$$

$$= \int_1^{\infty} \left(\frac{1}{\lfloor x \rfloor} - \frac{1}{x} \right) dx$$

$$= \int_1^{2} \left(\frac{1}{\lfloor x \rfloor} - \frac{1}{x} \right) dx + \int_2^{3} \left(\frac{1}{\lfloor x \rfloor} - \frac{1}{x} \right) dx + \int_3^{4} \left(\frac{1}{\lfloor x \rfloor} - \frac{1}{x} \right) dx + \cdots$$

$$= \int_1^{2} \left(\frac{1}{1} - \frac{1}{x} \right) dx + \int_2^{3} \left(\frac{1}{2} - \frac{1}{x} \right) dx + \int_3^{4} \left(\frac{1}{3} - \frac{1}{x} \right) dx + \cdots$$

$$= \sum_{k=1}^{\infty} \left(\int_k^{k+1} \left(\frac{1}{k} - \frac{1}{x} \right) dx \right)$$

$$= \sum_{k=1}^{\infty} \left(\frac{1}{k} - \log(k+1) + \log(k) \right)$$

$$= \sum_{k=1}^{\infty} \left(\frac{1}{k} \right) - \lim_{n \to \infty} \sum_{k=1}^{n} \left[\log(k+1) - \log(k) \right]$$

$$= \sum_{k=1}^{\infty} \left(\frac{1}{k} \right) - \lim_{n \to \infty} \left[\log(n+1) + \sum_{k=1}^{n-1} \log(k+1) - \sum_{k=2}^{n} \log(k) \right]$$

$$= \sum_{k=1}^{\infty} \left(\frac{1}{k} \right) - \lim_{n \to \infty} \left[\log(n+1) + \sum_{k=2}^{n} \log(k) - \sum_{k=2}^{n} \log(k) \right]$$

$$= \sum_{k=1}^{\infty} \left(\frac{1}{k} \right) - \lim_{n \to \infty} \left(\log(n+1) \right)$$

$$= \sum_{k=1}^{\infty} \left(\frac{1}{k} \right) - \lim_{n \to \infty} \left(\log(n) \right)$$

Implies That $I = \lim_{n \to \infty} \left[-\log(n) + \sum_{k=1}^{n} \left(\frac{1}{k} \right) \right] = \gamma = 0.57721...$ **Where** γ **Euler's constant**

Method 2

We Have $I = \int_1^{\infty} \left(\frac{1}{\lfloor x \rfloor} - \frac{1}{x} \right) dx$

$$= \lim_{n \to \infty} \sum_{k=1}^{n} \left(\int_k^{k+1} \left(\frac{1}{k} - \frac{1}{x} \right) dx \right)$$

$$= \lim_{n\to\infty} \sum_{k=1}^{n} \left(\frac{1}{k} - \log(k+1) + \log(k) \right)$$

$$= \lim_{n\to\infty} \left[\sum_{k=1}^{n} \left(\frac{1}{k} \right) - \log(n) \right]$$

$$= \gamma$$

SO, $\boxed{\int_1^{\infty} \frac{\{x\}}{x\lfloor x \rfloor} dx = \gamma}$

383. Calculate Integral $I = \int_0^{2025} \left\{ \sqrt{x} \right\} dx$

Solution; $I = \int_0^{2025} \left\{ \sqrt{x} \right\} dx$

Let $x = y^2 \Rightarrow dx = 2ydy$ **; If :** $x \in (0, 2025) \Rightarrow y \in (0, 45)$

We Get $I = 2\int_0^{45} \{y\} ydy$

$$= 2\int_0^{45} \left(y - \lfloor y \rfloor \right) ydy$$

$$= 2\int_0^{45} y^2 dy - \int_0^{25} \lfloor y \rfloor ydy$$

$$= 2\int_0^{45} y^2 dy - 2\sum_{k=0}^{44} \left(\int_k^{k+1} kydx \right)$$

$$= \frac{2}{3} y^3 \Big|_0^{45} - \sum_{k=0}^{44} \left(k \cdot y^2 \Big|_k^{k+1} \right)$$

$$= \frac{2}{3} \cdot 45^3 - \sum_{k=0}^{44} \left(2k^2 + k \right)$$

$$= \frac{2}{3} \cdot 45^3 - 2 \cdot \frac{44(44+1)(2 \cdot 44+1)}{6} + \frac{44(44+1)}{2} = 1020$$

SO, $\boxed{\int_0^{2025} \left\{ \sqrt{x} \right\} dx = 1020}$

384. Calculate Integral $I=\int_0^{2026}\left(x+\lfloor x\rfloor+\lceil x\rceil+[x]+\{x\}\right)dx$

Solution; $I=\int_0^{2026}\left(x+\lfloor x\rfloor+\lceil x\rceil+[x]+\{x\}\right)dx$

$$=\int_0^{2026}\left(x+\lfloor x\rfloor+\lceil x\rceil+\underbrace{[x]+\{x\}}_{x}\right)dx$$

$$=\int_0^{2026}\left(2x+\lfloor x\rfloor+\lceil x\rceil\right)dx$$

$$=\sum_{k=0}^{2025}\left[\int_k^{k+1}\left(2x+k+(k+1)\right)dx\right]$$

$$=\sum_{k=0}^{2025}\left[\left((2k+1)x+x^2\right)\Big|_k^{k+1}\right]$$

$$=2\sum_{k=0}^{2025}\left(2k+1\right)$$

$$=2\left(2\cdot\frac{2025\times 2026}{2}+2026\right)$$

$$=2\cdot 2026^2$$

SO, $\boxed{\int_0^{2026}\left(x+\lfloor x\rfloor+\lceil x\rceil+[x]+\{x\}\right)dx=2\cdot 2026^2}$

365. Calculate Integral $I=\int_0^{\log\left(1-\frac{\pi}{2}\right)}\frac{e^x\sin^3\left(1-e^x\right)}{\log\left[\cos\left(e^x-1\right)\right]}dx$

Solution; $I=\int_0^{\log\left(1-\frac{\pi}{2}\right)}\frac{e^x\sin^3\left(1-e^x\right)}{\log\left[\cos\left(e^x-1\right)\right]}dx$

$$=-\int_0^{\log\left(1-\frac{\pi}{2}\right)}\frac{1-\cos^2\left(e^x-1\right)}{\log\left[\cos\left(e^x-1\right)\right]}\cdot e^x\sin\left(e^x-1\right)dx$$

Let $u=\cos\left(e^x-1\right)\Rightarrow du=-e^x\sin\left(e^x-1\right)dx$ **; If** $x\in\left(0,\log\left(1-\frac{\pi}{2}\right)\right)\Rightarrow u\in(1,0)$

We Get $I=\int_1^0 \frac{1-u^2}{\log(u)}du$

$$=\int_0^1 \frac{u^2-1}{\log(u)}du$$
$$=\log(2+1)=\log(3)$$

SO, $\boxed{\int_0^{\log\left(1-\frac{\pi}{2}\right)} \frac{e^x\sin^3\left(1-e^x\right)}{\log\left[\cos\left(e^x-1\right)\right]}dx=\log(3)}$

386. Calculate Integral $I=\int_0^1 \frac{x^n}{1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\cdots+\frac{x^n}{n!}}dx$

Solution; $I=\int_0^1 \frac{x^n}{1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\cdots+\frac{x^n}{n!}}dx$

$$=n!\int_0^1 \frac{\frac{x^n}{n!}}{1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\cdots+\frac{x^n}{n!}}dx$$
$$=n!\int_0^1 \frac{\left(1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\cdots+\frac{x^n}{n!}\right)-\left(1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\cdots+\frac{x^{n-1}}{(n-1)!}\right)}{1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\cdots+\frac{x^n}{n!}}dx$$
$$=n!\int_0^1 \left(1-\frac{\left(1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\cdots+\frac{x^n}{n!}\right)'}{1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\cdots+\frac{x^n}{n!}}\right)dx$$
$$=n!\left(x-\log\left(1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\cdots+\frac{x^n}{n!}\right)\right)\Big|_0^1$$

$$= n!\left(1-\log\left(1+1+\frac{1}{2!}+\frac{1}{3!}+\cdots+\frac{1}{n!}\right)\right)$$

$$= n!\left[1-\log\left(1+H_n\right)\right]$$

SO, $\boxed{\int_0^1 \frac{x^n}{1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\cdots+\frac{x^n}{n!}}dx = n!\left[1-\log\left(1+H_n\right)\right]}$

387. Calculate Integral $I=\int_{-\infty}^{+\infty}\frac{\tan^{-1}\left(e^x\right)}{x^2+\pi^2}dx$

Solution; $I=\int_{-\infty}^{+\infty}\frac{\tan^{-1}\left(e^x\right)}{x^2+\pi^2}dx$

$$= \underbrace{\int_{-\infty}^{0}\frac{\tan^{-1}\left(e^x\right)}{x^2+\pi^2}dx}_{J}+\int_0^{+\infty}\frac{\tan^{-1}\left(e^x\right)}{x^2+\pi^2}dx \quad (*)$$

For $J=\int_{-\infty}^{0}\frac{\tan^{-1}\left(e^x\right)}{x^2+\pi^2}dx$

Let : $x=-y\Rightarrow dx=-dy$ **; If** $: x\in(-\infty,0)\Rightarrow y\in(+\infty,0)$

We Get $J=-\int_{+\infty}^{0}\frac{\tan^{-1}\left(e^{-x}\right)}{x^2+\pi^2}dx=\int_0^{+\infty}\frac{\tan^{-1}\left(e^{-x}\right)}{x^2+\pi^2}dx$

Take (*) Then $I=\int_0^{+\infty}\frac{\tan^{-1}\left(e^{-x}\right)+\tan^{-1}\left(e^x\right)}{x^2+\pi^2}dx$

$$= \frac{\pi}{2}\int_0^{+\infty}\frac{1}{x^2+\pi^2}dx$$

$$= \frac{\pi}{4}$$

SO, $\boxed{\int_{-\infty}^{+\infty}\frac{\tan^{-1}\left(e^x\right)}{x^2+\pi^2}dx=\frac{\pi}{4}}$

388. Calculate Integral $I=\int_0^{\frac{\pi}{4}}\frac{\frac{\pi}{4}-x\tan(x)}{\cos(2x)}dx$

Solution; $I=\int_0^{\frac{\pi}{4}}\frac{\frac{\pi}{4}-x\tan(x)}{\cos(2x)}dx$

$$=\int_0^{\frac{\pi}{4}}\frac{\frac{\pi}{4}-x\tan(x)}{1-\tan^2(x)}dx$$

Let : $u=\tan(x)\Rightarrow\frac{du}{1+u^2}=dx$; **If** : $x\in\left(0,\frac{\pi}{4}\right)\Rightarrow u\in(0,1)$

Implies That $I=\int_0^1\frac{\frac{\pi}{4}-u\tan^{-1}(u)}{1-u^2}du$

Let : $u=\frac{1-t}{1+t}\Rightarrow du=\frac{2}{(1+t)^2}dt$; **If** : $x\in(0,1)\Rightarrow u\in(0,1)$

We obtain $I=\int_0^1\frac{\frac{\pi}{4}-\frac{1-t}{1+t}\left(\tan^{-1}\left(\frac{1-t}{1+t}\right)\right)}{1-\left(\frac{1-t}{1+t}\right)^2}\cdot\frac{2}{(1+t)^2}dt$

$$=\frac{1}{2}\int_0^1\frac{\frac{\pi}{2}t-(1-t)\tan^{-1}(t)}{t(t+1)}dt$$

$$=\frac{\pi}{4}\int_0^1\left(\frac{1}{t+1}\right)dt+\frac{1}{2}\int_0^1\left(\frac{\tan^{-1}(t)}{t}\right)dt-\int_0^1\left(\frac{\tan^{-1}(t)}{t+1}\right)dt$$

$$=\frac{\pi}{4}\log(2)+\frac{1}{2}G-\frac{\pi}{8}\log(2)$$

$$=\frac{\pi}{8}\log(2)+\frac{1}{2}G$$

SO, $\boxed{\int_0^{\frac{\pi}{4}}\frac{\frac{\pi}{4}-x\tan(x)}{\cos(2x)}dx=\frac{\pi}{8}\log(2)+\frac{1}{2}G}$

389. Calculate Integral $I=\int_0^{\infty}\frac{\left[W(x)\right]^2}{x^2\sqrt{x}}dx$

Solution; $I=\int_0^{\infty}\frac{\left[W(x)\right]^2}{x^2\sqrt{x}}dx$

Let : $u=W(x)\equiv x=ue^u\Rightarrow dx=(u+1)e^u$ **; If** : $x\in(0,\infty)\Rightarrow u\in(0,\infty)$

We obtain $I=\int_0^{\infty}\frac{u^2(u+1)e^u}{\left(ue^u\right)^2\sqrt{ue^u}}du$

$$=\int_0^{\infty}\frac{u+1}{e^u\sqrt{ue^u}}du$$

$$=\int_0^{\infty}\left(u^{\frac{1}{2}}e^{-\frac{3}{2}u}+u^{-\frac{1}{2}}e^{\frac{-3}{2}u}\right)du$$

Let : $y=\frac{3}{2}t\Rightarrow\frac{2}{3}dy=dt$ **; If** : $u\in(0,\infty)\Rightarrow t\in(0,\infty)$

We Get $I=\frac{2}{3}\int_0^{\infty}\left[\left(\frac{2}{3}t\right)^{-\frac{1}{2}}e^{-t}+\left(\frac{2}{3}t\right)^{\frac{1}{2}}e^{-t}\right]dt$

$$=\frac{2}{3}\int_0^{\infty}\left[\left(\frac{2}{3}t\right)^{-\frac{1}{2}}e^{-t}+\left(\frac{2}{3}t\right)^{\frac{1}{2}}e^{-t}\right]dt$$

$$=\frac{2}{3}\left[\left(\frac{2}{3}\right)^{-\frac{1}{2}}\left(-\frac{1}{2}\right)!+\frac{2}{3}\left(\frac{2}{3}\right)^{\frac{1}{2}}\left(\frac{1}{2}\right)!\right]$$

$$=\sqrt{\frac{2}{3}}\cdot\sqrt{\pi}+\frac{2}{3}\cdot\sqrt{\frac{2}{3}}\cdot\frac{\sqrt{\pi}}{2}$$

$$=\frac{4\sqrt{6\pi}}{9}$$

SO, $\boxed{\int_0^{\infty}\frac{\left[W(x)\right]^2}{x^2\sqrt{x}}dx=\frac{4\sqrt{6\pi}}{9}}$

390. Calculate Integral $I=\int_{-1}^{+1}(-1)^{\varphi x}dx$

Solution; $I=\int_{-1}^{+1}(-1)^{\varphi x}dx$

We Have $e^{\pi i}=-1$ **Then** $(-1)^{\varphi x}=e^{i\pi\varphi x}$

We Get $I=\int_{-1}^{+1}e^{i\pi\varphi x}dx$

$$=\int_{-1}^{+1}e^{i\pi\varphi x}dx$$
$$=\frac{1}{i\pi\varphi}\left(e^{\pi\varphi i}-e^{-\pi\varphi i}\right)$$
$$=\frac{2}{\pi\varphi}\left(\frac{e^{\pi\varphi i}-e^{-\pi\varphi i}}{2i}\right)$$
$$=\frac{2}{\pi\varphi}\sin(\pi\varphi)$$

SO, $\boxed{\int_{-1}^{+1}(-1)^{\varphi x}dx=\frac{2}{\pi\varphi}\sin(\pi\varphi)}$

391. Calculate Integral $I=\int_{\frac{\pi}{6}}^{\frac{\pi}{4}}\frac{\sin(4x)}{\sin^4(x)+\cos^4(x)}dx$

Solution; $I=\int_{\frac{\pi}{6}}^{\frac{\pi}{4}}\frac{\sin(4x)}{\sin^4(x)+\cos^4(x)}dx$

$$=\int_{\frac{\pi}{6}}^{\frac{\pi}{4}}\frac{2\sin(2x)\cos(2x)}{\left(\sin^2(x)+\cos^2(x)\right)^2-2\sin^2(x)\cos^2(x)}dx$$
$$=\int_{\frac{\pi}{6}}^{\frac{\pi}{4}}\frac{4\sin(2x)\cos(2x)}{2-\sin^2(2x)}dx$$

Let : $u=\sin(2x)\Rightarrow du=2\cos(x)dx$ **; If** : $x\in\left(\frac{\pi}{6},\frac{\pi}{4}\right)\Rightarrow u\in\left(\frac{\sqrt{3}}{2},1\right)$

We Get $I=\int_{\frac{\sqrt{3}}{2}}^{1}\frac{2u}{2-u^2}du$

$$=\log\left|2-u^2\right|\Big|_{\frac{\sqrt{3}}{2}}^{1}$$

$$=\log\left(\frac{5}{4}\right)$$

SO, $\boxed{\int_{\frac{\pi}{6}}^{\frac{\pi}{4}}\frac{\sin(4x)}{\sin^4(x)+\cos^4(x)}dx=\log\left(\frac{5}{4}\right)}$

392. Calculate Integral $I=\int_0^{\frac{\pi}{4}}x\tan(x)dx$

Solution; $I=\int_0^{\frac{\pi}{4}}x\tan(x)dx$

Let : $u=x\Rightarrow du=dx$ **And** $dv=\tan(x)\Rightarrow v=-\log\left(\cos(x)\right)$

We obtain $I=-x\log\left(\cos(x)\right)\Big|_0^{\frac{\pi}{4}}+\int_0^{\frac{\pi}{4}}\log\left(\cos(x)\right)dx$

$$=\frac{\pi}{8}\log(2)+\int_0^{\frac{\pi}{4}}\left[-\log(2)-\sum_{k=1}^{\infty}\frac{(-1)^k\cos(2kx)}{k}\right]dx$$

$$=\frac{\pi}{8}\log(2)-\frac{\pi}{4}\log(2)-\sum_{k=1}^{\infty}\left(\frac{(-1)^k}{k}\int_0^{\frac{\pi}{4}}\cos(2kx)\right)dx$$

$$=-\frac{\pi}{8}\log(2)-\frac{1}{2}\sum_{k=1}^{\infty}\left(\frac{(-1)^k}{k^2}\cdot\sin(2kx)\Big|_0^{\frac{\pi}{4}}\right)$$

$$=-\frac{\pi}{8}\log(2)-\frac{1}{2}\sum_{k=1}^{\infty}\left(\frac{(-1)^k}{k^2}\sin\left(\frac{\pi}{2}k\right)\right)$$

$$=-\frac{\pi}{8}\log(2)+\frac{1}{2}G$$

SO, $\boxed{\int_0^{\frac{\pi}{4}}x\tan(x)dx=-\frac{\pi}{8}\log(2)+\frac{1}{2}G}$

393. Calculate Integral $I=\int_0^1 \frac{x^2\log(x^2+1)}{x^2+1}dx$

Solution; $I=\int_0^1 \frac{x^2\log(x^2+1)}{x^2+1}dx$

Let : $u=\log\left(x^2+1\right)\Rightarrow du=\frac{2x}{x^2+1}dx$ **And** $v=\frac{x^2}{x^2+1}dx\Rightarrow v=x-\tan^{-1}(x)$

We Get $I=\left(x-\tan^{-1}(x)\right)\log(x^2+1)\Big|_0^1-2\int_0^1\frac{x\left(x-\tan^{-1}(x)\right)}{x^2+1}dx$

$$=\left(1-\frac{\pi}{4}\right)\log(2)-2\int_0^1\frac{x^2}{x^2+1}dx+\int_0^1\frac{x\tan^{-1}(x)}{x^2+1}dx$$

$$=\left(1-\frac{\pi}{4}\right)\log(2)-2\int_0^1\frac{x^2}{x^2+1}dx+\int_0^1 x\tan^{-1}(x)d\left(\tan^{-1}(x)\right)$$

$$=\left(1-\frac{\pi}{4}\right)\log(2)-2\left(x-\tan^{-1}(x)\right)\Big|_0^1+\int_0^{\frac{\pi}{4}}x\tan^{-1}(x)\,dx$$

$$=\left(1-\frac{\pi}{4}\right)\log(2)-2\left(1-\frac{\pi}{4}\right)-\frac{\pi}{8}\log(2)+\frac{1}{2}G$$

$$=\left(1-\frac{3\pi}{8}\right)\log(2)+\frac{1}{2}(G+\pi)-2$$

SO, $\boxed{\int_0^1\frac{x^2\log(x^2+1)}{x^2+1}dx=\left(1-\frac{3\pi}{8}\right)\log(2)+\frac{1}{2}(G+\pi)-2}$

394. Calculate Integral $I=\int_0^1\frac{\sin(x)-x\cos(x)}{x\left(x+\sin(x)\right)}dx$

Solution; $I=\int_0^1\frac{\sin(x)-x\cos(x)}{x\left(x+\sin(x)\right)}dx$

$$=\int_0^1\frac{\sin(x)+x-x-x\cos(x)}{x\left(x+\sin(x)\right)}dx$$

$$=\int_0^1\left(\frac{1}{x}-\frac{1+\cos(x)}{x+\sin(x)}\right)dx$$

$$= \left(\log\left| \frac{x}{x+\sin(x)} \right| \right)\Big|_0^1$$

$$= -\lim_{x\to 1}\left(\log\left| \frac{x+\sin(x)}{x} \right| \right) + \lim_{x\to 0}\left(\log\left| \frac{x+\sin(x)}{x} \right| \right)$$

$$= -\left(\log\left|1+\sin(1)\right| \right) + \lim_{x\to 0}\left(\log\left|1+\frac{\sin(x)}{x}\right| \right)$$

$$= -\left(\log\left|1+\sin(1)\right| \right) + \log(2)$$

$$= \log\left(\frac{2}{1-\sin(1)} \right)$$

SO, $\boxed{\int_0^1 \frac{\sin(x)-x\cos(x)}{x\left(x+\sin(x)\right)}dx = \log\left(\frac{2}{1+\sin(1)}\right)}$

395. Calculate Integral $I = \int_0^n \frac{-\log(1-x)}{x}dx$

Solution; $I = \int_0^n \frac{-\log(1-x)}{x}dx$

$$= -\int_0^n \frac{\log(1-x)}{x}dx$$

$$= \sum_{k=1}^{\infty}\left(\frac{1}{k}\int_0^n \frac{x^k}{x}dx \right)$$

$$= \sum_{k=1}^{\infty}\left(\frac{1}{k}\int_0^n x^{k-1}dx \right)$$

$$= \sum_{k=1}^{\infty}\left(\frac{n^k}{k^2} \right)$$

$$= Li_2(n)$$

SO, $\boxed{\int_0^n \frac{-\log(1-x)}{x}dx = Li_2(n)}$

396. Calculate Integral $I=\int_{-6}^{\infty}\cos\left(x^2+12x+\frac{72+\pi}{2}\right)dx$

Solution; $I=\int_{-6}^{\infty}\cos\left(x^2+12x+\frac{72+\pi}{2}\right)dx$

$$=\int_{-6}^{\infty}\cos\left(x^2+12x+36+\frac{\pi}{2}\right)dx$$

$$=\int_{-6}^{\infty}\cos\left((x+6)^2+\frac{\pi}{2}\right)dx$$

$$=\int_{-6}^{\infty}\left[\cos(x+6)^2\cdot\cos\left(\frac{\pi}{2}\right)-\sin(x+6)^2\cdot\sin\left(\frac{\pi}{2}\right)\right]dx$$

$$=-\int_{-6}^{\infty}\sin(x+6)^2\,d(x+6)$$

$$=-\int_{0}^{\infty}\sin(x^2)dx$$

$$=-\mathrm{Im}\left[\int_{0}^{\infty}e^{ix^2}dx\right]$$

Let : $x=\sqrt{iy}\Rightarrow dx=\frac{\sqrt{i}}{2}y^{\frac{1}{2}-1}dy$ **; If :** $x\in(0,\infty)\Rightarrow y\in(0,\infty)$

We obtain $I=-\mathrm{Im}\left[\frac{\sqrt{i}}{2}\int_{0}^{\infty}e^{-y}y^{\frac{1}{2}-1}dx\right]$

$$=-\mathrm{Im}\left[\frac{\sqrt{\pi}}{2\sqrt{2}}(1+i)\right]$$

$$=-\frac{\sqrt{\pi}}{2\sqrt{2}}$$

$$=-\sqrt{\frac{\pi}{8}}$$

SO, $\boxed{\int_{-6}^{\infty}\cos\left(x^2+12x+\frac{72+\pi}{2}\right)dx=-\sqrt{\frac{\pi}{8}}}$

397. Calculate Integral $I=\int_{-\infty}^{+\infty}\cos(ax^2)\,dx \quad , a>0$

Solution; $I=\int_{-\infty}^{+\infty}\cos(ax^2)\,dx$

$$=2\int_{0}^{+\infty}\cos(ax^2)\,dx$$

$$=2\,\mathrm{Re}\left(\int_{0}^{+\infty}e^{iax^2}\,dx\right)$$

Let : $x=\sqrt{\frac{iu}{a}}\Rightarrow dx=\frac{(1+i)}{2\sqrt{2a}}\cdot u^{\frac{1}{2}-1}du$ **; If :** $x\in(0,\infty)\Rightarrow u\in(0,\infty)$

We obtain $I=2\,\mathrm{Re}\left(\frac{1+i}{2\sqrt{2a}}\int_{0}^{\infty}e^{-u}u^{\frac{1}{2}-1}dx\right)$

$$=2\,\mathrm{Re}\left(\frac{1+i}{2\sqrt{2a}}\cdot\sqrt{\pi}\right)$$

$$=2\left(\frac{\sqrt{\pi}}{2\sqrt{2a}}\right)$$

$$=\sqrt{\frac{\pi}{2a}}$$

SO, $\boxed{\int_{-\infty}^{+\infty}\cos(ax^2)\,dx=\sqrt{\frac{\pi}{2a}} \quad , a>0}$

398. Calculate Integral $I=\int_{0}^{1}\log^2(x^2+1)dx$

Solution; $I=\int_{0}^{1}\log^2(x^2+1)dx$

Let : $u=\log^2(x^2+1)\Rightarrow du=\frac{4x\log(x^2+1)}{x^2+1}dx$ **And :** $dv=dx\Rightarrow v=x$

We Get $I=x\log^2(x^2+1)\Big|_0^1-4\int_0^1\frac{x^2\log(x^2+1)}{x^2+1}dx$

$$= \log^2(2) - 4I_{293}$$

$$= \log^2(2) - 4\left[\left(1-\frac{3\pi}{8}\right)\log(2) + \frac{1}{2}(G+\pi) - 2\right]$$

$$= \log^2(2) - \left(4-\frac{3\pi}{2}\right)\log(2) - 2G - 2\pi + 8$$

SO, $\boxed{\int_0^1 \log^2(x^2+1)dx = \log^2(2) - \left(4-\frac{3\pi}{2}\right)\log(2) + 2G + 2\pi - 8}$

399. Calculate Integral $I = \int_0^1 \frac{\sin^2(x)}{x^2}dx$

Solution; $I = \int_0^1 \frac{\sin^2(x)}{x^2}dx$

Let : $u = \sin^2(x) \Rightarrow du = \sin(2x)dx$ **And** $dv = \frac{1}{x^2}dx \Rightarrow v = -\frac{1}{x}dx$

We obtain $I = -\frac{\sin^2(x)}{x}\Big|_0^\infty + \int_0^\infty \frac{\sin(2x)}{x}dx$

$$= -\lim_{x\to\infty}\frac{\sin^2(x)}{x} + \lim_{x\to 0}\frac{\sin^2(x)}{x} + \int_0^\infty \frac{\sin(2x)}{(2x)}d(2x)$$

$$= -0 + 0 + \int_0^\infty \frac{\sin(x)}{x}d(x)$$

$$= \int_0^\infty \frac{\sin(x)}{x}d(x)$$

$$= \frac{\pi}{2}$$

SO, $\boxed{\int_0^1 \frac{\sin^2(x)}{x^2}dx = \frac{\pi}{2}}$

400. Calculate Integral $I=\int_0^{\frac{\pi}{2}}\log^4\left(\tan(x)\right)dx$

Solution; $I=\int_0^{\frac{\pi}{2}}\log^4\left(\tan(x)\right)dx$

Let : $u=\tan(x)\Rightarrow dx=\dfrac{1}{1+u^2}du$; **If** : $x\in\left(0,\dfrac{\pi}{2}\right)\Rightarrow u\in(0,\infty)$

Implies That $I=\int_0^{\infty}\dfrac{\log^4(u)}{1+u^2}du$

$$=\int_0^1\frac{\log^4(u)}{1+u^2}du+\underbrace{\int_1^{\infty}\frac{\log^4(u)}{1+u^2}du}_{J}\quad (*)$$

For $J=\int_1^{\infty}\dfrac{\log^4(u)}{1+u^2}du$

Let : $u=\dfrac{1}{y}\Rightarrow du=-\dfrac{1}{y^2}dy$; **If:** $u\in(1,\infty)\Rightarrow y\in(1,0)$

We Get $J=-\int_1^0\dfrac{\log^4\left(y^{-1}\right)}{1+y^{-2}}\cdot\dfrac{1}{y^2}dy$

$$=\int_0^1\frac{\log^4(u)}{1+u^2}du$$

Take (*) Then $I=2\int_0^1\dfrac{\log^4(u)}{1+u^2}du$

Let : $u=e^{-z}\Rightarrow du=-e^{-z}dz$; **If** : $u\in(0,1)\Rightarrow z\in(\infty,0)$

We obtain $I=-2\int_{\infty}^0\dfrac{\log^4\left(e^{-z}\right)}{e^{-2z}+1}\cdot e^{-z}dz$

$$=2\int_0^{\infty}\frac{z^4e^{-z}}{e^{-2z}+1}dz$$

$$=2\sum_{k=0}^{\infty}\left[(-1)^k\int_0^{\infty}\left(z^4e^{-z}\cdot e^{-2kz}\right)dz\right]$$

$$= 2\sum_{k=0}^{\infty}\left[(-1)^k \int_0^{\infty}\left(z^4 e^{-(2k+1)z}\right)dz\right]$$

$$= 2\sum_{k=0}^{\infty}\left[\frac{(-1)^k}{(2k+1)^5}\int_0^{\infty}\left(z^4 e^{-z}\right)dz\right]$$

$$= 2\sum_{k=0}^{\infty}\left[\frac{(-1)^k \cdot 4!}{(2k+1)^5}\right]$$

$$= 2\cdot 4!\,\beta(5)$$

$$= 48\cdot\frac{5\pi^5}{1536} = \frac{5\pi^5}{32}$$

SO, $\boxed{\int_0^{\frac{\pi}{2}} \log^4\left(\tan(x)\right)dx = \frac{5\pi^5}{32}}$

401. Calculate Integral $I = \int_{-\infty}^{+\infty} \frac{\tan^{-1}\left(e^{ax}\right)}{x^2+b^2}dx \;\; ; a,b>0$

Solution; $I = \int_{-\infty}^{+\infty} \frac{\tan^{-1}\left(e^{ax}\right)}{x^2+b^2}dx \qquad (!)$

$$= \int_{-\infty}^{+\infty} \frac{\tan^{-1}\left(e^{-ax}\right)}{(-x)^2+b^2}dx$$

$$= \int_{-\infty}^{+\infty} \frac{\tan^{-1}\left(\frac{1}{e^{ax}}\right)}{x^2+b^2}dx \qquad (!!)$$

Take (!)+(!!) That $2I = \int_{-\infty}^{+\infty} \frac{\tan^{-1}\left(e^{ax}\right)+\tan^{-1}\left(\frac{1}{e^{ax}}\right)}{x^2+b^2}dx$

Implies That $I = \frac{\pi}{4}\int_{-\infty}^{+\infty}\frac{1}{x^2+b^2}dx = \frac{\pi^2}{4|b|}$

SO, $\boxed{\int_{-\infty}^{+\infty} \frac{\tan^{-1}\left(e^{ax}\right)}{x^2+b^2}dx = \frac{\pi^2}{4|b|} \;\; ; a,b>0}$

402. Calculate Integral $I=\int_0^{\infty} e^{-x}\log^2(x)dx$

Solution; $I=\int_0^{\infty} e^{-x}\log^2(x)dx$

By Contact $\Gamma(s)=\int_0^{\infty} x^{s-1}e^{-x}dx$

$$\Gamma'(s)=\int_0^{\infty} x^{s-1}e^{-x}\log(x)dx$$

$$\Gamma''(s)=\int_0^{\infty} x^{s-1}e^{-x}\log^2(x)dx$$

If $S=1$ **Then** $\Gamma''(1)=\int_0^{\infty} e^{-x}\log^2(x)dx=I$

We Get $I=\Gamma''(1)$

By Contact $\Gamma'(x)=\Gamma(x)\psi(x)$

$$\Gamma''(x)=\Gamma(x)\left[\psi^2(x)+\psi'(x)\right]$$

If $x=1$ **Then** $\Gamma'(1)=\Gamma(1)\psi(1)=-\gamma$

If $x=1$ **Then** $\Gamma''(1)=\Gamma(1)\left[\psi^2(1)+\psi'(1)\right]=\gamma^2+\frac{\pi^2}{6}$

SO, $\boxed{\int_0^{\infty} e^{-x}\log^2(x)dx=\gamma^2+\frac{\pi^2}{6}}$

403. Calculate Integral $I=\int_{-\infty}^{+\infty}\frac{x^2}{x^4+7x^2+1}dx$

Solution; $I=\int_{-\infty}^{+\infty}\frac{x^2}{x^4+7x^2+1}dx$

$$=\int_{-\infty}^{+\infty}\frac{1}{x^2+7+x^{-2}}dx$$

$$=\int_{-\infty}^{+\infty}\frac{1}{(x+x^{-1})^2+9}dx$$

$$= \int_{-\infty}^{+\infty} \frac{1}{x^2+9}\,dx \qquad \textbf{As per ‘‘ Glasser’s Master Theorem ”}$$

$$= \frac{1}{3}\tan^{-1}\left(\frac{1}{3}x\right)\Big|_{-\infty}^{+\infty}$$

$$= \frac{\pi}{3}$$

SO, $\boxed{\int_{-\infty}^{+\infty} \frac{x^2}{x^4+7x^2+1}\,dx = \frac{\pi}{3}}$

404. Calculate Integral $I = \int_0^{\infty} \frac{x^m}{1+x^n}\,dx$

Solution; $I = \int_0^{\infty} \frac{x^m}{1+x^n}\,dx$

Let : $x = u^{\frac{1}{n}} \Rightarrow dx = \frac{1}{n}u^{\frac{1}{n}-1}\,du$ **; If :** $x \in (0,\infty) \Rightarrow u \in (0,\infty)$

We Get $I = \frac{1}{n}\int_0^{\infty} \frac{u^{\frac{m}{n}}}{1+x} \cdot u^{\frac{1}{n}-1}\,dx$

$$= \frac{1}{n}\int_0^{\infty} \frac{u^{\frac{m+1}{n}-1}}{(1+x)^{\frac{m+1}{n}-\left(1-\frac{m+1}{n}\right)}}\,dx$$

$$= \frac{1}{n}\mathbf{B}\left(\frac{m+1}{n}, 1-\frac{m+1}{n}\right)$$

$$= \frac{1}{n}\cdot\Gamma\left(\frac{m+1}{n}\right)\Gamma\left(1-\frac{m+1}{n}\right)$$

$$= \frac{\pi}{n}\csc\left(\frac{m+1}{n}\right)$$

SO, $\boxed{\int_0^{\infty} \frac{x^m}{1+x^n}\,dx = \frac{\pi}{n}\csc\left(\frac{m+1}{n}\right)}$

405. Calculate Integral $I=\int_{-\infty}^{+\infty} e^{-ax^2}\cos(bx)dx \quad ,a>0$

Solution; $I=\int_{-\infty}^{+\infty} e^{-ax^2}\cos(bx)dx$

$$=\mathrm{Re}\left(\int_{-\infty}^{+\infty} e^{-ax^2}\cdot e^{bxi}dx\right)$$

$$=\mathrm{Re}\left(\int_{-\infty}^{+\infty} e^{-ax^2+bxi}dx\right)$$

$$=\mathrm{Re}\left(\int_{-\infty}^{+\infty} e^{-a\left(x-\frac{ib}{2a}\right)^2-\frac{b^2}{4a}}dx\right)$$

$$=e^{-\frac{b^2}{4a}}\mathrm{Re}\left(\int_{-\infty}^{+\infty} e^{-a\left(x-\frac{ib}{2a}\right)^2}dx\right)$$

Let $:u=x-\frac{ib}{2a}\Rightarrow du=dx$ **; If** $:x\in(-\infty,+\infty)\Rightarrow u\in(-\infty,+\infty)$

Implies That $I=e^{-\frac{b^2}{4a}}\mathrm{Re}\left(\int_{-\infty}^{+\infty} e^{-au^2}dx\right)$

Let $:y=\sqrt{a}x\equiv ax^2=y^2\Rightarrow dx=\frac{1}{\sqrt{a}}dy$ **; If** $:x\in(-\infty,+\infty)\Rightarrow u\in(-\infty,+\infty)$

We Get $I=e^{-\frac{b^2}{4a}}\mathrm{Re}\left(\frac{1}{\sqrt{a}}\int_{-\infty}^{+\infty} e^{-y^2}dy\right)$

$$=e^{-\frac{b^2}{4a}}\mathrm{Re}\left(\frac{1}{\sqrt{a}}\cdot\sqrt{\pi}\right)$$

$$=\sqrt{\frac{\pi}{a}}e^{-\frac{b^2}{4a}}$$

SO, $\boxed{\int_{-\infty}^{+\infty} e^{-ax^2}\cos(bx)dx=\sqrt{\frac{\pi}{a}}\,e^{-\frac{b^2}{4a}} \quad ,a>0}$

406. Calculate Integral $I=\int_{-1}^{1} Max\left\{\lfloor x\rfloor,\{x\},\lceil x\rceil\right\}dx$

Solution; $I=\int_{-1}^{1} Max\left\{\lfloor x\rfloor,\{x\},\lceil x\rceil\right\}dx$

$$=\int_{-1}^{1} Max\left\{\lfloor x\rfloor,x-\lfloor x\rfloor,\lceil x\rceil\right\}dx$$

$$=\int_{-1}^{0} Max\left\{\lfloor x\rfloor,x-\lfloor x\rfloor,\lceil x\rceil\right\}dx+\int_{0}^{1} Max\left\{\lfloor x\rfloor,x-\lfloor x\rfloor,\lceil x\rceil\right\}dx$$

$$=\int_{-1}^{0} Max\left\{-1,x+1,0\right\}dx+\int_{0}^{1} Max\left\{0,x-0,1\right\}dx$$

$$=\int_{-1}^{0}\left(x+1\right)dx+\int_{0}^{1}\left(1\right)dx$$

$$=\left(\frac{1}{2}x^2+x\right)\Big|_{-1}^{0}+x\Big|_{0}^{1}=\frac{3}{2}$$

$SO,$ $\boxed{\int_{-1}^{1} Max\left\{\lfloor x\rfloor,\{x\},\lceil x\rceil\right\}dx=\frac{3}{2}}$

407. Calculate Integral $I=\int_{-1}^{1} Max\left\{\lfloor x\rfloor^{\lfloor x\rfloor},\lceil x\rceil^{\lceil x\rceil}\right\}dx$

Solution; $I=\int_{-1}^{1} Max\left\{\lfloor x\rfloor^{\lfloor x\rfloor},\lceil x\rceil^{\lceil x\rceil}\right\}dx$

$$=\underbrace{\int_{-1}^{0} Max\left\{\lfloor x\rfloor^{\lfloor x\rfloor},\lceil x\rceil^{\lceil x\rceil}\right\}dx}_{J}+\underbrace{\int_{0}^{1} Max\left\{\lfloor x\rfloor^{\lfloor x\rfloor},\lceil x\rceil^{\lceil x\rceil}\right\}dx}_{K}\quad (*)$$

For : $J=\int_{-1}^{0} Max\left\{\lfloor x\rfloor^{\lfloor x\rfloor},\lceil x\rceil^{\lceil x\rceil}\right\}dx$

By : $\forall x\in(-1,0)$ **Then** $\lfloor x\rfloor^{\lfloor x\rfloor}=(-1)^{(-1)}=-1$ **And** $\lceil x\rceil^{\lceil x\rceil}=0^0=1$

Because : $\lim_{x\to 0}\left(\lceil x\rceil^{\lceil x\rceil}\right)\equiv\lim_{x\to 0}\left(x^{x}\right)=1$

We Get : $J=\int_{-1}^{0} Max\left\{-1,1\right\}dx$

$$=\int_{-1}^{0} 1dx=1$$

For : $K=\int_0^1 Max\left\{\lfloor x\rfloor^{\lfloor x\rfloor},\lceil x\rceil^{\lceil x\rceil}\right\}dx$

By : $\forall x\in(0,1)$ **Then** $\lfloor x\rfloor^{\lfloor x\rfloor}=0^0=1$ **And** $\lceil x\rceil^{\lceil x\rceil}=1^1=1$

We Get : $J=\int_0^1 Max\{1,1\}dx$

$=\int_0^1 1dx=1$

Take (*) That : $I=1+1=2$

SO, $\boxed{\int_{-1}^1 Max\left\{\lfloor x\rfloor^{\lfloor x\rfloor},\lceil x\rceil^{\lceil x\rceil}\right\}dx=2}$

408. Calculate Integral $I=\int_{-1}^1 Max\left\{\{x\}^{\lfloor x\rfloor},\lceil x+1\rceil^{\{x\}}\right\}dx$

Solution; $I=\int_{-1}^1 Max\left\{\{x\}^{\lfloor x\rfloor},\lceil x+1\rceil^{\{x\}}\right\}dx$

$=\int_{-1}^0 Max\left\{\left(x-\lfloor x\rfloor\right)^{\lfloor x\rfloor},\lceil x+1\rceil^{(x-\lfloor x\rfloor)}\right\}dx+\int_0^1 Max\left\{\left(x-\lfloor x\rfloor\right)^{\lfloor x\rfloor},\lceil x+1\rceil^{(x-\lfloor x\rfloor)}\right\}dx$

$=\int_{-1}^0 Max\left\{(x+1)^{-1},1^{(x+1)}\right\}dx+\int_0^1 Max\left\{(x-0)^0,2^{(x-0)}\right\}dx$

$=\int_{-1}^0 Max\left\{\frac{1}{x+1},1\right\}dx+\int_0^1 Max\left\{1,2^x\right\}dx$

$=\int_{-1}^0 1dx+\int_0^1 2^x dx$

$=x\Big|_{-1}^0+\frac{1}{\log(2)}\cdot 2^x\Big|_0^1$

$=1+\frac{1}{\log(2)}$

$=\log(e)+\log_2(e)$

SO, $\boxed{\int_{-1}^1 Max\left\{\{x\}^{\lfloor x\rfloor},\lceil x+1\rceil^{\{x\}}\right\}dx=\log(e)+\log_2(e)}$

409. Calculate Integral $I=\int_0^2 Max\left\{x,x^2\right\}dx$

Solution; $I=\int_0^2 Max\left\{x,x^2\right\}dx$

$\forall x\in(0,2)$, $x\cap x^2$ **Timing** $x=0,x=1$

If : $x\in(0,1)$ **That** $x^2\le x$ **They Have** $Max\left\{x,x^2\right\}=x$, $\forall x\in(0,1)$

If : $x\in(1,2)$ **That** $x^2\ge x$ **They Have** $Max\left\{x,x^2\right\}=x^2$, $\forall x\in(1,2)$

Implies That $I=\int_0^1 xdx+\int_1^2 x^2dx$

$$=\frac{1}{2}x^2\Big|_0^1+\frac{1}{3}x^3\Big|_1^2=\frac{17}{6}$$

SO, $\boxed{\int_0^2 Max\left\{x,x^2\right\}dx=\frac{17}{6}}$

But If $I=\int_0^2 Min\left\{x,x^2\right\}dx$

That $\forall x\in(0,2)$, $x\cap x^2$ **Timing** $x=0,x=1$

If : $x\in(0,1)$ **That** $x^2\le x$ **They Have** $Min\left\{x,x^2\right\}=x^2$, $\forall x\in(0,1)$

If : $x\in(1,2)$ **That** $x^2\ge x$ **They Have** $Min\left\{x,x^2\right\}=x$, $\forall x\in(1,2)$

Implies That $I=\int_0^1 x^2dx+\int_1^2 xdx$

$$=\frac{1}{3}x^3\Big|_0^1+\frac{1}{2}x^2\Big|_1^2=\frac{11}{6}$$

SO, $\boxed{\int_0^2 Min\left\{x,x^2\right\}dx=\frac{11}{6}}$

410. Calculate Integral $I=\int_{-2}^{2} Max\left\{2x^2, x^2+1\right\}dx$

Solution; $I=\int_{-2}^{2} Max\left\{2x^2, x^2+1\right\}dx$

$\forall x\in(-2,2)$, $2x^2\cap x^2+1$ **Timing** $x=-1,\ x=1$

If : $x\in(-2,-1)$ **That** $2x^2\geq x^2+1$ **They Have** $Max\left\{2x^2, x^2+1\right\}=2x^2$, $\forall x\in(-2,-1)$

If : $x\in(-1,1)$ **That** $2x^2\leq x^2+1$ **They Have** $Max\left\{2x^2, x^2+1\right\}=x^2+1$, $\forall x\in(-1,1)$

If : $x\in(1,2)$ **That** $2x^2\geq x^2+1$ **They Have** $Max\left\{2x^2, x^2+1\right\}=2x^2$, $\forall x\in(1,2)$

Implies That $I=\underbrace{\int_{-2}^{-1}2x^2dx}_{Let:x=-x}+\underbrace{\int_{-1}^{1}(x^2+1)dx}_{is\ an\ even\ function}+\int_{1}^{2}2x^2dx$

$$=2\int_{1}^{2}2x^2dx+2\int_{0}^{1}(x^2+1)dx=12$$

SO, $\boxed{\int_{-2}^{2} Max\left\{2x^2, x^2+1\right\}dx=12}$

But If $I=\int_{-2}^{2} Min\left\{2x^2, x^2+1\right\}dx$

That $\forall x\in(-2,2)$, $2x^2\cap x^2+1$ **Timing** $x=-1,\ x=1$

If : $x\in(-2,-1)$ **That** $2x^2\geq x^2+1$ **They Have** $Min\left\{2x^2, x^2+1\right\}=x^2+1$, $\forall x\in(-2,-1)$

If : $x\in(-1,1)$ **That** $2x^2\leq x^2+1$ **They Have** $Min\left\{2x^2, x^2+1\right\}=2x^2$, $\forall x\in(-1,1)$

If : $x\in(1,2)$ **That** $2x^2\geq x^2+1$ **They Have** $Min\left\{2x^2, x^2+1\right\}=x^2+1$, $\forall x\in(1,2)$

Implies That $I=\underbrace{\int_{-2}^{-1}(x^2+1)dx}_{Let:x=-x}+\underbrace{\int_{-1}^{1}2x^2dx}_{is\ an\ even\ function}+\int_{1}^{2}(x^2+1)dx$

$$=2\int_{1}^{2}(x^2+1)dx+2\int_{0}^{1}2x^2dx=\frac{20}{3}$$

SO, $\boxed{\int_{-2}^{2} Min\left\{2x^2, x^2+1\right\}dx=\frac{20}{3}}$

411. Calculate Integral $I=\int_{-1}^{1} Max\left\{x^2,\lfloor x\rfloor+1\right\}dx$

Solution; $I=\int_{-1}^{1} Max\left\{x^2,\lfloor x\rfloor+1\right\}dx$

$\forall x\in(-1,1)$, $x^2\cap\lfloor x\rfloor+1$ **No Junction**

If : $x\in(-2,-1)$ **That** $x^2\geq\lfloor x\rfloor+1$ **They Have** $Max\left\{x^2,\lfloor x\rfloor+1\right\}=x^2$ $,\forall x\in(-1,0)$

If : $x\in(-1,1)$ **That** $x^2\leq\lfloor x\rfloor+1$ **They Have** $Max\left\{x^2,\lfloor x\rfloor+1\right\}=\lfloor x\rfloor+1$ $,\forall x\in(0,1)$

Implies That $I=\int_{-1}^{0}x^2dx+\int_{0}^{1}\left(\lfloor x\rfloor+1\right)dx$

$$=\frac{1}{3}x\Big|_{-1}^{0}+\int_{0}^{1}(0+1)dx=\frac{4}{3}$$

SO, $\boxed{\int_{-2}^{2}\max\left\{x^2,\lfloor x\rfloor+1\right\}dx=\frac{4}{3}}$

412. Calculate Integral $I=\int_{1}^{e}\frac{1-\log(x)}{x\sqrt{x^2-\log^2(x)}}dx$

Solution; $I=\int_{1}^{e}\frac{1-\log(x)}{x\sqrt{x^2-\log^2(x)}}dx$

$$=\int_{1}^{e}\frac{1-\log(x)}{x^2\sqrt{1-\left(\frac{\log(x)}{x}\right)^2}}dx$$

$$=\int_{1}^{e}\frac{1}{\sqrt{1-\left(\frac{\log(x)}{x}\right)^2}}d\left(\frac{\log(x)}{x}\right)$$

$$=\sin^{-1}\left(\frac{\log(x)}{x}\right)\Big|_{1}^{e}=\sin^{-1}\left(\frac{1}{e}\right)$$

SO, $\boxed{\int_{1}^{e}\frac{1-\log(x)}{x\sqrt{x^2-\log^2(x)}}dx=\sin^{-1}\left(\frac{1}{e}\right)}$

413. Calculate Integral $I=\int_0^{2025}\frac{\lfloor x\rfloor}{\left\lceil\sqrt{x}\right\rceil}dx$

Solution; $I=\int_0^{2025}\frac{\lfloor x\rfloor}{\left\lceil\sqrt{x}\right\rceil}dx$

$$=\sum_{n=0}^{44}\left(\int_{n^2}^{(n+1)^2}\frac{\lfloor x\rfloor}{n+1}dx\right)$$

$$=\sum_{n=0}^{44}\left(\frac{1}{n+1}\int_{n^2}^{(n+1)^2}\lfloor x\rfloor dx\right)$$

$$=\sum_{n=0}^{44}\left[\frac{1}{n+1}\sum_{m=n^2}^{(n+1)^2-1}\left(\int_m^{m+1}m\,dx\right)\right]$$

$$=\sum_{n=0}^{44}\left[\frac{1}{n+1}\left(\sum_{m=n^2}^{(n+1)^2-1}(m)\right)\right]$$

$$=\sum_{n=0}^{44}\left[\frac{1}{n+1}\left(\sum_{m=1}^{(n+1)^2-1}(m)-\sum_{m=1}^{n^2-1}(m)\right)\right]$$

$$=\sum_{n=0}^{44}\left[\frac{1}{n+1}\left(\frac{1}{2}(n+1)^2[(n+1)^2-1]-\frac{1}{2}n^2(n^2-1)\right)\right]$$

$$=\sum_{n=0}^{44}\left[n(2n+1)\right]$$

$$=2\sum_{n=0}^{44}n^2+\sum_{n=0}^{44}n$$

$$=2\cdot\frac{44\cdot(44+1)\cdot(2\cdot44+1)}{6}+\frac{1}{2}\cdot44\cdot(44+1)$$

$$=59730$$

SO, $\boxed{\int_0^{2025}\frac{\lfloor x\rfloor}{\left\lceil\sqrt{x}\right\rceil}dx=59730}$

414. Calculate Integral $I=\int_0^{\infty}\frac{1}{\sqrt{6\left(1^2+2^2+3^2+\cdots+x^2\right)}}dx$

Solution; $I=\int_0^{\infty}\frac{1}{\sqrt{6\left(1^2+2^2+3^2+\cdots+x^2\right)}}dx$

$$=\int_0^{\infty}\frac{1}{\sqrt{x(x+1)(2x+1)}}dx$$

Let : $x=\frac{u-1}{2}\Rightarrow dx=\frac{1}{2}du$ **; If :** $x\in(0,\infty)\Rightarrow u\in(1,\infty)$

Implies That $I=\frac{1}{2}\int_1^{\infty}\frac{1}{\sqrt{\frac{(u-1)(u+1)u}{4}}}dx$

$$=\int_1^{\infty}\frac{u^{-\frac{1}{2}}}{\sqrt{(u^2-1)}}dx$$

Let : $u=v^{-1}\Rightarrow du=-v^{-2}dv$ **; If :** $u\in(1,\infty)\Rightarrow v\in(1,0)$

We Get $I=-\int_1^0\frac{v^{\frac{1}{2}}\cdot v^{-2}}{\sqrt{v^{-2}-1}}dv$

$$=\int_0^1 v^{-\frac{3}{2}}(v^{-2}-1)^{-\frac{1}{2}}\cdot(v^2)^{-\frac{1}{2}}dv$$

$$=\int_0^1(v^2)^{-\frac{3}{4}}(1-v^2)^{-\frac{1}{2}}d(v^2)$$

$$=\int_0^1(v^2)^{\frac{1}{4}-1}(1-v^2)^{\frac{1}{2}-1}d(v^2)$$

$$=\mathbf{B}\left(\frac{1}{4},\frac{1}{2}\right)$$

SO, $\boxed{\int_0^{\infty}\frac{1}{\sqrt{6\left(1^2+2^2+3^2+\cdots+x^2\right)}}dx=\mathbf{B}\left(\frac{1}{4},\frac{1}{2}\right)}$

415. Calculate Integral $I=\int_0^{\infty} xe^{-x}\log(x)\,dx$

Solution; $I=\int_0^{\infty} xe^{-x}\log(x)\,dx$

By Contact $\Gamma(s)=\int_0^{\infty} x^{s-1}e^{-x}dx$

$$\Gamma'(s)=\int_0^{\infty} x^{s-1}e^{-x}\log(x)dx$$

If $S=2$ **Then** $\Gamma'(2)=\int_0^{\infty} xe^{-x}\log^2(x)dx=I$

But $\Gamma'(z)=\Gamma(z)\,\psi(z)\equiv\Gamma'(2)=\Gamma(2)\,\psi(2)$

Which $\Gamma(2)=1$

And $\psi(n)=-\gamma+\sum_{k=1}^{n-1}\left(\frac{1}{k}\right)$ **Then** $\psi(2)=-\gamma+\sum_{k=1}^{1}\left(\frac{1}{k}\right)=1-\gamma$

We Get $\Gamma'(2)=1\cdot(1-\gamma)=1-\gamma$

SO, $\boxed{\int_0^{\infty} xe^{-x}\log(x)\,dx=1-\gamma}$

416. Calculate Integral $I=\int_{-1}^{1} Min\left\{1-x^2,x^2\right\}dx$

Solution; $I=\int_{-1}^{1} Min\left\{1-x^2,x^2\right\}dx$

$\forall x\in(-1,1)$, $1-x^2\cap x^2$ **Timing** $x=-\frac{1}{\sqrt{2}}$, $x=\frac{1}{\sqrt{2}}$

If : $x\in\left(-1,-\frac{1}{\sqrt{2}}\right)$ **That** $1-x^2\le x^2$ **They Have** $Min\left\{1-x^2,x^2\right\}=1-x^2$, $\forall x\in\left(-1,-\frac{1}{\sqrt{2}}\right)$

If : $x\in\left(-\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}}\right)$ **That** $1-x^2\ge x^2$ **They Have** $Min\left\{1-x^2,x^2\right\}=x^2$, $\forall x\in\left(-\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}}\right)$

If : $x\in\left(\frac{1}{\sqrt{2}},1\right)$ **That** $1-x^2\le x^2$ **They Have** $Min\left\{1-x^2,x^2\right\}=1-x^2$, $\forall x\in\left(\frac{1}{\sqrt{2}},1\right)$

Implies That $I=\underbrace{\int_{-1}^{-\frac{1}{\sqrt{2}}}(1-x^2)dx}_{Let:x=-x}+\underbrace{\int_{-\frac{1}{\sqrt{2}}}^{\frac{1}{\sqrt{2}}}x^2dx}_{is\,an\,even\,function}+\int_{\frac{1}{\sqrt{2}}}^{1}(1-x^2)dx$

$$=2\int_0^{\frac{1}{\sqrt{2}}}x^2dx+2\int_{\frac{1}{\sqrt{2}}}^{1}(1-x^2)dx=\frac{2}{3}\left(2-\sqrt{2}\right)$$

SO, $\boxed{\int_{-2}^{2}Min\left\{1-x^2,x^2\right\}dx=\frac{2}{3}\left(2-\sqrt{2}\right)}$

417. Calculate Integral $I=\int_0^{\frac{\pi}{2}}\sin^y(x)dx$

Solution; $I=\int_0^{\frac{\pi}{2}}\sin^y(x)dx$

$$=\frac{1}{2}\cdot 2\int_0^{\frac{\pi}{2}}\sin^{2\left(\frac{y+1}{2}\right)-1}(x)\cos^{2\left(\frac{1}{2}\right)-1}(x)dx$$

$$=\frac{1}{2}\mathbf{B}\left(\frac{y+1}{2},\frac{1}{2}\right)$$

$$=\frac{1}{2}\cdot\frac{\Gamma\left(\frac{y+1}{2}\right)\Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{y+1}{2}+\frac{1}{2}\right)}$$

$$=\frac{\sqrt{\pi}}{2}\cdot\frac{\Gamma\left(\frac{y+1}{2}\right)}{\Gamma\left(\frac{y}{2}+1\right)}=\frac{\sqrt{\pi}}{y}\cdot\frac{\Gamma\left(\frac{y+1}{2}\right)}{\Gamma\left(\frac{y}{2}\right)}$$

SO, $\boxed{\int_0^{\frac{\pi}{2}}\sin^y(x)dx=\frac{\sqrt{\pi}}{y}\cdot\frac{\Gamma\left(\frac{y+1}{2}\right)}{\Gamma\left(\frac{y}{2}\right)}=\begin{cases}\frac{\pi}{2}\cdot\frac{(y-1)!!}{y!!} & ,"y"\ is\,evan\\ \frac{(y-1)!!}{y!!} & ,"y"\ is\,odd\end{cases}}$

418. Calculate Integral $I=\int_0^1 (x^x)^{(x^x)^{(x^x)^{\cdot^{\cdot}}}}dx$

Solution; $I=\int_0^1 (x^x)^{(x^x)^{(x^x)^{\cdot^{\cdot}}}}dx$

Let : $y=(x^x)^{(x^x)^{(x^x)^{\cdot^{\cdot}}}} \equiv \log(y)=y\log(x^x)$

$$\equiv -\log(y)e^{-\log(y)}=-\log(x^x)$$
$$\equiv \mathbf{W}\left(-\log(y)e^{-\log(y)}\right)=\mathbf{W}\left[-x\log(x)\right]$$
$$\equiv \log(y)=-\mathbf{W}\left[-x\log(x)\right]$$
$$\equiv y=e^{-\mathbf{W}\left[-x\log(x)\right]}$$

By : $Z=\mathbf{W}(Z)e^{\mathbf{W}(Z)}$ **Take :** $Z=-\mathbf{W}\left[-x\log(x)\right]$

We Get : $-x\log(x)=\mathbf{W}\left(-x\log(x)\right)e^{\mathbf{W}(-x\log(x))}$

$$e^{\mathbf{W}[x\log(x)]}=\frac{-x\log(x)}{\mathbf{W}\left[-x\log(x)\right]} \equiv e^{-\mathbf{W}[x\log(x)]}=\frac{\mathbf{W}\left[-x\log(x)\right]}{-x\log(x)}$$

Implies That $y=\dfrac{\mathbf{W}\left[-x\log(x)\right]}{-x\log(x)}$

We obtain $I=\int_0^1 \dfrac{\mathbf{W}\left[-x\log(x)\right]}{-x\log(x)}dx$

But : $\mathbf{W}(x)=\sum_{n=1}^{\infty}\dfrac{(-n)^{n-1}x^n}{n!} \Rightarrow \mathbf{W}\left(-x\log(x)\right)=\sum_{n=1}^{\infty}\dfrac{(-n)^{n-1}\left(-x\log(x)\right)^n}{n!}$

We Get : $I=\int_0^1\left[\dfrac{1}{-x\log(x)}\sum_{n=1}^{\infty}\dfrac{(-n)^{n-1}\left(-x\log(x)\right)^n}{n!}\right]dx$

$$=\sum_{n=1}^{\infty}\left[\frac{(-n)^{n-1}}{n!}\int_0^1\left(\frac{\left(-x\log(x)\right)^n}{-x\log(x)}\right)dx\right]$$

$$=\sum_{n=1}^{\infty}\left[\frac{(-n)^{(n-1)}}{n!}\int_0^1\left(\left(-x\log(x)\right)^{n-1}\right)dx\right]$$

Let : $x=e^{-u}\Rightarrow dx=-e^{-u}du$; **If:** $x\in(0,1)\Rightarrow u\in(\infty,0)$

Implies That $I=\sum_{n=1}^{\infty}\left[\frac{(-n)^{(n-1)}}{n!}\int_{\infty}^{0}\left(\left(ue^{-u}\right)^{n-1}\right)\left(-e^{-u}\right)du\right]$

$$=\sum_{n=1}^{\infty}\left[\frac{(-n)^{(n-1)}}{n!}\int_0^{\infty}\left(u^{n-1}e^{-nu}\right)dx\right]$$

Let : $z=nu\Rightarrow du=\frac{1}{n}dz$; **If** : $u\in(0,\infty)\Rightarrow z\in(0,\infty)$

We Get $I=\sum_{n=1}^{\infty}\left[\frac{(-n)^{(n-1)}}{n!}\int_0^{\infty}\left(\left(\frac{z}{n}\right)^{n-1}e^{-z}\right)\frac{dz}{n}\right]$

$$=\sum_{n=1}^{\infty}\left[\frac{(-n)^{n-1}}{n!n^n}\int_0^{\infty}\left(z^{n-1}e^{-z}\right)dz\right]$$

$$=\sum_{n=1}^{\infty}\left[\frac{(-n)^{n-1}}{n!n^n}\cdot(n-1)!\right]$$

$$=\sum_{n=1}^{\infty}\left[\frac{(-n)^{n-1}}{n\cdot n^n}\right]$$

$$=\sum_{n=1}^{\infty}\left[\frac{(-1)^{n-1}}{n^2}\right]$$

$$=\eta(2)$$

$$=\left(1-2^{2-1}\right)\zeta(2)$$

$$=\frac{1}{2}\cdot\frac{\pi^2}{6}=\frac{\pi^2}{12}$$

SO, $\boxed{\int_0^1(x^x)^{(x^x)^{(x^x)^{\cdots}}}dx=\frac{\pi^2}{12}}$

419. Calculate Integral $I=\int_1^e \frac{\log(x)}{\left(1+\log(x)\right)^2}dx$

Solution; $I=\int_1^e \frac{\log(x)}{\left(1+\log(x)\right)^2}dx$

$$=\int_1^e \frac{e^{\log(x)}\log(x)}{\left(1+\log(x)\right)^2}d\left(\log(x)\right)$$

$$=\int_1^e \frac{e^{\log(x)}\left(1+\log(x)\right)-e^{\log(x)}}{\left(1+\log(x)\right)^2}d\left(\log(x)\right)$$

$$=\int_1^e \frac{\left(e^{\log(x)}\right)'\left(1+\log(x)\right)-\left(1+\log(x)\right)'e^{\log(x)}}{\left(1+\log(x)\right)^2}d\left(\log(x)\right)$$

$$=\int_1^e \left(\frac{e^{\log(x)}}{1+\log(x)}\right)'d\left(\log(x)\right)$$

$$=\frac{e^{\log(x)}}{1+\log(x)}\Bigg|_1^e=\frac{e}{2}-1$$

SO, $\boxed{\int_1^e \frac{\log(x)}{\left(1+\log(x)\right)^2}dx=\frac{e}{2}-1}$

420. Calculate Integral $I=\int_e^{e^2}\sqrt{x}\log(x)\sqrt{\log(x)-1}\,dx$

Solution; $I=\int_e^{e^2}\sqrt{x}\log(x)\sqrt{\log(x)-1}\,dx$

Let : $\log(x)=u\equiv x=e^u\Rightarrow dx=e^u du$ **; If** : $x\in(e,e^2)\Rightarrow u\in(1,2)$

We obtain $I=\int_1^2 u\sqrt{x-1}\,e^{\frac{1}{2}u}e^u\,du$

$$=\int_1^2 u\sqrt{u-1}\,e^{\frac{1}{2}u}e^u\,du$$

$$=\int_1^2 u\sqrt{u-1}\,e^{\frac{3}{2}u}\,du$$

Let : $v=u-1\Rightarrow dv=du$; **If** $u\in(1,2)\Rightarrow v\in(0,1)$

We Get $I=\int_0^1(v+1)v^{\frac{1}{2}}\cdot e^{\frac{3}{2}(v+1)}\,du$

$$=e^{\frac{3}{2}}\int_0^1\left(v^{\frac{3}{2}}+v^{\frac{1}{2}}\right)e^{\frac{3}{2}v}\,dv$$

Let : $z=v^{\frac{3}{2}}e^{\frac{3}{2}v}\Rightarrow\frac{2}{3}dz=\left(v^{\frac{3}{2}}-v^{\frac{1}{2}}\right)e^{\frac{3}{2}v}dv$; **If** : $u\in(0,1)\Rightarrow z\in\left(0,e^{\frac{3}{2}}\right)$

Implies That $I=\int_0^{e^{\frac{3}{2}}}\frac{2}{3}dz$

$$=\frac{2}{3}z\Big|_0^{e^{\frac{3}{2}}}=\frac{2}{3}e^3$$

SO, $\boxed{\int_e^{e^2}\sqrt{x}\log(x)\sqrt{\log(x)-1}\,dx=\frac{2}{3}e^3}$

421. Calculate Integral $I=\lim_{n\to\infty}\left(\frac{1^m+2^m+3^m+\ldots+n^m}{n^{m+1}}\right)$

Solution; $I=\lim_{n\to\infty}\left(\frac{1^m+2^m+3^m+\ldots+n^m}{n^{m+1}}\right)$

$$=\lim_{n\to\infty}\frac{1}{n}\left(\left(\frac{1}{n}\right)^m+\left(\frac{2}{n}\right)^m+\left(\frac{3}{n}\right)^m+\ldots+\left(\frac{n}{n}\right)^m\right)$$

$$=\lim_{n\to\infty}\frac{1}{n}\sum_{k=1}^{n}\left(\frac{k}{n}\right)^m$$

$$=\int_0^1x^m\,dx=\frac{1}{m+1}$$

SO, $\boxed{\lim_{n\to\infty}\left(\frac{1^m+2^m+3^m+\ldots+n^m}{n^{m+1}}\right)=\frac{1}{m+1}}$

422. Calculate Integral $I=\lim_{n\to\infty}\left(\frac{1}{n^2}\sec^2\left(\frac{1}{n^2}\right)+\frac{2}{n^2}\sec^2\left(\frac{4}{n^2}\right)+\ldots+\frac{n}{n^2}\sec^2\left(\frac{n^2}{n^2}\right)\right)$

Solution; $I=\lim_{n\to\infty}\left(\frac{1}{n^2}\sec^2\left(\frac{1}{n^2}\right)+\frac{2}{n^2}\sec^2\left(\frac{4}{n^2}\right)+\ldots+\frac{n}{n^2}\sec^2\left(\frac{n^2}{n^2}\right)\right)$

$$=\lim_{n\to\infty}\frac{1}{n}\left(\frac{1}{n}\sec^2\left(\frac{1}{n}\right)^2+\frac{2}{n}\sec^2\left(\frac{2}{n}\right)^2+\cdots+\frac{n}{n}\sec^2\left(\frac{n}{n}\right)^2\right)$$

$$=\lim_{n\to+\infty}\frac{1}{n}\sum_{k=1}^{n}\left(\left(\frac{k}{n}\right)\sec^2\left(\frac{k}{n}\right)^2\right)$$

$$=\int_0^1 x\sec^2(x^2)\,dx$$

$$=\frac{1}{2}\tan(x^2)\Big|_0^1=\frac{1}{2}\tan(1)$$

SO, $\boxed{\lim_{n\to\infty}\left(\frac{1}{n^2}\sec^2\left(\frac{1}{n^2}\right)+\frac{2}{n^2}\sec^2\left(\frac{4}{n^2}\right)+\ldots+\frac{n}{n^2}\sec^2\left(\frac{n^2}{n^2}\right)\right)=\frac{1}{2}\tan(1)}$

423. Calculate Integral $I=\lim_{n\to\infty}\left(\frac{\sqrt{1}+\sqrt{2}+\sqrt{3}+\ldots+\sqrt{n}}{n\sqrt{n}}\right)$

Solution; $I=\lim_{n\to\infty}\left(\frac{\sqrt{1}+\sqrt{2}+\sqrt{3}+\ldots+\sqrt{n}}{n\sqrt{n}}\right)$

$$=\lim_{n\to\infty}\frac{1}{n}\left(\sqrt{\frac{1}{n}}+\sqrt{\frac{2}{n}}+\sqrt{\frac{3}{n}}+\ldots+\sqrt{\frac{n}{n}}\right)$$

$$=\lim_{n\to+\infty}\frac{1}{n}\sum_{k=1}^{n}\left(\sqrt{\frac{k}{n}}\right)$$

$$=\int_0^1\sqrt{x}dx=\frac{2}{3}$$

SO, $\boxed{\lim_{n\to\infty}\left(\frac{\sqrt{1}+\sqrt{2}+\sqrt{3}+\ldots+\sqrt{n}}{n\sqrt{n}}\right)=\frac{2}{3}}$

424. Calculate Integral $I=\lim\limits_{n\to\infty}\left(\frac{n+1}{n^2+1^2}+\frac{n+2}{n^2+2^2}+\frac{n+3}{n^2+3^2}+...+\frac{n+n}{n^2+n^2}\right)$

Solution; $I=\lim\limits_{n\to\infty}\left(\frac{n+1}{n^2+1^2}+\frac{n+2}{n^2+2^2}+\frac{n+3}{n^2+3^2}+...+\frac{n+n}{n^2+n^2}\right)$

$$=\lim_{n\to\infty}\frac{1}{n}\left(\frac{1+\left(\frac{1}{n}\right)}{1+\left(\frac{1}{n}\right)^2}+\frac{1+\left(\frac{2}{n}\right)}{1+\left(\frac{2}{n}\right)^2}+\frac{1+\left(\frac{3}{n}\right)}{1+\left(\frac{3}{n}\right)^2}+...+\frac{1+\left(\frac{n}{n}\right)}{1+\left(\frac{n}{n}\right)^2}\right)$$

$$=\lim_{n\to+\infty}\frac{1}{n}\sum_{k=1}^{n}\left(\frac{1+\left(\frac{k}{n}\right)}{1+\left(\frac{k}{n}\right)^2}\right)$$

$$=\int_0^1\frac{1+x}{1+x^2}dx$$

$$=\frac{\pi}{4}+\frac{1}{2}\log(2)$$

SO, $\boxed{\lim\limits_{n\to\infty}\left(\frac{n+1}{n^2+1^2}+\frac{n+2}{n^2+2^2}+\frac{n+3}{n^2+3^2}+...+\frac{n+n}{n^2+n^2}\right)=\frac{\pi}{4}+\frac{1}{2}\log(2)}$

425. Calculate Integral $I=\lim\limits_{n\to\infty}\left(\frac{1^2}{1^3+n^3}+\frac{2^2}{2^3+n^3}+\frac{3^2}{3^3+n^3}+\cdots+\frac{n^2}{n^3+n^3}\right)$

Solution; $I=\lim\limits_{n\to\infty}\left(\frac{1^2}{1^3+n^3}+\frac{2^2}{2^3+n^3}+\frac{3^2}{3^3+n^3}+\cdots+\frac{n^2}{n^3+n^3}\right)$

$$=\lim_{n\to\infty}\frac{1}{n}\left(\frac{\left(\frac{1}{n}\right)^2}{1+\left(\frac{1}{n}\right)^3}+\frac{\left(\frac{2}{n}\right)^2}{1+\left(\frac{2}{n}\right)^3}+\frac{\left(\frac{3}{n}\right)^2}{1+\left(\frac{3}{n}\right)^3}+...+\frac{\left(\frac{n}{n}\right)^2}{1+\left(\frac{n}{n}\right)^3}\right)$$

$$= \lim_{n\to+\infty} \frac{1}{n} \sum_{k=1}^{n} \left(\frac{\left(\frac{k}{n}\right)^2}{1+\left(\frac{k}{n}\right)^3} \right)$$

$$= \int_0^1 \frac{x^2}{1+x^3}\,dx = \frac{1}{3}\log(2)$$

SO, $\boxed{\lim_{n\to\infty}\left(\frac{1^2}{1^3+n^3}+\frac{2^2}{2^3+n^3}+\frac{3^2}{3^3+n^3}+\ldots+\frac{n^2}{n^3+n^3}\right)=\frac{1}{3}\log(2)}$

426. Calculate Integral $I = \lim_{n\to\infty}\left[\left(1+\frac{1^2}{n^2}\right)^{\frac{2}{n^2}} \times \left(1+\frac{2^2}{n^2}\right)^{\frac{4}{n^2}} \times \ldots \times \left(1+\frac{n^2}{n^2}\right)^{\frac{2n}{n^2}}\right]$

Solution; $I = \lim_{n\to\infty}\left[\left(1+\frac{1^2}{n^2}\right)^{\frac{2}{n^2}} \times \left(1+\frac{2^2}{n^2}\right)^{\frac{4}{n^2}} \times \ldots \times \left(1+\frac{n^2}{n^2}\right)^{\frac{2n}{n^2}}\right]$

$$= \lim_{n\to\infty}\left[\prod_{k=0}^{n}\left(1+\frac{k^2}{n^2}\right)^{\frac{2k}{n^2}}\right]$$

$$= \exp\left[\lim_{n\to\infty}\log\left(\prod_{k=0}^{n}\left(1+\frac{k^2}{n^2}\right)^{\frac{2k}{n^2}}\right)\right]$$

$$= \exp\left[2\lim_{n\to\infty}\frac{1}{n}\left(\sum_{k=1}^{n}\frac{k}{n}\log\left(1+\left(\frac{k}{n}\right)^2\right)\right)\right]$$

$$= \exp\left(2\int_0^1 x\log(x^2+1)\,dx\right) = \frac{2}{e}$$

SO, $\boxed{\lim_{n\to\infty}\left[\left(1+\frac{1^2}{n^2}\right)^{\frac{2}{n^2}} \times \left(1+\frac{2^2}{n^2}\right)^{\frac{4}{n^2}} \times \ldots \times \left(1+\frac{n^2}{n^2}\right)^{\frac{2n}{n^2}}\right]=\frac{2}{e}}$

427. Calculate Integral $I=\lim\limits_{n\to\infty}\left(\frac{1}{(n+1)^2}+\frac{1}{(n+2)^2}+\frac{1}{(n+3)^2}+\cdots+\frac{1}{(n+n)^2}\right)$

Solution; $I=\lim\limits_{n\to\infty}\left(\frac{1}{(n+1)^2}+\frac{1}{(n+2)^2}+\frac{1}{(n+3)^2}+\cdots+\frac{1}{(n+n)^2}\right)$

$$=\lim_{n\to\infty}\frac{1}{n}\left(\frac{1}{\left(1+\frac{1}{n}\right)^2}+\frac{1}{\left(1+\frac{2}{n}\right)^2}+\frac{1}{\left(1+\frac{3}{n}\right)^2}+\cdots+\frac{1}{\left(1+\frac{n}{n}\right)^2}\right)$$

$$=\lim_{n\to\infty}\frac{1}{n}\left(\sum_{k=1}^{n}\frac{1}{\left(1+\frac{k}{n}\right)^2}\right)$$

$$=\int_0^1\frac{1}{(1+x)^2}dx=\frac{1}{2}$$

SO, $\boxed{\lim\limits_{n\to\infty}\left(\frac{1}{(n+1)^2}+\frac{1}{(n+2)^2}+\frac{1}{(n+3)^2}+\ldots+\frac{1}{(n+n)^2}\right)=\frac{1}{2}}$

428. Calculate Integral $I=\lim\limits_{n\to\infty}\left(\frac{1}{n\sqrt{n^2+1^2}}+\frac{2}{n\sqrt{n^2+2^2}}+\cdots+\frac{2n}{n\sqrt{n^2+(2n)^2}}\right)$

Solution; $I=\lim\limits_{n\to\infty}\left(\frac{1}{n\sqrt{n^2+1^2}}+\frac{2}{n\sqrt{n^2+2^2}}+\cdots+\frac{2n}{n\sqrt{n^2+(2n)^2}}\right)$

$$=\lim_{n\to\infty}\frac{1}{n}\left(\frac{\frac{1}{n}}{\sqrt{1+\left(\frac{1}{n}\right)^2}}+\frac{\frac{2}{n}}{\sqrt{1+\left(\frac{2}{n}\right)^2}}+\ldots+\frac{\frac{2n}{n}}{\sqrt{1+\left(\frac{2n}{n}\right)^2}}\right)$$

$$=\lim_{n\to\infty}\frac{1}{n}\left(\sum_{k=1}^{2n}\frac{\left(\frac{k}{n}\right)}{\sqrt{1+\left(\frac{k}{n}\right)^2}}\right)$$

$$=\int_0^2\frac{x}{\sqrt{1+x^2}}dx=\sqrt{5}-1$$

SO, $\boxed{\lim_{n\to\infty}\left(\frac{1}{n\sqrt{n^2+1^2}}+\frac{2}{n\sqrt{n^2+2^2}}+\cdots+\frac{2n}{n\sqrt{n^2+(2n)^2}}\right)=\sqrt{5}-1}$

429. Calculate Integral $I=\lim_{n\to\infty}\left[\left(1+\frac{1}{n^2}\right)\left(1+\frac{4}{n^2}\right)\left(1+\frac{9}{n^2}\right)...\left(1+\frac{n^2}{n^2}\right)\right]^{\frac{1}{n}}$

Solution; $I=\lim_{n\to\infty}\left[\left(1+\frac{1}{n^2}\right)\left(1+\frac{4}{n^2}\right)\left(1+\frac{9}{n^2}\right)...\left(1+\frac{n^2}{n^2}\right)\right]^{\frac{1}{n}}$

$$=\lim_{n\to\infty}\left(\prod_{k=1}^{n}\left(1+\frac{n^2}{n^2}\right)\right)^{\frac{1}{n}}$$

$$=\exp\left[\lim_{n\to\infty}\log\left(\prod_{k=1}^{n}\left(1+\frac{n^2}{n^2}\right)\right)^{\frac{1}{n}}\right]$$

$$=\exp\left[\lim_{n\to\infty}\frac{1}{n}\left(\sum_{k=1}^{n}\log\left(1+\frac{n^2}{n^2}\right)\right)\right]$$

$$=\exp\left(\underbrace{\int_0^1\log(1+x^2)\,dx}_{Use\ partial\ integral}\right)=2e^{\frac{\pi}{4}-2}$$

SO, $\boxed{\lim_{n\to\infty}\left[\left(1+\frac{1}{n^2}\right)\left(1+\frac{4}{n^2}\right)\left(1+\frac{9}{n^2}\right)...\left(1+\frac{n^2}{n^2}\right)\right]^{\frac{1}{n}}=2e^{\frac{\pi}{4}-2}}$

430. Calculate Integral $I=\lim_{n\to\infty}\left(\frac{1}{n}+\frac{1}{n+1}+\frac{1}{n+2}+...+\frac{1}{n+(a-1)n}\right)$

Solution; $I=\lim_{n\to\infty}\left(\frac{1}{n}+\frac{1}{n+1}+\frac{1}{n+2}+...+\frac{1}{n+(a-1)n}\right)$

$$=\lim_{n\to\infty}\frac{1}{n}\left(\frac{1}{1+\frac{0}{n}}+\frac{1}{1+\frac{1}{n}}+\frac{1}{1+\frac{2}{n}}+...+\frac{1}{1+\frac{a-1}{n}}\right)$$

$$=\lim_{n\to\infty}\frac{1}{n}\left(\sum_{k=0}^{a-1}\frac{1}{1+\left(\frac{k}{n}\right)}\right)=\int_0^{a-1}\frac{1}{(1+x)}dx=\log(a)$$

SO, $\boxed{\lim_{n\to\infty}\left(\frac{1}{n}+\frac{1}{n+1}+\frac{1}{n+2}+...+\frac{1}{n+(a-1)n}\right)=\log(a)}$

431. Calculate Integral $I=\lim_{n\to\infty}\left(\frac{n^2}{n^2+1}+\frac{n^2}{n^2+4}+\frac{n^2}{n^2+9}+...+\frac{n^2}{n^2+n^2}\right)$

Solution; $I=\lim_{n\to\infty}\left(\frac{n^2}{n^2+1}+\frac{n^2}{n^2+4}+\frac{n^2}{n^2+9}+...+\frac{n^2}{n^2+n^2}\right)$

$$=\lim_{n\to\infty}\frac{1}{n}\left(\frac{1}{1+\left(\frac{1}{n}\right)^2}+\frac{1}{1+\left(\frac{2}{n}\right)^2}+\frac{1}{1+\left(\frac{3}{n}\right)^2}+...+\frac{1}{1+\left(\frac{n}{n}\right)^2}\right)$$

$$=\lim_{n\to\infty}\frac{1}{n}\left(\sum_{k=1}^{n}\frac{1}{1+\left(\frac{k}{n}\right)^2}\right)=\int_0^1\frac{1}{1+x^2}dx=\frac{\pi}{4}$$

SO, $\boxed{\lim_{n\to\infty}\left(\frac{n^2}{n^2+1}+\frac{n^2}{n^2+4}+\frac{n^2}{n^2+9}+...+\frac{n^2}{n^2+n^2}\right)=\frac{\pi}{4}}$

432. Calculate Integral $I=\int_0^{\infty}\left(\frac{\mathbf{W}(x)}{x}\right)^2\cdot\frac{1}{\sqrt{x}}dx$

Solution; $I=\int_0^{\infty}\left(\frac{\mathbf{W}(x)}{x}\right)^2\cdot\frac{1}{\sqrt{x}}dx$

Let $x=ue^u\Rightarrow dx=(u+1)e^u du$ **; If** $x\in(0,\infty)\Rightarrow u\in(0,\infty)$

We Get $I=\int_0^{\infty}\frac{(u+1)u^2e^u}{(ue^u)^2\sqrt{ue^u}}dx$

$$=\int_0^{\infty}\left(u^{-1/2}e^{-3u/2}+u^{1/2}e^{-3u/2}\right)dx$$

Let $t=\frac{3}{2}u\Rightarrow du=\frac{2}{3}du$ **; If** $u\in(0,\infty)\Rightarrow t\in(0,\infty)$

Implies That $I=\frac{2}{3}\int_0^{\infty}\left(\left(\frac{2}{3}t\right)^{-1/2}e^{-t}+\left(\frac{2}{3}t\right)^{1/2}e^{-t}\right)dt$

$$=\sqrt{\frac{2}{3}}\cdot\Gamma\left(\frac{1}{2}\right)+\frac{2}{3}\sqrt{\frac{2}{3}}\cdot\Gamma\left(\frac{3}{2}\right)$$

$$=\frac{4\sqrt{6\pi}}{9}$$

SO, $\boxed{\int_0^{\infty}\left(\frac{\mathbf{W}(x)}{x}\right)^2\cdot\frac{1}{\sqrt{x}}dx=\frac{4\sqrt{6\pi}}{9}}$

433. Calculate Integral $I=\int_0^{\frac{\pi}{2}}\cos^2(x)\sqrt{1-\cos^2(x)}\log\left(\sin(x)\right)dx$

Solution; $I=\int_0^{\frac{\pi}{2}}\cos^2(x)\sqrt{1-\cos^2(x)}\log\left(\sin(x)\right)dx$

$$=\int_0^{\frac{\pi}{2}}\cos^2(x)\sin(x)\log\left(\sin(x)\right)dx$$

$$=-\int_0^{\frac{\pi}{2}}\cos^2(x)\log\left(\sqrt{1-\cos^2(x)}\right)d\left(\cos(x)\right)$$

Let : $\cos(x)=u \Rightarrow d\left(\cos(x)\right)=du$; **If** : $x\in\left(0,\frac{\pi}{2}\right)\Rightarrow u\in(0,1)$

We Get $I=-\frac{1}{2}\int_0^1 x^2\log(1-x^2)dx$

$$=-\frac{1}{2}\int_0^1\left(-x^2\sum_{k=1}^{\infty}\left(\frac{x^{2k}}{k}\right)\right)dx$$

$$=\frac{1}{2}\sum_{k=1}^{\infty}\left[\frac{1}{k}\left(\int_0^1 x^{2k+2}dx\right)\right]$$

$$=\frac{1}{2}\sum_{k=1}^{\infty}\left[\frac{1}{k(2k+3)}\right]$$

$$=\frac{1}{2}\sum_{k=1}^{\infty}\left[\frac{3/2}{k\left(k+3/2\right)}\right]$$

$$=\frac{1}{6}\left[\gamma+\psi\left(\frac{3}{2}+1\right)\right]$$

$$=\frac{1}{6}\left[\gamma+\frac{2}{3}+2+\psi\left(\frac{1}{2}\right)\right]$$

$$=\frac{1}{6}\left[\gamma+\frac{8}{3}-\gamma-\log(4)\right]$$

$$=\frac{4-\log(8)}{9}$$

SO, $\boxed{\int_0^{\frac{\pi}{2}}\cos^2(x)\sqrt{1-\cos^2(x)}\log\left(\sin(x)\right)dx=\frac{4-\log(8)}{9}}$

Note : $\sum_{n=1}^{\infty}\frac{m}{n(n+m)}=\gamma+\psi(m+1)$

$$:\psi(x+1)=\frac{1}{x}+\psi(x)$$

$$:\psi\left(\frac{1}{2}\right)=-\gamma-\log(4)$$

434. Calculate Integral $I=\int_0^{\frac{\pi}{2}}\frac{x\sin(2x)}{1+\cos^2(2x)}dx$

Solution; $I=\int_0^{\frac{\pi}{2}}\frac{x\sin(2x)}{1+\cos^2(2x)}dx$

$$=\int_0^{\frac{\pi}{2}}\frac{\left(\pi/2-x\right)\sin(2x)}{1+\cos^2(2x)}dx$$

$$=\frac{\pi}{4}\int_0^{\frac{\pi}{2}}\frac{\sin(2x)}{1+\cos^2(2x)}dx$$

$$=-\frac{\pi}{8}\int_0^{\frac{\pi}{2}}\frac{1}{1+\cos^2(2x)}d\left(\cos(2x)\right)$$

$$=-\frac{\pi}{8}\left[\tan^{-1}\left(\cos(2x)\right)\right]\Big|_0^{\frac{\pi}{2}}$$

$$=\frac{\pi^2}{16}$$

SO, $\boxed{\int_0^{\frac{\pi}{2}}\frac{x\sin(2x)}{1+\cos^2(2x)}dx=\frac{\pi^2}{16}}$

435. Calculate Integral $I=\int_0^{\infty}\frac{\arctan(x)}{x(x^2+1)}dx$

Solution; $I=\int_0^{\infty}\frac{\arctan(x)}{x(x^2+1)}dx$

Let : $x=\tan(t)\Rightarrow dx=\left(\tan^2(t)+1\right)dt$ **; If** : $x\in(0,\infty)\Rightarrow u\in\left(0,\frac{\pi}{2}\right)$

Implies That $I=\int_0^{\frac{\pi}{2}}\frac{t}{\tan(t)(\tan^2(t)+1)}\cdot\left(\tan^2(t)+1\right)dt$

$$=\int_0^{\frac{\pi}{2}}\frac{t}{\tan(t)}dt$$

$$= \int_0^{\frac{\pi}{2}} t\, d\left(\log(\sin t)\right)$$

Let $u = t \Rightarrow du = dt$ **And** $dv = d\left(\log(\sin(t))\right) \Rightarrow v = \log\left(\sin(t)\right)$

We Get $I = t\log\left(\sin(t)\right)\Big|_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} \log\left(\sin(t)\right) dt$

$$= -\int_0^{\frac{\pi}{2}} \log\left(\sin(t)\right) dt$$

$$= -\left(-\frac{\pi}{2}\log(2)\right)$$

$$= \frac{\pi}{2}\log(2)$$

SO, $\boxed{\int_0^{\infty} \frac{\arctan(x)}{x(x^2+1)}\, dx = \frac{\pi}{2}\log(2)}$

436. Calculate Integral $I = \int_{-\infty}^{+\infty} \frac{x^{2n}}{x^{2n} + \left(x^2-1\right)^{2n}}\, dx$

Solution; $I = \int_{-\infty}^{+\infty} \frac{x^{2n}}{x^{2n} + \left(x^2-1\right)^{2n}}\, dx$

$$= \int_{-\infty}^{+\infty} \frac{1}{1 + \left(x - \frac{1}{x}\right)^{2n}}\, dx$$

$$= \int_{-\infty}^{+\infty} \frac{1}{1+x^{2n}}\, dx$$

$$= 2\int_0^{\infty} \frac{1}{1+x^{2n}}\, dx$$

Let $x = t^{\frac{1}{2n}} \Rightarrow dx = \frac{1}{2n} t^{\frac{1}{2n}-1} dt$ **; If** $: x \in (0,\infty) \Rightarrow t \in (0,\infty)$

Implies That $I = 2 \cdot \frac{1}{2n} \int_0^{\infty} \frac{t^{\frac{1}{2n}-1}}{1+t}\, dt$

$$=\frac{1}{n}\int_0^{\infty}\frac{t^{\frac{1}{2n}-1}}{(1+t)^{\frac{1}{2n}+1-\frac{1}{2n}}}dt$$

$$=\frac{1}{n}\mathbf{B}\left(\frac{1}{2n},1-\frac{1}{2n}\right)$$

$$=\frac{\pi}{n}\csc\left(\frac{\pi}{2n}\right)$$

SO, $\boxed{\int_{-\infty}^{+\infty}\frac{x^{2n}}{x^{2n}+\left(x^2-1\right)^{2n}}dx=\frac{\pi}{n}\csc\left(\frac{\pi}{2n}\right)}$

437. Calculate Integral $I=\int_0^2\min\left\{x,\sqrt{4-x^2}\right\}dx$

Solution; $I=\int_0^2\min\left\{x,\sqrt{4-x^2}\right\}dx$

$\forall x\in(0,2)$, $x\cap\sqrt{4-x^2}$ **Timing** $x=\sqrt{2}$

If : $x\in(0,\sqrt{2})$ **That** $x\le\sqrt{4-x^2}$ **They Have** $\min\left\{x,\sqrt{4-x^2}\right\}=x$ $,\forall x\in(0,\sqrt{2})$

If : $x\in(\sqrt{2},2)$ **That** $x\ge\sqrt{4-x^2}$ **They Have** $\min\left\{x,x^2\right\}=\sqrt{4-x^2}$ $,\forall x\in(\sqrt{2},2)$

Implies That $I=\int_0^{\sqrt{2}}xdx+\int_{\sqrt{2}}^2\sqrt{4-x^2}dx$

By : $\int_0^{\sqrt{2}}xdx=1$

By : $\int_{\sqrt{2}}^2\sqrt{4-x^2}dx=\frac{\pi}{2}-1$

We Get $I=\frac{\pi}{2}$

SO, $\boxed{\int_0^2\min\left\{x,\sqrt{4-x^2}\right\}dx=\frac{\pi}{2}}$

438. Calculate Integral $I=\int_0^{\pi}\sin^2\left(x-\sqrt{\pi^2-x^2}\right)dx$

Solution; $I=\int_0^{\pi}\sin^2\left(x-\sqrt{\pi^2-x^2}\right)dx$

Let : $x=\pi\sin(\theta)\Rightarrow dx=\pi\cos(\theta)d\theta$ **; If :** $x\in(0,\pi)\Rightarrow\theta\in\left(0,\frac{\pi}{2}\right)$

Implies That $I=\pi\int_0^{\frac{\pi}{2}}\sin^2\left(\pi\sin(\theta)-\sqrt{\pi^2-\pi^2\sin^2(\theta)}\right)\cos(\theta)d\theta$

$$=\frac{\pi}{2}\int_0^{\frac{\pi}{2}}\left[1-\cos\left(2\pi\sin(\theta)-2\pi\cos(\theta)\right)\right]\cos(\theta)d\theta$$

$$=\frac{\pi}{2}\int_0^{\frac{\pi}{2}}\cos(\theta)\,d\theta-\frac{\pi}{2}\int_0^{\frac{\pi}{2}}\cos\left(2\pi\sin(\theta)-2\pi\cos(\theta)\right)\cos(\theta)d\theta$$

$$=\frac{\pi}{2}-\frac{\pi}{2}J\qquad(*)$$

For : $J=\int_0^{\frac{\pi}{2}}\cos\left(2\pi\sin(\theta)-2\pi\cos(\theta)\right)\cos(\theta)d\theta$ **(!)**

$$=\int_0^{\frac{\pi}{2}}\cos\left(2\pi\cos(\theta)-2\pi\sin(\theta)\right)\sin(\theta)d\theta$$ **(!!)**

Take (!) + (!!) Then $2J=\int_0^{\frac{\pi}{2}}\left(\cos(\theta)+\sin(\theta)\right)\cos\left(2\pi\cos(\theta)-2\pi\sin(\theta)\right)d\theta$

$$=\frac{1}{2\pi}\int_0^{\frac{\pi}{2}}\cos\left(2\pi\sin(\theta)-2\pi\cos(\theta)\right)d\left(2\pi\sin(\theta)-2\pi\cos(\theta)\right)$$

$$=\frac{1}{2\pi}\sin\left(2\pi\sin(\theta)-2\pi\cos(\theta)\right)\Big|_0^{\frac{\pi}{2}}$$

$$=0$$

That (*) We Get $I=\frac{\pi}{2}-\frac{\pi}{2}\times0=\frac{\pi}{2}$

SO, $\boxed{\int_0^{\pi}\sin^2\left(x-\sqrt{\pi^2-x^2}\right)dx=\frac{\pi}{2}}$

439. Calculate Integral $I=\int_0^{\frac{\pi}{3}}\frac{\sec(x)+\tan(x)}{\csc(x)+\cot(x)}\big(\sec(x)+\csc(x)\big)dx$

Solution; $I=\int_0^{\frac{\pi}{3}}\frac{\sec(x)+\tan(x)}{\csc(x)+\cot(x)}\big(\sec(x)+\csc(x)\big)dx$

$$=\int_0^{\frac{\pi}{3}}e^{\log\left(\frac{\sec(x)+\tan(x)}{\csc(x)+\cot(x)}\right)}\big(\sec(x)+\csc(x)\big)dx$$

$$=\int_0^{\frac{\pi}{3}}e^{\log\left(\frac{\sec(x)+\tan(x)}{\csc(x)+\cot(x)}\right)}d\left[\log\left(\frac{\sec(x)+\tan(x)}{\csc(x)+\cot(x)}\right)\right]$$

$$=e^{\log\left(\frac{\sec(x)+\tan(x)}{\csc(x)+\cot(x)}\right)}\Bigg|_0^{\frac{\pi}{3}}$$

$$=\frac{2+\sqrt{3}}{\sqrt{3}}$$

SO, $\boxed{\int_0^{\frac{\pi}{3}}\frac{\sec(x)+\tan(x)}{\csc(x)+\cot(x)}\big(\sec(x)+\csc(x)\big)dx=\frac{2+\sqrt{3}}{\sqrt{3}}}$

440. Calculate Integral $I=\int_0^1\frac{e^{\tan^{-1}(x)}}{(x^2+1)^{3/2}}dx$

Solution; $I=\int_0^1\frac{e^{\tan^{-1}(x)}}{(x^2+1)^{3/2}}dx$

Let : $x=\tan(\omega)\Rightarrow dx=\big(\tan^2(\omega)+1\big)d\omega$ **; If :** $x\in(0,1)\Rightarrow\omega\in\left(0,\frac{\pi}{4}\right)$

Implies That $I=\int_0^{\frac{\pi}{4}}\frac{e^{\omega}}{(\tan^2(\omega)+1)^{3/2}}\big(\tan^{-1}(\omega)+1\big)d\omega$

$$=\int_0^{\frac{\pi}{4}}\frac{e^{\omega}}{(\tan^2(\omega)+1)^{1/2}}d\omega$$

$$= \int_0^{\frac{\pi}{4}} e^{\omega}\cos(\omega)\,d\omega$$

$$= \frac{1}{2}e^{\omega}\left(\sin(\omega)+\cos(\omega)\right)\Big|_0^{\frac{\pi}{4}}$$

$$= \frac{1}{2}\left(\sqrt{2}e^{\frac{\pi}{4}}-1\right)$$

$SO,$ $\boxed{\int_0^1 \frac{e^{\tan^{-1}(x)}}{(x^2+1)^{3/2}}dx = \frac{1}{2}\left(\sqrt{2}e^{\frac{\pi}{4}}-1\right)}$

441. Calculate Integral $I = \int_0^1 \frac{x^{\alpha}-x^{\beta}}{\log(x)}dx \quad , (\alpha,\beta>-1)$

Solution; $I = \int_0^1 \frac{x^{\alpha}-x^{\beta}}{\log(x)}dx$

Then $I(\gamma) = \int_0^1 \frac{x^{\gamma}-x^{\beta}}{\log(x)}dx$ **Note:** $I(\alpha)=I(\gamma)=I\ ;\ I(\beta)=0$

That $I'(\gamma) = \int_0^1 \frac{x^{\gamma}\log(x)}{\log(x)}dx$

$$= \frac{1}{\gamma+1}$$

We Get $I(\gamma) = \log(\gamma+1)+C$

If : $\gamma=\beta$ **That** $I(\beta)=0=\log(\beta+1)+C$ **Then** $C=-\log(\beta+1)$

If : $\gamma=\alpha$ **That** $I(\alpha)=I=\log(\alpha+1)+C$

Implies That $I(\alpha)=I=\log(\alpha+1)-\log(\beta+1)$

$SO,$ $\boxed{\int_0^1 \frac{x^{\alpha}-x^{\beta}}{\log(x)}dx = \log\left(\frac{\alpha+1}{\beta+1}\right) \quad , (\alpha,\beta>-1)}$

442. Calculate Integral $I=\int_0^{\infty}\frac{\tan^{-1}(\alpha x)-\tan^{-1}(\beta x)}{x}dx \quad ,(\alpha,\beta>0)$

Solution; $I=\int_0^{\infty}\frac{\tan^{-1}(\alpha x)-\tan^{-1}(\beta x)}{x}dx$

Then $I(\gamma)=\int_0^{\infty}\frac{\tan^{-1}(\gamma x)-\tan^{-1}(\beta x)}{x}dx$ **Note:** $I(\alpha)=I(\gamma)=I\ ;\ I(\beta)=0$

That $I'(\gamma)=\int_0^{\infty}\frac{x}{x\left(1+(\gamma x)^2\right)}dx$

$$=\frac{1}{\gamma}\int_0^{\infty}\frac{1}{\left(1+(\gamma x)^2\right)}d(\gamma x)$$

$$=\frac{1}{\gamma}\tan^{-1}(\gamma x)\Big|_0^{\infty}$$

$$=\frac{\pi}{2\gamma}$$

We Get $I(\gamma)=\frac{\pi}{2}\log(\gamma)+C$

If : $\gamma=\beta$ **That** $I(\beta)=0=\frac{\pi}{2}\log(\beta)+C$ **Then** $C=-\frac{\pi}{2}\log(\beta)$

If : $\gamma=\alpha$ **That** $I(\alpha)=I=\frac{\pi}{2}\log(\alpha)+C$

Implies That $I(\alpha)=I=\frac{\pi}{2}\left[\log(\alpha)-\log(\beta)\right]$

$$=\frac{\pi}{2}\log\left(\frac{\alpha}{\beta}\right)$$

SO, $\boxed{\int_0^{\infty}\frac{\tan^{-1}(\alpha x)-\tan^{-1}(\beta x)}{x}dx=\frac{\pi}{2}\log\left(\frac{\alpha+1}{\beta+1}\right) \quad ,(\alpha,\beta>0)}$

443. Calculate Integral $I=\int_0^1 \frac{\sin\left(\log(x)\right)}{\log(x)}dx$

Solution; $I=\int_0^1 \frac{\sin\left(\log(x)\right)}{\log(x)}dx$

Then $I(\alpha)=\int_0^1 \frac{\sin\left(\alpha\log(x)\right)}{\log(x)}dx$ **Note:** $I(1)=I\ ;\ I(0)=0$

That $I'(\alpha)=\int_0^1 \cos\left(\alpha\log(x)\right)dx$

Let : $x=e^{-u}\Rightarrow dx=-e^{-u}du$ **; If** : $x\in(0,1)\Rightarrow u\in(\infty,0)$

We obtain $I'(\alpha)=-\int_\infty^0 e^{-u}\cos\left(\alpha u\right)du$

$$=\int_0^\infty e^{-u}\cos\left(\alpha u\right)du$$ **(Use Partial Integral)**

We Get $I'(\alpha)=\frac{1}{\alpha^2+1}\left(\alpha\sin(\alpha x)-\cos(\alpha x)\right)e^{-x}\Big|_0^\infty$

$$=\frac{1}{\alpha^2+1}$$

Implies That $I(\alpha)=\tan^{-1}(\alpha)+C$

If : $\alpha=0$ **That** $I(0)=0=\tan^{-1}(\alpha)+C$ **Then** $C=0$

If : $\alpha=1$ **That** $I(1)=I=\tan^{-1}(1)+C$

We obtain $I=\frac{\pi}{4}$

SO, $\boxed{\int_0^1 \frac{\sin\left(\log(x)\right)}{\log(x)}dx=\frac{\pi}{4}}$

444. Calculate Integral $I=\int_0^{\infty}\frac{\log(1+x^2)}{(1+x^2)}dx$

Solution; $I=\int_0^{\infty}\frac{\log(1+x^2)}{(1+x^2)}dx$

Then $I(\lambda)=\int_0^{\infty}\frac{\log(1+\lambda^2x^2)}{(1+x^2)}dx$ **Note:** $I(1)=I\ ;\ I(0)=0$

That $I'(\lambda)=\int_0^{\infty}\frac{2\lambda x^2}{(1+x^2)(1+\lambda^2x^2)}dx$

$$=\frac{2\lambda}{\lambda^2-1}\int_0^{\infty}\left(\frac{1}{1+x^2}-\frac{1}{1+\lambda^2x^2}\right)dx$$

$$=\frac{2\lambda}{\lambda^2-1}\left[\int_0^{\infty}\frac{1}{1+x^2}dx-\frac{1}{\lambda}\int_0^{\infty}\frac{1}{1+(\lambda x)^2}d(\lambda x)\right]$$

$$=\frac{2\lambda}{\lambda^2-1}\left[\tan^{-1}(x)-\frac{1}{\lambda}\tan^{-1}(\lambda x)\right]\Bigg|_0^{\infty}$$

$$=\frac{2\lambda}{\lambda^2-1}\times\frac{\pi}{2}\left(1-\frac{1}{\lambda}\right)$$

$$=\frac{\pi}{\lambda+1}$$

We obtain $I(\lambda)=\pi\log(\lambda+1)+C$

If : $\alpha=0$ **That** $I(0)=0=\tan^{-1}(\alpha)+C$ **Then** $C=0$

If : $\alpha=1$ **That** $I(1)=I=\pi\log(2)+C$

Implies That $I=\pi\log(2)$

SO, $\boxed{\int_0^{\infty}\frac{\log(1+x^2)}{(1+x^2)}dx=\pi\log(2)}$

445. Calculate Integral $I=\int_0^1 \frac{x\sin\left(\frac{\beta}{\alpha}\log(x)\right)}{\log(x)}dx \quad ,\alpha>0$

Solution; $I=\int_0^1 \frac{x\sin\left(\frac{\beta}{\alpha}\log(x)\right)}{\log(x)}dx$

Then $I(\gamma)=\int_0^1 \frac{x\sin\left(\frac{\gamma}{\alpha}\log(x)\right)}{\log(x)}dx$ **Note:** $I(\beta)=I\ ;\ I(0)=0$

That $I'(\gamma)=\frac{1}{\alpha}\int_0^1 x\cos\left(\frac{\gamma}{\alpha}\log(x)\right)dx$

Let : $x=e^{-\frac{\alpha}{\gamma}u}\Rightarrow dx=-\frac{\alpha}{\gamma}e^{-\frac{\alpha}{\gamma}u}du$ **; If** $x\in(0,1)\Rightarrow u\in(\infty,0)$

Implies That $I'(\gamma)=\frac{1}{\alpha}\int_\infty^0 e^{-\frac{\alpha}{\gamma}u}\cos(u)\left(-\frac{\alpha}{\gamma}e^{-\frac{\alpha}{\gamma}u}\right)du$

$$=\frac{1}{\gamma}\int_0^\infty e^{-\frac{2\alpha}{\gamma}u}\cos(u)\,du \qquad \textbf{(Use Partial Integral)}$$

We Get $I'(\gamma)=\frac{1}{\gamma}\left[\frac{\gamma}{\gamma^2+4\alpha^2}\left(\gamma\sin(u)-2\alpha\cos(u)\right)e^{-\frac{2\alpha}{\gamma}u}\right]_0^\infty$

$$=\frac{2\alpha}{\gamma^2+4\alpha^2}$$

We obtain $I(\gamma)=\tan^{-1}\left(\frac{\gamma}{2\alpha}\right)+C$

If : $\gamma=0$ **That** $I(0)=0=\tan^{-1}(0)+C$ **Then** $C=0$

If : $\gamma=\beta$ **That** $I(\beta)=I=\tan^{-1}\left(\frac{\beta}{2\alpha}\right)+C$

Implies That $I=\tan^{-1}\left(\frac{\beta}{2\alpha}\right)$

$$SO,\quad \boxed{\int_0^1 \frac{x\sin\left(\frac{\beta}{\alpha}\log(x)\right)}{\log(x)}dx=\tan^{-1}\left(\frac{\beta}{2\alpha}\right)\quad ,\alpha>0}$$

446. Calculate Integral $I=\int_0^{\infty}\frac{e^{-\alpha x}\sin(\beta x)}{x}dx \quad ,\ \alpha>0$

Solution; $I=\int_0^{\infty}\frac{e^{-\alpha x}\sin(\beta x)}{x}dx$

Then $I(\gamma)=\int_0^{\infty}\frac{e^{-\alpha x}\sin(\gamma x)}{x}dx$ **Note:** $I(\beta)=I\ ;\ I(0)=0$

That $I'(\gamma)=\int_0^{\infty}\frac{e^{-\alpha x}\sin(\gamma x)}{x}dx$

$$=\int_0^{\infty}e^{-\alpha x}\cos(\gamma x)dx \quad \textbf{(Use Partial Integral)}$$

$$=\frac{1}{\alpha^2+\gamma^2}e^{-\alpha x}\left(\gamma\sin(\gamma x)-\alpha\cos(\gamma x)\right)\Big|_0^{\infty}$$

$$=\frac{\alpha}{\alpha^2+\gamma^2}$$

We obtain $I(\gamma)=\int\frac{\alpha}{\alpha^2+\gamma^2}d\gamma=\tan^{-1}\left(\frac{\gamma}{\alpha}\right)+C$

If : $\gamma=0$ **That** $I(0)=0=\tan^{-1}(0)+C$ **Then** $C=0$

If : $\gamma=\beta$ **That** $I(\beta)=I=\tan^{-1}\left(\frac{\beta}{\alpha}\right)+C$

$$SO,\quad \boxed{\int_0^{\infty}\frac{e^{-\alpha x}\sin(\beta x)}{x}dx=\tan^{-1}\left(\frac{\beta}{\alpha}\right)\quad ,\alpha>0}$$

447. Calculate Integral $I=\int_0^1 \frac{\tan^{-1}(x)}{x\sqrt{1-x^2}}dx$

Solution; $I=\int_0^1 \frac{\tan^{-1}(x)}{x\sqrt{1-x^2}}dx$

Then $I(\theta)=\int_0^1 \frac{\tan^{-1}(\theta x)}{x\sqrt{1-x^2}}dx$ **Note:** $I(1)=I\ ;\ I(0)=0$

That $I'(\theta)=\int_0^1 \frac{1}{\left(1+(\theta x)^2\right)\sqrt{1-x^2}}dx$

Let : $x=\sin(y)\Rightarrow dx=\cos(y)dy$ **; If** : $x\in(0,1)\Rightarrow y\in\left(0,\frac{\pi}{2}\right)$

We Get $I'(\theta)=\int_0^{\frac{\pi}{2}} \frac{\cos(y)}{\left(1+\theta^2\sin^2(y)\right)\sqrt{1-\sin^2(y)}}dy$

$$=\int_0^{\frac{\pi}{2}} \frac{\sec^2(y)}{1+(1+\theta^2)\tan^2(y)}dy$$

$$=\int_0^{\frac{\pi}{2}} \frac{1}{1+(1+\theta^2)\tan^2(y)}d\left(\tan(y)\right)$$

$$=\frac{1}{\sqrt{1+\theta^2}}\tan^{-1}\left(y\sqrt{1+\theta^2}\right)\Big|_0^{\frac{\pi}{2}}$$

$$=\frac{\pi}{2}\cdot\frac{1}{\sqrt{1+\theta^2}}$$

Implies That $I(\theta)=\frac{\pi}{2}\int\frac{1}{\sqrt{1+\theta^2}}d\theta$

$$=\frac{\pi}{2}\sinh^{-1}(\theta)+C$$

$$=\frac{\pi}{2}\log\left(\theta+\sqrt{1+\theta^2}\right)+C$$

If : $\theta = 0$ **That** $I(0) = 0 = \frac{\pi}{2}\log\left(0+\sqrt{1+0^2}\right)+C$ **Then** $C = 0$

If : $\theta = 1$ **That** $I(1) = I = \frac{\pi}{2}\log(\sqrt{2}+1)$

SO, $\boxed{\int_0^1 \frac{\tan^{-1}(x)}{x\sqrt{1-x^2}}dx = \frac{\pi}{2}\log(\sqrt{2}+1)}$

448. Calculate Integral $I = \int_0^\infty \frac{e^{-ax^\gamma} - e^{-\beta x^\gamma}}{x}dx \quad (\alpha,\beta,\gamma > 0)$

Solution; $I = \int_0^\infty \frac{e^{-ax^\gamma} - e^{-\beta x^\gamma}}{x}dx$

Let : $u = x^\gamma \equiv \log(x) = \frac{1}{\gamma}\log(u) \Rightarrow \frac{1}{x}dx = \frac{1}{\gamma u}du$ **; If** : $x \in (0,\infty) \Rightarrow u \in (0,\infty)$

Implies That $I = \frac{1}{\gamma}\int_0^\infty \frac{e^{-au} - e^{-\beta u}}{u}du$

Then $I(\lambda) = \frac{1}{\gamma}\int_0^\infty \frac{e^{-\lambda u} - e^{-\beta u}}{u}du$ **Note:** $I(\alpha) = I$; $I(\beta) = 0$

That $I'(\lambda) = -\frac{1}{\gamma}\int_0^\infty e^{-\lambda u}\,du = -\frac{1}{\gamma\lambda}$

We Get $I(\lambda) = -\frac{1}{\gamma}\log(\lambda)+C$

If : $\lambda = \beta$ **That** $I(\beta) = 0 = -\frac{1}{\gamma}\log(\beta)+C$ **Then** $C = \frac{1}{\gamma}\log(\beta)$

If : $\lambda = \alpha$ **That** $I(\alpha) = I = -\frac{1}{\gamma}\log(\alpha)+\frac{1}{\gamma}\log(\beta)$ **Then** $I = \frac{1}{\gamma}\log\left(\frac{\beta}{\alpha}\right)$

SO, $\boxed{\int_0^\infty \frac{e^{-ax^\gamma} - e^{-\beta x^\gamma}}{x}dx = \frac{1}{\gamma}\log\left(\frac{\beta}{\alpha}\right)}$

449. Calculate Integral $I=\int_0^{\infty}\frac{\sin(nx)\cos(n\theta)+\sin(mx)\cos(mx)}{x}dx \quad (n,m>0)$

Solution; $I=\int_0^{\infty}\frac{\sin(nx)\cos(n\theta)+\sin(mx)\cos(mx)}{x}dx$

$$=\frac{1}{2}\int_0^{\infty}\frac{\sin(2nx)+\sin(2mx)}{x}dx \qquad \textbf{Take}: \begin{cases} k=2n=2m \\ k>0 \end{cases}$$

We obtain $I=\int_0^{\infty}\frac{\sin(kx)}{x}dx$ **Note**: $\begin{cases} \int_0^{\infty}\frac{\sin(kx)}{x}dx=\frac{\pi}{2} & ,k>0 \\ \int_0^{\infty}\frac{\sin(kx)}{x}dx=-\frac{\pi}{2} & ,k<0 \end{cases}$

Implies That $I=\int_0^{\infty}\frac{\sin(kx)}{x}dx=\frac{\pi}{2}$

SO, $\boxed{\int_0^{\infty}\frac{\sin(nx)\cos(n\theta)+\sin(mx)\cos(mx)}{x}dx=\frac{\pi}{2} \quad (n,m>0)}$

450. Calculate Integral $I=\int_{\frac{1}{4}}^{\frac{1}{2}}\left(\psi(1-x)-\psi(1+x)\right)dx$

Solution; $I=\int_{\frac{1}{4}}^{\frac{1}{2}}\left(\psi(1-x)-\psi(1+x)\right)dx$

$$=\int_{\frac{1}{4}}^{\frac{1}{2}}\left(\psi(1-x)-\psi(x)-\frac{1}{x}\right)dx \qquad \textbf{Note}: \psi(1+z)=\psi(z)+\frac{1}{z}$$

$$=\pi\int_{\frac{1}{4}}^{\frac{1}{2}}\cot(\pi x)\,dx-\int_{\frac{1}{4}}^{\frac{1}{2}}\left(\frac{1}{x}\right)dx \qquad \textbf{Note}: \psi(1-z)-\psi(z)=\pi\cot(\pi x)$$

$$=\frac{1}{2}\log\left(\frac{1}{2}\right)$$

SO, $\boxed{\int_{\frac{1}{4}}^{\frac{1}{2}}\left(\psi(1-x)-\psi(1+x)\right)dx=\frac{1}{2}\log\left(\frac{1}{2}\right)}$

451. Calculate Integral $I=\int_0^{\infty}\frac{\cos^2(\beta x)-\cos^2(\alpha x)}{x}dx \quad (\alpha,\beta>0)$

Solution; $I=\int_0^{\infty}\frac{\cos^2(\beta x)-\cos^2(\alpha x)}{x}dx$

$$=\frac{1}{2}\int_0^{\infty}\frac{\cos(2\alpha x)-\cos(2\beta x)}{x}dx$$

Then $I(\gamma)=\frac{1}{2}\int_0^{\infty}\frac{\cos(2\alpha x)-\cos(2\beta x)}{x}\cdot e^{-\gamma x}dx$ **Note :** $I(0)=I\ ,I(\infty)=0$

That $I'(\gamma)=\frac{1}{2}\int_0^{\infty}\left(\cos(2\beta x)-\cos(2\alpha x)\right)e^{-\gamma x}dx$

$$=\frac{1}{2}\int_0^{\infty}\left(\cos(2\beta x)-\cos(2\alpha x)\right)e^{-\gamma x}dx$$

$$=\underbrace{\frac{1}{2}\int_0^{\infty}\cos(2\beta x)e^{-\gamma x}dx}_{J}-\underbrace{\frac{1}{2}\int_0^{\infty}\cos(2\alpha x)e^{-\gamma x}dx}_{K}$$

For : $J=\frac{1}{2}\int_0^{\infty}\cos(2\beta x)e^{-\gamma x}dx$ **(Use Partial Integral)**

$$=\frac{1}{2}\cdot\frac{1}{4\beta^2+\gamma^2}\left(2\beta\sin(2\beta x)-\gamma\cos(2\beta x)\right)e^{-\gamma x}\Big|_0^{\infty}$$

$$=\frac{1}{2}\cdot\frac{\gamma}{4\beta^2+\gamma^2}$$

For : $K=\frac{1}{2}\int_0^{\infty}\cos(2\alpha x)e^{-\gamma x}dx$ **(Use Partial Integral)**

$$=\frac{1}{2}\cdot\frac{1}{4\alpha^2+\gamma^2}\left(\gamma\cos(2\alpha x)-2\alpha\sin(2\alpha x)\right)e^{-\gamma x}\Big|_0^{\infty}$$

$$=\frac{1}{2}\cdot\frac{\gamma}{4\alpha^2+\gamma^2}$$

We Get $I'(\gamma)=\frac{1}{2}\left(\frac{\gamma}{4\beta^2+\gamma^2}-\frac{\gamma}{4\alpha^2+\gamma^2}\right)$

Implies That $I(\gamma)=\frac{1}{4}\int\left(\frac{1}{4\beta^2+\gamma^2}-\frac{1}{4\alpha^2+\gamma^2}\right)d(\gamma^2)$

$$=\frac{1}{4}\log\left(\frac{\gamma^2+2\beta^2}{\gamma^2+2a^2}\right)+C$$

If : $\gamma=\infty$ **That** $I(\infty)=0=\frac{1}{4}\lim_{\gamma\to\infty}\log\left(\frac{\gamma^2+2\beta^2}{\gamma^2+2a^2}\right)+C$ **Then** $C=0$

If : $\gamma=0$ **That** $I(0)=I=\frac{1}{4}\log\left(\frac{2\beta^2}{2a^2}\right)$ **Then** $I=\frac{1}{2}\log\left(\frac{\beta}{\alpha}\right)$

SO, $\boxed{\int_0^\infty\frac{\cos^2(\beta x)-\cos^2(\alpha x)}{x}dx=\frac{1}{2}\log\left(\frac{\beta}{\alpha}\right)}$

452. Calculate Integral $I=\int_0^\pi x\log\big(\sin(x)\big)dx$

Solution; $I=\int_0^\pi x\log\big(\sin(x)\big)dx$

$$=\int_0^\pi(\pi-x)\log\big(\sin(\pi-x)\big)dx$$

We Get $I=\frac{\pi}{2}\int_0^\pi\log\big(\sin(x)\big)dx$

$$=\frac{\pi}{2}\cdot 2\int_0^{\frac{\pi}{2}}\log\big(\sin(x)\big)dx$$
$$=\pi\left(\frac{\pi}{2}\log\left(\frac{1}{2}\right)\right)$$
$$=\frac{\pi^2}{2}\log\left(\frac{1}{2}\right)$$

SO, $\boxed{\int_0^\pi x\log\big(\sin(x)\big)dx=\frac{\pi^2}{2}\log\left(\frac{1}{2}\right)}$

453. Calculate Integral $I=\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\log\left(\cos(x)+\sin(x)\right)dx$

Solution; $I=\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\log\left(\cos(x)+\sin(x)\right)dx$ (*)

$$=\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\log\left(\cos(x)-\sin(x)\right)dx \quad (**)$$

Take (*) + (**) Then $2I=\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\log\left(\cos^2(x)-\sin^2(x)\right)dx$

$$=\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\log\left(\cos(2x)\right)dx$$

$$=2\int_{0}^{\frac{\pi}{4}}\log\left(\cos(2x)\right)dx$$

$$=\int_{0}^{\frac{\pi}{2}}\log\left(\cos(x)\right)dx$$

$$=\frac{\pi}{2}\log\left(\frac{1}{2}\right)$$

Implies That $I=\frac{\pi}{4}\log\left(\frac{1}{2}\right)$

$SO,$ $\boxed{\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\log\left(\cos(x)+\sin(x)\right)dx=\frac{\pi}{4}\log\left(\frac{1}{2}\right)}$

454. Calculate Integral $I=\int_{0}^{2}x\sqrt{x^{\log(x)}\sqrt[3]{x^{\log^2(x)}\sqrt[4]{x^{\log^3(x)}\cdots}}}\,dx$

Solution; $I=\int_{0}^{2}x\sqrt{x^{\log(x)}\sqrt[3]{x^{\log^2(x)}\sqrt[4]{x^{\log^3(x)}\cdots}}}\,dx$

$$=\int_{0}^{2}x\sqrt{x^{\log(x)}\sqrt[3]{x^{\log^2(x)}\sqrt[4]{x^{\log^3(x)}\cdots}}}\,dx$$

$$=\int_{0}^{2}x\cdot x^{\frac{\log(x)}{2!}+\frac{\log^2(x)}{3!}+\frac{\log^3(x)}{4!}+\cdots}dx$$

$$= \int_0^2 x^{1+\frac{\log(x)}{2}+\frac{\log^2(x)}{3}+\frac{\log^3(x)}{4}+\cdots} dx$$

$$= \int_0^2 x^{\frac{x-1}{\log(x)}} dx$$

$$= \int_0^2 e^{\frac{x-1}{\log(x)}\cdot\log(x)} dx$$

$$= \int_0^2 e^{x-1} dx$$

$$= e - e^{-1}$$

SO, $\boxed{\int_0^2 x\sqrt{x^{\log(x)}\sqrt[3]{x^{\log^2(x)}\sqrt[4]{x^{\log^3(x)}\cdots}}}\, dx = e - e^{-1}}$

455. Calculate Integral $I = \int_0^\infty \frac{\log(x)}{x^2+x+1} dx$

Solution; $I = \int_0^\infty \frac{\log(x)}{x^2+x+1} dx$

Let : $x = y^{-1} \Rightarrow dx = -y^{-2}dy$ **; If:** $x \in (0,\infty) \Rightarrow y \in (\infty, 0)$

We Get $I = -\int_\infty^0 \frac{\log(y^{-1})}{y^{-2}+y^{-1}+1} y^{-2} dy$

$$I = -\int_0^\infty \frac{\log(y)}{(y^{-2}+y^{-1}+1)y^2} dy$$

$$= -\int_0^\infty \frac{\log(y)}{y^2+y+1} dy \qquad \textbf{Note: } f(x) = f(y)$$

$$= -\int_0^\infty \frac{\log(x)}{x^2+x+1} dx$$

$$= -I$$

We obtain $I = -I$ **Then** $I = 0$

SO, $\boxed{\int_0^\infty \frac{\log(x)}{x^2+x+1} dx = 0}$

456. Calculate Integral $I=\int_0^{\infty}\frac{\log(x)}{x^2+2x+4}dx$

Solution; $I=\int_0^{\infty}\frac{\log(x)}{x^2+2x+4}dx$

Let : $x=2u\Rightarrow dx=2du$ **; If:** $x\in(0,\infty)\Rightarrow u\in(0,\infty)$

Implies That $I=2\int_0^{\infty}\frac{\log(2u)}{4u^2+4u+4}du$

$$=2\int_0^{\infty}\frac{\log(2)}{(2u+1)^2+3}du+\frac{1}{2}\underbrace{\int_0^{\infty}\frac{\log(u)}{u^2+u+1}du}_{0}$$

$$=\frac{2\log(4)}{\sqrt{3}}\tan^{-1}\left(\frac{2u+1}{\sqrt{3}}\right)\Bigg|_0^{\infty}$$

$$=\frac{\pi}{3\sqrt{3}}\log(2)$$

SO, $\boxed{\int_0^{\infty}\frac{\log(x)}{x^2+2x+4}dx=\frac{\pi}{3\sqrt{3}}\log(2)}$

457. Calculate Integral $I=\int_{-\pi}^{\pi}\tan^{-1}\left(\pi^x\right)dx$

Solution; $I=\int_{-\pi}^{\pi}\tan^{-1}\left(\pi^x\right)dx$ **(*)**

$$=\int_{-\pi}^{\pi}\tan^{-1}\left(\pi^{-x}\right)dx \quad \textbf{(**)}$$

Take (*) + (**) Then $2I=\int_{-\pi}^{\pi}\left(\tan^{-1}\left(\pi^x\right)+\tan^{-1}(\pi^{-x)}\right)dx$

$$=\int_{-\pi}^{\pi}\frac{\pi}{2}dx$$

$$=\pi^2 \qquad \textbf{Note}:\tan^{-1}(x)+\tan^{-1}(x^{-1})=\frac{\pi}{2} \quad ,\forall x>0$$

Implies That $I=\frac{\pi^2}{2}$

SO, $\boxed{\int_{-\pi}^{\pi}\tan^{-1}\left(\pi^x\right)dx=\frac{\pi^2}{2}}$

458. Calculate Integral $I=\int_{-1}^{1}\cos^{-1}(x^{2025})dx$

Solution; $I=\int_{-1}^{1}\cos^{-1}(x^{2025})dx$

$$=\int_{-1}^{1}\cos^{-1}(-x^{2025})dx$$

$$=\int_{-1}^{1}\left(\pi-\cos^{-1}(x^{2025})\right)dx$$

We Get $I=\frac{1}{2}\int_{-1}^{1}\pi dx=\frac{\pi}{2}$

SO, $\boxed{\int_{-1}^{1}\cos^{-1}(x^{2025})dx=\frac{\pi}{2}}$

459. Calculate Integral $I=\int_{0}^{1}\frac{\tan^{-1}(x)}{x+1}dx$

Solution; $I=\int_{0}^{1}\frac{\tan^{-1}(x)}{x+1}dx$

Let $:u=\tan^{-1}(x)\Rightarrow du=\frac{1}{x^2+1}dx$ **And** $dv=\frac{1}{x+1}dx\Rightarrow v=\log(x+1)$

Implies That $I=\tan^{-1}(x)\log(x+1)\Big|_0^1-\int_0^1\frac{\log(x+1)}{x^2+1}dx$

$$=\frac{\pi}{4}\log(2)-J \quad \textbf{(!)}$$

For $:J=\int_{0}^{1}\frac{\log(x+1)}{x^2+1}dx$

Let $x=\tan(u)\Rightarrow du=\frac{1}{x^2+1}dx$ **; If** $:x\in(0,1)\Rightarrow u\in\left(0,\frac{\pi}{4}\right)$

We Get $:J=\int_{0}^{\frac{\pi}{4}}\log\left(\tan(u)+1\right)du$

$$=\int_{0}^{\frac{\pi}{4}}\log\left[\left(\tan\left(\frac{\pi}{4}-x\right)+1\right)\right]du$$

$$= \int_0^{\frac{\pi}{4}} \log\left[\frac{1-\tan(x)}{1+\tan(x)}+1\right] du$$

$$= \int_0^{\frac{\pi}{4}} \log(2)\, du - \int_0^{\frac{\pi}{4}} \log\left(\tan(x)+1\right) du$$

We obtain $J = \frac{1}{2}\int_0^{\frac{\pi}{4}} \log(2)\, du = \frac{\pi}{8}\log(2)$

Take (!) Then $I = \frac{\pi}{4}\log(2) - \frac{\pi}{8}\log(2) = \frac{\pi}{8}\log(2)$

SO, $\boxed{\int_0^1 \frac{\tan^{-1}(x)}{x+1} dx = \frac{\pi}{8}\log(2)}$

460. Calculate Integral $I = \int_0^1 \frac{1}{(x^2-x+1)(e^{2x}+e)} dx$

Solution; $I = \int_0^1 \frac{1}{(x^2-x+1)(e^{2x}+e)} dx$ **(*)**

$$= \int_0^1 \frac{1}{[(1-x)^2-(1-x)+1](e^{2(1-x)}+e)} dx$$

$$= \int_0^1 \frac{1}{(x^2-x+1)(e^{-2x+2}+e)} \times \frac{e^{2x-1}}{e^{2x-1}} dx$$

$$= \int_0^1 \frac{e^{2x-1}}{(x^2-x+1)(e^{2x}+e)} dx \quad \textbf{(**)}$$

Take (*) + (**) That $I = \int_0^1 \frac{e^{2x-1}+1}{(x^2-x+1)(e^{2x}+e)} dx$

$$= \frac{1}{e}\int_0^1 \frac{1}{(x^2-x+1)} dx$$

$$= \frac{2\pi}{3e\sqrt{3}}$$

SO, $\boxed{\int_0^1 \frac{1}{(x^2-x+1)(e^{2x}+e)} dx = \frac{2\pi}{3e\sqrt{3}}}$

461. Calculate Integral $I=\int_0^{\alpha+1}\beta^{\lfloor x\rfloor}dx \quad (\alpha\in\mathbb{N})$

Solution; $I=\int_0^{\alpha+1}\beta^{\lfloor x\rfloor}dx$

$$=\sum_{\gamma=0}^{\alpha}\left(\int_{\gamma}^{\gamma+1}\beta^{\gamma}dx\right)$$

$$=\sum_{\gamma=0}^{\alpha}\left(\beta^{\gamma}\right)$$

$$=\frac{\beta^{\alpha+1}-1}{\beta-1}$$

SO, $\boxed{\int_0^{\alpha+1}\beta^{\lfloor x\rfloor}dx=\frac{\beta^{\alpha+1}-1}{\beta-1} \quad (\alpha\in\mathbb{N})}$

462. Calculate Integral $I=\int_{-4}^{4}\sqrt{4-4x+x^2}\,dx$

Solution; $I=\int_{-4}^{4}\sqrt{4-4x+x^2}\,dx$

$$I=\int_{-4}^{4}\sqrt{(2-x)^2}\,dx$$

$$=\int_{-4}^{4}|2-x|\,dx$$

$$=\int_{-4}^{2}|2-x|\,dx+\int_{2}^{4}|2-x|\,dx$$

$$=\int_{-4}^{2}(2-x)\,dx-\int_{2}^{4}(2-x)\,dx$$

$$=\left(2x-\frac{1}{2}x^2\right)\Big|_{-4}^{2}-\left(2x-\frac{1}{2}x^2\right)\Big|_{2}^{4}$$

$$=20$$

SO, $\boxed{\int_{-4}^{4}\sqrt{4-4x+x^2}\,dx=20}$

463. Calculate Integral $I=\int_0^{\pi}\frac{1}{1+\sin^{\cos(x)}(x)}dx$

Solution; $I=\int_0^{\pi}\frac{1}{1+\sin^{\cos(x)}(x)}dx$ **(!)**

$$=\int_0^{\pi}\frac{1}{1+\sin^{\cos(\pi-x)}(\pi-x)}dx$$

$$=\int_0^{\pi}\frac{1}{1+\sin^{-\cos(x)}(x)}dx$$

$$=\int_0^{\pi}\frac{\sin^{\cos(x)}(x)}{1+\sin^{\cos(x)}(x)}dx \quad \textbf{(!!)}$$

Take (!) + (!!) Then $2I=\int_0^{\pi}\frac{1+\sin^{\cos(x)}(x)}{1+\sin^{\cos(x)}(x)}dx$

Implies That $I=\frac{\pi}{2}$

SO, $\boxed{\int_0^{\pi}\frac{1}{1+\sin^{\cos(x)}(x)}dx=\frac{\pi}{2}}$

464. Calculate Integral $I=\int_0^{e}x\pi^{\log(x)}dx$

Solution; $I=\int_0^{e}x\pi^{\log(x)}dx$

$$=\int_0^{e}x\cdot x^{\log(\pi)}dx$$

$$=\frac{x^{\log(\pi)+2}}{\log(\pi)+2}\Big|_0^{e}$$

$$=\frac{\pi e^2}{\log(\pi)+2}$$

SO, $\boxed{\int_0^{e}x\pi^{\log(x)}dx=\frac{\pi e^2}{\log(\pi)+2}}$

465. Calculate Integral $I=\int_0^{\infty}\left(\frac{1-e^{-x}}{x}\right)^2 dx$

Solution; $I=\int_0^{\infty}\left(\frac{1-e^{-x}}{x}\right)^2 dx$

Then $I(\lambda)=\int_0^{\infty}\left(\frac{1-e^{-\lambda x}}{x}\right)^2 dx$

That $I'(\lambda)=2\int_0^{\infty}\left(\frac{e^{-\lambda x}-e^{-2\lambda x}}{x}\right)dx$

$$=2\log\left(\frac{2\lambda}{\lambda}\right)$$

$$=\log(4)$$

Implies That $I(\lambda)=\lambda\log(4)+C$

If : $\lambda=0$ **That** $I(0)=0=0\log(4)+C$ **Then** $C=0$

If : $\lambda=1$ **That** $I(1)=I=\log(4)+C$ **We obtain** $I=\log(4)$

$SO,$ $\boxed{\int_0^{\infty}\left(\frac{1-e^{-x}}{x}\right)^2 dx=\log(4)}$

466. Calculate Integral $I=\int_0^{\infty}\frac{1-e^{-x^2}}{x^2}dx$

Solution; $I=\int_0^{\infty}\frac{1-e^{-x^2}}{x^2}dx$

Then $I(\alpha)=\int_0^{\infty}\frac{1-e^{-\alpha x^2}}{x^2}dx$

That $I'(\alpha)=\int_0^{\infty}e^{-\alpha x^2}\,dx$

$$=\frac{1}{\sqrt{\alpha}}\int_0^{\infty}e^{-(\sqrt{\alpha}x)^2}\,d(\sqrt{\alpha}x)$$

$$=\frac{1}{2\sqrt{\alpha}}\cdot\sqrt{\pi}$$

Implies That $I(\alpha)=\sqrt{\alpha\pi}+C$

If : $\alpha=0$ **That** $I(0)=0=\sqrt{0\cdot\pi}+C$ **Then** $C=0$

If : $\alpha=1$ **That** $I(1)=I=\sqrt{\pi}+C$ **We obtain** $I=\sqrt{\pi}$

SO, $\boxed{\int_0^\infty \frac{1-e^{-x^2}}{x^2}dx=\sqrt{\pi}}$

467. Calculate Integral $I=\int_0^\infty \frac{e^{-\alpha x}\cos(\beta x)-e^{-\gamma x}\cos(\mu x)}{x}dx \quad (\alpha,\gamma>0)$

Solution; $I=\int_0^\infty \frac{e^{-\alpha x}\cos(\beta x)-e^{-\gamma x}\cos(\mu x)}{x}dx$

$$=\underbrace{\int_0^\infty \frac{e^{-\alpha x}\cos(\beta x)}{x}dx}_{J}-\underbrace{\int_0^\infty \frac{e^{-\gamma(x)}\cos(\mu x)}{x}dx}_{K} \quad (@)$$

For : $J=\int_0^\infty \frac{e^{\alpha x}\cos(\beta x)}{x}dx$

Then $J(\omega)=\int_0^\infty \frac{e^{\omega x}\cos(\beta x)}{x}dx$ **Note :** $J(\alpha)=J$, $J(\infty)=0$

That $J'(\omega)=-\int_0^\infty e^{-\omega x}\cos(\beta x)\,dx$

$$=-\frac{\omega}{\omega^2+\beta^2}$$

We Get $J(\omega)=-\frac{1}{2}\log(\omega^2+\beta^2)+C_J$

If : $\omega=\infty$ **That** $J(\infty)=0=-\frac{1}{2}\lim_{\omega\to\infty}\log(\omega^2+\beta^2)+C_J$ **Then** $C_J=\frac{1}{2}\lim_{\omega\to\infty}\log(\omega^2+\beta^2)$

If : $\omega=\alpha$ **That** $J(\alpha)=J=-\frac{1}{2}\log(\alpha^2+\beta^2)+\frac{1}{2}\lim_{\omega\to\infty}\log(\omega^2+\beta^2)$

For $K=\int_0^{\infty}\frac{e^{-\gamma(x)}\cos(\mu x)}{x}dx$

Then $K(\theta)=\int_0^{\infty}\frac{e^{-\theta(x)}\cos(\mu x)}{x}dx$ **Note :** $K(\alpha)=K\ ,K(\infty)=0$

That $K'(\theta)=-\int_0^{\infty}e^{-\theta(x)}\cos(\mu x)\,dx$

$$=-\frac{\theta}{\theta^2+\mu^2}$$

We Get $K(\theta)=-\frac{1}{2}\log(\theta^2+\mu^2)+C_K$

If : $\theta=\infty$ **That** $K(\infty)=0=-\frac{1}{2}\lim_{\theta\to\infty}\log(\theta^2+\mu^2)+C_K$ **Then** $C_K=\frac{1}{2}\lim_{\theta\to\infty}\log(\theta^2+\mu^2)$

If : $\theta=\gamma$ **That** $K(\gamma)=K=-\frac{1}{2}\log(\gamma^2+\mu^2)+\frac{1}{2}\lim_{\theta\to\infty}\log(\theta^2+\mu^2)$

Take (@) : $I=-\frac{1}{2}\log(\alpha^2+\beta^2)+\frac{1}{2}\lim_{\omega\to\infty}\log(\omega^2+\beta^2)+\frac{1}{2}\log(\gamma^2+\mu^2)-\frac{1}{2}\lim_{\theta\to\infty}\log(\theta^2+\mu^2)$

$$=\frac{1}{2}\log\left(\frac{\gamma^2+\mu^2}{\alpha^2+\beta^2}\right)+\frac{1}{2}\lim_{\lambda\to\infty}\log\left(\frac{\lambda^2+\beta^2}{\lambda^2+\mu^2}\right)$$

$$=\frac{1}{2}\log\left(\frac{\gamma^2+\mu^2}{\alpha^2+\beta^2}\right)$$

SO, $\boxed{\int_0^{\infty}\frac{e^{-\alpha x}\cos(\beta x)-e^{-\gamma x}\cos(\mu x)}{x}dx=\frac{1}{2}\log\left(\frac{\gamma^2+\mu^2}{\alpha^2+\beta^2}\right)\quad(\alpha,\gamma>0)}$

468. Calculate Integral $I=\int_0^{\infty}\frac{\sin(\lambda x)\sin(\mu x)}{x}dx$

Solution; $I=\int_0^{\infty}\frac{\sin(\lambda x)\sin(\mu x)}{x}dx$

Then $I(\omega)=\int_0^{\infty}e^{-\omega x}\frac{\sin(\lambda x)\sin(\mu x)}{x}dx$ **Note :** $I(0)=I\ \ ,I(\infty)=0$

That $I'(\omega)=-\int_0^{\infty}e^{-\omega x}\sin(\lambda x)\sin(\mu x)\,dx$

$$=\frac{1}{2}\left[\int_0^{\infty}e^{-\omega x}\cos[(\lambda+\mu)x]dx-\int_0^{\infty}e^{-\omega x}\cos[(\lambda-\mu)x]dx\right]$$

$$=\frac{1}{2}\left[\frac{\omega}{\omega^2+(\lambda+\mu)^2}-\frac{\omega}{\omega^2+(\lambda-\mu)^2}\right]$$

Implies That $I(\omega)=\frac{1}{4}\log\left(\frac{\omega^2+(\lambda+\mu)^2}{\omega^2+(\lambda-\mu)^2}\right)+C$

If : $\omega=\infty$ **That** $I(\infty)=0=\frac{1}{4}\lim_{\omega\to\infty}\log\left(\frac{\omega^2+(\lambda+\mu)^2}{\omega^2+(\lambda-\mu)^2}\right)+C$ **Then** $C=0$

If : $\omega=0$ **That** $I(0)=I=\frac{1}{4}\log\left(\frac{(\lambda+\mu)^2}{(\lambda-\mu)^2}\right)+C$ **We obtain** $I=\frac{1}{2}\log\left(\frac{\lambda+\mu}{\lambda-\mu}\right)$

$SO,$ $\boxed{\int_0^{\infty}\frac{\sin(\lambda x)\sin(\mu x)}{x}dx=\frac{1}{2}\log\left(\frac{\lambda+\mu}{\lambda-\mu}\right)\quad,(\lambda>\mu>0)}$

469. Calculate Integral $I=\int_0^1\frac{\log^n(1-x)}{x}dx\quad(n>0,n\neq1)$

Solution; $I=\int_0^1\frac{\log^n(1-x)}{x}dx$

Let : $e^{-z}=1-x\Rightarrow dx=e^{-z}dz$ **; If :** $x\in(0,1)\Rightarrow z\in(0,\infty)$

Implies That $I=\int_0^{\infty}\frac{\log^n(e^{-z})}{1-e^{-z}}\cdot e^{-z}\,dz$

Let : $z=1-x\Rightarrow dx=-dz$ **; If :** $x\in(0,1)\Rightarrow z\in(1,0)$

$$=(-1)^n\int_0^{\infty}\frac{z^n}{e^z-1}dz$$

$$=(-1)^n\Gamma(n-1)\zeta(n-1)$$

$SO,$ $\boxed{\int_0^1\frac{\log^n(1-x)}{x}dx=(-1)^n\Gamma(n-1)\zeta(n-1)\quad(n>0,n\neq1)}$

470. Calculate Integral $I=\int_0^1 \frac{\log(x)\log(1-\alpha x)}{x}dx \quad \left(|\alpha|\le 1\right)$

Solution; $I=\int_0^1 \frac{\log(x)\log(1-\alpha x)}{x}dx$

$$=\int_0^1 \frac{\log(x)}{x}\left(-\sum_{k=1}^{\infty}\frac{(\alpha x)^k}{k}\right)dx$$

$$=-\sum_{k=1}^{\infty}\left(\frac{\alpha^k}{k}\int_0^1 x^{k-1}\log(x)dx\right)$$

Let : $u=\log(x)\Rightarrow du=\frac{1}{x}dx$ **And** $dv=x^{k-1}dx\Rightarrow v=\frac{x^k}{k}$

We obtain $I=-\sum_{k=1}^{\infty}\left[\frac{\alpha^k}{k^2}\left(\underbrace{x^k\log(x)\Big|_0^1}_{0}-\int_0^1 x^{k-1}dx\right)\right]$

$$=\sum_{k=1}^{\infty}\left(\frac{\alpha^k}{k^3}\cdot x^k\Big|_0^1\right)$$

$$=\sum_{k=1}^{\infty}\left(\frac{\alpha^k}{k^3}\right)$$

$$=Li_3(\alpha)$$

SO, $\boxed{\int_0^1 \frac{\log(x)\log(1-\alpha x)}{x}dx=Li_3(\alpha) \quad \left(|\alpha|\le 1\right)}$

471. Calculate Integral $I=\int_0^1 \log\left(\Gamma\left(\frac{x}{2}\right)\Gamma\left(\frac{x+1}{x}\right)\right)dx$

Solution; $I=\int_0^1 \log\left(\Gamma\left(\frac{x}{2}\right)\Gamma\left(\frac{x+1}{x}\right)\right)dx$

$$=\int_0^1 \log\left(2^{1-x}\sqrt{\pi}\Gamma(x)\right)dx$$

Note $\Gamma\left(\frac{x}{2}\right)\Gamma\left(\frac{x+1}{x}\right)=2^{1-x}\sqrt{\pi}\Gamma(x)$

$$=\int_0^1\left[\log\left(2^{1-x}\right)+\log\left(\sqrt{\pi}\right)\right]dx+\int_0^1\log\left(\Gamma(x)\right)dx$$

$$= \int_0^1 \left[(1-x)\log(2) + \frac{1}{2}\log(\pi) \right] dx + \int_0^1 \log\left(\Gamma(x)\right) dx$$

$$= \left(\frac{1}{2}\log(2) + \frac{1}{2}\log(\pi) \right) + \left(\frac{1}{2}\log(2\pi) \right)$$

$$= \log(2\pi)$$

SO, $\boxed{\int_0^1 \log\left(\Gamma\left(\frac{x}{2}\right)\Gamma\left(\frac{x+1}{x}\right)\right) dx = \log(2\pi)}$

472. Calculate Integral $I = \int_0^1 \frac{\log\left(1-(\omega x)^2\right)}{x^2\sqrt{1-x^2}} dx \quad , \left(|\omega| \le 1\right)$

Solution; $I = \int_0^1 \frac{\log\left(1-(\omega x)^2\right)}{x^2\sqrt{1-x^2}} dx$

Then $I(\alpha) = \int_0^1 \frac{\log(1-\alpha^2 x^2)}{x^2\sqrt{1-x^2}} dx$ **Note** $I(\omega) = I \ , I(0) = 0$

That $I'(\alpha) = -\int_0^1 \frac{2\alpha}{(1-\alpha^2 x^2)\sqrt{1-x^2}} dx$

Let : $x = \sin(u) \Rightarrow dx = \cos(u)du$ **; If :** $x \in (0,1) \Rightarrow u \in \left(0, \frac{\pi}{2}\right)$

Implies That $I'(\alpha) = -\int_0^{\frac{\pi}{2}} \frac{2\alpha\cos(u)}{(1-\alpha^2\sin^2(u))\sqrt{1-\sin^2(u)}} du$

$$= -\int_0^{\frac{\pi}{2}} \frac{2\alpha}{1-\alpha^2\sin^2(u)} du$$

$$= -\int_0^{\frac{\pi}{2}} \frac{2\alpha}{1+(1-\alpha^2)\tan^2(u)} d\left(\tan(u)\right)$$

$$= -\frac{2\alpha}{\sqrt{1-\alpha^2}} \tan^{-1}\left(\sqrt{1-\alpha^2}\tan(u)\right)\Big|_0^{\frac{\pi}{2}}$$

$$=-\frac{\alpha\pi}{\sqrt{1-\alpha^2}}$$

We Get $I(\alpha)=\pi\sqrt{1-\alpha^2}+C$

If : $\alpha=0$ **That** $I(0)=0=\pi+C$ **Then** $C=-\pi$

If : $\alpha=\omega$ **That** $I(\omega)=I=\pi\sqrt{1-\omega^2}-\pi$

We obtain $I=\pi\left(\sqrt{1-\omega^2}-1\right)$

$$SO,\quad \boxed{\int_0^1\frac{\log\left(1-(\omega x)^2\right)}{x^2\sqrt{1-x^2}}dx=\pi\left(\sqrt{1-\omega^2}-1\right)\quad ,\left(|\omega|\le 1\right)}$$

473. Calculate Integral $I=\int_0^\infty\frac{1}{\sqrt{1+x^2}}\tan^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right)dx$

Solution; $I=\int_0^\infty\frac{1}{\sqrt{1+x^2}}\tan^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right)dx$

Then $I(\omega)=\int_0^\infty\frac{1}{\sqrt{1+x^2}}\tan^{-1}\left(\frac{\omega}{\sqrt{1+x^2}}\right)dx$ **Note:** $I(1)=I\ ,I(0)=0$

That $I'(\omega)=\int_0^\infty\frac{1}{\sqrt{1+x^2}}\times\frac{1}{\sqrt{1+x^2}}\times\frac{1+x^2}{1+\omega^2+x^2}dx$

$$=\int_0^\infty\frac{1}{1+\omega^2+x^2}dx$$

$$=\frac{1}{\sqrt{1+\omega^2}}\tan^{-1}\left(\frac{x}{\sqrt{1+\omega^2}}\right)\Bigg|_0^\infty$$

$$=\frac{\pi}{2\sqrt{1+\omega^2}}$$

Implies That $I(\omega)=\frac{\pi}{2}\log\left(\omega+\sqrt{1+\omega^2}\right)+C$

If : $\omega = 0$ **That** $I(0) = 0 = 0 + C$ **Then** $C = 0$

If : $\omega = 1$ **That** $I(1) = I = \frac{\pi}{2}\log\left(1+\sqrt{2}\right)$

We obtain $I = \frac{\pi}{2}\log\left(1+\sqrt{2}\right)$

SO, $\boxed{\int_0^\infty \frac{1}{\sqrt{1+x^2}}\tan^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right)dx = \frac{\pi}{2}\log\left(1+\sqrt{2}\right)}$

474. Calculate Integral $I = \int_0^\infty \frac{x - \arctan(x)}{x^3}dx$

Solution; $I = \int_0^\infty \frac{x - \arctan(x)}{x^3}dx$

Then $I(\alpha) = \int_0^\infty \frac{\alpha x - \arctan(\alpha x)}{x^3}dx$ **Note :** $I(1) = I$, $I(0) = 0$

Then $I'(\alpha) = \int_0^\infty \frac{1}{x^3}\left(x - \frac{x}{1+\alpha^2x^2}\right)dx$

$$= \int_0^\infty \left(\frac{\alpha^2}{1+\alpha^2x^2}\right)dx$$

$$= \alpha\arctan(\alpha x)\Big|_0^\infty$$

$$= \frac{\pi\alpha}{2}$$

Implies That $I(\alpha) = \frac{\pi}{4}\alpha^2 + C$

If : $\alpha = 0$ **That** $I(0) = 0 = 0 + C$ **Then** $C = 0$

If : $\alpha = 1$ **That** $I(1) = I = \frac{\pi}{4}$ **We obtain** $I = \frac{\pi}{4}$

SO, $\boxed{\int_0^\infty \frac{x - \arctan(x)}{x^3}dx = \frac{\pi}{4}}$

475. Calculate Integral $I=\int_0^{\infty}\frac{\cos(\alpha x)}{\beta^2+x^2}dx \quad ,(\alpha>0,\beta\geq 0)$

Solution; $I=\int_0^{\infty}\frac{\cos(\alpha x)}{\beta^2+x^2}dx$

Then $I(\gamma)=\int_0^{\infty}\frac{\cos(\gamma x)}{\beta^2+x^2}dx$ **Note :** $I(\alpha)=I \ , I(0)=\int_0^{\infty}\frac{1}{\beta^2+x^2}=\frac{\pi}{2\beta}$

That $I'(\gamma)=\int_0^{\infty}\frac{\cos(\gamma x)}{\beta^2+x^2}dx$

$$=-\int_0^{\infty}\frac{x\sin(\gamma x)}{\beta^2+x^2}dx$$

$$=-\int_0^{\infty}\frac{\sin(\gamma x)}{x^2}dx+\beta^2\int_0^{\infty}\frac{\sin(\gamma x)}{x(\beta^2+x^2)}dx$$

$$=-\frac{\pi}{2}+\beta^2\int_0^{\infty}\frac{\sin(\gamma x)}{x(\beta^2+x^2)}dx$$

We Get $I''(\gamma)=\beta^2\int_0^{\infty}\frac{\cos(\gamma x)}{\beta^2+x^2}dx$

$$I''(\gamma)=\beta^2 I(\gamma)$$

Differentiating again we come to the second order differential equation

Implies That $I(\gamma)=Ae^{\beta\gamma}+Be^{-\beta\gamma}$ **(*) Where** $A,B\in\mathbb{R}$

We obtain $I'(\gamma)=A\beta e^{\beta\gamma}-B\beta e^{-\beta\gamma}$ **(**)**

Take (*) Then $I(0)=\frac{\pi}{2\beta}=A+B$

And (*) Then $I'(0)=-\frac{\pi}{2}=A\beta-B\beta$

Take (*) & (**) That $\begin{cases}A+B=\frac{\pi}{2\beta}\\ A-B=-\frac{\pi}{2\beta}\end{cases}\Rightarrow A=0\ ,\ B=\frac{\pi}{2\beta}$

Take (*) Then $I(\gamma)=\frac{\pi}{2\beta}e^{-\beta\gamma}$ **But :** $I(\alpha)=I$

We obtain $I=\frac{\pi}{2\beta}e^{-\beta\alpha}$

SO, $\boxed{\int_0^\infty \frac{\cos(\alpha x)}{\beta^2+x^2}dx=\frac{\pi}{2\beta}e^{-\beta\alpha} \quad ,(\alpha>0,\beta\geq 0)}$

476. Calculate Integral $I=\int_0^\infty \exp\left(-x-\frac{1}{x}\right)\frac{1}{\sqrt{x}}dx$

Solution; $I=\int_0^\infty \exp\left(-x-\frac{1}{x}\right)\frac{1}{\sqrt{x}}dx$

Then $I(\alpha)=\int_0^\infty \exp\left(-x-\frac{\alpha}{x}\right)\frac{1}{\sqrt{x}}dx$ **Note :** $I(1)=I \ , I(0)=\int_0^\infty \frac{\exp(-x)}{\sqrt{x}}dx=\sqrt{\pi}$

That $I'(\alpha)=-\int_0^\infty \exp\left(-x-\frac{\alpha}{x}\right)\frac{1}{x\sqrt{x}}dx$

Let : $x=\frac{\alpha}{u}\Rightarrow dx=-\frac{\alpha}{u^2}du$ **; If :** $x\in(0,\infty)\Rightarrow u\in(\infty,0)$

We obtain $I'(\alpha)=-\int_0^\infty \exp\left(-u-\frac{\alpha}{u}\right)\frac{1}{\frac{a}{u}\sqrt{\frac{a}{u}}}\times\frac{a}{u^2}dx$

$$=-\frac{1}{\sqrt{\alpha}}\int_0^\infty \exp\left(-u-\frac{\alpha}{u}\right)\frac{1}{\sqrt{u}}dx$$

$$I'(\alpha)=-\frac{1}{\sqrt{\alpha}}I(\alpha)$$

So we have the separable differential equation

Implies That $I(\alpha)=A\exp(-2\sqrt{\alpha})$

If : $\alpha=0$ **That** $I(0)=\sqrt{\pi}=A$

If : $\alpha = 1$ **That** $I(1) = I = \sqrt{\pi}\exp(-2)$

We Get $I = \sqrt{\pi}\exp(-2)$

$$SO,\ \boxed{\int_0^{\infty} \exp\left(-x - \frac{1}{x}\right)\frac{1}{\sqrt{x}}dx = \sqrt{\pi}e^{-2}}$$

477. Calculate Integral $I = \int_0^1 \frac{Ti_n(x)}{x}dx \quad ,(n \geq 0)$

Solution; $I = \int_0^1 \frac{Ti_n(x)}{x}dx$

$$= \int_0^1 \frac{1}{x}\left(\sum_{k=0}^{\infty}\frac{(-1)^k x^{2k+1}}{(2k+1)^n}\right)dx$$

$$= \sum_{k=0}^{\infty}\left(\frac{(-1)^k}{(2k+1)^n}\int_0^1 x^{2k}dx\right)$$

$$= \sum_{k=0}^{\infty}\left(\frac{(-1)^k}{(2k+1)^{n+1}}\right)$$

$$= \mathbf{B_{n+1}}$$

$$SO,\ \boxed{\int_0^1 \frac{Ti_n(x)}{x}dx = \mathbf{B_{n+1}} \quad ,(n \geq 0)}$$

478. Calculate Integral $I = \int_0^1 \frac{x}{e^x + e^{1-x}}dx$

Solution; $I = \int_0^1 \frac{x}{e^x + e^{1-x}}dx$

$$= \int_0^1 \frac{1-x}{e^{1-x} + e^{1-(1-x)}}dx$$

$$= \int_0^1 \frac{1}{e^x + e^{1-x}}dx - \int_0^1 \frac{x}{e^x + e^{1-x}}dx$$

We obtain $I = \frac{1}{2}\int_0^1 \frac{e^x}{e^{2x} + e}dx$

$$= \frac{1}{2}\int_0^1 \frac{1}{e^{2x}+e} d(e^x)$$

$$= \frac{1}{2\sqrt{e}} \tan^{-1}\left(\frac{e^x}{\sqrt{e}}\right)\Bigg|_0^1$$

$$= \frac{1}{2\sqrt{e}}\left[\tan^{-1}\left(\frac{e}{\sqrt{e}}\right) - \tan^{-1}\left(\frac{1}{\sqrt{e}}\right)\right]$$

$$= \frac{1}{2\sqrt{e}} \tan^{-1}\left(\frac{e-1}{2\sqrt{e}}\right)$$

SO, $\boxed{\int_0^1 \frac{x}{e^x + e^{1-x}} dx = \frac{1}{2\sqrt{e}} \tan^{-1}\left(\frac{e-1}{2\sqrt{e}}\right)}$

479. Calculate Integral $I = \int_0^\infty Li_n(e^{-x})\, dx \quad ,(n \geq 0)$

Solution; $I = \int_0^\infty Li_n(e^{-x})\, dx \quad ,(n \geq 0)$

$$= \int_0^\infty \left(\sum_{k=1}^\infty \frac{e^{-kx}}{k^n}\right) dx$$

$$= \sum_{k=1}^\infty \left[\int_0^\infty \left(\frac{e^{-kx}}{k^n}\right) dx\right]$$

$$= \sum_{k=1}^\infty \left[\int_0^\infty \left(-\frac{1}{k^{n+1}} \cdot e^{-kx}\Big|_0^\infty\right) dx\right]$$

$$= \sum_{k=1}^\infty \left(\frac{1}{k^{n+1}}\right)$$

$$= \zeta(n+1) \ \vee \ \zeta(n+1,0)$$

SO, $\boxed{\int_0^\infty Li_n(e^{-x})\, dx = \zeta(n+1,0) \quad ,(n \geq 0)}$

480. Calculate Integral $I=\int_0^1 \log(x)\log(1-\alpha x)dx \ , \left(|\alpha|<1\right)$

Solution; $I=\int_0^1 \log(x)\log(1-\alpha x)dx \ , \left(|\alpha|<1\right)$

$$=\int_0^1 \log(x)\left(\sum_{k=1}^{\infty}\frac{\alpha^k x^k}{k}\right)dx$$

$$=\sum_{k=1}^{\infty}\left(\frac{\alpha^k}{k}\int_0^1 x^k\log(x)dx\right)$$

Let : $x=e^{-u}\Rightarrow dx=-e^{-u}du$ **; If :** $x\in(0,1)\Rightarrow u\in(\infty,0)$

Implies That $I=\sum_{k=1}^{\infty}\left(\frac{\alpha^k}{k}\int_0^{\infty} xe^{-(k+1)x}dx\right)$

$$=\sum_{k=1}^{\infty}\left(\frac{\alpha^k}{k}\times\frac{1}{(k+1)^2}\right)$$

$$=\sum_{k=1}^{\infty}\alpha^k\left(\frac{1}{k}-\frac{1}{k+1}-\frac{1}{(k+1)^2}\right)$$

$$=\sum_{k=1}^{\infty}\left(\frac{\alpha^k}{k}\right)-\sum_{k=1}^{\infty}\left(\frac{\alpha^k}{k+1}\right)-\sum_{k=1}^{\infty}\left(\frac{\alpha^k}{(k+1)^2}\right)$$

$$=\sum_{k=1}^{\infty}\left(\frac{\alpha^k}{k}\right)-\frac{1}{\alpha}\sum_{k=1}^{\infty}\left(\frac{\alpha^{k+1}}{k+1}\right)-\frac{1}{\alpha}\sum_{k=1}^{\infty}\left(\frac{\alpha^{k+1}}{(k+1)^2}\right)$$

$$=-\log(1-\alpha)-\frac{1}{\alpha}\left(-\log(1-\alpha)-\alpha\right)-\frac{1}{\alpha}\left(Li_2(\alpha)-\alpha\right)$$

$$=\frac{1-\alpha}{\alpha}\log(1-\alpha)-\frac{1}{\alpha}Li_2(\alpha)+2$$

$SO,$ $\boxed{\int_0^{\infty} Li_n(e^{-x})\,dx=\frac{1-\alpha}{\alpha}\log(1-\alpha)-\frac{1}{\alpha}Li_2(\alpha)+2 \quad ,\left(|\alpha|<1\right)}$

481. Calculate Integral $I=\int_0^1 x^{n-1}\log\left(\frac{1}{x}\right)dx \quad ,\left(n>0\right)$

Solution; $I=\int_0^1 x^{n-1}\log\left(\frac{1}{x}\right)dx$

$$=-\int_0^1 x^{n-1}\log(x)\,dx$$

We Have : $\mathbf{B}(n,m)=\int_0^1 x^{n-1}(1-x)^{m-1}dx$

We obtain $\mathbf{B}(n,1)=\int_0^1 x^{n-1}dx$

Then $-\frac{d}{dn}\left(\mathbf{B}(n,1)\right)=-\int_0^1 x^{n-1}\log(x)dx$

Implies That $I=-\frac{d}{dn}\left(\mathbf{B}(n,1)\right)$

$$=-\frac{d}{dn}\left(\frac{\Gamma(n)\Gamma(1)}{\Gamma(n+1)}\right)$$

$$=\frac{1}{n^2}$$

$SO,$ $\boxed{\int_0^1 x^{n-1}\log\left(\frac{1}{x}\right)dx=\frac{1}{n^2} \quad ,\left(n>0\right)}$

482. Calculate Integral $I=\int_1^\infty \frac{\{x\}}{x^2}dx$

Solution; $I=\int_1^\infty \frac{\{x\}}{x^2}dx$

$$=\int_1^\infty \frac{x-\lfloor x\rfloor}{x^2}dx$$

$$=\sum_{n=1}^{\infty}\left(\int_n^{n+1}\frac{x-n}{x^2}dx\right)$$

$$= \lim_{k\to\infty} \sum_{n=1}^{k} \left[\left(\log(x) + \frac{n}{x} \right) \Big|_{n}^{n+1} \right]$$

$$= \lim_{k\to\infty} \sum_{n=1}^{k} \left(\log\left(\frac{n+1}{n} \right) - \frac{1}{n+1} \right)$$

$$= \lim_{k\to\infty} \left[\log(k+1) - \sum_{n=1}^{k} \frac{1}{n+1} \right]$$

$$= 1 - \lim_{k\to\infty} \left[\left(\mathrm{H}_{k+1} - \log(1+k) \right) \right]$$

$$= 1 - \gamma$$

SO, $\boxed{\int_1^{\infty} \frac{\{x\}}{x^2} dx = -\gamma + 1}$

483. Calculate Integral $I = \int_1^{\infty} \frac{\{x\}}{x^n} dx \quad , \left(n \in \mathbb{N} - \{1,2\} \right)$

Solution; $I = \int_1^{\infty} \frac{\{x\}}{x^n} dx$

$$= \int_1^{\infty} \frac{x - \lfloor x \rfloor}{x^n} dx$$

$$= \int_1^{\infty} x^{-n+1} dx - \sum_{k=1}^{\infty} \left(\int_k^{k+1} k x^{-n} \, dx \right)$$

$$= -\frac{x^{-n+2}}{n-2} \Big|_1^{\infty} + \sum_{k=1}^{\infty} k \left(\frac{x^{-n+1}}{n-1} \Big|_k^{k+1} \right)$$

$$= \frac{1}{n-2} + \frac{1}{n-1} \sum_{k=1}^{\infty} k \left((k+1)^{-n+1} - k^{-n+1} \right)$$

$$= \frac{1}{n-2} + \frac{1}{n-1} \left(-\zeta(n-1) \right)$$

$$= \frac{1}{n-2} - \frac{\zeta(n-1)}{n-1}$$

SO, $\boxed{\int_1^{\infty} \frac{\{x\}}{x^n} dx = \frac{1}{n-2} - \frac{\zeta(n-1)}{n-1} \quad , \left(n \in \mathbb{N} - \{1,2\} \right)}$

484. Calculate Integral $I=\int_0^1 \tan^{-1}(x^2-x+1)dx$

Solution; $I=\int_0^1 \tan^{-1}(x^2-x+1)dx$

Let : $u=\tan^{-1}(x^2-x+1)\Rightarrow du=\dfrac{2x-1}{x^2-2x+2}dx$ **And** $dv=dx\Rightarrow v=x$

Implies That $I=x\tan^{-1}(x^2-x+1)\Big|_0^1-\int_0^1\dfrac{(2x-1)x}{(x^2-x+1)^2+1}dx$

$$=\frac{\pi}{4}-\int_0^1\left(\frac{x-1}{(x-1)^2+1}+\frac{1}{(x-1)^2+1}-\frac{x}{x^2+1}\right)dx$$

$$=\frac{\pi}{4}-\left(\frac{1}{2}\log\left|\frac{x^2-2x+2}{x^2+1}\right|+\tan^{-1}(x-1)\right)\Big|_0^1$$

$$=\log(2)$$

SO, $\boxed{\int_0^1 \tan^{-1}(x^2-x+1)dx=\log(2)}$

485. Calculate Integral $I=\int_0^\infty\dfrac{2x-1}{x^4-2x^3+3x^2-2x+2}dx$

Solution; $I=\int_0^\infty\dfrac{2x-1}{x^4-2x^3+3x^2-2x+2}dx$

$$=\int_0^\infty\frac{2x-1}{\underbrace{x^4-2x^3+3x^2-2x+1}_{(x^2-x+1)^2}+1}dx$$

$$=\int_0^\infty\frac{2x-1}{(x^2-x+1)^2+1}dx$$

$$=\tan^{-1}(x^2-x+1)\Big|_0^\infty$$

$$=\frac{\pi}{2}-\frac{\pi}{4}=\frac{\pi}{2}$$

SO, $\boxed{\int_0^\infty\dfrac{2x-1}{x^4-2x^3+3x^2-2x+2}dx=\dfrac{\pi}{2}}$

486. Calculate Integral $I=\int_0^{\infty}\frac{\log^2(x)}{1+x^2}dx$

Solution; $I=\int_0^{\infty}\frac{\log^2(x)}{1+x^2}dx$

$$=\int_0^{1}\frac{\log^2(x)}{1+x^2}dx+\int_1^{\infty}\frac{\log^2(x)}{1+x^2}dx$$

For : $\int_1^{\infty}\frac{\log^2(x)}{1+x^2}dx$ **Let :** $x=u^{-1}\Rightarrow dx=-u^{-2}du$ **; If :** $x\in(1,\infty)\Rightarrow u\in(1,0)$

We Get : $\int_1^{\infty}\frac{\log^2(x)}{1+x^2}dx=-\int_1^{0}\frac{\log^2(u^{-1})}{1+u^{-2}}\cdot u^{-2}du$

$$=\int_0^{1}\frac{\log^2(u)}{1+u^2}du$$

Implies That $I=2\int_0^{1}\frac{\log^2(x)}{1+x^2}dx$

$$=2\sum_{n=0}^{\infty}\left((-1)^n\int_0^1 x^{2n}\log^2(x)dx\right)$$

Let : $x=e^{-z}\Rightarrow dx=-e^{-z}dz$ **; If :** $x\in(0,1)\Rightarrow z\in(\infty,0)$

We obtain $I=2\sum_{n=0}^{\infty}\left((-1)^n\int_0^{\infty}z^2e^{-(2n+1)z}dz\right)$

$$=2\sum_{n=0}^{\infty}\left(\frac{(-1)^n}{(2n+1)^3}\int_0^{\infty}z^2e^{-z}dz\right)$$

$$=4\sum_{n=0}^{\infty}\left(\frac{(-1)^n}{(2n+1)^3}\right)$$

$$=4\beta(3)=\frac{\pi^3}{8}$$

SO, $\boxed{\int_0^{\infty}\frac{\log^2(x)}{1+x^2}dx=\frac{\pi^3}{8}}$

487. Calculate Integral $I=\int_0^{\infty}\frac{\sin^2\left(\tan(x)\right)}{x^2}dx$

Solution; $I=\int_0^{\infty}\frac{\sin^2\left(\tan(x)\right)}{x^2}dx$

$$=\int_0^{\infty}\frac{\sin^2(x)}{x^2}\times\frac{\sin^2\left(\tan(x)\right)}{\sin^2(x)}dx$$

$$=\int_0^{\frac{\pi}{2}}\frac{\sin^2\left(\tan(x)\right)}{\sin^2(x)}dx$$

$$=\int_0^{\frac{\pi}{2}}\frac{\sin^2\left(\tan(x)\right)}{\tan^2(x)}d\left(\tan(x)\right)$$

$$=\int_0^{\infty}\frac{\sin^2(x)}{x^2}d(x)$$

$$=\frac{\pi}{2}$$

SO, $\boxed{\int_0^{\infty}\frac{\sin^2\left(\tan(x)\right)}{x^2}dx=\frac{\pi}{2}}$

488. Calculate Integral $I=\int_0^1\frac{\log(1+x^2)}{1+x^2}dx$

Solution; $I=\int_0^1\frac{\log(1+x^2)}{1+x^2}dx$

Let: $x=\tan(u)\Rightarrow du=\frac{1}{1+x^2}dx$ **; If :** $x\in(0,1)\Rightarrow u\in\left(0,\frac{\pi}{4}\right)$

Implies That $I=\int_0^{\frac{\pi}{4}}\log\left(1+\tan^2(u)\right)du$

$$=\int_0^{\frac{\pi}{4}}\log\left(\sec^2(x)\right)du$$

$$=-2\int_0^{\frac{\pi}{4}}\log\left(\cos(x)\right)du$$

$$=-2\left(-\frac{\pi}{4}\log(2)+\frac{1}{2}G\right)$$

$$=\frac{\pi}{2}\log(2)-G$$

SO, $\boxed{\int_0^1\frac{\log(1+x^2)}{1+x^2}dx=\frac{\pi}{2}\log(2)-G}$

489. Calculate Integral $I=\int_0^1\left(\frac{\tan^{-1}(x)}{x}\right)^2dx$

Solution; $I=\int_0^1\left(\frac{\tan^{-1}(x)}{x}\right)^2dx$ **(Use Partial integral)**

We Get $I=-\frac{\left(\tan^{-1}(x)\right)^2}{x}\Big|_0^1+2\int_0^1\frac{\tan^{-1}(x)}{x(1+x^2)}dx$

$$=-\frac{\pi^2}{16}+2\int_0^1\frac{\tan^{-1}(x)}{x}dx-2\int_0^1\frac{x\tan^{-1}(x)}{1+x^2}dx \quad (*)$$

For : $2\int_0^1\frac{\tan^{-1}(x)}{x}dx=2G$

For : $2\int_0^1\frac{x\tan^{-1}(x)}{1+x^2}dx=\tan^{-1}(x)\log(1+x^2)\Big|_0^1-\int_0^1\frac{\log(1+x^2)}{1+x^2}dx$ **(Use Partial integral)**

$$=\left(\frac{\pi}{4}\log(2)\right)-\left(\frac{\pi}{2}\log(2)-G\right)$$

$$=-\frac{\pi}{4}\log(2)+G$$

Take (*) Then $I=-\frac{\pi^2}{16}+2G+\frac{\pi}{4}\log(2)-G$

SO, $\boxed{\int_0^1\left(\frac{\tan^{-1}(x)}{x}\right)^2dx=\frac{\pi}{4}\log(2)-\frac{\pi^2}{16}+G}$

490. Calculate Integral $I=\int_0^1 \log\left(\frac{1+x^2}{x}\right)\frac{dx}{1+x^2}$

Solution; $I=\int_0^1 \log\left(\frac{1+x^2}{x}\right)\frac{dx}{1+x^2}$

$$=\int_0^1 \Big(\log(1+x^2)-\log(x)\Big)\frac{dx}{1+x^2}$$
$$=\int_0^1 \frac{\log(1+x^2)}{1+x^2}dx-\int_0^1 \frac{\log(x)}{1+x^2}dx$$
$$=\frac{\pi}{2}\log(2)-G-(-G)$$
$$=\frac{\pi}{2}\log(2)$$

$SO,$ $\boxed{\int_0^1 \log\left(\frac{1+x^2}{x}\right)\frac{dx}{1+x^2}=\frac{\pi}{2}\log(2)}$

491. Calculate Integral $I=\int_0^1 \log\left(\frac{1-x^2}{x}\right)\frac{dx}{1+x^2}$

Solution; $I=\int_0^1 \log\left(\frac{1-x^2}{x}\right)\frac{dx}{1+x^2}$

$$I=\int_0^1 \Big(\log(1-x^2)-\log(x)\Big)\frac{dx}{1+x^2}$$
$$=\int_0^1 \frac{\log(1-x^2)}{1+x^2}dx-\int_0^1 \frac{\log(x)}{1+x^2}dx$$
$$=\frac{\pi}{4}\log(2)-G-(-G)$$
$$=\frac{\pi}{4}\log(2)$$

$SO,$ $\boxed{\int_0^1 \log\left(\frac{1-x^2}{x}\right)\frac{dx}{1+x^2}=\frac{\pi}{4}\log(2)}$

492. Calculate Integral $I=\int_0^{\frac{\pi}{2}}\log\left(1+\tan(x)\right)dx$

Solution; $I=\int_0^{\frac{\pi}{2}}\log\left(1+\tan(x)\right)dx$

$$=\int_0^{\frac{\pi}{2}}\log\left(1+\cot(x)\right)dx$$

$$=\underbrace{\int_0^{\frac{\pi}{2}}\log\left(\sin(x)+\cos(x)\right)dx}_{J}-\underbrace{\int_0^{\frac{\pi}{2}}\log\left(\sin(x)\right)dx}_{K}$$

For : $J=\int_0^{\frac{\pi}{2}}\log\left(\sin(x)+\cos(x)\right)dx$

$$=\int_0^{\frac{\pi}{2}}\log\left(\cos\left(\frac{\pi}{4}-x\right)\right)dx+\int_0^{\frac{\pi}{2}}\log(\sqrt{2})dx$$

$$=2\int_0^{\frac{\pi}{4}}\log\left(\cos(x)\right)dx+\int_0^{\frac{\pi}{2}}\log(\sqrt{2})dx$$

$$=2\left(-\frac{\pi}{4}\log(2)+\frac{1}{2}G\right)+\frac{\pi}{4}\log(2)$$

$$=-\frac{\pi}{4}\log(2)+G$$

For : $J=\int_0^{\frac{\pi}{2}}\log\left(\sin(x)\right)dx$

$$=\frac{\pi}{2}\log(2)$$

We obtain $I=-\frac{\pi}{4}\log(2)+G+\frac{\pi}{2}\log(2)$

$$=\frac{\pi}{4}\log(2)+G$$

$SO,$ $\boxed{\int_0^{\frac{\pi}{2}}\log\left(1+\tan(x)\right)dx=\frac{\pi}{4}\log(2)+G}$

493. Calculate Integral $I=\int_0^{\infty}\frac{\tan^{-1}(x)}{x^2-x+1}dx$

Solution; $I=\int_0^{\infty}\frac{\tan^{-1}(x)}{x^2-x+1}dx$ **(!)**

Let : $x=u^{-1}\Rightarrow dx=-u^{-2}du$ **; If :** $x\in(0,\infty)\Rightarrow u\in(\infty,0)$

We obtain $I=\int_0^{\infty}\frac{\tan^{-1}(u^{-1})}{u^{-2}-u^{-1}+1}\times\frac{1}{u^2}du$

$$=\int_0^{\infty}\frac{\tan^{-1}(x^{-1})}{x^2-x+1}dx \quad \textbf{(!!)}$$

Take (!)+(!!) Then $2I=\int_0^{\infty}\frac{\tan^{-1}(x)+\tan^{-1}(x^{-1})}{x^2-x+1}dx$

Implies That $I=\frac{\pi}{4}\int_0^{\infty}\frac{1}{x^2-x+1}dx$

$$=\frac{\pi}{4}\left(\frac{4\pi}{3\sqrt{3}}\right)$$

$$=\frac{\pi^2}{3\sqrt{3}}$$

SO, $\boxed{\int_0^{\infty}\frac{\tan^{-1}(x)}{x^2-x+1}dx=\frac{\pi^2}{3\sqrt{3}}}$

494. Calculate Integral $I=\int_0^1\frac{1-x}{(1+x)(1+x^2)}\tan^{-1}(x)dx$

Solution; $I=\int_0^1\frac{1-x}{(1+x)(1+x^2)}\tan^{-1}(x)dx$

$$=\underbrace{\int_0^1\frac{\tan^{-1}(x)}{1+x}dx}_{J}-\underbrace{\int_0^1\frac{x\tan^{-1}(x)}{1+x^2}dx}_{K} \quad \textbf{(*)}$$

For : $J=\int_0^1 \frac{\tan^{-1}(x)}{1+x}dx$

Let : $x=\tan(y)\Rightarrow dx=\sec^2(y)dy$ **; If :** $x\in(0,1)\Rightarrow y\in\left(0,\frac{\pi}{4}\right)$

Implies That : $J=\int_0^{\frac{\pi}{4}} \frac{y}{1+\tan(y)}\sec^2(y)\,dy$ **(Use Partial integral)**

$$=y\log\left(1+\tan(y)\right)\Big|_0^{\frac{\pi}{4}}-\int_0^{\frac{\pi}{4}}\log\left(1+\tan(y)\right)dy$$

$$=\frac{\pi}{4}\log(2)-\frac{\pi}{8}\log(2)$$

$$=\frac{\pi}{8}\log(2)$$

For : $J=\int_0^1 \frac{x\tan^{-1}(x)}{1+x^2}dx=\mathrm{I}_{489}=\frac{1}{2}\left(-\frac{\pi}{4}\log(2)+G\right)$

Take (*) That $I=\frac{\pi}{8}\log(2)-\frac{1}{2}\left(-\frac{\pi}{4}\log(2)+G\right)$

$$=\frac{1}{2}\left(\frac{\pi}{2}\log(2)-G\right)$$

SO, $\boxed{\int_0^1 \frac{1-x}{(1+x)(1+x^2)}\tan^{-1}(x)dx=\frac{1}{2}\left(\frac{\pi}{2}\log(2)-G\right)}$

495. Calculate Integral $I=\int_0^1 \tan^{-1}(x^2-x+1)dx$

Solution; $I=\int_0^1 \tan^{-1}(x^2-x+1)dx$

$$=\int_0^1\left(\frac{\pi}{2}-\tan^{-1}\left(\frac{1}{x^2-x+1)}\right)\right)dx$$

$$=\frac{\pi}{2}-\int_0^1 \tan^{-1}\left(\frac{x-(x-1)}{1+x(x-1)}\right)dx$$

$$=\frac{\pi}{2}-\int_0^1 \tan^{-1}\left(\frac{(1-x)+x}{1+(1-x)x}\right)dx$$

$$=\frac{\pi}{2}-\int_0^1 \tan^{-1}(x)dx-\int_0^1 \tan^{-1}(1-x)dx$$

$$=\frac{\pi}{2}-2\int_0^1 \tan^{-1}(x)dx$$

$$=\frac{\pi}{2}-2\left(\frac{\pi}{4}-\frac{1}{2}\log(2)\right)$$

$$=\log(2)$$

SO, $\boxed{\int_0^1 \tan^{-1}(x^2-x+1)dx=\log(2)}$

496. Calculate Integral $I=\int_0^{\frac{\pi}{2}}\frac{\tan(x)}{4\log^2\left(\tan(x)\right)+\pi^2}dx$

Solution; $I=\int_0^{\frac{\pi}{2}}\frac{\tan(x)}{4\log^2\left(\tan(x)\right)+\pi^2}dx$ **(!)**

$$=\int_0^{\frac{\pi}{2}}\frac{\cot(x)}{4\log^2\left(\tan(x)\right)+\pi^2}dx \quad \textbf{(!!)}$$

Take (!) + (!!) That $2I=\int_0^{\frac{\pi}{2}}\frac{\tan(x)+\cot(x)}{4\log^2\left(\tan(x)\right)+\pi^2}dx$

Implies That $I=\frac{1}{8}\int_0^{\frac{\pi}{2}}\frac{1}{\log^2\left(\tan(x)\right)+\left(\pi/2\right)^2}d\left(\log\left(\tan(x)\right)\right)$

$$=\frac{1}{8}\times\frac{2}{\pi}\tan^{-1}\left[\log\left(\tan(x)\right)\right]_0^{\frac{\pi}{2}}$$

$$=\frac{1}{8}\times\frac{2}{\pi}\times\pi=\frac{1}{4}$$

SO, $\boxed{\int_0^{\frac{\pi}{2}}\frac{\tan(x)}{4\log^2\left(\tan(x)\right)+\pi^2}dx=\frac{1}{4}}$

497. Calculate Integral $I=\int_{\frac{\sqrt{e}}{2}}^{\infty} e^{-W^2(x)}dx$

Solution; $I=\int_{\frac{\sqrt{e}}{2}}^{\infty} e^{-W^2(x)}dx$

Let : $x=ue^u \Rightarrow dx=(u+1)e^u du$ **; If :** $x\in\left(\frac{\sqrt{e}}{2},\infty\right)\Rightarrow u\in\left(\frac{1}{2},\infty\right)$

Implies That $I=\int_{\frac{1}{2}}^{\infty}(u+1)e^{u-u^2}dx$

$$=e^{\frac{1}{4}}\int_{\frac{1}{2}}^{\infty}(u+1)e^{-\frac{1}{4}+u-u^2}dx$$

$$=e^{\frac{1}{4}}\int_{\frac{1}{2}}^{\infty}(u+1)e^{-\left(u-\frac{1}{2}\right)^2}dx$$

Let : $y=u-\frac{1}{2}\Rightarrow dy=du$ **; If :** $u\in\left(\frac{1}{2},\infty\right)\Rightarrow y\in(0,\infty)$

We obtain $I=\sqrt[4]{e}\int_{\frac{1}{2}}^{\infty}\left(y+\frac{3}{2}\right)e^{-y^2}dx$

$$=\sqrt[4]{e}\int_{0}^{\infty}\left(y+\frac{3}{2}\right)e^{-y^2}dx$$

$$=\sqrt[4]{e}\left(\frac{1}{2}+\frac{3}{2}\cdot\frac{\sqrt{\pi}}{2}\right)$$

$$=\frac{\sqrt[4]{e}}{4}\left(2+3\sqrt{\pi}\right)$$

SO, $\boxed{\int_{\frac{\sqrt{e}}{2}}^{\infty} e^{-W^2(x)}dx=\frac{\sqrt[4]{e}}{4}\left(2+3\sqrt{\pi}\right)}$

498. Calculate Integral $I=\int_0^1 \frac{1}{\sqrt{1-x^2}}\log\left(\frac{\sqrt{1+x}+1}{\sqrt{1+x}-1}\right)dx$

Solution; $I=\int_0^1 \frac{1}{\sqrt{1-x^2}}\log\left(\frac{\sqrt{1+x}+1}{\sqrt{1+x}-1}\right)dx$

Let : $x=\cos(2y)\Rightarrow dx=-2\sin(2x)dx$ **; If** : $x\in(0,1)\Rightarrow y\in\left(\frac{\pi}{4},0\right)$

Implies That $I=2\int_{\frac{\pi}{4}}^0 \frac{\sin(2y)}{\sqrt{1-\cos^2(2y)}}\log\left(\frac{\sqrt{1+\cos(2y)}+1}{\sqrt{1+\cos(2y)}-1}\right)dy$

$$=2\int_0^{\frac{\pi}{4}}\log\left(\frac{\sqrt{2}\cos(y)+1}{\sqrt{2}\cos(y)-1}\right)dy$$

$$=2\int_0^{\frac{\pi}{4}}\log\left(\frac{\cos(y)+\cos\left(\pi/4\right)}{\cos(y)-\cos\left(\pi/4\right)}\right)dy$$

$$=2\int_0^{\frac{\pi}{4}}\log\left[\cot\left(\frac{\pi}{8}+\frac{y}{2}\right)\cot\left(\frac{\pi}{8}-\frac{y}{2}\right)\right]dy$$

Let : $z=\frac{y}{2}\Rightarrow dy=2dz$ **; If** : $y\in\left(0,\frac{\pi}{4}\right)\Rightarrow z\in\left(0,\frac{\pi}{8}\right)$

We Get $I=4\left(\underbrace{\int_0^{\frac{\pi}{8}}\log\cot\left(\frac{\pi}{8}+z\right)dz}_{J}+\underbrace{\int_0^{\frac{\pi}{8}}\log\cot\left(\frac{\pi}{8}-z\right)dz}_{K}\right)$

For : $J=\int_0^{\frac{\pi}{8}}\log\cot\left(\frac{\pi}{8}+z\right)dz$

Let : $u=\frac{\pi}{8}+z\Rightarrow du=dz$ **; If** : $z\in\left(0,\frac{\pi}{8}\right)\Rightarrow u\in\left(\frac{\pi}{8},\frac{\pi}{4}\right)$

We obtain : $J=\int_{\frac{\pi}{8}}^{\frac{\pi}{4}}\log\left(\cot(u)\right)du$ **(*)**

For : $K=\int_0^{\frac{\pi}{8}}\log\cot\left(\frac{\pi}{8}-z\right)dz$

Let : $u=\frac{\pi}{8}-z\Rightarrow du=-dz$; **If** : $z\in\left(0,\frac{\pi}{8}\right)\Rightarrow u\in\left(\frac{\pi}{8},0\right)$

We obtain : $K=-\int_{\frac{\pi}{8}}^{0}\log\left(\cot(u)\right)du$

$$=\int_0^{\frac{\pi}{8}}\log\left(\cot(u)\right)du \quad (**)$$

Take (*) + (**) Then $I=4\left(\int_{\frac{\pi}{8}}^{\frac{\pi}{4}}\log\left(\cot(u)\right)du+\int_0^{\frac{\pi}{8}}\log\left(\cot(u)\right)du\right)$

$$=4\int_0^{\frac{\pi}{4}}\log\left(\cot(u)\right)du$$

$$=4G$$

SO, $\boxed{\int_0^1\frac{1}{\sqrt{1-x^2}}\log\left(\frac{\sqrt{1+x}+1}{\sqrt{1+x}-1}\right)dx=4G}$

499. Calculate Integral $I=\int_0^1\frac{\tan^{-1}(x)\log(x)}{x}dx$

Solution; $I=\int_0^1\frac{\tan^{-1}(x)\log(x)}{x}dx$

$$=\int_0^1\frac{\log(x)}{x}d\left(\frac{1}{x^2+1}\right) \quad \textbf{(Use Partial integral)}$$

$$=\log^2(x)\tan^{-1}(x)\Big|_0^1-\frac{1}{2}\int_0^1\frac{\log^2(x)}{x^2+1}dx$$

$$=-\frac{1}{2}\sum_{n=0}^{\infty}\left((-1)^n\int_0^1x^{2n}\log^2(x)\,dx\right)$$

Let : $x=e^{-y}\Rightarrow dx=-e^{-y}dy$; **If** : $x\in(0,1)\Rightarrow y\in(\infty,0)$

We obtain $I=-\frac{1}{2}\sum_{n=0}^{\infty}\left((-1)^n\int_0^{\infty}y^2e^{-(2n+1)y}\,dy\right)$

$$=-\frac{1}{2}\sum_{n=0}^{\infty}\frac{(-1)^n}{(2n+1)^3}$$

$$=-\frac{1}{2}\beta(3)$$

$$=-\frac{\pi^3}{32}$$

SO, $\boxed{\int_0^1 \frac{\tan^{-1}(x)\log(x)}{x}dx=-\frac{\pi^3}{32}}$

500. Calculate Integral $I=\int_0^{\frac{\pi}{2}}\frac{1}{(1+\sin(x))^2\sqrt{\log(\sec(x)+\tan(x))}}dx$

Solution; $I=\int_0^{\frac{\pi}{2}}\frac{1}{(1+\sin(x))^2\sqrt{\log(\sec(x)+\tan(x))}}dx$

Let : $u=\sec(x)+\tan(x)\Rightarrow dx=\cos(x)\frac{du}{u}$ **; If** : $x\in\left(0,\frac{\pi}{2}\right)\Rightarrow u\in(1,\infty)$

By $(1+\sin(x))^2=u^2\cos^2(x)$ **And** $2\cos(x)=\frac{u}{u^2+1}$

Implies That $I=\int_1^{\infty}\frac{1}{u^2\cos^2(x)\sqrt{\log(u)}}\cdot\frac{\cos(x)}{u}du$

$$=\int_1^{\infty}\frac{1}{u^3\cos(x)\sqrt{\log(u)}}du$$

$$=\int_1^{\infty}\frac{1}{u^3\left(\frac{1}{2}\times\frac{u}{u^2+1}\right)\sqrt{\log(u)}}du$$

$$=\frac{1}{2}\int_1^{\infty}\frac{u^2+1}{u^4\sqrt{\log(u)}}du$$

Then $I(\alpha)=\frac{1}{2}\int_1^{\infty}\frac{\alpha u^2+1}{u^4\sqrt{\log(u)}}du$ **Note:** $I(1)=I$

That $I'(\alpha)=\frac{1}{2}\int_1^{\infty}\frac{1}{u^2\sqrt{\log(u)}}du$

Let : $u=e^{z^2}\Rightarrow du=2ze^{z^2}dz$; **If** : $u\in(1,\infty)\Rightarrow z\in(0,\infty)$

We Get $I'(\alpha)=\frac{1}{2}\int_0^{\infty}\frac{2ze^{z^2}}{ze^{2z^2}}dz$

$$=\int_0^{\infty}e^{-z^2}dz$$

$$=\frac{\sqrt{\pi}}{2}$$

They Give $I(\alpha)=\frac{\sqrt{\pi}}{2}\alpha+C$ **(*)**

If : $\alpha=1\Rightarrow I(1)=I=\frac{\sqrt{\pi}}{2}+C$ **(#)**

If : $\alpha=0\Rightarrow I(0)=\frac{1}{2}\int_1^{\infty}\frac{1}{u^4\sqrt{\log(u)}}du$

$$=\int_0^{\infty}e^{-3u^2}du=\frac{\sqrt{\pi}}{2\sqrt{3}}\quad \textbf{(**)}$$

Take (*) Then $I(0)=0+C=C$

Take (**) That $I(0)=C=\frac{\sqrt{\pi}}{2\sqrt{3}}$

Take (#) We Have $I=\frac{\sqrt{\pi}}{2}+\frac{\sqrt{\pi}}{2\sqrt{3}}=\frac{\sqrt{\pi}}{2\sqrt{3}}\left(\sqrt{3}+1\right)$

SO, $\boxed{\int_0^{\frac{\pi}{2}}\frac{1}{\left(1+\sin(x)\right)^2\sqrt{\log\left(\sec(x)+\tan(x)\right)}}dx=\frac{\sqrt{\pi}}{2\sqrt{3}}\left(\sqrt{3}+1\right)}$

លំហាត់ស្រាវជ្រាវបន្ថែម

01. $\mathrm{I}=\int_0^{\frac{\pi}{2}}\log\left(\sec^2(x)+\tan^4(x)\right)dx$

02. $\mathrm{I}=\int_0^{\frac{\pi}{2}}\log\left(2+\tan^2(x)\right)dx$

03. $\mathrm{I}=\int_0^{\frac{\pi}{2}}\arctan\left(\frac{\sin(x)}{\sqrt{2}}\right)dx$

04. $\mathrm{I}=\int_0^{\frac{\pi}{2}}\log\left(\frac{2+\sin(x)}{2-\sin(x)}\right)dx$

05. $\mathrm{I}=\int_0^{\frac{\pi}{2}}\frac{\arctan\left(\sin(x)\right)}{\sin(x)}dx$

06. $\mathrm{I}=\int_0^{\frac{\pi}{2}}x\sqrt{\tan(x)}\,dx$

07. $\mathrm{I}=\int_0^{\frac{\pi}{2}}x\cot(x)\,dx$

08. $\mathrm{I}=\int_0^{\infty}\frac{\log\left(x^2+x+1\right)}{x^2+1}dx$

09. $\mathrm{I}=\int_0^{\infty}\frac{\log\left(x+\sqrt{x^2+1}\right)}{\left(x+\sqrt{x^2+1}\right)^2}dx$

10. $\mathrm{I}=\int_0^{\infty}\frac{x-\sin(x)}{x^3(x^2+4)}dx$

11. $\mathrm{I}=\int_0^{\infty}\frac{x^{4/5}-x^{2/3}}{(x^2+1)\log(x)}dx$

12. $\mathrm{I}=\int_0^{\infty}e^{-\left(\frac{x^2}{y^2}+y^2\right)}dx$

13. $\mathrm{I}=\int_0^{\infty}\frac{1-\cos(xy)}{x}dx$

14. $\mathrm{I}=\int_0^{\infty}\frac{1}{(x^2+q)^{p+1}}dx$

15. $\mathrm{I}=\int_0^{\infty}\exp\left(-x^2\right)dx$

16. $\mathrm{I}=\int_0^{\infty}\cos(x^2)dx$

17. $\mathrm{I}=\int_0^{\infty}\sin(x^2)\,dx$

18. $\mathrm{I}=\int_0^{\infty}\frac{\sin^2(x)}{x^2(x^2+1)}dx$

19. $\mathrm{I}=\int_{-\infty}^{+\infty}\frac{\log\left(x^2+1\right)}{x^4+x^2+1}dx$

20. $\mathrm{I}=\int_{-\infty}^{+\infty}\frac{\log\left(x^4+x^2+1\right)}{x^4+1}dx$

21. $\mathrm{I}=\int_{-\infty}^{+\infty}\left(x^2+\frac{1}{x^2}\right)e^{-\left(x^2+\frac{1}{x^2}\right)}dx$

22. $\mathrm{I}=\int_{-\infty}^{+\infty}\frac{x^2+2\cos(x)-2}{x^4}dx$

23. $\mathrm{I}=\int_{-\infty}^{+\infty}\left(\frac{1-\cos(x)}{x^2}\right)^2dx$

24. $\mathrm{I}=\int_{-\pi}^{\pi}\frac{\cos(x)}{1+e^{x}}dx$

25. $\mathrm{I}=\int_{0}^{1}\frac{\log\left(x^{3}+1\right)}{x^{2}+1}dx$

26. $\mathrm{I}=\int_{0}^{\frac{\pi}{2}}\frac{\log\left(\sin(2x)\right)}{\left(\alpha+\beta\sin^{2}(x)\right)\left(\beta+\alpha\cos^{2}(x)\right)}dx$

27. $\mathrm{I}=\int_{\phi}^{\phi^{2}}\frac{\log^{n}\left(\log(x)+\gamma_{n}\right)}{\log^{n}\left(\log(\phi^{3}-x)+\gamma_{n}\right)+\log^{n}\left(\log(x)+\gamma_{n}\right)}dx \quad ,\phi=\frac{1+\sqrt{5}}{2}\;;\gamma_{n}=\mathrm{H}_{n}-\log(n)$

28. $\mathrm{I}=\int_{0}^{\frac{\pi}{4}}\frac{x\left[\log\left(\cos(x)+\sin(x)\right)-\log\left(\cos(x)-\sin(x)\right)\right]}{\cos(x)\left(\cos(x)+\sin(x)\right)}dx=\boxed{\frac{G\log(2)}{2}+\frac{\pi^{2}}{192}}$

29. $\mathrm{I}=\int_{0}^{\frac{\pi}{4}}\log\left(\cos(x)-\sin(x)\right)\cos(2x)\,dx=\boxed{-\frac{1}{4}}$

30. $\mathrm{I}=\int_{0}^{\frac{\pi}{4}}\tan(x)\log\left(\tan(x)\right)dx=\boxed{-\frac{\pi^{2}}{48}}$

31. $\mathrm{I}=\int_{0}^{\frac{\pi}{4}}\tan^{-1}\left(\alpha\sqrt{1-\tan^{2}(x)}\right)dx=\boxed{\frac{\pi}{\sqrt{2}(2\alpha^{2}+1)}\left(1-\frac{\pi}{\sqrt{2}\sqrt{\alpha^{2}+1}}\right)}$

32. $\mathrm{I}=\int_{0}^{\frac{\pi}{4}}x\log\left(1+\cot(x)\right)dx=\boxed{\frac{8\pi C-7\zeta(3)+\pi^{2}\log(2)}{64}}$

33. $\mathrm{I}=\int_{0}^{\frac{\pi}{2}}x\log\left(1+\cot(x)\right)dx=\boxed{\frac{4\pi C-7\zeta(3)+\pi^{2}\log(2)}{16}}$

34. $\mathrm{I}=\int_{0}^{\frac{\pi}{2}}\arctan\left(\alpha\sin(x)\right)dx=\boxed{Li_{2}\left(\frac{\sqrt{1+\alpha^{2}}-1}{\alpha}\right)-Li_{2}\left(\frac{1-\sqrt{1+\alpha^{2}}}{\alpha}\right)}$

35. $\mathrm{I}=\int_0^{\frac{\pi}{2}}\sin^m(x)\cos^n(x)\,dx=\boxed{\dfrac{\Gamma\left(\dfrac{m+1}{2}\right)\Gamma\left(\dfrac{m+1}{2}\right)}{2\Gamma\left(\dfrac{m+n+2}{2}\right)}}$

36. $\mathrm{I}=\int_0^{\frac{\pi}{2}}\log^2\big(\cos(x)\big)\,dx=\boxed{\dfrac{\pi}{2}\log^2(2)+\dfrac{\pi^3}{24}}$

37. $\mathrm{I}=\int_0^{\frac{\pi}{2}}x\cos(8x)\log\left(\dfrac{1+\tan(x)}{1-\tan(x)}\right)dx=\boxed{\dfrac{\pi}{12}}$

38. $\mathrm{I}=\int_0^{\frac{\pi}{2}}\dfrac{\log\big(\cos(x)\big)}{\tan(x)}\cdot\log\Big(\log_{\cos(x)}\sin(x)\Big)\,dx=\boxed{\dfrac{\pi^2}{24}}$

39. $\mathrm{I}=\int_0^{\frac{\pi}{2}}\dfrac{x}{8\cos^2(2x)+1}\,dx=\boxed{\dfrac{\pi^2}{24}}$

40. $\mathrm{I}=\int_0^{\frac{\pi}{2}}\dfrac{1}{\sqrt{1+\sin(x)}}\,dx=\boxed{\sqrt{2}\log(\sqrt{2}+1)}$

41. $\mathrm{I}=\int_0^{\pi}\log\big(1-\cos(x)\big)\cdot\log\big(1+\cos(x)\big)\,dx=\boxed{\pi\log^2(2)-\dfrac{\pi^2}{6}}$

42. $\mathrm{I}=\int_0^{\pi}\sin^2\big(3x+\cos^4(5x)\big)\,dx=\boxed{\dfrac{\pi}{2}}$

43. $\mathrm{I}=\int_0^{\pi}\dfrac{x\sin(x)}{\sqrt{1+\sin^2(x)}}\,dx=\boxed{\dfrac{\pi^2}{8}}$

44. $\mathrm{I}=\int_0^{\infty}\dfrac{x}{(x^2+1)(x^4+x^2+1)}\,dx=\boxed{\dfrac{\pi}{6\sqrt{3}}}$

45. $\mathrm{I}=\int_0^{\infty}\frac{\log(1+x^2)}{\sqrt{1+x^4}}dx=\boxed{\frac{\left[\left(-3/4\right)!\right]^2}{16}\left(\frac{2\log(2)}{\sqrt{\pi}}+\sqrt{\pi}\right)}$

46. $\mathrm{I}=\int_0^{\infty}\frac{x}{x^2+4}\tan^{-1}\left(\frac{2x}{x^2+1}\right)dx=\boxed{\frac{\pi}{2}\log\left(\frac{\sqrt{2}+3}{\sqrt{2}+1}\right)}$

47. $\mathrm{I}=\int_0^{\infty}\frac{x}{x^2+\alpha^2}\tan^{-1}\left(\frac{2\beta x}{\gamma^2+\beta^2}\right)dx=\boxed{\frac{\pi}{2}\log\left(\frac{\alpha+\beta+\sqrt{\gamma^2+\beta^2}}{\alpha-\beta+\sqrt{\gamma^2+\beta^2}}\right)}$

48. $\mathrm{I}=\int_0^{\infty}\frac{x}{x^2+4}\tan^{-1}\left(\frac{2x\tan(\theta)}{x^2+1}\right)dx=\boxed{\frac{\pi}{2}\left[\log\left(\sec(\theta)+\tan(\theta)-\log\left(\frac{3+\tan\left(\theta/2\right)}{3-\tan\left(\theta/2\right)}\right)\right)\right]}$

49. $\mathrm{I}=\int_0^{n}\frac{1}{(x^2+1)\sqrt{n^2+x^2}}dx=\boxed{\frac{\pi}{2\sqrt{n^2+1}}}\quad,(n>0)$

50. $\mathrm{I}=\int_1^{\infty}\frac{1}{x+x^2\tan^{-1}(x)+x^3+x^4\tan^{-1}(x)}dx=\boxed{\log\left(\frac{\pi+4}{2\pi}\right)}$

51. $\mathrm{I}=\int_0^{\infty}\frac{(x^4-x^3)}{(1+x^2)(1+x^3)(1+x^4)}dx=\boxed{0}$

52. $\mathrm{I}=\int_0^{\infty}\frac{\log^2(x)}{1+x^2}dx=\boxed{\frac{\pi^3}{8}}$

53. $\mathrm{I}=\int_0^{\infty}\frac{\log(x+1)\cdot\log x}{x^{2n-1}}dx=\boxed{\frac{1}{4}}$

54. $\mathrm{I}=\int_0^{\infty}\frac{\log(x^4+\omega^4)}{x^2+\lambda^2}dx=\boxed{\frac{\pi}{|\lambda|}\log\left(\omega^2+\lambda^2+|\lambda||\omega|\sqrt{2}\right)}$

55. $\mathrm{I}=\int_0^{\infty}\frac{\log(x^2+\omega^2)}{x^2+\lambda^2}dx=\boxed{\frac{\pi}{\lambda}\log\left(\omega+\lambda\right)}$

56. $\mathrm{I}=\int_0^{\infty}\frac{1}{x}\log\left(\frac{x^3-x^2-x+1}{x^3+x^2+x+1}\right)dx=\boxed{-\frac{3\pi^2}{4}}$

57. $\mathrm{I}=\int_0^{\infty}\frac{\log(x)}{x}\left(e^{-x}-\frac{1}{\sqrt[4]{1+8x}}\right)dx=\boxed{\frac{1}{8}\left(\gamma^2-32G-5\pi^2\right)}$

58. $\mathrm{I}=\int_0^{\infty}x^ne^{-\alpha x}\,dx=\boxed{\frac{n!}{\alpha^{n+1}}\ ,\left(n\in(-1,\infty)\ \ ;\ \mathrm{Re}(\alpha)>0\right)}$

59. $\mathrm{I}=\int_0^{\beta}x^ne^{-\alpha x}\,dx=\boxed{\frac{n!}{\alpha^{n+1}}\left[1-e^{-\alpha\beta}\sum_{\gamma=0}^{n}\frac{(\alpha\beta)^{\gamma}}{\gamma!}\right]}$

60. $\mathrm{I}=\int_0^{\infty}e^{-\alpha x^{\beta}}\,dx=\boxed{\frac{1}{\beta}\alpha^{-\frac{1}{\beta}}\Gamma\left(\frac{1}{\beta}\right)}$

61. $\mathrm{I}=\int_0^{\infty}x^ne^{-\alpha x^{\beta}}\,dx=\boxed{\frac{1}{\beta}\alpha^{-\frac{n+1}{\beta}}\Gamma\left(\frac{n+1}{\beta}\right)}$

62. $\mathrm{I}=\int_0^{\infty}e^{-\alpha x}\sin(\beta x)\,dx=\boxed{\frac{\beta}{\alpha^2+\beta^2}}\quad,\alpha>0$

63. $\mathrm{I}=\int_0^{\infty}e^{-\alpha x}\cos(\beta x)\,dx=\boxed{\frac{\alpha}{\alpha^2+\beta^2}}\quad,\alpha>0$

64. $\mathrm{I}=\int_0^{\infty} xe^{-\alpha x}\sin(\beta x)\,dx=\boxed{\dfrac{2\alpha\beta}{\left(\alpha^2+\beta^2\right)^2}}\quad,\alpha>0$

65. $\mathrm{I}=\int_0^{\infty} xe^{-\alpha x}\cos(\beta x)\,dx=\boxed{\dfrac{\alpha^2-\beta^2}{\left(\alpha^2+\beta^2\right)^2}}\quad,\alpha>0$

66. $\mathrm{I}=\int_0^{\infty} \dfrac{e^{-\alpha x}\cos(\beta x)}{x}\,dx=\boxed{\arctan\left(\dfrac{\beta}{\alpha}\right)}$

67. $\mathrm{I}=\int_0^{\infty} \dfrac{e^{-\alpha x}-e^{-\beta x}}{x}\,dx=\boxed{\log\left|\dfrac{\beta}{\alpha}\right|}$

68. $\mathrm{I}=\int_0^{\infty} \dfrac{e^{-\alpha x}-e^{-\beta x}}{x}\sin(\gamma x)\,dx=\boxed{\tan^{-1}\left(\dfrac{\beta}{\gamma}\right)-\tan^{-1}\left(\dfrac{\alpha}{\gamma}\right)}$

69. $\mathrm{I}=\int_0^{\infty} \dfrac{e^{-\alpha x}-e^{-\beta x}}{x}\cos(\gamma x)\,dx=\boxed{\dfrac{1}{2}\log\left(\dfrac{\beta^2+\gamma^2}{\alpha^2+\gamma^2}\right)}$

70. $\mathrm{I}=\int_0^{\infty} \dfrac{\left(1-\cos(x)\right)}{x^2}e^{-\alpha x}\,dx=\boxed{\cot^{-1}(\alpha)-\dfrac{\alpha}{2}\log\left(\dfrac{1}{\alpha^2}+1\right)}$

71. $\mathrm{I}=\int_0^{\infty} \dfrac{\log(x+1)}{x(x^2+1)}\,dx=\boxed{\dfrac{5\pi^2}{48}}$

72. $\mathrm{I}=\int_0^{\infty} \dfrac{\log(x^2+1)}{x(x^2+1)}\,dx=\boxed{\dfrac{\pi^2}{12}}$

73. $\mathrm{I}=\int_0^{\infty} \dfrac{\log(x^2+1)}{x(x+1)}\,dx=\boxed{\dfrac{5\pi^2}{24}}$

74. $\mathrm{I}=\int_0^{\infty}\frac{\log(x^2+1)}{x^2(x^2+1)}dx=\boxed{\pi\left(1-\log(2)\right)}$

75. $\mathrm{I}=\int_{-\infty}^{+\infty}e^{\alpha x^4+\beta x^3+\gamma x^2+\theta x+\lambda}\,dx=\boxed{e^{\lambda}\sum_{n,m,p=0}^{\infty}\frac{\beta^{4n}}{(4n)!}\cdot\frac{\gamma^{4n}}{(4m)!}\cdot\frac{\theta^{4n}}{(4p)!}\cdot\frac{\Gamma\left(3n+m+n+\frac{1}{4}\right)}{\alpha^{3n+m+n+\frac{1}{4}}}}$

76. $\mathrm{I}=\int_0^1\frac{\cos\left(\log((x))\right)\cdot\log x}{x+1}dx=\boxed{1}$

77. $\mathrm{I}=\int_0^1\frac{\log(x)\cdot\log(x^2-x+1)}{x+1}dx=\boxed{\frac{2}{3}\zeta(3)}$

78. $\mathrm{I}=\int_0^1\frac{\log(x^2+1)}{x+1}dx=\boxed{\frac{3\log^2(2)}{4}-\frac{\pi^2}{48}}$

79. $\mathrm{I}=\int_0^1\frac{x+x^2+x^3+\cdots+x^{2n}-2nx}{(x+1)\log(x)}dx=\boxed{\log\left[\left(\frac{2}{\pi}\right)^n(2n)!!\right]}$

80. $\mathrm{I}=\int_0^1\frac{x+x^2+x^3+x^4+x^5-5x}{(x+1)^3\log(x)}dx=\boxed{\log\left(\frac{3\pi^3}{32}\right)-\frac{7\zeta(3)}{2\pi^2}}$

81. $\mathrm{I}=\int_{-1}^1\frac{\cos(x)}{e^{1/x}+1}dx=\boxed{\sin(1)}$

82. $\mathrm{I}=\int_0^1\frac{\log(x)}{x^2+1}dx=\boxed{-G}$

83. $\mathrm{I}=\int_0^1\frac{\log(x+1)}{x(x^2+1)}dx=\boxed{\frac{7\pi^2}{96}-\frac{\log^2(2)}{8}}$

84. $\mathrm{I}=\int_0^1 \frac{\log(1+x)\cdot\log(x^3+1)}{x}dx=\boxed{\frac{2\pi}{3}CI_2\left(\frac{2\pi}{3}\right)-CI_3\left(\frac{2\pi}{3}\right)-\frac{3}{2}\zeta(3)}$

85. $\mathrm{I}=\int_0^4 \frac{\log(x)}{\sqrt{4x-x^2}}dx=\boxed{0}$

86. $\mathrm{I}=\int_0^{\frac{1}{2}}\left(\frac{1+x^2}{1-x^2}\right)\log\left(\frac{1+x}{1-x}\right)dx=\boxed{\frac{1}{2}\left(\log^2(3)-\log(3^3)+\log(2^4)\right)}$

87. $\mathrm{I}=\int_0^{\frac{1}{2}} \frac{x\sin^{-1}(x)}{\sqrt{1-x^2}}dx=\boxed{\frac{6-\sqrt{3}\pi}{12}}$

88. $\mathrm{I}=\int_0^1 \frac{x^{p-1}}{x^q+1}dx=\boxed{\sum_{n=0}^{\infty}\frac{(-1)^n}{p+nq}}$

89. $\mathrm{I}=\int_0^1 x^n\log^m(x)\,dx=\boxed{\frac{(-1)^m\Gamma(m+1)}{(n+1)^{m+1}}}$

90. $\mathrm{I}=\int_0^1 \frac{x^{p-1}\log^m(x)}{(1+x^q)^n}dx=\boxed{(-1)^m\Gamma(m+1)\sum_{k=0}^{\infty}\frac{(-1)^k\Gamma(m+1)}{(p+kq)^{m+1}}\cdot\binom{n+k-1}{k}}$

91. $\mathrm{I}=\int_0^1 \log^n\left(\frac{1-x}{1+x}\right)dx=\boxed{2(-1)^n\Gamma(n+1)\eta(n)}$

92. $\mathrm{I}=\int_0^1 \frac{\log^n(x)}{(x+1)^5}dx=\boxed{\frac{(-1)^n\Gamma(n+1)}{24}\cdot\left[\eta(n-3)+6\eta(n-2)+11\eta(n-1)+6\eta(n)\right]}$

93. $\mathrm{I}=\int_0^1 \frac{\log^n(1-x^\alpha)}{x}dx=\boxed{\frac{(-1)^n}{\alpha}\Gamma(n+1)\zeta(n+1)}$

94. $\mathrm{I}=\int_0^1 \frac{\cos\left(\log((x))\cdot\log x\right.}{x+1}dx=\boxed{1}$

95. $\mathrm{I}=\int_0^1 \frac{\frac{\pi}{2}-\tan^{-1}\left(\sqrt{x^2+2}\right)}{(x^2+1)\sqrt{x^2+2}}dx=\boxed{\frac{\pi^2}{32}}$

96. $\mathrm{I}=\int_\alpha^\beta \frac{\log x}{\sqrt{(\alpha-x)(\beta-x)}}dx=\boxed{2\pi\log\left(\frac{\sqrt{\alpha}+\sqrt{\beta}}{2}\right)}$

97. $\mathrm{I}=\int_0^1\int_0^1\cdots\int_0^1 \frac{(x_1x_2\cdots x_n)^{m-1}}{1-kx_1x_2\cdots x_n}dx_1dx_2\cdots dx_n=\Phi(k,n,m)$ Where Φ Lerch Transcendent

98. $\mathrm{I}=\int_0^1\int_0^1\cdots\int_0^1 \frac{(x_1x_2\cdots x_n)^{m-1}\log(x_1x_2\cdots x_n)}{1-kx_1x_2\cdots x_n}dx_1dx_2\cdots dx_n=-n\Phi(k,n+1,m)$

99. $\mathrm{I}=\int_0^1\int_0^1\cdots\int_0^1 \frac{(x_1x_2\cdots x_n)^{m-1}\log^2(x_1x_2\cdots x_n)}{1-kx_1x_2\cdots x_n}dx_1dx_2\cdots dx_n=n(n+1)\Phi(k,n+2,m)$

100. $\mathrm{I}=\int_0^1 \frac{x^{\beta-1}\log^n(x)}{1+x^\alpha+x^{2\alpha}+\cdots+x^{\alpha m}}dx=\frac{1}{\left(\alpha(m+1)\right)^{n+1}}\left(\psi^n\left(\frac{\beta+\alpha}{(m+1)\alpha}\right)-\psi^n\left(\frac{\beta}{(m+1)\alpha}\right)\right)$

ឯកសារយោងនិងឯកសារស្ទើរអានបន្ថែម

1. ដកស្រង់ចេញពីការប្រលង Massachusetts Institute of Technology (MIT)
2. ដកស្រង់ចេញពីអ៊ីនធើណេត (YouTube Google Facebook) និងការនិពន្ធពីខ្ញុំផ្ទាល់
3. ដកស្រង់ចេញពីរសៀវភៅអាំងតេក្រាល វៀតណាម បារាំង និងឯកសារបរទេសជាច្រើនទៀត
4. Advanced Integration Techniques
5. Almost Impossible Integrals, Sums and Series.pdf
6. GURU 250+ SOLUTIONS.pdf
7. Improper Riemann Integrals, 2nd Editi... (Z-Library).pdf
8. Improper Riemann Integrals (Ioannis M... (Z-Library).pdf
9. Integral calculus book course.pdf
10. Ryabushko_P2_1991_352.pdf
11. A_S_Demidov_Equations_of_Mathematical_Physics_Generalized_Functions.pdf
12. Almost Impossible Integrals, Sums and Series.pdf
13. G_Plonka,_D_Potts,_G_Steidl,_M_Tasche.pdf
14. Integrals and Series [Vol 2 - Spl Functions] - PDF Room.pdf
15. Series Involving Euler's Eta Function.pdf
16. Special Techniques For Solving Integral Examples And Problems
17. Gamma function - Wikipedia
18. Digamma function - Wikipedia
19. Ramanujan tau function – Wikipe

